

Block

1

SAMPLING DISTRIBUTIONS

UNIT 1

Introduction to Sampling Distribution	7
--	----------

UNIT 2

Sampling Distribution(s) of Statistic(s)	29
---	-----------

UNIT 3

Standard Sampling Distributions-I	53
--	-----------

UNIT 4

Standard Sampling Distributions-II	69
---	-----------

Appendix	91
-----------------	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathasan
Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Rahul Roy
Math. and Stat. Unit
Indian Statistical Institute, New Delhi

Prof. Parvin Sinclair
Pro-Vice Chancellor
IGNOU, New Delhi

Dr. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Geeta Kaicker
Director, School of Sciences
IGNOU, New Delhi

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. R. M. Pandey
Department of Bio-Statistics
All India Institute of Medical Sciences
New Delhi

Dr. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Faculty members of School of Sciences, IGNOU

Statistics

Dr. Neha Garg
Dr. Nitin Gupta
Mr. Rajesh Kaliraman
Dr. Manish Trivedi

Mathematics

Dr. Deepika
Prof. Poornima Mital
Prof. Sujatha Varma
Dr. S. Venkataraman

Block Preparation Team

Prof. G. K. Shukla (**Editor**)
Decision Science Group
Indian Institute of Management
Lucknow (UP)

Mr. Prabhat Kumar Sangal (**Units 1 - 4**)
School of Sciences, IGNOU

Dr. Parmod Kumar (**Language Editor**)
School of Humanities, IGNOU

Dr. Manish Trivedi (**Units 3 & 4**)
School of Sciences, IGNOU

Course Coordinator: Mr. Prabhat Kumar Sangal

Programme Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P), School of Sciences, IGNOU
CRC prepared by Mr. Prabhat Kumar Sangal, School of Sciences, IGNOU

Acknowledgement: I gratefully acknowledge my colleagues Dr. Manish Trivedi and Mr. Rajesh Kaliraman, Statistics Discipline, School of Sciences for their great support.

July, 2013

© Indira Gandhi National Open University, 2013

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University.

Further information on the Indira Gandhi National Open University may be obtained from University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Printed at:

STATISTICAL INFERENCE

In MST-003, the basic rules of probabilities and various probability distributions such as Bernoulli, Binomial, Poisson, Uniform, Normal, Exponential, etc. have been discussed. Now, the concepts of probability theory and probability distributions are used in drawing inference about the population parameters.

Generally, population parameters are unknown and when the population is too large or the units of the population are destructive in nature or there is a limited resources and manpower available then it is not possible practically to examine each and every unit of the population to obtain the population parameters. In such situations, one can draw sample from the population under study and utilize sample observations to draw reliable conclusions about the population parameters. The results obtained from the sample are projected in such a way that they are valid for the entire population. This technique is known as **statistical inference**. The statistical inference may be divided into two areas or parts:

- (i) The population parameters are unknown and we may want to guess the true value of the unknown parameters on the basis of a random sample drawn from the population. This type of problem is known as **“Estimation”**.
- (ii) Some information is available about the population and we may like to verify whether the information is true or not on the basis of a random sample drawn from the population. This type of problem is known as **“Testing of hypothesis”**.

This course is divided into four blocks and each block consists of four units.

In **Block 1**, we shall be discussing the concept of sampling distribution of a statistic. The sampling distributions of sample mean, sample proportion, sample variance, difference of two sample means, etc. are also discussed in this block. Here, we also discuss some important sampling distributions which are widely used in statistical inference and known as exact sampling distributions such as χ^2 , t and F.

In **Block 2**, we discuss first part of statistical inference called estimation, through which we find the estimate of unknown parameter on the basis of sample data. Two types of estimation as point estimation and interval estimation are discussed in this block. The criteria of good estimator and different methods of estimation for the unknown parameters are also discussed in this block.

In **Block 3**, the second part of statistical inference, that is, testing of hypothesis is discussed. We discuss how the estimated or assumed or a hypothetical value of the population parameter can be tested and how we can take the decision about rejection or non-rejection of the hypothesis or assumed value. Here, different parametric tests as Z-test, t-test, F-test and χ^2 -test for testing of hypothesis about population mean, difference of two population means, population proportion, difference of two population proportions, population variance, etc. are also discussed.

Most of the parametric tests are based on the assumptions that the parent population is normal and the data are measured in interval or ratio scale and concerned with testing the population mean, proportion, variance, etc. But if the assumption(s) is (are) not fulfilled then we cannot use the parametric test.

In such situations, non-parametric tests do the same job for us. In **Block 4**, we discussed different non-parametric tests i.e. sign test, Wilcoxon signed-rank test, chi-square test for goodness of fit and independence of attributes, Kolmogorov-Smirnov test for goodness of fit, Kruskal -Wallis test for one way analysis of variance with completed randomised design, Friedman test for randomised block design, etc.

Although the material is self contained and self explained in nature, even though, if the learners are interested to gain more and want to study the contents in more detail, you may consult the following books:

- Goon, A.M., Gupta, M.K. and Dasgupta, B.; *An Outline of Statistical Theory*, Vol II, World Press, Calcutta.
- Rohatgi, V.K.; *An Introduction to Probability and Statistics*, Wiley-India.
- Mood, A.M.; *Introduction to the Theory of Statistics*, Tata McGraw-Hill Book Company, New Delhi.
- Gupta, S. C. and Kapoor, V. K.; *Fundamentals of Mathematical Statistics*, Sultan Chand & Sons.
- Makhopadhyay. P.; *Applied Statistics*, New Central Book Agency, Calcutta.
- Gibbons, J.D.; *Nonparametric Statistical Inference*, McGraw-Hill Book Company, New York.
- Daniel Wayne W.; *Applied Nonparametric Statistics*, Houghton Mifflin Company.
- Conover, W.J.; *Practical Nonparametric Statistics (3rd Ed.)*, Wiley-India.

SAMPLING DISTRIBUTIONS

One of the major objectives of statistical analysis is to draw reliable conclusions about the population on the basis of the analysis of the sample data. This process is known as statistical inference. For drawing the inference about the population parameter, we use a function of sample values such as sample mean, sample proportion, sample standard deviation, etc. which is known as statistic. If we draw all possible samples of same size and for each sample we calculate a statistic then the value of statistic may or may not be varying sample to sample. If we arrange all possible values of that statistic with its corresponding probabilities then this arrangement is known as sampling distribution of that statistic.

This is the first block of the course MST-004. The aim of this block is to put a foundation stone for the next blocks of this course and will provide a platform to the learners to understand the statistical inference. It comprises four units.

Unit 1: Introduction to Sampling Distribution

This unit explains the basis concept of sampling distribution of a statistic with examples. Here, we also discussed the most important theorem of Statistics “Central Limit Theorem” according to this theorem, the sampling distributions tend to be approximately normal for sample sizes greater than 30. In this unit, we also discussed the law of large numbers. According to law of large numbers, we come to know “how we draw a reliable inference about the population parameter by using a sample of finite size?”

Unit 2: Sampling Distribution(s) of Statistic(s)

This unit explores the sampling distributions of sample mean, sample proportion, sample variance, difference of two sample means, and difference of two sample proportions and ratio of two sample variances.

Unit 3: Standard Sampling Distributions-I

Sometimes, statistic may follow a particular sampling distribution such as χ^2 (read as chi-square), t, F, etc. which are known as exact sampling distributions. This unit provides the brief introduction of chi-square and t-distributions. Properties and applications of these distributions also explained in this unit.

Unit 4: Standard Sampling Distributions-II

Last unit of this block is devoted to the rest important sampling distribution which is widely used in Statistics i.e. F-distribution. Here, we discuss the properties and applications of this distribution. The standard sampling distributions are interacted with each other, therefore, the relations between them are also explored in this unit. In this unit, we also describe the procedure of finding the tabulated (critical) value of a variate which follows the exact sampling distribution such as χ^2 , t and F.

Notations and Symbols

$X_1, X_2, \dots, X_n$	:	Random sample of size n
SRSWR	:	Simple random sampling with replacement
SRSWOR	:	Simple random sampling without replacement
$\bar{X}$	:	Sample mean
S^2	:	Sample variance
SE	:	Standard error
$N(\mu, \sigma^2)$	:	Normal distribution with mean μ and variance σ^2
$Z \sim N(0, 1)$	:	Standard normal variate
P	:	Population proportion
p	:	Sample proportion
v	:	Degrees of freedom(df)
$\chi^2_{(v)}$	:	Chi-square with v df
Γ	:	Gamma function
$B(a, b)$	:	Beta function
$\log x$	:	Logarithm base e
α	:	Area under the curve
$t_{(v), \alpha}$	:	Tabulated value of t -statistic with v df such that the area under the curve of t -distribution to its right (upper) tail is equal to α
$\chi^2_{(v), \alpha}$	:	Tabulated value of χ^2 -statistic with v df such that the area under the curve of χ^2 -distribution to its right (upper) tail is equal to α
$F_{(v_1, v_2), \alpha}$	:	Tabulated value of F -statistic with (v_1, v_2) df such that the area under the curve of F -distribution to its right (upper) tail is equal to α

UNIT 1 INTRODUCTION TO SAMPLING DISTRIBUTION

Structure

- 1.1 Introduction
 - Objectives
- 1.2 Basic Terminology
- 1.3 Introduction to Sampling Distribution
- 1.4 Standard Error
- 1.5 Central Limit Theorem
- 1.6 Law of Large Numbers
- 1.7 Summary
- 1.8 Solutions /Answers

1.1 INTRODUCTION

In many situations, if we extract some information from all the elements or items of a large population in general, it may be time consuming and expensive. Also, if the elements or items of population are destroyed under investigation then the information gathering from all the elements is not making a sense because all the elements will be destroyed. Therefore, in most of the cases in daily life, business and industry the information is gathered by means of **sampling**. The results of a properly taken sample enable the investigator to arrive at generalisations that are valid for entire population. The process of generalising sample results to the population is called **Statistical Inference**.

For drawing inference about the population parameters, we draw all possible samples of same size and determine a function of sample values, which is called **statistic**, for each sample. The values of statistic are generally varied from one sample to another sample. Therefore, the sample statistic is a random variable and follows a distribution. The distribution of the statistic is called **sampling distribution of the statistic**.

This unit is divided into 8 sections. Section 1.1 is introductive in nature. In Section 1.2, we defined the basic terminology used in statistical inference. The introduction and needs of sampling distributions are described in the Section 1.3. In Section 1.4, we explored the concept of standard error. The most important theorem of Statistics “Central Limit Theorem” is described in Section 1.5. In Section 1.6, we explored the concept of law of large numbers. Unit ends by providing summary of what we have discussed in this unit in Section 1.7 and solution of exercises in Section 1.8.

Objectives

After studying this unit, you should be able to:

- define statistical inference;
- define the basic terms as population, sample, parameter, statistic, estimator, estimate, etc. used in statistical inference;
- explain the concept of sampling distribution;
- explore the importance and needs of sampling distribution;

- explain the concept of standard error;
- describe the most important theorem of Statistics “Central Limit Theorem”;
- explain the concept of law of large numbers; and
- determine adequate sample size required for sample survey.

1.2 BASIC TERMINOLOGY

Before discussing the sampling distribution of a statistic, we shall be discussing basic definitions of some of the important terms which are very helpful to understand the fundamentals of statistical inference.

Population

In general sense “population” means a large group of people who live in a particular geographical area. For example, the group of people who live in New Delhi, the group of people working in IGNOU, students enrolled in PGDAST programme in IGNOU, etc.

In Statistics, population has a much broader meaning. It does not only refer to people but also the group of elements or units under consideration by the analyst. Thus, population is the collection or group of individuals /items /units/ observations under study. For example, the collection of books in a library, the particles in a salt bag, the rivers in India, the students in a classroom, etc. are considered as populations in Statistics.

The total number of elements / items / units / observations in a population is known as population size and denoted by N . The characteristic under study may be denoted by X or Y .

We may classify the population into two types as:

(1) Finite and Infinite Population

If a population contains finite number of units or observations, it is called a **finite population**. For example, the population of students in a class, the population of bolts produced in a factory in a day, the population of electric bulbs in a lot, the population of books in a library, etc. are the finite populations because in these examples the number of units in the population is finite in numbers therefore these are all the examples of finite population.

If a population contains an infinite (uncountable) number of units or observations, it is called an **infinite population**. For example, the population of particles in a salt bag, the population of stars in the sky, the population of hairs on human body, etc. are infinite populations because in these examples the number of units in the population is not finite therefore these are all the examples of infinite population.

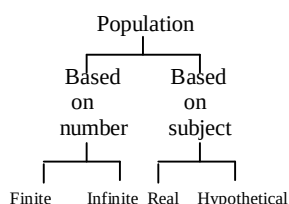
But theoretically sometimes, populations of too large in size are assumed infinite.

(2) Real and Hypothetical Population

The population can also be classified as real and hypothetical. A population comprising the items or units which are all physically present is known as **real population**. All of the examples given above are examples of a real population.

If a population consists the items or units which are not physically present but the existence of them can only be imagined or conceptualised then it is known

The group of individuals / items / units/ observations under study is known as population.



as **hypothetical population**. For example, the population of heads or tails in successive tosses of a coin a large number of times is considered as hypothetical population.

Sample

To extract the information from all the elements or units or items of a large population may be, in general, time consuming, expensive and difficult. And also if the elements or units of population are destroyed under investigation then gathering the information from all the units is not make a sense. For example, to test the strength of the chalk, to test the quality of crackers, etc. the chalks and crackers are destroyed under testing procedure. In such situations, a small part of population is selected from the population which is called a **sample**. Thus, the sample can be defined as below:

“A sample is a part / fraction / subset of the population.”

The procedure of drawing a sample from the population is called **sampling**. You will learn about sampling in detail in Block 1 of course MST-005. The number of units selected in the sample is known as sample size and it is denoted by n .

Complete Survey and Sample Survey

Statistical data may be collected by two methods:

- (1) Complete Survey or Complete Enumeration or Census
- (2) Sample Survey or Sample Enumeration

(1) Complete Survey or Complete Enumeration or Census

When each and every element or unit of the population is investigated or studied for the characteristics under study then we call it **complete survey** or **census**. For example, suppose we want to find out the average height of the students of a study centre then if we measure the height of each and every student of this study centre to find the average height of the students then such type of survey is called complete survey.

(2) Sample Survey or Sample Enumeration

When only a part or a small number of elements or units (i.e. sample) of population are investigated or studied for the characteristics under study then we call it **sample survey** or **sample enumeration**. In the above example, if we select some students of this study centre and measure the height to find average height of the students then such type of survey is called sample survey.

Simple Random Sampling or Random Sampling

The simplest and most common method of sampling is simple random sampling. In simple random sampling, the sample is drawn in such a way that each element or unit of the population has an equal and independent chance of being included in the sample. If a sample is drawn by this method then it is known as a **simple random sample or random sample**. The random sample of size n is denoted by $X_1, X_2, \dots, X_n$ or $Y_1, Y_2, \dots, Y_n$ and the observed value of this sample is denoted by $x_1, x_2, \dots, x_n$ or $y_1, y_2, \dots, y_n$. Some author use $x_1, x_2, \dots, x_n$ to represent the random sample instead of $X_1, X_2, \dots, X_n$. But throughout the course, we use capital letters to represent the random sample, that is, $X_1, X_2, \dots, X_n$ and $x_1, x_2, \dots, x_n$ simply be assumed that these are the independent identically distributed(iid) random variables.

In simple random sampling elements or units are selected or drawn one by one therefore it may be classified into two types as:

(1) Simple Random Sampling without Replacement (SRSWOR)

In simple random sampling, if the elements or units are selected or drawn one by one in such a way that an element or unit drawn at a time is not replaced back to the population before the subsequent draws is called **SRSWOR**. If we draw a sample of size n from a population of size N without replacement then total number of possible samples is ${}^N C_n$. For example, consider a population that consists of three elements, A, B and C. Suppose we wish to draw a random sample of two elements then $N = 3$ and $n = 2$. The total number of possible random samples without replacement is ${}^N C_n = {}^3 C_2 = 3$ as (A, B), (A, C) and (B, C).

(2) Simple Random Sampling with Replacement (SRSWR)

In simple random sampling, if the elements or units are selected or drawn one by one in such a way that a unit drawn at a time is replaced back to the population before the subsequent draw is called **SRSWR**. In this method, the same element or unit can appear more than once in the sample and the probability of selection of a unit at each draw remains same i.e. $1/N$. In this method, total number of possible samples is N^n . In above example, the total number of possible random samples with replacement is $N^n = 3^2 = 9$ as (A, A), (A, B), (A, C), (B, A), (B, B), (B, C), (C, A), (C, B) and (C, C).

Note1: The statistical inference is based on the theoretical distribution so in whole statistical inference, we assume that the random sample is selected from the infinite population or from a finite population with replacement so that removal of the sample unit has no appreciable effect on the composition of the population. That is, we draw the random sample such that:

- (a) The n successive sample observations are independent and
- (b) The population composition remains constant from draw to draw.

For example, if we draw a random sample $X_1, X_2, \dots, X_n$ from normal population with mean μ and variance σ^2 then X_i 's are independent and follow same normal distribution.

Parameter

A group of elements or units under study can be regarded as a population and the characteristics of population can be described with some measures such as total numbers of items, average, variance, etc. These measures are known as parameters of the population. Thus, we can define parameter as:

A parameter is a function of population values which is used to represent the certain characteristic of the population. For example, population mean, population variance, population coefficient of variation, population correlation coefficient, etc. are all parameters. Population parameter mean usually denoted by μ and population variance denoted by σ^2 .

We know that the population can be described with the help of distribution and the distribution is fully determined with the help of its constants such as, in case of normal distribution, we need to know μ and σ^2 to determine the normal distribution, in case of Poisson distribution, we need to know λ , etc. These constants are known as parameter.

In SRSWR each of the $1^{\text{st}}, 2^{\text{nd}}, \dots, n^{\text{th}}$ draw the elements are remain same N due to replacement so by rule of multiplication total number of possible samples is $N \times N \times \dots n \text{ times} = N^n$.

A parameter is a function of population values which is used to represent the certain characteristic of the population.

Sample Mean and Sample Variance

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density(mass) function $f(x, \theta)$ then sample mean is defined as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

And sample variance is defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Here, we divide $\sum_{i=1}^n (X_i - \bar{X})^2$ by $(n-1)$ rather than n as our definition of the variance described in Unit 2 of MST-002. The reason for taking $(n-1)$ in place of n will become clear in the Section 5.4 of Unit 5 of this course.

Statistic

Any quantity calculated from sample values and does not contain any unknown parameter is known as **statistic**. For example, if $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 (both are unknown) then sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is a statistic whereas

$\bar{X} - \mu$ and $\bar{X} / \sigma$ are not statistics because both are function of unknown parameters.

Estimator and Estimate

Generally, population parameters are unknown and the whole population is too large to find out the parameters. Since the sample drawn from a population always contains some or more information about the population, therefore in such situations, we guess or estimate the value of the parameter under study based on a random sample drawn from that population.

So if a statistic is used to estimate an unknown population parameter then it is known as **estimator** and the value of the estimator based on observed value of the sample is known as **estimate** of parameter. For example, suppose the parameter λ of the Poisson population $f(x, \lambda)$ is unknown so we draw a random sample $X_1, X_2, \dots, X_n$ of size n from this Poisson population to estimate λ . If

we use sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ to estimate λ then $\bar{X}$ is called estimator and

any particular value of this estimator, say, $\bar{x}$ is called estimate of unknown parameter λ . Consider another example, if we want to estimate the average height (μ) of students in a college with the help of sample mean $\bar{X}$ then $\bar{X}$ is the estimator and its particular value, say, 165 cm is the estimate of the population average height (μ).

In this course, we use capital letters for estimator and small letters for estimated value. In general, the estimator is denoted by $T_n = t(X_1, X_2, \dots, X_n)$ where 'n' denotes the sample size and the estimator T is a function of the random sample $X_1, X_2, \dots, X_n$.

Now, it is right place to do some examples.

Any statistic used to estimate an unknown population parameter is known as **estimator** and the particular value of the estimator is known as **estimate** of parameter. The estimated value of sample mean and sample variance are denoted by $\bar{x}$ and s^2 respectively.

Example 1: Categories the following populations as finite and infinite:

- (i) Population of newly born babies during a year in a particular hospital.
- (ii) Population of real numbers between 1 and 2.
- (iii) Population of number of words on this page.

Solution: (i) and (iii) are the finite populations because the number of elements (babies or words) in both are finite whereas (ii) is infinite because there are infinite real numbers are possible between 1 and 2.

Example 2: If population size is 6 then how many samples of size of 4 are possible with replacement?

Solution: Here, we are given that

Population size = $N = 6$,

Sample size = $n = 4$

Since we know that all possible samples of size n taken from a population of size N with replacement are N^n so in our case $N^n = 6^4 = 1296$.

Now, you check your understanding of above discussion by answering the following exercises.

E1) In which situation we use only sample survey:

- (i) Blood testing.
- (ii) To know the average income of the people living in a colony.
- (iii) To know the number of students enrolled in IGNOU.
- (iv) To know the average life of electric bulbs.

E2) If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from normal population with mean μ and variance σ^2 both are unknown then find the statistic in the following:

- (i) $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)$
- (ii) $\sum_{i=1}^n X_i$
- (iii) $\sum_{i=1}^n X_i / \sigma$
- (iv) $\sum_{i=1}^n X_i^2$

E3) The weights (in kg.) of 5 workers in a factory are 56, 62, 74, 45 and 50. How many samples of size of 2 are possible with replacement? Also write all possible samples of size 2.

1.3 INTRODUCTION TO SAMPLING DISTRIBUTION

As we have discussed in Section 1.1 that if the population is too large or the units or items of the population are destructive in nature or there is a limited resources such as men power, money, etc., then it is not possible practically to examine each and every unit of the population to obtain the necessary information about the population. For example, Suppose we want to know the average life of electric bulbs of a certain brand which are manufactured by a company. The company manufactures a lot of bulbs say 5,000 per day. Then gathering information about average life of all the bulbs is not making a sense

because the bulbs are destroyed under investigation. In another example, if we want to know the average income of the persons living in a big city then collecting the information from each and every person is very much time and manpower consuming. In such situations, one can draw sample from the population under study and utilize sample observations to extract the necessary information about the population. The results obtained from the sample are projected in such a way that they are valid for the entire population. Therefore, the sample works like a **“Vehicle”** to reach (drawing) at valid conclusions about the population. Thus, statistical inference can be defined as:

“The process of projecting the sample results for the whole population is known as **statistical inference**.”

For drawing the inference about the population, we analyse the sample data, that is, we calculate the sample statistic as sample mean, sample proportion, sample variance, etc. Generally, if we want to draw the inference about the population mean we use sample mean, about population variance we use sample variance, etc.

For example, suppose we want to study the average life of electric bulbs produced by the company discussed in above example and we decide to estimate the average life of electric bulbs on the basis of a sample. So we take a sample, say, 10 bulbs and measure the life of each bulb selected in the sample. The observed values of life of bulbs are given in column 2 of the Table 1.1 given below:

Table 1.1: Life Times of 10 Bulbs in Sample-I and Sample-II

S.No.	Life of Bulbs (in hours)	
	Sample-I	Sample-II
1	2000	400
2	500	0
3	1000	1400
4	1500	1200
5	1200	800
6	200	700
7	700	1200
8	1400	800
9	800	600
10	1100	1500
Total	10400	8600
$\bar{X}$	1040	860

From the above table, the average (mean) life of these selected bulbs is 1040 hours. Now, if we are using this sample to make inference about the population average life then we say that, the sample mean is 1040 hours give an estimate of the average life of electric bulbs for the whole population. But if we take another sample of same size 10 from the population and calculate the sample mean as shown in the above table. The average life of bulbs selected in sample-II is 860 hours. This is different from the mean of sample-I. Thus, if we are using this sample mean to estimate the population mean then we gets a different estimate of the average life of bulbs for the whole population. Similarly, we can take many samples of same size 10 and for each sample we get sample mean which may or may not be distinct. From all of the sample means, we try to estimate the average life of bulbs of the whole population.

For better understanding of the process of generalising these sample results to the whole population, we consider the example in which size of the population is very small.

Consider a population comprising four typists who type the sample page of a manuscript. The number of errors made by each typist is shown in Table 1.2 given below:

Table 1.2: Number of Error per Typist

Typist	Number of Errors
A	4
B	2
C	3
D	1

The population mean (average number of errors) can be obtained as

$$\mu = \frac{4 + 2 + 3 + 1}{4} = 2.5$$

Now, let us assume that we do not know the average number of errors made by typists. So we decide to estimate the population mean on the basis of sample of size $n = 2$. There are $N^n = 4^2 = 16$ possible simple random samples with replacement of size 2. All possible samples of size $n = 2$ are given below and for each sample the sample mean is calculated as shown in Table 1.3 given below:

Table 1.3: Calculation of Sample Mean

Sample Number	Sample in Term of Typist	Sample Observation	Sample Mean ($\bar{X}$)
1	(A, A)	(4, 4)	4.0
2	(A, B)	(4, 2)	3.0
3	(A, C)	(4, 3)	3.5
4	(A, D)	(4, 1)	2.5
5	(B, A)	(2, 4)	3.0
6	(B, B)	(2, 2)	2.0
7	(B, C)	(2, 3)	2.5
8	(B, D)	(2, 1)	1.5
9	(C, A)	(3, 4)	3.5
10	(C, B)	(3, 2)	2.5
11	(C, C)	(3, 3)	3.0
12	(C, D)	(3, 1)	2.0
13	(D, A)	(1, 4)	2.5
14	(D, B)	(1, 2)	1.5
15	(D, C)	(1, 3)	2.0
16	(D, D)	(1, 1)	1.0

From the above table, we can see that the value of the sample statistic (sample mean) is varying from sample to sample and out of 16 samples only 4 samples(4,7,10 and 13) have their means equal to population mean whereas other 12 samples result in some error in the estimation process. The error is the difference between the population mean and the sample mean used for estimate.

Now, consider all possible values of sample mean and calculate their probabilities (using relative frequency approach of probability described in Unit 2 of MST-003) of occurrence. Then we arrange every possible value of sample mean with their respective probabilities in the following Table 1.4 given below:

Table 1.4: Sampling Distribution of Sample Means

S. No.	$\bar{X}$	Frequency(f)	Probability(p)
1	1.0	1	1/16 = 0.0625
2	1.5	2	2/16 = 0.1250
3	2.0	3	3/16 = 0.1875
4	2.5	4	4/16 = 0.2500
5	3.0	3	3/16 = 0.1875
6	3.5	2	2/16 = 0.1250
7	4.0	1	1/16 = 0.0625

So the arrangement of all possible values of sample mean with their corresponding probabilities is called the sampling distribution of mean. Thus, we can define the sampling distribution of a statistic as:

“The probability distribution of all possible values of a sample statistic that would be obtained by drawing all possible samples of the same size from the population is called sampling distribution of that statistic.”

The sampling distribution of sample means itself has mean, variance, etc. Therefore, the mean of sample means can be obtained by the formula

$$\begin{aligned}\text{Mean of sample means} = \bar{\bar{X}} &= \frac{1}{K} \sum_{i=1}^k \bar{X}_i f_i \quad \text{where, } K = \sum_{i=1}^k f_i \\ &= \frac{1}{16} (1.0 \times 1 + 1.5 \times 2 + \dots + 4.0 \times 1) = 2.5 = \mu\end{aligned}$$

The mean of sample means can also be calculated as

$$E(\bar{X}) = \bar{\bar{X}} = \sum_{i=1}^k \bar{X}_i p_i = 1.0 \times \frac{1}{16} + 1.5 \times \frac{2}{16} + \dots + 4.0 \times \frac{1}{16} = 2.5$$

Thus, we have seen for this population that mean of sample means is equal to the population mean, that is, $\bar{\bar{X}} = \mu = 2.5$. The fact that these two means are equal is not a chance as it can be shown that $\bar{\bar{X}}$ equals to population mean for any population and any given sample size.

But if the population is large say having 1,000 elements and we want to draw all samples of size 2 then there are $(1000)^2 = 1000000$ possible simple random samples with replacement. Then the listing of all such samples and finding the sampling distribution would be a difficult task empirically. Therefore, we may consider a theoretical experiment in which we draw all possible sample of a fixed size and obtain a sampling distribution.

Although in practice only a random sample is actually selected and the concept of sampling distribution is used to draw the inference about the population parameters.

It is now time for you to try the following exercise to make sure that you understand about sampling distribution.

- E4)** If lives of 3 Televisions of certain company are 8, 6 and 10 years then construct the sampling distribution of average life of Televisions by taking all samples of size 2.

1.4 STANDARD ERROR

As we have seen in the previous section that the values of sample statistic may vary from sample to sample and all the sample values are not equal to the population parameter. Now, one can be interested to measure how much the values of sample statistic vary from the population parameter on average. As we have already studied in Unit 2 of MST-002 that the standard deviation is used as a measure of variation.

Thus, for measuring the variation in the values of sample statistic around the population parameter we calculate the standard deviation of the sampling distribution. This is known as standard error of that statistic. Thus, standard error of a statistic can be defined as:

“The standard deviation of a sampling distribution of a statistic is known as standard error and it is denoted by SE.”

Therefore, the standard error of sample mean is given by

$$SE(\bar{X}) = \sqrt{\frac{1}{K} \sum_{i=1}^k f_i (\bar{x}_i - \mu)^2} \text{ where, } K = \sum_{i=1}^k f_i$$

In the previous example, we can calculate the standard error of sample mean as

$$\begin{aligned} SE(\bar{X}) &= \sqrt{\frac{1}{16} [1 \times (1.0 - 2.5)^2 + 2 \times (1.5 - 2.5)^2 + \dots + 1 \times (4.0 - 2.5)^2]} \\ &= \sqrt{\frac{1}{16} (2.25 + 2 + \dots + 2.25)} = \sqrt{\frac{10}{16}} = 0.791 \end{aligned}$$

The computation of the standard error is a tedious process. There is an alternative method to compute standard error of the mean from a single sample as:

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 then the standard errors of sample mean ($\bar{X}$) is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

The standard error is used to express the accuracy or precision of the estimate of population parameter because the reciprocal of the standard error is the measure of reliability or precision of the statistic. Standard error also determines the probable limits or confidence limits (described in Units 7 & 8 of this course) within which the population parameter may be expected to lie with certain level of confidence. Standard error is also applicable in testing of hypothesis (described in Block 3 of this course).

Standard deviation of sampling distribution of a statistic is called standard error.

The standard errors of some other well known statistics are given below:

1. The standard error of sample proportion (p) is given by

$$SE(p) = \sqrt{\frac{PQ}{n}}$$

where, P is population proportion and Q = 1 - P.

2. The standard error of sample median ($\tilde{X}$) is given by

$$SE(\tilde{X}) = \frac{\pi}{2} \frac{\sigma}{\sqrt{n}} = 1.25331 SE(\bar{X})$$

3. The standard error of difference of two sample means is given by

$$SE(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where, σ_1^2 and σ_2^2 are the population variances and n_1 and n_2 are the sample sizes of two independent samples.

4. The standard error of difference between two sample proportions is given by

$$SE(p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

where, P_1 and P_2 are population proportions of two different populations and $Q_1 = 1 - P_1$ & $Q_2 = 1 - P_2$.

From all the above formulae, we can understand that standard error is inversely proportional to the sample size. Therefore, as sample size increases the standard error decreases.

Note 2: These standard errors discussed above were calculated under the assumption that sampling is done either from an infinite population or from a finite population with replacement. But, in real life sampling problems, most sampling plans do not permit an element to be selected twice in a given sample (i.e. sampling with replacement). Consequently, if the population is not large in relation to the size of the sample and sampling is done without replacement

then we multiply the above standard errors by the correction factor $\sqrt{\frac{N-n}{N-1}}$.

where, N is the population size. This correction factor is known as finite population correction factor.

Therefore, in this case, the standard error of the sample mean is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Similarly,

$$SE(p) = \sqrt{\frac{N-n}{N-1} \frac{PQ}{n}}$$

In practice, if the sample size n is less than or equal to 10% of the population size N, this correction factor may be ignored for all practical purposes.

Let us do one example relating to standard error.

Example 3: Diameter of a steel ball bearing produced by a semi-automatic machine is known to be distributed normally with mean 12 cm and standard deviation 0.1 cm. If we take a random sample of size 10 with replacement then find standard error of sample mean for estimating the population mean of diameter of steel ball bearing for whole population.

Solution: Here, we are given that

$$\mu = 12, \sigma = 0.1, n = 10$$

Since the sampling is done with replacement therefore the standard error of sample mean for estimating population mean is given by

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{0.1}{\sqrt{10}} = 0.03$$

In the same way, you can try the following exercise.

-
- E5)** The average weight of certain type of tyres is 200 pounds and standard deviation is 4 pounds. A sample of 50 tyres is selected. Obtain the standard error of sample mean.
- E6)** A machine produces a large number of items of which 15% are found to be defective. If a random sample of 200 items is taken from the population, then find the standard error of sampling distribution of proportion.
-

1.5 CENTRAL LIMIT THEOREM

The central limit theorem is the most important theorem of Statistics. It was first introduced by De Moivre in the early eighteenth century.

According to the central limit theorem, if $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 then the sampling distribution of the sample mean tends to normal distribution with mean μ and variance σ^2/n as sample size tends to large ($n > 30$) whatever the form of parent population, that is,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and the variate

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

follows the normal distribution with mean 0 and variance unity, that is, the variate Z follows standard normal distribution.

We do not intend to prove this theorem here, but merely show graphical evidence of its validity in Fig. 1.1. Here, we will also try to show that how large must the sample size be for which we can assume that the central limit theorem applies?

According to the central limit theorem, the sampling distribution of the sample means tends to normal distribution as sample size tends to large ($n > 30$).

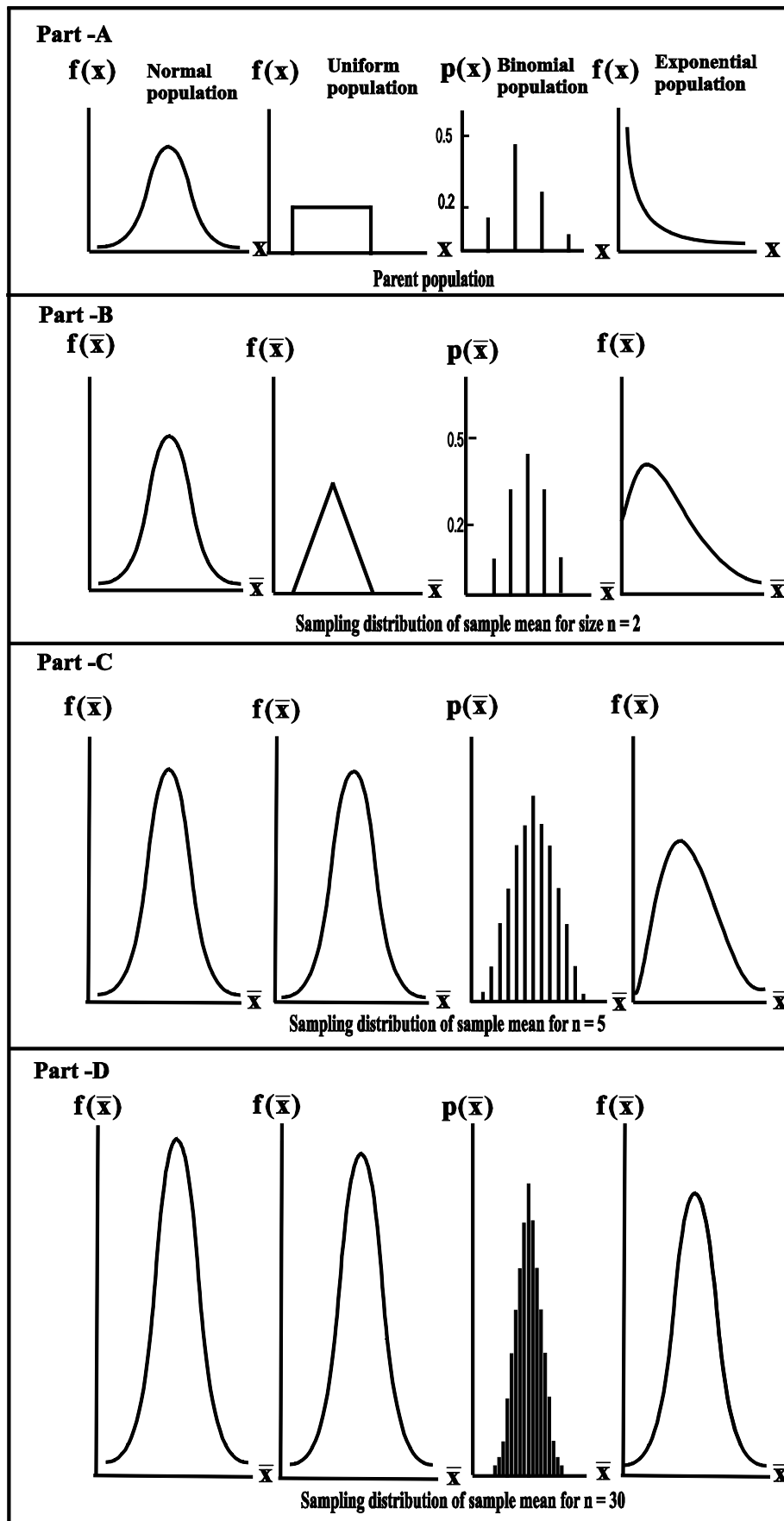


Fig. 1.1: Sampling distribution of sample means for various populations when $n = 2$, $n = 5$ and $n = 30$

Sampling Distributions

In this figure, we are trying to understand the sampling distribution of sample mean $\bar{X}$ for different populations and for varying sample sizes. We divide this figure into four parts A, B, C and D. The part 'A' of this figure shows four different populations as normal, uniform, binomial and exponential. The rest parts B, C and D represent the shape of sampling distribution of mean of sizes $n = 2$, $n = 5$ and $n = 30$ respectively drawn from the populations shown in first row (Part-A). From the first column of this figure, we observed that when the parent population is normal then all the sampling distributions for varying sample sizes are also normal, having same mean but their variances decrease as n increases.

The second column of this figure represents the uniform population. Here, we observe that the sampling distribution of $\bar{X}$ is symmetrical when $n = 2$ and tends to normal when $n = 30$.

However, the third column of this figure represents the binomial population (discrete). Again when $n = 2$, the sampling distribution of $\bar{X}$ is symmetrical and for $n = 5$ it is quite bell shaped and tends to normal when $n = 30$. The last column of this figure represents the exponential population which is highly skewed. Here, we observe that for $n = 30$, the distribution of $\bar{X}$ is symmetrical bell shaped normal.

Here, we conclude that if we draw a random sample of large size $n > 30$ from the population then the sampling distribution of $\bar{X}$ can be approximated by a normal probability distribution, whatever the form of parent population.

Note 3: The central limit theorem can also be applicable in the same way for the sampling distribution of sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc. that is, if we take a random sample of large size ($n = 36 > 30$) from the population then the sampling distribution of sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc. approximated by a normal probability distribution, whatever the form of parent population. The mean and variance of these sampling distributions shall be discussed in the next unit.

Let us do one example based on central limit theorem.

Example 4: Suppose in IGNOU, the average (mean) weight of all male students is 60 kg and standard deviation is 25 kg. If a sample of 36 male students is selected at random, find the probability that the male students having average weight

- (i) more than 70 kg
- (ii) less than 55 kg
- (iii) between 50 kg and 65 kg

Solution: Here, we are given that

$$\mu = 60 \text{ kg}, \sigma = 25 \text{ kg}, n = 36$$

Since, we draw a large sample ($n > 30$), therefore by central limit theorem the sampling distribution of the sample means will also follow a normal distribution with mean

$$E(\bar{X}) = \mu = 60$$

and variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(25)^2}{36} = 17.36$$

- (i) The probability that the male students having average weight more than 70 kg is given by

$$P[\bar{X} > 70] \quad [\text{See Fig. 1.2}]$$

To get the value of this probability, we convert the variate $\bar{X}$ into a standard normal variate Z by the transformation

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 60}{\sqrt{17.36}} = \frac{\bar{X} - 60}{4.17}$$

Therefore, subtract 60 from each term and then divide each term by 4.17 in the above inequality. Thus, the probability becomes

$$\begin{aligned} P\left[\frac{\bar{X} - 60}{4.17} > \frac{70 - 60}{4.17}\right] &= P[Z > 2.40] \\ &= 0.5 - P[0 \leq Z \leq 2.40] \\ &= 0.5 - 0.4918 \quad \left[\text{Using table area under normal curve}\right] \\ &= 0.0082 \end{aligned}$$

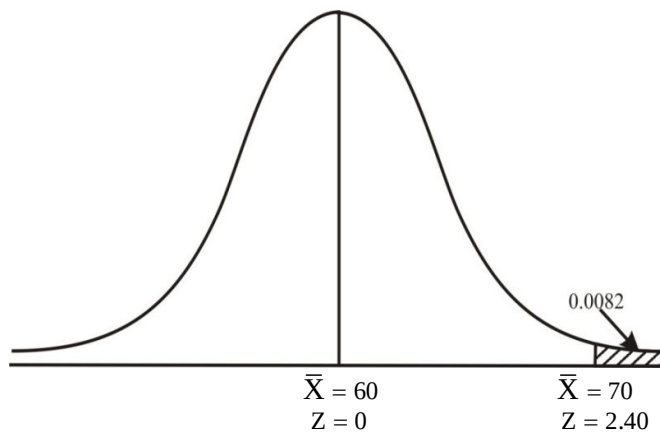


Fig. 1.2

- (ii) The probability that the male students having average weight less than 55 kg is given by

$$\begin{aligned} P[\bar{X} < 55] &= P\left[\frac{\bar{X} - 60}{4.17} < \frac{55 - 60}{4.17}\right] \\ &= P[Z < -1.20] \quad [\text{See Fig. 1.3}] \\ &= P[Z > 1.20] \quad \left[\because \text{normal curve is symmetrical about the line } Z = 0.\right] \\ &= 0.5 - P[0 \leq Z \leq 1.20] \\ &= 0.5 - 0.3849 \quad \left[\text{Using table area under normal curve}\right] \\ &= 0.1151 \end{aligned}$$

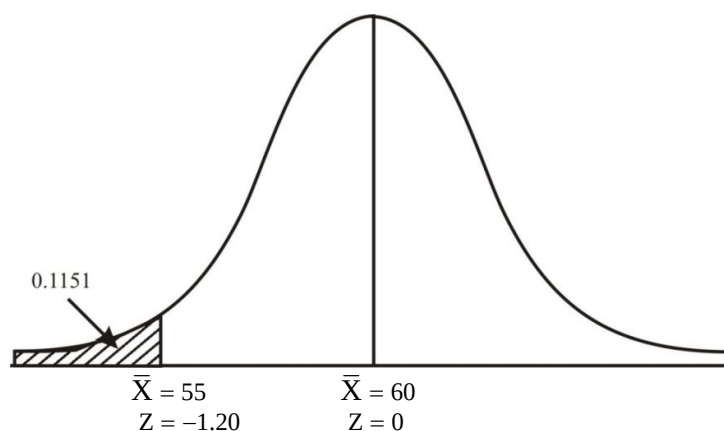


Fig. 1.3

- (iii) The probability that the male students having average weight between 50 kg and 65 kg is given by

$$\begin{aligned}
 P[50 < \bar{X} < 65] &= P\left[\frac{50 - 60}{4.17} < \frac{\bar{X} - 60}{4.17} < \frac{65 - 60}{4.17}\right] \\
 &= P[-2.40 < Z < 1.20] \quad \quad \quad [\text{See Fig. 1.4}] \\
 &= P[-2.40 < Z < 0] + P[0 < Z < 1.20] \\
 &= P[0 < Z < 2.40] + P[0 < Z < 1.20] \\
 &= 0.4918 + 0.3849 \quad \quad \quad \left[\begin{array}{l} \text{Using table area} \\ \text{under normal curve} \end{array} \right] \\
 &= 0.8767
 \end{aligned}$$

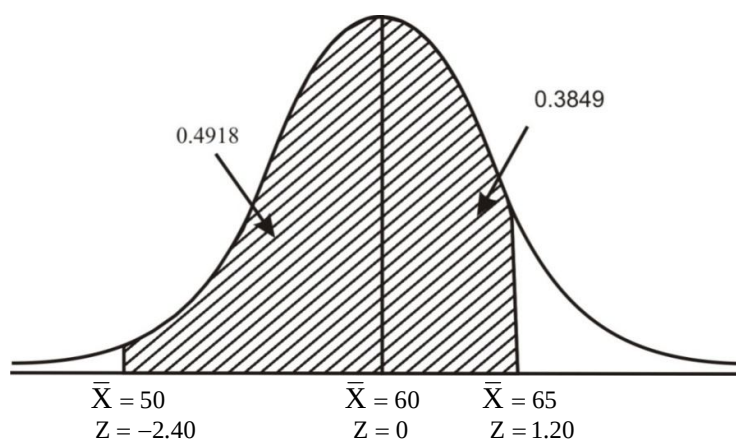


Fig. 1.4

Now, you can try the following exercise which will make you more user friendly with the use of central limit theorem.

-
- E7)** Average height of the students of science group in a college is 65 inches with a standard deviation of 2.2 inches. If a sample of 40 students is selected at random, what is the probability that the average height of these 40 students lies between 64 and 65.5 inches?
-

We have already discussed in Section 1.2 of this unit that the population parameters are generally unknown and for estimating parameters, we draw all possible random samples of same size from the population and calculate the values of sample statistic such as sample mean, sample proportion, sample variance, etc. for all samples and with the help of these values we form sampling distribution of that statistic. Then we draw inference about the population parameters.

But in real-world the sampling distributions are never really observed. The process of finding sampling distribution would be very tedious because it would involve very large number of samples. So in real-world problems, we draw a random sample from the population to draw inference about the population parameters. A very crucial question then arises: “Using a random sample of finite size say n , can we make a reliable inference about population parameter?” The answer is “yes”, reliable inference about population parameter can be made by using only a finite sample and we shall demonstrate this by “**law of large numbers**”. This law of large numbers can be stated in words as:

“A positive integer can be determined such that if a random sample of size n or larger is taken from a population with parameter, say, population mean (μ), the probability that the sample mean $\bar{X}$ will deviate from population mean μ can be made to be as close to 1 as desired.”

This law states that for any two arbitrary small numbers ε and η where $\varepsilon > 0$ and $0 < \eta < 1$, there exists an integer n such that if a random sample of size n or larger is drawn from the population and calculate the sample mean $\bar{X}$ for this sample, then the probability that $\bar{X}$ deviates from μ by less than ε is greater than $1 - \eta$ (as close to 1 as desired), that is, $\bar{X}$ arbitrarily close to μ .

Symbolically this can be written as

For any $\varepsilon > 0$ and $0 < \eta < 1$ there exists an integer n such that $n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$, where σ^2 is the finite variance of population then

$$P[|\bar{X} - \mu| < \varepsilon] \geq 1 - \eta$$

$$\text{or } P[-\varepsilon < \bar{X} - \mu < \varepsilon] \geq 1 - \eta \quad \left[\because |X| < a \Rightarrow -a < X < a \right]$$

The proof of “law of large numbers” is beyond the scope of this course.

Here, with an example we have to try to understand the law of large numbers in action, as we take more observations the sample mean approaches to the population mean.

Suppose that the distribution of the weight of all the young men living in a city is close to normal distribution with mean 65 kg and standard deviation 25 kg. To understand the law of large numbers, we calculate the sample mean weight ($\bar{X}$) for varying sample size $n = 1, 2, 3, \dots$. Fig. 1.5 shows the behaviour of the sample mean weight $\bar{X}$ of men chosen at random from this city. The graph plots the values of $\bar{X}$ (along vertical axis and sample size along horizontal axis) as sample size varying from $n = 1, 2, 3, \dots$

Sampling Distributions

First we start a sample of size $n = 1$, that is, we select a man randomly from the city. Suppose selected man had weight 70 kg, therefore, the line of the graph starts from this point. Now, select second man randomly and suppose his weight is 55 kg. So for $n = 2$ the sample mean is

$$\bar{X} = \frac{70 + 55}{2} = 62.5 \text{ kg}$$

This is the second point on the graph. Now, select third man randomly from the city and suppose his weight is 83 kg. Therefore, for $n = 3$ the sample mean is

$$\bar{X} = \frac{70 + 55 + 83}{3} = 69.3 \text{ kg}$$

This is the third point on the graph.

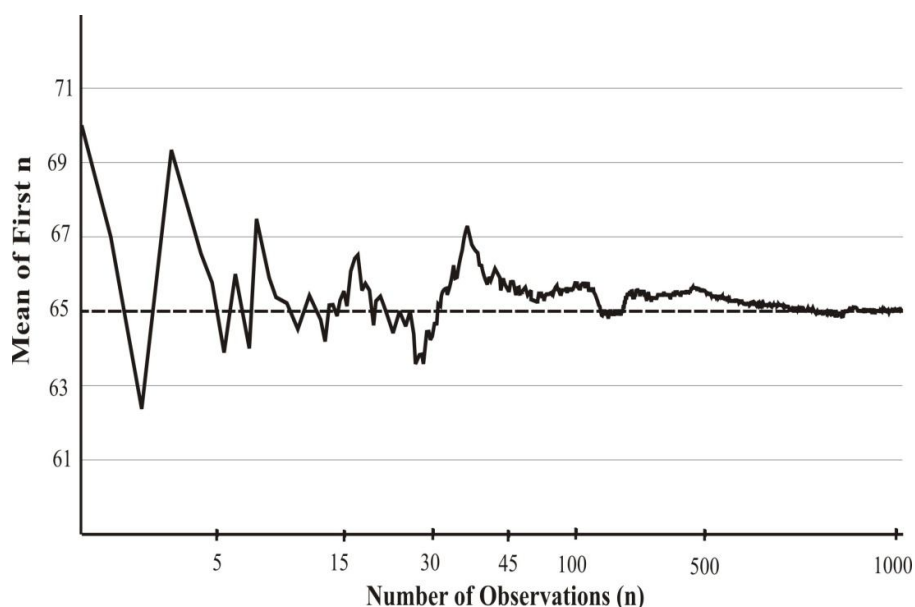


Fig. 1.5

This process will continue. From the graph we can see that the mean of the sample changes as we make more observations, eventually, however, the mean of the observations get close and close to the population mean 65 kg as the sample size increases.

Note 4: The law of large numbers can also be applied for the sample proportion, sample standard deviation, difference of two sample means, difference of two sample proportions, etc.

Example 5: A pathologist wants to estimate the mean time required to complete a certain analysis on the basis of sample study so that he may be 99% confident that the mean time may remain with ± 2 days of the mean. As per the available records, the population variance is 5 days². What must be the size of the sample for this study?

Solution: Here, we are given

$$1 - \eta = 0.99 \Rightarrow \eta = 0.01, \quad \varepsilon = 2, \quad \sigma^2 = 5$$

By the law of large numbers, we have

$$n \geq \frac{\sigma^2}{\varepsilon^2 \eta}$$

$$= \frac{5}{2^2 \times 0.01} = 125$$

That is, $n \geq 125$

Hence, at least 125 units must be drawn in a sample.

Example 6: An investigator wishes to estimate the mean of a population using a sample large enough that the probability will be 0.95 that the sample mean will not differ from the population mean by more than 25 percent of the standard deviation. How large a sample should be taken?

Solution: We have

$$1 - \eta = 0.95 \Rightarrow \eta = 0.05, \varepsilon = 0.25\sigma$$

By the law of large numbers, we have

$$\begin{aligned} n &\geq \frac{\sigma^2}{\varepsilon^2 \eta} \\ &= \frac{\sigma^2}{(0.25\sigma)^2 \times 0.05} = 320 \end{aligned}$$

That is, $n \geq 320$

Therefore, at least 320 units must be taken in the sample so that the required probability will be 0.95.

Now, try the following exercise for your practice.

E8) The mean of a population is unknown and having a variance equal to 2. Find out that how large a sample must be taken, so that the probability will be at least 0.95 that the sample mean will lie within the range of 0.5 of the population mean?

We now end this unit by giving a summary of what we have covered in it.

1.7 SUMMARY

In this unit, we have covered the following points:

1. The statistical procedure which is used for drawing conclusions about the population parameter on the basis of the sample data is called “**statistical inference**”.
2. The group of units or items under study is known as “**Population**” whereas a part or a fraction of population is known as “**sample**”.
3. A “**parameter**” is a function of population values which is used to represent the certain characteristic of the population and any quantity calculated from sample values and does not contain any unknown population parameter is known as “**statistic**”.
4. Any statistic used to estimate an unknown population parameter is known as “**estimator**” and the particular value of the estimator is known as “**estimate**” of parameter.
5. The probability distribution of all possible values of a sample statistic that would be obtained by drawing all possible samples of the same size from the population is called “**sampling distribution**” of that statistic.

6. The standard deviation of the sampling distribution of a statistic is known as “**standard error**”.
7. The most important theorem of Statistics is “**central limit theorem**” which state that the sampling distribution of the sample means tends to normal distribution as sample size tends to large ($n > 30$).
8. According to law of large numbers, a positive integer can be determined such that if a random sample of size n or larger is taken from a population with parameter, say, population mean (μ), the probability that the sample mean will be very closed to population mean, can be made as close to 1 as desired.

1.8 SOLUTIONS / ANSWERS

- E1)** In (i) and (iv) we use only sample survey because the units (persons or blubs) are destroyed under investigation and the information gathering from all the elements is not making a sense whereas in (ii) and (iii) we can use sample as well as complete survey.
- E2)** (ii) and (iv) are the statistics because both do not contain the unknown parameters (i.e. μ and σ^2) whereas (i) and (iii) are not statistics because they contain unknown parameters.
- E3)** Here, we are given that

$$N = 5, n = 2$$

All possible samples (with replacement) are $N^n = 5^2 = 25$ which are shown in the Table 1.5 given below:

Table 1.5: Possible samples

Sample Number	Sample Observation	Sample Number	Sample Observation
1	56, 56	14	74, 45
2	56, 62	15	74, 50
3	56, 74	16	45, 56
4	56, 45	17	45, 62
5	56, 50	18	45, 74
6	62, 56	19	45, 45
7	62, 62	20	45, 50
8	62, 74	21	50, 56
9	62, 45	22	50, 62
10	62, 50	23	50, 74
11	74, 56	24	50, 45
12	74, 62	25	50, 50
13	74, 74		

- E4)** Here, we are given that

$$N = 3, n = 2$$

Since we have to estimate the average life of Televisions on the basis of samples of size $n = 2$ therefore, all possible samples (with replacement) are $N^n = 3^2 = 9$ and for each sample we calculate the sample mean as shown in Table 1.6 given below:

Table 1.6: Calculation of Sample Mean

Sample Number	Sample Observation	Sample Mean ($\bar{X}$)
1	8, 8	8
2	8, 6	7
3	8, 10	9
4	6, 8	7
5	6, 6	6
6	6, 10	8
7	10, 8	9
8	10, 6	8
9	10, 10	10

Since the arrangement of all possible values of sample mean with their corresponding probabilities is called the sampling distribution of mean thus, we arrange every possible value of sample mean with their respective probabilities in the following Table 1.7 given below:

Table 1.7: Sampling distribution of sample means

S.No.	Sample Mean ($\bar{X}$)	Frequency	Probability
1	6	1	1/9
2	7	2	2/9
3	8	3	3/9
4	9	2	2/9
5	10	1	1/9

E5) Here, we are given that

$$\mu = 200, \sigma = 4, n = 50$$

Since the sampling is done with replacement therefore we know that the standard error of sample mean for estimating population mean is given by

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{50}} = 0.57$$

E6) Here, we are given that

$$P = \frac{15}{100} = 0.15, n = 200$$

Therefore, we know that the standard error of sample proportion is given by

$$SE(p) = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.15 \times (1 - 0.15)}{200}} = 0.025$$

E7) Here, we are given that

$$\mu = 65, \sigma = 2.2, n = 40$$

Since we draw a large sample $n = 40 > 30$, therefore, by central limit theorem, the sampling distribution of the sample mean will follow a normal distribution with mean

$$E(\bar{X}) = \mu = 65$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(2.2)^2}{40} = 0.12$$

Therefore, the probability that the average height of these 40 students will lie between 64 and 65.5 inches is given by

$$P[64 < \bar{X} < 65.5] \quad [\text{See Fig.1.6}]$$

To get the value of this probability, we convert the normal variate $\bar{X}$ into a standard normal variate Z by the transformation

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 65}{\sqrt{0.12}} = \frac{\bar{X} - 65}{0.35}$$

Therefore, subtract 65 from each term and then divide each term by 0.35 in the above inequality. Thus, the probability becomes

$$\begin{aligned} P\left[\frac{64 - 65}{0.35} < \frac{\bar{X} - 65}{0.35} < \frac{65.5 - 65}{0.35}\right] &= P[-2.86 < Z < 1.43] \\ &= P[-2.86 < Z < 0] + P[0 < Z < 1.43] \\ &= P[0 < Z < 2.86] + P[0 < Z < 1.43] \\ &= 0.4979 + 0.4236 = 0.9215 \end{aligned}$$

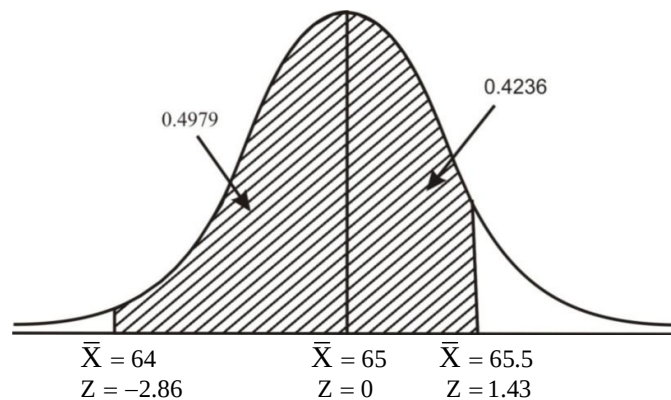


Fig. 1.6

E8) Here, we are given that

$$\sigma^2 = 2, \varepsilon = 0.5, 1 - \eta = 0.95 \Rightarrow \eta = 0.05$$

By the law of large numbers, we have

$$\begin{aligned} n &\geq \frac{\sigma^2}{\varepsilon^2 \eta} \\ &= \frac{2}{(0.5)^2 \times (0.05)} = 160 \end{aligned}$$

That is, $n \geq 160$

Therefore, at least 160 units must be taken in the sample so that the probability will be at least 0.95 that the sample mean will lie within 0.5 of the population mean.

UNIT 2 SAMPLING DISTRIBUTION(S) OF STATISTIC(S)

Structure

- 2.1 Introduction
 - Objectives
- 2.2 Sampling Distribution of Sample Mean
- 2.3 Sampling Distribution of Difference of Two Sample Means
- 2.4 Sampling Distribution of Sample Proportion
- 2.5 Sampling Distribution of Difference of Two Sample Proportions
- 2.6 Sampling Distribution of Sample Variance
- 2.7 Sampling Distribution of Ratio of Two Sample Variances
- 2.8 Summary
- 2.9 Solutions/Answers

2.1 INTRODUCTION

In previous unit, we have discussed the concept of sampling distribution of a statistic. There are so many problems in business and economics where it becomes necessary to draw inference about population parameters such as population mean, population proportion, population variance, difference of two population means, etc. For example, one may want to estimate the average height of the students in a college, a businessman may want to estimate the proportion of defective items in a production lot, a manufacturer of car tyres may want to estimate the variance of the diameter of tyres, a pharmacist may want to estimate the difference of effect of two types of drugs, etc.

Generally, sample mean is used to draw inference about the population mean. Similarly, sample proportion and sample variance are used to draw inference about the population proportion and population variance respectively. Therefore, it becomes necessary to know the sampling distribution of sample mean, sample proportion and sample variance, etc.

This unit is divided in 9 sections. Section 2.1 is introductive in nature. One of the most important sample statistics which is used to draw conclusion about population mean is sample mean. So in Section 2.2, the sampling distribution of mean is described, whereas in Section 2.3, the sampling distribution of difference of two sample means is explored. Second useful statistic generally used in Social and Business problems is sample proportion. The sampling distribution of sample proportion is described in Section 2.4 and in Section 2.5 sampling distribution of difference of two sample proportions is explored. Sometimes, it is useful to estimate the variation of population. For this we use sampling distribution of sample variance. The sampling distribution of sample variance is described in Section 2.6 whereas the sampling distribution of ratio of two sample variances is given in Section 2.7. Unit ends by providing summary of what we have discussed in this unit in Section 2.8 and solution of exercises in Section 2.9.

Objectives

After studying this unit, you should be able to:

- describe the need of sampling distributions of different statistics;

- explain the sampling distribution of sample mean;
- explore the sampling distribution of difference of two sample means;
- describe the sampling distribution of sample proportion;
- explore the sampling distribution of difference of two sample proportions;
- explain the sampling distribution of sample variance; and
- describe the sampling distribution of ratio of two sample variances.

2.2 SAMPLING DISTRIBUTION OF SAMPLE MEAN

One of the most important sample statistics which is used to draw conclusion about the population mean is sample mean. For example, an investigator may want to estimate average income of the peoples living in a particular geographical area, a product manager may want to estimate the average life of electric bulbs manufactured by a company, a pathologist may want to estimate the mean time required to complete a certain analysis, etc. In the above cases an estimate of the population mean is required and one may estimate this on the basis of a sample taken from that population. For this, sampling distribution of sample mean is required.

We have already given you the flavor of sampling distribution of sample mean with the help of an example in Section 1.3 of previous unit in which we draw all possible samples of same size from the population and calculate the sample mean for each sample. After calculating the value of sample mean for each sample we observed that the values of sample mean vary from sample to sample. Then the sample mean is treated as random variable and a probability distribution is constructed for the values of sample mean. This probability distribution is known as sampling distribution of sample mean. Therefore, the sampling distribution of sample mean can be defined as:

“The probability distribution of all possible values of sample mean that would be obtained by drawing all possible samples of the same size from the population is called sampling distribution of sample mean or simply says sampling distribution of mean.”

Now, the question may arise in your mind “what is the shape of sampling distribution of mean, it is normal, exponential, binomial, Poisson, etc.?”

With the help of Fig. 1.1 of previous unit, we have described the shape of the sampling distribution of mean for different populations and for varying sample sizes. From this figure, we observed that when the parent population is normal then all the sampling distributions for varying sample sizes are also normal whereas when parent population is uniform, binomial, exponential then the shapes of the sampling distributions of mean are not in the form of specify distribution when sample size is small ($n = 2$ or 5).

Generally, when samples are drawn non-normal populations then it is not possible to specify the shape of the sampling distribution of mean when the sample size is small. Although when sample size is large (> 30) then we observed that sampling distribution of mean converges to normal distribution whatever the form of the population i.e. normal or non-normal.

After knowing the shapes of the sampling distribution of mean in different situations, you may be interested to know the mean and variance of the sampling distribution of mean when samples are drawn from normal population.

The sampling distribution of sample means is a theoretical probability distribution of sample means that would be obtained by drawing all possible samples of the same size from the population.

In practice, only one random sample is actually selected and the concept of sampling distribution is used to draw the inference about the population parameters. If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a normal population with mean μ and variance σ^2 then it has also been established that sampling distribution of sample mean $\bar{X}$ is also normal. The mean and variance of sampling distribution of $\bar{X}$ can be obtained as

$$\begin{aligned}\text{Mean of } \bar{X} &= E(\bar{X}) = E\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] \quad [\text{By definition of } \bar{X}] \\ &= \frac{1}{n}[E(X_1) + E(X_2) + \dots + E(X_n)]\end{aligned}$$

Since $X_1, X_2, \dots, X_n$ are randomly drawn from same population so they also follow the same distribution as the population. Therefore,

$$E(X_1) = E(X_2) = \dots = E(X_n) = E(X) = \mu$$

and

$$\text{Var}(X_1) = \text{Var}(X_2) = \dots = \text{Var}(X_n) = \text{Var}(X) = \sigma^2$$

Thus,

$$\begin{aligned}E(\bar{X}) &= \frac{1}{n} \left(\underbrace{\mu + \mu + \dots + \mu}_{n\text{-times}} \right) \\ &= \frac{1}{n}(n\mu) = \mu\end{aligned}$$

$$E(\bar{X}) = \mu$$

and variance

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Var}\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{n^2} \left(\underbrace{\sigma^2 + \sigma^2 + \dots + \sigma^2}_{n\text{-times}} \right) = \frac{1}{n^2}(n\sigma^2) \\ \text{Var}(\bar{X}) &= \frac{\sigma^2}{n}\end{aligned}$$

Hence, we conclude that if the samples are drawn from normal population with mean μ and variance σ^2 then the sampling distribution of mean $\bar{X}$ is also normal distribution with mean μ and variance σ^2/n , that is,

$$\text{If } X_i \sim N(\mu, \sigma^2) \text{ then } \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

The standard error (SE) of sample mean can be obtained by the definition of SE as

$$\text{SE}(\bar{X}) = \text{SD}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$$

Note 1: The statistical inference is based on the theoretical distribution so in whole statistical inference, we assume that the random sample is selected from the infinite population or from a finite population with replacement.

Sampling Distribution(s) of Statistic(s)

The statistical inference is based on the theoretical distribution so if we say that a random sample, say, $X_1, X_2, \dots, X_n$ is taken from a population its means that the random sample is selected from the infinite population or from a finite population with replacement such that X_i 's are independent and followed same population distribution.

If X and Y are two independent random variables and a & b are two constant then by the addition theorem of expectation, we have

$$E(aX + bY) = aE(X) + bE(Y)$$

and

$$\begin{aligned}\text{Var}(aX + bY) &= a^2\text{Var}(X) + b^2\text{Var}(Y)\end{aligned}$$

If the sampling is done without replacement from a finite normal population of size N then the sample mean is distributed normally with mean

$$E(\bar{X}) = \mu$$

and variance

$$\text{Var}(\bar{X}) = \frac{N-n}{N-1} \frac{\sigma^2}{n}$$

The proof of this formula is beyond the scope of this course.

Therefore, standard error of sample mean is given by

$$\text{SE}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}$$

If the sample is drawn from any population other than normal then by central limit theorem, the sampling distribution of $\bar{X}$ tends to normal as the sample size n increases, that means, the sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n when the sample size is large (> 30).

Let us see an application of the sampling distribution of mean with the help of an example.

Example 1: Diameter of a steel ball bearing produced on a semi-automatic machine is known to be distributed normally with mean 12 cm and standard deviation 0.1 cm. If we take a random sample of size 10 then find

- (i) Mean and variance of sampling distribution of mean.
- (ii) The probability that the sample mean lies between 11.95 cm and 12.05 cm.

Solution: Here, we are given that

$$\mu = 12, \sigma = 0.1, n = 10$$

- (i) Since the population is normally distributed therefore the sampling distribution of the sample mean also follows a normal distribution with mean

$$E(\bar{X}) = \mu = 12$$

and variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(0.1)^2}{10} = 0.001$$

- (ii) The probability that the sample mean lies between 11.95 cm and 12.05 cm is given by

$$P[11.95 < \bar{X} < 12.05] \quad [\text{See Fig.2.1}]$$

To get the value of this probability, we convert $\bar{X}$ into a standard normal variate Z by the transformation

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 12}{\sqrt{0.001}} = \frac{\bar{X} - 12}{0.03}$$

Therefore, by subtracting 12 from each term and then dividing each term by 0.03 in the above inequality, we get required probability as

$$\begin{aligned}
 P\left[\frac{11.95-12}{0.03} < \frac{\bar{X}-12}{0.03} < \frac{12.05-12}{0.03}\right] \\
 &= P[-1.67 < Z < 1.67] \\
 &= P[-1.67 < Z < 0] + P[0 < Z < 1.67] \\
 &= P[0 < Z < 1.67] + P[0 < Z < 1.67] \\
 &= 2P[0 < Z < 1.67] \\
 &= 2 \times 0.4525 \quad \left[\begin{array}{l} \text{Using table for area} \\ \text{under normal curve} \end{array} \right] \\
 &= 0.9050
 \end{aligned}$$

Therefore, the probability that the sample mean lies between 11.95 and 12.05 cm is 0.9050.

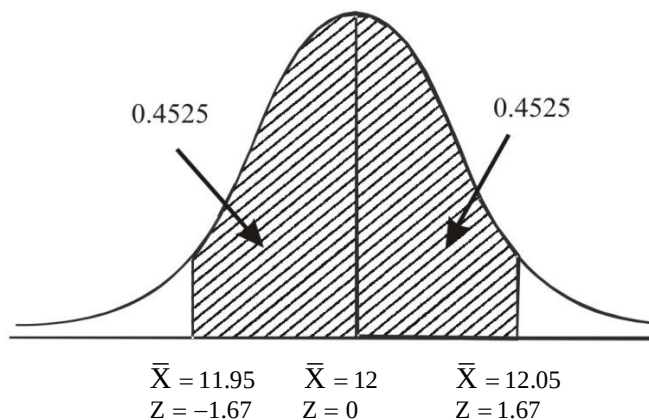


Fig. 2.1

Now, continuing our discussion about the sampling distribution of mean, we consider another situation.

In the cases described in last two pages, we assume that variance σ^2 of the normal population is known but in general it is not known and in such a situation the only alternative left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the σ^2 where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

Thus, in this case, it has been established that the sampling distribution of mean is not normal and the variate $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is known as t-variate and follows t-distribution with $(n-1)$ df instead of normal distribution. This distribution has wide applications in Statistics therefore it will be described in detail in Unit 3 of this course.

If n is sufficiently large (> 30) then we know that almost all the distributions are very closely approximated by normal distribution. Thus in this case, t-distribution is also approximated normal distribution. So for large sample size (> 30), the sampling distribution of mean also follows normal distribution with mean μ and variance S^2/n .

Now, try some exercises for your practice.

-
- E1)** If parent population is normal with mean μ and variance σ^2 then, what is sampling distribution of sample means?
- E2)** The weight of certain type of a truck tyre is known to be distributed normally with mean 200 pounds and standard deviation 4 pounds. A random sample of 10 tyres is selected.
- What is the sampling distribution of sample mean? Also obtain the mean and variance of this distribution.
 - Find the probability that the mean of this sample is greater than or equal to 202 pounds.
- E3)** The mean life of electric bulbs produced by a company is 2550 hours. A random sample of 100 bulbs is selected and the standard deviation is found to be 54 hours. Find the mean and variance of the sampling distribution of mean.
-

2.3 SAMPLING DISTRIBUTION OF DIFFERENCE OF TWO SAMPLE MEANS

There are so many problems in business and economics where someone may be interested to draw the inference about the difference of two population means. For example, two manufacturing companies of bulbs are produced same type of bulbs and one may be interested to know which one is better than the other, an investigator may want to know the difference of average income of the peoples living in a two cities, say, A and B, two different types of drugs, were tried on certain number of patients for controlling blood pressure and one may be interested to know which one has better effect on controlling blood pressure, etc. Therefore, in such situation, to draw the inference we require the sampling distribution of difference of two sample means.

Let the same characteristic measures from two populations be represented by X and Y variables and the variation in the values of these constitute two population, say, population-I for variation in X and population-II for variation in Y. Suppose population-I having mean μ_1 and variance σ_1^2 and population-II having mean μ_2 and variance σ_2^2 . Then we take all possible samples of same size n_1 from population-I and then the sample mean, say, $\bar{X}$ is calculated for each sample. Similarly, all possible samples of same size n_2 are taken from the population-II and the sample mean, say, $\bar{Y}$ is calculated for each sample. Then we consider all possible differences of means $\bar{X}$ and $\bar{Y}$. The difference of these means may or may not differ from sample to sample and so we construct the probability distribution of these differences. The probability distribution thus obtained is known as sampling distribution of the difference of sample means. Therefore, the sampling distribution of difference of sample means can be defined as:

“The probability distribution of all values of the difference of two sample means would be obtained by drawing all possible samples from both the populations is called sampling distribution of difference of two sample means.”

The sa
of diff
means
sampli
differe
($\bar{x} - \bar{y}$)
obtain
possibl
the pop

If both the parent populations are normal, that is,

$$X \sim N(\mu_1, \sigma_1^2) \text{ and } Y \sim N(\mu_2, \sigma_2^2)$$

then as we discussed in previous section that

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \text{ and } \bar{Y} \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

Thus, as we have discussed in Unit 4 of MST-003 that if two independent random variables X and Y are normally distributed then the difference $(X - Y)$ of these also normally distributed. Therefore, in our case, the sampling distribution of difference of two sample means $(\bar{X} - \bar{Y})$ also follows normal distribution with mean

$$E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2$$

and variance

$$\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Therefore, standard error of difference of two sample means is given by

$$\text{SE}(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

If the parent populations are not normal but n_1 and n_2 are large (> 30), then by central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely

normally distributed with mean $(\mu_1 - \mu_2)$ and variance $\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$.

Let us see an application of the sampling distribution of difference of two sample means with the help of an example.

Example 2: Electric CFL manufactured by company A have mean lifetime of 2400 hours with standard deviation 200 hours, while CFL manufactured by company B have mean lifetime of 2200 hours with standard deviation of 100 hours. If random samples of 125 electric CFL of each company are tested, find

- The mean and standard error of the sampling distribution of the difference of mean lifetime of electric CFLs.
- The probability that the CFLs of company A will have a mean lifetime at least 160 hours more than the mean lifetime of the CFLs of company B.

Solution: Here, we are given that

$$\mu_1 = 2400, \sigma_1 = 200, \mu_2 = 2200, \sigma_2 = 100$$

$$n_1 = n_2 = 125$$

- Let $\bar{X}$ and $\bar{Y}$ denote the mean lifetime of CFLs taken from companies A and B respectively. Since n_1 and n_2 are large ($n_1, n_2 > 30$) therefore, by the central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ follows normal distribution with mean

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2 = 2400 - 2200 = 200$$

and variance

$$\text{Var}(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If X and Y are independent random variables then

$$E(X - Y) = E(X) - E(Y)$$

and

$$\text{Var}(X - Y)$$

$$= \text{Var}(X) + \text{Var}(Y)$$

Sampling Distributions

$$= \frac{(200)^2}{125} + \frac{(100)^2}{125} = 400$$

Therefore, the standard error is given by

$$SE(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{400} = 20$$

(ii) Here, we want to find out the probability which is given by

$$P[\bar{X} \geq 160 + \bar{Y}] = P[(\bar{X} - \bar{Y}) \geq 160] \quad [\text{See Fig. 2.2}]$$

To get the value of this probability, we convert variate $(\bar{X} - \bar{Y})$ into a standard normal variate Z by the transformation

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{\sqrt{\text{Var}(\bar{X} - \bar{Y})}} = \frac{(\bar{X} - \bar{Y}) - 200}{\sqrt{400}} = \frac{(\bar{X} - \bar{Y}) - 200}{20}$$

Therefore, by subtracting 200 from each term and then dividing each term by 20 in the above inequality, we get the required probability

$$\begin{aligned} P\left[\frac{(\bar{X} - \bar{Y}) - 200}{20} \geq \frac{160 - 200}{20}\right] &= P[Z \geq -2] \\ &= P[-2 < Z < 0] + P[0 < Z < \infty] \\ &= P[0 < Z < 2] + P[0 < Z < \infty] \\ &= 0.4772 + 0.5 = 0.9772 \end{aligned}$$

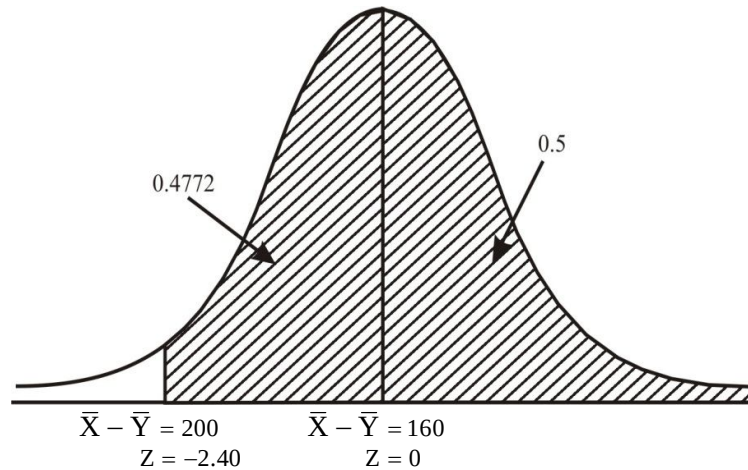


Fig. 2.2

Now, continuing our discussion about the sampling distribution of difference of two means, we consider another situation.

If population variances σ_1^2 & σ_2^2 are unknown then we estimate σ_1^2 & σ_2^2 by the values of the sample variances S_1^2 & S_2^2 of the samples taken from the first and second population respectively. And for large sample sizes n_1 and n_2 (> 30), the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely normally distributed with

mean $(\mu_1 - \mu_2)$ and variance $\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)$.

If population variances σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 = \sigma_2^2 = \sigma^2$ then σ^2 is estimated by pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [n_1 S_1^2 + n_2 S_2^2]$$

and variate

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1 + n_2 - 2)}$$

follows t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom. As similar as sampling distribution of mean, for large sample sizes n_1 and n_2 (> 30) the sampling distribution of $(\bar{X} - \bar{Y})$ is very closely normally distributed with mean $(\mu_1 - \mu_2)$ and variance $S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$.

Now, you can try the following exercise.

-
- E4)** The average height of all the workers in a hospital is known to be 68 inches with a standard deviation of 2.3 inches whereas the average height of all the workers in a company is known to be 65 inches with a standard deviation of 2.5 inches. If a sample of 35 hospital workers and a sample of 50 company workers are selected at random, what is the probability that the sample mean of height of hospital workers is at least 2 inch greater than that of company workers?
-

2.4 SAMPLING DISTRIBUTION OF SAMPLE PROPORTION

In Section 2.2, we have discussed the sampling distribution of sample mean. But in many real word situations, in business and other areas where the data collected in form of counts or the collected data classified into two categories or groups according to an attribute. For example, the peoples living in a colony may be classified into two groups (male and female) with respect to the characteristic sex, the patients in a hospital may be classified into two groups as cancer and non-cancer patients, the lot of articles may be classified as defective and non-defective, etc.

Generally, such types of data are considered in terms of proportion of elements / individuals / units / items possess (success) or not possess (failure) a given characteristic or attribute. For example, the proportion of female in the population, proportion of cancer patents in a hospital, proportion of defective articles in a lot, etc.

In such situations, we deal with population proportion instead of population mean.

When population proportion is unknown and the whole population is too large to find out the proportion. In such a situation, to draw the inference about the population proportion, we require the sampling distribution of sample proportion.

Sampling Distributions

For sampling distribution of sample proportion, we draw all possible samples from the population and for each sample we calculate the sample proportion p as

$$p = \frac{X}{n} \leq 1$$

where, X is the number of observations / individuals / items / units in the sample which have the particular characteristic under study and n is the total number of observations in the sample i.e. sample size.

For better understanding of the process, we consider the following example in which size of the population is very small:

Suppose, there is a lot of 3 cartons A, B & C of electric bulbs and each carton contains 20 bulbs. The number of defective bulbs in each carton is given below:

Table 2.1: Number of Defective Bulbs per Carton

Carton	Number of Defectives Bulbs
A	2
B	4
C	1

The population proportion of defective bulbs can be obtained as

$$P = \frac{2 + 4 + 1}{20 + 20 + 20} = \frac{7}{60}$$

Now, let us assume that we do not know the population proportion of defective bulbs. So we decide to estimate population proportion of defective bulbs on the basis of samples of size $n = 2$. There are $N^n = 3^2 = 9$ possible samples of size 2 with replacement. The all possible samples and their respective proportion defectives are given in the following table:

Table 2.2: Calculation of Sample Proportion

Sample	Sample Carton	Sample Observation	Sample Proportion(p)
1	(A, A)	(2, 2)	4/40
2	(A, B)	(2, 4)	6/40
3	(A, C)	(2, 1)	3/40
4	(B, A)	(4, 2)	6/40
5	(B, B)	(4, 4)	8/40
6	(B, C)	(4, 1)	5/40
7	(C, A)	(1, 2)	3/40
8	(C, B)	(1, 4)	5/40
9	(C, C)	(1, 1)	2/40

From the above table, we can see that value of the sample proportion is varying from sample to sample. So we consider all possible sample proportions and calculate their probability of occurrence. Since there are 9 possible samples therefore the probability of selecting a sample is $1/9$. Then we arrange the possible sample proportion with their respective probability in Table 2.3 given in next page:

Table 2.3: Sampling Distribution of Sample Proportion

S.No.	Sample Proportion(p)	Frequency	Probability
1	2/40	1	1/9
2	3/40	2	2/9
3	4/40	1	1/9
4	5/40	2	2/9
5	6/40	2	2/9
6	8/40	1	1/9

**Sampling Distribution(s)
of Statistic(s)**

This distribution is called the sampling distribution of sample proportion.

Thus, we can define the sampling distribution of sample proportion as:

“The probability distribution of all values of the sample proportion that obtained by drawing all possible samples of same size from the population is called the sampling distribution of the sample proportion.”

This distribution also has its mean. Therefore, the mean of sampling distribution of sample proportion can be obtained as

$$\bar{p} = \frac{1}{K} \sum_{i=1}^k p_i f_i \quad \text{where, } K = \sum_{i=1}^k f_i$$

$$= \frac{1}{9} \left(\frac{2}{40} \times 1 + \frac{3}{40} \times 2 + \dots + \frac{8}{40} \times 1 \right) = \frac{7}{60}$$

The sampling distribution of the sample proportion is a theoretical probability distribution of sample proportions that would be obtained by drawing all possible samples of the same size from the population.

Thus, we have seen that mean of sample proportion is equal to the population proportion.

As we have already mentioned in the previous unit that finding mean, variance and standard error from this process is tedious so we calculate these by another short-cut method when population proportion is known.

As we discussed in Unit 5 of course MST-003 that if a population whose elements are divided into two mutually exclusive groups– one containing the elements which possess a certain attribute (success) and other containing elements which do not possess the attribute(failure), then number of successes(elements possess a certain attribute) follows a binomial distribution with mean

$$E(X) = nP$$

and variance

$$\text{Var}(X) = nPQ \quad \text{where } Q = 1 - P$$

where, P is the probability or proportion of success in the population.

Now, we can easily find the mean and variance of the sampling distribution of sample proportion by using the above expression as

$$E(p) = E\left(\frac{X}{n}\right)$$

$$= \frac{1}{n} E(X) = \frac{1}{n} nP$$

$$E(p) = P$$

and variance

$$\begin{aligned}
 \text{Var}(p) &= \text{Var}\left(\frac{X}{n}\right) \\
 &= \frac{1}{n^2} \text{Var}(X) \quad \left[\because \text{Var}(aX) = a^2 \text{Var}(X) \right] \\
 &= \frac{1}{n^2} nPQ \quad \left[\because \text{Var}(X) = nPQ \right] \\
 \text{Var}(p) &= \frac{PQ}{n}
 \end{aligned}$$

Also standard error of sample proportion can be obtained as

$$SE(p) = \sqrt{\text{Var}(p)} = \sqrt{\frac{PQ}{n}}$$

If the sampling is done without replacement from a finite population then the mean and variance of sample proportion is given by

$$E(p) = P$$

and variance

$$\text{Var}(p) = \frac{N-n}{N-1} \frac{PQ}{n}$$

where, N is the population size and the factor $(N-n) / (N-1)$ is called finite population correction.

If sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean P and variance PQ/n where, $Q = 1 - P$.

Let us see an application of the sampling distribution proportion with the help of an example.

Example 3: A machine produces a large number of items of which 15% are found to be defective. If a random sample of 200 items is taken from the population and sample proportion is calculated then find

- (i) Mean and standard error of sampling distribution of proportion.
- (ii) The probability that less than or equal to 12% defectives are found in the sample.

Solution: Here, we are given that

$$P = \frac{15}{100} = 0.15, n = 200$$

- (i) We know that when sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then sample proportion p is approximately normally distributed with mean P and variance PQ/n where, $Q = 1 - P$. But here the sample proportion is not given so we assume that the conditions of normality hold, that is, $np > 5$ and $nq > 5$. So mean of sampling distribution of sample proportion is given by

$$E(p) = P = 0.15$$

and variance

**Sampling Distribution(s)
of Statistic(s)**

$$\text{Var}(p) = \frac{PQ}{n} = \frac{0.15 \times 0.85}{200} = 0.0006$$

Therefore, the standard error is given by

$$\text{SE}(p) = \sqrt{\text{Var}(p)} = \sqrt{0.0006} = 0.025$$

- (ii) The probability that the sample proportion will be less than or equal to 12% defectives is given by

$$P[p \leq 0.12] \quad [\text{See Fig. 2.3}]$$

To get the value of this probability, we can convert the random variate p into standard normal variate Z by the transformation

$$Z = \frac{p - E(p)}{\sqrt{\text{Var}(p)}} = \frac{p - 0.15}{\sqrt{0.0006}} = \frac{p - 0.15}{0.025}$$

Therefore, by subtracting 0.15 from each term and then dividing each term by 0.025 in the above inequality, we get required probability

$$\begin{aligned} P\left[\frac{p - 0.15}{0.025} \leq \frac{0.12 - 0.15}{0.025}\right] &= P[Z \leq -1.20] \\ &= 0.5 - P[-1.20 < Z \leq 0] \\ &= 0.5 - P[0 \leq Z < 1.20] \\ &= 0.5 - 0.3849 \\ &= 0.1151 \end{aligned}$$

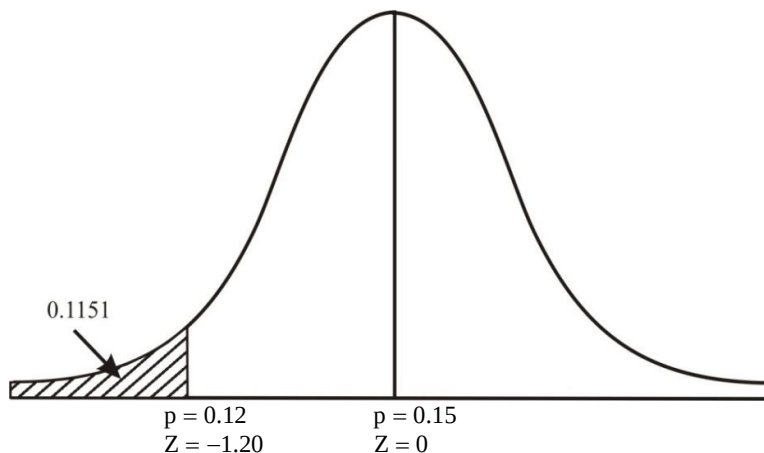


Fig. 2.3

Now, it is time for you to try the following exercise to make sure that you have learnt the sampling distribution of sample proportion.

-
- E5)** A state introduced a policy to give loan to unemployed doctors to start own clinic. Out of 10000 unemployed doctors 7000 accept the policy and got the loan. A sample of 100 unemployed doctors is taken at the time of allotment of loan. What is the probability that sample proportion would have exceeded 60% acceptance.
-

2.5 SAMPLING DISTRIBUTION OF DIFFERENCE OF TWO SAMPLE PROPORTIONS

In Section 2.3, we have discussed the sampling distribution of difference of two sample means. In some cases, we are interested to comparative study that the proportions of an attributes in the two different populations or group are equal or not. For example, one may wish to test whether the proportions of alcohol drinkers in the two cities are same, one may wish to compare the proportions of literates between two groups of people, one may estimate the decrease in the proportions of consumption of tea after the increase in excise duty on it, etc. Therefore, we require the sampling distribution of difference of two sample proportions for getting the inference about two populations.

Suppose, there are two populations, say, population-I and population-II under study and the population-I having population proportion P_1 and population-II having population proportion P_2 according to an attribute. For describing the sampling distribution of difference of two sample proportions, we consider all possible samples of same size n_1 taken from population-I and for each sample calculate the sample proportion p_1 of success. Similarly, determine the sample proportion p_2 of success by considering all possible sample of same size n_2 from population-II. Then we consider all possible differences of proportions p_1 and p_2 . The difference of these proportions may or may not be differ so we construct the probability distribution of these differences. The probability distribution thus obtained is called the sampling distribution of the difference of sample proportions. Therefore, the sampling distribution of difference of two sample proportions can be defined as:

“The probability distribution of all values of the difference of two sample proportions that have been obtained by drawing all possible samples of same sizes from both the populations is called sampling distribution of difference between two sample proportions.”

As we have seen in case of single proportion described in previous section that if sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean P and variance PQ/n where, $Q = 1 - P$. Therefore, if n_1 and n_2 are sufficiently large, such that $n_1p_1 > 5$, $n_1q_1 > 5$, $n_2p_2 > 5$ and $n_2q_2 > 5$ then

$$p_1 \sim N\left(P_1, \frac{P_1Q_1}{n_1}\right) \text{ and } p_2 \sim N\left(P_2, \frac{P_2Q_2}{n_2}\right)$$

where, $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$.

Also, by the property of normal distribution described in Unit 13 of MST 003, the sampling distribution of the difference of sample proportions follows normal distribution with mean

$$E(p_1 - p_2) = E(p_1) - E(p_2) = P_1 - P_2$$

and variance

$$\text{Var}(p_1 - p_2) = \text{Var}(p_1) + \text{Var}(p_2) = \frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}$$

That is,

The sa
differ
propor
sampli
differ
propor
would
drawin
from b

$$p_1 - p_2 \sim N\left(P_1 - P_2, \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}\right)$$

Thus, standard error is given by

$$SE(p_1 - p_2) = \sqrt{\text{Var}(p_1 - p_2)} = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Let us see an application of the sampling distribution of difference of two sample proportions with the help of an example.

Example 4: In one population, 30% persons had blue-eyed and in second population 20% had the same blue-eye. A random sample of 200 persons is taken from each population independently and calculate the sample proportion for both samples, then find the probability that the difference in sample proportions is less than or equal to 0.02.

Solution: Here, we are given that

$$P_1 = 0.30, P_2 = 0.20,$$

$$n_1 = n_2 = 200$$

Let p_1 and p_2 be the sample proportions of blue-eye persons in the samples taken from both the populations respectively. We know that when n_1 and n_2 are sufficiently large, such that $n_1 p_1 > 5$, $n_1 q_1 > 5$, $n_2 p_2 > 5$ and $n_2 q_2 > 5$ then sampling distribution of the difference between two sample proportions is approximately normally distributed. But here the sample proportions are not given so we assume that the conditions of normality hold. So mean of sampling distribution of the difference between two sample proportions is given by

$$E(p_1 - p_2) = P_1 - P_2 = 0.30 - 0.20 = 0.10$$

and variance

$$\begin{aligned} \text{Var}(p_1 - p_2) &= \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2} \\ &= \frac{0.30 \times 0.70}{200} + \frac{0.20 \times 0.80}{200} \\ &= 0.0019 \end{aligned}$$

The probability that the difference in sample proportions is less than or equal to 0.02 is given by

$$P[p_1 - p_2 \leq 0.02] \quad [\text{See Fig. 2.4}]$$

To get the value of this probability, we convert variate $(p_1 - p_2)$ into a standard normal variate Z by the transformation

$$Z = \frac{(p_1 - p_2) - E(p_1 - p_2)}{\sqrt{\text{Var}(p_1 - p_2)}} = \frac{(p_1 - p_2) - 0.10}{\sqrt{0.0019}} = \frac{(p_1 - p_2) - 0.10}{0.04}$$

Therefore, by subtracting 0.10 from each term and then dividing each term by 0.04 in the above inequality, we get the required probability as

$$\begin{aligned}
 P\left[\frac{(p_1 - p_2) - 0.10}{0.04} \leq \frac{0.02 - 0.10}{0.04}\right] &= P[Z \leq -2.0] \\
 &= 0.5 - P[-2.0 \leq Z \leq 0] \\
 &= 0.5 - P[0 \leq Z \leq 2.0] \\
 &= 0.5 - 0.4772 = 0.0228
 \end{aligned}$$

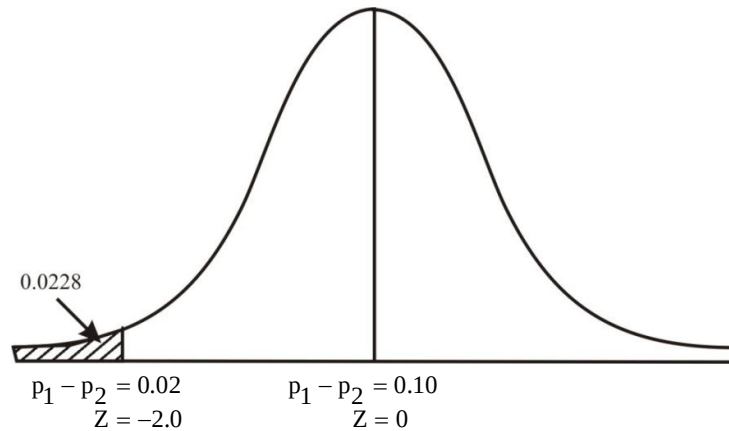


Fig. 2.4

Now, try to solve the following exercises to ensure that you understand this section properly.

-
- E6)** In city A, 12% persons were found to be alcohol drinkers and in another city B, 10% persons were found alcohol drinkers. If 100 persons of city A and 150 persons of city B are selected randomly, what is the probability that the difference in sample proportions lies between 0.01 and 0.08?
-

2.6 SAMPLING DISTRIBUTION OF SAMPLE VARIANCE

In Sections 2.2 and 2.4, we have discussed the sampling distributions of sample mean and sample proportion respectively. But many practical situations concerned with the variability. For example, a manufacturer of steel ball bearings may want to know about the variation of diameter of steel ball bearing, a life insurance company may be interested in the variation of the number of policies in different years, etc. Therefore, we need information about the sampling distribution of sample variance.

For describing the sampling distribution of the sample variance, we consider all possible sample of same size, say, n taken from the population having variance σ^2 and for each sample we calculate sample variance S^2 . The values of S^2 may vary from sample to sample so we construct the probability distribution of sample variances. The probability distribution thus obtained is known as sampling distribution of the sample variance. Therefore, the sampling distribution of sample variance can be defined as:

“The probability distribution of all values of the sample variance would be obtained by drawing all possible sample of same size from the parent population is called the sampling distribution of the sample variance.”

Let the characteristic measures from the population be represented by X variable and the variation in the values of X constitute a population. Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population with mean μ and variance σ^2 . W.S. Gosset first described the sampling distribution of sample variance S^2 . Here, it is to be noted that S^2 always be positive, therefore S^2 does not follow normal distribution because the normal variate takes positive as well as negative values. The distribution of sample variance can not be obtained directly therefore in this case, some transformation is made by multiplying S^2 by $(n-1)$ and then dividing the product by σ^2 . The obtained new variable is denoted by the symbol χ^2 read as chi-square and follows the chi-square distribution with $(n-1)$ degrees of freedom as,

For degrees of freedom, please see Section 3.2 of next unit.

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2$$

where, S^2 is called sample variance and given below

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{for all } i=1, 2, \dots, n$$

The chi-square distribution has wild applications in statistics, therefore, it will be described in detail in the next unit of this course.

For solving the problems relating to sampling distribution of variance, we convert the variate S^2 in a χ^2 -variate by above transformation. For better understanding, we explain it by taking an example given below.

Example 5: A manufacturer of steel ball bearings has found the variance of diameter of ball bearings 0.0018 inches². What is the probability that a random sample of 25 ball bearings will result in a sample variance at least 0.002 inches²?

Solution: Here, we are given that

$$\sigma^2 = 0.0018, n = 25$$

The probability that the sample variance is at least 0.002 inches² is given by

$$P[S^2 \geq 0.002]$$

To get the value of this probability, we convert variate S^2 into χ^2 -variate which follows the chi-square distribution with $n-1 = 24$ degrees of freedom by the transformation

$$\chi_{(n-1)}^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{24S^2}{0.0018}$$

Therefore, multiplying each term by 24 and then dividing each term by 0.0018 in the above inequality. We get the required probability as

$$P\left[\frac{24S^2}{0.0018} \geq \frac{24 \times 0.002}{0.0018}\right] = P\left[\chi_{(24)}^2 \geq 26.67\right]$$

From the chi-square table, we see that $P[\chi_{(24)}^2 \geq 15.66] = 0.90$ and

$$P[\chi_{(24)}^2 \geq 33.20] = 0.10.$$

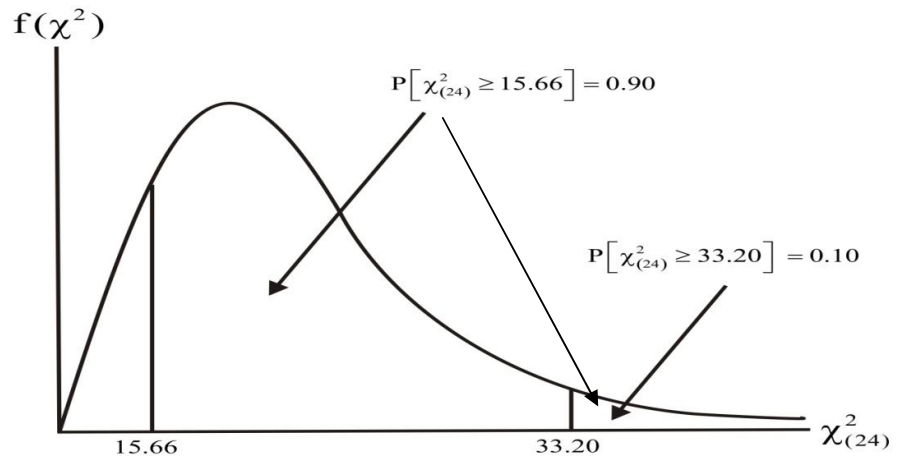


Fig. 2.5

Therefore, the above probability must lie between 0.10 and 0.90, that is,

$$0.10 < P[\chi^2 \geq 26.67] = P(S^2 \geq 0.002) < 0.90$$

Now, you can try the following exercises.

-
- E7)** The weight of certain type of truck tyres has a variance of 11 pounds². A random sample of 20 tyres is selected. What is the probability that the variance of this sample is greater than or equal to 16 pounds²?
-

2.7 SAMPLING DISTRIBUTION OF RATIO OF TWO SAMPLE VARIANCES

In previous section, we have discussed about the sampling distribution of sample variance where we were dealing with one population. Now, one may be interested to pursue for the comparative study of the variances of the two normal populations as we have done for the means and the proportions. For example, suppose a quality control engineer wants to determine whether or not the variance of the quality of the product is changing over the time, an economist may wish to know whether the variability in incomes differs across the two populations, etc. For solving such types of problems we require the sampling distribution of the ratio of sample variances.

Suppose there are two normal populations, say, population-I and population-II under study, the population-I has variance σ_1^2 and population-II has variance σ_2^2 . For describing the sampling distribution for ratio of population variances, we consider all possible samples of same size n_1 from population-I and for each sample we calculate sample variance S_1^2 . Similarly, calculate sample variance S_2^2 from each sample of same size n_2 drawn from population-II.

Then we consider all possible values of the ratio of the variances S_1^2 and S_2^2 , The values of S_1^2 / S_2^2 may vary from sample to sample so we construct the probability distribution of the ratio of the sample variances. The probability distribution thus obtained is known as sampling distribution of the ratio of sample variances. Therefore, the sampling distribution of ratio of sample variances can be defined as:

“The probability distribution of all values of the ratio of two sample variances would be obtained by drawing all possible samples from both the populations is called sampling distribution of ratio of sample variances.”

The sampling distribution of ratio of variances is given by Prof. R. A. Fisher in 1924. According to Prof. R. A. Fisher, the ratio of two independent chi-square variates when divided by their respective degrees of freedom follows F-distribution as

$$F = \frac{\chi^2_{(n_1-1)} / (n_1-1)}{\chi^2_{(n_2-1)} / (n_2-1)}$$

Since $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$ therefore,

$$F = \frac{(n_1-1)S_1^2 / \sigma_1^2 (n_1-1)}{(n_2-1)S_2^2 / \sigma_2^2 (n_2-1)}$$

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F_{(n_1-1, n_2-1)}$$

If $\sigma_1^2 = \sigma_2^2$ then

$$F = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)}$$

Therefore, the sampling distribution of ratio of sample variances follows F-distribution with $(n_1 - 1, n_2 - 1)$ degrees of freedom. The F-distribution is described in detail in fourth unit of this block.

With this we end this unit. We now summarise our discussion.

2.8 SUMMARY

In this unit, we have discussed following points:

1. The need of sampling distributions of different statistics.
2. The sampling distribution of sample mean.
3. The sampling distribution of difference between two sample means.
4. The sampling distribution of sample proportion.
5. The sampling distribution of difference of two sample proportions.
6. The sampling distribution of sample variance.
7. The sampling distribution of ratio of two sample variances.

2.9 SOLUTIONS / ANSWERS

E1) Since parent population is normal so sampling distribution of sample means is also normal with mean μ and variance σ^2/n .

E2) Here, we are given that

$$\mu = 200, \sigma = 4, n = 10$$

- (i) Since parent population is normal so sampling distribution of sample means is also normal. Therefore, the mean of this

Sampling Distributions

distribution is given by

$$E(\bar{X}) = \mu = 200$$

and variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{(4)^2}{10} = \frac{16}{10} = 1.6$$

(ii) Now, the probability that the sample mean $\bar{X}$ is greater than or equal to 202 pounds is given by

$$P[\bar{X} \geq 202] \quad [\text{See Fig. 2.6}]$$

To get the value of this probability, we convert $\bar{X}$ into a standard normal variate Z by the transformation

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 200}{\sqrt{1.6}} = \frac{\bar{X} - 200}{1.26}$$

Therefore, by subtracting 200 from each term and then dividing each term by 1.26 in the above inequality, we get required probability as

$$\begin{aligned} P\left[\frac{\bar{X} - 200}{1.26} \geq \frac{202 - 200}{1.26}\right] &= P[Z \geq 1.59] \\ &= 0.5 - P[0 \leq Z \leq 1.59] \\ &= 0.5 - 0.4441 = 0.0559 \end{aligned}$$

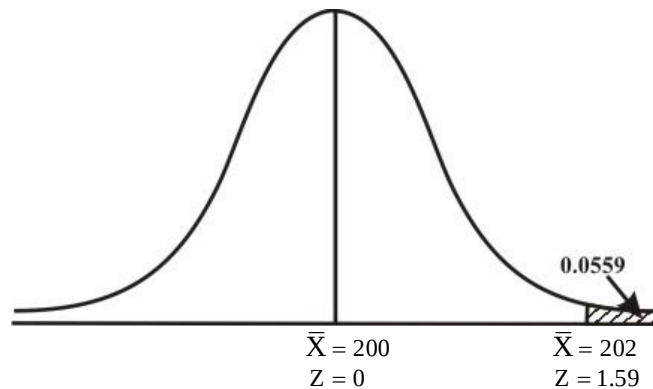


Fig. 2.6

E3) Here, we are given that

$$\mu = 2550, n = 100, S = 54$$

First of all, we find the sampling distribution of sample mean. Since sample size is large ($n = 100 > 30$) therefore, by the central limit theorem, the sampling distribution of sample mean follows normal distribution. Therefore, the mean of this distribution is given by

$$E(\bar{X}) = \mu = 2550$$

and variance

$$\text{Var}(\bar{X}) = \frac{S^2}{n} = \frac{(54)^2}{100} = \frac{2916}{100} = 29.16$$

E4) Here, we are given that

$$\mu_1 = 68, \sigma_1 = 2.3, n_1 = 35$$

$$\mu_2 = 65, \sigma_2 = 2.5, n_2 = 50$$

To find the required probability, first of all we find the sampling distribution of difference of two sample means. Let $\bar{X}$ and $\bar{Y}$ denote the mean height of workers taken from hospital and company respectively. Since n_1 and n_2 are large ($n_1, n_2 > 30$) therefore, by the central limit theorem, the sampling distribution of $(\bar{X} - \bar{Y})$ follows normal distribution with mean

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2 = 68 - 65 = 3$$

and variance

$$\begin{aligned} \text{Var}(\bar{X} - \bar{Y}) &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{(2.3)^2}{35} + \frac{(2.5)^2}{50} \\ &= 0.1511 + 0.1250 = 0.2761 \end{aligned}$$

The probability that the sample mean of height of hospital workers is 2 inch less than that of company workers is given by

$$P[(\bar{X} - \bar{Y}) < 2] \quad [\text{See Fig. 2.7}]$$

To get the value of this probability, we convert variate $(\bar{X} - \bar{Y})$ into a standard normal variate Z by the transformation

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{\sqrt{\text{Var}(\bar{X} - \bar{Y})}} = \frac{(\bar{X} - \bar{Y}) - 3}{\sqrt{0.2761}} = \frac{(\bar{X} - \bar{Y}) - 3}{0.53}$$

Therefore, by subtracting 3 from each term and then dividing each term by 0.53 in the above inequality, we get required probability

$$\begin{aligned} P\left[\frac{(\bar{X} - \bar{Y}) - 3}{0.53} < \frac{2 - 3}{0.53}\right] &= P[Z < -1.89] = 0.5 - P[-1.89 \leq Z \leq 0] \\ &= 0.5 - P[0 \leq Z \leq 1.89] \\ &= 0.5 - 0.4706 = 0.0294 \end{aligned}$$

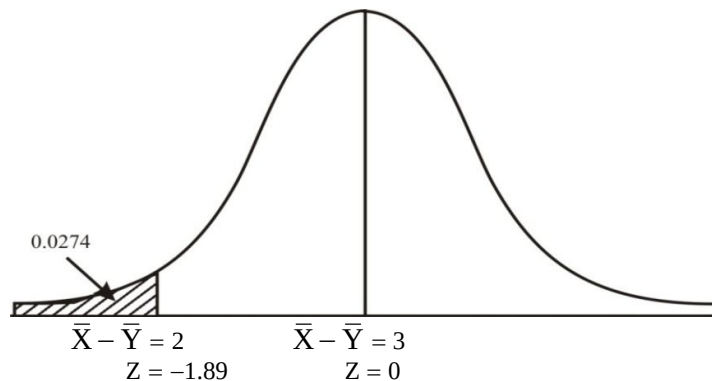


Fig. 2.7

**Sampling Distribution(s)
of Statistic(s)**

E5) Here, we are given that

$$N = 10000, X = 7000 \Rightarrow P = \frac{X}{N} = \frac{7000}{10000} = 0.70 \text{ \& } n = 100$$

First of all, we find the sampling distribution of sample proportion. Here, the sample proportion is not given and n is large so we can assume that the conditions of normality hold. So the sampling distribution is approximately normally distributed with mean

$$E(p) = P = 0.70$$

and variance

$$\text{Var}(p) = \frac{PQ}{n} = \frac{0.70 \times 0.30}{100} = 0.0021 \quad [\because Q = 1 - P]$$

Since the sample size n is less than 10% of the population size N ,

therefore, correction factor $\sqrt{\frac{N-n}{N-1}}$ is ignored.

The probability that the sample proportion would have exceeded 60% acceptance is given by

$$P[p > 0.60] \quad [\text{See Fig. 2.8}]$$

To get the value of this probability, we can convert the random variate p into standard normal variate Z by the transformation

$$Z = \frac{p - E(p)}{\sqrt{\text{Var}(p)}} = \frac{p - 0.70}{\sqrt{0.0021}} = \frac{p - 0.70}{0.046}$$

Therefore, by subtracting 0.70 from each term and then dividing each term by 0.046 in the above inequality, we get the required probability as

$$\begin{aligned} P\left[\frac{p - 0.70}{0.046} > \frac{0.60 - 0.70}{0.046}\right] &= P[Z > -2.17] \\ &= P[-2.17 < Z < 0] + 0.5 \\ &= P[0 < Z < 2.17] + 0.5 \\ &= 0.4850 + 0.5 \\ &= 0.9850 \end{aligned}$$

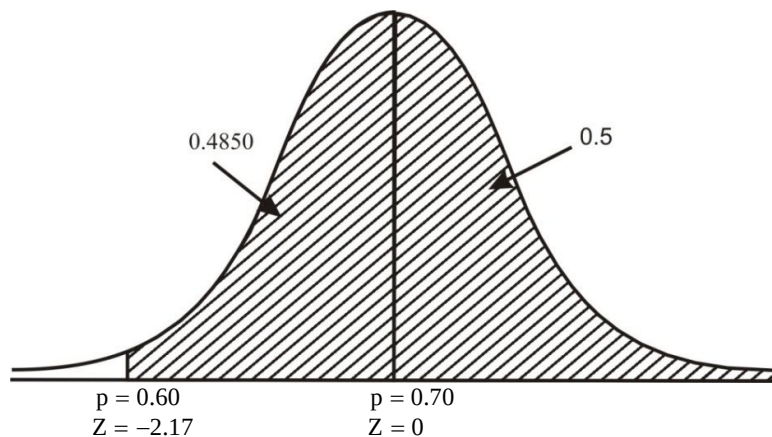


Fig. 2.8

E6) Here, we are given that

**Sampling Distribution(s)
of Statistic(s)**

$$P_1 = \frac{12}{100} = 0.12, P_2 = \frac{10}{100} = 0.10, n_1 = 100, n_2 = 150$$

Let p_1 and p_2 be the sample proportions of alcohol drinkers in two cities A and B respectively. Here the sample proportions are not given and n_1 and n_2 are large ($n_1, n_2 > 30$) so we can assume that conditions of normality hold. So the sampling distribution of difference of proportions is approximately normally distributed with mean

$$E(p_1 - p_2) = P_1 - P_2 = 0.12 - 0.10 = 0.02$$

and variance

$$\begin{aligned} \text{Var}(p_1 - p_2) &= \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2} \\ &= \frac{0.12 \times 0.78}{100} + \frac{0.10 \times 0.90}{150} \quad [\because Q = 1 - P] \\ &= 0.0009 + 0.0006 = 0.0015 \end{aligned}$$

The probability that the difference in sampling proportions lies between 0.01 and 0.08 is given by

$$P[0.01 < (p_1 - p_2) < 0.08] \quad [\text{See Fig. 2.9}]$$

To get the value of this probability, we convert variate $(p_1 - p_2)$ into a standard normal variate Z by the transformation

$$Z = \frac{(p_1 - p_2) - E(p_1 - p_2)}{\sqrt{\text{Var}(p_1 - p_2)}} = \frac{(p_1 - p_2) - 0.02}{\sqrt{0.0015}} = \frac{(p_1 - p_2) - 0.02}{0.039}$$

Therefore, by subtracting 0.02 from each term and then dividing each term by 0.039 in the above inequality, we have required probability

$$\begin{aligned} P\left[\frac{0.01 - 0.02}{0.039} < \frac{(p_1 - p_2) - 0.02}{0.039} < \frac{0.08 - 0.02}{0.039}\right] \\ &= P[-0.26 < Z < 1.54] \\ &= P[-0.26 < Z < 0] + P[0 < Z < 1.54] \\ &= P[0 < Z < 0.26] + P[0 < Z < 1.54] \\ &= 0.1026 + 0.4382 = 0.5408 \end{aligned}$$

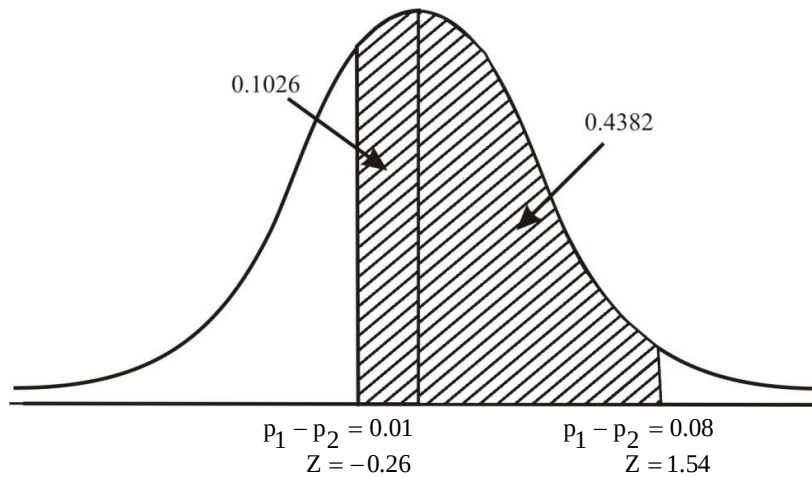


Fig. 2.9

E7) Here, we are given that

$$\sigma^2 = 11, n = 20$$

The probability that the variance of the given sample is greater than 16 pounds² is given by

$$P[S^2 \geq 16]$$

To get the value of this probability, we convert variate S^2 into χ^2 -variate which follows the chi-square distribution with $n - 1 = 19$ degrees of freedom by the transformation

$$\chi_{(n-1)}^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{19S^2}{11}$$

Therefore, multiplying each term by 19 and then dividing each term by 11 in the above inequality, we get required probability as

$$P\left[\frac{19S^2}{11} \geq \frac{19 \times 16}{11}\right] = P[\chi_{(19)}^2 \geq 27.64]$$

From the chi-square table, we see that $P[\chi_{(19)}^2 \geq 27.20] = 0.10$ and

$$P[\chi_{(19)}^2 \geq 30.14] = 0.05.$$

$f(\chi^2)$

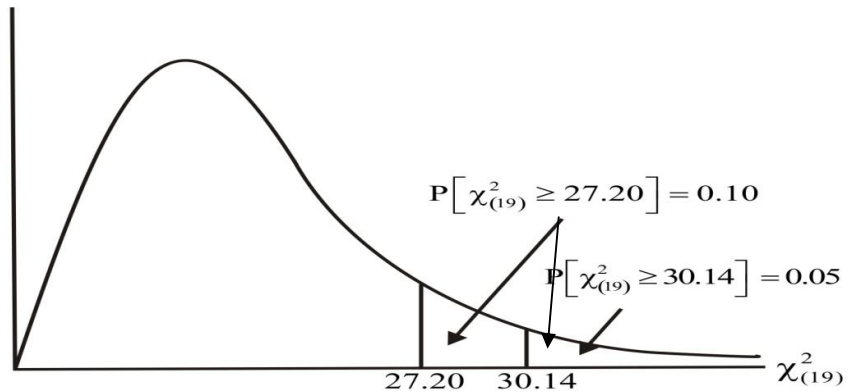


Fig. 2.10

Therefore, the above probability must lie between 0.05 and 0.10, that is,

$$0.05 < P[\chi^2 \geq 29.09] = P(S^2 \geq 16) < 0.10.$$

UNIT 3 STANDARD SAMPLING DISTRIBUTIONS-I

Structure

- 3.1 Introduction
 - Objectives
- 3.2 Introduction to χ^2 -distribution
- 3.3 Properties of χ^2 -distribution
 - Probability Curve of χ^2 -distribution
 - Mean and Variance of χ^2 -distribution
- 3.4 Applications of χ^2 -distribution
- 3.5 Introduction to t-distribution
- 3.6 Properties of t-distribution
 - Probability Curve of t-distribution
 - Mean and Variance of t-distribution
- 3.7 Applications of t-distribution
- 3.8 Summary
- 3.9 Solutions /Answers

3.1 INTRODUCTION

In Unit 1 of this block, we have discussed the fundamentals of sampling distributions and some basic definitions related to it. In Unit 2, we have discussed the sampling distributions of various sample statistics such as sample mean, sample proportion, sample variance, etc. Sometimes statistics such as sample mean, sample proportion, sample variance, etc. may follow a particular sampling distribution. Based on this fact Prof. R.A. Fisher, Prof. G. Snedecor and some other statisticians worked in this area and found some exact sampling distributions which are often followed by some of the statistics. In present and next unit of this block, we shall discuss some important sampling distributions such as χ^2 (read as chi-square), t and F. Generally, these sampling distributions are named on the name of the originator, that is, Fisher's F-distribution is named on Prof. R.A. Fisher.

This unit is divided in 9 sections. Section 3.1 is introductive in nature. In Section 3.2 the brief introduction of χ^2 -distribution is given. The different properties and probability curves for various degrees of freedom $n = 1, 4, 10$ and 22 are described along with the mean and variance in Section 3.3. The distribution of χ^2 is used in testing the goodness of fit, independence of attributes, etc. Therefore, in Section 3.4 the important applications of χ^2 -distribution are listed. In Section 3.5, the brief introduction of t-distribution is described. Properties of t-distribution, probability curve, the general formula for central moment, mean and variance are described in Section 3.6. Different applications of t-distribution are listed in Section 3.7. Unit ends by providing summary of what we have discussed in this unit in Section 3.8 and solution of exercises in Section 3.9.

Objectives

After studying this unit, you should be able to:

- explain the χ^2 -distribution;

- describe the properties of χ^2 -distribution;
- explain the probability curve of χ^2 -distribution;
- find the mean and variance of χ^2 -distribution;
- explore the applications of χ^2 -distribution;
- explain the t-distribution;
- describe the properties of t-distribution;
- describe the probability curve of t-distribution;
- find the mean and variance of t-distribution; and
- explore the applications of t-distribution.

3.2 INTRODUCTION TO χ^2 -DISTRIBUTION

Before describing the chi-square (χ^2) distribution, first we will discuss very useful concept “degrees of freedom”. The exact sampling distributions are described with the help of degrees of freedom.

Degrees of Freedom (df)

The term degree of freedom (df) is related to the independency of sample observations. In general, the number of degree of freedom is the total number of observations minus the number of independent constraints or restrictions imposed on the observations.

For a sample of n observations, if there are k restrictions among observations ($k < n$), then the degrees of freedom will be $(n-k)$.

For example, suppose there are two observations X_1, X_2 and a restriction (condition) is imposed that their total should be equal to 100. Then we can arbitrarily assign value to any one of these two as 30, 98, 52, etc. but the value of rest is automatically determined by a simple adjustment like $X_2 = (100 - X_1)$. Therefore, we can say that only one observation is independent or there is one degree of freedom.

Consider another example, suppose there are 50 observations as $X_1, X_2, \dots, X_{50}$ and the restriction (condition) are imposed on these like as:

$$(i) \sum_{i=1}^{50} X_i = 2000 \quad (ii) \sum_{i=1}^{50} X_i^2 = 10000 \quad (iii) X_5 = 2X_4$$

The restriction (i) reveals sum of observations equal to 2000 (ii) reveals sum of square of observations equal to 10000 and restriction (iii) implies fifth observation double to the fourth observation.

With these three restrictions (i), (ii) and (iii), there shall be only 47 observations independently chosen and rest three could be obtained by solving equations or by adjustments. The degrees of freedom of this set of observations of size 50 is now $(50 - 3) = 47$.

Chi-square Distribution

The chi-square distribution is first discovered by Helmer in 1876 and later independently explained by Karl- Pearson in 1900. The chi-square distribution was discovered mainly as a measure of goodness of fit in case of frequency.

If a random sample $X_1, X_2, \dots, X_n$ of size n is drawn from a normal population having mean μ and variance σ^2 then the sample variance can be defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

or

$$\sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)S^2 = \nu S^2$$

where, $\nu = n-1$ and the symbol ν read as 'nu'.

Thus, the variate $\chi^2 = \frac{\nu S^2}{\sigma^2}$, which is the ratio of sample variance multiplied by its degrees of freedom and the population variance follows the χ^2 -distribution with ν degrees of freedom.

The probability density function of χ^2 -distribution with ν df is given by

$$f(\chi^2) = \frac{1}{2^{\nu/2} \Gamma(\frac{\nu}{2})} e^{-\chi^2/2} (\chi^2)^{(\nu/2)-1}; \quad 0 < \chi^2 < \infty \quad \dots (1)$$

where, $\nu = n-1$.

The chi-square distribution can also be we described as:

If random variable X follows normal distribution with mean μ and variance σ^2 , then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$, known as standard normal variate, follows normal distribution with mean 0 and variance 1. The square of a standard normal variate Z follows chi-square distribution with 1 degree of freedom (df), that is,

If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ and

$$Z^2 = \left(\frac{X-\mu}{\sigma} \right)^2 \sim \chi_{(1)}^2$$

where, $\chi_{(1)}^2$ read as chi-square with one degree of freedom.

In general, if X_i 's ($i=1, 2, \dots, n$) are n independent normal variates with means μ_i and variances σ_i^2 ($i=1, 2, \dots, n$) then the sum of squares of n standard normal variate follows chi-square distribution with n df i.e.

If $X_i \sim N(\mu_i, \sigma_i^2)$, then $Z_i = \frac{X_i - \mu_i}{\sigma_i} \sim N(0,1)$

Therefore, $\chi^2 = \sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2 \sim \chi_{(n)}^2$

The probability density function of χ^2 -distribution with n df is given by

Generally, ν symbol is used to represent the general degrees of freedom. Its value may be $n, n-1$, etc.

$$f(\chi^2) = \frac{1}{2^{n/2} \Gamma\left(\frac{n}{2}\right)} e^{-\chi^2/2} (\chi^2)^{(n/2)-1}; \quad 0 < \chi^2 < \infty \quad \dots (2)$$

Try the following exercise to make sure that you have understood the chi-square distribution.

E1) Write down the pdf of chi-square distribution in each of the following cases:

- (i) 6 degrees of freedom
- (ii) 10 degrees of freedom

E2) Below, in each case, the pdf of chi-square distribution is given. Obtain the degrees of freedom of each chi-square distribution:

(i) $f(\chi^2) = \frac{1}{96} e^{-\chi^2/2} (\chi^2)^3; \quad 0 < \chi^2 < \infty$

(ii) $f(\chi^2) = \frac{1}{2} e^{-\chi^2/2}; \quad 0 < \chi^2 < \infty$

3.3 PROPERTIES OF χ^2 -DISTRIBUTION

In previous section, we have discussed the χ^2 -distribution and its probability density function. Now, in this section, we shall discuss some important properties of χ^2 -distribution as given below:

1. The probability curve of the chi-square distribution lies in the first quadrant because the range of χ^2 -variate is from 0 to ∞ .
2. Chi-square distribution has only one parameter n, that is, the degrees of freedom.
3. Chi-square probability curve is highly positive skewed.
4. Chi-square-distribution is a uni-modal distribution, that is, it has single mode.
5. The mean and variance of chi-square distribution with n df are n and 2n respectively.

After deliberating the properties of χ^2 -distribution above, some of these are discussed in detailed in the subsequent sub-sections.

3.3.1 Probability Curve of χ^2 -distribution

The probability distribution of all possible values of a statistic is known as the sampling distribution of that statistic. The shape of probability distribution of a statistic can be shown by the probability curve. In this sub-section, we will discuss the probability curve of chi-square distribution.

A rough sketch of the probability curve of chi-square distribution is shown in Fig. 3.1 at different values of degrees of freedom as n = 1, 4, 10 and 22.

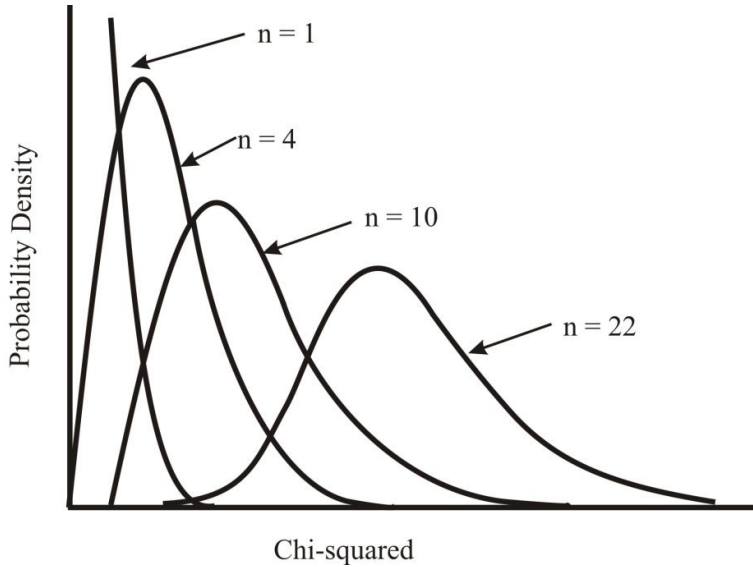


Fig. 3.1: Chi-square probability curves for $n = 1, 4, 10$ and 22

After looking the probability curves of χ^2 -distribution for $n = 1, 4, 10$ and 22 , one can understand that probability curve takes shape of inverse J for $n = 1$. The probability curve of chi-square distribution gets skewed more and more to the right as n becomes smaller and smaller. It becomes more and more symmetrical as n increases because as n tends to ∞ , the χ^2 -distribution becomes normal distribution.

3.3.2 Mean and Variance of χ^2 -distribution

In previous sub-section, the probability curve of χ^2 -distribution is discussed with some of its properties. Now in this sub-section, we will discuss the mean and variance of the χ^2 -distribution. The mean and variance of χ^2 -distribution is derived with the help of the moment about origin. As we have discussed in Unit 3 of MST-002 that the first order moment about origin is known as mean and central second order moment is known as variance of the distribution. So we first obtain the r^{th} order moment about origin of χ^2 -distribution as

$$\begin{aligned}
 \mu'_r &= E[(\chi^2)^r] = \int_0^\infty (\chi^2)^r f(\chi^2) d\chi^2 \\
 &= \int_0^\infty (\chi^2)^r \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-\chi^2/2} (\chi^2)^{(n/2)-1} d\chi^2 \\
 &= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \int_0^\infty e^{-\frac{1}{2}\chi^2} (\chi^2)^{\left(\frac{n}{2}+r\right)-1} d\chi^2 \\
 &= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \left[\frac{\frac{n}{2} + r}{\frac{n}{2}} \right] \left[\because \int_0^\infty e^{-ax} x^{n-1} dx = \frac{\Gamma(n)}{a^n} \right] \dots (3)
 \end{aligned}$$

The Gamma function is discussed in Unit 15 of MST-003, according to this $\int_0^\infty e^{-ax} x^{n-1} dx = \frac{\Gamma(n)}{a^n}$

Now, if we put $r = 1$ in the above expression given in equation (3), we get the value of first order moment about origin which is known as mean i.e.

$$\begin{aligned}\text{Mean} = \mu'_1 &= \frac{1}{2^{n/2}} \frac{\sqrt{\frac{n}{2} + 1}}{\sqrt{\frac{n}{2}} \left(\frac{1}{2}\right)^{\frac{n}{2}+1}} \\ &= \frac{1}{2^{n/2}} \frac{\frac{n}{2} \sqrt{\frac{n}{2}}}{\sqrt{\frac{n}{2}} \left(\frac{1}{2}\right)^{\frac{n}{2}+1}} = n \quad \left[\because \sqrt{n+1} = n \sqrt{n} \right]\end{aligned}$$

Similarly, if we put $r = 2$ in the formula of μ'_r given in equation (3) then we get the value of second order moment about origin i.e.

$$\begin{aligned}\mu'_2 &= \frac{1}{2^{n/2}} \frac{\sqrt{\frac{n}{2} + 2}}{\sqrt{\frac{n}{2}} \left(\frac{1}{2}\right)^{\frac{n}{2}+2}} \\ &= \frac{1}{\sqrt{\frac{n}{2}}} \frac{\left(\frac{n}{2} + 1\right) \left(\frac{n}{2}\right) \sqrt{\frac{n}{2}}}{\left(\frac{1}{2}\right)^2} \quad \left[\because \sqrt{n+2} = (n+1) \sqrt{n+1} \right. \\ &\quad \left. = (n+1) n \sqrt{n} \right] \\ &= 2^2 \left(\frac{n}{2} + 1\right) \left(\frac{n}{2}\right) = n(n+2)\end{aligned}$$

Now, we obtain the value of variance by putting the value of first order and second order moments about origin in the formula given below

$$\begin{aligned}\text{Variance} &= \mu'_2 - (\mu'_1)^2 \\ &= n(n+2) - (n)^2 = n^2 + 2n - n^2 = 2n\end{aligned}$$

So the mean and variance of the chi-square distribution with n df are given by

$$\text{Mean} = n \text{ and Variance} = 2n$$

Hence, we have observed that the variance is twice of the mean of chi-square distribution.

Similarly, on putting $r = 3, 4, \dots$ in the formula given in equation (3) one may get higher order moments about origin such as $\mu'_3, \mu'_4, \dots$

Example 1: What are the mean and variance of chi-square distribution with 5 degrees of freedom?

Solution: We know that the mean and variance of chi-square distribution with n degrees of freedom are

$$\text{Mean} = n \text{ and Variance} = 2n$$

In our case, $n = 5$, therefore,

$$\text{Mean} = 5 \text{ and Variance} = 10$$

To evaluate your understanding try to answer the following exercises.

- E3)** What are the mean and variance of chi-square distribution with 10 degrees of freedom?
- E4)** What are the mean and variance of chi-square distribution with pdf given below

$$f(\chi^2) = \frac{1}{96} e^{-\chi^2/2} (\chi^2)^3; \quad 0 < \chi^2 < \infty$$

3.4 APPLICATIONS OF χ^2 -DISTRIBUTION

The applications of chi-square distribution are very wide in Statistics. Some of them are listed below. The chi-square distribution is used:

1. To test the hypothetical value of population variance.
2. To test the goodness of fit, that is, to judge whether there is a discrepancy between theoretical and experimental observations.
3. To test the independence of two attributes, that is, to judge whether the two attributes are independent.

The first application listed above shall be discussed in detail in Unit 12 of this course and the remaining two applications shall be discussed in detail in Unit 16 of this course.

Now, try to write down the applications of chi-square distribution by answering the following exercise.

- E5)** List the applications of chi-square distribution.

3.5 INTRODUCTION TO t-DISTRIBUTION

The t-distribution was discovered by W.S. Gosset in 1908. He was better known by the pseudonym '**Student**' and hence t-distribution is called '**Student's t-distribution**'.

If a random sample $X_1, X_2, \dots, X_n$ of size n is drawn from a normal population having mean μ and variance σ^2 then we know that the sample mean $\bar{X}$ is distributed normally with mean μ and variance σ^2/n , that is, if $X_i \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \sigma^2/n)$, and also the variate

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is distributed normally with mean 0 and variance 1, i.e. $Z \sim N(0, 1)$.

In general, the standard deviation σ is not known and in such a situation the only alternative left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the unknown σ^2 where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

Thus, in this case the variate $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is not normally distributed whereas it

follows t-distribution with (n-1) df i.e.

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)} \quad \dots (4)$$

$$\text{where, } S^2 = \frac{1}{(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2$$

The t-variate is a widely used variable and its distribution is called student's t-distribution on the pseudonym name 'Student' of W.S. Gosset. The probability density function of variable t with (n-1) = v degrees of freedom is given by

$$f(t) = \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right) \left(1 + \frac{t^2}{v}\right)^{(v+1)/2}}; \quad -\infty < t < \infty \quad \dots (5)$$

where, $B\left(\frac{1}{2}, \frac{v}{2}\right)$ is known as beta function and

$$B\left(\frac{1}{2}, \frac{v}{2}\right) = \frac{\sqrt{\frac{1}{2}} \sqrt{\frac{v}{2}}}{\sqrt{\frac{v+1}{2}}} \quad \left[\because B(a, b) = \frac{\sqrt{a} \sqrt{b}}{\sqrt{a+b}} \right]$$

Later on Prof. R.A. Fisher found that t-distribution can also be applied to test the regression coefficient and other practical problems. He proposed t-variate as the ratio of standard normal variate to the square root of an independent chi-square variate divided by its degrees of freedom. Therefore, if Z is a standard normal variate with mean 0 and variance 1 and χ^2 is an independent chi-square variate with n df i.e.

$$Z \sim N(0,1) \quad \text{and} \quad \chi^2 \sim \chi_{(n)}^2$$

Then Fisher's t-variate is given by

$$t = \frac{Z}{\sqrt{\chi^2/n}} \sim t_{(n)} \quad \dots (6)$$

That is, the t-variate follows the t-distribution with n df and has the same probability density function as student t-distribution with n df. Therefore,

$$f(t) = \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right) \left(1 + \frac{t^2}{n}\right)^{(n+1)/2}}; \quad -\infty < t < \infty \quad \dots (7)$$

After describing the t-distribution, we try to calculate the value of t-variate as in the given example.

Example 2: The life of light bulbs manufactured by the company A is known to be normally distributed. The CEO of the company claims that an average life time of the light bulbs is 300 days. A researcher randomly selects 25 bulbs for testing the life time and he observed the average life time of the sampled bulbs is 290 days with standard deviation of 50 days. Calculate value of t-variate.

Solution: Here, we are given that

$$\mu = 300, n = 25, \bar{X} = 290 \text{ and } S = 50$$

The value of t-variate can be calculated by the formula given below

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Therefore, we have

$$t = \frac{290 - 300}{50/\sqrt{25}} = \frac{-10}{10} = -1$$

Now, try the following exercises for your practice.

E6) Write down the pdf of t-distribution in each of the following cases:

- (i) 3 degrees of freedom
- (ii) 9 degrees of freedom

E7) Obtain the degrees of freedom of t-distribution whose pdf is given below:

$$f(t) = \frac{1}{\sqrt{5} B\left(\frac{1}{2}, \frac{5}{2}\right) \left(1 + \frac{t^2}{5}\right)^3}; \quad -\infty < t < \infty$$

E8) If the scores on an IQ test of the students of a class are assumed to be normally distributed with a mean of 60. From the class, 15 students are randomly selected and an IQ test of the similar level is conducted. The average test score and standard deviation of test scores in the sample group are found to be 65 and 12 respectively. Calculate the value of t-variate.

3.6 PROPERTIES OF t-DISTRIBUTION

In previous section, we have discussed the t-distribution briefly. Now, we shall discuss some of the important properties of the t-distribution.

The t-distribution has the following properties:

1. The t-distribution is a uni-modal distribution, that is, t-distribution has single mode.
2. The mean and variance of the t-distribution with n df are zero and $\frac{n}{n-2}$ if $n > 2$ respectively.
3. The probability curve of t-distribution is similar in shape to the standard normal distribution and is symmetric about $t = 0$ line but flatter than normal curve.
4. The probability curve is bell shaped and asymptotic to the horizontal axis.

The properties deliberated above are some of the important properties of the t-distribution. We shall discuss some of these properties such as probability curve, mean and variance of the t-distribution in detail in the subsequent sub-sections.

3.6.1 Probability Curve of t-distribution

The probability curve of t-distribution is bell shaped and symmetric about $t = 0$ line. The probability curves of t-distribution is shown in Fig. 3.2 at two different values of degrees of freedom as at $n = 4$ and $n = 12$.

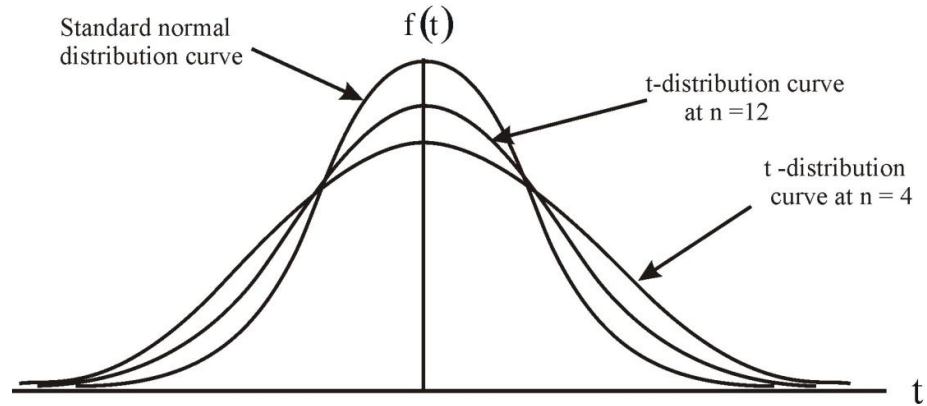


Fig. 3.2: Probability curves for t-distribution at $n = 4, 12$ along with standard normal curve

In the figure given above, we have drawn the probability curves of t-distribution at two different values of degrees of freedom with probability curve of standard normal distribution. By looking at the figure, one can easily understand that the probability curve of t-distribution is similar in shape to that of normal distribution and asymptotic to the horizontal-axis whereas it is flatter than standard normal curve. The probability curve of the t-distribution is tending to the normal curve as the value of n increases. Therefore, for sufficiently large value of sample size $n(> 30)$, the t-distribution tends to the normal distribution.

3.6.2 Mean and Variance of t-distribution

In previous sub-section, we have discussed the probability curve of t-distribution. After discussing the probability curve it came to our knowledge that t-distribution is symmetrical about $t = 0$ line. Since $f(t)$ is symmetrical about $t = 0$ line, therefore, all moments of odd order about origin vanish or become zero, i.e.

$$\mu'_{2r+1} = 0 \quad \text{for all } r = 0, 1, 2, 3, \dots \quad \dots (8)$$

In particular, for $r = 0$, we get

$$\mu'_1 = 0 = \text{Mean}$$

That is, the mean of the t-distribution is zero. Similarly, one can show that other odd moments about origin are zero by putting $r = 1, 2, \dots$

The moments of even order are given by

$$\mu'_{2r} = \int_{-\infty}^{\infty} t^{2r} f(t) dt$$

Now, we can also write

$$\begin{aligned} \mu'_{2r} &= 2 \int_0^{\infty} t^{2r} f(t) dt \\ &= 2 \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^{\infty} \frac{t^{2r}}{\left(1 + \frac{t^2}{n}\right)^{(n+1)/2}} dt \end{aligned}$$

$$= \frac{2}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^\infty \frac{(t^2)^r}{\left(1 + \frac{t^2}{n}\right)^{(n+1)/2}} dt \quad \dots (9)$$

Putting $1 + \frac{t^2}{n} = \frac{1}{y} \Rightarrow t^2 = n\left(\frac{1}{y} - 1\right)$

Therefore, $2t dt = -\frac{n}{y^2} dy \Rightarrow 2dt = -\frac{n}{ty^2} dy = \frac{n}{\sqrt{n\left(\frac{1}{y} - 1\right)} y^2} dy$

Also $t = 0 \Rightarrow y = 1$ and $t \rightarrow \infty \Rightarrow y \rightarrow 0$

By putting these values in equation (9), we get

$$\mu'_{2r} = -\frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_1^0 \frac{\left[n\left(\frac{1}{y} - 1\right)\right]^r}{\left(\frac{1}{y}\right)^{(n+1)/2}} \cdot \frac{n}{\sqrt{n\left(\frac{1}{y} - 1\right)} y^2} dy$$

$$\left[\because \int_a^0 f(x) dx = -\int_0^a f(x) dx \right]$$

$$= -\frac{\sqrt{n}}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^1 \left[n\left(\frac{1}{y} - 1\right)\right]^r \cdot \frac{1}{y^{\frac{n+1}{2}-2}} dy$$

$$= -\frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^1 \left(\frac{1-y}{y}\right)^{r-\frac{1}{2}} y^{\frac{n+1}{2}-2} dy$$

$$= -\frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^1 (1-y)^{r-\frac{1}{2}} y^{\frac{n+1}{2}-2-r+\frac{1}{2}} dy$$

$$= -\frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^1 y^{\frac{n}{2}-r-1} (1-y)^{r+\frac{1}{2}-1} dy$$

$$= -\frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} B\left(\frac{n}{2}-r, r+\frac{1}{2}\right) \left[\because \text{For } a > 0 \& b > 0 \right. \\ \left. \int_0^1 x^{a-1} (1-x)^{b-1} dx = B(a, b) \right]$$

$$= n^r \frac{\overline{\frac{1}{2} + \frac{n}{2}} \overline{\frac{n}{2} - r} \overline{r + \frac{1}{2}}}{\overline{\frac{1}{2}} \overline{\frac{n}{2}} \overline{\frac{n}{2} - r + r + \frac{1}{2}}}; \quad \frac{n}{2} - r > 0 \left[\because B(a, b) = \frac{\overline{a} \overline{b}}{\overline{a+b}} \right]$$

$$\mu'_{2r} = n^r \frac{\overline{\frac{n}{2} - r} \overline{r + \frac{1}{2}}}{\overline{\frac{1}{2}} \overline{\frac{n}{2}}}; \quad n > 2r \quad \dots (10)$$

Now, if we put $r = 1$ in the above expression given in equation (10), we get the value of second order moment about origin i.e.

$$\begin{aligned}\mu'_2 &= n \frac{\left[\frac{n}{2} - 1 \right] \left[1 + \frac{1}{2} \right]}{\left[\frac{1}{2} \right] \left[\frac{n}{2} \right]}; \quad n > 2 \\ &= n \frac{\left[\frac{n}{2} - 1 \right] \left[\frac{1}{2} \right]}{\left[\frac{1}{2} \left(\frac{n}{2} - 1 \right) \right] \left[\frac{n}{2} - 1 \right]} \quad \left[\because \overline{n+1} = n \overline{n} \right] \\ &= \frac{n}{(n-2)}; \quad n > 2\end{aligned}$$

Now, we obtain the value of variance by putting the value of first order and second order moments about origin in the formula given below

$$\begin{aligned}\text{Variance} &= \mu'_2 - (\mu'_1)^2 \\ &= \frac{n}{(n+2)} - (0)^2 \\ &= \frac{n}{(n+2)}; \quad n > 2\end{aligned}$$

So the mean and variance of the t-distribution with n df are given by

$$\text{Mean} = 0 \text{ and Variance} = \frac{n}{(n+2)}; \quad n > 2$$

Now, you can try an exercise to see how much you learn about the properties of t-distribution.

E9) What are the mean and variance of t-distribution with 20 degrees of freedom?

3.7 APPLICATIONS OF t-DISTRIBUTION

In previous section of this unit, we have discussed some important properties of t-distribution and derived its mean and variance. Now, you may be interested to know the applications of the t-distribution. In this section, we shall discuss the important applications of the t-distribution. The t-distribution has wide number of applications in Statistics. The t-distribution is used:

1. To test the hypothesis about the population mean.
2. To test the hypothesis about the difference of two population means of two normal populations.
3. To test the hypothesis that population correlation coefficient is zero.

The applications of the t-distribution listed above are discussed further in Unit 12 of this course.

Now, it is time to write down the main applications of t-distribution by answering the following exercise.

E10) Write any three applications of t-distribution.

We now end this unit by giving a summary of what we have covered in it.

3.8 SUMMARY

In this unit, we have covered the following points:

1. The chi-square distribution.
 2. The properties of χ^2 -distribution.
 3. The probability curve of χ^2 -distribution.
 4. Mean and variance of χ^2 -distribution.
 5. Applications of χ^2 -distribution.
 6. Student's-t and Fisher's t-distributions.
 7. Properties of t-distribution.
 8. The probability curve of t-distribution.
 9. Mean and variance of t-distribution.
 10. Applications of t-distribution.
-

3.9 SOLUTIONS / ANSWERS

E1) We know that the probability density function of χ^2 -distribution with n df is given by

$$f(\chi^2) = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-\chi^2/2} (\chi^2)^{(n/2)-1}; \quad 0 < \chi^2 < \infty$$

(i) In our case, n = 6, therefore, for n = 6 df the pdf of χ^2 -distribution is given by

$$\begin{aligned} f(\chi^2) &= \frac{1}{2^{6/2} \sqrt{\frac{6}{2}}} e^{-\chi^2/2} (\chi^2)^{(6/2)-1}; \quad 0 < \chi^2 < \infty \\ &= \frac{1}{8 \times \sqrt{3}} e^{-\chi^2/2} (\chi^2)^2 \\ &= \frac{1}{16} e^{-\chi^2/2} (\chi^2)^2; \quad 0 < \chi^2 < \infty \quad [\because \sqrt{3} = \underline{2} = 2] \end{aligned}$$

(ii) Similarly, for n = 10 df the pdf of χ^2 -distribution is given by

$$\begin{aligned} f(\chi^2) &= \frac{1}{2^{10/2} \sqrt{\frac{10}{2}}} e^{-\chi^2/2} (\chi^2)^{(10/2)-1}; \quad 0 < \chi^2 < \infty \\ &= \frac{1}{32 \times \sqrt{5}} e^{-\chi^2/2} (\chi^2)^4 \end{aligned}$$

$$= \frac{1}{768} e^{-\chi^2/2} (\chi^2)^4; \quad 0 < \chi^2 < \infty \quad \left[\because \overline{5} = \underline{4} = 24 \right]$$

E2) We know that the probability density function of χ^2 -distribution with n df is given by

$$f(\chi^2) = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-\chi^2/2} (\chi^2)^{(n/2)-1}; \quad 0 < \chi^2 < \infty \quad \dots (11)$$

(i) Here, we are given that

$$f(\chi^2) = \frac{1}{96} e^{-\chi^2/2} (\chi^2)^3; \quad 0 < \chi^2 < \infty$$

The given pdf can be arranged as

$$f(\chi^2) = \frac{1}{2^{8/2} \sqrt{\frac{8}{2}}} e^{-\chi^2/2} (\chi^2)^{\frac{8}{2}-1} \quad \left[\because 96 = 16 \times 6 = 2^{8/2} \sqrt{\frac{8}{2}} \right]$$

Comparing with equation (11), we get n = 8. So degree of freedom of chi-square distribution is 8.

(ii) Here, we are given that

$$f(\chi^2) = \frac{1}{2} e^{-\chi^2/2}; \quad 0 < \chi^2 < \infty$$

The given pdf can be arranged as

$$f(\chi^2) = \frac{1}{2^{2/2} \sqrt{\frac{2}{2}}} e^{-\chi^2/2} (\chi^2)^{\frac{2}{2}-1} \quad \left[\because 2 = 2 \times 1 = 2^{2/2} \sqrt{\frac{2}{2}} \right]$$

Comparing with equation (11), we get n = 1. So degree of freedom of chi-square distribution is 1.

E3) We know that the mean and variance of chi-square distribution with n degrees of freedom are

$$\text{Mean} = n \text{ and Variance} = 2n$$

In our case, n = 10, therefore,

$$\text{Mean} = 10 \text{ and Variance} = 20$$

E4) Here, we are given that

$$f(\chi^2) = \frac{1}{96} e^{-\chi^2/2} (\chi^2)^3; \quad 0 < \chi^2 < \infty$$

As similar as (i) of **E2)**, we get n = 8, therefore,

$$\text{Mean} = n = 8 \text{ and Variance} = 2n = 16.$$

E5) Refer Section 3.4.

E6) We know that the probability density function of t-distribution with n df is given by

$$f(t) = \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right) \left(1 + \frac{t^2}{n}\right)^{(n+1)/2}}; \quad -\infty < t < \infty$$

(i) In our case, $n = 3$, therefore, for $n = 3$ df the pdf of t-distribution is given by

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{3} B\left(\frac{1}{2}, \frac{3}{2}\right) \left(1 + \frac{t^2}{3}\right)^{(3+1)/2}}; \quad -\infty < t < \infty \\ &= \frac{\frac{\sqrt{1 + \frac{3}{2}}}{2}}{\sqrt{3} \left(\frac{1}{2} \frac{3}{2}\right) \left(1 + \frac{t^2}{3}\right)^2} \\ &= \frac{\sqrt{2}}{\sqrt{3} \left(\frac{1}{2} \times \frac{1}{2} \frac{3}{2}\right) \left(1 + \frac{t^2}{3}\right)^2} \quad \left[\because \frac{n}{2} = \left(\frac{n}{2} - 1\right) \frac{n}{2} - 1 \right] \\ &= \frac{2}{\sqrt{3} \pi \left(1 + \frac{t^2}{3}\right)^2}; \quad -\infty < t < \infty \quad \left[\because \frac{1}{2} = \sqrt{\pi} \text{ and } \sqrt{2} = 1 \right] \end{aligned}$$

(ii) Similarly, for $n = 9$ df the pdf of t-distribution is given by

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{9} B\left(\frac{1}{2}, \frac{9}{2}\right) \left(1 + \frac{t^2}{9}\right)^{(9+1)/2}}; \quad -\infty < t < \infty \\ &= \frac{\frac{\sqrt{1 + \frac{9}{2}}}{2}}{3 \left(\frac{1}{2} \frac{9}{2}\right) \left(1 + \frac{t^2}{9}\right)^5} \\ &= \frac{\sqrt{5}}{3 \left(\frac{1}{2} \times \frac{7}{2} \frac{5}{2} \frac{3}{2} \frac{1}{2} \frac{1}{2}\right) \left(1 + \frac{t^2}{9}\right)^5} \quad \left[\because \frac{n}{2} = \left(\frac{n}{2} - 1\right) \frac{n}{2} - 1 \right] \\ &= \frac{24}{3 \left(\frac{105}{16}\right) \pi \left(1 + \frac{t^2}{9}\right)^5} \quad \left[\because \frac{1}{2} = \sqrt{\pi} \text{ and } \sqrt{5} = \sqrt{4} = 2 \right] \\ &= \frac{128}{105 \pi \left(1 + \frac{t^2}{9}\right)^5}; \quad -\infty < t < \infty \end{aligned}$$

E7) We know that the probability density function of t-distribution with n df is given by

$$f(t) = \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right) \left(1 + \frac{t^2}{n}\right)^{(n+1)/2}}; \quad -\infty < t < \infty \quad \dots (12)$$

Here, we are given that

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{5} B\left(\frac{1}{2}, \frac{5}{2}\right) \left(1 + \frac{t^2}{5}\right)^3}; \quad -\infty < t < \infty \\ &= \frac{1}{\sqrt{5} B\left(\frac{1}{2}, \frac{5}{2}\right) \left(1 + \frac{t^2}{5}\right)^{(5+1)/2}}; \quad -\infty < t < \infty \end{aligned}$$

Comparing with equation (12), we get $n = 5$. So degrees of freedom of t-distribution are 5.

E8) Here, we are given that

$$\mu = 60, n = 15, \bar{X} = 65 \text{ and } S = 12$$

We know that the t-variate is

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Therefore, we have

$$t = \frac{65 - 60}{12/\sqrt{15}} = \frac{5}{4.39} = 1.14$$

E9) We know that the mean and variance of t-distribution with n degrees of freedom are

$$\text{Mean} = 0 \text{ and Variance} = \frac{n}{(n+2)}; \quad n > 2$$

In our case, $n = 20$, therefore,

$$\text{Mean} = 0 \text{ and Variance} = \frac{20}{(20+2)} = \frac{10}{11}$$

E10) Refer Section 3.7.

UNIT 4 STANDARD SAMPLING DISTRIBUTIONS-II

Structure

- 4.1 Introduction
 - Objectives
- 4.2 Introduction to F-distribution
- 4.3 Properties of F-distribution
 - Probability Curve of F-distribution
 - Mean and Variance of F-distribution
- 4.4 Applications of F-distribution
- 4.5 Relation between t, χ^2 and F-distributions
- 4.6 Tabulated Values of t, χ^2 and F
 - Tabulated Value of t-variate
 - Tabulated Value of χ^2 -variate
 - Tabulated Value of F-variate
- 4.7 Summary
- 4.8 Solutions /Answers

4.1 INTRODUCTION

In Unit 3, we have discussed chi-square and t-distributions in detail with their properties and applications. As we know that the sampling distributions χ^2 , t and F are mostly in use. In this unit, we shall discuss the F-distribution in brief with their properties and applications. As we have said that F-distribution was introduced by Prof. R. A. Fisher and it is the ratio of two independent chi-square variates when divided by its respective degrees of freedom. Later on the F-distribution was also described by the square of t-variate.

As usual after giving a brief introduction of this unit in this Section 4.1, we start our discussion by giving an over view of the F-distribution in Section 4.2. Different properties of F-distribution are explained in Section 4.3. The probability curves of F-distribution for various degrees of freedom ($v_1 = 5, v_2 = 5$), ($v_1 = 5, v_2 = 20$) and ($v_1 = 20, v_2 = 5$) are described along with the r^{th} moment about origin, mean and variance. The distribution of F is used in testing the equality of two variances, equality of several population means, etc. Therefore in Section 4.4 the different applications of F-distribution are explored. As mentioned above, all the standard sampling distribution discussed in this unit and previous unit are very useful and inter-related. Therefore, in Section 4.5 the relations between the t, χ^2 and F -distributions are obtained. In Section 4.6, the method of obtaining the tabulated value of a variate is explained which follows either of t, χ^2 and F-distributions. Unit ends by providing summary of what we have discussed in this unit in Section 4.7 and solution of exercises in Section 4.8.

Objectives

After studying this unit, you should be able to:

- introduce the F-distribution;
- explain the properties of F-distribution;

- describe the probability curve of F-distribution;
- derive the mean and variance of F-distribution;
- explore the applications of F-distribution;
- describe the relations between t, χ^2 and F-distributions; and
- explain the method of obtaining the tabulated value of a variate which follows either of t, χ^2 and F-distributions.

4.2 INTRODUCTION TO F-DISTRIBUTION

As we have said in previous unit that F-distribution was introduced by Prof. R. A. Fisher and defined as the ratio of two independent chi-square variates when divided by their respective degrees of freedom. If we draw a random sample $X_1, X_2, \dots, X_{n_1}$ of size n_1 from a normal population with mean μ_1 and variance σ_1^2 and another independent random sample $Y_1, Y_2, \dots, Y_{n_2}$ of size n_2 from another normal population with mean μ_2 and variance σ_2^2 respectively then as we have studied in Unit 3 that $v_1 S_1^2 / \sigma_1^2$ is distributed as chi-square variate with v_1 df i.e.

$$\chi_1^2 = \frac{v_1 S_1^2}{\sigma_1^2} \sim \chi_{(v_1)}^2 \quad \dots (1)$$

$$\text{where, } v_1 = n_1 - 1, \bar{X} = \frac{1}{n_1} \sum_{i=1}^{n_1} X_i \text{ and } S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

Similarly, $v_2 S_2^2 / \sigma_2^2$ is distributed as chi-square variate with v_2 df i.e.

$$\chi_2^2 = \frac{v_2 S_2^2}{\sigma_2^2} \sim \chi_{(v_2)}^2 \quad \dots (2)$$

$$\text{where, } v_2 = n_2 - 1, \bar{Y} = \frac{1}{n_2} \sum_{i=1}^{n_2} Y_i \text{ and } S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

Now, if we take the ratio of the above chi-square variates given in equations (1) and (2), then we get

$$\begin{aligned} \frac{\chi_1^2}{\chi_2^2} &= \frac{v_1 S_1^2 / \sigma_1^2}{v_2 S_2^2 / \sigma_2^2} \\ \Rightarrow \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} &= \frac{\chi_1^2 / v_1}{\chi_2^2 / v_2} \sim F_{(v_1, v_2)} \quad \dots (3) \end{aligned}$$

By observing the above form given in equation (3) we reveal that the ratio of two independent chi-square variates when divided by their respective degrees of freedom follows F-distribution where, v_1 and v_2 are called the degrees of freedom of F-distribution.

Now, if variances of both the populations are equal i.e. $\sigma_1^2 = \sigma_2^2$ then F-variate is written in the form of ratio of two sample variances i.e.

$$F = \frac{S_1^2}{S_2^2} \sim F_{(v_1, v_2)} \quad \dots (4)$$

By observing the above form of F-variate given in equation (4) we reveal that under the assumption $\sigma_1^2 = \sigma_2^2$, the ratio of two independent random sample variances follows F-distribution with v_1 and v_2 df. The probability density function of F-variate is given as

$$f(F) = \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \frac{F^{(v_1/2)-1}}{\left(1 + \frac{v_1}{v_2} F\right)^{(v_1+v_2)/2}}; \quad 0 < F < \infty \quad \dots (5)$$

After describing the F-distribution, we try to calculate the value of F-variate as in the given example.

Example 1: A statistician selects 7 women randomly from the population of women, and 12 men from a population of men. The table given below shows the standard deviation of each sample and population:

Population	Population Standard Deviation	Sample Standard Deviation
Women	40	45
Men	80	75

Compute the value of F-variate.

Solution: Here, we are given that

$$n_1 = 7, n_2 = 12, \sigma_1 = 40, \sigma_2 = 80, S_1 = 45, S_2 = 75$$

The value of F-variate can be calculated by the formula given below

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

where, S_1^2 & S_2^2 are the sample variances.

Therefore, we have

$$F = \frac{(45)^2 / (40)^2}{(75)^2 / (80)^2} = \frac{1.27}{0.88} = 1.93$$

For the above calculation, the degrees of freedom v_1 for women's data are $7 - 1 = 6$ and the degrees of freedom v_2 for men's data are $12 - 1 = 11$.

Now, try to answer the following exercises to see how much you learn about F-distribution.

E1) Write down the pdf of F-distribution in each of the following cases:

- (i) (4, 2) degrees of freedom
- (ii) (2, 4) degrees of freedom

E2) If random variable X follows the F-distribution whose pdf is given by

$$f(x) = \frac{1}{(1+x)^2}; \quad 0 < x < \infty$$

Obtain the degrees of freedom of this distribution.

- E3)** For the purpose of a survey 15 students are selected randomly from class A and 10 students are selected randomly from class B. At the stage of the analysis of the sample data, the following information is available:

Class	Population Standard Deviation	Sample Standard Deviation
Class A	65	60
Class B	45	50

Calculate the value of F-variate.

After introducing the F-distribution, one may be interested in knowing the properties of this distribution. Now, we shall discuss some of the important properties of F-distribution in the next section.

4.3 PROPERTIES OF F-DISTRIBUTION

The F-distribution has wide properties in Statistics. Some of them are as follow:

1. The probability curve of F-distribution is positively skewed curve. The curve becomes highly positive skewed when v_2 is smaller than v_1 .
2. F-distribution curve extends on abscissa from 0 to ∞ .
3. F-distribution is a uni-modal distribution, that is, it has single mode.
4. The square of t-variate with v df follows F-distribution with 1 and v degrees of freedom.
5. The mean of F-distribution with (v_1, v_2) df is $\frac{v_2}{v_2 - 2}$ for $v_2 > 2$.
6. The variance of F-distribution with (v_1, v_2) df is

$$\frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \text{ for } v_2 > 4.$$

7. If we interchange the degrees of freedom v_1 and v_2 then there exists a very useful relation as

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}}$$

Here, we have deliberated some of the important properties of the F-distribution and some of them have discussed such as probability curve, mean and variance of F-distribution in Sub-sections 4.3.1 and 4.3.2 respectively.

4.3.1 Probability Curve of F-distribution

In Section 4.2, we have defined the F-variate as the ratio of two sample variances when population variances are equal and follows F-distribution with v_1 and v_2 df i.e.

$$F = \frac{S_1^2}{S_2^2} \sim F_{(v_1, v_2)} \quad \dots (6)$$

where, S_1^2 and S_2^2 are the sample variances of two normal populations. As you have studied that the F-variate varies from 0 to ∞ , therefore, it is always positive so probability curve of F-distribution wholly lies in the first quadrant. These v_1 and v_2 are the degrees of freedom and are called the parameters of F-distribution. Hence, the shape of probability curve F-distribution depends on v_1 and v_2 . The probability curves of F-distribution for various pairs of degrees of freedom (v_1, v_2) i.e. (5, 5), (5, 20) and (20, 5) are shown in Fig. 4.1 given below:

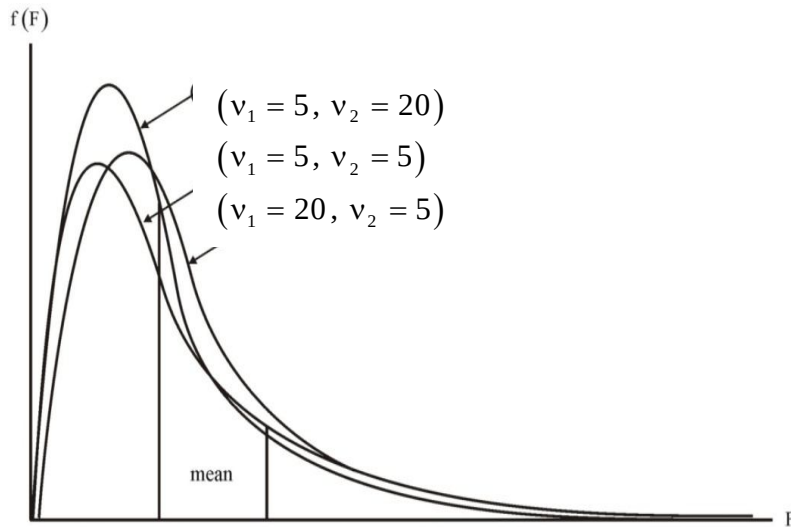


Fig. 4.1: Probability curves of F-distribution for (5, 5), (5, 20) and (20, 5) degrees of freedom.

After looking on the probability curves of F-distribution drawn above, we can observe that the probability curve of the F-distribution is uni-model curve. And by increasing in the first degrees of freedom from $v_1 = 5$ to $v_1 = 20$ the mean of the distribution (shown by vertical line) does not change but probability curve shifted from the tail to the centre of the distribution whereas increasing in the second degrees of freedom from $v_2 = 5$ to $v_2 = 20$ the mean of the distribution (shown by vertical line) decrease and the probability curve shifted from the tail to the centre of the distribution. One can also get an idea about the skewness of the F-distribution, we have observed from the probability curve of the F-distribution that it is positive skewed curve and it becomes very highly positive skewed if v_2 becomes small.

4.3.2 Mean and Variance of F-distribution

In previous sub-section, the probability curve of F-distribution is discussed with some of its features. Now in this sub-section, we shall describe the mean and variance of the F-distribution. The mean and variance of F-distribution is derived with the help of the moment about origin. As we have discussed in Unit 3 of MST-002 that the first order moment about origin is known as mean and central second order moment of is known as variance of

the distribution. The r^{th} order moment about origin of F-distribution is obtained as

$$\begin{aligned}\mu'_r &= E(F^r) = \int_0^\infty F^r f(F) dF \\ &= \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \int_0^\infty F^r \frac{F^{(v_1/2)-1}}{\left(1 + \frac{v_1}{v_2} F\right)^{(v_1+v_2)/2}} dF \quad \dots (7)\end{aligned}$$

If we take $\frac{v_1}{v_2} F = x \Rightarrow F = \frac{v_2}{v_1} x$ then $dF = \frac{v_2}{v_1} dx$. Also when $F = 0 \Rightarrow x = 0$ and when $F \rightarrow \infty \Rightarrow x \rightarrow \infty$.

Now, by putting these values in above equation (7), we get

$$\begin{aligned}\mu'_r &= \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \int_0^\infty \frac{\left\{\left(\frac{v_2}{v_1}\right)x\right\}^{\frac{v_1}{2}+r-1}}{(1+x)^{(v_1+v_2)/2}} \left(\frac{v_2}{v_1}\right) dx \\ &= \frac{\left(\frac{v_2}{v_1}\right)^r}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \int_0^\infty \frac{x^{(v_1/2)+r-1}}{(1+x)^{(v_1/2)+r+\{(v_2/2)-r\}}} dx \\ &= \frac{\left(\frac{v_2}{v_1}\right)^r}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} B\left(\frac{v_1}{2}+r, \frac{v_2}{2}-r\right) \left[\begin{array}{l} \because \text{For } a > 0 \text{ \& } b > 0 \\ \int_0^\infty \frac{x^{a-1}}{(1+x)^{a+b}} dx \\ = B(a, b) \end{array} \right]\end{aligned}$$

Thus,

$$\mu'_r = \left(\frac{v_2}{v_1}\right)^r \frac{\left|\frac{v_1}{2}+r\right| \left|\frac{v_2}{2}-r\right|}{\left|\frac{v_1}{2}\right| \left|\frac{v_2}{2}\right|} \text{ for } v_2 > 2r \left[B(a, b) = \frac{|a+b|}{|a| |b|} \right] \quad \dots (8)$$

Now, if we put $r = 1$ in the above expression given in equation (8), we get the value of first order moment about origin which is known as mean i.e.

$$\begin{aligned}\text{Mean} = \mu'_1 &= \left(\frac{v_2}{v_1}\right) \cdot \frac{\left|\frac{v_1}{2}+1\right| \left|\frac{v_2}{2}-1\right|}{\left|\frac{v_1}{2}\right| \left|\frac{v_2}{2}\right|} \text{ for } v_2 > 2 \\ &= \left(\frac{v_2}{v_1}\right) \cdot \frac{\frac{v_1}{2} \left|\frac{v_1}{2}\right| \left|\frac{v_2}{2}-1\right|}{\left|\frac{v_1}{2}\right| \left(\frac{v_2}{2}-1\right) \left|\frac{v_2}{2}-1\right|} \quad (\because |r| = (r-1)|r-1|) \\ &= \frac{v_2}{v_2-2} \text{ for } v_2 > 2 \quad \dots (9)\end{aligned}$$

Similarly, if we put $r = 2$ in the formula of μ'_r given in equation (8) then we get the value of second order moment about origin i.e.

$$\begin{aligned}\mu'_2 &= \left(\frac{v_2}{v_1}\right)^2 \cdot \frac{\left|\frac{v_1}{2} + 2\right| \left|\frac{v_2}{2} - 2\right|}{\left|\frac{v_1}{2}\right| \left|\frac{v_2}{2}\right|} \quad \text{for } v_2 > 4 \\ &= \left(\frac{v_2}{v_1}\right)^2 \frac{\left(\frac{v_1}{2} + 1\right) \left(\frac{v_1}{2}\right) \left|\frac{v_1}{2}\right| \left|\frac{v_2}{2} - 2\right|}{\left|\frac{v_1}{2}\right| \left(\frac{v_2}{2} - 1\right) \left(\frac{v_2}{2} - 2\right) \left|\frac{v_2}{2} - 2\right|} \quad \text{for } v_2 > 4 \\ &= \frac{v_2^2(v_1 + 2)}{v_1(v_2 - 2)(v_2 - 4)} \quad \text{for } v_2 > 4 \quad \dots (10)\end{aligned}$$

Now, we obtain the value of variance by putting the value of first order and second order moments about origin in the formula given below

$$\begin{aligned}\text{Variance} &= \mu'_2 - (\mu'_1)^2 \\ &= \frac{v_2^2(v_1 + 2)}{v_1(v_2 - 2)(v_2 - 4)} - \frac{v_2^2}{(v_2 - 2)^2} \\ &= \frac{v_2^2[(v_1 + 2)(v_2 - 2) - v_1(v_2 - 4)]}{v_1(v_2 - 2)^2(v_2 - 4)} \\ &= \frac{v_2^2(v_1v_2 - 2v_1 + 2v_2 - 4 - v_1v_2 + 4v_1)}{v_1(v_2 - 2)^2(v_2 - 4)} \\ &= \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \quad \text{for } v_2 > 4 \quad \dots (11)\end{aligned}$$

From the above derivation of mean and variance, we can conclude that mean of F-distribution is independent of v_1 .

Similarly, on putting $r = 3$ and 4 in the formula given in equation (8) one may get μ'_3 and μ'_4 which are the third and fourth order moments about origin respectively.

Now, you can try some exercises to see how much you learn about the properties of F-distribution.

E4) Write any five properties of F-distribution.

E5) What are the mean and variance of F-distribution with $v_1 = 5$ and $v_2 = 10$ degrees of freedom?

4.4 APPLICATIONS OF F-DISTRIBUTION

After discussing the main properties of F-distribution in previous section, we are now discussing some of the important applications of F-distribution.

Though F-distribution has many applications but some of them are listed here.

The F-distribution has the following applications:

1. F-distribution is used to test the hypothesis about equality of the variances of two normal populations.
2. F-distribution is used to test the hypothesis about multiple correlation coefficients.
3. F-distribution is used to test the hypothesis about correlation ratio.
4. F-distribution is used to test the equality of means of k-populations, when one characteristic of the population is considered i.e. F-distribution is used in one-way analysis of variance.
5. F-distribution is used to test the equality of k-population means for two characteristics at a time i.e. F-distribution is used in two-way analysis of variance.

The first application listed above will be discussed in Unit 12 of this course, fourth and fifth applications will be discussed in Block-2 of MST-005, second application will be discussed in detail in specialisation courses and the remaining third application is beyond the scope of this course.

Now, try the following exercise.

E6) Write four applications of F-distribution.

4.5 RELATION BETWEEN t , χ^2 AND F-DISTRIBUTIONS

After describing the exact sampling distributions χ^2 , t & F , we shall explain relationship between t & F -distributions and χ^2 & F -distributions. These relations can be described in terms of the following two statements:

Statement 1: If a variate t follows Student's t -distribution with v df, then square of t follows F -distribution with $(1, v)$ df i.e. if $t \sim t_{(v)}$ then $t^2 \sim F(1, v)$.

Statement 2: If F -variate follows F -distribution with (v_1, v_2) df and if

$v_2 \rightarrow \infty$ then $\chi^2 = v_1 F$ follows a χ^2 -distribution with v_1 df.

The proofs of these statements are beyond the scope of this course.

Now, try to answer the following exercises.

E7) If a variate t follows Student's t -distribution with 4 df, then what is the distribution of square of t . Also write the pdf of that distribution.

4.6 TABULATED VALUES OF t , χ^2 AND F

As we have seen in Unit 14 of MST-003 that the area or probability of normal distribution can be calculated with the help of tables. Similarly, we can also find the area or probability of a variate which follows either of t , χ^2 and F -distributions. In this section, we shall discuss the method of obtaining the tabulated values of t , χ^2 and F -variates.

4.6.1 Tabulated Values of t-Variate

A variate which follows the t-distribution is described with the help of degrees of freedom (v). **Table-II** in the Appendix contains the tabulated (critical) values of t-variate for different degrees of freedom (v) such that the area under the probability curve of t-distribution to its right tail (upper tail) is equal to α as shown in Fig. 4.2. The critical value of t-variate is represented by $t_{(v),\alpha}$ where, v represents the df and α area under the right tail or level of significance.

In this table, the first column of left hand side contains values of degrees of freedom (v) and the column heading represents the values of α on the right hand side (tail) of the probability curve of the t-distribution and the entries in the body of the table are the values of t-variate for each particular value of v and α . We discuss the process of finding the values of t-variate by taking an example as:

If we want to find out the value of t-variate for which the area on the right tail is 0.05 and the degrees of freedom is 8. Then we start with the first column of the t-table and downward headed ' v ' until entry 8 is reached. Then proceed right to the column headed $\alpha = 0.05$. For your convenient, part of the t-table is shown in Table 4.1 given below:

Table 4.1: t-table for Tabulated (Critical) Values of t-variate

One- tail $\alpha =$	0.10	0.05	0.025	0.01	0.005
$v=1$	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355

Thus, we get the required value of t-variate as $t_{(v)} = t_{(v),\alpha} = t_{(8),0.05} = 1.860$. The value of t-variate equal to 1.860 means, the probability that t-variate would exceed 1.860 is 0.05 as shown in Fig. 4.3.

Since t-distribution is symmetrical about $t = 0$ line, therefore, it is sufficient to tabulate the values of t-variate for right tail only. Therefore, the value of t-variate corresponding to a left hand tail for α is $-t_{(v),\alpha}$ as shown in Fig. 4.4. Thus, in above example, if we want to find the value of t-variate such that left area is 0.05 for 8 df then due to symmetry of t-distribution the value of t-variate is -1.860 .

If we want to find the values of t-variate when the total area on both tails is α then by symmetry the half area i.e. $\alpha/2$ lies in both tails as shown in Fig. 4.5. In such situation, we find the value of t-variate corresponding to $\alpha/2$ area and there are two values of t-variate as $\pm t_{(v),\alpha/2}$. For example, if we want to find out the values of t-variate for which the area on the both tails is 0.05 and the degrees of freedom is 8. Then by symmetry, we find the value of variate t corresponding to $\alpha/2 = 0.05/2 = 0.025$ and 8 df. Therefore, from the t-table we get $t_{(v),\alpha/2} = t_{(8),0.025} = 2.571$. So required values of t-variate are $\pm t_{(v),\alpha/2} = \pm t_{(8),0.025} = \pm 2.571$.

Example 2: Find the value of t-variate in each case for which the area on the right tail (α) and the degrees of freedom (v) are given in next page:

Standard Sampling Distributions-II

For level of significance, please see Unit 9 of this course.

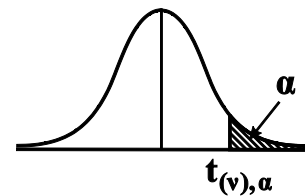


Fig. 4.2

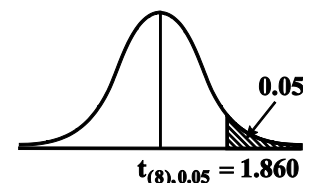


Fig. 4.3

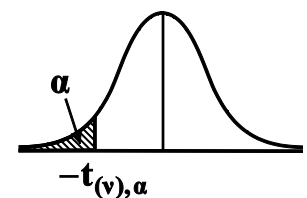


Fig. 4.4

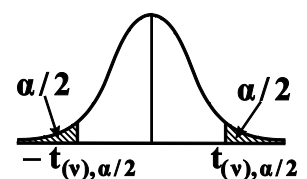


Fig. 4.5

(i) $\alpha = 0.01$ and $v = 12$ (ii) $\alpha = 0.10$ and $v = 15$ **Solution:**

(i) Here, we want to find the value of t-variate for

$$\alpha = 0.01 \text{ and } v = 12$$

Therefore, we start with the first column of t-table given in the Appendix and downward headed v until entry 12 is reached. Then proceed right to the column headed $\alpha = 0.01$. So we get the required value of t-variate as $t_{(v), \alpha} = t_{(12), 0.01} = 2.681$.

(ii) Here, we are given that

$$\alpha = 0.10 \text{ and } v = 15$$

By proceeding same way as above, from t-table, we get the required value of t-variate as $t_{(v), \alpha} = t_{(15), 0.10} = 1.341$. For your convenient, part of the t-table is shown in Table 4.2 given below:

Table 4.2: t-Table for Tabulated (Critical) Values of t-variate

One- tail $\alpha =$	0.10	0.05	0.025	0.01	0.005
v = 10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921

Example 3: Find the values of t-variate in each part for which the area on the both tails and the degrees of freedom are

(i) $\alpha = 0.01$ and $v = 12$ (ii) $\alpha = 0.10$ and $v = 29$ **Solution:**

(i) Here, we want to find the value of t-variate for two-tails and for

$$\alpha = 0.01 \text{ and } v = 12$$

Since total area on the both tails is 0.01 therefore, by symmetry the area on the right tail is $0.01/2 = 0.005$. Thus, we start with the first column of t-table and downward headed v until entry 12 is reached. Then we proceed right to the column headed $\alpha = 0.005$. We get $t_{(v), \alpha/2} = t_{(12), 0.005} = 3.055$. So the required values of t-variate are $\pm t_{(v), \alpha/2} = \pm t_{(12), 0.005} = \pm 3.055$.

(ii) Here, we are given that

$$\alpha = 0.10 \text{ and } v = 29$$

By proceeding same way as above, from the t-table, we get the required values of t-variate as $\pm t_{(v), \alpha/2} = \pm t_{(29), 0.05} = \pm 1.699$.

Method of Finding the Values of t-variate for Degrees of Freedom which are not listed in the Table

The t-table given in the Appendix does not list values for every possible degree of freedom. Therefore, it becomes necessary to know how to find values of t-variate for degrees of freedom not listed in the table.

We discuss the process of finding the values which are not listed in the table with the help of an example as:

Suppose we want to find out the value of t-variate for 34 degrees of freedom such that the area on the right side is equal to 0.05. The t-table in the Appendix does not list a tabulated value for this degree of freedom, so we need to interpolate it.

First of all, we read the tabulated values of t-variate that are just greater and just smaller than the degrees of freedom for our interest. Thus, from the t-table, we get the values of t-variate for 40 and 30 df and area $\alpha = 0.05$ as

$$t_{(40), 0.05} = 1.684 \quad t_{(30), 0.05} = 1.697$$

Note that larger the degrees of freedom smaller the tabulated value of t-variate.

Now, we calculate how much the t-value changes for each degree of freedom between these two tabulated values. Here, there is a difference of 10(40 – 30) degrees of freedom and t-value change of 0.013(1.697 – 1.684).

Thus, to determine how much it changes for each degree of freedom, divide the difference between the t-values by the difference between the degrees of freedom as

$$\frac{0.013}{10} = 0.0013$$

Since, we have to obtain the value for 34 degrees of freedom, this is either 4 more than 30 or 6 less than 40. Therefore, we can interpolate from either values. To get from 30 to 34 degrees of freedom there is a difference of 4 (34 – 30). So multiply this difference by the amount by which the t-value change per degree of freedom i.e. 0.0013. This result as

$$4 \times 0.0013 = 0.0052$$

Since larger the degrees of freedom smaller the tabulated value of t-variate therefore, subtracting this value 0.0052 from $t_{(30), 0.05} = 1.697$ to get the required value. Thus,

$$t_{(34), 0.05} = 1.697 - 0.0052 = 1.6918$$

Now, if we interpolate from 40 degrees of freedom then the difference 6(40 – 34) multiply by 0.0013 and adding this to 1.684, we get

$$t_{(34), 0.05} = 1.684 + 6 \times 0.0013 = 1.6918$$

Thus, we get the same value.

Example 4: Find the value of t-variate in each part for which the area on the right tail (α) and the degrees of freedom (v) are given below:

- (i) $\alpha = 0.01$ and $v = 42$ (ii) $\alpha = 0.01$ and $v = 35$

Solution:

- (i) Here, we want to find the value of t-variate for

$$\alpha = 0.01 \text{ and } v = 42$$

Since t-table does not have the tabulated value for 42 degrees of freedom so we need to interpolate it. For this, we find the tabulated

values of t-variate that are just greater and just less than the degree of freedom 42. Thus, we get

$$t_{(60), 0.01} = 2.390 \quad t_{(40), 0.01} = 2.423$$

There is a difference of 20 degrees of freedom between these two and a difference 0.033(2.423-2.390) in the t-values.

Thus, each degree of freedom has an approximate change in the value of t-variate is

$$\frac{0.033}{20} = 0.00165$$

To get 42 degrees of freedom, multiply 0.00165 by 2 (42-40), we get

$$0.00165 \times 2 = 0.003$$

Since larger the degrees of freedom smaller the tabulated value of t-variate, therefore, subtracting this from 2.423, we get required values as

$$t_{(42), 0.01} = 2.423 - 0.003 = 2.420$$

- (ii) Similarly, we find the tabulated values of t-variate that are just greater and just less than the degree of freedom 35. Thus, we have

$$t_{(40), 0.01} = 2.423 \quad t_{(30), 0.01} = 2.457$$

There is a difference of 10 degrees of freedom between these two and a difference 0.034(2.457-2.423) in the t-values.

Thus, each degree of freedom has an approximate change in the value of t-variate is

$$\frac{0.034}{10} = 0.0034$$

To get 35 degrees of freedom, multiplying 0.0034 by 5 (35-30), we have

$$0.0034 \times 5 = 0.017$$

Since larger the degrees of freedom smaller the tabulated value of t-variate, therefore, subtracting this from 2.457, we get required value as

$$t_{(35), 0.01} = 2.457 - 0.017 = 2.440$$

Now, you can try the following exercise.

E8) Find the values of t-variate for which the area and the degrees of freedom are

- (i) $\alpha = 0.01$ (one-right tail) and $v = 11$ (ii) $\alpha = 0.05$ (one-left tail) and $v = 16$
 (iii) $\alpha = 0.05$ (two-tails) and $v = 19$ (iv) $\alpha = 0.05$ (two-tails) and $v = 45$

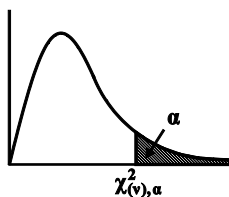


Fig. 4.6

4.6.2 Tabulated Value of χ^2 -Variate

As similar to t-table, the **Table-III** in the Appendix contains the tabulated (critical) values of chi-square variate for different degrees of freedom (v) such that the area under the probability curve of χ^2 -distribution to its right (upper) tail is equal to α as shown in Fig. 4.6.

In this table, the column headings indicates the area on the upper portion (right tail) of the probability curve of chi-square distribution and the first column on the left hand side indicates the values of degrees of freedom (ν). Also the entry in the body of the table indicates the value of chi-square variate. Therefore, $\chi^2_{(\nu),\alpha}$ is the value such that the area under the probability curve of chi-square distribution with ν df to its right tail is equal to α i.e. the probability that any value of χ^2 -variate is greater than or equal to $\chi^2_{(\nu),\alpha}$ is α . We discuss the process of finding the values of χ^2 -variate by taking an example as:

If we want to find out the value of chi-square variate with 7 df for which the area on the right tail is 0.01 then we start with first column i.e. degrees of freedom (ν) and proceed downward until 7 is reached then proceed right to the column headed $\alpha = 0.01$ which gives the required value of χ^2 as shown in the Table 4.3 given below and is also indicate in Fig. 4.7 as

$$\chi^2_{\alpha} = \chi^2_{(\nu),\alpha} = \chi^2_{(7),0.01} = 18.48.$$

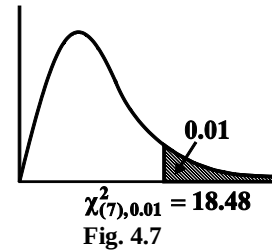


Table 4.3: Chi-square Table

$\alpha =$	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
$\nu = 1$	---	---	---	---	0.02	2.71	3.84	5.02	6.63	7.88
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21	10.60
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34	12.84
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28	14.86
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28

The chi-square distribution is a skewed (not symmetrical) distribution. Therefore, it is necessary to tabulate values of χ^2 for $\alpha > 0.50$ also.

Note 1: Values of chi-square variate for degrees of freedom which are not listed in the table are obtained in the similar manner as discussed in case of t-variate. This is explained in part (iv) of the Example 5 given below.

Now, let us do one example based on the above discussion.

Example 5: Find the values of chi-square variate for which the area and degrees of freedom are

- (i) $\alpha = 0.05$ (one-tail) and $\nu = 2$
- (ii) $\alpha = 0.99$ (one-tail) and $\nu = 15$
- (iii) $\alpha = 0.05$ (two-tails) and $\nu = 6$
- (iv) $\alpha = 0.01$ (one-tail) and $\nu = 64$

Solution:

- (i) Here, we want to find the value of χ^2 -variate for one-tail and for $\alpha = 0.05$ and $\nu = 2$

Thus, we first start with the first column of χ^2 -table given in the Appendix and downward headed ν until entry 2 is reached. Then proceed right to the column headed $\alpha = 0.05$. We get the required value of χ^2 -variate as $\chi^2_{(\nu),\alpha} = \chi^2_{(2),0.05} = 5.99$.

- (ii) Here, we want to find the value of χ^2 -variate for one-tail and for

$$\alpha = 0.99 \text{ and } v = 15$$

By proceeding same way as above, from the χ^2 -table, we get the required value of χ^2 -variate as $\chi^2_{(v),\alpha} = \chi^2_{(15),0.99} = 5.23$.

(iii) Here, we want to find the value of χ^2 -variate for two-tails and for

$$\alpha = 0.05 \text{ and } v = 6$$

Since total area on the both tails is 0.05 therefore, half area lies on both tails as $0.05/2 = 0.025$. In this case, χ^2 -variate has two values one on right-tail as $\chi^2_{\alpha/2} = \chi^2_{(v),\alpha/2} = \chi^2_{(6),0.025}$ and one on left-tail as

$$\chi^2_{(1-\alpha/2)} = \chi^2_{(v),(1-\alpha/2)} = \chi^2_{(6),0.975}$$

So by proceeding same way as above we get required values of χ^2 -variate as $\chi^2_{(6),0.025} = 14.45$ and $\chi^2_{(6),0.975} = 1.24$ as shown in Fig.4.8.

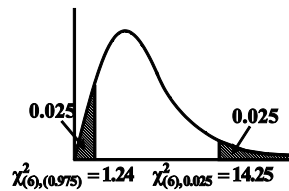


Fig. 4.8

(iv) Here, we want to find the value of χ^2 -variate for one-tail and for

$$\alpha = 0.01 \text{ and } v = 64$$

Since χ^2 -table does not have the tabulated value for 64 degrees of freedom so we need to interpolate it. For this we find the tabulated values of χ^2 -variate that are just greater and just less than the degree of freedom 64 with $\alpha = 0.01$. Thus, we have

$$\chi^2_{(70), 0.01} = 100.42 \quad \chi^2_{(60), 0.01} = 88.38$$

There is a difference of 10 degrees of freedom between these two and a difference 12.04(100.42–88.38) in the χ^2 -value. Also note that larger the degrees of freedom larger the tabulated value of χ^2 -variate.

Thus, each degree of freedom has an approximate change in the value of χ^2 -variate as

$$\frac{12.04}{10} = 1.204$$

To get the value of χ^2 -variate for 64 degrees of freedom, multiplying 1.204 by 4 (64–60), we get

$$1.204 \times 4 = 4.816$$

Since larger the degrees of freedom larger the tabulated value of χ^2 -variate so adding 4.816 in 88.38, we get required value as

$$\chi^2_{(64),0.01} = 88.38 + 4.816 = 93.196$$

Now, you can try the following exercise.

-
- E9)** Find the value of χ^2 -variate for which the area on the right tail and the degrees of freedom are
- (i) $\alpha = 0.01$ and $v = 11$
 - (ii) $\alpha = 0.90$ and $v = 19$
 - (iii) $\alpha = 0.05$ and $v = 44$
-

4.6.3 Tabulated Value of F-Variate

F-distribution is described with the help of two degrees of freedom, one for numerator and other for denominator. For each combination of degrees of freedom, F-variate has different tabulated value. **Table-IV** in the Appendix contains the tabulated (critical) values of F-variate for various degrees of freedom such that the area under the probability curve of F-distribution to its right (upper) tail is equal to $\alpha = 0.10, 0.05, 0.025$ and 0.01 . A general case is shown in Fig. 4.9.

The first row of F-table indicates the values of degrees of freedom (v_1) for numerator and first column on the left hand side indicates the values of degrees of freedom (v_2) for denominator. We discuss the process of finding the tabulated values of F-variate by taking an example as:

Suppose we want to find the value of F-variate for which the area on the right tail is 0.05 and the degrees of freedom for numerator and denominator are $v_1 = 10$ and $v_2 = 15$ respectively. Since $\alpha = 0.05$ so we select the F-table for $\alpha = 0.05$ and start with the first column of this table and proceed down to this column headed v_2 until entry 15 is reached then proceed right to the column headed $v_1 = 10$ as shown in the Table 4.4. In this way, we get the required value of F as $F_{(v_1, v_2), \alpha} = F_{(10, 15), 0.05} = 2.54$ which is also shown in Fig. 4.10.

Table 4.4: F-Table for $\alpha = 0.05$

Degrees of freedom for numerator (v_1)	Degrees of freedom for denominator (v_2)											
	1	2	3	4	5	6	7	8	9	10	11	12
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46	2.42

Note 2: The value of F-variate for v_1 degrees of freedom for numerator and v_2 for denominator is not same for v_2 degrees of freedom for numerator and v_1 for denominator, that is,

$$F_{(v_1, v_2)} \neq F_{(v_2, v_1)}$$

The F-distribution is also a skewed (not symmetrical) distribution. Therefore, it becomes necessary to tabulate values of F-distribution for $\alpha > 0.50$ also. The tabulated values of F-distribution for $\alpha > 0.50$ can be calculated by the relation as

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}} \quad [\text{Recall property listed in Section 4.3}]$$

Note 3: Values of F-variate for degrees of freedom which are not listed in the table are obtained in the similar manner as discussed in case of t-variate. This is explained in part (iv) of the Example 6 given below.

Now, let us do one example based on the above discussion.

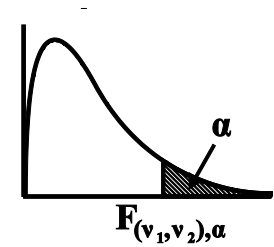


Fig. 4.9

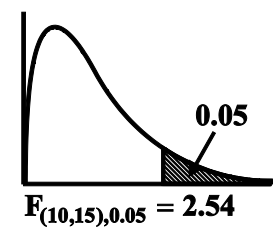


Fig. 4.10

Example 6: Find the values of F-variate in each part for which the area (α) and the degrees of freedom(v_1, v_2) are

- (i) $\alpha = 0.01$ (one-tail) and ($v_1 = 5, v_2 = 8$)
- (ii) $\alpha = 0.95$ (one-tail) and ($v_1 = 12, v_2 = 16$)
- (iii) $\alpha = 0.05$ (two-tails)and ($v_1 = 4, v_2 = 10$)
- (iv) $\alpha = 0.01$ (one-tail) and ($v_1 = 10, v_2 = 32$)

Solution:

- (i) Here, we want to find the value of F-variate for

$$\alpha = 0.01 \text{ and } (v_1 = 5, v_2 = 8)$$

Thus, we select the F-table for $\alpha = 0.01$ and start with the first column of this table and downward headed v_2 until entry 8 is reached. Then proceed right to the column headed $v_1 = 5$. We get the required value of F-variate as 6.63.

- (ii) Here, we want to find the value of F-variate for

$$\alpha = 0.95 \text{ and } (v_1 = 12, v_2 = 16)$$

Since we have F-tables for $\alpha = 0.10, 0.05, 0.025$ and 0.01 only therefore, for $\alpha = 0.95$, we can use the relation

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}}$$

From the F-table for $\alpha = 0.05$, we have

$$F_{(v_2, v_1), \alpha} = F_{(16, 12), 0.05} = 2.60$$

Thus,

$$F_{(v_1, v_2), (1-\alpha)} = F_{(12, 16), 0.95} = \frac{1}{F_{(16, 12), 0.05}} = \frac{1}{2.60} = 0.3846$$

- (iii) Here, we want to find the value of F-variate for two-tails and for

$$\alpha = 0.05 \text{ and } v_1 = 4, v_2 = 10$$

Since total area on the both tails is 0.05 therefore, half area lies on both tails as $0.05/2 = 0.025$. In this case, F-variate has two values one on right-tail as $F_{\alpha/2} = F_{(v_1, v_2), \alpha/2} = F_{(4, 10), 0.025}$ and one on left-tail as

$F_{(1-\alpha/2)} = F_{(v_1, v_2), (1-\alpha/2)} = F_{(4, 10), 0.975}$. So by proceeding same way as above we get required values of F-variate as $F_{(4, 10), 0.025} = 4.47$ and

$$F_{(4, 10), 0.975} = \frac{1}{F_{(10, 4), 0.025}} = \frac{1}{8.84} = 0.11 \text{ as shown in Fig.4.11.}$$

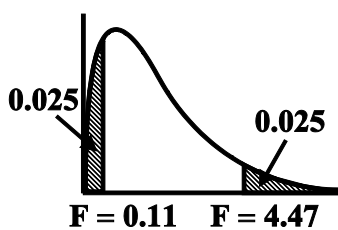


Fig.4.11

- (iv) Here, we want to find the value of F-variate for

$$\alpha = 0.01 \text{ and } (v_1 = 10, v_2 = 32)$$

Since F-table for $\alpha = 0.01$ does not have the tabulated value for these degrees of freedom so we need to interpolate it. For this we find the tabulated values of F that are just greater and just less than the degree of freedom 32. Thus we have

$$F_{(10, 40), 0.01} = 2.80 \quad F_{(10, 30), 0.01} = 2.98$$

There is a difference of 10 degrees of freedom between these two and a difference 0.18(2.98-2.80) in the F-values. Also note that larger the degrees of freedom smaller the tabulated value of F-variate.

Thus, each degree of freedom has an approximate change in the value of F as

$$\frac{0.18}{10} = 0.018$$

To get 32 degrees of freedom, multiplying 0.018 by 2 (32-30), we get

$$0.018 \times 2 = 0.036$$

Since larger the degrees of freedom smaller the tabulated value of F-variate, therefore, subtracting this from 2.98, we get required values as

$$F_{(10, 32)} = 2.98 - 0.036 = 2.944$$

Now, try to answer the following exercise.

E10) Find the value of F-variate in each part for which the area on the right side (α) and the degrees of freedom (v_1, v_2) are

- (i) $\alpha = 0.05$ and ($v_1 = 8, v_2 = 12$)
 - (ii) $\alpha = 0.99$ and ($v_1 = 5, v_2 = 10$)
 - (iii) $\alpha = 0.05$ and ($v_1 = 23, v_2 = 5$)
-

Let us end with a brief look at what we have covered in this unit.

4.7 SUMMARY

In this unit, we have covered the following points:

1. The F-distribution.
 2. The properties of F-distribution.
 3. The probability curve of F-distribution.
 4. Mean and variance of F-distribution.
 5. The applications of F-distribution.
 6. The relation between t, F and χ^2 -distributions.
 7. The method of obtaining the tabulated value of a variate which follows either of t, χ^2 and F-distributions.
-

4.8 SOLUTIONS / ANSWERS

E1) The probability density function of F-distribution with v_1 and v_2 df is given as

$$f(F) = \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \frac{F^{(v_1/2)-1}}{\left(1 + \frac{v_1}{v_2} F\right)^{(v_1+v_2)/2}}; \quad 0 < F < \infty$$

- (i) Therefore, for $v_1 = 4$ and $v_2 = 2$ df the pdf of F-distribution is given by

$$\begin{aligned}
 f(F) &= \frac{(4/2)^{4/2}}{B\left(\frac{4}{2}, \frac{2}{2}\right)} \frac{F^{(4/2)-1}}{\left(1 + \frac{4}{2}F\right)^{(4+2)/2}} \\
 &= \frac{4\sqrt{2+1}}{2\sqrt{1}} \frac{F}{(1+2F)^3} \quad \left[\because B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \right] \\
 &= \frac{8F}{(1+2F)^3}; \quad 0 < F < \infty \quad \left[\begin{array}{l} \because \overline{3} = \underline{2} = 2 \\ \& \overline{2} = \underline{1} = 1 \end{array} \right]
 \end{aligned}$$

(ii) Similarly, for $v_1 = 2$ and $v_2 = 4$ df the pdf of F-distribution is given by

$$\begin{aligned}
 f(F) &= \frac{(2/4)^{2/2}}{B\left(\frac{2}{2}, \frac{4}{2}\right)} \frac{F^{(2/2)-1}}{\left(1 + \frac{2}{4}F\right)^{(2+4)/2}} \\
 &= \frac{\sqrt{2+1}}{4\sqrt{2}\sqrt{1}} \frac{1}{\left(1 + \frac{F}{2}\right)^3} \quad \left[\because B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \right] \\
 &= \frac{1}{2\left(1 + \frac{F}{2}\right)^3}; \quad 0 < F < \infty \quad \left[\begin{array}{l} \because \overline{3} = \underline{2} = 2 \\ \& \overline{2} = \underline{1} = 1 \end{array} \right]
 \end{aligned}$$

Hence, from (i) and (ii) parts, we observed that

$$F_{(v_1, v_2)} \neq F_{(v_2, v_1)}$$

E2) If random variable X follows the F-distribution with v_1 and v_2 df then probability density function of X is given as

$$f(x) = \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \frac{x^{(v_1/2)-1}}{\left(1 + \frac{v_1}{v_2}x\right)^{(v_1+v_2)/2}}; \quad 0 < x < \infty \dots (12)$$

Here, we are given that

$$f(x) = \frac{1}{(1+x)^2}; \quad 0 < x < \infty$$

This can be written as

$$f(x) = \frac{(2/2)^{2/2}}{B\left(\frac{2}{2}, \frac{2}{2}\right)} \frac{x^{(2/2)-1}}{\left(1 + \frac{2}{2}x\right)^{(2+2)/2}}; \quad 0 < x < \infty \left[\because B(1, 1) = 1 \right]$$

Comparing with equation (12), we get $v_1 = 2$ and $v_2 = 2$.

E3) Here, we are given that

$$n_1 = 15, n_2 = 10, \sigma_1 = 65, \sigma_2 = 45, S_1 = 60, S_2 = 50$$

The value of F-variate can be calculated as follows:

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

where, S_1^2 & S_2^2 are the values of sample variances.

Therefore, we have

$$\begin{aligned} F &= \frac{(60)^2 / (65)^2}{(50)^2 / (45)^2} \\ &= \frac{0.85}{1.23} = 0.69 \end{aligned}$$

E4) Refer Section 4.3.

E5) We know that the mean and variance of F-distribution with v_1 and v_2 degrees of freedom are

$$\text{Mean} = \frac{v_2}{v_2 - 2} \quad \text{for } v_2 > 2$$

and

$$\text{Variance} = \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \quad \text{for } v_2 > 4.$$

In our case, $v_1 = 5$ and $v_2 = 10$, therefore,

$$\text{Mean} = \frac{v_2}{v_2 - 2} = \frac{10}{10 - 2} = \frac{5}{4}$$

$$\text{Variance} = \frac{2(10)^2(5 + 10 - 2)}{5(10 - 2)^2(10 - 4)} = \frac{65}{48}$$

E6) Refer Section 4.4.

E7) We know that if a variate t follows Student's t -distribution with v df, then square of t follows F-distribution with $(1, v)$ df i.e. if $t \sim t_{(v)}$ then $t^2 \sim F_{(1, v)}$. In our case, $v = 4$, therefore square of t follows F-distribution with $(1, 4)$ df whose pdf is given by

$$f(F) = \frac{(v_1/v_2)^{v_1/2}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \frac{F^{(v_1/2)-1}}{\left(1 + \frac{v_1}{v_2} F\right)^{(v_1+v_2)/2}}; \quad 0 < F < \infty$$

Therefore, for $v_1 = 1$ and $v_2 = 4$ df the pdf of F-distribution is given by

$$f(F) = \frac{(1/4)^{2/2}}{B\left(\frac{1}{2}, \frac{4}{2}\right)} \frac{F^{(1/2)-1}}{\left(1 + \frac{1}{4} F\right)^{(1+4)/2}}$$

$$\begin{aligned}
&= \frac{\sqrt{\frac{1}{2} + 2}}{4 \sqrt{\frac{1}{2}}} \frac{1}{\sqrt{F} \left(1 + \frac{F}{4}\right)^{5/2}} \\
&= \frac{\frac{3}{2} \sqrt{\frac{1}{2}}}{4 \sqrt{\frac{1}{2}}} \frac{1}{\sqrt{F} \left(1 + \frac{F}{4}\right)^{5/2}} \quad \left[\because \sqrt{2} = 1\right] \\
&= \frac{3}{16} \frac{1}{\sqrt{F} \left(1 + \frac{F}{4}\right)^{5/2}}; \quad 0 < F < \infty
\end{aligned}$$

E8)

- (i) Here, we want to find the value of t-variate for one-right tail corresponding

$$\alpha = 0.01 \text{ and } v = 11$$

Therefore, we start with the first column of t-table given in the Appendix and downward headed v until entry 11 is reached. Then proceed right to the column headed $\alpha = 0.01$. We get the required value of t-variate as $t_{(v), \alpha} = t_{(11), 0.01} = 2.718$.

- (ii) Here, we want to find the value of t-variate for one-left tail corresponding

$$\alpha = 0.05 \text{ and } v = 16$$

By proceeding same way as above, from the t-table, we get the value of t-variate as 1.746. So the required values of t-variate for one left-tail is $-t_{(v), \alpha} = -t_{(16), 0.05} = -1.746$.

- (iii) Here, we want to find the values of t-variate for two-tails corresponding

$$\alpha = 0.05 \text{ and } v = 19$$

Since total area on the both tails is 0.05 therefore, by symmetry, the area on the right tail is $0.05/2 = 0.025$. Thus, we first start with the first column of t-table and downward headed v until entry 19 is reached. Then proceed right to the column headed 0.025. We get $t_{(v), \alpha/2} = t_{(19), 0.025} = 2.093$. So the required values of t-variate are $\pm t_{(v), \alpha/2} = \pm t_{(19), 0.025} = \pm 2.093$.

- (iv) Here, we want to find the values of t-variate for two-tails corresponding

$$\alpha = 0.05 \text{ and } v = 45$$

Since total area on the both tails is 0.05 therefore, by symmetry the area on the right tail is $0.05/2 = 0.025$ and also t-table does not have the tabulated value for 45 degrees of freedom with $\alpha = 0.025$, so we need to interpolate it. For this, we find the tabulated values of t-variate that are just greater and just less than the degrees of freedom 45 at $\alpha = 0.025$. Thus, we get

$$t_{(60), 0.025} = 2.000 \quad t_{(40), 0.025} = 2.021$$

There is a difference of 20 degrees of freedom between these two and a difference 0.021(2.021–2.000) in the t-values.

Thus, each degree of freedom has an approximate change in the value of t as

$$\frac{0.021}{20} = 0.00105$$

To get 45 degrees of freedom, multiply 0.00105 by 5 (45–40), we get

$$0.00105 \times 5 = 0.00525$$

Now subtracting this from 2.021, we get t-value as

$$t_{(45), 0.025} = 2.021 - 0.00525 = 2.01575$$

So the required values of t-variate are $\pm t_{(v), \alpha/2} = \pm t_{(45), 0.025} = \pm 2.01575$.

E9) (i) Here, we want to find the value of χ^2 -variate for

$$\alpha = 0.01 \text{ and } v = 11$$

Thus, we start with the first column of χ^2 -table given in the Appendix and downward headed v until entry 11 is reached. Then proceed right to the column headed $\alpha = 0.01$. We get the required value of χ^2 -variate as 24.72.

(ii) Here, we want to find the value of χ^2 -variate for

$$\alpha = 0.90 \text{ and } v = 19$$

By proceeding same way as above, from the χ^2 -table, we get the required value of χ^2 as 11.65.

(iii) Here, we want to find the value of χ^2 -variate for

$$\alpha = 0.05 \text{ and } v = 44$$

Since χ^2 -table does not have the tabulated value for 44 degrees of freedom so we need to interpolate it. For this we find the tabulated values of χ^2 -variate that are just greater and just less than the degrees of freedom 44 with $\alpha = 0.05$. Thus, we have

$$\chi^2_{(50), 0.05} = 67.50 \quad \chi^2_{(40), 0.05} = 55.76$$

There is a difference of 10 degrees of freedom between these two and a difference 11.74(67.50 – 55.76) in the χ^2 -values. Also note that larger the degrees of freedom larger the tabulated value of χ^2 .

Thus, each degree of freedom has an approximate change in the value of χ^2 as

$$\frac{11.74}{10} = 1.174$$

To get 44 degrees of freedom, multiplying 1.174 by 4 (44–40), we get

$$1.174 \times 4 = 4.696$$

Now adding this in 55.76, we get required value as

$$\chi_{(44),0.05}^2 = 55.76 + 4.696 = 60.456$$

E10) (i) Here, we want to find the value of F-variate for

$$\alpha = 0.05 \text{ and } (v_1 = 8, v_2 = 12)$$

Thus, we select the F-table for $\alpha = 0.05$ and start with the first column of this table and downward headed v_2 until entry 12 is reached. Then proceed right to the column headed $v_1 = 8$. We get the required value of F as 2.85.

(ii) Here, we are given that

$$\alpha = 0.99 \text{ and } (v_1 = 5, v_2 = 10)$$

Since we have F-tables for $\alpha = 0.10, 0.05, 0.025$ and 0.01 only therefore, for $\alpha = 0.99$, we can use the relation

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}}$$

From the F-table for $\alpha = 0.01$, we have

$$F_{(10,5),0.01} = 4.47$$

Thus,

$$F_{(v_1, v_2), (1-\alpha)} = F_{(5,10),0.99} = \frac{1}{F_{(10,5),0.01}} = \frac{1}{4.47} = 0.2237$$

(iii) Here, we want to find the value of F-variate for

$$\alpha = 0.05 \text{ and } (v_1 = 23, v_2 = 5)$$

Since F-table for $\alpha = 0.05$ does not have the tabulated value for these degrees of freedom so we need to interpolate it. For this we find the tabulated values of F that are just greater and just less than the degrees of freedom 23. Thus we have

$$F_{(24,5),0.05} = 4.53 \quad F_{(20,5),0.05} = 4.56$$

Here, there is a difference of 4(24-20) degrees of freedom and F value changes by 0.03(4.56 - 4.53). Thus, to determine how much it approximately changes for each degree of freedom divide the difference between the F values by the difference between the degrees of freedom, we get

$$\frac{0.03}{4} = 0.0075$$

Since, we have 23 degrees of freedom, this is either 3 more than 20 or 1 less than 24. Therefore, we can interpolate from either values. To get from 20 to 23 degrees of freedom there is a difference of 3(23-20). So multiplying this difference by the amount by which the F-value approximate change per degrees of freedom i.e. 0.0075. These results in

$$3 \times 0.0075 = 0.0225$$

Since larger the degrees of freedom smaller the tabulated value of F-variate, therefore, subtracting this value 0.0225 from 4.56 to get required value. Thus,

$$F_{(23, 5), 0.05} = 4.56 - 0.0225 = 4.5375$$

**Standard Sampling
Distributions-II**

Block

2

ESTIMATION

UNIT 5

Introduction to Estimation	5
-----------------------------------	----------

UNIT 6

Point Estimation	37
-------------------------	-----------

UNIT 7

Interval Estimation for One Population	55
---	-----------

UNIT 8

Interval Estimation for Two Populations	85
--	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathasan
Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Rahul Roy
Math. and Stat. Unit
Indian Statistical Institute, New Delhi

Prof. Parvin Sinclair
Pro-Vice Chancellor
IGNOU, New Delhi

Dr. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Geeta Kaicker
Director, School of Sciences
IGNOU, New Delhi

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. R. M. Pandey
Department of Bio-Statistics
All India Institute of Medical Sciences
New Delhi

Dr. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Faculty members of School of Sciences, IGNOU

Statistics

Dr. Neha Garg
Dr. Nitin Gupta
Mr. Rajesh Kaliraman
Dr. Manish Trivedi

Mathematics

Dr. Deepika
Prof. Poornima Mital
Prof. Sujatha Varma
Dr. S. Venkataraman

Block Preparation Team

Dr. Ramkishan (**Editor**)
Department of Statistics
D. A. V. (PG) College
C.C. S. University, Merrut

Mr. Prabhat Kumar Sangal
School of Sciences, IGNOU

Dr. Parmod Kumar (**Language Editor**)
School of Humanities, IGNOU

Course Coordinator: Mr. Prabhat Kumar Sangal

Programme Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P), School of Sciences, IGNOU
CRC prepared by Mr. Prabhat Kumar Sangal, School of Sciences, IGNOU

Acknowledgement: I gratefully acknowledge my colleagues Mr. Rajesh Kaliraman and Dr. Neha Garg, Statistics Discipline, School of Sciences for their great support.

July, 2013

© Indira Gandhi National Open University, 2013

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University.

Further information on the Indira Gandhi National Open University may be obtained from University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Printed at:

ESTIMATION

In Block 1 of this course, you have studied the sampling distributions of different statistics as sample mean, sample proportion, sample variance, etc. and standard sampling distributions as χ^2 , t, F and Z which provide a platform to the learners how to draw the inference about the population parameter(s) on the basis of the sample(s).

In present block, we shall be studying the estimation theory, through which we estimate the unknown parameter on the basis of sample data. Two types of estimation i.e. point estimation and interval estimation are discussed in this block. This block comprises four units.

Unit 5: Introduction to Estimation

Estimation admits two problems; the first is to select some criteria or properties such that if an estimator possesses these properties it is said to be the best estimator among all possible estimators and the second is to derive some methods or techniques through which we obtain an estimator which possesses such properties. This unit is devoted to explain the criteria of good estimator. This unit also explains different properties of good estimator such as unbiasedness, consistency, efficiency and sufficiency with different examples.

Unit 6: Point Estimation

This unit explores the basic concepts of point estimation. In point estimation, we determine a single statistic whose value is used to estimate the value of unknown parameter. In this unit, we shall discuss some frequently used methods of finding point estimate such as method of maximum likelihood, method of moments and method of least squares.

Unit 7: Interval Estimation for One Population

Instead of estimating the population parameter by a single value, an interval is used for estimating the population parameter within which we can be reasonably sure that the true value of parameter will lie. This technique is known as interval estimation. In this unit, we shall discuss the method of obtaining the interval estimates of population mean, population proportion and population variance of normal population. Also we shall explore the interval estimation for population parameters of non-normal populations.

Unit 8: Interval Estimation for Two Populations

This unit is devoted to describe the method of obtaining the confidence interval for difference of population means, difference of population proportions and ratio of population variances of two normal populations.

Notations and Symbols

$X_1, X_2, \dots, X_n$	:	Random sample
$x_1, x_2, \dots, x_n$	:	Observed value of random sample
Θ	:	Parameter space and read as big theta
$f(x, \theta)$	:	Probability density (mass) function
$f(x_1, x_2, \dots, x_n, \theta)$	:	Joint probability density (mass) function of sample values
$L(\theta)$	:	Likelihood function of parameter θ
$T = t(X_1, X_2, \dots, X_n)$	:	Estimator
$E(T)$	:	Expectation of T
$\text{Var}(T)$	:	Variance of T
SE	:	Standard error
e	:	Efficiency
T^*	:	Most efficient estimator
MVUE	:	Minimum variance unbiased estimator
$\hat{\theta}$	:	Estimate of θ
$\frac{\partial}{\partial \theta}$	:	Partial derivative with respect to θ
$X_{(1)}, X_{(2)}, \dots, X_{(n)}$	:	Ordered statistic
ML	:	Maximum Likelihood
M'_r	:	r^{th} sample moment about origin
M_r	:	r^{th} sample moment about mean
μ'_r	:	r^{th} population moment about origin
μ_r	:	r^{th} population moment about mean
$1 - \alpha$	:	Confidence coefficient or Confidence level
$Q = q(X_1, X_2, \dots, X_n, \theta)$	:	Pivotal quantity
E	:	Sampling error or margin of error

UNIT 5 INTRODUCTION TO ESTIMATION

Structure

- 5.1 Introduction
 - Objectives
- 5.2 Basic Terminology
- 5.3 Characteristics of Estimators
- 5.4 Unbiasedness
- 5.5 Consistency
- 5.6 Efficiency
 - Most Efficient Estimator
 - Minimum Variance Unbiased Estimator
- 5.7 Sufficiency
- 5.8 Summary
- 5.9 Solutions /Answers

5.1 INTRODUCTION

In many real-life problems, the population parameter(s) is (are) unknown and someone is interested to obtain the value(s) of parameter(s). But, if the whole population is too large to study or the units of the population are destructive in nature or there is a limited resources and manpower available then it is not practically convenient to examine each and every unit of the population to find the value(s) of parameter(s). In such situations, one can draw sample from the population under study and utilize sample observations to estimate the parameter(s).

Every one of us makes estimate(s) in our day to day life. For example, a house wife estimates the monthly expenditure on the basis of particular needs, a sweet shopkeeper estimates the sale of sweets on a day, etc. So the technique of finding an estimator to produce an estimate of the unknown parameter on the basis of a sample is called estimation.

There are two methods of estimation:

1. Point Estimation
2. Interval Estimation

In point estimation, we determine a appropriate single statistic whose value is used to estimate the unknown parameter whereas in interval estimation, we determine an interval that contains true value of the unknown parameter with certain confidence. The point estimation and interval estimation are briefly described in Unit 6 and Unit 7 respectively of this block.

Estimation admits two problems; the first is to select some criteria or properties such that if an estimator possesses these properties it is said to be the best estimator among possible estimators and the second is to derive some methods or techniques through which we obtain an estimator which possesses such properties. This unit is devoted to explain the criteria of good estimator. This unit is divided into nine sections. Section 5.1 is introductory in nature. The basic terms used in estimation are defined in Section 5.2. Section 5.3 is devoted to criteria of good estimator which are explained one by one in subsequent

sections. Section 5.4 is explored the concept of unbiasedness with examples. Unbiasedness is based on fixed sample size whereas the concept based on varying sample size, that is, consistency is described in Section 5.5. There exists more than one consistent estimator of a parameter, therefore in Section 5.6 is explained the next property efficiency. Section 5.7 is devoted to describe sufficiency. Unit ends by providing summary of what we have discussed in this unit in Section 5.8 and solution of exercises in Section 5.9.

Objectives

After studying this unit, you should be able to:

- define the parameter space and joint probability density (mass) function;
- describe the characteristics of an estimator;
- explain the unbiasedness of an estimator;
- explain the consistency of an estimator;
- explain the efficiency of an estimator;
- explain the most efficient estimator;
- explain the sufficiency of an estimator; and
- describe the minimum variance unbiased estimator.

5.2 BASIC TERMINOLOGY

Before discussing the properties of a good estimator, we discuss basic definitions of some important terms. These terms are very useful in understanding the fundamentals of theory of estimation discussed in this block.

Discrete and Continuous Distributions

In Units 12 and 13 of MST-003, we have discussed standard discrete and continuous distributions as binomial, Poisson, normal, exponential, etc. We know that the populations can be described with the help of distributions, therefore, standard discrete and continuous distributions are used in statistical inference. Here, we discuss some standard discrete and continuous distributions in brief as in tabular form:

S. No.	Distribution	Parameter(s)	Mean	Variance
1	Bernoulli (discrete) $P[X = x] = p^x (1-p)^{1-x}; x = 0, 1$	p	p	pq
2	Binomial (discrete) $P[X = x] = {}^n C_x p^x q^{n-x}; x = 0, 1, \dots, n$	n & P	np	npq
3	Poisson (discrete) $P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, \dots \& \lambda > 0$	λ	λ	λ
4	Uniform (discrete) $P[X = x] = \frac{1}{n}; x = 0, 1, \dots, n$	n	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
5	Hypergeometric (discrete) $P[X = x] = \frac{{}^M C_x {}^{N-M} C_{n-x}}{{}^N C_n}; x = 0, 1, \dots, \min\{M, n\}$	N, M & n	$\frac{nM}{N}$	$\frac{NM(N-M)(N-n)}{N^2(N-1)}$

6	Geometric (discrete) $P[X = x] = pq^x; x = 0, 1, 2, \dots$	p	$\frac{p}{q}$	$\frac{p}{q^2}$
7	Negative Binomial (discrete) $P[X = x] = {}^{x+r-1}C_{r-1} p^r q^x; x = 0, 1, 2, \dots$	$r \text{ \& } p$	$\frac{rp}{q}$	$\frac{rp}{q^2}$
8	Normal (continuous) $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty$ & $\sigma > 0, -\infty < \mu < \infty$	$\mu \text{ \& } \sigma^2$	μ	σ^2
9	Standard Normal (continuous) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}; -\infty < x < \infty$	--	0	1
10	Uniform (continuous) $f(x) = \frac{1}{b-a}; a < x < b, b > a$	$a \text{ \& } b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
11	Exponential (continuous) $f(x) = \theta e^{-\theta x}; x \geq 0 \text{ \& } \theta > 0$	θ	$\frac{1}{\theta}$	$\frac{1}{\theta^2}$
12	Gamma (continuous) $f(x) = \frac{a^b}{\Gamma(b)} e^{-ax} x^{b-1}; x > 0 \text{ \& } a > 0$	$a \text{ \& } b$	$\frac{b}{a}$	$\frac{b}{a^2}$
13	Beta First Kind (continuous) $f(x) = \frac{1}{B(a,b)} x^{a-1} (1-x)^{b-1}; 0 < x < 1$ & $a > 0, b > 0$	$a \text{ \& } b$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2 (a+b+1)}$
14	Beta Second Kind (continuous) $f(x) = \frac{1}{B(a,b)} \frac{x^{a-1}}{(1+x)^{a+b}}; x > 0$ & $a > 0, b > 0$	$a \text{ \& } b$	$\frac{a}{b-1}$	$\frac{a(a+b+1)}{(b-1)^2 (b-2)}$

Parameter Space

The set of all possible values that the parameter θ or parameters $\theta_1, \theta_2, \dots, \theta_k$ can assume is called the parameter space. It is denoted by Θ and is read as “**big theta**”. For example, if parameter θ represents the average life of electric bulbs manufactured by a company then parameter space of θ is $\Theta = \{\theta : \theta \geq 0\}$, that is, the parameter average life θ can take all possible values greater than or equal to 0. Similarly, in normal distribution (μ, σ^2) , the parameter space of parameters μ and σ^2 is $\Theta = \{(\mu, \sigma^2) : -\infty < \mu < \infty; 0 < \sigma < \infty\}$.

Joint Probability Density (Mass) Function

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$ where, θ is the population parameter then the joint probability density (mass) function of sample values is denoted by $f(x_1, x_2, \dots, x_n, \theta)$ and defined as

For discrete case,

$$f(x_1, x_2, \dots, x_n, \theta) = P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n]$$

Estimation

since $X_1, X_2, \dots, X_n$ are independent, therefore,

$$f(x_1, x_1, \dots, x_n, \theta) = P[X_1 = x_1]P[X_2 = x_2] \dots P[X_n = x_n]$$

In this case, the function $f(x_1, x_1, \dots, x_n, \theta)$ represents the probability that the particular sample $x_1, x_2, \dots, x_n$ has been drawn for a fixed (given) value of parameter θ .

For continuous case,

$$f(x_1, x_1, \dots, x_n, \theta) = f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta)$$

In this case, the function $f(x_1, x_1, \dots, x_n, \theta)$ represents the probability density function of the random sample $X_1, X_2, \dots, X_n$.

The process of finding the joint probability density (mass) function is described by taking some examples as:

If a random sample $X_1, X_2, \dots, X_n$ of size n is taken from Poisson distribution whose pdf is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

then joint probability mass function of $X_1, X_2, \dots, X_n$ can be obtained as

$$\begin{aligned} f(x_1, x_1, \dots, x_n, \lambda) &= P[X_1 = x_1]P[X_2 = x_2] \dots P[X_n = x_n] \\ &= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} \\ &= \frac{e^{\underbrace{-\lambda - \lambda - \dots - \lambda}_{n\text{-times}}} \lambda^{x_1 + x_2 + \dots + x_n}}{x_1! x_2! \dots x_n!} \\ &= \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \end{aligned}$$

Similarly, if $X_1, X_2, \dots, X_n$ is a random sample of size n taken from exponential population whose pdf is given by

$$f(x, \theta) = \theta e^{-\theta x}; \quad x > 0, \theta > 0$$

then the joint probability density function of sample values can be obtained as

$$\begin{aligned} f(x_1, x_1, \dots, x_n, \theta) &= f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta) \\ &= \theta e^{-\theta x_1} \cdot \theta e^{-\theta x_2} \dots \theta e^{-\theta x_n} \\ &= \theta^{\underbrace{1+1+\dots+1}_{n\text{-times}}} e^{-\theta(x_1 + x_2 + \dots + x_n)} \\ f(x_1, x_1, \dots, x_n, \theta) &= \theta^n e^{-\theta \sum_{i=1}^n x_i} \end{aligned}$$

Let us check your understanding of above by answering the following exercises.

-
- E1)** What is the pmf of Poisson distribution with parameter $\lambda = 5$. Also find the mean and variance of this distribution.
- E2)** If θ represents the average marks of IGNOU's students in a paper of 50 marks. Find the parameter space of θ .
- E3)** A random sample $X_1, X_2, \dots, X_n$ of size n is taken from Poisson distribution whose pdf is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

Obtain joint probability mass function of $X_1, X_2, \dots, X_n$.

5.3 CHARACTERISTICS OF ESTIMATORS

It is to be noted that a large number of estimators can be proposed for an unknown parameter. For example, if we want to estimate the average income of the persons living in a city then the sample mean, sample median, sample mode, etc. can be used to estimate the average income. Now, the question arises, "Are some of possible estimators better, in some sense, than the others?" Generally, an estimator can be called good for two different situations:

- (i) **When the true value of parameter is being estimated is known**— An estimator might be called good if its value close to the true value of the parameter to be estimated. In other words, the estimator whose sampling distribution concentrates as closely as possible near the true value of the parameter may be regarded as the good estimator.
- (ii) **When the true value of the parameter is unknown**— An estimator may be called good if the data give good reason to believe that the estimate will be closed to the true value.

In the whole estimation, we estimate the parameter when the true value of the parameter is unknown. Hence, we must choose estimates not because they are certainly close to the true value, but because there is a good reason to believe that the estimated value will be close to the true value of parameter. In this unit, we shall describe certain properties, which help us in deciding whether an estimator is better than others.

Prof. Ronald A. Fisher was the man who pushed ahead the theory of estimation and introduced these concepts and gave some properties of good estimator as follows:

1. Unbiasedness
2. Consistency
3. Efficiency
4. Sufficiency

We shall discuss these properties one by one in the subsequent sections.

Now, give the answer of the following exercise.

-
- E4)** Write the four properties of good estimator.
-



*Prof. Ronald A. Fisher,
a Great English
Mathematical
Statistician.
(1890-1962)*

5.4 UNBIASEDNESS

An estimator is said to be unbiased if the expected value of the estimator is equal to the true value of the parameter being estimated.

Generally, population parameter(s) is (are) unknown and if the whole population is too large to study to find the value of unknown parameter(s) then one can estimate the population parameter(s) with the help of estimator(s) which is(are) always a function of sample values.

An estimator is said to be unbiased for the population parameter such as population mean, population variance, population proportion, etc. if and only if the average or mean of the sampling distribution of the estimator is equal to the true value of the parameter.

Mathematically,

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$ where, θ is the population parameter then an estimator $T = t(X_1, X_2, \dots, X_n)$ is said to be unbiased estimator of the parameter θ if and only if

$$E(T) = \theta; \quad \text{for all } \theta \in \Theta$$

This property of estimator is called unbiasedness.

Normally, it is preferable that the expected value of the estimator should be exactly equal to the true value of the parameter being estimated. But if the expected value of the estimator does not equal to the true value of parameter, then the estimator is said to be “biased estimator”, that is, if

$$E(T) \neq \theta$$

then estimator T is called biased estimator of θ .

The amount of biasness is given by

$$b(\theta) = E(T) - \theta$$

If $b(\theta) > 0$ or $E(T) > \theta$, then the estimator T is said to be positively biased for parameter θ .

If $b(\theta) < 0$ or $E(T) < \theta$, then the estimator T is said to be negatively biased for parameter θ .

If $E(T) \rightarrow \theta$ as $n \rightarrow \infty$ i.e. if an estimator T is unbiased for a large sample only then estimator T is said to be asymptotically unbiased for θ .

Now, we explain the procedure how to show that a statistic is unbiased or not for a parameter with the help of some examples:

Example 1: Show that sample mean ($\bar{X}$) is an unbiased estimator of the population mean (μ) if it exists.

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from any population with mean μ . Then for unbiasedness we have to show that

$$E(\bar{X}) = \mu$$

Consider,

$$E(\bar{X}) = E\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] \quad [\text{By definition of sample mean}]$$

$$= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] [\because E(aX + bY) = aE(X) + bE(Y)]$$

Since $X_1, X_2, \dots, X_n$ are randomly drawn from same population so they also follow the same distribution as the population. Therefore,

$$E(X_1) = E(X_2) = \dots = E(X_n) = E(X) = \mu$$

Thus,

$$E(\bar{X}) = \frac{1}{n} \left(\underbrace{\mu + \mu + \dots + \mu}_{n\text{-times}} \right)$$

$$= \frac{1}{n} (n\mu) = \mu$$

$$E(\bar{X}) = \mu$$

Hence, sample mean $(\bar{X})$ is an unbiased estimator of the population mean μ .

Also if $x_1, x_2, \dots, x_n$ are the observed values of the random sample

$X_1, X_2, \dots, X_n$ then $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is unbiased estimate of population mean.

Example 2: A random sample of 10 cadets of a centre is selected and measures their weights (in kg) which are given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 78

Determine an unbiased estimate of the average weight of cadets of the centre.

Solution: We know that sample mean $(\bar{X})$ is an unbiased estimator of the population mean and its particular value is the unbiased estimate of population mean, therefore,

$$\begin{aligned} \bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i = \frac{X_1 + X_2 + \dots + X_n}{n} \\ &= \frac{48 + 50 + 62 + 75 + 80 + 60 + 70 + 56 + 52 + 78}{10} = 63.10 \end{aligned}$$

Hence, an unbiased estimate of the average weight of cadets of the centre is 63.10 kg.

Example 3: A random sample $X_1, X_2, \dots, X_n$ of size n taken from a population whose pdf is given by

$$f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}; \quad x > 0, \theta > 0$$

Show that sample mean $(\bar{X})$ is an unbiased estimator of parameter θ .

Solution: For unbiasedness, we have to show that

$$E(\bar{X}) = \theta$$

Here, we are given that

If X and Y are two random variables and a & b are two constants then by the addition theorem of expectation, we have $E(aX + bY) = aE(X) + bE(Y)$

Estimation

$$f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}; \quad x > 0, \theta > 0$$

Since we do not know the mean of this distribution therefore first of all we find the mean of this distribution. So we consider,

$$\begin{aligned} E(X) &= \int_0^{\infty} x f(x, \theta) dx \\ &= \int_0^{\infty} x \frac{1}{\theta} e^{-x/\theta} dx = \frac{1}{\theta} \int_0^{\infty} x e^{-x/\theta} dx \\ &= \frac{1}{\theta} \int_0^{\infty} x^{2-1} e^{-x/\theta} dx = \frac{1}{\theta} \times \frac{\sqrt{2}}{(1/\theta)^2} = \theta \quad \left[\because \int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma(n)}{a^n} \right] \end{aligned}$$

Since $X_1, X_2, \dots, X_n$ are randomly drawn from same population having mean θ , therefore,

$$E(X_1) = E(X_2) = \dots = E(X_n) = E(X) = \theta$$

Consider,

If X and Y are two random variables and a & b are two constants then by the addition theorem of expectation, we have
 $E(aX + bY) = aE(X) + bE(Y)$

$$\begin{aligned} E(\bar{X}) &= E\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] \quad [\text{By definition of sample mean}] \\ &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \quad \left[\because E(aX + bY) = aE(X) + bE(Y) \right] \\ &= \frac{1}{n} (\underbrace{\theta + \theta + \dots + \theta}_{n\text{-times}}) \\ &= \frac{1}{n} (n\theta) = \theta \end{aligned}$$

Thus, $\bar{X}$ is an unbiased estimator of θ .

Note 1: If $X_1, X_2, \dots, X_n$ is a random sample taken from a population with mean μ and variance σ^2 , then

$$S'^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is a biased estimator of } \sigma^2.$$

whereas, $S^2 = \frac{n}{n-1} S'^2$ i.e. $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of σ^2 .

The proof of the above result is beyond the scope of this course. But for your convenience, we will show this result with the help of the following example.

Example 4: Consider a population comprising three televisions of certain company. If lives of televisions are 8, 6 and 10 years then construct the sampling distribution of average life of Televisions by taking samples of size 2 and show that sample mean is an unbiased estimator of population mean life. Also show that S'^2 is not an unbiased estimator of population variance whereas S^2 is an unbiased estimator of population variance where,

$$S'^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ and } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Solution: Here, population consists three televisions whose lives are 8, 6 and 10 years so we can find the population mean and variance as

$$\mu = \frac{8+6+10}{3} = 8$$

$$\sigma^2 = \frac{1}{3} [(8-8)^2 + (6-8)^2 + (10-8)^2] = \frac{8}{3} = 2.67$$

Here, we are given that

Population size = $N = 3$ and sample size = $n = 2$

Therefore, possible numbers of samples (with replacement) that can be drawn from this population are $N^n = 3^2 = 9$. For each of these 9 samples, we will calculate the values of $\bar{X}$, S'^2 and S^2 by the formulae given below:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, S'^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ and } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

and necessary calculations for these results are shown in Table 5.1 given below:

Table 5.1: Calculation for $\bar{X}$, S'^2 and S^2

Sample	Sample Observation	$\bar{X}$	$\sum_{i=1}^2 (X_i - \bar{X})^2$	S'^2	S^2
1	8, 8	8	0	0	0
2	8, 6	7	2	1	2
3	8, 10	9	2	1	2
4	6, 8	7	2	1	2
5	6, 6	6	0	0	0
6	6, 10	8	8	4	8
7	10, 8	9	2	1	2
8	10, 6	8	8	4	8
9	10, 10	10	0	0	0
Total		72		12	24

We calculate $\bar{X}$, S'^2 and S^2 as

$$\bar{X}_1 = \frac{1}{2}(8+8) = 8, \bar{X}_2 = \frac{1}{2}(8+6) = 7, \dots, \bar{X}_9 = \frac{1}{2}(10+10) = 10$$

$$S_1'^2 = \frac{1}{2} [(8-8)^2 + (8-8)^2] = 0, S_2'^2 = \frac{1}{2} [(8-7)^2 + (6-7)^2] = 1, \dots, \\ S_9'^2 = \frac{1}{2} [(10-10)^2 + (10-10)^2] = 0$$

Estimation

$$S_1^2 = \frac{1}{2-1}[(8-8)^2 + (8-8)^2] = 0, S_2^2 = \frac{1}{2-1}[(8-7)^2 + (6-7)^2] = 2, \dots, \\ S_9^2 = \frac{1}{2-1}[(10-10)^2 + (10-10)^2] = 0$$

Form the Table 5.1, we have

$$E(\bar{X}) = \frac{1}{k} \sum_{i=1}^k \bar{X}_i = \frac{1}{9} \times 72 = 8 = \mu$$

Hence, sample mean is unbiased estimator of population mean.

Also

$$E(S'^2) = \frac{1}{k} \sum_{i=1}^k S_i'^2 = \frac{1}{9} \times 12 = 1.33 \neq \sigma^2$$

Therefore, S'^2 is not an unbiased estimator of σ^2 whereas,

$$E(S^2) = \frac{1}{k} \sum_{i=1}^k S_i^2 = \frac{1}{9} \times 24 = 2.67 = \sigma^2$$

S^2 is unbiased estimator of σ^2 .

Remark 1:

1. Unbiased estimators may not be unique. For example, sample mean and sample median are unbiased estimators of population mean of normal population.
2. Unbiased estimators do not always exist for all the parameters. For example, for a Bernoulli distribution (θ), there is no unbiased estimator for θ^2 . Similarly, for a Poisson distribution (λ), there exists no unbiased estimator for $1/\lambda$.
3. If an estimator is unbiased for all types of distribution, then it is called an absolutely unbiased estimator. For example, sample mean is an absolutely unbiased estimator of population mean, if population mean exists.
4. If T_n and T_n^* are two unbiased estimator of parameter θ then $aT_n + (1-a)T_n^*$ is also unbiased estimator of θ where, 'a' ($0 \leq a \leq 1$) is any constant.

For the better understanding of the unbiasedness try some exercises.

E5) If $X_1, X_2, \dots, X_n$ is a random sample taken from Poisson distribution whose probability mass function is given by

$$P[X=x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

then show that the sample mean $\bar{X}$ is unbiased estimator of λ .

E6) If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose pdf is

$$f(x, \theta) = e^{-(x+\theta)}; \quad \theta \leq x < \infty; \quad -\infty < \theta < \infty$$

then show that the sample mean is an unbiased estimator of $(1 + \theta)$.

One weakness of unbiasedness is that it requires only the average value of the estimator equals to the true value of population parameter. It does not require those values of the estimator to be reasonably close to the population parameter. For this reason, we require some other properties of good estimator as consistency, efficiency and sufficiency which are described in subsequent sections.

5.5 CONSISTENCY

In previous section, we have learnt about the unbiasedness. An estimator T is said to be unbiased estimator of parameter, say, θ if the mean of sampling distribution of estimator T is equal to the true value of the parameter θ . This concept was defined for a fixed sample size. In this section, we will learn about consistency which is defined for increasing sample size.

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$ where, θ is the population parameter then consider a sequence of estimators, say, $T_1 = t_1(X_1)$, $T_2 = t_2(X_1, X_2)$, $T_3 = t_3(X_1, X_2, X_3), \dots, T_n = t_n(X_1, X_2, \dots, X_n)$. A sequence of estimators is said to be consistent for parameter θ if the deviation of the values of estimator from the parameter tends to zero as the sample size increases. That means values of estimators tend to get closer to the parameter θ as sample size increases.

In other words, a sequence $\{T_n\}$ of estimators is said to be consistent sequence of estimators of θ if T_n converges to θ in probability, that is

$$T_n \xrightarrow{P} \theta \text{ as } n \rightarrow \infty \text{ for every } \theta \in \Theta \quad \dots (3)$$

or for every $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} P[|T_n - \theta| < \varepsilon] = 1 \quad \dots (4)$$

or for every $\varepsilon > 0$ and $\eta > 0$ there exist $n \geq m$ such that

$$P[|T_n - \theta| < \varepsilon] > 1 - \eta ; \quad n \geq m \quad \dots (5)$$

where, m is some very large value of n . Expressions (3), (4) and (5) are to mean the same thing.

Generally, to show that an estimator is consistent with the help of above definition is slightly difficult, therefore, we use sufficient conditions for consistency which are given below:

Sufficient conditions for consistency

If $\{T_n\}$ is a sequence of estimators such that for all $\theta \in \Theta$

- (i) $E(T_n) \rightarrow \theta$ as $n \rightarrow \infty$, that is, estimator T_n is either unbiased or asymptotically unbiased estimator of θ and
- (ii) $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$, that is, variance of estimator T_n converges to 0 as $n \rightarrow \infty$.

Estimation

Then estimator T_n is a consistent estimator of θ .

Now, we explain the procedure based on both the criteria (definition and sufficient condition) to show that a statistic is consistent or not for a parameter with the help of some examples:

Example 5: Prove that sample mean is always a consistent estimator of the population mean provided that the population has a finite variance.

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample taken from a population having mean μ and finite variance σ^2 . By the definition of consistency, we have

$$\begin{aligned}\lim_{n \rightarrow \infty} P[|T_n - \theta| < \varepsilon] &= \lim_{n \rightarrow \infty} P[|\bar{X} - \mu| < \varepsilon] \quad [\text{Here, } T_n = \bar{X}] \\ &= \lim_{n \rightarrow \infty} P\left[\left|\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right| < \frac{\varepsilon\sqrt{n}}{\sigma}\right]\end{aligned}$$

By central limit theorem (described in Unit 1 of this course), we know that the variate $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is a standard normal variate for large sample size n .

Therefore,

$$\begin{aligned}\lim_{n \rightarrow \infty} P[|T_n - \theta| < \varepsilon] &= \lim_{n \rightarrow \infty} P\left[|Z| < \frac{\varepsilon\sqrt{n}}{\sigma}\right] \\ &= \lim_{n \rightarrow \infty} P\left[-\frac{\varepsilon\sqrt{n}}{\sigma} < Z < \frac{\varepsilon\sqrt{n}}{\sigma}\right] \quad [\because |X| < a \Rightarrow -a < X < a] \\ &= \lim_{n \rightarrow \infty} \int_{-\frac{\varepsilon\sqrt{n}}{\sigma}}^{\frac{\varepsilon\sqrt{n}}{\sigma}} f(z) dz \quad \left[\because P[a < U < b] = \int_a^b f(u) du\right] \\ &= \lim_{n \rightarrow \infty} \int_{-\frac{\varepsilon\sqrt{n}}{\sigma}}^{\frac{\varepsilon\sqrt{n}}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \quad \left[\because f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\right] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz\end{aligned}$$

Since $\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the pdf of a standard normal variate Z therefore, the integration of this in whole range $-\infty$ to ∞ is unity.

Thus,

$$\lim_{n \rightarrow \infty} P[|T_n - \theta| < \varepsilon] = \lim_{n \rightarrow \infty} P[|\bar{X} - \mu| < \varepsilon] = 1 \text{ as } n \rightarrow \infty$$

Hence, sample mean is a consistent estimator of population mean.

Note 2: This example can also be proved with the help of sufficient conditions for consistency as shown in next example.

Example 6: If $X_1, X_2, \dots, X_n$ is a random sample taken from Poisson distribution(λ), then show that sample mean ($\bar{X}$) is consistent estimator of λ .

Solution: We know that the mean and variance of Poisson distribution (λ) are

$$E(X) = \lambda \text{ and } \text{Var}(X) = \lambda$$

Since $X_1, X_2, \dots, X_n$ are independent and come from same Poisson distribution, therefore,

$$E(X_i) = E(X) = \lambda \text{ and } \text{Var}(X_i) = \text{Var}(X) = \lambda \quad \text{for all } i = 1, 2, \dots, n$$

Now consider,

$$\begin{aligned} E(\bar{X}) &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \quad [\text{By definition of sample mean}] \\ &= \frac{1}{n}[E(X_1) + E(X_2) + \dots + E(X_n)] \quad \left[\because E(aX + bY) = aE(X) + bE(Y)\right] \\ &= \frac{1}{n}\left(\underbrace{\lambda + \lambda + \dots + \lambda}_{n\text{-times}}\right) = \frac{1}{n}(n\lambda) = \lambda \end{aligned}$$

Thus, sample mean ($\bar{X}$) is unbiased estimator of λ .

Now consider,

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \quad \left[\begin{array}{l} \text{If } X \text{ and } Y \text{ are two} \\ \text{independent} \\ \text{random variable then} \\ \text{Var}(aX + bY) \\ = a^2\text{Var}(X) + b^2\text{Var}(Y) \end{array}\right] \\ &= \frac{1}{n^2}\left(\underbrace{\lambda + \lambda + \dots + \lambda}_{n\text{-times}}\right) \\ &= \frac{1}{n^2}(n\lambda) \\ \text{Var}(\bar{X}) &= \frac{\lambda}{n} \rightarrow 0 \text{ as } n \rightarrow \infty \end{aligned}$$

Hence, by sufficient condition of consistency, it follows that sample mean ($\bar{X}$) is consistent estimator of λ .

Remark 2:

1. Consistent estimators may not be unique. For example, sample mean and sample median are consistent estimators of population mean of normal population.
2. An unbiased estimator may or may not be consistent.
3. A consistent estimator may or may not be unbiased.

Estimation

Continuous function is described in Sections 5.7 and 5.8 of Unit 5 of MST- 001.

4. If T_n is a consistent estimator of θ and f is a continuous function of θ then $f(T_n)$ is consistent estimator of $f(\theta)$. This property is known as invariance property. For example, if $\bar{X}$ is a consistent estimator of population mean θ then $e^{\bar{X}}$ also consistent estimator of e^θ because e^θ is a continuous function of θ .

It is now time for you to try the following exercises to make sure that you have understood consistency.

E7) If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from pdf

$$f(x, \theta) = \begin{cases} 1; & \theta \leq x \leq \theta + 1 \\ 0; & \text{elsewhere} \end{cases}$$

then show that the sample mean is an unbiased as well as consistent estimator of $\left(\theta + \frac{1}{2}\right)$.

E8) If $X_1, X_2, \dots, X_n$ are n observations taken from geometric distribution with parameter θ , then show that $\bar{X}$ is consistent estimator of $1/\theta$. Also find consistent estimator of $e^{1/\theta}$.

5.6 EFFICIENCY

In some situations, we see that there are more than one estimators of a parameter which are unbiased as well as consistent. For example, sample mean and sample median both are unbiased and consistent for the parameter μ when sampling is done from normal population with mean μ and known variance σ^2 . In such situations, there arises a necessity of some other criterion which will help us to choose 'best estimator' among them. A criterion which is based on the concept of variance of the sampling distribution of the estimator is termed as efficiency.

If T_1 and T_2 are two estimators of a parameter θ . Then T_1 is said to be more efficient than T_2 for all sample sizes if

$$\text{Var}(T_1) < \text{Var}(T_2) \quad \text{for all } n$$

Let us do some examples about efficiency:

Example 7: Show that sample mean is more efficient estimator than sample median for estimating mean of normal population.

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample taken from normal population with mean μ and variance σ^2 . Also let $\bar{X}$ and $\tilde{X}$ be the sample mean and sample median respectively. We have seen in Unit 2 that the sampling distribution of the mean from a normal population follows normal distribution with means μ and variance σ^2/n . Similarly, it can be shown that the sampling distribution of the median from a normal population also follows normal distribution with mean μ and variance $\frac{\pi \sigma^2}{2n}$. Therefore,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$\text{Var}(\tilde{X}) = \frac{\pi\sigma^2}{2n}$$

But $\frac{\pi\sigma^2}{2n} > \frac{\sigma^2}{n} \left[\because \frac{\pi}{2} \text{ and } \frac{\sigma^2}{n} > 1 \right]$ therefore, $\text{Var}(\bar{X}) < \text{Var}(\tilde{X})$. Thus, we

conclude that sample mean is more efficient estimator than sample median.

Example 8: If X_1, X_2, X_3, X_4 and X_5 is a random sample of size 5 taken from a population with mean μ and variance σ^2 . The following two estimators are suggested to estimate μ

$$T_1 = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}, \quad T_2 = \frac{X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5}{15}$$

Are both estimators unbiased? Which one is more efficient?

Solution: Since $X_1, X_2, \dots, X_5$ are independent and taken from same population with mean μ and variance σ^2 therefore,

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2 \quad \text{for all } i = 1, 2, \dots, 5$$

Consider,

$$\begin{aligned} E(T_1) &= E\left[\frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}\right] \\ &= \frac{1}{5} [E(X_1) + E(X_2) + E(X_3) + E(X_4) + E(X_5)] \\ &= \frac{1}{5} [\mu + \mu + \mu + \mu + \mu] \end{aligned}$$

$$E(T_1) = \mu$$

Similarly,

$$\begin{aligned} E(T_2) &= E\left[\frac{X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5}{15}\right] \\ &= \frac{1}{15} [E(X_1) + 2E(X_2) + 3E(X_3) + 4E(X_4) + 5E(X_5)] \\ &= \frac{1}{15} [\mu + 2\mu + 3\mu + 4\mu + 5\mu] \\ &= \frac{1}{15} (15\mu) \end{aligned}$$

$$E(T_2) = \mu$$

Hence, both the estimators T_1 and T_2 are unbiased estimators of μ .

Now for efficiency, we consider

$$\text{Var}(T_1) = \text{Var}\left[\frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}\right]$$

Estimation

$$\begin{aligned} &= \frac{1}{25} [\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \text{Var}(X_4) + \text{Var}(X_5)] \\ &= \frac{1}{25} [\sigma^2 + \sigma^2 + \sigma^2 + \sigma^2 + \sigma^2] \\ &= \frac{1}{25} (5\sigma^2) \end{aligned}$$

$$\text{Var}(T_1) = \frac{1}{5} \sigma^2$$

Similarly,

$$\begin{aligned} \text{Var}(T_2) &= \text{Var} \left[\frac{X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5}{15} \right] \\ &= \frac{1}{225} \left[\text{Var}(X_1) + 4\text{Var}(X_2) + 9\text{Var}(X_3) \right. \\ &\quad \left. + 16\text{Var}(X_4) + 25\text{Var}(X_5) \right] \\ &= \frac{1}{225} [\sigma^2 + 4\sigma^2 + 9\sigma^2 + 16\sigma^2 + 25\sigma^2] = \frac{55\sigma^2}{225} \\ \text{Var}(T_2) &= \frac{11\sigma^2}{45} \end{aligned}$$

Since, $\text{Var}(T_1) < \text{Var}(T_2)$, therefore, we conclude that estimator T_1 is more efficient than T_2 .

5.6.1 Most Efficient Estimator

In a class of estimators of a parameter, if there exists one estimator whose variance is minimum (least) among the class, then it is called most efficient estimator of that parameter. For example, suppose T_1 , T_2 and T_3 are three estimators of parameter θ having variance $1/n$, $1/(n+1)$ and $5/n$ respectively. Since variance of estimator T_2 is minimum, therefore, estimator T_2 is most efficient estimator in that class.

The efficiency of an estimator measured with respect to the most efficient estimator is called “**Absolute Efficiency**”. If T^* is the most efficient estimator having variance $\text{Var}(T^*)$ and T is any other estimator having variance $\text{Var}(T)$, then efficiency of T is defined as

$$e = \frac{\text{Var}(T^*)}{\text{Var}(T)}$$

Since variance of most efficient estimator is minimum, therefore,

$$e = \frac{\text{Var}(T^*)}{\text{Var}(T)} < 1$$

5.6.2 Minimum Variance Unbiased Estimator

An estimator T of parameter θ is said to be minimum variance unbiased estimator (MVUE) of θ if and only if

- (i) $E(T) = \theta$ for all $\theta \in \Theta$, that is, estimator T is unbiased estimator of θ and
- (ii) $\text{Var}(T) \leq \text{Var}(T')$ for all $\theta \in \Theta$, that is, variance of estimator T is less than or equal to variance of any other unbiased estimator T' .

The minimum variance unbiased estimator (MVUE) is the most efficient unbiased estimator of parameter θ in the sense that it has minimum variance in class of unbiased estimators. Some authors used uniformly minimum variance unbiased estimator (UMVUE) in place of minimum variance unbiased estimator (MVUE).

Now, you can try the following exercises.

E9) If $X_1, X_2, \dots, X_n$ is a random sample taken from a population having

mean μ and variance σ^2 , then show that the statistic $T' = \frac{1}{n+1} \sum_{i=1}^n X_i$ is biased but more efficient than sample mean for estimating the population mean.

E10) Suppose $X_1, X_2, \dots, X_n$ is a random sample taken from normal population with mean μ and variance σ^2 . The following two estimators are suggested to estimate μ as

$$T_1 = \frac{X_1 + X_2 + X_3}{3} \quad \text{and} \quad T_2 = \frac{X_1 + X_2}{2} + X_3$$

Are both estimators unbiased? Which one of them is more efficient?

E11) Define most efficient estimator and minimum variance unbiased estimator.

5.7 SUFFICIENCY

In statistical inference, the aim of the investigator or statistician may be to make a decision about the value of the unknown parameter (θ). The information that guides the investigator in making a decision is supplied by the random sample $X_1, X_2, \dots, X_n$. However, in most of the cases the observations would be too numerous and too complicated. Direct use of these observations is complicated or cumbersome, therefore, a simplification or condensation would be desirable. The technique of condensing or reducing the random sample $X_1, X_2, \dots, X_n$ into a statistic such that it contains all the information about parameter θ that is contained in the sample is known as sufficiency. So prior to continuing our search of finding best estimator, we introduce the concept of sufficiency.

A sufficient statistic is a particular kind of statistic that condenses random sample $X_1, X_2, \dots, X_n$ in a statistic $T = t(X_1, X_2, \dots, X_n)$ in such a way that no information about parameter θ is lost. That means, it contains all the information about θ that is contained in the sample and if we know the value of sufficient statistic, then the sample values themselves are not needed and can nothing tell you more about θ . In other words,

Estimation

A statistic T is said to be sufficient statistic for estimating a parameter θ if it contains all the information about θ which are available in the sample. This property of an estimator is called sufficiency. In other words,

An estimator T is sufficient for parameter θ if and only if the conditional distribution of $X_1, X_2, \dots, X_n$ given $T = t$ is independent of θ .

Mathematically,

$$f(x_1, x_2, \dots, x_n / T = t) = g(x_1, x_2, \dots, x_n)$$

where, the function $g(x_1, x_2, \dots, x_n)$ does not depend on the parameter θ .

Note 3: Generally, the above definition is used to show that a particular statistic is not a sufficient statistic because it may be very tedious task to obtain the conditional distribution. Hence, we use factorization theorem which facilitates us to find sufficient statistic without any difficulty which is given below.

Theorem (Factorization Theorem): Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from the probability density (mass) function $f(x, \theta)$. A statistic or estimator T is said to be a sufficient for parameter θ if and only if the joint density (mass) function of $X_1, X_2, \dots, X_n$ can be factored as

$$f(x_1, x_2, \dots, x_n, \theta) = g[t(x), \theta] \cdot h(x_1, x_2, \dots, x_n)$$

where, the function $g[t(x), \theta]$ is a non-negative function of parameter θ and observed sample values $(x_1, x_2, \dots, x_n)$ only through the function $t(x)$ and the function $h(x_1, x_2, \dots, x_n)$ is non-negative function of $(x_1, x_2, \dots, x_n)$ and does not involve the parameter θ .

For applying factorization theorem, we try to factor the joint density (mass) function as the product of two functions, one of which is function of parameter(s) and another is independent of parameter(s).

The proof of this theorem is beyond the scope of this course.

Note 4: The factorization theorem should not be used to show that a given statistic or estimator T is not sufficient.

Now, do some examples to show that a statistic is sufficient estimator for a parameter by using the factorization theorem:

Example 9: Show that the sample mean is sufficient for the parameter λ of the Poisson distribution.

Solution: We know that the probability mass function of Poisson distribution with parameter λ is

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

Let $X_1, X_2, \dots, X_n$ be a random sample taken from Poisson distribution with parameter λ . Then joint mass function of $X_1, X_2, \dots, X_n$ can be obtained as

$$f(x_1, x_2, \dots, x_n, \lambda) = P[X_1 = x_1] \cdot P[X_2 = x_2] \dots P[X_n = x_n]$$

$$\begin{aligned}
&= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \cdots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} \\
&= \frac{e^{\overbrace{-\lambda - \lambda - \dots - \lambda}^{n \text{ times}}} \lambda^{x_1 + x_2 + \dots + x_n}}{x_1! x_2! \dots x_n!} \\
&= \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \quad \left[\prod_{i=1}^n x_i! \text{-- represents the product of } x_i! \right]
\end{aligned}$$

The joint mass function can be factored as

$$\begin{aligned}
f(x_1, x_2, \dots, x_n, \lambda) &= (e^{-n\lambda} \lambda^{n\bar{x}}) \times \left(\frac{1}{\prod_{i=1}^n x_i!} \right) \quad \left[\because \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \right] \\
&= g(t(x), \lambda) \cdot h(x_1, x_2, \dots, x_n)
\end{aligned}$$

where, $g(t(x), \lambda) = e^{-n\lambda} \lambda^{n\bar{x}}$ is a function of parameter λ and the observed sample values $x_1, x_2, \dots, x_n$ only through $t(x) = \bar{x}$ and $h(x_1, x_2, \dots, x_n) = \frac{1}{\prod_{i=1}^n x_i!}$ is

a function of sample values $x_1, x_2, \dots, x_n$ and is independent of parameter λ .

Hence, by factorization theorem of sufficiency, $\bar{X}$ is sufficient statistic for λ .

Note 4: Since throughout the course we are using capital letter for statistic or estimator therefore in the last line of above example we use $\bar{X}$ in place of $\bar{x}$. Thus, in all the examples and exercises relating to sufficient statistic, we are using similar approach.

Example 10: A random sample $X_1, X_2, \dots, X_n$ of size n is taken from gamma distribution whose pdf is given by

$$f(x, a, b) = \frac{a^b}{\Gamma(b)} e^{-ax} x^{b-1}$$

Obtain sufficient statistic for

- (i) 'b' when 'a' is known
- (ii) 'a' when 'b' is known
- (iii) 'a' and 'b' both are unknown.

Solution: The joint density function can be obtained as

$$\begin{aligned}
f(x_1, x_2, \dots, x_n, a, b) &= f(x_1, a, b) f(x_2, a, b) \dots f(x_n, a, b) \\
&= \frac{a^b}{\Gamma(b)} e^{-ax_1} x_1^{b-1} \cdot \frac{a^b}{\Gamma(b)} e^{-ax_2} x_2^{b-1} \dots \frac{a^b}{\Gamma(b)} e^{-ax_n} x_n^{b-1} \\
&= \frac{a^{nb}}{(\Gamma(b))^n} e^{-a \sum_{i=1}^n x_i} \left(\prod_{i=1}^n x_i \right)^{b-1}
\end{aligned}$$

Estimation

Case I: When 'a' is known then we find sufficient statistic for 'b'

The joint density function can be factored as

$$f(x_1, x_2, \dots, x_n, b) = \left[\frac{a^{nb}}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1} \right] \cdot e^{-a \sum_{i=1}^n x_i}$$
$$= g[t(x), b] \cdot h(x_1, x_2, \dots, x_n)$$

where, $g[t(x), b] = \frac{a^{nb}}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1}$ is a function of parameter 'b' and sample

values $x_1, x_2, \dots, x_n$ only through $t(x) = \prod_{i=1}^n x_i$ and $h(x_1, x_2, \dots, x_n) = e^{-a \sum_{i=1}^n x_i}$ is a function of sample values $x_1, x_2, \dots, x_n$ and is independent of parameter 'b'.

Hence, by factorization theorem of sufficiency, $\prod_{i=1}^n X_i$ is sufficient statistic for parameter 'b'

Case II: When 'b' is known then we find sufficient statistic for 'a'

The joint density function can be factored as

$$f(x_1, x_2, \dots, x_n, a) = \left[a^{nb} e^{-a \sum_{i=1}^n x_i} \right] \cdot \frac{1}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1}$$
$$= g[t(x), a] \cdot h(x_1, x_2, \dots, x_n)$$

Since 'b' is known so 'b' is treated as a constant.

where, $g[t(x), a] = a^{nb} e^{-a \sum_{i=1}^n x_i}$ is a function of parameter 'a' and sample values

$x_1, x_2, \dots, x_n$ only through $t(x) = \sum_{i=1}^n x_i$ and $h(x_1, x_2, \dots, x_n) = \frac{1}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1}$ is a

function of sample values $x_1, x_2, \dots, x_n$ and is independent of parameter 'a'.

Hence by factorization theorem of sufficiency, $\sum_{i=1}^n X_i$ is sufficient statistic for 'a'.

Case III: When 'a' and 'b' are unknown then we find jointly sufficient statistics for 'a' and 'b'

The joint density function can be factored as

$$f(x_1, x_2, \dots, x_n, a, b) = \left[a^{nb} e^{-a \sum_{i=1}^n x_i} \frac{1}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1} \right] \cdot 1$$
$$= g[t_1(x), t_2(x), a, b] \cdot h(x_1, x_2, \dots, x_n)$$

Since 'a' is known so 'a' is treated as a constant.

where, $g[t_1(x), t_2(x), a, b] = a^{nb} e^{-a \sum_{i=1}^n x_i} \frac{1}{(\overline{b})^n} \left(\prod_{i=1}^n x_i \right)^{b-1}$ is a function of

parameters 'a' & 'b' and sample values $x_1, x_2, \dots, x_n$ only through $t_1(x) = \sum_{i=1}^n x_i$

and $t_2(x) = \prod_{i=1}^n x_i$ whereas, $h(x_1, x_2, \dots, x_n) = 1$ and independent of parameters 'a' and 'b'.

Hence, by factorization theorem, $\sum_{i=1}^n X_i$ and $\prod_{i=1}^n X_i$ are jointly sufficient for parameters 'a' & 'b'.

Example 11: If $X_1, X_2, \dots, X_n$ is a random sample taken from uniform distribution $U(\alpha, \beta)$, find the sufficient statistics for α and β .

Solution: The probability density function of $U(\alpha, \beta)$ is given by

$$f(x, \alpha, \beta) = \frac{1}{\beta - \alpha}; \quad \alpha \leq x \leq \beta$$

The joint density function can be obtained as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \alpha, \beta) &= f(x_1, \alpha, \beta) \cdot f(x_2, \alpha, \beta) \dots f(x_n, \alpha, \beta) \\ &= \frac{1}{\beta - \alpha} \cdot \frac{1}{\beta - \alpha} \dots \frac{1}{\beta - \alpha} \\ &= \frac{1}{(\beta - \alpha)^n} \end{aligned}$$

Since the range of variable depends upon the parameters so we consider ordered statistics $X_{(1)}, X_{(2)}, \dots, X_{(n)}$. Therefore, the joint density function can be factored as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \alpha, \beta) &= \frac{1}{(\beta - \alpha)^n}; \quad \alpha \leq x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} \leq \beta \\ &= \left[\frac{1}{(\beta - \alpha)^n} I_1(x_{(1)}, \alpha) I_2(x_{(n)}, \beta) \right] \cdot 1 \end{aligned}$$

where, $x_{(1)}$ and $x_{(n)}$ are the minimum and maximum sample observations respectively and

$$\begin{aligned} I_1(x_{(1)}, \alpha) &= \begin{cases} 1; & \text{if } x_{(1)} \geq \alpha \\ 0; & \text{otherwise} \end{cases} \\ I_2(x_{(n)}, \beta) &= \begin{cases} 1; & \text{if } x_{(n)} \leq \beta \\ 0; & \text{otherwise} \end{cases} \end{aligned}$$

Therefore,

$$f(x_1, x_2, \dots, x_n, \alpha, \beta) = g[t_1(x), t_2(x), \alpha, \beta] \cdot h(x_1, x_2, \dots, x_n)$$

where, $g[t_1(x), t_2(x), \alpha, \beta] = \left[\frac{1}{(\beta - \alpha)^n} I_1(x_{(1)}, \alpha) I_2(x_{(n)}, \beta) \right]$ is a function of parameters (α, β) and sample values $x_1, x_2, \dots, x_n$ only through $t_1(x) = x_{(1)}$ and $t_2(x) = x_{(n)}$ whereas, $h(x_1, x_2, \dots, x_n) = 1$ and independent of parameters ' α ' and ' β '.

Hence, by factorization theorem of sufficiency, $X_{(1)}$ and $X_{(n)}$ are jointly sufficient for α and β .

Remark 3:

1. A sufficient estimator is always a consistent estimator.
2. A sufficient estimator may be unbiased.
3. A sufficient estimator is the most efficient estimator if an efficient estimator exists.
4. The random sample $X_1, X_2, \dots, X_n$ and order statistics $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ are always sufficient estimators because both contain all the information about the parameter(s) of the population.
5. If T is a sufficient statistic for the parameter θ and $\phi(T)$ is a one to one function of T then $\phi(T)$ is also sufficient for θ . For example, if $T = \sum X_i$ is sufficient statistic for parameter θ then $\bar{X} = \frac{1}{n} \sum X_i = \frac{T}{n}$ is also sufficient for θ because $\bar{X} = \frac{T}{n}$ is a one to one function of T .

Now, you will understand more clearly about the sufficiency, when you try the following exercises.

-
- E12)** If $X_1, X_2, \dots, X_n$ is a random sample taken from $\exp(\theta)$ then find sufficient statistic for θ .
- E13)** If $X_1, X_2, \dots, X_n$ is a random sample taken from normal population $N(\mu, \sigma^2)$, then obtain sufficient statistic for μ and σ^2 or both according as other parameter is known or unknown.
- E14)** If $X_1, X_2, \dots, X_n$ is a random sample from uniform population over the interval $[0, \theta]$. Find sufficient estimator of θ .
-

We now end this unit by giving a summary of what we have covered in it.

5.8 SUMMARY

In this unit, we have covered the following points:

1. The parameter space and joint probability density (mass) function.
2. The basic characteristics of an estimator.
3. Unbiasedness of an estimator.
4. Consistency of an estimator.
5. Efficiency of an estimator.

6. The most efficient estimator.
7. Minimum variance unbiased estimator.
8. The sufficiency of an estimator.

5.9 SOLUTIONS / ANSWERS

E1) We know that the pmf of Poisson distribution is

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

and mean and variance of this distribution are

$$\text{Mean} = \text{Variance} = \lambda$$

In our case, $\lambda = 5$, therefore, pmf of Poisson distribution is

$$P[X = x] = \frac{e^{-5} 5^x}{x!}; \quad x = 0, 1, 2, \dots$$

Also mean and variance of this distribution are

$$\text{Mean} = \text{Variance} = \lambda = 5$$

E2) Since parameter θ represents the average marks of IGNOU's students in a paper of 50 marks, therefore, a student can take minimum 0 marks and maximum 50 marks. Thus, the parameter space of θ is $\Theta = \{\theta : 0 \leq \theta \leq 50\}$.

E3) The probability mass function of Poisson distribution is

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

The joint probability mass function of $X_1, X_2, \dots, X_n$ can be obtained as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \lambda) &= P[X_1 = x_1] \cdot P[X_2 = x_2] \dots P[X_n = x_n] \\ &= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} \\ &= \frac{e^{\underbrace{-\lambda - \lambda - \dots - \lambda}_{n \text{ times}}} \lambda^{x_1 + x_2 + \dots + x_n}}{x_1! x_2! \dots x_n!} \\ &= \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \end{aligned}$$

E4) Refer Section 5.3.

E5) We know that the mean of Poisson distribution with parameter λ is λ i.e.

$$E(X) = \lambda$$

Estimation

Since $X_1, X_2, \dots, X_n$ are independent and come from same Poisson distribution, therefore,

$$E(X_i) = E(X) = \lambda \quad \text{for all } i = 1, 2, \dots, n$$

Now consider,

$$\begin{aligned} E(\bar{X}) &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \quad \left[\begin{array}{l} \text{By definition of} \\ \text{sample mean} \end{array} \right] \\ &= \frac{1}{n}[E(X_1) + E(X_2) + \dots + E(X_n)] \quad \left[\begin{array}{l} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{array} \right] \\ &= \frac{1}{n}\left(\underbrace{\lambda + \lambda + \dots + \lambda}_{n\text{-times}}\right) = \frac{1}{n}(n\lambda) = \lambda \end{aligned}$$

Hence, sample mean ($\bar{X}$) is unbiased estimator of parameter λ .

E6) Here, we have to show that

$$E(\bar{X}) = 1 + \theta$$

We are given that

$$f(x, \theta) = e^{-(x-\theta)}; \quad \theta \leq x < \infty, \quad -\infty < \theta < \infty$$

Since we do not know the mean of this distribution therefore first of all we find the mean of this distribution. So we consider,

$$\begin{aligned} E(X) &= \int_{\theta}^{\infty} x f(x, \theta) dx \\ &= \int_{\theta}^{\infty} x e^{-(x-\theta)} dx \end{aligned}$$

Putting $x - \theta = y \Rightarrow dx = dy$. Also when $x = \theta \Rightarrow y = 0$ & when $x \rightarrow \infty \Rightarrow y \rightarrow \infty$. Therefore,

$$\begin{aligned} E(X) &= \int_0^{\infty} (y + \theta) e^{-y} dy \\ &= \int_0^{\infty} y e^{-y} dy + \theta \int_0^{\infty} e^{-y} dy \\ &= \int_0^{\infty} y^{2-1} e^{-y} dy + \theta \int_0^{\infty} y^{1-1} e^{-y} dy \\ &= \overline{2} + \theta \times \overline{1} = 1 + \theta \quad \left[\begin{array}{l} \because \int_0^{\infty} x^{n-1} e^{-x} dx = \overline{n} \\ \text{and } \overline{2} = \overline{1} = 1 \end{array} \right] \end{aligned}$$

Since $X_1, X_2, \dots, X_n$ are independent and come from same population, so

$$E(X_i) = E(X) = 1 + \theta \quad \text{for all } i = 1, 2, \dots, n$$

Now consider,

$$\begin{aligned}
 E(\bar{X}) &= E\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] \quad \left[\begin{array}{l} \text{By definition of} \\ \text{sample mean} \end{array}\right] \\
 &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \quad \left[\begin{array}{l} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{array}\right] \\
 &= \frac{1}{n} \left[\underbrace{(1 + \theta) + (1 + \theta) + \dots + (1 + \theta)}_{n\text{-times}} \right] \\
 &= \frac{1}{n} [n(1 + \theta)] \\
 &= 1 + \theta
 \end{aligned}$$

Thus, sample mean is an unbiased estimator of $(1 + \theta)$.

E7) We have

$$f(x, \theta) = 1; \quad \theta \leq x \leq \theta + 1$$

This is the pdf of uniform distribution $U[\theta, \theta + 1]$ and we know that for $U[a, b]$

$$E(X) = \frac{a + b}{2} \text{ and } \text{Var}(X) = \frac{(b - a)^2}{12}$$

In our case, $a = \theta$ and $b = \theta + 1$, therefore

$$E(X) = \frac{\theta + \theta + 1}{2} = \theta + \frac{1}{2} \text{ and } \text{Var}(X) = \frac{(\theta + 1 - \theta)^2}{12} = \frac{1}{12}$$

Since $X_1, X_2, \dots, X_n$ are independent and come from same population, therefore,

$$E(X_i) = E(X) = \theta + \frac{1}{2} \text{ and } \text{Var}(X_i) = \text{Var}(X) = \frac{1}{12} \quad \forall i = 1, 2, \dots, n$$

To show that $\bar{X}$ is unbiased estimator of $\left(\theta + \frac{1}{2}\right)$, we consider

$$\begin{aligned}
 E(\bar{X}) &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \quad \left[\begin{array}{l} \text{By definition of} \\ \text{sample mean} \end{array}\right] \\
 &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \quad \left[\begin{array}{l} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{array}\right] \\
 &= \frac{1}{n} \left[\underbrace{\left(\theta + \frac{1}{2}\right) + \left(\theta + \frac{1}{2}\right) + \dots + \left(\theta + \frac{1}{2}\right)}_{n\text{-times}} \right] \\
 &= \frac{1}{n} \left[n \left(\theta + \frac{1}{2} \right) \right]
 \end{aligned}$$

Estimation

$$= \theta + \frac{1}{2}$$

Therefore, $\bar{X}$ is unbiased estimator of $\left(\theta + \frac{1}{2}\right)$.

For consistency, we have to show that

$$E(\bar{X}) \rightarrow \theta + \frac{1}{2} \quad \text{and} \quad \text{Var}(\bar{X}) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

If X and Y are two independent random variables and a & b are two constants then

$$\begin{aligned} \text{Var}(aX + bY) \\ = a^2 \text{Var}(X) + b^2 \text{Var}(Y) \end{aligned}$$

Now consider,

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{n^2} \left(\underbrace{\frac{1}{12} + \frac{1}{12} + \dots + \frac{1}{12}}_{n\text{-times}} \right) \\ &= \frac{1}{n^2} \left(\frac{n}{12} \right) \end{aligned}$$

$$\text{Now, } \text{Var}(\bar{X}) = \frac{1}{12n} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

$$\text{Thus, } E(\bar{X}) = \theta + \frac{1}{2} \quad \text{and} \quad \text{Var}(\bar{X}) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Hence, sample mean $\bar{X}$ is also consistent estimator of $\left(\theta + \frac{1}{2}\right)$.

E8) We know that the mean and variance of geometric distribution(θ) are given by

$$E(X) = \frac{1}{\theta} \quad \text{and} \quad \text{Var}(X) = \frac{1-\theta}{\theta^2}$$

Since $X_1, X_2, \dots, X_n$ are independent and come from same geometric distribution, therefore,

$$E(X_i) = E(X) \quad \text{and} \quad \text{Var}(X_i) = \text{Var}(X) \quad \text{for all } i = 1, 2, \dots, n$$

First, we show sample mean $\bar{X}$ is consistent estimator of $1/\theta$.

Therefore, we consider,

$$E(\bar{X}) = E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \quad \left[\begin{array}{l} \text{By definition of} \\ \text{sample mean} \end{array} \right]$$

$$\begin{aligned}
&= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \quad \begin{bmatrix} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{bmatrix} \\
&= \frac{1}{n} \left(\underbrace{\frac{1}{\theta} + \frac{1}{\theta} + \dots + \frac{1}{\theta}}_{n\text{-times}} \right) \\
&= \frac{1}{n} \left(\frac{n}{\theta} \right) = \frac{1}{\theta}
\end{aligned}$$

Now consider,

$$\begin{aligned}
\text{Var}(\bar{X}) &= \text{Var} \left[\frac{1}{n} (X_1 + X_2 + \dots + X_n) \right] \\
&= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\
&= \frac{1}{n^2} \left[\underbrace{\left(\frac{1-\theta}{\theta^2} \right) + \left(\frac{1-\theta}{\theta^2} \right) + \dots + \left(\frac{1-\theta}{\theta^2} \right)}_{n\text{-times}} \right] \\
&= \frac{1}{n^2} \left[n \left(\frac{1-\theta}{\theta^2} \right) \right] = \frac{1}{n} \left(\frac{1-\theta}{\theta^2} \right) \\
\text{Var}(\bar{X}) &= \frac{1}{n} \left(\frac{1-\theta}{\theta^2} \right) \rightarrow 0 \text{ as } n \rightarrow \infty
\end{aligned}$$

Since $E(\bar{X}) = \frac{1}{\theta}$ and $\text{Var}(\bar{X}) \rightarrow 0$ as $n \rightarrow \infty$

Hence, sample mean $\bar{X}$ is consistent estimator of $1/\theta$.

Since $e^{1/\theta}$ is continuous function of $1/\theta$ therefore, by invariance property of consistency $e^{\bar{X}}$ is consistent estimator of $e^{1/\theta}$.

E9) Since $X_1, X_2, \dots, X_n$ is a random sample taken from a population having mean μ and variance σ^2 .

Therefore,

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2 \quad \text{for all } i = 1, 2, \dots, n$$

Consider,

$$\begin{aligned}
E(T') &= E \left(\frac{1}{n+1} \sum_{i=1}^n X_i \right) \\
&= \frac{1}{n+1} E(X_1 + X_2 + \dots + X_n) \\
&= \frac{1}{n+1} [E(X_1) + E(X_2) + \dots + E(X_n)] \quad \begin{bmatrix} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{bmatrix}
\end{aligned}$$

Estimation

$$= \frac{1}{n+1} \left(\underbrace{\mu + \mu + \dots + \mu}_{n\text{-times}} \right)$$

$$= \frac{1}{n+1} (n\mu) \neq \mu$$

Therefore, T' is biased estimator of population mean μ .

For efficiency, we find variances of estimator T' and sample mean $\bar{X}$ as

$$\begin{aligned} \text{Var}(T') &= \text{Var} \left[\frac{1}{n+1} (X_1 + X_2 + \dots + X_n) \right] \\ &= \frac{1}{(n+1)^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{(n+1)^2} \left(\underbrace{\sigma^2 + \sigma^2 + \dots + \sigma^2}_{n\text{-times}} \right) \\ &= \frac{1}{(n+1)^2} (n\sigma^2) = \frac{n\sigma^2}{(n+1)^2} \end{aligned}$$

If X and Y are two independent random variables and a & b are two constants then

$$\begin{aligned} \text{Var}(aX + bY) \\ &= a^2 \text{Var}(X) + b^2 \text{Var}(Y) \end{aligned}$$

Now consider,

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var} \left[\frac{1}{n} (X_1 + X_2 + \dots + X_n) \right] \\ &= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{n^2} \left(\underbrace{\sigma^2 + \sigma^2 + \dots + \sigma^2}_{n\text{-times}} \right) \\ &= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n} \end{aligned}$$

Since, $\text{Var}(T') < \text{Var}(\bar{X})$, therefore, T' is more efficient estimator than sample mean.

E10) Since $X_1, X_2, \dots, X_n$ are independent taken from normal population with mean μ and variance σ^2 therefore,

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2 \quad \text{for all } i = 1, 2, 3$$

To check estimators T_1 and T_2 are unbiased, we find expectations of T_1 and T_2 as

$$\begin{aligned} E(T_1) &= E \left(\frac{X_1 + X_2 + X_3}{3} \right) \\ &= \frac{1}{3} [E(X_1) + E(X_2) + E(X_3)] \\ E(T_1) &= \frac{1}{3} [\mu + \mu + \mu] \end{aligned}$$

If X and Y are two independent random variables and a & b are two constants then

$$\begin{aligned} E(aX + bY) \\ &= aE(X) + bE(Y) \text{ and} \end{aligned}$$

$$\begin{aligned} \text{Var}(aX + bY) \\ &= a^2 \text{Var}(X) + b^2 \text{Var}(Y) \end{aligned}$$

$$= \frac{1}{\mu} (3\mu) = \mu$$

Now consider,

$$\begin{aligned} E(T_2) &= E\left(\frac{X_1 + X_2}{4} + \frac{X_3}{2}\right) \\ &= \frac{1}{4}[E(X_1) + E(X_2)] + \frac{1}{2}E(X_3) \\ &= \frac{1}{4}[\mu + \mu] + \frac{\mu}{2} = \frac{\mu}{2} + \frac{\mu}{2} \\ E(T_2) &= \mu \end{aligned}$$

Hence, T_1 and T_2 both are unbiased estimators of μ .

For efficiency, we find the variances of T_1 and T_2 as

$$\begin{aligned} \text{Var}(T_1) &= \text{Var}\left(\frac{X_1 + X_2 + X_3}{3}\right) \\ &= \frac{1}{9}[\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)] \\ &= \frac{1}{9}[\sigma^2 + \sigma^2 + \sigma^2] = \frac{3\sigma^2}{9} \end{aligned}$$

$$\text{Var}(T_1) = \frac{\sigma^2}{3}$$

Now consider,

$$\begin{aligned} \text{Var}(T_2) &= \text{Var}\left(\frac{X_1 + X_2}{4} + \frac{X_3}{2}\right) \\ &= \frac{1}{16}[\text{Var}(X_1) + \text{Var}(X_2)] + \frac{1}{4}\text{Var}(X_3) \\ &= \frac{1}{16}[\sigma^2 + \sigma^2] + \frac{1}{4}\sigma^2 \\ &= \frac{\sigma^2}{8} + \frac{\sigma^2}{4} = \frac{\sigma^2 + 2\sigma^2}{8} \\ &= \frac{3\sigma^2}{8} \end{aligned}$$

Since $\text{Var}(T_1) < \text{Var}(T_2)$ therefore, T_1 is more efficient estimator of μ than T_2 .

E11) Refer Sub-sections 5.6.1 and 5.6.2.

E12) Here, we take random sample from $\exp(\theta)$ whose probability density function is given by

$$f(x, \theta) = \theta e^{-\theta x}; \quad x > 0 \text{ \& } \theta > 0$$

Estimation

The joint density function of $X_1, X_2, \dots, X_n$ can be obtained and can be factored as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \theta) &= f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta) \\ &= \theta e^{-\theta x_1} \cdot \theta e^{-\theta x_2} \dots \theta e^{-\theta x_n} \\ &= \theta^n e^{-\theta \sum_{i=1}^n x_i} \\ &= \left[\theta^n e^{-\theta \sum_{i=1}^n x_i} \right] \cdot 1 \end{aligned}$$

$$L(\theta) = g[t(x), \theta] \cdot h(x_1, x_2, \dots, x_n)$$

where, $g[t(x), \theta] = \theta^n e^{-\theta \sum_{i=1}^n x_i}$ is a function of parameter θ and sample values $x_1, x_2, \dots, x_n$ only through $t(x) = \sum_{i=1}^n x_i$ and $h(x_1, x_2, \dots, x_n) = 1$, is independent of θ . Hence by factorization theorem of sufficiency, $\sum_{i=1}^n X_i$ is sufficient estimator of θ .

E13) Here, we take random sample from $N(\mu, \sigma^2)$ whose probability density function is

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

The joint density function of $X_1, X_2, \dots, X_n$ can be obtained as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \mu, \sigma^2) &= f(x_1, \mu, \sigma^2) \cdot f(x_2, \mu, \sigma^2) \dots f(x_n, \mu, \sigma^2) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_1-\mu)^2} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_2-\mu)^2} \dots \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_n-\mu)^2} \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n [(x_i - \bar{x}) + (\bar{x} - \mu)]^2} = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2} \dots (1) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2} \quad [\text{add and subtract } \bar{x}] \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n [(x_i - \bar{x})^2 + (\bar{x} - \mu)^2 - 2(x_i - \bar{x})(\bar{x} - \mu)]} \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (\bar{x} - \mu)^2 - 2(\bar{x} - \mu) \sum_{i=1}^n (x_i - \bar{x}) \right]} \end{aligned}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 - 0 \right]} \left[\sum_{i=1}^n (x_i - \bar{x}) = 0, \text{ by the property of mean} \right]$$

$$f(x_1, x_2, \dots, x_n, \mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 - \frac{n}{2\sigma^2} (\bar{x} - \mu)^2} \dots (2)$$

Case I: Sufficient statistic for μ when σ^2 is known

The joint density function given in equation (2) can be factored as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \mu) &= \left[e^{-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2} \cdot \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \right] \cdot e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2} \\ &= g(t(x), \mu) h(x_1, x_2, \dots, x_n) \end{aligned}$$

where, $g[t(x), \mu] = e^{-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2} \cdot \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n$ is a function of parameter μ

and sample values $x_1, x_2, \dots, x_n$ only through $t(x) = \bar{x}$, whereas

$h(x_1, x_2, \dots, x_n) = e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2}$ is independent of μ . Hence, $\bar{X}$ is sufficient estimator for μ when σ^2 is known.

Case II: Sufficient statistic for σ^2 when μ is known

The joint density function given in equation (1) can be factored as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \sigma^2) &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \cdot 1 \\ &= g(t(x), \mu) \cdot h(x_1, x_2, \dots, x_n) \end{aligned}$$

where, $g[t(x), \mu] = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$ is a function of parameter

σ^2 and sample values $x_1, x_2, \dots, x_n$ only through $t(x) = \sum_{i=1}^n (x_i - \mu)^2$,

whereas $h(x_1, x_2, \dots, x_n) = 1$ is independent of σ^2 . Hence by

factorization theorem of sufficiency, $\sum_{i=1}^n (X_i - \mu)^2$ is sufficient estimator for σ^2 when μ is known.

Case III: When both μ and σ^2 are unknown

The joint density function given in equation (2) can be factored as

$$f(x_1, x_2, \dots, x_n, \mu, \sigma^2) = \left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} [(n-1)s^2 - n(\bar{x} - \mu)^2]} \right] \cdot 1$$

Estimation

$$\text{where, } s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2$$

$$f(x_1, x_2, \dots, x_n, \mu, \sigma^2) = g(t_1(x), t_2(x), \mu, \sigma^2) \cdot h(x_1, x_2, \dots, x_n)$$

where, $g(t_1(x), t_2(x), \mu, \sigma^2)$ is a function of (μ, σ^2) and sample values $x_1, x_2, \dots, x_n$ only through $t_1(x) = \bar{x}$, $t_2(x) = s^2$, whereas $h(x_1, x_2, \dots, x_n)$ is independent of (μ, σ^2) . Hence by factorization theorem, $\bar{X}$ and S^2 are jointly sufficient for μ and σ^2 .

Note 5: Here, it is remembered that $\bar{X}$ is not sufficient statistic for μ if σ^2 is unknown and S^2 is not sufficient for σ^2 if μ is unknown.

E14) The sample is taken from $U[0, \theta]$ whose probability density function is

$$f(x, \theta) = \frac{1}{\theta} \quad ; \quad 0 \leq x \leq \theta \quad \theta > 0$$

The joint density function of $X_1, X_2, \dots, X_n$ can be obtained as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \theta) &= f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta) \\ &= \frac{1}{\theta} \cdot \frac{1}{\theta} \dots \frac{1}{\theta} = \frac{1}{\theta^n} \end{aligned}$$

Since range of variable depends upon the parameter θ , so we consider ordered statistics $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$.

Therefore, the joint density function can be factored as

$$\begin{aligned} f(x_1, x_2, \dots, x_n, \theta) &= \frac{1}{\theta^n}; \quad 0 \leq x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} \leq \theta \\ &= \left[\frac{1}{\theta^n} I(x_{(n)}, \theta) \right] \cdot 1 \end{aligned}$$

where,

$$I(x_{(n)}, \theta) = \begin{cases} 1; & \text{if } x_{(n)} \leq \theta \\ 0; & \text{otherwise} \end{cases}$$

Therefore,

$$f(x_1, x_2, \dots, x_n, \theta) = g[t(x), \theta] \cdot h(x_1, x_2, \dots, x_n)$$

where, $g[t(x), \theta]$ is a function of θ and sample values only through $t(x) = x_{(n)}$, whereas $h(x_1, x_2, \dots, x_n)$ is independent of θ .

Hence, by factorization theorem $X_{(n)}$ is sufficient statistic for θ .

UNIT 6 POINT ESTIMATION

Structure

- 6.1 Introduction
 - Objectives
- 6.2 Point Estimation
 - Methods of Point Estimation
- 6.3 Method of Maximum Likelihood
 - Properties of Maximum Likelihood Estimators
- 6.4 Method of Moments
 - Properties of Moment Estimators
 - Drawbacks of Moment Estimators
- 6.5 Method of Least Squares
 - Properties of Least Squares Estimators
- 6.6 Summary
- 6.7 Solutions / Answers

6.1 INTRODUCTION

In previous unit, we have discussed some important properties of an estimator such as unbiasedness, consistency, efficiency, sufficiency, etc. And according to Prof. Ronald A. Fisher, if an estimator possess these properties then it is said to be a good estimator. Now, our point is to search such estimators which possess as many of these properties as possible. In this unit, we shall discuss some frequently used methods of finding point estimate such as method of maximum likelihood, method of moments and method of least squares.

This unit is divided into seven sections. Section 6.1 is introductory in nature. The point estimation and frequently used methods of point estimation are explored in Section 6.2. The most important method of point estimation i.e. method of maximum likelihood and the properties of its estimators are described in Section 6.3. The method of moments with properties and drawbacks of moment estimators are described in Section 6.4. Section 6.5 is devoted to the method of least squares and its properties. Unit ends by providing summary of what we have discussed in this unit in Section 6.6 and solution of exercises in Section 6.7.

Objectives

After going through this unit, you should be able to:

- define and obtain the point estimation;
- define and obtain the likelihood function;
- explore the different methods of point estimation;
- explain the method of maximum likelihood;
- describe the properties of maximum likelihood estimators;
- discuss the method of moments;
- describe the properties of moment estimators;
- explain the method of least squares; and
- explore the properties of least squares estimators.

The technique of estimating the unknown parameter with a single value is known as point estimation.

6.2 POINT ESTIMATION

There are so many situations in our day to day life where we need to estimate the some unknown parameter(s) of the population on the basis on the sample observations. For example, a house wife may want to estimate the monthly expenditure, a sweet shopkeeper may want to estimate the sale of sweets on a day, a student may want to estimates the study hours for reading of a particular unit of this course, etc. This need is fulfilled by the technique of estimation. So the technique of finding an estimator to produce an estimate of the unknown parameter is called estimation.

We have already said that estimation is broadly divided into two categories namely:

- Point estimation and
- Interval estimation

If, we find a single value with the help of sample observations which is taken as the estimated value of unknown parameter then this value is known as point estimate and the technique of estimating the unknown parameter with a single value is known as “**point estimation**”.

If instead of finding a single value to estimate the unknown parameter if we find two values between which the parameter may be considered to lie with certain probability(confidence) is known as interval estimate of the parameter and this technique of estimating is known as “**interval estimation**”. For example, if we estimate the average weight of men living in a colony on the basis of sample mean, say, 62 kg then 62 kg is called point estimate of average weight of men in the colony and this procedure is called as point estimation. If we estimate the average weight of men by an interval, say, [40,110] with 90% confidence that true value of the weight lie in this interval then this interval is called interval estimate and this procedure is called as interval estimation.

Now, the question may arise in your mind that “how point and interval estimates are obtained?” So we will describe some of the important and frequently used methods of point estimation in the subsequent sections of this unit and methods of interval estimation in the next unit.

6.2.1 Methods of Point Estimation

Some of the important and frequently used methods of point estimation are:

1. Method of maximum likelihood
2. Method of moments
3. Method of least squares
4. Method of minimum chi-square
5. Method of minimum variance

The method of maximum likelihood, method of moments and method of least squares will be discussed in detail in subsequent sections one by one and other methods are beyond the of scope of this course.

Now, try the following exercises.

E1) Find which technique of estimation (point estimation or interval estimation) is used in each case given below:

- (i) An investigator estimates average income Rs. 1.5 lack per annum of the people living in a particular geographical area, on the basis of a sample of 50 people taken from that geographical area.
- (ii) A product manager of a company estimates the average life of electric bulbs in the range 800 hours and 1000 hours, with certain confidence, on the basis of a sample of 20 bulbs.
- (iii) A pathologist estimates the mean time required to complete a certain analysis in the range 30 minutes to 45 minutes, with certain confidence, on the basis of a random sample of 25.

E2) List any three methods of point estimation.

6.3 METHOD OF MAXIMUM LIKELIHOOD

For describing the method of maximum likelihood, first we have to define likelihood function.

Likelihood Function

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with joint probability density (mass) function $f(x_1, x_2, \dots, x_n, \theta)$ of sample values then likelihood function is denoted by $L(\theta)$ and is defined as follows:

$$L(\theta) = f(x_1, x_2, \dots, x_n, \theta)$$

For discrete case,

$$L(\theta) = P[X_1 = x_1]P[X_2 = x_2] \dots P[X_n = x_n]$$

For continuous case,

$$L(\theta) = f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta)$$

The main difference between the joint probability density (mass) function and the likelihood function is that in joint probability density (mass) function we consider the X 's as variables and the parameter θ as fixed and consider as a function of sample observations while in the likelihood function we consider the parameter θ as the variable and the X 's as fixed and consider as a function of parameter θ .

The process of finding the likelihood function is described by taking an example.

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from exponential distribution (θ) whose pdf is given by

$$f(x, \theta) = \theta e^{-\theta x}; \quad x > 0, \theta > 0$$

Then the likelihood function of parameter θ can be obtained as

$$L(\theta) = f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta)$$

$$= \theta e^{-\theta x_1} \cdot \theta e^{-\theta x_2} \dots \theta e^{-\theta x_n}$$

$$= \theta^{\overbrace{1+1+\dots+1}^{n\text{-times}}} e^{-\theta(x_1+x_2+\dots+x_n)}$$

$$L(\theta) = \theta^n e^{-\theta \sum_{i=1}^n x_i}$$

Estimation

The likelihood principle states that all the information in a sample to draw the inference about the value of unknown parameter θ is found in the corresponding likelihood function. Therefore, the likelihood function gives the relative likelihoods for different values of the parameters, given the sample data.

From theoretical point of view, one of the most important methods of point estimation is method of maximum likelihood because it generally gives very good estimators as judged from various criteria. It was initially given by Prof. C.F. Gauss but later on it was used as a general method of estimation by Prof. Ronald A. Fisher in 1912. The principle of maximum likelihood estimation is to find /estimate /choose the value of unknown parameter which would most likely generate the observed data. We know that the likelihood function gives the relative likelihoods for different values of the parameters for the observed data. Therefore, we search the value of unknown parameter for which the likelihood function is maximum corresponding to the observed data. The concept of maximum likelihood estimation is explained with a simple example given below:

Suppose, we toss a coin 5 times and we observe 3 heads and 2 tails. Instead of assuming that the probability of getting head is $p = 0.5$, we want to find / estimate the value of p that makes the observed data most likely. Since number of heads follows the binomial distribution, therefore, the probability (likelihood function) of getting 3 heads in 5 tosses is given by

$$P[X = 3] = {}^5C_3(p)^3(1-p)^2$$

Imagine that p was 0.1 then

$$P[X = 3] = {}^5C_3(0.1)^3(0.1)^2 = 0.0081$$

Similarly, for different values of p the probability of getting 3 heads in 5 tosses is given in Table 6.1 given below:

Table 6.1: Probability/Likelihood Function Corresponding to Different Values of p

S. No.	p	Probability/ Likelihood Function
1	0.1	0.0081
2	0.2	0.0512
3	0.3	0.1323
4	0.4	0.2304
5	0.5	0.3125
6	0.6	0.3456
7	0.7	0.3087
8	0.8	0.2048
9	0.9	0.0729

From Table 6.1, we can conclude that p is more likely to be 0.6 because at $p = 0.6$ the probability is maximum or the likelihood function is maximum.

Therefore, principle of maximum likelihood (ML) consists in finding an estimate for the unknown parameter θ within the admissible range of θ , i.e. within parameter space Θ , which makes the likelihood function as large as possible, that is, maximize the likelihood function. Such an estimate is known as maximum likelihood estimate for unknown parameter θ . Thus, if there exists

an estimate, say, $\hat{\theta}(x_1, x_2, \dots, x_n)$ of the sample values which maximizes likelihood function $L(\theta)$ is known as “**maximum likelihood estimate**”. That is,

$$L(\hat{\theta}) = \sup_{\theta \in \Theta} L(\theta)$$

For maximising the likelihood function, the theory of maxima or minima (discussed in Unit 6 of MST-002) is applied i.e. we differentiate $L(\theta)$ or L partially with respect to the parameter θ and put it equal to zero. The equation so obtain is known as likelihood equation, that is,

$$\frac{\partial}{\partial \theta} L = 0$$

Then we solve the likelihood equation for parameter θ which gives the ML estimate if its second derivative is negative at $\theta = \hat{\theta}$, that is,

$$\left. \frac{\partial^2}{\partial \theta^2} L \right|_{\theta=\hat{\theta}} < 0$$

Since likelihood function (L) is the product of n functions so differentiating L as such is very difficult. Also L is always non-negative and $\log L$ remains a finite value. So $\log L$ attains maximum when L is maximum. Hence, we can consider $\log L$ in place of L and we find ML estimate as

$$\frac{\partial}{\partial \theta} (\log L) = 0$$

provided,

$$\left. \frac{\partial^2}{\partial \theta^2} (\log L) \right|_{\theta=\hat{\theta}} < 0$$

When there are more than one parameter, say, $\theta_1, \theta_2, \dots, \theta_k$ then ML estimates of these parameters can be obtained as the solution of k simultaneous likelihood equations

$$\frac{\partial}{\partial \theta_i} (\log L) = 0; \quad \text{for all } i = 1, 2, \dots, k$$

provided, the matrix of derivatives

$$\left. \frac{\partial^2}{\partial \theta_i \partial \theta_j} (\log L) \right|_{\theta_i=\hat{\theta}_i \text{ \& \; } \theta_j=\hat{\theta}_j} < 0; \quad \text{for all } i \neq j = 1, 2, \dots, k$$

Let us explain the procedure of ML estimation with the help of some examples.

Example 1: If the number of weekly accidents occurring on a mile stretch of a particular road follows Poisson distribution with parameter λ then find the maximum likelihood estimate of parameter λ on the basis of the following data:

Number of Accidents	0	1	2	3	4	5	6
Frequency	10	12	12	9	5	3	1

Solution: Here, the number of weekly accidents occurring on a mile stretch of a particular road follows Poisson distribution with parameter λ so pmf of the Poisson distribution is given by

Point Estimation

$\hat{\theta}$ - is used to represent the estimate / estimator of parameter and read as cap i.e.

$\hat{\theta}$ - means it is the estimate of parameter θ and read as theta cap.

Estimation

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

First we find the theoretical ML estimate of the parameter λ as:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from this Poisson distribution, therefore, likelihood function of parameter λ can be obtained as

$$\begin{aligned} L(\lambda) &= L = P[X = x_1] \cdot P[X = x_2] \dots P[X = x_n] \\ &= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} \\ &= \frac{e^{\underbrace{-\lambda - \lambda - \dots - \lambda}_{n\text{-times}}} \lambda^{x_1 + x_2 + \dots + x_n}}{x_1! x_2! \dots x_n!} \\ &= \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \end{aligned} \quad \dots (1)$$

Taking log on both sides

$$\log L = -n\lambda + \sum_{i=1}^n x_i \log \lambda - \log \prod_{i=1}^n x_i! \quad \dots (2)$$

Differentiating equation (2) partially with respect to λ , we get

$$\frac{\partial}{\partial \lambda} (\log L) = -n + \sum_{i=1}^n x_i \left(\frac{1}{\lambda} \right) - 0 \quad \dots (3)$$

For maxima or minima, that is, for finding ML estimate, we put

$$\begin{aligned} \frac{\partial}{\partial \lambda} (\log L) &= 0 \\ \Rightarrow -n + \frac{1}{\lambda} \sum_{i=1}^n x_i &= 0 \\ \Rightarrow \hat{\lambda} &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{x} \end{aligned}$$

Now, we obtain the second derivative, that is, differentiating equation (3) partially with respect to λ , we get

$$\frac{\partial^2}{\partial \lambda^2} (\log L) = 0 + \sum_{i=1}^n x_i \left(-\frac{1}{\lambda^2} \right)$$

Put $\lambda = \bar{x}$, we have

$$\left. \frac{\partial^2}{\partial \lambda^2} (\log L) \right|_{\lambda=\bar{x}} = -\sum_{i=1}^n x_i \left(\frac{1}{\bar{x}^2} \right) < 0 \quad \text{for all values of } x_i \text{'s}$$

Therefore, maximum likelihood estimate of parameter λ of Poisson distribution is sample mean.

Since we observed that the sample mean is the maximum likelihood estimate of parameter λ so we calculate the sample mean of the given data as

S. No.	Number of Accidents(X)	Frequency(f)	fX
1	0	10	0
2	1	12	12
3	2	12	24
4	3	9	27
5	4	5	20
6	5	3	15
7	6	1	6
		N = 52	$\sum fX = 104$

The formula for calculating mean is

$$\bar{X} = \frac{1}{N} \sum fX \quad \text{where, N is the total number of accidents}$$

$$= \frac{1}{52} \times 104 = 2$$

Hence, maximum likelihood estimate of λ is 2.

Example 2: For random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimators for μ and σ^2 .

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population $N(\mu, \sigma^2)$, whose probability density function is given by

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

Therefore, the likelihood function for parameters μ and σ^2 can be obtained as

$$\begin{aligned} L(\mu, \sigma^2) &= L = f(x_1, \mu, \sigma^2) \cdot f(x_2, \mu, \sigma^2) \dots f(x_n, \mu, \sigma^2) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_1-\mu)^2} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_2-\mu)^2} \dots \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_n-\mu)^2} \\ &= \left(\frac{1}{2\pi\sigma^2} \right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \quad \dots (4) \end{aligned}$$

Taking log on both sides of equation (4), we get

$$\log L = \frac{n}{2} \log 1 - \frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \quad \dots (5)$$

Differentiating equation (5) partially with respect to μ and σ^2 respectively, we get

$$\frac{\partial}{\partial \mu}(\log L) = -0 - 0 - 0 - \frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1) \quad \dots (6)$$

$$\frac{\partial}{\partial \sigma^2}(\log L) = -0 - 0 - \frac{n}{2} \frac{1}{\sigma^2} - \frac{1}{2(\sigma^2)^2} (-1) \sum_{i=1}^n (x_i - \mu)^2 \quad \dots (7)$$

For finding ML estimate of μ , we put

Estimation

$$\begin{aligned}\frac{\partial}{\partial \mu}(\log L) &= 0 \\ \Rightarrow -\frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1) &= 0 \\ \Rightarrow \sum_{i=1}^n (x_i - \mu) &= 0 \\ \Rightarrow \sum_{i=1}^n x_i - n\mu = 0 \Rightarrow \hat{\mu} &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}\end{aligned}$$

Thus, the ML estimate for μ is the observed sample mean $\bar{x}$.

For ML estimate of σ^2 , we put

$$\begin{aligned}\frac{\partial}{\partial \sigma^2} \log L &= 0 \\ \Rightarrow -\frac{n}{2\sigma^2} - (-1) \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 &= 0 \\ \Rightarrow \frac{-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2}{2\sigma^4} &= 0 \\ \Rightarrow -n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 &= 0 \\ \Rightarrow \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 \\ \Rightarrow \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = s'^2\end{aligned}$$

Thus, the ML estimates for μ and σ^2 are $\bar{x}$ and s'^2 respectively.

Hence, ML estimators for μ and σ^2 are $\bar{X}$ and S'^2 respectively.

Note 1: Since throughout the course we are using capital letters for estimators therefore in the last line of above example we use capital letters for ML estimators for μ and σ^2 .

Note 2: Here, the maxima and minima method is used to obtain the ML estimates when the range of random variable is independent of parameter θ . Whereas when the range of random variable is involved or depends on parameter θ then this method fails to find the ML estimates. In such cases, we use order statistics to maximize the likelihood function. Following example will explain this concept.

Example 3: Obtain the ML estimators for α and β for the uniform or rectangular population whose pdf is given by

$$f(x, \alpha, \beta) = \begin{cases} \frac{1}{\beta - \alpha}; & \alpha \leq x \leq \beta \\ 0; & \text{elsewhere} \end{cases}$$

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from uniform population $U(\alpha, \beta)$. Therefore, likelihood function can be obtained as

$$\begin{aligned}
L &= f(x_1, \alpha, \beta) \cdot f(x_2, \alpha, \beta) \dots f(x_n, \alpha, \beta) \\
&= \underbrace{\left(\frac{1}{\beta - \alpha}\right) \left(\frac{1}{\beta - \alpha}\right) \dots \left(\frac{1}{\beta - \alpha}\right)}_{n\text{-times}} \\
&= \left(\frac{1}{\beta - \alpha}\right)^n \\
\Rightarrow \log L &= -n \log(\beta - \alpha) \quad \dots (8)
\end{aligned}$$

Differentiating equation (8) partially with respect to α and β respectively, we get likelihood equations for α and β as

$$\frac{\partial}{\partial \alpha}(\log L) = 0 \Rightarrow \frac{n}{\beta - \alpha} = 0$$

and

$$\frac{\partial}{\partial \beta}(\log L) = 0 \Rightarrow \frac{-n}{\beta - \alpha} = 0$$

Both the equations give an inadmissible solution for α & β as $\beta = \alpha = \infty$. So the method of differentiation fails. Thus, we have to use another method to obtain the desired result.

In such situations, we use basic principal of maximum likelihood, that is, we choose the value of the parameters α & β which maximize the likelihood function. If $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ is an ascending ordered arrangement of the observed sample, then $\alpha \leq x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} \leq \beta$. Also, it can be seen $\beta \geq x_{(n)}$ and $\alpha \leq x_{(1)}$. $\beta \geq x_{(n)}$ means, β takes values greater than or equal to $x_{(n)}$ and least value of β is $x_{(n)}$. Similarly, $\alpha \leq x_{(1)}$ means, α takes values less than or equal to $x_{(1)}$ and maximum value of α is $x_{(1)}$. Now, likelihood function will be maximum when α is maximum and β is minimum. Thus, the minimum possible value of β consistent with the sample is $x_{(n)}$ and the maximum possible value of α consistent with the sample is $x_{(1)}$. Hence, L is maximum if $\beta = x_{(n)}$ and $\alpha = x_{(1)}$.

Thus, ML estimates for α and β are given by
 $\hat{\alpha} = x_{(1)} = \text{Smallest sample observation}$

and

$$\hat{\beta} = x_{(n)} = \text{Largest sample observation}$$

Hence, ML estimators for α and β are $X_{(1)}$ and $X_{(n)}$ respectively.

6.3.1 Properties of Maximum Likelihood Estimators

The following are the properties of maximum likelihood estimators:

1. A ML estimator is not necessarily unique.
2. A ML estimator is not necessarily unbiased.
3. A ML estimator may not be consistent in rare case.
4. If a sufficient statistic exists, it is a function of the ML estimators.

The order statistics of a random sample

$X_1, X_2, \dots, X_n$ are the sample values placed in ascending order of magnitude. These are denoted by

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

5. If $T = t(X_1, X_2, \dots, X_n)$ is a ML estimator of θ and $\gamma(\theta)$ is a one to one function of θ , then $\gamma(T)$ is a ML estimator of $\gamma(\theta)$. This is known as invariance property of ML estimator.
6. When ML estimator exists, then it is most efficient in the group of such estimators.

It is now time for you to try the following exercises to make sure that you get the concept of ML estimators.

E3) Prove that for the binomial population with density function

$$P[X = x] = {}^nC_x p^x q^{n-x}; \quad x = 1, 2, \dots, n, q = 1 - p$$

the maximum likelihood estimator for p is $\bar{X} / n$.

E4) Obtain the ML estimate of θ for the following distribution

$$f(x, \theta) = \frac{1}{\theta}; \quad 0 \leq x \leq \theta, \theta > 0$$

If the sample values are 1.5, 1.0, 0.7, 2.2, 1.3 and 1.2.

E5) List any five properties of maximum likelihood estimators.

6.4 METHOD OF MOMENTS

The method of moments is the oldest but simple method for determining the point estimate of the unknown parameters. It was discovered by Karl Pearson in 1894. The application of this method was invariable continued until Prof. Ronald A. Fisher introduced maximum likelihood estimation method. The principle of this method consists of equating the sample moments to the corresponding moments of the population, which are the function of unknown population parameter(s). We equate as many sample moments as there are unknown parameters and solve these simultaneous equations for estimating unknown parameter(s). This method of obtaining the estimate(s) of unknown parameter(s) is called **“Method of Moments”**.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$ with k unknown parameters, say, $\theta_1, \theta_2, \dots, \theta_k$. Then the r^{th} sample moment about origin is

$$M'_r = \frac{1}{n} \sum_{i=1}^n X_i^r$$

and about mean is

$$M_r = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^r$$

While the r^{th} population moment about origin is

$$\mu'_r = E(X)^r$$

and about mean is

$$\mu_r = E(X - \mu)^r$$

Generally, the first moment about origin (zero) and rest central moments (about mean) are equated to the corresponding sample moments. Thus, the equations are

$$\mu'_1 = M'_1$$

$$\mu_r = M_r \quad ; r = 2, 3, \dots, k$$

By solving these k equations for unknown parameters, we get the moment estimators.

Let us explain the concept of method of moments with the help of some examples.

Example 4: Find the estimator for λ by the method of moments for the exponential distribution whose probability density function is given by

$$f(x, \lambda) = \frac{1}{\lambda} e^{-x/\lambda}; \quad x > 0, \lambda > 0$$

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from exponential distribution whose probability density function is given by

$$f(x, \lambda) = \frac{1}{\lambda} e^{-x/\lambda}; \quad x > 0, \lambda > 0$$

We know that the first moment about origin, that is, population mean of exponential distribution with parameter λ is

$$\mu'_1 = \lambda$$

and the corresponding sample moment about origin is

$$M'_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Therefore, by the method of moments, we equate population moment with corresponding sample moment. Thus,

$$\mu'_1 = M'_1$$

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Hence, moment estimator for λ is $\bar{X}$.

Example 5: If $X_1, X_2, \dots, X_m$ is a random sample taken from binomial distribution (n, p) where, n and p are unknown, obtain moment estimators for both n and p.

Solution: We know that the mean and variance of binomial distribution (n, p) are given by

$$\mu'_1 = np \text{ and } \mu_2 = npq$$

Also the corresponding first sample moment about origin & second sample moment about mean (central moment) are

$$M'_1 = \frac{1}{m} \sum_{i=1}^m X_i = \bar{X} \text{ and } M_2 = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2$$

Therefore, by the method of moments, we equate population moments with corresponding sample moments. Thus,

Estimation

$$\mu'_1 = M'_1$$

$$np = \frac{1}{m} \sum_{i=1}^m X_i = \bar{X} \quad \dots (9)$$

and

$$\mu_2 = M_2$$

$$npq = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2 = S'^2 \quad \dots (10)$$

We solve above equations (9) & (10) for n and p by dividing equation (10) by equation (9), we get

$$\hat{q} = \frac{S'^2}{\bar{X}}$$

Since $p = 1 - q$, therefore the estimator of p is

$$\hat{p} = 1 - \hat{q} = 1 - \frac{S'^2}{\bar{X}} = \frac{\bar{X} - S'^2}{\bar{X}}$$

Put the value of p in equation (9), we get

$$n \left(\frac{\bar{X} - S'^2}{\bar{X}} \right) = \bar{X}$$

$$\hat{n} = \frac{\bar{X}^2}{\bar{X} - S'^2}$$

Hence, moment estimators for p and n are $\frac{\bar{X} - S'^2}{\bar{X}}$ and $\frac{\bar{X}^2}{\bar{X} - S'^2}$ respectively where,

$$\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i \text{ and } S'^2 = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2.$$

Example 6: Show that moment estimator and maximum likelihood estimator are same of the parameter θ of the geometric distribution $G(\theta)$ whose pmf is

$$P[X = x] = \theta(1-\theta)^x; \quad \theta > 0, x = 0, 1, 2, \dots$$

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from $G(\theta)$ whose probability mass function is given by

$$P[X = x] = \theta(1-\theta)^x; \quad x = 0, 1, 2, \dots$$

Here, we first find the maximum likelihood estimator of θ . The likelihood function for parameter θ can be obtained as

$$\begin{aligned} L(\theta) &= P[X = x_1] \cdot P[X = x_2] \dots P[X = x_n] \\ &= \theta(1-\theta)^{x_1} \cdot \theta(1-\theta)^{x_2} \dots \theta(1-\theta)^{x_n} \\ &= \theta^n (1-\theta)^{\sum_{i=1}^n x_i} \end{aligned}$$

Taking log on both sides, we get

$$\log L = n \log \theta + \sum_{i=1}^n x_i \log(1-\theta) \quad \dots (11)$$

Differentiating equation (11) partially with respect to θ and equating to zero, we get

Point Estimation

$$\begin{aligned}\frac{\partial}{\partial \theta}(\log L) &= \frac{n}{\theta} + \sum_{i=1}^n x_i \cdot \frac{1}{1-\theta}(-1) = 0 \\ \Rightarrow \frac{n}{\theta} - \frac{\sum_{i=1}^n x_i}{1-\theta} &= 0 \Rightarrow \frac{n}{\theta} = \frac{\sum_{i=1}^n x_i}{1-\theta} \\ \Rightarrow n(1-\theta) &= \theta \sum_{i=1}^n x_i \Rightarrow n - n\theta - \theta \sum_{i=1}^n x_i = 0 \\ \Rightarrow \theta \left(\sum_{i=1}^n x_i + n \right) &= n \\ \Rightarrow \hat{\theta} &= \frac{n}{\sum_{i=1}^n x_i + n} = \frac{1}{\bar{x} + 1}\end{aligned}$$

Also, it can be seen that the second derivative, i.e.

$$\left. \frac{\partial^2}{\partial \theta^2}(\log L) \right|_{\theta = \frac{1}{\bar{x}+1}} < 0$$

Therefore, the ML estimator of θ is $\frac{1}{\bar{X} + 1}$.

Now, we find the moment estimator of θ .

We know that the first moment about origin, that is, mean of geometric distribution is

$$\mu'_1 = \frac{1-\theta}{\theta}$$

and the corresponding sample moment is

$$M'_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Therefore, by the method of moments, we have

$$\begin{aligned}\mu'_1 &= M'_1 \\ \Rightarrow \frac{1-\theta}{\theta} &= \bar{X} \\ \Rightarrow \theta \bar{X} &= 1-\theta \Rightarrow \theta \bar{X} + \theta = 1 \\ \Rightarrow \theta(\bar{X} + 1) &= 1 \Rightarrow \hat{\theta} = \frac{1}{(\bar{X} + 1)}\end{aligned}$$

Thus, moment estimator of θ is $\frac{1}{\bar{X} + 1}$.

Hence, the maximum likelihood estimator and moment estimator both are same in case of geometric distribution.

6.4.1 Properties of Moment Estimators

The following are the properties of moment estimators:

1. The moment estimators can be obtained easily.
2. The moment estimators are not necessarily unbiased.
3. The moment estimators are consistent because by the law of large numbers a sample moment (raw or central) is a consistent estimator for the corresponding population moment.
4. The moment estimators are generally less efficient than maximum likelihood estimators.
5. The moment estimators are asymptotically normally distributed.
6. The moment estimators may not be function of sufficient statistics.
7. The moment estimators are not unique.

6.4.2 Drawbacks of Moment Estimators

The following are the drawbacks of moment estimators:

1. This method is based on equating population moments with sample moments. But in some situations, like as cauchy distribution, the population moment does not exist therefore in such situations this method cannot be used.
2. This method does not, in general, give estimators with all the desirable properties of a good estimator.
3. The property of efficiency is not possessed by these estimators.
4. The moment estimators are not unbiased in general.
5. Generally, the moment estimators and the maximum likelihood estimators are identical. But if they do differ, then ML estimates are usually preferred.

Now, try to solve the following exercises to ensure that you have understood method of moments properly.

-
- E6)** Obtain the estimator of parameter λ when sample is taken from a Poisson population by the method of moments.
- E7)** Obtain the moment estimators of the parameters μ and σ^2 when the sample is drawn from normal population.
- E8)** Describe the properties and drawbacks of moment estimators.
-

6.5 METHOD OF LEAST SQUARES

The idea of least squares estimation emerges from the concept of method of maximum likelihood. Consider the maximum likelihood of the parameter μ when σ^2 is known on the basis of a random sample $Y_1, Y_2, \dots, Y_n$ of size n taken from a normal population (μ, σ^2) . The density function of normal population is given by

$$f(y, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}; -\infty < y < \infty, -\infty < \mu < \infty, \sigma > 0$$

Then likelihood function for μ and σ^2 is

$$L(\mu, \sigma^2) = L = f(y_1, \mu, \sigma^2) \cdot f(y_2, \mu, \sigma^2) \dots f(y_n, \mu, \sigma^2)$$

$$= \left(\frac{1}{2\pi\sigma^2} \right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2}$$

Taking log on both sides, we have

$$\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2$$

By principle of maximum likelihood estimation, we have to maximize $\log L$ with respect to μ , and $\log L$ is maximum when $\sum_{i=1}^n (y_i - \mu)^2$ is minimum, i.e. sum of squares $\sum_{i=1}^n (y_i - \mu)^2$ must be least.

The method of least squares is mostly used to estimate the parameter of linear function. Now, suppose that the population mean μ is itself a linear function of parameters $\theta_1, \theta_2, \dots, \theta_k$, that is,

$$\mu = x_1\theta_1 + x_2\theta_2 + \dots + x_k\theta_k$$

$$= \sum_{j=1}^k x_j\theta_j$$

where, x_i 's are not random variables but known constant coefficients of unknown parameter θ_i 's for forming a linear function of θ_i 's.

We have to minimize

$$E = \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_j\theta_j \right)^2 \text{ with respect to } \theta_i.$$

Hence, method of least squares gets its name from the minimization of a sum of squares. The principle of least squares states that we choose the values of unknown population parameters $\theta_1, \theta_2, \dots, \theta_k$, say, $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k$ on the basis of observed sample observations $y_1, y_2, \dots, y_n$ which minimize the sum of squares of deviations $\sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_j\theta_j \right)^2$.

Note: The method of least squares has already been discussed in Unit 5 of MSL-002 and further application of this method in estimating the parameters of regression models is discussed in specialisation courses.

6.5.1 Properties of Least Squares Estimators

Least squares estimators are not so popular. They possess some properties which are as follows:

1. Least squares estimators are unbiased in case of linear models.
2. Least squares estimators are minimum variance unbiased estimators (MVUE) in case of linear models.

Now, try the one exercise.

E9) Describe the two properties of least squares estimators.

We now end this unit by giving a summary of what we have covered in it.

6.6 SUMMARY

After studying this unit, you must have learnt about:

1. The point estimation.
2. Different methods of finding point estimators.
3. The method of maximum likelihood and its properties.
4. The method of moments, its properties and drawbacks.
5. The method of least squares and its properties.

6.7 SOLUTIONS /ANSWERS

E1)

- (i) Here, investigator estimates the average income Rs. 1.5 lack i.e. he / she estimate average income with a single value, therefore, the investigator used point estimation technique.
- (ii) The product manager estimates the average life of electric bulbs with the help of two values 800 hours and 1000 hours, therefore, the product manager used interval estimation technique.
- (iii) A pathologist estimates the mean time required completing a certain analysis with the help of two values 30 minutes and 45 minutes, therefore, he/she used interval estimation technique.

E2) Refer Sub-section 6.2.1.

E3) Let $X_1, X_2, \dots, X_m$ be a random sample of size m taken from $B(n, p)$ whose pmf is given by

$$P[X = x] = {}^n C_x p^x q^{n-x}; \quad x = 0, 1, \dots, n \text{ \& } q = 1 - p$$

The likelihood function for p can be obtained as

$$\begin{aligned} L(p) &= L = P[X = x_1] \cdot P[X = x_2] \dots P[X = x_m] \\ &= {}^n C_{x_1} p^{x_1} q^{n-x_1} \cdot {}^n C_{x_2} p^{x_2} q^{n-x_2} \dots {}^n C_{x_m} p^{x_m} q^{n-x_m} \\ &= \prod_{i=1}^m ({}^n C_{x_i}) p^{\sum_{i=1}^m x_i} q^{\sum_{i=1}^m (n-x_i)} \end{aligned}$$

Taking log on both sides, we have

$$\log L = \log \prod_{i=1}^m ({}^n C_{x_i}) + \sum_{i=1}^m x_i \log p + \sum_{i=1}^m (n - x_i) \log(1 - p) \quad [\because q = 1 - p]$$

Differentiate partially with respect to p and equating to zero, we have

$$\begin{aligned} \frac{\partial}{\partial p} (\log L) &= 0 + \sum_{i=1}^m x_i \left(\frac{1}{p} \right) + \sum_{i=1}^m (n - x_i) \left(\frac{1}{1-p} \right) (-1) = 0 \\ \Rightarrow \frac{\sum_{i=1}^m x_i}{p} &= \frac{\sum_{i=1}^m (n - x_i)}{1-p} \end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{\sum_{i=1}^m x_i}{p} = \frac{nm - \sum_{i=1}^m x_i}{1-p} \\
&\Rightarrow \sum_{i=1}^m x_i - p \sum_{i=1}^m x_i = nmp - p \sum_{i=1}^m x_i \\
&\Rightarrow \hat{p} = \frac{\sum_{i=1}^m x_i}{nm} = \frac{\bar{X}}{n} \quad \left[\because \bar{X} = \frac{1}{m} \sum_{i=1}^m x_i \right]
\end{aligned}$$

Also, it can be seen that the second derivative, i.e.

$$\left. \frac{\partial^2}{\partial p^2} (\log L) \right|_{p=\frac{\bar{X}}{n}} < 0$$

Hence, ML estimator for parameter p is $\bar{X}/n$.

- E4)** Let $X_1, X_2, \dots, X_n$ be a random sample taken from given $U(0, \theta)$ distribution. The likelihood function for θ is given by

$$\begin{aligned}
L(\theta) &= L = f(x_1, \theta) \cdot f(x_2, \theta) \dots f(x_n, \theta) \\
&= \left(\frac{1}{\theta}\right) \left(\frac{1}{\theta}\right) \dots \left(\frac{1}{\theta}\right) = \left(\frac{1}{\theta}\right)^n
\end{aligned}$$

Taking log on both sides, we have

$$\log L = -n \log \theta \quad \dots (12)$$

Differentiate equation (12) partially with respect to θ , we get

$$\frac{\partial}{\partial \theta} (\log L) = \frac{-n}{\theta} = 0, \text{ has no solution for } \theta.$$

So ML estimate cannot be found by differentiation. Therefore, by the principle of ML estimation, we choose the value of θ which maximize likelihood function. Hence, we choose θ as small as possible.

If $x_{(1)}, x_{(2)}, \dots, x_{(n)}$, is an ordered sample from this population, then $0 \leq x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} \leq \theta$. Also, it can be seen that $\theta \geq x_{(n)}$ which means that θ takes values greater than or equal to $x_{(n)}$ and minimum value of θ is $x_{(n)}$. Now, likelihood function will be maximum when θ is minimum. The minimum value of θ is $x_{(n)}$. Therefore, ML estimate of θ is the maximum observation of the sample, that is, $\hat{\theta} = x_{(n)}$.

Here, the given random sample is 1.5, 1.0, 0.7, 2.2, 1.3 and 1.2. Therefore, the ordered sample is $0.7 < 1.0 < 1.1 < 1.3 < 1.5 < 2.2$. Here, the maximum observation of this sample is 2.2, therefore maximum likelihood estimate of θ is 2.2.

- E5)** Refer Section 6.3.1.
- E6)** Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from Poisson population whose probability mass function is given by

Estimation

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

We know that for Poisson distribution

$$\mu'_1 = \lambda$$

and the corresponding sample moment is

$$M'_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Therefore, by the method of moments, we equate population moment with corresponding sample moment. Thus,

$$\mu'_1 = M'_1$$

$$\lambda = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Hence, moment estimator for λ is $\bar{X}$.

- E7)** Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population $N(\mu, \sigma^2)$, whose probability density function is given by

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; \quad -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

We know that for $N(\mu, \sigma^2)$

$$\mu'_1 = \mu \quad \text{and} \quad \mu'_2 = \sigma^2$$

and the corresponding sample moments are

$$M'_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \quad \text{and} \quad M'_2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Therefore by the method of moments, we equate population moments with corresponding sample moments. Thus,

$$\mu'_1 = M'_1$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

and

$$\mu'_2 = M'_2$$

$$\Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = S'^2$$

Hence, moment estimators for μ and σ^2 are $\bar{X}$ and S'^2 respectively.

- E8)** Refer Sub-sections 6.4.1 and 6.4.2.

- E9)** Refer as Section 6.5.1.

UNIT 7 INTERVAL ESTIMATION FOR ONE POPULATION

Structure

- 7.1 Introduction
 - Objectives
- 7.2 Interval Estimation
 - Confidence Interval and Confidence Coefficient
 - One-Sided Confidence Intervals
- 7.3 Method of Obtaining Confidence Interval
- 7.4 Confidence Interval for Population Mean
 - Confidence Interval for Population Mean when Population Variance is Known
 - Confidence Interval for Population Mean when Population Variance is Unknown
- 7.5 Confidence Interval for Population Proportion
- 7.6 Confidence Interval for Population Variance
 - Confidence Interval for Population Variance when Population Mean is known
 - Confidence Interval for Population Variance when Population Mean is Unknown
- 7.7 Confidence Interval for Non-Normal Populations
- 7.8 Shortest Confidence Interval
- 7.9 Determination of Sample Size
- 7.10 Summary
- 7.11 Solutions / Answers

7.1 INTRODUCTION

In the previous unit, we have discussed the point estimation, under which we learn how one can obtain point estimate(s) of the unknown parameter(s) of the population using sample observations. Everything is fine with point estimation but it has one major drawback that it does not specify how confident we can be that the estimated close to the true value of the parameter.

Hence, point estimate may have some possible error of the estimation and it does not give us an idea of how these estimates deviate from the true value of the parameter being estimated. This limitation of point estimation is over come by the technique of interval estimation. Therefore, instead of making the inference of estimating the true value of the parameter through a point estimate one should make the inference of estimating the true value of parameter by a pair of estimate values which are constituted an interval in which true value of parameter expected to lie with certain confidence. The technique of finding such interval is known as “**Interval Estimation**”.

For example, suppose that we want to estimate the average income of persons living in a colony. If 50 persons are selected at random from that colony and the annual average income is found to be Rs. 84240 then the statement that the average annual income of the persons in the colony is between Rs. 80000 and Rs. 90000 definitely more likely to be correct than the statement that the annual average income is Rs. 84240.

This unit is divided into eleven sections. Section 7.1 is introductory in nature. The confidence interval and confidence coefficient are defined in Section 7.2. The general method of finding the confidence interval is explored in Section

Estimation

7.3. Confidence interval for population mean in different cases as population variance is known and unknown are described in Section 7.4 whereas in Section 7.5, the confidence interval for population proportion is explained. The confidence interval for population variance in different cases when population mean is known and unknown are described in Section 7.6. Section 7.7 is devoted to explain the confidence interval for non-normal populations. The concept of shortest confidence interval and determination of sample size are explored in Sections 7.8 and 7.9 respectively. Unit ends by providing summary of what we have discussed in this unit in Section 7.10 and solution of exercises in Section 7.11.

Objectives

After studying this unit, you should be able to:

- need of interval estimation;
- define the interval estimation;
- describe the method of obtaining the confidence interval;
- obtain the confidence interval for population mean of a normal population when population variance is known and unknown;
- obtain the confidence interval for population proportion;
- obtain the confidence interval for population variance of a normal population when population mean is known and unknown;
- obtain the confidence intervals for population parameters of a non-normal populations;
- explain the concept of the shortest interval; and
- determination of sample size.

7.2 INTERVAL ESTIMATION

In previous section, we introduced you with the interval estimation and after that we can conclude that if we find two values with the help of sample observations and constitute an interval such that it contain the true value of parameter with certain probability then it is known as interval estimate of the parameter. This technique of estimation is known as “Interval Estimation”.

In this section, we will formally define:

- Confidence Interval and Confidence Coefficient
- One-sided Confidence Intervals

in the following two sub-sections.

7.2.1 Confidence Interval and Confidence Coefficient

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population whose probability density (mass) function is $f(x, \theta)$. Let $T_1 = t_1(X_1, X_2, \dots, X_n)$ and $T_2 = t_2(X_1, X_2, \dots, X_n)$ (where $T_1 \leq T_2$) be two statistics such that the probability that the random interval $[T_1, T_2]$ includes the true value of population parameter θ is $(1 - \alpha)$, that is,

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha$$

as shown in the Fig.7.1, where, α does not depend on θ .

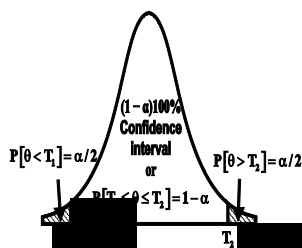


Fig. 7.1

Then the random interval $[T_1, T_2]$ is known as $(1 - \alpha)$ 100% confidence interval for unknown population parameter θ and $(1 - \alpha)$ is known as confidence coefficient or confidence level. The above probability statement may be explained with the help of an example:

Suppose we say that the probability that the random interval contains the true value of parameter θ is 0.95 then by this statement we simply mean that if 100 samples of same size, say, 'n' are drawn from the given population $f(x, \theta)$ and the random interval $[T_1, T_2]$ is computed for each sample, then 95 times out of 100 intervals, the random interval $[T_1, T_2]$ contains the true value of parameter θ . Hence, higher the probability $(1 - \alpha)$, more the confident we will have that the random interval $[T_1, T_2]$ will actually include the true value of θ . The statistics T_1 and T_2 are known as lower and upper confidence limits or confidence bounds or fiducial limits, respectively for θ . And the interval is known as two-sided confidence interval. The length of confidence interval is defined as

$$L = \text{Upper confidence limit} - \text{Lower confidence limit}$$

i.e. $L = T_2 - T_1$

The confidence interval may also be one-sided so in the next sub-section we define one-sided confidence intervals.

7.2.2 One-Sided Confidence Intervals

In some situations, we may be interested in finding an upper bound or lower bound but not both for population parameter with a given confidence. For example, one may be interested to obtain a bound such that he/she is 95% confident that the average life of the electric bulbs of a company is no less than one year. In such cases, we construct one-sided confidence intervals.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from the population having probability density(mass) function $f(x, \theta)$ and also let T_1 be a statistic such that

$$P[\theta \geq T_1] = 1 - \alpha$$

as shown in Fig. 7.2, then statistic T_1 is called a lower confidence bound for parameter θ with confidence coefficient $(1 - \alpha)$ and $[T_1, \infty)$ is called a lower one-sided $(1 - \alpha)$ 100% confidence interval for parameter θ .

Similarly, let T_2 be a statistic such that

$$P[\theta \leq T_2] = 1 - \alpha$$

as shown in Fig. 7.3, then statistic T_2 is called an upper confidence bound for parameter θ with confidence coefficient $(1 - \alpha)$ and $(-\infty, T_2]$ is called an upper one-sided $(1 - \alpha)$ 100% confidence interval for parameter θ .

Note 1: Generally, one-sided confidence intervals are rarely used so we focus on two-sided confidence interval in this course. **Generally, confidence interval means two-sided confidence interval unless or otherwise it is stated as one-sided.**

Now, you can try the following exercises.

E1) Find the length of the following confidence intervals:

- (i) $P[-1.65 \leq \mu \leq 3.0] = 0.95$ (ii) $P[-1.68 \leq \mu \leq 2.70] = 0.95$
 (iii) $P[-1.70 \leq \mu \leq 2.54] = 0.95$ (iv) $P[-1.96 \leq \mu \leq 1.96] = 0.95$

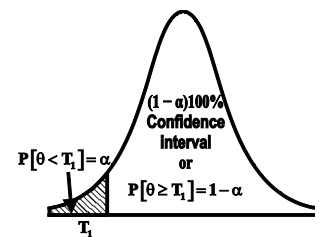


Fig. 7.2

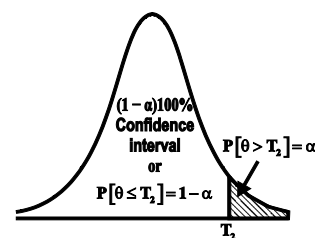


Fig. 7.3

E2) Find the lower and upper confidence limits and also confidence coefficient of the following confidence intervals:

- (i) $P[0 \leq \theta \leq 1.5] = 0.90$ (ii) $P[-1 \leq \theta \leq 2] = 0.95$
 (iii) $P[-2 \leq \theta \leq 2] = 0.98$ (iv) $P[-2.5 \leq \mu \leq 2.5] = 0.99$

7.3 METHOD OF OBTAINING CONFIDENCE INTERVAL

After knowing about the confidence interval, the question may arise in your mind that “how confidence intervals are obtained?” There are following two methods are generally used for obtaining confidence intervals:

1. Pivotal quantity method
2. Statistical method

The statistical method is beyond of scope of this course so we will keep our focus only on pivotal-quantity method which is also known as general method of interval estimation. Before going to describe this method, we first define the pivotal quantity as:

Pivotal Quantity

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population having probability density (mass) function $f(x, \theta)$. If quantity $Q = q(X_1, X_2, \dots, X_n, \theta)$ is a function of $X_1, X_2, \dots, X_n$ and parameter θ such that its distribution does not dependent on unknown parameter θ then the quantity Q is known as a pivotal quantity.

For example, if $X_1, X_2, \dots, X_n$ is a random sample taken from normal population with mean μ and variance 4, i.e. $N(\mu, 4)$ where, the parameter μ is unknown then we know that the sampling distribution of sample mean is also normal with mean μ and variance $4/n$, that is, $\bar{X} \sim N(\mu, 4/n)$ and the sampling

distribution of variate $Z = \frac{\bar{X} - \mu}{\sqrt{4/n}}$ is $N(0, 1)$. Since distribution of $\bar{X}$ depends

on the parameter μ to be estimated, therefore, it is not a pivotal quantity whereas the distribution of variate Z is independent of parameter μ so it is a pivotal quantity.

Pivotal Quantity Method

The pivotal quantity method for confidence interval has following steps:

Step 1: First of all, we search the statistic for unknown parameter which can be used to estimate the parameter, say, θ , preferably a sufficient statistic, whose distribution is completely known. After that we find the function based on that statistic whose distribution does not dependent on parameter θ which is to be estimated i.e. we find the pivotal quantity Q .

Step 2: Introduce two constants, say, ‘a’ and ‘b’, depending on α but not on unknown parameter θ , such that

$$P[a \leq Q \leq b] = 1 - \alpha$$

Step 3: Since pivotal quantity is a function of parameter, therefore, we convert above interval for parameter θ as

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha$$

where, T_1 and T_2 are functions of sample values and a & b .

Step 4: Determine constants 'a' and 'b' by minimizing the length of the interval

$$L = T_2 - T_1$$

With the help of the pivotal quantity method, we will find the confidence intervals for population mean, proportion, variance which will describe one by one in subsequent sections.

Now, you can try the following exercise.

E3) Describe the general method of constructing confidence interval for population parameter.

7.4 CONFIDENCE INTERVAL FOR POPULATION MEAN

There are so many problems in real life where it becomes necessary to obtain the confidence interval of population mean. For example, an investigator may interested to find the interval estimate of average income of the people living in a particular geographical area, a product manager may want to find the interval estimate of average life of electric bulbs manufactured by a company, a pathologist may want to obtain the interval estimate of the mean time required to complete a certain analysis, etc.

For describing confidence interval for population mean, let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population having mean μ and variance σ^2 . We can determine confidence interval for population mean μ under following two cases:

1. When population variance σ^2 is known
2. When population variance σ^2 is unknown.

These two cases are discussed one by one in subsequent Sub-sections 7.4.1 and 7.4.2 respectively.

7.4.1 Confidence Interval for Population Mean when Population Variance is Known

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population $N(\mu, \sigma^2)$ when σ^2 is known, that is, σ^2 has a specify value, say, σ_0^2 .

To find out the confidence interval for population mean, first of all we search the statistic for estimating μ whose distribution is completely known.

Generally, we use the value of statistic (sample mean) $\bar{X}$ to estimate the population mean μ and also it is a sufficient statistic for parameter θ . Therefore, we use $\bar{X}$ to make the pivotal quantity.

We know that when parent population is normal $N(\mu, \sigma^2)$ then sampling distribution of sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n , that is, if

Estimation

$$X_i \sim N(\mu, \sigma^2)$$

then

$$\bar{X} \sim N(\mu, \sigma^2 / n)$$

and the variate

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

follows the normal distribution with mean 0 and variance unity. Therefore, the probability density function of Z is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; -\infty < z < \infty$$

Since distribution of Z is independent of the parameter to be estimated i.e. μ , therefore, Z can be taken as pivotal quantity. So we introduce two constants, say, $z_{\alpha/2}$ and $z_{1-\alpha/2} = -z_{\alpha/2}$ (since distribution of Z is symmetrical about Z = 0 line see Fig. 7.4) such that

$$P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 1 - \alpha$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z as shown in Fig. 7.4.

By putting the value of Z in above equation / probability statement, we get

$$P\left[-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \leq z_{\alpha/2}\right] = 1 - \alpha$$

Now, for converting this interval for parameter μ , we multiply each term in above inequality by $\sigma / \sqrt{n}$, we get

$$P\left[-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq +z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

Now, subtracting $\bar{X}$ from each term in above inequality then we get

$$P\left[-\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

Multiplying each term by (-1) in above inequality, we get

$$P\left[\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \geq \mu \geq \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha \left[\because \text{by multiplying } (-1) \text{ the inequality is reversed} \right]$$

This can be rewritten as

$$P\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

Hence, $(1 - \alpha)$ 100% confidence interval of population mean is given by

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] \quad \dots (1)$$

and corresponding limits are given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (2)$$

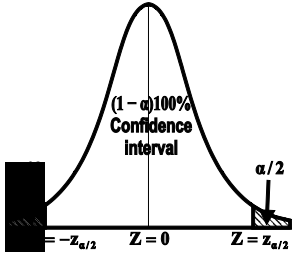


Fig. 7.4

Note 2: The value of $z_{\alpha/2}$ can be obtained by the method described in Unit 14 of MST-003. In interval estimation generally, we have to find the 90%, 95%, 98% and 99% confidence intervals therefore, the corresponding values of $z_{\alpha/2}$ are summaries in the Table 7.1 given below and we will use these values directly when needed:

Table 7.1: Communally Used Values of Standard Normal Variate Z

$1 - \alpha$	0.90	0.95	0.98	0.99
α	0.10	0.05	0.02	0.01
$\alpha/2$	0.05	0.025	0.01	0.005
$z_{\alpha/2}$	1.645	1.96	2.33	2.58

For example, if we want to find the 99% confidence interval (two-sided) for μ then

$$1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$$

For $\alpha = 0.01$, the value of $z_{\alpha/2} = z_{0.005}$ is 2.58 therefore, 99% confidence interval for μ is given by

$$\left[\bar{X} - 2.58 \frac{\sigma}{\sqrt{n}}, \bar{X} + 2.58 \frac{\sigma}{\sqrt{n}} \right]$$

Application of the above discussion can be seen in the following example.

Example 1: The mean life of the tyres manufactured by a company follows normal distribution with standard deviation 3200 kms. A sample of 250 tyres is taken and it is found that the average life of the tyres is 50000 kms with a standard deviation of 3500 kms. Establish the 99% confidence interval within which the mean life of tyres of the company is expected to lie.

Solution: Here, we are given that

$$n = 250, \sigma = 3200, \bar{X} = 50000, S = 3500$$

Since population standard deviation i.e. population variance σ^2 is known, therefore, we use $(1 - \alpha)$ 100% confidence limits for population mean when population variance is known which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z and for 99% confidence interval, we have

$$1 - \alpha = 0.99 \Rightarrow \alpha = 0.01. \text{ For } \alpha = 0.01 \text{ we have, } z_{\alpha/2} = z_{0.005} = 2.58.$$

Therefore, the 99% confidence limits are

$$\bar{X} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$

By putting the values of n , $\bar{X}$ and σ , the 99% confidence limits are

$$50000 \pm 2.58 \times \frac{3200}{\sqrt{250}}$$

$$50000 \pm 522.20 = 49477.80 \text{ and } 50522.20$$

Hence, 99% confidence interval within which the mean life of tyres of the company is expected to lie is

$$[49477.80, 50522.20]$$

7.4.2 Confidence Interval for Population Mean when Population Variance is Unknown

In the cases, described in previous sub-section we assume that variance σ^2 of the normal population is known but in general it is not known and in such a situation the only alternative left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the σ^2 where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

In this situation, we know that the variate

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$$

follows t-distribution (described in Unit 3 of this course) with $(n-1)$ df.

Therefore, probability density function of statistic t is given by

$$f(t) = \frac{1}{B\left(\frac{n-1}{2}, \frac{1}{2}\right) \sqrt{n-1}} \left(1 + \frac{t^2}{n-1}\right)^{-\frac{n}{2}} ; -\infty < t < \infty$$

Since distribution of statistic t is independent of the parameter to be estimated, therefore, t can be taken as pivotal quantity. So we introduce two constants

$t_{(n-1), \alpha/2}$ and $t_{(n-1), (1-\alpha/2)} = -t_{(n-1), \alpha/2}$ (since t-distribution is symmetrical about $t = 0$ line see Fig. 7.5) such that

$$P\left[-t_{(n-1), \alpha/2} \leq t \leq t_{(n-1), \alpha/2}\right] = 1 - \alpha \quad \dots (3)$$

where, $t_{(n-1), \alpha/2}$ is the value of the variate t with $n-1$ df having an area of $\alpha/2$ under the right tail of the probability curve of variate t as shown in Fig. 7.5.

By putting the value of variate t in equation (3), we get

$$P\left[-t_{(n-1), \alpha/2} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{(n-1), \alpha/2}\right] = 1 - \alpha$$

Now, for converting the above interval for parameter μ , we multiply each term in above inequality by $S/\sqrt{n}$ then we get

$$P\left[-t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \leq \bar{X} - \mu \leq t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}\right] = 1 - \alpha$$

After subtracting $\bar{X}$ from each term in above inequality, we get

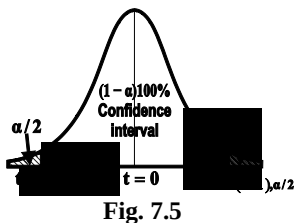
$$P\left[-\bar{X} - t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \leq -\mu \leq -\bar{X} + t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}\right] = 1 - \alpha$$

Now, by multiplying each term by (-1) in above inequality, we get

$$P\left[\bar{X} + t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \geq \mu \geq \bar{X} - t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}\right] = 1 - \alpha \quad \left[\because \text{by multiplying } (-1) \text{ the inequality is reversed} \right]$$

This can be rewritten as

$$P\left[\bar{X} - t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}\right] = 1 - \alpha$$



Hence, when variance σ^2 is unknown then $(1 - \alpha)$ 100% confidence interval for population mean of normal population is given by

$$\left[\bar{X} - t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \right] \quad \dots (4)$$

and corresponding limits are given by

$$\bar{X} \mp t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}} \quad \dots (5)$$

Note 2: For different confidence intervals and different degrees of freedom, the values of $t_{(n-1), \alpha/2}$ are different. Therefore, for given values of α and n we read the tabulated value of t-statistic from the table of t-distribution (t-table) given in Appendix (at the end of Block 1 of this course) by using the method described in Unit 4 of this course.

For example, if we want to find the 95% confidence interval for μ when $n = 8$ then we have

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$$

From t-table, for $\alpha = 0.05$ and $v = n - 1 = 7$, we have the value of

$$t_{(n-1), \alpha/2} = t_{(7), 0.025} = 2.365.$$

As we have seen in t-table of the Appendix that when sample size is greater than 30 ($n > 30$) then all values of variate t are not given in this table so for convenient as we have discussed in Unit 2 of this course that when n is sufficiently large (> 30) then we know that almost all the distributions are very closely approximated by normal distribution. Thus in this case t-distribution is also approximated normal distribution. So the variate

$$Z = \frac{\bar{X} - \mu}{S / \sqrt{n}} \sim N(0, 1)$$

also follows the normal distribution with mean 0 and variance unity. Therefore, when population variance is unknown and sample size is large then the $(1 - \alpha)$ 100% confidence interval for population mean may be obtained by using the same procedure as we have followed in case when σ^2 is known by taking S^2 in place of σ^2 which is given as

$$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right] \quad \dots (6)$$

and corresponding limits are given by

$$\bar{X} \mp z_{\alpha/2} \frac{S}{\sqrt{n}} \quad \dots (7)$$

Following example will explain the application of the above discussion:

Example 2: It is known that the average weight of students of a Study Centre of IGNOU follows normal distribution. To estimate the average weight, a sample of 10 students is taken from this Study Centre and measured their weights (in kg) which are given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Compute the 95% confidence interval for the average weight of students of Study Centre of IGNOU.

Estimation

Solution: Since population variance is unknown, therefore, $(1 - \alpha)$ 100% confidence limits for the average weight of students of Study Centre are given by

$$\bar{X} \pm t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}$$

$$\text{where, } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Calculation for $\bar{X}$ and S:

S. No.	Weight (X)	$(X - \bar{X})$	$(X - \bar{X})^2$
1	48	-15	225
2	50	-13	169
3	62	-1	1
4	75	12	144
5	80	17	289
6	60	-3	9
7	70	7	49
8	56	-7	49
9	52	-11	121
10	77	14	196
Sum	$\sum X = 630$		$\sum (X - \bar{X})^2 = 1252$

From the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{10} \times 630 = 63$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 = \frac{1}{9} \times 1252 = 139.11$$

$$\Rightarrow S = \sqrt{139.11} = 11.79$$

For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$. Also from t-table, we have, $t_{(n-1), \alpha/2} = t_{(9), 0.025} = 2.306$.

Thus, the 95% confidence limits are

$$\begin{aligned} \bar{X} \pm t_{(n-1), 0.025} \frac{S}{\sqrt{n}} &= 63 \pm 2.306 \times \frac{11.79}{\sqrt{10}} \\ &= 63 \pm 8.60 = 54.40 \text{ and } 71.60 \end{aligned}$$

Hence, required 95% confidence interval for the average weight of students of Study Centre of IGNOU is given by

$$[54.4, 71.60]$$

Example 3: The mean life of 100 electric blubs produced by a company is 2550 hours with a standard deviation 54 hours. Find 95% confidence limits for population mean life of electric blubs produced by the company.

Solution: Here, we are given that

$$n = 100, \bar{X} = 2550, S = 54$$

Since population variance is unknown and sample size is large (> 30), therefore, we can use $(1-\alpha)$ 100 % confidence limits for population mean which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}}$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z . For 95% confidence limits, we have

$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, the 95% confidence limits for mean life of electric bulbs are

$$\begin{aligned}\bar{X} \pm 1.96 \frac{S}{\sqrt{n}} &= 2550 \pm 1.96 \frac{54}{\sqrt{100}} \\ &= 2550 \pm 1.96 \times 5.4 \\ &= 2550 \pm 10.58 = 2539.42 \text{ and } 2560.58\end{aligned}$$

Now, it is time for you to try the following exercises to make sure that you have learnt about the confidence interval for population mean in different cases.

-
- E4)** Certain refined oil is packed in tins holding 15 kg each. The filling machine maintains this but have a standard deviation 0.30 kg. A sample of 200 tins is taken from the production line. If sample mean is 15.25 kg then find the 95% confidence interval for the average weight of oil tins.
- E5)** Sample mean of weights (in kg) of 150 students of IGNOU is found to be 65 kg with standard deviation 12 kg. Find the 95% confidence limits in which the average weight of all students of IGNOU expected to lie.
- E6)** It is known that the average height of cadets of a centre follows normal distribution. A sample of 6 cadets of the centre was taken and measured their heights (in inch) which are given below:

70 72 80 82 78 80

From this data, estimate the 95% confidence limits for the average height of cadets of the particular centre.

7.5 CONFIDENCE INTERVAL FOR POPULATION PROPORTION

In Section 7.4, we have discussed the confidence interval for population mean. But in many real word situations, in business and other areas, the data are collected in form of counts or the collected data classified into two categories or groups according to an attribute or characteristic under study. Generally, such types of data are considered in terms of proportion of elements / individuals / units / items possess or not possess a given characteristic or attribute. For example, the proportion of female in the population, proportion of diabetes patients in a hospital, proportion of Science books in a library, proportion of defective articles in a lot, etc.

In such situations, we deal population proportion instead of population mean and one may want to obtain the confidence interval for population proportion.

Estimation

For example, a sociologist may want to know the confidence interval for proportion of female in the population of a state. A doctor may want to know the confidence interval for proportion of diabetes patients in a hospital, a product manager may want to know the confidence interval for proportion of defective articles in a lot, etc.

Generally, population proportion is estimated by sample proportion.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population with population proportion P . Also let X denotes the number of observations or elements possess a certain attribute (successes) out of n observations of the sample then sample proportion p can be defined as

$$p = \frac{X}{n} \leq 1$$

As we have seen in Section 2.4 of the Unit 2 of this course that mean and variance of the sampling distribution of sample proportion are

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n}$$

where, $Q = 1 - P$.

But sample proportion is generally considered for large sample so if sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean P and variance PQ/n . Therefore, the variate

$$Z = \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}} \sim N(0,1)$$

is approximately normally distributed with mean 0 and variance unity. Since distribution of Z is independent of parameter P so it can be taken as pivotal quantity, therefore, we introduce two constants $z_{\alpha/2}$ and $z_{(1-\alpha/2)} = -z_{\alpha/2}$ such that

$$P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 1 - \alpha$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z as shown in Fig. 7.6.

Putting the value of Z , we get

$$P\left[-z_{\alpha/2} \leq \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}} \leq z_{\alpha/2}\right] = 1 - \alpha \quad \dots (8)$$

For large sample, the variance $P(1-P)/n$ can be estimated by $p(1-p)/n$ therefore, putting $p(1-p)/n$ in place of $P(1-P)/n$ in equation (8), we get

$$P\left[-z_{\alpha/2} \leq \frac{p - P}{\sqrt{\frac{p(1-p)}{n}}} \leq z_{\alpha/2}\right] = 1 - \alpha$$

Now, for converting the above interval for parameter P , we multiplying each term by $\sqrt{\frac{p(1-p)}{n}}$ and then subtracting p from each term in the above inequality, we get

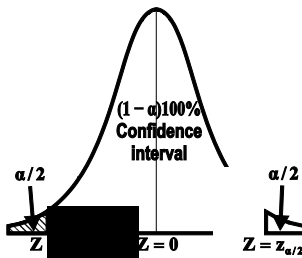


Fig. 7.6

$$P\left[-p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq -P \leq -p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] = 1 - \alpha$$

Now, by multiplying each term by (-1) in above inequality, we get

$$P\left[p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \geq P \geq p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] = 1 - \alpha \quad \left[\begin{array}{l} \because \text{by multiplying} \\ (-1) \text{ the inequality} \\ \text{is reversed} \end{array}\right]$$

This can be written as

$$P\left[p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq P \leq p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] = 1 - \alpha$$

Hence, $(1 - \alpha)$ 100% confidence interval for population proportion is given by

$$\left[p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}, p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] \quad \dots (9)$$

Therefore, corresponding confidence limits are

$$p \mp z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \quad \dots (10)$$

Following example will explain the application of the above discussion:

Example 4: A sample of 200 voters is chosen at random from all voters in a given city. 60% of them were in favour of a particular candidate. If large number of voters cast their votes then find 99% and 95% confidence intervals for the proportion of voters in favour of a particular candidate.

Solution: Here, we are given

$$n = 200, p = 0.60$$

First we check the condition of normality as

$\because np = 200 \times 0.60 = 120 > 5$ and $nq = 200 \times (1 - 0.60) = 200 \times 0.40 = 80 > 5$ so $(1 - \alpha)$ 100% confidence limits for the proportion are given by

$$p \mp z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

For 99% confidence interval, we have $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$. For $\alpha = 0.01$, we have $z_{0.005} = 2.58$ and for $\alpha = 0.05$, $z_{0.025} = 1.96$.

Therefore, 99% confidence limits of voters in favour of a particular candidate are

$$\begin{aligned} p \mp z_{0.005}\sqrt{\frac{p(1-p)}{n}} &= 0.60 \mp 2.58 \times \sqrt{\frac{0.60 \times 0.40}{200}} \\ &= 0.60 \mp 2.58 \times 0.03 \\ &= 0.60 \mp 0.08 = 0.52 \text{ and } 0.68 \end{aligned}$$

Hence, required 99% confidence interval for the proportion of voters in favour of a particular candidate is given by

$$[0.52, 0.68]$$

Similarly, 95% confidence limits are given by

$$p \mp z_{0.025} \sqrt{\frac{p(1-p)}{n}} = 0.60 \mp 1.96 \times 0.03$$

$$= 0.60 \mp 0.06 = 0.54 \text{ and } 0.66$$

Hence, 95% confidence interval for the proportion of voters in favour of a particular candidate is given by

$$[0.54, 0.66]$$

It is your time to try the following exercise.

E7) A random sample of 400 apples was taken from a large consignment and 80 were found to be bad. Obtain the 99% confidence limits for the proportion of bad apples in the consignment.

7.6 CONFIDENCE INTERVAL FOR POPULATION VARIANCE

In Sections 7.4 and 7.5, we discussed the confidence interval for population mean and proportion respectively. But there are many practical situations where one may be interested to obtain the interval estimate of the population variance. For example, a manufacturer of steel ball bearings may want to obtain the interval estimate of the variation of diameter of steel ball bearing, an economist may wish to know the interval estimate for the variability in income of the person living in a city, etc.

Similar to confidence interval for the population mean we can determine confidence interval for population variance into two following cases:

1. When population mean is known and
2. When population mean is unknown.

These two cases are described one by one, in Sub-sections 7.6.1 and 7.6.2 respectively.

7.6.1 Confidence Interval for Population Variance when Population Mean is Known

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population having mean μ and variance σ^2 where μ is known, that is, μ has a specified value. In this case, we know that the variate

$$\chi^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi^2_{(n)}$$

follows the chi-square distribution with n degrees of freedom whose probability density function is given by

$$f(\chi^2) = \frac{1}{2^{n/2} \Gamma(n/2)} e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1}; \quad 0 < \chi^2 < \infty$$

Since distribution of χ^2 is independent of parameter σ^2 therefore, χ^2 can be

taken as pivotal quantity, therefore, we introduce two constants $\chi^2_{(n), \alpha/2}$ and $\chi^2_{(n), (1-\alpha/2)}$ such that

$$P\left[\chi^2_{(n), (1-\alpha/2)} \leq \chi^2 \leq \chi^2_{(n), \alpha/2}\right] = 1 - \alpha \quad \dots (11)$$

where, $\chi^2_{(n), \alpha/2}$ and $\chi^2_{(n), (1-\alpha/2)}$ are the value of the χ^2 variate at n df having area of $\alpha/2$ under the right tail and $\alpha/2$ under the left tail respectively of the probability curve of χ^2 as shown in Fig. 7.7.

Putting the value of χ^2 in equation (11), we get

$$P\left[\chi^2_{(n), (1-\alpha/2)} \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \leq \chi^2_{(n), \alpha/2}\right] = 1 - \alpha$$

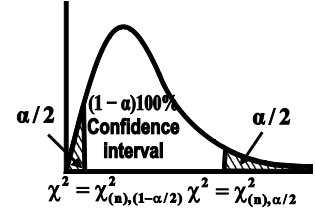


Fig. 7.7

Now, for converting this interval for σ^2 , we divide each term by $\sum_{i=1}^n (X_i - \mu)^2$ in above inequality then we get

$$P\left[\frac{\chi^2_{(n), (1-\alpha/2)}}{\sum_{i=1}^n (X_i - \mu)^2} \leq \frac{1}{\sigma^2} \leq \frac{\chi^2_{(n), \alpha/2}}{\sum_{i=1}^n (X_i - \mu)^2}\right] = 1 - \alpha$$

Reciprocal each term of the above inequality

$$P\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), (1-\alpha/2)}} \geq \sigma^2 \geq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), \alpha/2}}\right] = 1 - \alpha \quad \left[\because \text{by reciprocating, the inequality is reversed} \right]$$

This can be written as

$$P\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), \alpha/2}} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), (1-\alpha/2)}}\right] = 1 - \alpha$$

Hence, $(1 - \alpha)100\%$ confidence interval for population variance when population mean is known in normal population is given by

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), \alpha/2}}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), (1-\alpha/2)}} \right] \quad \dots (12)$$

and the corresponding $(1 - \alpha) 100\%$ confidence limits are given by

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), \alpha/2}} \quad \text{and} \quad \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), (1-\alpha/2)}} \quad \dots (13)$$

Note 3: For different confidence interval and degrees of freedom the values of $\chi^2_{(n), \alpha/2}$ and $\chi^2_{(n), (1-\alpha/2)}$ are different. Therefore, for given values of α and n we read the tabulated value of these from the table of χ^2 -distribution (**χ^2 -table**) (given in

Appendix at the end of Block 1 of this course) by the method described in Unit 4 of this course.

For example, if we want to find the 95% confidence interval for σ^2 then

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$$

From the χ^2 -table, for $\alpha = 0.05$ and if $n=10$, we have

$$\chi^2_{(n), \alpha/2} = \chi^2_{(10), 0.025} = 20.48 \text{ and } \chi^2_{(n), (1-\alpha/2)} = \chi^2_{(10), 0.975} = 3.25$$

Therefore, 95% confidence interval for variance is given by

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{20.48}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{3.25} \right]$$

Following example will explain the application of the above discussion.

Example 5: Diameter of steel ball bearing produced by a company is known to be normally distributed. To know the variation in the diameter of steel ball bearings, the product manager takes a random sample of 10 ball bearings from the lot having average diameter 5.0 cm and measures diameter (in cm) of each selected ball bearing. The results are given below:

S. No.	1	2	3	4	5	6	7	8	9	10
Diameter	5.0	5.1	5.0	5.2	4.9	5.0	5.0	5.1	5.1	5.2

Find the 95% confidence interval for variance in the diameter of steel ball bearings of the lot from which the sample is drawn.

Solution: Here, we are given that

$$n = 10, \mu = 5.0$$

Since population mean is given, therefore, we use $(1 - \alpha)100\%$ confidence interval for population variance when population mean is known which is given by

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), \alpha/2}}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{(n), (1-\alpha/2)}} \right]$$

Calculation for $\sum_{i=1}^n (X_i - \mu)^2$:

S. No.	Diameter (X)	$(X - \mu)$	$(X - \mu)^2$
1	5.0	0	0
2	5.1	0.1	0.01
3	5.0	0	0
4	5.2	0.2	0.04
5	4.9	-0.1	0.01
6	5.0	0	0
7	5.0	0	0
8	5.1	0.1	0.01
9	5.1	0.1	0.01
10	5.2	0.2	0.04
Total			$\sum (X - \mu)^2 = 0.12$

From the above calculation, we have

$$\sum (X_i - \mu)^2 = 0.12$$

For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ then from χ^2 -table, we have

$$\chi^2_{(n), \alpha/2} = \chi^2_{(10), 0.025} = 20.48 \text{ and } \chi^2_{(n), (1-\alpha/2)} = \chi^2_{(10), 0.975} = 3.25$$

Thus, 95% confidence interval for variance in the diameter of steel ball bearings of the lot is given by

$$\left[\frac{0.12}{20.48}, \frac{0.12}{3.25} \right]$$

or $[0.0059, 0.0369]$

7.6.2 Confidence Interval for Population Variance when Population Mean is Unknown

Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from normal population with unknown mean μ and variance σ^2 . In this case, the value of sample mean $\bar{X}$ is used to estimate μ . As we have seen in Section 4.2 of Unit 4 of this course that the variate

$$\chi^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)} \quad \text{where, } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

follows the chi-square distribution with $(n-1)$ degrees of freedom whose probability density function is given by

$$f(\chi^2) = \frac{1}{2^{\frac{(n-1)}{2}} \Gamma\left(\frac{n-1}{2}\right)} e^{-\chi^2/2} (\chi^2)^{\frac{n-1}{2}-1}; \quad 0 < \chi^2 < \infty$$

Since distribution of χ^2 is independent of parameter to be estimated, therefore, χ^2 can be taken as pivotal quantity. So we introduce two constants $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ such that

$$P[\chi^2_{(n-1), (1-\alpha/2)} \leq \chi^2 \leq \chi^2_{(n-1), \alpha/2}] = 1 - \alpha \quad \dots (14)$$

where, $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ are the value of the χ^2 -variate at $(n-1)$ df having area of $\alpha/2$ under the right tail and $\alpha/2$ under the left tail of the probability curve of χ^2 as shown in Fig. 7.8.

Putting the value of χ^2 in equation (14), we get

$$P\left[\chi^2_{(n-1), (1-\alpha/2)} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{(n-1), \alpha/2}\right] = 1 - \alpha$$

Now, for converting this interval for σ^2 we dividing each term in above inequality by $(n-1)S^2$ then we get

Interval Estimation for One Population

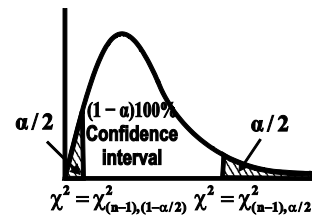


Fig. 7.8

Estimation

$$P\left[\frac{\chi^2_{(n-1), (1-\alpha/2)}}{(n-1)S^2} \leq \frac{1}{\sigma^2} \leq \frac{\chi^2_{(n-1), \alpha/2}}{(n-1)S^2}\right] = 1 - \alpha$$

By taking reciprocal of each term of above inequality, we get

$$P\left[\frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \geq \sigma^2 \geq \frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}}\right] = 1 - \alpha \left[\because \text{by reciprocating, the inequality is reversed} \right]$$

This can be written as

$$P\left[\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}}\right] = 1 - \alpha$$

Hence, the $(1-\alpha)$ 100% confidence interval for population variance when population mean is unknown is given by

$$\left[\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}}, \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \right] \quad \dots (15)$$

and corresponding confidence limits are

$$\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}} \quad \text{and} \quad \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \quad \dots (16)$$

where, $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ are the values of χ^2 -variate at $(n-1)$ degrees of freedom and the values of these can be read from χ^2 -table. For example, if we want to find the 95% confidence interval for σ^2 then

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$$

From χ^2 -table for $\alpha = 0.05$ and $n = 10$ degrees of freedom, we have

$$\chi^2_{(n-1), \alpha/2} = \chi^2_{(9), 0.025} = 19.02 \quad \text{and} \quad \chi^2_{(9), 0.975} = 2.70.$$

Therefore, 95% confidence interval for variance is given by

$$\left[\frac{9S^2}{19.02}, \frac{9S^2}{2.70} \right]$$

Let us do an example based on above discussion.

Example 6: A random sample of 10 workers is taken from a factory. The wages (in hundreds) per months of these workers are given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Obtain 95% confidence interval for the variance of wages of all the workers of the factory.

Solution: Here, the population mean is unknown therefore, we use $(1-\alpha)$ 100% confidence interval for population variance when population mean is unknown which is given by

$$\left[\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}}, \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \right]$$

where, $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ are the values of χ^2 variate at $(n-1)$ degrees of freedom, whereas

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 \text{ and } \bar{X} = \frac{1}{n} \sum X$$

Calculation for $\bar{X}$ and S^2 :

S. No.	Weight (X)	$(X - \bar{X})$	$(X - \bar{X})^2$
1	48	-15	225
2	50	-13	169
3	62	-1	1
4	75	12	144
5	80	17	289
6	60	-3	9
7	70	7	49
8	56	-7	49
9	52	-11	121
10	77	14	196
Sum	$\sum X = 630$		$\sum (X - \bar{X})^2 = 1252$

From the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{10} \times 630 = 63$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 = \frac{1}{9} \times 1252 = 139.11$$

For 95% confidence interval

$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$
and $\alpha/2 = 0.025$ &
 $1 - \alpha/2 = 0.975$.

From χ^2 -table, we have $\chi^2_{(n-1), \alpha/2} = \chi^2_{(9), 0.025} = 19.02$ and

$$\chi^2_{(n-1), (1-\alpha/2)} = \chi^2_{(9), 0.975} = 2.70$$

Thus, 95% confidence interval for the variance of wages of all the workers of the factory is given by

$$\left[\frac{10 \times 139.11}{19.02}, \frac{10 \times 139.11}{2.70} \right] \text{ or } [73.14, 515.22]$$

Now, you can try following exercises to see how much you have followed.

-
- E8)** A study of variation in weights of soldiers was made and it is known that the mean weight of soldiers follows the normal distribution. A sample of 12 soldiers is taken from the soldier's population and sample variance is found 60 pound². Estimate the 95% confidence interval for the variance of soldier's weight of the population from which the sample was drawn.
- E9)** If $X_1 = -5$, $X_2 = 4$, $X_3 = 2$, $X_4 = 6$, $X_5 = -1$, $X_6 = 4$, $X_7 = 0$, $X_8 = 10$ and $X_9 = 7$ are the sample observations taken from normal population $N(\mu, \sigma^2)$, obtain confidence interval for σ^2 .
-

7.7 CONFIDENCE INTERVAL FOR NON-NORMAL POPULATIONS

So far in this unit, we have kept our discussion on the confidence interval of the normal population except population proportion. But one may be interested to find out the confidence interval when the population under study is not normal. The aim of this section is to give an idea how we can obtain confidence interval for non-normal populations. For example, one may be interested to estimate, say, 95% confidence interval of parameter θ when population under study follows exponential distribution(θ).

We know that when the sample size is large then almost all the sampling distributions of the statistics $\bar{X}$, S^2 , etc. follow normal distribution. So when the sample is large we can also obtain the confidence interval as follows:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n (sufficiently large i.e. $n > 30$) taken from $f(x, \theta)$ then according to the central limit theorem the sampling distribution of sample mean $\bar{X}$ is normal, that is,

$$\bar{X} \sim N[E(\bar{X}), \text{Var}(\bar{X})]$$

Then the variate

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} \sim N(0,1)$$

is approximately normally distributed with mean 0 and variance unity. Since distribution of Z is independent of parameter so it can be taken as pivotal quantity therefore we introduce two constants $z_{\alpha/2}$ and $z_{(1-\alpha/2)} = -z_{\alpha/2}$ such that

$$P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 1 - \alpha \quad \dots (17)$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z .

By putting the value of Z in equation (17), we get

$$P\left[-z_{\alpha/2} \leq \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} \leq z_{\alpha/2}\right] = 1 - \alpha$$

After this, we have to convert this interval for parameter θ as discussed in Section 7.3.

Following example will explain the procedure more clearly.

Example 7: Obtain 95% confidence interval to estimate θ when a large sample is taken from exponential population whose probability density function is given by

$$f(x, \theta) = \theta e^{-\theta x}; \quad x > 0, \theta > 0$$

Solution: Let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from exponential population whose probability distribution is given by

$$f(x, \theta) = \theta e^{-\theta x}; \quad x > 0, \theta > 0$$

We know that for exponential distribution

$$E(X) = \frac{1}{\theta} \text{ and } \text{Var}(X) = \frac{1}{\theta^2}$$

Since $X_1, X_2, \dots, X_n$ are independent and come from same exponential distribution, therefore,

$$E(X_i) = E(X) = 1/\theta \text{ and } \text{Var}(X_i) = \text{Var}(X) = 1/\theta^2 \text{ for all } i = 1, 2, \dots, n$$

Now consider,

$$\begin{aligned} E(\bar{X}) &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \text{ [By definition of sample mean]} \\ &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \left[\begin{array}{l} \because E(aX + bY) \\ = aE(X) + bE(Y) \end{array} \right] \\ &= \frac{1}{n} \left(\underbrace{\frac{1}{\theta} + \frac{1}{\theta} + \dots + \frac{1}{\theta}}_{n\text{-times}} \right) \\ &= \frac{1}{n} \left(n \frac{1}{\theta} \right) = \frac{1}{\theta} \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \left[\begin{array}{l} \text{If } X \text{ and } Y \text{ are} \\ \text{two independent} \\ \text{random variable} \\ \text{then } \text{Var}(aX + bY) \\ = a^2 \text{Var}(X) + b^2 \text{Var}(Y) \end{array} \right] \\ &= \frac{1}{n^2} \left(\underbrace{\frac{1}{\theta^2} + \frac{1}{\theta^2} + \dots + \frac{1}{\theta^2}}_{n\text{-times}} \right) \\ &= \frac{1}{n^2} \left(n \frac{1}{\theta^2} \right) \\ \text{Var}(\bar{X}) &= \frac{1}{n\theta^2} \end{aligned}$$

Thus, the variate

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - 1/\theta}{\sqrt{1/n\theta^2}} \sim N(0,1)$$

is approximately normally distributed with mean 0 and variance unity. Since distribution of Z is independent of parameter so it can be taken as pivotal quantity, therefore we introduce two constants $z_{\alpha/2}$ and $z_{(1-\alpha/2)} = -z_{\alpha/2}$ such that

$$P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 1 - \alpha$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z .

For 95% confidence interval, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$. So confidence interval for θ is

$$P[-1.96 \leq Z \leq 1.96] = 1 - \alpha$$

Putting the value of Z, we have

$$\begin{aligned}
 & P \left[-1.96 \leq \frac{\bar{X} - \frac{1}{\theta}}{\sqrt{\frac{1}{n\theta^2}}} \leq 1.96 \right] = 0.95 \\
 & \Rightarrow P \left[-1.96 \leq \frac{\frac{1}{\theta}(\theta\bar{X} - 1)}{\frac{1}{\theta}\sqrt{\frac{1}{n}}} \leq 1.96 \right] = 0.95 \\
 & \Rightarrow P \left[-1.96 \leq \sqrt{n}(\theta\bar{X} - 1) \leq 1.96 \right] = 0.95 \\
 & \Rightarrow P \left[\frac{-1.96}{\sqrt{n}} \leq (\theta\bar{X} - 1) \leq \frac{1.96}{\sqrt{n}} \right] = 0.95 \\
 & \Rightarrow P \left[1 - \frac{1.96}{\sqrt{n}} \leq \theta\bar{X} \leq 1 + \frac{1.96}{\sqrt{n}} \right] = 0.95 \\
 & \Rightarrow P \left[\frac{1}{\bar{X}} \left(1 - \frac{1.96}{\sqrt{n}} \right) \leq \theta \leq \frac{1}{\bar{X}} \left(1 + \frac{1.96}{\sqrt{n}} \right) \right] = 0.95
 \end{aligned}$$

Hence, 95% confidence interval for parameter θ is

$$\left[\frac{1 - 1.96/\sqrt{n}}{\bar{X}}, \frac{1 + 1.96/\sqrt{n}}{\bar{X}} \right]$$

7.8 SHORTEST CONFIDENCE INTERVAL

It may be noted that for a confidence coefficient, there are many confidence intervals for a parameter are possible. For example, from normal table (given in the end of the Block 1 of this course) we can have many sets of a's and b's to give 95% confidence interval for μ some of them are given below:

$$P[-1.65 \leq Z \leq 3.0] = 0.95, \quad P[-1.68 \leq Z \leq 2.70] = 0.95$$

$$P[-1.70 \leq Z \leq 2.54] = 0.95, \quad P[-1.96 \leq Z \leq 1.96] = 0.95, \text{ etc.}$$

Therefore, we need some criterion with the help of which we may choose the best (best in the sense of minimum length) confidence interval among these set of confidence intervals.

An obvious criterion (method) of selecting the shortest one out of these is that, we chose a's and b's in such a way that the length of interval is minimum. In above case, the lengths of these intervals are

$$L = T_2 - T_1$$

$$\text{So, } L_1 = 3.0 - (-1.65) = 4.65, \quad L_2 = 2.70 - (-1.68) = 4.38$$

$$L_3 = 2.54 - (-1.70) = 4.24, \quad L_4 = 1.96 - (-1.96) = 3.92$$

Hence, the last one has the minimum length. Therefore, it is best confidence interval for μ in all the above intervals on the basis of the minimum length criterion.

So, far you have become familiar with the main goal of this block. The discussion of whole block centered on the theme of estimate some population parameter of interest. To estimate some population parameter we have to draw a random sample. A natural question which may arise in your mind “how large should my sample be?” And this is very important question which is commonly asked. From statistical point of view, the best answer of this question is “take as large sample as you can afford. That is, if possible ‘sample’ the entire population that means study all units of the population under study because by taking all units of the population we will have all the information about the parameter and we will know the exact value of population parameter which is better than any estimate of that parameter. Generally, this is impractical to take entire population to be sampled due to economic constraints, time constraints and other limitations. So the answer, “take as large sample as you can afford is best if we ignore all costs because as you have studied in Section 1.4 of Unit 1 of this course that the larger the sample size smaller the standard error of the statistic that means less the uncertainty.

When the recourses in terms of money and time are limited then the question is “how to found the minimum sample size which will satisfy some precision requirements” In such cases, we should require first the answers of the following three questions about his / her requirement about the survey:

1. How close do you want your sample estimate to be to the unknown parameter? That means what should be the allowable difference between the sample estimate and true value of population parameter. This difference is known as sampling error or margin of error and represented by E.
2. The next question is, what do you want the confidence level to be so that the difference between the estimate and the parameter is less than or equal to E? That is, 90%, 95%, 99%, etc.
3. The last question is what is the population variance or population proportion as may be the case?

When we have the answers of these three questions then we will get an answer of the minimum required sample size.

In this section, we will describe the procedure of determining of sample size for estimating population mean and population proportion.

Determination of Minimum Sample Size for Estimating Population Mean

For determining minimum sample size for estimating population mean we use confidence interval. Since as you seen in Section 7.4 of this unit that confidence interval for population mean depends upon the nature of population and population variance (σ^2) is known or unknown so following cases may arise:

Case I: Population is normal and population variance (σ^2) is known

If σ^2 is known and population is normal then we know that $(1 - \alpha)100\%$ confidence interval for population mean is given by

$$P\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

Estimation

$$\text{Also } P\left[-Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha \quad \dots (18)$$

Since the normal distribution is symmetric so we can concentrate on the right-hand equality so

$$\bar{X} - \mu \leq Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (19)$$

This inequality implies that the largest value that the difference $\bar{X} - \mu$ can assume is $Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.

Also the difference between the estimator (sample mean $\bar{X}$) and the population parameter (population mean μ) is called the sampling error so

$$E = \bar{X} - \mu = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \dots (20)$$

Solving this equation for n, we have

$$n = \frac{Z_{\alpha/2}^2 \sigma^2}{E^2} \quad \dots (21)$$

When population is finite of size N and sampling is to be done without replacement then finite population correction $\sqrt{\frac{N-n}{N-1}}$ is required so equation (20) becomes

$$E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

which gives

$$n = \frac{N Z_{\alpha/2}^2 \sigma^2}{E^2 (N-1) + Z_{\alpha/2}^2 \sigma^2} \quad \dots (22)$$

If the finite population correction is ignored then equation (22) is reduced to equation (21).

Case II: Population is non-normal and population variance (σ^2) is known

If the population is not assumed to be normal and the population variance σ^2 is known then by central limit theorem we know that sampling distribution of mean approximate normally distributed as sample size increases. So the above method can be used determining minimum sample size. Once the required sample size is obtained, we can check to see it that sample size greater than 30 and if it does we may be confident that our method of solution was appropriate.

Case III: Population is normal or non-normal and population variance (σ^2) is unknown

In this case, we use value of sample variance (S^2) to estimate the population variance but S^2 is also calculated from a sample and we have not take a sample yet. So in this case, determination of sample size is not directly obtained. The most frequently methods for estimating σ^2 are as follows:

1. A pilot or preliminary sample may be drawn from the population under study and the variance computed from this sample may be used as an estimate of σ^2 .
2. The variance of previous or similar studies may used to estimate σ^2 .
3. If the population is assumed to be normal then we may use the fact that the

range is approximately equal to six times of standard deviation i.e.
 $\sigma = R / 6$. This approximate require only knowledge of largest and smallest
 value of the variable under study because range may be defined as

$$R = \text{largest value} - \text{smallest value}$$

Determination of Minimum Sample Size for Estimating Population Proportion

The method of determination of minimum sample size for estimating population proportion is similar as that described in estimating population mean. So the formula for minimum sample size is given by

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} \quad \dots (23)$$

where, P is the population proportion and $E = p - P$ is the sampling error or margin of error.

When population is finite of size N and sampling is to be done without

replacement then finite population correction $\sqrt{\frac{N-n}{N-1}}$ is required so

$$n = \frac{N z_{\alpha/2}^2 P(1-P)}{E^2 (N-1) + z_{\alpha/2}^2 P(1-P)} \quad \dots (24)$$

Here, sample size is depended upon the population proportion and it is generally unknown so the most frequently methods for estimating P are as follows:

1. A pilot or preliminary sample may be drawn from the population under study and the sample proportion computed from this sample may be used as an estimate of P.
2. The proportion of previous or similar studies may used to estimate P.
3. Absence of any information, we may use $P = 0.5$.

Now, it is time to do some examples based on determination of sample size.

Example 8: A hospital administrator wishes to estimate the mean weight of babies born in her hospital. How large a sample of birth records should be taken if she wants a 99 percent confidence that the estimate within the range of 0.4 pound? Assume that a reasonable estimate of σ is 0.5 pound.

Solution: Here, we are given that

$$E = \text{margin of error} = 0.4, \text{ confidence level} = 0.99 \text{ and } \sigma = 0.5$$

Also for 99% confidence, $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$ and $\alpha/2 = 0.005$, we have

$$z_{\alpha/2} = z_{0.005} = 2.58.$$

Hospital administrator wishes to obtain the minimum sample size for estimating the mean weight of babies born in her hospital so we have

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2} = \frac{(2.58)^2 (0.5)^2}{(0.4)^2} = 10.40 \sim 11$$

Hence, hospital administrator should obtain a random sample of at least 11 babies.

Example 9: The manufacturers of a car want to estimate the proportion of people who are interested in a certain model. The company wants to know the population proportion, P , to within 0.05 with 95% confidence. Current company records indicate that the proportion P may be around 0.20. What is the minimum required sample size for this survey?

Solution: Here, we are given that

$$E = \text{margin of error} = 0.05, \text{ confidence level} = 0.95 \text{ and } P = 0.20$$

Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

The manufacturers of a car interested to obtain the minimum sample size for estimating the population proportion so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} = \frac{(1.96)^2 (0.20)(0.80)}{(0.05)^2} = 245.86 \sim 246$$

Hence, the company should require a random sample of at least 246 people.

You will be more cleared about this, when you try the following exercises.

-
- E10)** A survey is planned to determine the average annual family medical expenses of employees of a large company. The management of the company wishes to be 95% confident that the sample average is correct to within ± 100 Rs of the true average family expenses. A pilot study indicates that the standard deviation can be estimated as 400 Rs. How large a sample size is necessary?
- E 11)** The manager of a bank in a small city would like to determine the proportion of the depositors per week. The manager wants to be 95% confident of being correct to within ± 0.10 of the true proportion of depositors per week. A guess is that the parentage of such depositors is about 8%, what sample size is needed?
-

With this we have reached end this unit. Let us summarise what we have discussed in this unit.

7.10 SUMMARY

In this unit, we have covered the following points:

1. The interval estimation.
2. The method of obtaining the confidence intervals.
3. The method of obtaining confidence interval for population mean of a normal population when variance is known and unknown
4. The method of obtaining confidence interval for population proportion of a population.
5. The method of obtaining confidence interval for population variance of a normal population when population mean is known and unknown.
6. The method of obtaining confidence interval for population parameters of non-normal populations.
7. The concept of the shortest interval.
8. Determination of sample size.

7.11 SOLUTIONS / ANSWERS

Interval Estimation for One Population

E1) We know that the length of confidence interval $[T_1, T_2]$ is given by

$$L = T_2 - T_1$$

Therefore, in our case we have

(i) $L = 3.0 - (-1.65) = 4.65$

(ii) $L = 2.70 - (-1.68) = 4.38$

(iii) $L = 2.54 - (-1.70) = 4.24$

(iv) $L = 1.96 - (-1.96) = 3.92$

E2) We know that if we have confidence interval for parameter θ is

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha$$

then

$$\text{Lower confidence limit (LCL)} = T_1$$

$$\text{Upper confidence limit (UCL)} = T_2$$

$$\text{Confidence coefficient (CC)} = 1 - \alpha$$

Therefore in our cases, we have

(i) $LCL = 0, UCL = 1.5 \text{ \& } CC = 0.90$

(ii) $LCL = -1, UCL = 2 \text{ \& } CC = 0.95$

(iii) $LCL = -2, UCL = 2 \text{ \& } CC = 0.98$

(iv) $LCL = -2.5, UCL = 2.5 \text{ \& } CC = 0.99$

E3) Refer Section 7.3.

E4) Here, we are given that

$$n = 200, \sigma = 0.30, \bar{X} = 15.25$$

Since population standard deviation or variance is known so we use $(1 - \alpha)$ 100% confidence limits for population mean when population variance is known which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where, $z_{\alpha/2}$ is the value of the variate Z having an area of $\alpha/2$ under the right tail of the probability curve of Z . For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have,

$$z_{\alpha/2} = z_{0.025} = 1.96.$$

Thus, the 95% confidence limits for the average weight of oil tins are

$$\begin{aligned} \bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}} &= 15.25 \pm 1.96 \times \frac{0.30}{\sqrt{200}} \\ &= 15.25 \pm 0.04 = 15.21 \text{ and } 15.29 \end{aligned}$$

E5) Here, we are given that

$$n = 150, \bar{X} = 65, S = 12$$

Since population variance is unknown and sample size is large sample,

Estimation

so we use $(1 - \alpha)$ 100% confidence limits for population mean when population variance is unknown which are given by

$$\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}}$$

For 95% confidence interval, we have $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and for $\alpha = 0.05$, we have, $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, the 95% confidence limits in which the average weight of all students of a Study centre of IGNOU expected to lie are given by

$$\begin{aligned}\bar{X} \pm 1.96 \frac{S}{\sqrt{n}} &= 65 \pm 1.96 \times \frac{12}{\sqrt{150}} \\ &= 65 \pm 1.92 = 63.08 \text{ and } 66.92\end{aligned}$$

Hence, required confidence limits are

63.08 and 66.92.

- E6)** Since population variance is unknown and sample size is small ($n < 30$), therefore, $(1 - \alpha)$ 100% confidence limits for average height of credits of particular centre are given by

$$\bar{X} \pm t_{(n-1), \alpha/2} \frac{S}{\sqrt{n}}$$

$$\text{where, } \bar{X} = \frac{1}{n} \sum X, S = \sqrt{\frac{1}{n-1} \sum (X - \bar{X})^2}$$

Calculation for $\bar{X}$ and S :

S. No	X	$(X - \bar{X})$	$(X - \bar{X})^2$
1	70	-7	49
2	72	-5	25
3	80	3	9
4	82	5	25
5	78	1	1
6	80	3	9
	$\sum X = 462$		$\sum (X - \bar{X})^2 = 118$

Form the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{6} \times 462 = 77$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 = \frac{1}{6-1} \times 118 = 23.6$$

$$\Rightarrow S = 4.86$$

For 95% confidence interval
 $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$
and $\alpha/2 = 0.025$.

From t-table, we have $t_{(n-1), \alpha/2} = t_{(5), 0.025} = 2.571$.

Thus, 95% confidence limits are given by

$$\begin{aligned}\bar{X} \pm t_{(n-1), 0.025} \frac{S}{\sqrt{n}} &= 77 \pm 2.571 \times \frac{4.86}{\sqrt{6}} \\ &= 77 \pm 2.571 \times 1.98 \\ &= 77 \pm 5.09 = 71.91 \text{ and } 82.09\end{aligned}$$

E7) We have

$$n = 400, X = 80$$

$$p = \frac{X}{n} = \frac{80}{400} = \frac{1}{5} = 0.20$$

$\therefore np = 400 \times 0.20 = 80 > 5$ and $nq = 400 \times (1 - 0.20) = 400 \times 0.80 = 320 > 5$ so $(1 - \alpha)100\%$ confidence limits for the proportion are given by

$$p \mp z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

For $\alpha = 0.01$, we have, $z_{\alpha/2} = z_{0.005} = 2.58$.

Therefore, 99% confidence limits are

$$\begin{aligned} & 0.20 \mp 2.58 \sqrt{\frac{0.20 \times 0.80}{400}} \\ & = 0.2 \mp 2.58 \times 0.02 \\ & = 0.2 \mp 0.05 = 0.15 \text{ and } 0.25 \end{aligned}$$

Interval Estimation for One Population

For 99% confidence interval

$1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$ and $\alpha/2 = 0.005$.

E8) Here, we are given that

$$n = 12, S^2 = 60$$

Since population mean is unknown, therefore, we use $(1 - \alpha) 100\%$ confidence interval for population variance when population mean unknown which is given by

$$\left[\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}}, \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \right]$$

where, $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ are the values of χ^2 variate at $(n-1)$ degrees of freedom. From χ^2 -table, we have

$$\chi^2_{(n-1), \alpha/2} = \chi^2_{(11), 0.025} = 21.92 \text{ and } \chi^2_{(n-1), (1-\alpha/2)} = \chi^2_{(11), 0.975} = 3.82$$

Thus, 95% confidence interval for the variance of soldier's weight is given by

$$\left[\frac{11 \times 60}{21.92}, \frac{11 \times 60}{3.82} \right] \Rightarrow [30.11, 172.77]$$

For 95% confidence interval

$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$.

E9) Since population mean is unknown, therefore, we use $(1 - \alpha) 100\%$ confidence interval for population variance which is given by

$$\left[\frac{(n-1)S^2}{\chi^2_{(n-1), \alpha/2}}, \frac{(n-1)S^2}{\chi^2_{(n-1), (1-\alpha/2)}} \right]$$

where, $\chi^2_{(n-1), \alpha/2}$ and $\chi^2_{(n-1), (1-\alpha/2)}$ are the values of χ^2 variate at $(n-1)$ degrees of freedom, whereas

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2 \text{ and } \bar{X} = \frac{1}{n} \sum X$$

EstimationCalculation for $\bar{X}$ and S^2 :

X	$(X - \bar{X})$	$(X - \bar{X})^2$
-5	-8	64
4	1	1
2	-1	1
6	3	9
-1	-4	16
4	1	1
0	-3	9
10	7	49
7	4	16
$\sum X = 27$		$\sum (X - \bar{X})^2 = 166$

Therefore,

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{9}(27) = 3$$

Also, $(n-1)S^2 = \sum (X - \bar{X})^2 = 166$ and from χ^2 -table, we have

$$\chi^2_{(n-1), \alpha/2} = \chi^2_{(8), 0.005} = 21.96 \text{ and } \chi^2_{(n-1), (1-\alpha/2)} = \chi^2_{(8), 0.995} = 1.34$$

Thus, 99% confidence interval for σ^2 is

$$\left[\frac{166}{21.96}, \frac{166}{1.34} \right] \text{ or } [7.56, 123.88]$$

For 99% confidence interval
 $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$
and $\alpha/2 = 0.005$

E10) Here, we are given thatE = margin of error = 100, confidence level = 0.95 and $\sigma = 400$ Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

The management of the company wishes to obtain the minimum sample size for estimating the average annual family medical expenses of employees of the company so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2} = \frac{(1.96)^2 (400)^2}{(100)^2} = 61.46 \sim 62$$

Hence, the management of the company should require a random sample of at least 62 employees.

E 11) Here, we are given thatE = margin of error = 0.1, confidence level = 0.95 and $P = 0.08$ Also for 95% confidence, $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$ and $\alpha/2 = 0.025$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

The manager wants to obtain the minimum sample size for determining the proportion so the required formula is given below

$$n = \frac{z_{\alpha/2}^2 P(1-P)}{E^2} = \frac{(1.96)^2 (0.08)(0.94)}{(0.05)^2} = 115.56 \sim 116$$

Hence, the manager should require a random sample of at least 116 depositors.

UNIT 8 INTERVAL ESTIMATION FOR TWO POPULATIONS

Structure

- 8.1 Introduction
 - Objectives
- 8.2 Confidence Interval for Difference of Two Population Means
- 8.3 Confidence Interval for Difference of Two Population Proportions
- 8.4 Confidence Interval for Ratio of Two Population Variances
- 8.5 Summary
- 8.6 Solutions / Answers

8.1 INTRODUCTION

In the previous unit, we have discussed the method of obtaining confidence interval for population mean, population proportion and population variance for a population under study. There are so many situations where two populations exist and one wants to obtain the interval estimate for the difference or ratio of two parameters as means, proportions, variances, etc. For example, a company manufacturing two types of bulbs and product manager may be interested to obtain the confidence interval for difference of average life of two types of bulbs, one may wish to obtain the interval estimate of the difference of proportions of alcohol drinkers in the two cities, a quality control engineer wants to obtain the interval estimate for the ratio of variances of the quality of the product, etc.

Therefore, it becomes necessary to construct the confidence interval for difference of means, proportions and ratio of variances of two populations. In this unit, we shall discuss how we construct confidence intervals for difference or ratio of the above mentioned parameters of two populations.

This unit comprises the following six sections. Section 8.1 introduces the need of confidence intervals for the difference or ratio of the parameters of two normal populations. Section 8.2 is devoted to method of obtaining the confidence interval for difference of two population means when population variances are known and unknown. Section 8.3 described the method of obtaining the confidence intervals for difference of two population proportions with examples, whereas the method of obtaining the confidence interval for ratio of population variances is explored in Section 8.4. Unit ends by providing summary of what we have discussed in this unit in Section 8.5 and solution of exercises in Section 8.6.

Objectives

After studying this unit, you should be able to:

- introduce the confidence intervals in case of two populations;
- describe the method of obtaining the confidence interval for difference of means of two normal populations when variances are known and unknown;
- describe the method of obtaining the confidence interval for difference of means of two normal populations when observations are paired;

Block

3

TESTING OF HYPOTHESIS

UNIT 9

Concepts of Testing of Hypothesis	5
--	----------

UNIT 10

Large Sample Tests	25
---------------------------	-----------

UNIT 11

Small Sample Tests	63
---------------------------	-----------

UNIT 12

Chi-Square and F-Tests	91
-------------------------------	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathasan
Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Parvin Sinclair
Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Geeta Kaicker
Director, School of Sciences
IGNOU, New Delhi

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. R. M. Pandey
Department of Bio-Statistics
All India Institute of Medical Sciences
New Delhi

Prof. Rahul Roy
Math. and Stat. Unit
Indian Statistical Institute, New Delhi

Dr. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Dr. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Faculty members of School of Sciences, IGNOU

Statistics

Dr. Neha Garg
Dr. Nitin Gupta
Mr. Rajesh Kaliraman
Dr. Manish Trivedi

Mathematics

Dr. Deepika
Prof. Poornima Mital
Prof. Sujatha Varma
Dr. S. Venkataraman

Block Preparation Team

Dr. Ramkishan (**Editor**)
Department of Statistics
D. A. V. (PG) College
C.C. S. University, Merrut

Dr. Parmod Kumar (**Language Editor**)
School of Humanities, IGNOU

Dr. Diwakar Shukla (**Units 1 - 4**)
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Mr. Prabhat Kumar Sangal (**Units 1 - 4**)
School of Sciences, IGNOU

Course Coordinator: Mr. Prabhat Kumar Sangal

Programme Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P), School of Sciences, IGNOU
CRC prepared by Mr. Prabhat Kumar Sangal, School of Sciences, IGNOU

Acknowledgement: I gratefully acknowledge my colleague Mr. Rajesh Kaliraman, Statistics Discipline, School of Sciences for their great support.

July, 2013

© Indira Gandhi National Open University, 2013

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University.

Further information on the Indira Gandhi National Open University may be obtained from University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Printed at:

TESTING OF HYPOTHESIS

In previous block of this course, we have discussed one part of statistical inference, that is, estimation and we have learnt how we estimate the unknown population parameter(s) by using **point estimation** and **interval estimation**. In this block we will focus on the second part of statistical inference which is known as **testing of hypothesis**.

In this block, you will study the basic concept of testing of hypothesis and different kinds of well-known tests. At the end of this block you will be aware of the idea, procedure and applications of the hypothesis testing. This block contains four units.

Unit 9: Concepts of Testing of Hypothesis

This unit explains the fundamental aspect relating to testing of hypothesis with examples. It describes basic concepts and methodologies as hypothesis, critical region, types of errors, level of significance, general procedure of testing a hypothesis, concept of p-value, relation between confidence interval and testing of hypothesis.

Unit 10: Large Sample Tests

This unit explores the procedure of testing the hypothesis based on one sample and two samples when sample size is large and taken from normal or non-normal population(s). In this unit, you will learn about Z-test for testing of hypothesis about mean, difference of two means, proportion, difference of two proportions, variance and equality of two variances.

Unit 11: Small Sample Tests

This unit is devoted to test a hypothesis based on one sample and two samples when sample sizes are small. In this unit, you will learn about t-test for testing of hypothesis about mean and difference of two means. This unit also explores the idea for testing the hypothesis for equality of two means when samples are dependent and testing of hypothesis about population correlation coefficient.

Unit 12: Chi-square and F Tests

The last unit of this block describes the chi-square test for population variance and F-test for equality of two population variances.

Notations and Symbols

$X_1, X_2, \dots, X_n$	:	Random sample of size n
H_0	:	Null hypothesis
H_1 or H_A	:	Alternative hypothesis
ω	:	Critical (rejection) region
$\bar{\omega}$	:	Non-rejection region
α	:	Size of critical region or type-I error or level of significance
β	:	Type-II error
$1-\beta$	:	Power of the test
z_α	:	Critical value of Z-test at α level of significance
$t_{(v), \alpha}$	:	Critical value of t-test with v degrees of freedom at α level of significance
$\chi^2_{(v), \alpha}$	:	Critical value of χ^2 -test with v degrees of freedom at α level of significance
$F_{(v_1, v_2), \alpha}$	:	Critical value of F-test with (v_1, v_2) degrees of freedom at α level of significance

UNIT 9 CONCEPTS OF TESTING OF HYPOTHESIS

Structure

- 9.1 Introduction
 - Objectives
- 9.2 Hypothesis
 - Simple and Composite Hypotheses
 - Null and Alternative Hypotheses
- 9.3 Critical Region
- 9.4 Type-I and Type-II Errors
- 9.5 Level of Significance
- 9.6 One-Tailed and Two-Tailed Tests
- 9.7 General Procedure of Testing a Hypothesis
- 9.8 Concept of p-Value
- 9.9 Relation between Confidence Interval and Testing of Hypothesis
- 9.10 Summary
- 9.11 Solutions /Answers

9.1 INTRODUCTION

In previous block of this course, we have discussed one part of statistical inference, that is, estimation and we have learnt how we estimate the unknown population parameter(s) by using **point estimation** and **interval estimation**. In this block, we will focus on the second part of statistical inference which is known as **testing of hypothesis**.

In our day-to-day life, we see different commercials advertisements in television, newspapers, magazines, etc. such as

- (i) The refrigerator of certain brand saves up to 20% electric bill,
- (ii) The motorcycle of certain brand gives 60 km/liter mileage,
- (iii) A detergent of certain brand produces the cleanest wash,
- (iv) Ninety nine out of hundred dentists recommend brand A toothpaste for their patients to save the teeth against cavity, etc.

Now, the question may arise in our mind “can such types of claims be verified statistically?” Fortunately, in many cases the answer is “yes”.

The technique of testing such type of claims or statements or assumptions is known as testing of hypothesis. The truth or falsity of a claim or statement is never known unless we examine the entire population. But practically it is not possible in mostly situations so we take a random sample from the population under study and use the information contained in this sample to take the decision whether a claim is true or false.

This unit is divided into 11 sections. Section 9.1 is introductory in nature. In Section 9.2, we defined the hypothesis. The concept and role of critical region in testing of hypothesis is described in Section 9.3. In Section 9.4, we explored the types of errors in testing of hypothesis whereas level of significance is explored in Section 9.5. In Section 9.6, we explored the types of tests in testing

of hypothesis. The general procedure of testing a hypothesis is discussed in Section 9.7. In Section 9.8, the concept of p-value in decision making about the null hypothesis is discussed whereas the relation between confidence interval and testing of hypothesis is discussed in Section 9.9. Unit ends by providing summary of what we have discussed in this unit in Section 9.10 and solution of exercises in Section 9.11.

Objectives

After reading this unit, you should be able to:

- define a hypothesis;
- formulate the null and alternative hypotheses;
- explain what we mean by type-I and type-II errors;
- explore the concept of critical region and level of significance;
- define one-tailed and two-tailed tests;
- describe the general procedure of testing a hypothesis;
- concept of p-value; and
- test a hypothesis by using confidence interval.

Before coming to the procedure of testing of hypothesis, we will discuss the basis terms used in this procedure one by one in subsequent sections.

9.2 HYPOTHESIS

As we have discussed in previous section that in our day-to-day life, we see different commercials advertisements in television, newspapers, magazines, etc. and if someone may be interested to test such type of claims or statement then we come across the problem of testing of hypothesis. For example,

- (i) a customer of motorcycle wants to test whether the claim of motorcycle of certain brand gives the average mileage 60 km/liter is true or false,
- (ii) the businessman of banana wants to test whether the average weight of a banana of Kerala is more than 200 gm,
- (iii) a doctor wants to test whether new medicine is really more effective for controlling blood pressure than old medicine,
- (iv) an economist wants to test whether the variability in incomes differ in two populations,
- (v) a psychologist wants to test whether the proportion of literates between two groups of people is same, etc.

In all the cases discussed above, the decision maker is interested in making inference about the population parameter(s). However, he/she is not interested in estimating the value of parameter(s) but he/she is interested in testing a claim or statement or assumption about the value of population parameter(s). Such claim or statement is postulated in terms of hypothesis.

In statistics, a hypothesis is a statement or a claim or an assumption about the value of a population parameter (e.g., mean, median, variance, proportion, etc.).

Similarly, in case of two or more populations a hypothesis is comparative statement or a claim or an assumption about the values of population parameters. (e.g., means of two populations are equal, variance of one population is greater than other, etc.). The plural of hypothesis is hypotheses.

In hypothesis testing problems first of all we should be identifying the claim or statement or assumption or hypothesis to be tested and write it in the words. Once the claim has been identified then we write it in symbolical form if possible. As in the above examples,

- (i) Customer of motorcycle may write the claim or postulate the hypothesis “the motorcycle of certain brand gives the average mileage 60 km/liter.” Here, we are concerning the **average** mileage of the motorcycle so let μ represents the average mileage then our hypothesis becomes $\mu = 60 \text{ km / liter}$.
- (ii) Similarly, the businessman of banana may write the statement or postulate the hypothesis “the average weight of a banana of Kerala is greater than 200 gm.” So our hypothesis becomes $\mu > 200 \text{ gm}$.
- (iii) Doctor may write the claim or postulate the hypothesis “ the new medicine is really more effective for controlling blood pressure than old medicine.” Here, we are concerning the **average** effect of the medicines so let μ_1 and μ_2 represent the average effect of new and old medicines respectively on controlling blood pressure then our hypothesis becomes $\mu_1 > \mu_2$.
- (iv) Economist may write the statement or postulate the hypothesis “ the variability in incomes differ in two populations.” Here, we are concerning the **variability** in income so let σ_1^2 and σ_2^2 represent the variability in incomes in two populations respectively then our hypothesis becomes $\sigma_1^2 \neq \sigma_2^2$.
- (v) Psychologist may write the statement or postulate the hypothesis “the proportion of literates between two groups of people is same.” Here, we are concerning the **proportion** of literates so let P_1 and P_2 represent the proportions of literates of two groups of people respectively then our hypothesis becomes $P_1 = P_2$ or $P_1 - P_2 = 0$.

The hypothesis is classified according to its nature and usage as we will discuss in subsequent subsections.

9.2.1 Simple and Composite Hypotheses

In general sense, if a hypothesis specifies only one value or exact value of the population parameter then it is known as simple hypothesis. And if a hypothesis specifies not just one value but a range of values that the population parameter may assume is called a composite hypothesis.

As in the above examples, the hypothesis postulated in (i) $\mu = 60 \text{ km/liter}$ is simple hypothesis because it gives a single value of parameter ($\mu = 60$) whereas the hypothesis postulated in (ii) $\mu > 200 \text{ gm}$ is composite hypothesis because it does not specify the exact average value of weight of a banana. It may be 260, 350, 400 gm or any other.

Similarly, (iii) $\mu_1 > \mu_2$ or $\mu_1 - \mu_2 > 0$ and (iv) $\sigma_1^2 \neq \sigma_2^2$ or $\sigma_1^2 - \sigma_2^2 \neq 0$ are not simple hypotheses because they specify more than one value as $\mu_1 - \mu_2 = 4$, $\mu_1 - \mu_2 = 7$, $\sigma_1^2 - \sigma_2^2 = 2$, $\sigma_1^2 - \sigma_2^2 = -5$, etc. and (v) $P_1 = P_2$ or $P_1 - P_2 = 0$ is simple hypothesis because it gives a single value of parameter as $P_1 - P_2 = 0$.

9.2.2 Null and Alternative Hypotheses

As we have discussed in last page that in hypothesis testing problems first of all we identify the claim or statement to be tested and write it in symbolical

A hypothesis which completely specifies parameter(s) of a theoretical population (probability distribution) is called a simple hypothesis otherwise called composite hypothesis.

Testing of Hypothesis

We state the null and alternative hypotheses in such a way that they cover all possibility of the value of population parameter.

form. After that we write the complement or opposite of the claim or statement in symbolical form. In our example of motorcycle, the claim is $\mu = 60$ km/liter then its complement is $\mu \neq 60$ km/liter. In (ii) the claim is $\mu > 200$ gm then its complement is $\mu \leq 200$ gm. If the claim is $\mu < 200$ gm then its complement is $\mu \geq 200$ gm. The claim and its complement are formed in such a way that they cover all possibility of the value of population parameter.

Once the claim and its complement have been established then we decide of these two which is the null hypothesis and which is the alternative hypothesis. The thumb rule is that the statement containing equality is the null hypothesis. That is, the hypothesis which contains symbols $=$ or $\leq$ or $\geq$ is taken as null hypothesis and the hypothesis which does not contain equality i.e. contains $\neq$ or $<$ or $>$ is taken as alternative hypothesis. The null hypothesis is denoted by H_0 and alternative hypothesis is denoted by H_1 or H_A .

In our example of motorcycle, the claim is $\mu = 60$ km/liter and its complement is $\mu \neq 60$ km/liter. Since claim $\mu = 60$ km/liter contains equality sign so we take it as a null hypothesis and complement $\mu \neq 60$ km/liter as an alternative hypothesis, that is,

$$H_0: \mu = 60 \text{ km/liter and } H_1: \mu \neq 60 \text{ km/liter}$$

In our second example of banana, the claim is $\mu > 200$ gm and its complement is $\mu \leq 200$ gm. Since complement $\mu \leq 200$ gm contains equality sign so we take complement as a null hypothesis and claim $\mu > 200$ gm as an alternative hypothesis, that is,

$$H_0: \mu \leq 200 \text{ gm and } H_1: \mu > 200 \text{ gm}$$

Formally these hypotheses are defined as

The hypothesis which we wish to test is called as the null hypothesis.

According to Prof. R.A. Fisher,

“A null hypothesis is a hypothesis which is tested for possible rejection under the assumption that it is true.”

The hypothesis which complements to the null hypothesis is called alternative hypothesis.

Note 1: Some authors use equality sign in null hypothesis instead of $\geq$ and $\leq$ signs.

The alternative hypothesis has two types:

- (i) Two-sided (tailed) alternative hypothesis
- (ii) One-sided (tailed) alternative hypothesis

If the alternative hypothesis gives the alternate of null hypothesis in both directions (less than and greater than) of the value of parameter specified in null hypothesis then it is known as two-sided alternative hypothesis and if it gives an alternate only in one direction (less than or greater than) only then it is known as one-sided alternative hypothesis. For example, if our alternative hypothesis is $H_1: \theta \neq 60$ then it is a two-sided alternative hypothesis because it means that the value of parameter θ is greater than or less than 60. Similarly, if $H_1: \theta > 60$ then it is a right-sided alternative hypothesis because it means that the value of parameter θ is greater than 60 and if $H_1: \theta < 60$ then it is a left-sided alternative hypothesis because it means that the value of parameter θ is less than 60.

In testing procedure, we assume that the null hypothesis is true until there is sufficient evidence to prove that it is false. Generally, the hypothesis is tested

with the help of a sample so evidence in testing of hypothesis comes from a sample. If there is enough sample evidence to suggest that the null hypothesis is false then we reject the null hypothesis and support the alternative hypothesis. If the sample fails to provide us sufficient evidence against the null hypothesis we are not saying that the null hypothesis is true because here, we take the decision on the basis of a random sample which is a small part of the population. To say that null hypothesis is true we must study all observations of the population under study. For example, if someone wants to test that the person of India has two hands then to prove that this is true we must check all the persons of India whereas to prove that it is false we require a person he / she has one hand or no hand. So we can only say that there is not enough evidence against the null hypothesis.

Note 2: When we assume that null hypothesis is true then we are actually assuming that the population parameter is equal to the value in the claim. In our example of motorcycle, we assume that $\mu = 60$ km/liter whether the null hypothesis is $\mu = 60$ km/liter or $\mu \leq 60$ km/liter or $\mu \geq 60$ km/liter.

Now, you can try the following exercises.

-
- E1)** A company manufactures car tyres. Company claims that the average life of its tyres is 50000 miles. To test the claim of the company, formulate the null and alternative hypotheses.
- E2)** Write the null and alternative hypotheses in case (iii), (iv) and (v) of our example given in Section 9.2.
- E3)** A businessman of orange formulates different hypotheses about the average weight of the orange which are given below:
- (i) $H_0: \mu = 100$ (ii) $H_1: \mu > 100$ (iii) $H_0: \mu \leq 100$ (iv) $H_1: \mu \neq 100$
(v) $H_1: \mu > 150$ (vi) $H_0: \mu = 130$ (vii) $H_1: \mu \neq \mu_0$

Categorize the above cases into simple and composite hypotheses.

After describing the hypothesis and its types our next point in the testing of hypothesis is critical region which will be described in next section.

9.3 CRITICAL REGION

As we have discussed in Section 9.1 that generally, null hypothesis is tested by the sample data. Suppose $X_1, X_2, \dots, X_n$ be a random sample drawn from a population having unknown population parameter θ . The collection of all possible values of $X_1, X_2, \dots, X_n$ is a set called sample space(S) and a particular value of $X_1, X_2, \dots, X_n$ represents a point in that space.

In order to test a hypothesis, the entire sample space is partitioned into two disjoint sub-spaces, say, ω and $S - \omega = \bar{\omega}$. If calculated value of the test statistic lies in ω , then we reject the null hypothesis and if it lies in $\bar{\omega}$, then we do not reject the null hypothesis. The region ω is called a “**rejection region or critical region**” and the region $\bar{\omega}$ is called a “**non-rejection region**”.

Therefore, we can say that

“A region in the sample space in which if the calculated value of the test statistic lies, we reject the null hypothesis then it is called critical region or rejection region.”

This can be better understood with the help of an example. Suppose, 100 students appear in total 10 papers two of each in English, Physics, Chemistry, Mathematics and Computer Science of a Programme. Suppose the scores in

A statistic is a function of sample observations (not including parameter). Basically, a test statistic is a statistic which is used in decision making about the null hypothesis.

Testing of Hypothesis

these papers are denoted by $X_1, X_2, \dots, X_{10}$ and maximum marks =100 for each paper. For obtaining the distinction award in this Programme, a student needs to have total score equal to or more than 750 which is a rule.

Suppose, we select one student randomly out of 100 students and we want to test that the selected student is a distinction award holder. So we can take the null and alternative hypotheses as

H_0 : Selected student is a distinction award holder

H_1 : Selected student is not a distinction award holder

For taking the decision about the selected student, we define a statistic

$T_{10} = \sum_{i=1}^{10} X_i$ as the sum of the scores in all the 10 papers of the student. The

range of T_{10} is $0 \leq T_{10} \leq 1000$. Now, we divide the whole space (0-1000) into two regions as no-distinction awarded region (less than 750) and distinction awarded region (greater than or equal to 750) as shown in Fig. 9.1. Here, 750 is the critical value which separates the no-distinction and distinction awarded regions.

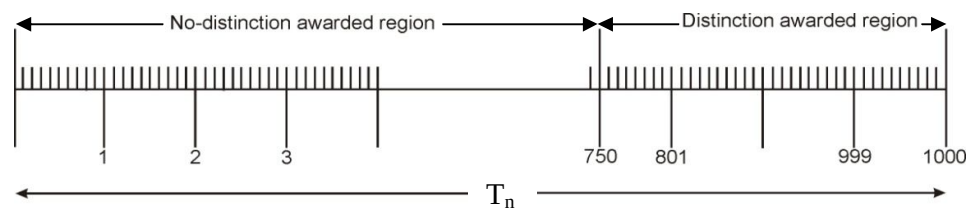


Fig. 9.1: Non-rejection and critical regions for distinction award

On the basis of scores in all the papers of the selected student, we calculate the value of the statistic $T_{10} = \sum_{i=1}^{10} X_i$. And calculated value may fall in distinction awarded region or not, depending upon the observed value of test statistic.

For making a decision to reject or do not reject H_0 , we use test statistic

$T_{10} = \sum_{i=1}^{10} X_i$ (sum of scores of 10 papers). If calculated value of test statistic T_{10}

lies in no-distinction awarded region (critical region), that is, $T_{10} < 750$ then we reject H_0 and if calculated value of test statistic T_{10} lies in distinction awarded region (non-rejection region), that is, $T_{10} \geq 750$ then we do not reject H_0 . It is a basic structure of the procedure of testing of hypothesis which needs two regions like:

(i) Region of rejection of null hypothesis H_0

(ii) Region of non-rejection of null hypothesis H_0

The point of discussion in this test procedure is “**how to fix the cut off value 750**”? What is the justification for this value? The distinction awarded region may be like $T_{10} \geq 800$ or at $T_{10} \geq 850$ or at $T_{10} \geq 900$. So, there must be a scientific justification for the cut-off value 750. In a statistical test procedure it is obtained by using the probability distribution of the test statistic.

The region of rejection is called critical region. It has a pre-fixed area generally denoted by α , corresponding to a cut-off value in a probability distribution of test statistic.

The rejection (critical) region lies in one-tail or two-tails on the probability curve of sampling distribution of the test statistic its depends upon the

alternative hypothesis. Therefore, three cases arise:

Case I: If the alternative hypothesis is right-sided such as $H_1: \theta > \theta_0$ or $H_1: \theta_1 > \theta_2$ then the entire critical or rejection region of size α lies on right tail of the probability curve of sampling distribution of the test statistic as shown in Fig. 9.2.

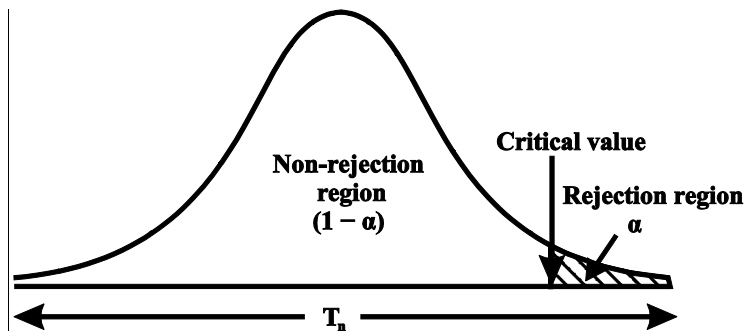


Fig. 9.2

Case II: If the alternative hypothesis is left-sided such as $H_1: \theta < \theta_0$ or $H_1: \theta_1 < \theta_2$ then the entire critical or rejection region of size α lies on left tail of the probability curve of sampling distribution of the test statistic as shown in Fig. 9.3.

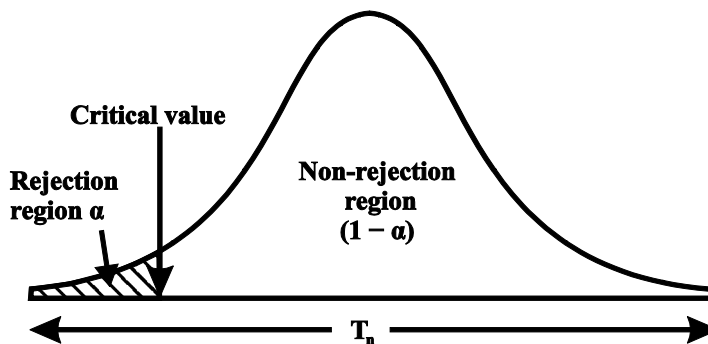


Fig. 9.3

Case III: If the alternative hypothesis is two sided such as $H_1: \theta \neq \theta_0$ or $H_1: \theta_1 \neq \theta_2$ then critical or rejection regions of size $\alpha/2$ lies on both tails of the probability curve of sampling distribution of the test statistic as shown in Fig. 9.4.

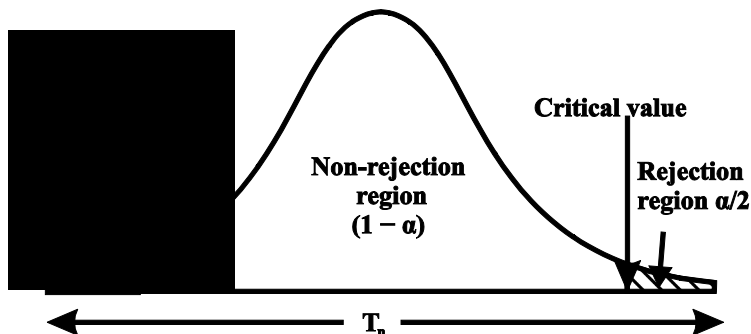


Fig. 9.4

Now, you can try the following exercise.

E4) If $H_0: \theta = 60$ and $H_1: \theta \neq 60$ then critical region lies in one-tail or two-tails.

Critical value is a value or values that separate the region of rejection from the non-rejection region.

9.4 TYPE-I AND TYPE-II ERRORS

In Section 9.3, we have discussed a rule that if the value of test statistic falls in rejection (critical) region then we reject the null hypothesis and if it falls in the non-rejection region then we do not reject the null hypothesis. A test statistic is calculated on the basis of observed sample observations. But a sample is a small part of the population about which decision is to be taken. A random sample may or may not be a good representative of the population.

A faulty sample misleads the inference (or conclusion) relating to the null hypothesis. For example, an engineer infers that a packet of screws is sub-standard when actually it is not. It is an error caused due to poor or inappropriate (faulty) sample. Similarly, a packet of screws may infer good when actually it is sub-standard. So we can commit two kinds of errors while testing a hypothesis which are summarised in Table 9.1 which is given below:

Table 9.1: Type of Errors

Decision	H_0 True	H_1 True
Reject H_0	Type-I Error	Correct Decision
Do not reject H_0	Correct Decision	Type-II Error

Let us take a situation where a patient suffering from high fever reaches to a doctor. And suppose the doctor formulates the null and alternative hypotheses as

H_0 : The patient is a malaria patient

H_1 : The patient is not a malaria patient

Then following cases arise:

Case I: Suppose that the hypothesis H_0 is really true, that is, patient actually a malaria patient and after observation, pathological and clinical examination, the doctor rejects H_0 , that is, he / she declares him / her a non-malaria-patient. It is not a correct decision and he / she commits an error in decision known as type-I error.

Case II: Suppose that the hypothesis H_0 is actually false, that is, patient actually a non-malaria patient and after observation, the doctor rejects H_0 , that is, he / she declares him / her a non-malaria-patient. It is a correct decision.

Case III: Suppose that the hypothesis H_0 is really true, that is, patient actually a malaria patient and after observation, the doctor does not reject H_0 , that is, he / she declares him / her a malaria-patient. It is a correct decision.

Case IV: Suppose that the hypothesis H_0 is actually false, that is, patient actually a non-malaria patient and after observation, the doctor does not reject H_0 , that is, he / she declares him / her a malaria-patient. It is not a correct decision and he / she commits an error in decision known as type-II error.

Thus, we formally define type-I and type-II errors as below:

Type-I Error:

The decision relating to rejection of null hypothesis H_0 when it is true is called type-I error. The probability of committing the type-I error is called size of test, denoted by α and is given by

$$\alpha = P [\text{Reject } H_0 \text{ when } H_0 \text{ is true}] = P [\text{Reject } H_0 / H_0 \text{ is true}]$$

We reject the null hypothesis if random sample / test statistic falls in rejection region, therefore,

$$\alpha = P [X \in \omega / H_0]$$

where $X = (X_1, X_2, \dots, X_n)$ is a random sample and ω is the rejection region and

$$1 - \alpha = 1 - P[\text{Reject } H_0 / H_0 \text{ is true}]$$

$$= P[\text{Do not reject } H_0 / H_0 \text{ is true}] = P[\text{Correct decision}]$$

The $(1 - \alpha)$ is the probability of correct decision and it correlates to the concept of $100(1 - \alpha)\%$ confidence interval used in estimation.

Type-II Error:

The decision relating to non-rejection of null hypothesis H_0 when it is false (i.e. H_1 is true) is called type-II error. The probability of committing type-II error is generally denoted by β and is given by

$$\beta = P[\text{Do not reject } H_0 \text{ when } H_0 \text{ is false}]$$

$$= P[\text{Do not reject } H_0 \text{ when } H_1 \text{ is true}]$$

$$= P[\text{Do not reject } H_0 / H_1 \text{ is true}]$$

$$= P[X \in \bar{\omega} / H_1] \text{ where, } \bar{\omega} \text{ is the non-rejection region.}$$

and

$$1 - \beta = 1 - P[\text{Do not reject } H_0 / H_1 \text{ is true}]$$

$$= P[\text{Reject } H_0 / H_1 \text{ is true}] = P[\text{Correct decision}]$$

The $(1 - \beta)$ is the probability of correct decision and also known as “**power of the test**”. Since it indicates the ability or power of the test to recognize correctly that the null hypothesis is false, therefore, we wish a test that yields a large power.

We say that a statistical test is ideal if it minimizes the probability of both types of errors and maximizes the probability of correct decision. But for fix sample size, α and β are so interrelated that the decrement in one results into the increment in other. So minimization of both probabilities of type-I and type-II errors simultaneously for fixed sample size is not possible without increasing sample size. Also both types of errors will be at zero level (i.e. no error in decision) if size of the sample is equal to the population size. But it involves huge cost if population size is large. And it is not possible in all situations such as testing of blood.

Depending on the problem in hand, we have to choose the type of error which has to minimize. For this, we have to look at a situation, suppose there is a decision making problem and there is a rule that if we make type-I error, we lose 10 rupees and if we make type-II error we lose 1000 rupees. In this case, we try to eliminate the type-II error, since it is more expensive.

In another situation, suppose the Delhi police arrests a person whom they suspect is a murderer. Now, policemen have to test hypothesis:

H_0 : Arrested person is innocent (not murderer)

H_1 : Arrested person is a murderer

The type-I error is

$$\alpha = P [\text{Reject } H_0 \text{ when it is true}]$$

That is, suspected person who is actually an innocent will be sent to jail when H_0 rejects, although H_0 being a true.

The type-II error is

$$\beta = P [\text{Do not reject } H_0 \text{ when } H_1 \text{ is true}]$$

That is, when arrested person truly a murderer but released by the police. Now, we see that in this case type-I error is more serious than type-II error because a murderer may be arrested / punished later on but sending jail to an innocent person is serious.

Consider another situation, suppose we want to test the null hypothesis $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$ on the basis of tossing a coin once, where p is the probability of getting a head in a single toss (trial). And we reject the null hypothesis if a head appears and do not reject otherwise. The type-I error, that is, the probability of Reject H_0 when it is true can be calculated easily (as shown in Example 1) but the computation of type-II error is not possible because there are infinitely many alternatives for p such as $p = 0.6$, $p = 0.1$, etc.

Generally, strong control on α is necessary. It should be kept as low as possible. In test procedure, we prefix it at very low level like $\alpha = 0.05$ (5%) or 0.01 (1%) .

Now, it is time to do some examples relating to α and β .

Example 1: It is desired to test a hypothesis $H_0 : p = p_0 = 1/2$ against the alternative hypothesis $H_1 : p = p_1 = 1/4$ on the basis of tossing a coin once, where p is the probability of “getting a head” in a single toss (trial) and agreeing to reject H_0 if a head appears and accept H_0 otherwise. Find the value of α and β .

Solution: In such type of problems, first of all we search for critical region.

Here, we have critical region $\omega = \{\text{head}\}$

Therefore, the probability of type-I error can be obtained as

$$\begin{aligned}\alpha &= P[\text{Reject } H_0 \text{ when } H_0 \text{ is true}] \\ &= P[X \in \omega / H_0] = P[\text{Head appears} / H_0] \\ &= P[\text{Head appears}]_{p=\frac{1}{2}} = \frac{1}{2} \quad \left[\because H_0 \text{ is true so we take value of parameter } p \text{ given in } H_0 \right]\end{aligned}$$

Also,

$$\begin{aligned}\beta &= P[\text{Do not reject } H_0 \text{ when } H_1 \text{ is true}] \\ &= P[X \notin \omega / H_1] = P[\text{Tail appears} / H_1] \\ &= P[\text{Tail appears}]_{p=\frac{1}{4}} \quad \left[\because H_1 \text{ is true so we take value of parameter } p \text{ given in } H_1 \right] \\ &= 1 - P[\text{Head appears}]_{p=\frac{1}{4}} = 1 - \frac{1}{4} = \frac{3}{4}\end{aligned}$$

Example 2: For testing $H_0 : \theta = 1$ against $H_1 : \theta = 2$, the pdf of the variable is given by

$$f(x, \theta) = \begin{cases} \frac{1}{\theta}; & 0 \leq x \leq \theta \\ 0; & \text{elsewhere} \end{cases}$$

Obtain type-I and type-II errors when critical region is $X \geq 0.4$. Also obtain power function of the test.

Solution: Here, we have critical (rejection) and non-rejection regions as

$$\omega = \{X : X \geq 0.4\} \text{ and } \bar{\omega} = \{X : X < 0.4\}$$

We have to test the null hypothesis

$$H_0 : \theta = 1 \text{ against } H_1 : \theta = 2$$

The size of type-I error is given by

$$\alpha = P[X \in \omega / H_0] = P[X \geq 0.4 / \theta = 1]$$

$$= \left[\int_{0.4}^{\theta} f(x, \theta) dx \right]_{\theta=1} \left[\because P[X \geq a] = \int_a^{\infty} f(x, \theta) dx \right] \quad \dots (1)$$

Now, by using $f(x, \theta) = \frac{1}{\theta}; 0 \leq x \leq \theta$, we get from equation (1)

$$\alpha = \left[\int_{0.4}^{\theta} \frac{1}{\theta} dx \right]_{\theta=1} = \int_{0.4}^1 dx = (x)_{0.4}^1 = 1 - 0.4 = 0.6$$

Similarly, the size of type-II error is given by

$$\beta = P[X \in \bar{\omega} / H_1] = P[X < 0.4 / \theta = 2]$$

$$\beta = \left[\int_0^{0.4} \frac{1}{\theta} dx \right]_{\theta=2} = \int_0^{0.4} \frac{1}{2} dx = \frac{1}{2} (x)_0^{0.4} = \frac{1}{2} (0.4 - 0) = 0.2$$

The power function of the test = $1 - \beta = 1 - 0.2 = 0.8$.

Now, you can try the following exercise.

-
- E5)** An urn contains either 4 white and 2 black balls or 2 white and 4 black balls. Two balls are to be drawn from the urn. If less than two white balls are obtained, it will be decided that this urn contains 2 white and 4 black balls. Calculate the values of α and β .
-

9.5 LEVEL OF SIGNIFICANCE

So far in this unit, we have discussed the hypothesis, types of hypothesis, critical region and types of errors. In this section, we shall discuss very useful concept “**level of significance**”, which play an important role in decision making while testing a hypothesis.

The probability of type-I error is known as level of significance of a test. It is also called the size of the test or size of critical region, denoted by α . Generally, it is pre-fixed as 5% or 1% level ($\alpha = 0.05$ or 0.01). As we have discussed in Section 9.3 that if calculated value of the test statistic lies in rejection(critical) region, then we reject the null hypothesis and if it lies in non-rejection region, then we do not reject the null hypothesis. Also we note that when H_0 is rejected then automatically the alternative hypothesis H_1 is accepted. Now, one point of our discussion is that how to decide critical value(s) or cut-off value(s) for a known test statistic.

If distribution of test statistic could be expressed into some well-known distributions like Z , χ^2 , t , F etc. then our problem will be solved and using the probability distribution of test statistic, we can find the cut-off value(s) that provides us critical area equal to 5% (or 1%).

Another viewpoint about the level of significance relates to the trueness of the conclusion. If H_0 do not reject at level, say, $\alpha = 0.05$ (5% level) then a person will be confident that “concluding statement about H_0 ” is true with 95% assurance. But even then it may false with 5% chance. There is no cent-percent assurance about the trueness of statement made for H_0 .

As an example, if among 100 scientists, each draws a random sample and use the same test statistic to test the same hypothesis H_0 conducting same experiment, then 95 of them will reach to the same conclusion about H_0 . But still 5 of them may differ (i.e. against the earlier conclusion).

Similar argument can be made for, say, $\alpha = 0.01$ (=1%). It is like when H_0 is rejected at $\alpha = 0.01$ by a scientist, then out of 100 similar researchers who work together at the same time for the same problem, but with different random samples, 99 of them would reach to the same conclusion however, one may differ.

Now, you can try the following exercise.

E6) If probability of type-I error is 0.05 then what is the level of significance?

9.6 ONE-TAILED AND TWO-TAILED TESTS

We have seen in Section 9.3 that rejection (critical) region lies at one-tail or two-tails on the probability curve of sampling distribution of the test statistic its depend upon the form of alternative hypothesis. Similarly, the test of testing the null hypothesis also depends on the alternative hypothesis.

A test of testing the null hypothesis is said to be two-tailed test if the alternative hypothesis is two-tailed whereas if the alternative hypothesis is one-tailed then a test of testing the null hypothesis is said to be one-tailed test.

For example, if our null and alternative hypothesis are

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0$$

then the test for testing the null hypothesis is two-tailed test because the alternative hypothesis is two-tailed that means, the parameter θ can take value greater than θ_0 or less than θ_0 .

If the null and alternative hypotheses are

$$H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0$$

then the test for testing the null hypothesis is right-tailed test because the alternative hypothesis is right-tailed.

Similarly, if the null and alternative hypotheses are

$$H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0$$

then the test for testing the null hypothesis is left-tailed test because the alternative hypothesis is left-tailed.

The above discussion can be summarised in Table 9.2.

Table 9.2: Null and Alternative Hypotheses and Corresponding One-tailed and Two-tailed Tests

Null Hypothesis	Alternative Hypothesis	Types of Critical Region / Test
$H_0 : \theta = \theta_0$	$H_1 : \theta \neq \theta_0$	Two-tailed test having critical regions under both tails.
$H_0 : \theta \leq \theta_0$	$H_1 : \theta > \theta_0$	Right-tailed test having critical region under right tail only.
$H_0 : \theta \geq \theta_0$	$H_1 : \theta < \theta_0$	Left- tailed test having critical region under left tail only.

Let us do one example based on type of tests.

Example 3: A company has replaced its original technology of producing electric bulbs by CFL technology. The company manager wants to compare the average life of bulbs manufactured by original technology and new technology CFL. Write appropriate null and alternate hypotheses. Also say about one tailed and two tailed tests.

Solution: Suppose the average lives of original and CFL technology bulbs are denoted by μ_1 and μ_2 respectively.

If company manager is interested just to know whether any significant difference exists in average-life time of two types of bulbs then null and alternative hypotheses will be:

$$H_0: \mu_1 = \mu_2 \quad [\text{average lives of two types of bulbs are same}]$$

$$H_1: \mu_1 \neq \mu_2 \quad [\text{average lives of two types of bulbs are different}]$$

Since alternative hypothesis is two-tailed therefore corresponding test will be two-tailed.

If company manager is interested just to know whether average life of CFL is greater than original technology bulbs then our null and alternative hypotheses will be

$$H_0: \mu_1 \geq \mu_2$$

$$H_1: \mu_1 < \mu_2 \quad \left[\begin{array}{l} \text{average life of CFL technology bulbs} \\ \text{is greater than average life of original technology} \end{array} \right]$$

Since alternative hypothesis is left-tailed therefore corresponding test will be left-tailed test.

Now, you can try the following exercises.

E7) If we have null and alternative hypotheses as

$$H_0: \theta = \theta_0 \text{ and } H_1: \theta \neq \theta_0$$

then corresponding test will be

- (i) left-tailed test (ii) right-tailed test (iii) both-tailed test

Write the correct option.

E8) The test whether one-tailed or two-tailed depends on

- (i) Null hypothesis (H_0) (ii) Alternative hypothesis (H_1)
(iii) Neither H_0 nor H_1 (iv) Both H_0 and H_1
-

9.7 GENERAL PROCEDURE OF TESTING A HYPOTHESIS

Testing of hypothesis is a huge demanded statistical tool by many discipline and professionals. It is a step by step procedure as you will see in next three units through a large number of examples. The aim of this section is just give you flavor of that sequence which involves following steps:

Step I: First of all, we have to setup null hypothesis H_0 and alternative hypothesis H_1 . Suppose, we want to test the hypothetical / claimed /

assumed value θ_0 of parameter θ . So we can take the null and alternative hypotheses as

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0 \\ H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0 \end{array} \right\} \quad [\text{for one-tailed test}]$$

In case of comparing same parameter of two populations of interest, say, θ_1 and θ_2 , then our null and alternative hypotheses would be

$$H_0 : \theta_1 = \theta_2 \text{ and } H_1 : \theta_1 \neq \theta_2 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta_1 \leq \theta_2 \text{ and } H_1 : \theta_1 > \theta_2 \\ H_0 : \theta_1 \geq \theta_2 \text{ and } H_1 : \theta_1 < \theta_2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

Step II: After setting the null and alternative hypotheses, we establish a criteria for rejection or non-rejection of null hypothesis, that is, decide the level of significance (α), at which we want to test our hypothesis. Generally, it is taken as 5% or 1% ($\alpha = 0.05$ or 0.01).

Step III: The third step is to choose an appropriate test statistic under H_0 for testing the null hypothesis as given below:

$$\text{Test statistic} = \frac{\text{Statistic} - \text{Value of the parameter under } H_0}{\text{Standard error of statistic}}$$

After that, specify the sampling distribution of the test statistic preferably in the standard form like Z (standard normal), χ^2 , t, F or any other well-known in literature.

Step IV: Calculate the value of the test statistic described in Step III on the basis of observed sample observations.

Step V: Obtain the critical (or cut-off) value(s) in the sampling distribution of the test statistic and construct rejection (critical) region of size α . Generally, critical values for various levels of significance are putted in the form of a table for various standard sampling distributions of test statistic such as Z-table, χ^2 -table, t-table, etc.

Step VI: After that, compare the calculated value of test statistic obtained from Step IV, with the critical value(s) obtained in Step V and locates the position of the calculated test statistic, that is, it lies in rejection region or non-rejection region.

Step VII: In testing of hypothesis ultimately we have to reach at a conclusion. It is done as explained below:

- (i) If calculated value of test statistic lies in rejection region at α level of significance then we reject null hypothesis. It means that the sample data provide us sufficient evidence against the null hypothesis and there is a significant difference between hypothesized value and observed value of the parameter.
- (ii) If calculated value of test statistic lies in non-rejection region at α level of significance then we do not reject null hypothesis. It means that the sample data fails to provide us sufficient evidence against the null hypothesis and the difference between hypothesized value and observed value of the parameter due to fluctuation of sample.

Note 3: Nowadays the decision about the null hypothesis is taken with the help of p-value. The concept of p-value is very important, because computer packages and statistical software such as SPSS, SAS, STATA, MINITAB, EXCEL, etc. all provide p-value. So, Section 9.8 is devoted to explain the concept of p-value.

Now, with the help of an example we explain the above procedure.

Example 4: Suppose, it is found that average weight of a potato was 50 gm and standard deviation was 5.1 gm nearly 5 years ago. We want to test that due to advancement in agricultural technology, the average weight of a potato has been increased. To test this, a random sample of 50 potatoes is taken and calculate the sample mean ($\bar{X}$) as 52gm. Describe the procedure to carry out this test.

Solution: Here, we are given that

Specified value of population mean = $\mu_0 = 50$ gm,

Population standard deviation = $\sigma = 5.1$ gm,

Sample size = $n = 50$,

Sample mean = $\bar{X} = 52$ gm

To carry out the above test, we have to follow up the following steps:

Step I: First of all, we setup null and alternative hypotheses. Here, we want to test that the average weight of potato is increased. So our claim is “average weight of potato has increased” i.e. $\mu > 50$ and its complement is $\mu \leq 50$. Since complement contains equality sign so we can take the complement as the null hypothesis and claim as the alternative hypothesis, that is,

$$H_0: \mu \leq 50 \text{ gm and } H_1: \mu > 50 \text{ gm [Here, } \theta = \mu \text{]}$$

Since the alternative hypothesis is right-tailed, so our test is right-tailed.

Step II: After setting the null and alternative hypotheses, we fix level of significance α . Suppose, $\alpha = 0.01$ (= 1 % level).

Step III: Define a test statistic to test the null hypothesis as

$$\text{Test statistic} = \frac{\text{Statistic} - \text{Value of the parameter under } H_0}{\text{Standard error of statistic}}$$

$$T = \frac{\bar{X} - 50}{\sigma / \sqrt{n}}$$

Since sample size is large ($n = 50 > 30$) so by central limit theorem the sampling distribution of test statistic approximately follows standard normal distribution (as explained in Unit 1 of this course), i.e. $T \sim N(0,1)$

Step IV: Calculate the value of test statistic on the basis of sample observations as

$$T = \frac{52 - 50}{5.1 / \sqrt{50}} = \frac{2}{0.72} = 2.78$$

Step V: Now, we find the critical value. The critical value or cut-off value for standard normal distribution is given in **Table I (Z-table)** in the Appendix at the end of Block 1 of this course. So from this table, the critical value for right-tailed test at $\alpha = 0.01$ is $z_\alpha = 2.33$.

Testing of Hypothesis

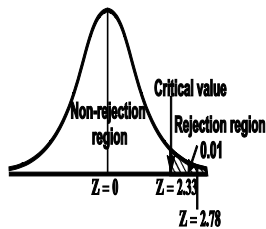


Fig. 9.5

Step IV: Now, to take the decision about the null hypothesis, we compare the calculated value of test statistic with the critical value.

Since calculated value of test statistic (= 2.78) is greater than critical value (= 2.33), that means calculated value of test statistic lies in rejection region at 1% level of significance as shown in Fig. 9.5. So we reject null hypothesis and support the alternative hypothesis. Since alternative hypothesis is our claim, so we support the claim.

Thus, we conclude that sample does not provide us sufficient evidence against the claim so we may assume that the average weight of potato has increased.

Now, you can try the following exercise.

E9) What is the first step in testing of hypothesis?

9.8 CONCEPT OF p-VALUE

In Note 3 of Section 9.6, we promised that p-value will be discussed in Section 9.8. So, it is the time to keep our promise. Nowadays use of p-value is becoming more and more popular because of the following two reasons:

- most of the statistical software provides p-value rather than critical value.
- p-value provides more information compared to critical value as far as rejection or do not rejection of H_0 .

The first point listed above needs no explanation. But second point lies in the heart of p-value and needs to explain more clearly. Moving in this direction, we note that in scientific applications one is not only interested simply in rejecting or not rejecting the null hypothesis but he/she is also interested to assess how strong the data has the evidence to reject H_0 . For example, as we have seen in Example 4 based on general procedure of testing a hypothesis where we tested the null hypothesis

$$H_0: \theta \leq 50 \text{ gm against } H_1: \theta > 50 \text{ gm}$$

To test the null hypothesis, we calculated the value of test statistic as 2.78 and the critical value (z_α) at $\alpha = 0.01$ was $z_\alpha = 2.33$.

Since calculated value of test statistic (= 2.78) is greater than critical (tabulated) value (= 2.33), therefore, we reject the null hypothesis at 1% level of significance.

Now, if we reject the null hypothesis at this level (1%) surely we have to reject it at higher level because at $\alpha = 0.05$, $z_\alpha = 1.645$ and at $\alpha = 0.10$, $z_\alpha = 1.28$. However, the calculated value of test statistic is much higher than 1.645 and 1.28, therefore, the question arises “could the null hypothesis also be rejected at values of α smaller than 0.01?” The answer is “yes” and we can compute the smallest level of significance (α) at which a null hypothesis can be rejected. This smallest level of significance (α) is known as “p-value”.

The p-value is the smallest value of level of significance(α) at which a null hypothesis can be rejected using the obtained value of the test statistic and can be defined as:

The p-value is the probability of obtaining a test statistic equal to or more extreme (in the direction of sporting H_1) than the actual value obtained when null hypothesis is true.

The p-value also depends on the type of the test. If test is one-tailed then the p-value is defined as:

For right-tailed test:

$$p\text{-value} = P[\text{Test Statistic (T)} \geq \text{observed value of the test statistic}]$$

For left-tailed test:

$$p\text{-value} = P[\text{Test Statistic (T)} \leq \text{observed value of the test statistic}]$$

If test is two-tailed then the p-value is defined as:

For two-tailed test:

$$p\text{-value} = 2P[T \geq |\text{observed value of the test statistic}|]$$

Procedure of taking the decision about the null hypothesis on the basis of p-value:

To take the decision about the null hypothesis based on p-value, the p-value is compared with level of significance (α) and if p-value is equal or less than α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

The p-value for various tests can be obtained with the help of the tables given in the Appendix of the Block 1 of this course. But unless we are dealing with the standard normal distribution, the exact p-value is not obtained with the tables as mentioned above. But if we test our hypothesis with the help of computer packages or softwares such as SPSS, SAS, MINITAB, STATA, EXCEL, etc. then these types of computer packages or softwares present the p-value as part of the output for each hypothesis testing procedure. Therefore, in this block we will also describe the procedure to take the decision about the null hypothesis on the basis of critical value as well as p-value concepts.

9.9 RELATION BETWEEN CONFIDENCE INTERVAL AND TESTING OF HYPOTHESIS

In Units 7 and 8 of this course, we have learned about confidence intervals which were used to estimate the unknown population parameters with certain confidence. In Section 9.7, we have discussed the general procedure of testing a hypothesis which has been used in making decision about the specified/ assumed/ hypothetical values of population parameters. Both confidence interval and hypothesis testing have been used for different purposes but have been based on the same set of concepts. Therefore, there is an extremely close relationship between confidence interval and hypothesis testing.

In confidence interval, if we construct $(1-\alpha)$ 100% confidence interval for an unknown parameter then this interval contained all probable values for the parameter being estimated and relatively improbable values are not contained by this interval.

So this concept can also be used in hypothesis testing. For this, we construct an appropriate $(1-\alpha)$ 100% confidence interval for the parameter specified by the null hypothesis as we have discussed in Units 7 and 8 of this course and if the value of the parameter specified by the null hypothesis lies in this confident interval then we do not reject the null hypothesis and if this specified value does not lie in this confidence interval then we reject the null hypothesis.

Therefore, three cases may arise:

Case I: If we want to test the null hypothesis $H_0: \theta = \theta_0$ against the alternative hypothesis $H_1: \theta \neq \theta_0$ at 5% or 1% level of significance then we construct two-sided $(1 - \alpha)100\% = 95\%$ or 99% confidence interval for the parameter θ . And we have 95% or 99% (as may be the case) confidence that this interval will include the parameter value θ_0 . If the value of the parameter specified by the null hypothesis i.e. θ_0 lies in this confidence interval then we do not reject the null hypothesis otherwise we reject the null hypothesis.

Case II: If we want to test the null hypothesis $H_0: \theta \leq \theta_0$ against the alternative hypothesis $H_1: \theta > \theta_0$ then we construct the lower one-sided confidence bound for parameter θ . If the value of the parameter specified by the null hypothesis i.e. θ_0 is greater than or equal to this lower bound then we do not reject the null hypothesis otherwise we reject the null hypothesis.

Case III: If we want to test the null hypothesis $H_0: \theta \geq \theta_0$ against the alternative hypothesis $H_1: \theta < \theta_0$ then we construct the upper one-sided confidence bound for parameter θ . If the value of the parameter specified by the null hypothesis i.e. θ_0 is less than or equal to this upper bound then we do not reject the null hypothesis otherwise we reject the null hypothesis.

For example, referring back to Example 4 of this unit, here we want to test the null hypothesis

$$H_0 : \mu \leq 50 \text{ gm against } H_1 : \mu > 50 \text{ gm}$$

This was tested with the help of test statistic

$$T = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \quad [\text{Here, } \theta = \mu]$$

and we reject the null hypothesis at $\alpha = 0.01$.

This problem could also have been solved by obtaining confidence interval estimate of population mean which is described in Section 7.4 of Unit 7.

Here, we are given that

$$n = 50, \bar{X} = 50 \text{ and } \sigma = 5.1$$

Since alternative hypothesis is right-tailed, therefore, we construct lower one-sided confidence bound for population mean.

Since population variance is known so lower one-sided $(1 - \alpha) 100\%$ confidence bound for population mean when population variance is known is given by

$$\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

Since we test our null hypothesis at $\alpha = 0.01$ therefore, we contract 99% lower confidence bound and for $\alpha = 0.01$, we have $z_{\alpha} = z_{0.01} = 2.33$.

Thus, lower one-sided 99% confidence bound for average weight of potatoes is

$$52 - 2.33 \frac{5.1}{\sqrt{50}} = 52 - 1.68 = 50.32$$

Since the value of the parameter specified by the null hypothesis i.e. $\mu = 50$ is less than lower bound for average weight of potato so we reject the null

hypothesis.

Thus, we can use three approaches (critical value, p-value and confidence interval) for taking decision about null hypothesis.

With this we end this unit. Let us summarise what we have discussed in this unit.

9.10 SUMMARY

In this unit, we have covered the following points:

1. Statistical hypothesis, null hypothesis, alternative hypothesis, simple & composite hypotheses.
2. Type-I and Type-II errors.
3. Critical region.
4. One-tailed and two-tailed tests.
5. General procedure of testing a hypothesis.
6. Level of significance.
7. Concept of p-value.
8. Relation between confidence interval and testing of hypothesis.

9.11 SOLUTIONS / ANSWERS

- E1)** Here, we wish to test the claim of the company that the average life of its car tyres is 50000 miles so

Claim: $\mu = 50000$ miles and complement: $\mu \neq 50000$ miles

Since claim contains equality sign so we take claim as the null hypothesis and complement as the alternative hypothesis i.e.

$$H_0: \mu = 50000 \text{ miles}$$

$$H_1: \mu \neq 50000 \text{ miles}$$

- E2)** (iii) Here, doctor wants to test whether new medicine is really more effective for controlling blood pressure than old medicine so

Claim: $\mu_1 > \mu_2$ and complement: $\mu_1 \leq \mu_2$

Since complement contains equality sign so we take complement as the null hypothesis and claim as the alternative hypothesis i.e.

$$H_0: \mu_1 \leq \mu_2$$

$$H_1: \mu_1 > \mu_2$$

- (iv) Here, economist wants to test whether the variability in incomes differ in two populations so

Claim: $\sigma_1^2 \neq \sigma_2^2$ and complement: $\sigma_1^2 = \sigma_2^2$

Since complement contains equality sign so we take complement as the null hypothesis and claim as the alternative hypothesis i.e.

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Testing of Hypothesis

(v) Here, psychologist wants to test whether the proportion of literates between two groups of people is same so

Claim: $P_1 = P_2$ and complement: $P_1 \neq P_2$

Since claim contains equality sign so we take claim as the null hypothesis and complement as the alternative hypothesis i.e.

$$H_0: P_1 = P_2$$

$$H_1: P_1 \neq P_2$$

- E3)** Here, (i) and (vi) represent the simple hypotheses because these hypotheses tell us the exact values of parameter average weight of orange μ as $\mu = 100$ and $\mu = 130$.

The rest (ii), (iii), (iv), (v) and (vii) represent the composite hypotheses because these hypotheses do not tell us the exact values of parameter μ .

- E4)** Since alternative hypothesis $H_1: \theta \neq 60$ is two tailed so critical region lies in two-tails.

- E5)** Let A and B denote the number of white balls and black balls in the urn respectively. Further, let X be the number of white balls drawn among the two balls from the urn then we can take the null and alternative hypotheses as

$$H_0: A = 4 \text{ \& } B = 2 \text{ and } H_1: A = 2 \text{ \& } B = 4$$

The critical region is given by

$$w = \{X : X < 2\} = \{X : X = 0, 1\}$$

Thus,

$$\alpha = P [\text{Reject } H_0 \text{ when } H_0 \text{ is true}]$$

$$= P[X \in w / H_0] = P[X = 0 / H_0] + P[X = 1 / H_0]$$

$$= \frac{{}^4C_0 {}^2C_0}{{}^6C_2} + \frac{{}^4C_1 {}^2C_1}{{}^6C_2} = \frac{1 \times 1}{15} + \frac{4 \times 2}{15} = \frac{1}{15} + \frac{8}{15}$$

$$\alpha = \frac{9}{15} = \frac{3}{5}$$

Similarly,

$$\beta = P [\text{Do not reject } H_0 \text{ when } H_1 \text{ is true}]$$

$$= P[X \notin w / H_1] = P[X = 2 / H_1] = \frac{{}^2C_2 {}^4C_0}{{}^6C_2} = \frac{1 \times 1}{15} = \frac{1}{15}$$

- E6)** Since level of significance is the probability of type-I error so in this case level of significance is 0.05 or 5%.

- E7)** Here, the alternative hypothesis is two-tailed therefore, the test will be two-tailed test.

- E8)** Whether the test of testing a hypothesis is one-tailed or two-tailed depends on the alternative hypothesis. So correct option is (ii).

- E9)** First step in testing of hypothesis is to setup null and alternative hypotheses.

UNIT 10 LARGE SAMPLE TESTS

Structure

- 10.1 Introduction
 - Objectives
- 10.2 Procedure of Testing of Hypothesis for Large Samples
- 10.3 Testing of Hypothesis for Population Mean Using Z-Test
- 10.4 Testing of Hypothesis for Difference of Two Population Means Using Z-Test
- 10.5 Testing of Hypothesis for Population Proportion Using Z-Test
- 10.6 Testing of Hypothesis for Difference of Two Population Proportions Using Z-Test
- 10.7 Testing of Hypothesis for Population Variance Using Z-Test
- 10.8 Testing of Hypothesis for Two Population Variances Using Z-Test
- 10.9 Summary
- 10.10 Solutions /Answers

10.1 INTRODUCTION

In previous unit, we have defined basic terms used in testing of hypothesis. After providing you necessary material required for any test, we can move towards discussing particular tests one by one. But before doing that let us tell you the strategy we are adopting here.

First we categorise the tests under two heads:

- Large sample tests
- Small sample tests

After that, their unit wise distribution is done. In this unit, we will discuss large sample tests whereas in Units 11 and 12 we will discuss small sample tests. The tests which are described in these units are known as “**parametric tests**”.

Sometimes in our studies in the fields of economics, psychology, medical, etc. we take a sample of objects / units / participants / patients, etc. such as 70, 500, 1000, 10,000, etc. This situation comes under the category of large samples.

As a thumb rule, a sample of size n is treated as a large sample only if it contains more than 30 units (or observations, $n > 30$). And we know that, for large sample ($n > 30$), one statistical fact is that almost all sampling distributions of the statistic(s) are closely approximated by the normal distribution. Therefore, the test statistic, which is a function of sample observations based on $n > 30$, could be assumed follow the normal distribution approximately (or exactly).

But story does not end here. There are some other issues which need to be taken care of. Some of these issues have been highlighted by making different cases in each test as you will see when go through Sections 10.3 to 10.8 of this unit.

This unit is divided into ten sections. Section 10.1 is introductory in nature. General procedure of testing of hypothesis for large samples is described in

Section 10.2. In Section 10.3, testing of hypothesis for population mean is discussed whereas in Section 10.4, testing of hypothesis for difference of two population means with examples is described. Similarly, in Sections 10.5 and 10.6, testing of hypothesis for population proportion and difference of two population proportions are explained respectively. Testing of hypothesis for population variance and two population variances are described in Sections 10.7 and 10.8 respectively. Unit ends by providing summary of what we have discussed in this unit in Section 10.9 and solution of exercises in Section 10.10.

Objectives

After studying this unit, you should be able to:

- judge for a given situation whether we should go for large sample test or not;
- Applying the Z-test for testing the hypothesis about the population mean and difference of two population means;
- Applying the Z-test for testing the hypothesis about the population proportion and difference of two population proportions; and
- Applying the Z-test for testing the hypothesis about the population variance and two population variances.

10.2 PROCEDURE OF TESTING OF HYPOTHESIS FOR LARGE SAMPLES

As we have described in previous section that for large sample size ($n > 30$), one statistical fact is that almost all sampling distributions of the statistic(s) are closely approximated by the normal distribution. Therefore, when sample size is large one can apply the normal distribution based test procedures to test the hypothesis.

In previous unit, we have given the procedure of testing of hypothesis in general. Let us now discuss the procedure of testing a hypothesis for large samples in particular.

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size $n (> 30)$ selected from a population having unknown parameter θ and we want to test the hypothesis about the hypothetical / claimed / assumed value θ_0 of parameter θ . For this, a test procedure is required. We discuss it step by step as follows:

Step I: First of all, we have to setup null hypothesis H_0 and alternative hypothesis H_1 . Here, we want to test the hypothetical / claimed / assumed value θ_0 of parameter θ . So we can take the null and alternative hypotheses as

$$\begin{array}{ll} H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0 & \text{[for two-tailed test]} \\ \text{or} & \\ \left. \begin{array}{l} H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0 \\ H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0 \end{array} \right\} & \text{[for one-tailed test]} \end{array}$$

In case of comparing same parameter of two populations of interest, say, θ_1 and θ_2 , then our null and alternative hypotheses would be

$$H_0 : \theta_1 = \theta_2 \text{ and } H_1 : \theta_1 \neq \theta_2 \quad \text{[for two-tailed test]}$$

$$\text{or } \left. \begin{array}{l} H_0 : \theta_1 \leq \theta_2 \text{ and } H_1 : \theta_1 > \theta_2 \\ H_0 : \theta_1 \geq \theta_2 \text{ and } H_1 : \theta_1 < \theta_2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

Step II: After setting the null and alternative hypotheses, we have to choose level of significance. Generally, it is taken as 5% or 1% ($\alpha = 0.05$ or 0.01). And accordingly rejection and non-rejection regions will be decided.

Step III: Third step is to determine an appropriate test statistic, say, Z in case of large samples. Suppose T_n is the sample statistic such as sample mean, sample proportion, sample variance, etc. for the parameter θ then for testing the null hypothesis, test statistic is given by

$$Z = \frac{T_n - E(T_n)}{SE(T_n)} = \frac{T_n - E(T_n)}{\sqrt{\text{Var}(T_n)}} \quad \left[\begin{array}{l} \because \text{we know that SE of a statistic is} \\ \text{the SD of the sampling distribution} \\ \text{of that statistic} \\ \therefore SE(T_n) = SD(T_n) = \sqrt{\text{Var}(T_n)} \end{array} \right]$$

where, $E(T_n)$ is the expectation (or mean) of T_n and $\text{Var}(T_n)$ is variance of T_n .

Step IV: As already mentioned for large samples, statistical fact is that almost all sampling distributions of the statistic(s) are closely approximated by the normal distribution as the parent population is normal or non-normal. So, the test statistic Z will assumed to be approximately normally distributed with mean 0 and variance 1 as

$$Z = \frac{T_n - E(T_n)}{\sqrt{\text{Var}(T_n)}} \sim N(0,1)$$

By putting the values of T_n , $E(T_n)$ and $\text{Var}(T_n)$ in above formula we calculate the value of test statistic Z . Let z be the calculated value of test statistic Z .

Step V: After that, we obtain the critical (cut-off or tabulated) value(s) in the sampling distribution of the test statistic Z corresponding to α assumed in Step II. These critical values are given in **Table-I (Z-table)** at the Appendix of Block 1 of this course corresponding to different level of significance (α). For convenient some useful critical values at $\alpha = 0.01, 0.05$ for Z -test are given in **Table 10.1** in this section. After that, we construct rejection (critical) region of size α in the probability curve of the sampling distribution of test statistic Z .

Step VI: Take the decision about the null hypothesis based on the calculated and critical values of test statistic obtained in Step IV and Step V. Since critical value depends upon the nature of the test that it is one-tailed test or two-tailed test so following cases arise:

In case of one-tailed test:

Case I: When $H_0 : \theta \leq \theta_0$ and $H_1 : \theta > \theta_0$ (right-tailed test)

In this case, the rejection (critical) region falls under the right tail of the probability curve of the sampling distribution of test statistic Z . Suppose z_α is the critical value at α level of significance so entire region greater than or equal to z_α is the rejection region and less than z_α is the non-rejection region as shown in Fig. 10.1.

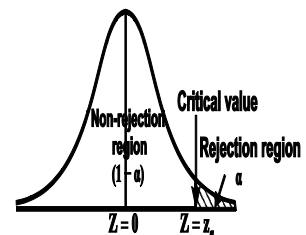


Fig. 10.1

Testing of Hypothesis

If z (calculated value) $\geq z_\alpha$ (tabulated value), that means the calculated value of test statistic Z lies in the rejection region, then we reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that sample data provides us sufficient evidence against the null hypothesis and there is a significant difference between hypothesized or specified value and observed value of the parameter.

If $z < z_\alpha$, that means the calculated value of test statistic Z lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that the sample data fails to provide us sufficient evidence against the null hypothesis and the difference between hypothesized value and observed value of the parameter due to fluctuation of sample.

so the population parameter θ may be θ_0 .

Case II: When $H_0 : \theta \geq \theta_0$ and $H_1 : \theta < \theta_0$ (left-tailed test)

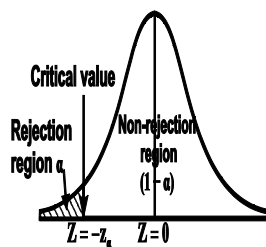


Fig. 10.2

In this case, the rejection (critical) region falls under the left tail of the probability curve of the sampling distribution of test statistic Z . Suppose $-z_\alpha$ is the critical value at α level of significance then entire region less than or equal to $-z_\alpha$ is the rejection region and greater than $-z_\alpha$ is the non-rejection region as shown in Fig. 10.2.

If $z \leq -z_\alpha$, that means the calculated value of test statistic Z lies in the rejection region, then we reject the null hypothesis H_0 at α level of significance.

If $z > -z_\alpha$, that means the calculated value of test statistic Z lies in the non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

In case of two-tailed test: When $H_0 : \theta = \theta_0$ and $H_1 : \theta \neq \theta_0$

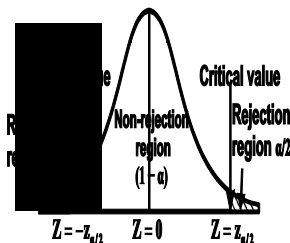


Fig. 10.3

In this case, the rejection region falls under both tails of the probability curve of sampling distribution of the test statistic Z . Half the area (α) i.e. $\alpha/2$ will lie under left tail and other half under the right tail. Suppose $-z_{\alpha/2}$ and $z_{\alpha/2}$ are the two critical values at the left-tailed and right-tailed respectively. Therefore, entire region less than or equal to $-z_{\alpha/2}$ and greater than or equal to $z_{\alpha/2}$ are the rejection regions and between $-z_{\alpha/2}$ and $z_{\alpha/2}$ is the non-rejection region as shown in Fig. 10.3.

If $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$, that means the calculated value of test statistic Z lies in the rejection region, then we reject the null hypothesis H_0 at α level of significance.

If $-z_{\alpha/2} < z < z_{\alpha/2}$, that means the calculated value of test statistic Z lies in the non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Table 10.1 shows some commonly used critical (cut-off or tabulated) values for one-tailed test and two-tailed test at different level of significance (α) for Z -test.

Table 10.1: Critical Values for Z-test

Level of Significance (α)	Two-Tailed Test	One-Tailed Test	
		Right-Tailed Test	Left- Tailed Test
$\alpha = 0.05$ (= 5%)	$\pm z_{\alpha/2} = \pm 1.96$	$z_{\alpha} = 1.645$	$-z_{\alpha} = -1.645$
$\alpha = 0.01$ (= 1%)	$\pm z_{\alpha/2} = \pm 2.58$	$z_{\alpha} = 2.33$	$-z_{\alpha} = -2.33$

Note 1: As we have discussed in Step IV of this procedure that when sample size is large then test statistic follows the normal distribution as the parent population is normal or non-normal so we do not require any assumption of the form of the parent population for large sample size but when sample size is small ($n < 30$) then for applying parametric test we must require the assumption that the population is normal as we shall in Units 11 and 12. If this assumption is not fulfilled then we apply the non-parametric tests which will be discussed in Block 4 of this course.

Decision making procedure about the null hypothesis using the concept of p-value:

To take the decision about the null hypothesis on the basis of p-value, the p-value is compared with given level of significance (α) and if p-value is less than or equal to α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

Since test statistic Z follows approximately normal distribution with mean 0 and variance unity, i.e. standard normal distribution and we also know that standard normal distribution is symmetrical about $Z = 0$ line therefore, if z represents the calculated value of Z then p-value can be calculated as follows:

For one-tailed test:

For $H_1: \theta > \theta_0$ (right-tailed test)

$$\text{p-value} = P[Z \geq z]$$

For $H_1: \theta < \theta_0$ (left-tailed test)

$$\text{p-value} = P[Z \leq z]$$

For two-tailed test:

For $H_1: \theta \neq \theta_0$

$$\text{p-value} = 2P[Z \geq |z|]$$

These p-values for Z-test can be obtained with the help of **Table-I (Z-table)** given in the Appendix at the end of Block 1 of this course (which gives the probability $[0 \leq Z \leq z]$ for different value of z) as discussed in Unit 14 of MST-003.

For example, if test is right-tailed and calculated value of test statistic Z is 1.23 then

$$\begin{aligned}
 \text{p-value} &= P[Z \geq z] = P[Z \geq 1.23] = 0.5 - P[0 < Z < 1.23] \\
 &= 0.5 - 0.3907 \left[\begin{array}{l} \text{From Z-table given in Appendix} \\ \text{of Block 1 of this course.} \end{array} \right] \\
 &= 0.1093
 \end{aligned}$$

Now, you can try the following exercises.

E1) If an investigator observed that the calculated value of test statistic lies in non-rejection region then he/she will

- (i) reject the null hypothesis
- (ii) accept the null hypothesis
- (iii) not reject the null hypothesis

Write the correct option.

E2) If we have null and alternative hypotheses as

$$H_0: \theta = \theta_0 \text{ and } H_1: \theta \neq \theta_0$$

then the rejection (critical) region lies in

- (i) left tail
- (ii) right tail
- (iii) both tails

Write the correct option.

E3) If test is two-tailed and calculated value of test statistic Z is 2.42 then calculate the p-value for the Z -test.

10.3 TESTING OF HYPOTHESIS FOR POPULATION MEAN USING Z-TEST

In previous section, we have discussed the general procedure for Z -test. Now we are discussing the Z -test for testing the hypothesis or claim about the population mean when sample size is large. Let population under study has mean μ and variance σ^2 , where μ is unknown and σ^2 may be known or unknown. We will consider both cases under this heading. For testing a hypothesis about population mean we draw a random sample $X_1, X_2, \dots, X_n$ of size $n > 30$ from this population. As we know that for drawing the inference about the population mean we generally use sample mean and for test statistic, we require the mean and standard error of sampling distribution of the statistic (mean). Here, we are considering large sample so we know by central limit theorem that sample mean is asymptotically normally distributed with mean μ and variance σ^2/n whether parent population is **normal or non-normal**. That is, if $\bar{X}$ is the sample mean of the random sample then

$$E(\bar{X}) = \mu, \text{ Var}(\bar{X}) = \frac{\sigma^2}{n} \quad \dots (1)$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore \text{SE}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}} \quad \dots (2)$$

Now, follow the same procedure as we have discussed in previous section, that is, first of all we have to setup null and alternative hypotheses. Since here we want to test the hypothesis about the population mean so we can take the null and alternative hypotheses as

$$H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0 \left[\text{for two-tailed test} \right] \left[\begin{array}{l} \text{Here, } \theta = \mu \text{ and } \theta_0 = \mu_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right]$$

or

$$\left. \begin{array}{l} H_0 : \mu \leq \mu_0 \text{ and } H_1 : \mu > \mu_0 \\ H_0 : \mu \geq \mu_0 \text{ and } H_1 : \mu < \mu_0 \end{array} \right\} \left[\text{for one-tailed test} \right]$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{\bar{X} - E(\bar{X})}{SE(\bar{X})}$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \quad \left[\begin{array}{l} \text{Using equations (1) and (2) and} \\ \text{under } H_0 \text{ we assume that } \mu = \mu_0. \end{array} \right]$$

The sampling distribution of the test statistic depends upon σ^2 that it is known or unknown. Therefore, two cases arise:

Case I: When σ^2 is known

In this case, the test statistic follows the normal distribution with mean 0 and variance unity when the sample size is the large as the population under study is normal or non-normal. If the sample size is small then test statistic Z follows the normal distribution only when population under study is normal. Thus,

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

Case II: When σ^2 is unknown

In this case, we estimate σ^2 by the value of sample variance (S^2) where,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Then become test statistic $\frac{\bar{X} - \mu_0}{S / \sqrt{n}}$ follows the t-distribution with

(n-1) df as the sample size is large or small provided the population under study follows normal as we have discussed in Unit 2 of this course. But when population under study is not normal and sample size is large then this test statistic approximately follows normal distribution with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \sim N(0, 1)$$

After that, we calculate the value of test statistic as may be the case (σ^2 is known or unknown) and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in the previous section.

From above discussion of testing of hypothesis about population mean, we note following point:

- (i) When σ^2 is known then we apply the Z-test as the population under study is normal or non-normal for the large sample. But when sample size is

When we assume that the null hypothesis is true then we are actually assuming that the population parameter is equal to the value in the null hypothesis. For example, we assume that $\mu = 60$ whether the null hypothesis is $\mu = 60$ or $\mu \leq 60$ or $\mu \geq 60$.

Testing of Hypothesis

small then we apply the Z-test only when population under study is normal.

- (ii) When σ^2 is unknown then we apply the t-test only when the population under study is normal as sample size is large or small. But when the assumption of normality is not fulfilled and sample size is large then we can apply the Z-test.
- (iii) When sample is small and σ^2 is known or unknown and the form of the population is not known then we apply the non-parametric test as we will be discussed in Block 4 of this course.

Following examples will help you to understand the procedure more clearly.

Example 1: A light bulb company claims that the 100-watt light bulb it sells has an average life of 1200 hours with a standard deviation of 100 hours. For testing the claim 50 new bulbs were selected randomly and allowed to burn out. The average lifetime of these bulbs was found to be 1180 hours. Is the company's claim is true at 5% level of significance?

Solution: Here, we are given that

Specified value of population mean = $\mu_0 = 1200$ hours,

Population standard deviation = $\sigma = 100$ hours,

Sample size = $n = 50$

Sample mean = $\bar{X} = 1180$ hours.

In this example, the population parameter being tested is population mean i.e. average life of a bulb (μ) and we want to test the company's claim that average life of a bulb is 1200 hours. So our claim is $\mu = 1200$ and its complement is $\mu \neq 1200$. Since claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. So

$$H_0 : \mu = \mu_0 = 1200 \text{ [average life of a bulb is 1200 hours]}$$

$$H_1 : \mu \neq 1200 \text{ [average life of a bulb is not 1200 hours]}$$

Also the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding mean when population SD (variance) is known and sample size $n = 50 (> 30)$ is large. So we will go for Z-test.

Thus, for testing the null hypothesis the test statistic is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \\ &= \frac{1180 - 1200}{100 / \sqrt{50}} = \frac{-20}{14.14} = -1.41 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of test statistic $Z (= -1.41)$ is greater than critical value ($= -1.96$) and less than the critical value ($= 1.96$), that means it lies in non-rejection region as shown in Fig. 10.4, so we do not reject the null hypothesis. Since the null hypothesis is the claim so we support the claim at 5% level of significance.

Decision according to p-value:

The test is two-tailed, therefore,

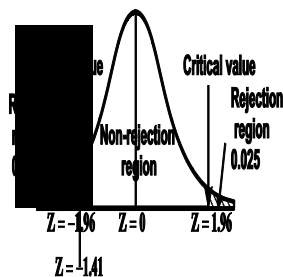


Fig. 10.4

$$p\text{-value} = 2P[Z \geq |z|] = 2P[Z \geq 1.41]$$

$$= 2[0.5 - P[0 \leq Z \leq 1.41]] = 2(0.5 - 0.4207) = 0.1586$$

Since p-value (= 0.1586) is greater than α (= 0.05) so we do not reject the null hypothesis at 5% level of significance.

Decision according to confidence interval:

Here, test is two-tailed, therefore, we contract two-sided confidence interval for population mean.

Since population standard deviation is known, therefore, we can use $(1-\alpha)$ 100 % confidence interval for population mean when population variance is known which is given by

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

Here, $\alpha = 0.05$, we have $z_{\alpha/2} = z_{0.025} = 1.96$.

Thus, 95% confidence interval for average life of a bulb is given by

$$\left[1180 - 1.96 \frac{100}{\sqrt{50}}, 1180 + 1.96 \frac{100}{\sqrt{50}} \right]$$

or $[1180 - 27.71, 1180 + 27.71]$

or $[1152.29, 1207.71]$

Since 95% confidence interval for average life of a bulb contains the value of the parameter specified by the null hypothesis, that is, $\mu = \mu_0 = 1200$ so we do not reject the null hypothesis.

Thus, we conclude that sample does not provide us sufficient evidence against the claim so we may assume that the company's claim that the average life of a bulb is 1200 hours is true.

Note 2: Here, we note that the decisions about null hypothesis based on three approaches (critical value or classical, p-value and confidence interval) are same. The learners are advised to make the decision about the claim or statement by using only one of the three approaches in the examination. Here, we used all these approaches only to give you an idea how they can be used in a given problem. Those learners who will opt biostatistics specialisation will see and realize the importance of confidence interval approach in Unit 16 of MSTE-004.

Example 2: A manufacturer of ball point pens claims that a certain pen manufactured by him has a mean writing-life at least 460 A-4 size pages. A purchasing agent selects a sample of 100 pens and put them on the test. The mean writing-life of the sample found 453 A-4 size pages with standard deviation 25 A-4 size pages. Should the purchasing agent reject the manufacturer's claim at 1% level of significance?

Solution: Here, we are given that

Specified value of population mean = $\mu_0 = 460$,

Testing of Hypothesis

Sample size = $n = 100$,

Sample mean = $\bar{X} = 453$,

Sample standard deviation = $S = 25$

Here, we want to test the manufacturer's claim that the mean writing-life (μ) of pen is at least 460 A-4 size pages. So our claim is $\mu \geq 460$ and its complement is $\mu < 460$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. So

$$H_0 : \mu \geq \mu_0 = 460 \text{ and } H_1 : \mu < 460$$

Also the alternative hypothesis is left-tailed so the test is left-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. So we should use t-test for if writing-life of pen follows normal distribution. But it is not the case. Since sample size is $n = 100$ ($n > 30$) large so we go for Z-test. The test statistic of Z-test is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \\ &= \frac{453 - 460}{25 / \sqrt{100}} = \frac{-7}{2.5} = -2.8 \end{aligned}$$

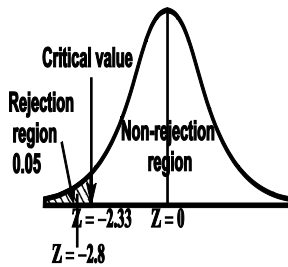


Fig. 10.5

We have from Table 10.1 the critical (tabulated) value for left-tailed Z test at 1% level of significance is $z_\alpha = -2.33$.

Since calculated value of test statistic $Z (= -2.8)$ is less than the critical value ($= -2.33$), that means calculated value of test statistic Z lies in rejection region as shown in Fig. 10.5, so we reject the null hypothesis. Since the null hypothesis is the claim so we reject the manufacturer's claim at 1% level of significance.

Decision according to p-value:

The test is left-tailed, therefore,

$$p\text{-value} = P[Z \leq z] = P[Z \leq -2.8] = P[Z \geq 2.8] \quad \left[\because Z \text{ is symmetrical about } Z = 0 \text{ line} \right]$$

$$= 0.5 - P[0 \leq Z \leq 2.8] = 0.5 - 0.4974 = 0.0026$$

Since p-value ($= 0.0026$) is less than $\alpha (= 0.01)$ so we reject the null hypothesis at 1% level of significance.

Therefore, we conclude that the sample provide us sufficient evidence against the claim so the purchasing agent rejects the manufacturer's claim at 1% level of significance.

Now, you can try the following exercises.

-
- E4)** A sample of 900 bolts has a mean length 3.4 cm. Is the sample regarded to be taken from a large population of bolts with mean length 3.25 cm and standard deviation 2.61 cm at 5% level of significance?
- E5)** A big company uses thousands of CFL lights every year. The brand that the company has been using in the past has average life of 1200 hours. A new brand is offered to the company at a price lower than they are paying

for the old brand. Consequently, a sample of 100 CFL light of new brand is tested which yields an average life of 1220 hours with standard deviation 90 hours. Should the company accept the new brand at 5% level of significance?

10.4 TESTING OF HYPOTHESIS FOR DIFFERENCE OF TWO POPULATION MEANS USING Z-TEST

In previous section, we have learnt about the testing of hypothesis about the population mean. But there are so many situations where we want to test the hypothesis about difference of two population means or two population means. For example, two manufacturing companies of bulbs are produced same type of bulbs and one may be interested to test that one is better than the other, an investigator may want to test the equality of the average incomes of the peoples living in two cities, etc. Therefore, we require an appropriate test for testing the hypothesis about the difference of two population means.

Let there be two populations, say, population-I and population-II under study. Also let μ_1, μ_2 and σ_1^2, σ_2^2 denote the means and variances of population-I and population-II respectively where both μ_1 and μ_2 are unknown but σ_1^2 and σ_2^2 may be known or unknown. We will consider all possible cases here. For testing the hypothesis about the difference of two population means, we draw a random sample of large size n_1 from population-I and a random sample of large size n_2 from population-II. Let $\bar{X}$ and $\bar{Y}$ be the means of the samples selected from population-I and II respectively.

These two populations may or may not be normal but according to the central limit theorem, the sampling distribution of difference of two large sample means asymptotically normally distributed with mean $(\mu_1 - \mu_2)$ and variance $(\sigma_1^2/n_1 + \sigma_2^2/n_2)$ as described in Unit 2 of this course.

Thus,

$$E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 \quad \dots (3)$$

and

$$\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore SE(\bar{X} - \bar{Y}) = \sqrt{\text{Var}(\bar{X} - \bar{Y})} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \dots (4)$$

Now, follow the same procedure as we have discussed in Section 10.2, that is, first of all we have to setup null and alternative hypotheses. Here, we want to test the hypothesis about the difference of two population means so we can take the null hypothesis as

$$H_0 : \mu_1 = \mu_2 \text{ (no difference in means) } \left[\begin{array}{l} \text{Here, } \theta_1 = \mu_1 \text{ and } \theta_2 = \mu_2 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right]$$

or $H_0 : \mu_1 - \mu_2 = 0$ (difference in two means is 0)

and the alternative hypothesis as

$$H_1 : \mu_1 \neq \mu_2 \quad \text{[for two-tailed test]}$$

$$\text{or } \left. \begin{array}{l} H_0 : \mu_1 \leq \mu_2 \text{ and } H_1 : \mu_1 > \mu_2 \\ H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2 \end{array} \right\} \quad \text{[for one-tailed test]}$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{(\bar{X} - \bar{Y}) - E(\bar{X} - \bar{Y})}{SE(\bar{X} - \bar{Y})}$$

$$\text{or } Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \text{[using equations (3) and (4)]}$$

Since under null hypothesis we assume that $\mu_1 = \mu_2$, therefore, we have

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Now, the sampling distribution of the test statistic depends upon σ_1^2 and σ_2^2 that both are known or unknown. Therefore, four cases arise:

Case I: When σ_1^2 & σ_2^2 are known and $\sigma_1^2 = \sigma_2^2 = \sigma^2$

In this case, the test statistic follows normal distribution with mean 0 and variance unity when the sample sizes are large as both the populations under study are normal or non-normal. But when sample sizes are small then test statistic Z follows normal distribution only when populations under study are normal, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, 1)$$

Case II: When σ_1^2 & σ_2^2 are known and $\sigma_1^2 \neq \sigma_2^2$

In this case, the test statistic also follows the normal distribution as described in case I, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Case III: When σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 = \sigma_2^2 = \sigma^2$

In this case, σ^2 is estimated by value of pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [n_1 S_1^2 + n_2 S_2^2]$$

and

$$S_1^2 = \frac{1}{(n_1 - 1)} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \text{ and } S_2^2 = \frac{1}{(n_2 - 1)} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

and test statistic follows t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom as the sample sizes are large or small provided populations under study follow normal distribution as described in Unit 2 of this course. But when the populations under study are not normal and sample sizes n_1 and n_2 are large (> 30) then by central limit theorem, test statistic approximately normally distributed with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, 1)$$

Case IV: When σ_1^2 & σ_2^2 are unknown and $\sigma_1^2 \neq \sigma_2^2$

In this case, σ_1^2 & σ_2^2 are estimated by the values of the sample variances S_1^2 & S_2^2 respectively and the exact distribution of test statistic is difficult to derive. But when sample sizes n_1 and n_2 are large (> 30) then central limit theorem, the test statistic approximately normally distributed with mean 0 and variance unity, that is,

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0, 1)$$

After that, we calculate the value of test statistic and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2.

From above discussion of testing of hypothesis about population mean, we note following point:

- (i) When σ_1^2 & σ_2^2 are known then we apply the Z-test as both the population under study are normal or non-normal for the large sample. But when sample sizes are small then we apply the Z-test only when populations under study are normal.
- (ii) When σ_1^2 & σ_2^2 are unknown then we apply the t-test only when the populations under study are normal as sample sizes are large or small. But when the assumption of normality is not fulfilled and sample sizes are large then we can apply the Z-test.
- (iii) When samples are small and σ_1^2 & σ_2^2 are known or unknown and the form of the population is not known then we apply the non-parametric test as we will be discussed in Block 4 of this course.

Let us do some examples based on above test.

Example 3: In two samples of women from Punjab and Tamilnadu, the mean height of 1000 and 2000 women are 67.6 and 68.0 inches respectively. If population standard deviation of Punjab and Tamilnadu are same and equal to 5.5 inches then, can the mean heights of Punjab and Tamilnadu women be regarded as same at 1% level of significance?

Testing of Hypothesis

Solution: We are given

$$n_1 = 1000, n_2 = 2000, \bar{X} = 67.6, \bar{Y} = 68.0 \text{ and } \sigma_1 = \sigma_2 = \sigma = 5.5$$

Here, we wish to test that the mean height of Punjab and Tamilnadu women is same. If μ_1 and μ_2 denote the mean heights of Punjab and Tamilnadu women respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding two population means. The standard deviations of both populations are known and sample sizes are large, so we should go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\ &= \frac{67.6 - 68.0}{\sqrt{\left(\frac{(5.5)^2}{1000} + \frac{(5.5)^2}{2000}\right)}} = \frac{-0.4}{5.5 \sqrt{\left(\frac{1}{1000} + \frac{1}{2000}\right)}} \\ &= \frac{-0.4}{5.5 \times 0.0387} = -1.88 \end{aligned}$$

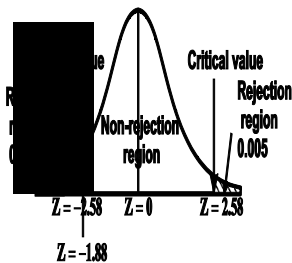


Fig. 10.6

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of Z ($= -1.88$) is greater than the critical value ($= -2.58$) and less than the critical value ($= 2.58$), that means it lies in non-rejection region as shown in Fig. 10.6, so we do not reject the null hypothesis i.e. we fail to reject the claim.

Decision according to p-value:

The test is two-tailed, therefore,

$$p\text{-value} = 2P[Z \geq |z|] = 2P[Z \geq 1.88]$$

$$= 2[0.5 - P[0 \leq Z \leq 1.88]] = 2(0.5 - 0.4699) = 0.0602$$

Since p-value ($= 0.0602$) is greater than $\alpha (= 0.01)$ so we do not reject the null hypothesis at 1% level of significance.

Thus, we conclude that the samples do not provide us sufficient evidence against the claim so we may assume that the average height of women of Punjab and Tamilnadu is same.

Example 4: A university conducts both face to face and distance mode classes for a particular course intended both to be identical. A sample of 50 students of face to face mode yields examination results mean and SD respectively as:

$$\bar{X} = 80.4, \quad S_1 = 12.8$$

and other sample of 100 distance-mode students yields mean and SD of their examination results in the same course respectively as:

$$\bar{Y} = 74.3, \quad S_2 = 20.5$$

Are both educational methods statistically equal at 5% level?

Solution: Here, we are given that

$$n_1 = 50, \quad \bar{X} = 80.4, \quad S_1 = 12.8;$$

$$n_2 = 100, \quad \bar{Y} = 74.3, \quad S_2 = 20.5$$

We wish to test that both educational methods are statistically equal. If μ_1 and μ_2 denote the average marks of face to face and distance mode students respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

We want to test the null hypothesis regarding two population means when standard deviations of both populations are unknown. So we should go for t-test if population of difference is known to be normal. But it is not the case. Since sample sizes are large (n_1 , and $n_2 > 30$) so we go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \\ &= \frac{80.4 - 74.3}{\sqrt{\frac{(12.8)^2}{50} + \frac{(20.5)^2}{100}}} = \frac{6.1}{\sqrt{3.28 + 4.20}} = 2.23 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= 2.23) is greater than the critical values ($= \pm 1.96$), that means it lies in rejection region as shown in Fig. 10.7, so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Decision according to p-value:

The test is two-tailed, therefore,

$$\begin{aligned} \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 2.23] \\ &= 2[0.5 - P[0 \leq Z \leq 2.23]] = 2(0.5 - 0.4871) = 0.0258 \end{aligned}$$

Since p-value (= 0.0258) is less than $\alpha (= 0.05)$ so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so both methods of education, i.e. face-to-face and distance-mode, are not statistically equal.

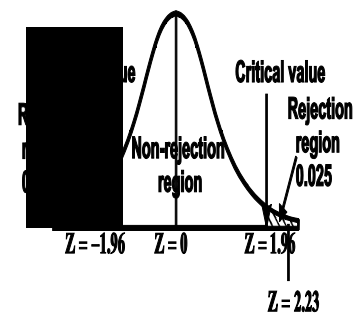


Fig. 10.7

Now, you can try the following exercises.

- E6)** Two brands of electric bulbs are quoted at the same price. A buyer was tested a random sample of 200 bulbs of each brand and found the following information:

	Mean Life (hrs.)	SD(hrs.)
Brand A	1300	41
Brand B	1280	46

Is there significant difference in the mean duration of their lives of two brands of electric bulbs at 1% level of significance?

- E7)** Two research laboratories have identically produced drugs that provide relief to BP patients. The first drug was tested on a group of 50 BP patients and produced an average 8.3 hours of relief with a standard deviation of 1.2 hours. The second drug was tested on 100 patients, producing an average of 8.0 hours of relief with a standard deviation of 1.5 hours. Does the first drug provide a significant longer period of relief at a significant level of 5%?

10.5 TESTING OF HYPOTHESIS FOR POPULATION PROPORTION USING Z-TEST

In Section 10.3, we have discussed the procedure of testing of hypothesis for population mean when sample size is large. But in many real world situations, in business and other areas where collected data are in form of counts or the collected data are classified into two categories or groups according to an attribute or a characteristic. For example, the peoples living in a colony may be classified into two groups (male and female) with respect to the characteristic sex, the patients in a hospital may be classified into two groups as cancer and non-cancer patients, the lot of articles may be classified as defective and non-defective, etc. Here, collected data are available in dichotomous or binary outcomes which is a special case of nominal scale and the data categorized into two mutually exclusive and exhaustive classes generally known as success and failure out comes. For example, the characteristic sex can be measured as success if male and failure if female or vice versa. So in such situations, proportion is suitable measure to apply.

In such situations, we require a test for testing a hypothesis about population proportion.

For this purpose, let $X_1, X_2, \dots, X_n$ be a random sample of size n taken from a population with population proportion P . Also let X denotes the number of observations or elements possess a certain attribute (number of successes) out of n observations of the sample then sample proportion p can be defined as

$$p = \frac{X}{n} \leq 1$$

As we have seen in Section 2.4 of the Unit 2 of this course that mean and variance of the sampling distribution of sample proportion are

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n}$$

where, $Q = 1 - P$.

Now, two cases arise:

Case I: When sample size is not sufficiently large i.e. either of the conditions $np > 5$ or $nq > 5$ does not meet, then we use exact binomial test. But exact binomial test is beyond the scope of this course.

Case II: When sample size is sufficiently large, such that $np > 5$ and $nq > 5$ then by central limit theorem, the sampling distribution of sample proportion p is approximately normally distributed with mean and variance as

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n} \quad \dots (5)$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore \text{SE}(p) = \sqrt{\frac{PQ}{n}} \quad \dots (6)$$

Now, follow the same procedure as we have discussed in Section 10.2, first of all we setup null and alternative hypotheses. Since here we want to test the hypothesis about specified value P_0 of the population proportion so we can take the null and alternative hypotheses as

$$H_0 : P = P_0 \text{ and } H_1 : P \neq P_0 \left[\text{for two-tailed test} \right] \left[\begin{array}{l} \text{Here, } \theta = P \text{ and } \theta_0 = P_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : P \leq P_0 \text{ and } H_1 : P > P_0 \\ H_0 : P \geq P_0 \text{ and } H_1 : P < P_0 \end{array} \right\} \left[\text{for one-tailed test} \right]$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{p - E(p)}{\text{SE}(p)}$$

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \sim N(0, 1) \text{ under } H_0 \left[\text{using equations (5) and (6)} \right]$$

After that, we calculate the value of test statistic and compare it with the critical value(s) given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2.

Let us do some examples of testing of hypothesis about population proportion.

Example 5: A machine produces a large number of items out of which 25% are found to be defective. To check this, company manager takes a random sample of 100 items and found 35 items defective. Is there an evidence of more deterioration of quality at 5% level of significance?

Solution: The company manager wants to check that his machine produces 25% defective items. Here, attribute under study is defectiveness. And we define our success and failure as getting a defective or non defective item.

Let P = Population proportion of defectives items = $0.25 (= P_0)$

p = Observed proportion of defectives items in the sample = $35/100 = 0.35$

Here, we want to test that machine produces more defective items, that is, the proportion of defective items (P) greater than 0.25. So our claim is $P > 0.25$

Testing of Hypothesis

and its complement is $P \leq 0.25$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. So

$$H_0 : P \leq P_0 = 0.25 \text{ and } H_1 : P > 0.25$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore np = 100 \times 0.35 = 35 > 5$$

$$nq = 100 \times (1 - 0.35) = 100 \times 0.65 = 65 > 5$$

We see that condition of normality meets, so we can go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\ &= \frac{0.35 - 0.25}{\sqrt{\frac{0.25 \times 0.75}{100}}} = \frac{0.10}{0.0433} = 2.31 \end{aligned}$$

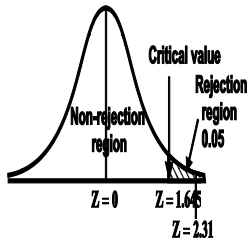


Fig. 10.8

Since test is right-tailed so the critical value at 5% level of significance is $z_{\alpha} = z_{0.05} = 1.645$.

Since calculated value of test statistic Z (= 2.31) is greater than the critical value (= 1.645), that means it lies in the rejection region as shown in Fig. 10.8. So we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Decision according to p-value:

The test is right-tailed, therefore,

$$\begin{aligned} \text{p-value} &= P[Z \geq z] = P[Z \geq 2.31] \\ &= 0.5 - P[0 < Z < 2.31] = 0.5 - 0.4896 \\ &= 0.0104 \end{aligned}$$

Since p-value (= 0.0104) is less than α (= 0.05) so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that deterioration in quality exists at 5% level of significance.

Example 6: A die is thrown 9000 times and draw of 2 or 5 is observed 3100 times. Can we regard that die is unbiased at 5% level of significance.

Solution: Let getting a 2 or 5 be our success, and getting a number other than 2 or 5 be a failure then in usual notions, we have

$$n = 9000, X = \text{number of successes} = 3100, p = 3100/9000 = 0.3444$$

Here, we want to test that the die is unbiased and we know that if die is unbiased then proportion or probability of getting 2 or 5 is

$$\begin{aligned} P &= \text{Probability of getting a 2 or 5} \\ &= \text{Probability of getting 2} + \text{Probability of getting 5} \\ &= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} = 0.3333 \end{aligned}$$

So our claim is $P = 0.3333$ and its complement is $P \neq 0.3333$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : P = P_0 = 0.3333 \quad \text{and} \quad H_1 : P \neq 0.3333$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore np = 9000 \times 0.3444 = 3099.6 > 5$$

$$nq = 9000 \times (1 - 0.3444) = 9000 \times 0.6556 = 5900.4 > 5$$

We see that condition of normality meets, so we can go for Z-test.

So, for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\ &= \frac{0.3444 - 0.3333}{\sqrt{\frac{0.3333 \times 0.6667}{9000}}} = \frac{0.0111}{0.005} = 2.22 \end{aligned}$$

Since test is two-tailed so the critical values at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= 2.22) is greater than the critical value (= 1.96), that means it lies in rejection region, so we reject the null hypothesis i.e. we reject our claim.

Decision according to p-value:

Since test is two-tailed so

$$\begin{aligned} \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 2.22] \\ &= 2[0.5 - P[0 < Z < 2.22]] = 2(0.5 - 0.4868) = 0.0264 \end{aligned}$$

Since p-value (= 0.0264) is less than α (= 0.05), so we reject the null hypothesis at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so die cannot be considered as unbiased.

Now, you can try the following exercises.

E8) In a sample of 100 MSc. Economics first year students of a University, it was seen that 54 students came from Science background and the rest are

from other background. Can we assume that 50% of the students are from Science background in MSc. Economics first year students in the University at 1% level of significance?

- E9)** Out of 200 patients who are given a particular injection 180 survived. Test the hypothesis that the survival rate is more than 80% at 5% level of significance?

10.6 TESTING OF HYPOTHESIS FOR DIFFERENCE OF TWO POPULATION PROPORTIONS USING Z-TEST

In Section 10.5, we have discussed the testing of hypothesis about the population proportion. In some cases, we are interested to test the hypothesis about difference of two population proportions of an attributes in the two different populations or groups. For example, one may wish to test whether the proportions of alcohol drinkers in the two cities are same, one may wish to test proportion of literates in a group of people is greater than the proportion of literates in other group of people, etc. Therefore, we require the test for testing the hypothesis about the difference of two population proportions.

Let there be two populations, say, population-I and population-II under study. And also let we draw a random sample of size n_1 from population-I with population proportion P_1 and a random sample of size n_2 from population-II with population proportion P_2 . If X_1 and X_2 are the number of observations / individuals / items / units possessing the given attribute in the sample of sizes n_1 and n_2 respectively then sample proportions can be defined as

$$p_1 = \frac{X_1}{n_1} \text{ and } p_2 = \frac{X_2}{n_2}$$

As we have seen in Section 2.5 of the Unit 2 of this course that mean and variance of the sampling distribution of difference of sample proportions are

$$E(p_1 - p_2) = P_1 - P_2$$

and variance

$$\text{Var}(p_1 - p_2) = \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}$$

where, $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$.

Now, two cases arise:

Case I: When sample sizes are not sufficiently large i.e. any of the conditions $n_1 p_1 > 5$ or $n_1 q_1 > 5$ or $n_2 p_2 > 5$ or $n_2 q_2 > 5$ does not meet, then we use exact binomial test. But exact binomial test is beyond the scope of this course.

Case II: When sample sizes are sufficiently large, such that $n_1 p_1 > 5$, $n_1 q_1 > 5$, $n_2 p_2 > 5$ and $n_2 q_2 > 5$ then by central limit theorem, the sampling distribution of sample proportions p_1 and p_2 are approximately normally as

$$p_1 \sim N\left(P_1, \frac{P_1 Q_1}{n_1}\right) \text{ and } p_2 \sim N\left(P_2, \frac{P_2 Q_2}{n_2}\right)$$

Also, by the property of normal distribution described in Unit 13 of MST-003, the sampling distribution of the difference of sample proportions follows normal distribution with mean

$$E(p_1 - p_2) = E(p_1) - E(p_2) = P_1 - P_2 \quad \dots (7)$$

and variance

$$\text{Var}(p_1 - p_2) = \text{Var}(p_1) + \text{Var}(p_2) = \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}$$

That is,

$$p_1 - p_2 \sim N\left(P_1 - P_2, \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}\right)$$

Thus, standard error is given by

$$\text{SE}(p_1 - p_2) = \sqrt{\text{Var}(p_1 - p_2)} = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}} \quad \dots (8)$$

Now, follow the same procedure as we have discussed in Section 10.2, first of all we have to setup null and alternative hypotheses. Here, we want to test the hypothesis about the difference of two population proportions so we can take the null hypothesis as

$$H_0 : P_1 = P_2 \text{ (no difference in proportions)} \left[\begin{array}{l} \text{Here, } \theta_1 = P_1 \text{ and} \\ \theta_2 = P_2 \text{ if we compare} \\ \text{it with general} \\ \text{procedure.} \end{array} \right]$$

or $H_0 : P_1 - P_2 = 0$ (difference in two proportions is 0)

and the alternative hypothesis may be

$$\begin{array}{ll} H_1 : P_1 \neq P_2 & \text{[for two-tailed test]} \\ \text{or} & \\ \left. \begin{array}{l} H_0 : P_1 \leq P_2 \text{ and } H_1 : P_1 > P_2 \\ H_0 : P_1 \geq P_2 \text{ and } H_1 : P_1 < P_2 \end{array} \right\} & \text{[for one-tailed test]} \end{array}$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{(p_1 - p_2) - E(p_1 - p_2)}{\text{SE}(p_1 - p_2)}$$

$$\text{or } Z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} \quad \text{[using equations (7) and (8)]}$$

Since under null hypothesis we assume that $P_1 = P_2 = P$, therefore, we have

$$Z = \frac{P_1 - P_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where, $Q = 1 - P$.

Generally, P is unknown then it is estimated by the value of pooled proportion $\hat{P}$, where

$$\hat{P} = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2} \text{ and } \hat{Q} = 1 - \hat{P}$$

After that, we calculate the value of test statistic and compare it with the critical value(s) given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2.

Now, it is time for doing some examples for testing of hypothesis about the difference of two population proportions.

Example 7: In a random sample of 100 persons from town A, 60 are found to be high consumers of wheat. In another sample of 80 persons from town B, 40 are found to be high consumers of wheat. Do these data reveal a significant difference between the proportions of high wheat consumers in town A and town B (at $\alpha = 0.05$)?

Solution: Here, attribute under study is high consuming of wheat. And we define our success and failure as getting a person of high consumer of wheat and not high consumer of wheat respectively.

We are given that

n_1 = total number of persons in the sample of town A = 100

n_2 = total number of persons in the sample of town B = 80

X_1 = number of persons of high consumer of wheat in town A = 60

X_2 = number of persons of high consumer of wheat in town B = 40

The sample proportion of high wheat consumers in town A is

$$p_1 = \frac{X_1}{n_1} = \frac{60}{100} = 0.60$$

and the sample proportion of wheat consumers in town B is

$$p_2 = \frac{X_2}{n_2} = \frac{40}{80} = 0.50$$

Here, we want to test that the proportion of high consumers of wheat in two towns, say, P_1 and P_2 , is not same. So our claim is $P_1 \neq P_2$ and its complement is $P_1 = P_2$. Since the complement contains the equality sign, so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : P_1 = P_2 = P \text{ and } H_1 : P_1 \neq P_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore n_1p_1 = 100 \times 0.60 = 60 > 5, n_1q_1 = 100 \times 0.40 = 40 > 5$$

$$n_2p_2 = 80 \times 0.50 = 40 > 5, n_2q_2 = 80 \times 0.50 = 40 > 5$$

We see that condition of normality meets, so we can go for Z-test.

The estimate of the combined proportion (P) of high wheat consumers in two towns is given by

$$\hat{p} = \frac{n_1p_1 + n_2p_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{60 + 40}{100 + 80} = \frac{5}{9}$$

$$\hat{q} = 1 - \hat{p} = 1 - \frac{5}{9} = \frac{4}{9}$$

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{0.60 - 0.50}{\sqrt{\frac{5}{9} \times \frac{4}{9} \left(\frac{1}{100} + \frac{1}{80}\right)}} = \frac{0.10}{0.0745} = 1.34 \end{aligned}$$

The critical values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2}$
 $= \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (=1.34) is less than the critical value (= 1.96) and greater than critical value (= -1.96), that means calculated value of Z lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim.

Decision according to p-value:

Since the test is two-tailed, therefore

$$\begin{aligned} \text{p-value} &= 2 P[Z \geq z] = 2P[Z \geq 1.34] \\ &= 2[0.5 - P[0 \leq Z \leq 1.34]] = 2(0.5 - 0.4099) = 0.1802 \end{aligned}$$

Since p-value (= 0.1802) is greater than α (= 0.05) so we do not reject the null hypothesis at 5% level of significance.

Thus, we conclude that the samples provide us the sufficient evidence against the claim so we may assume that the proportion of high consumers of wheat in two towns A and B is same.

Example 8: A machine produced 60 defective articles in a batch of 400. After overhauling it produced 30 defective in a batch of 300. Has the machine improved due to overhauling? (Take $\alpha = 0.01$).

Solution: Here, the machine produced articles and attribute under study is defectiveness. And we define our success and failure as getting a defective or non defective article. Therefore, we are given that

X_1 = number of defective articles produced by the machine before overhauling
 $= 60$

X_2 = number of defective articles produced by the machine after overhauling
 $= 30$

Testing of Hypothesis

and $n_1 = 400$, $n_2 = 300$,

Let p_1 = Observed proportion of defective articles in the sample before the overhauling

$$= \frac{X_1}{n_1} = \frac{60}{400} = 0.15$$

and p_2 = Observed proportion of defective articles in the sample after the overhauling

$$= \frac{X_2}{n_2} = \frac{30}{300} = 0.10$$

Here, we want to test that machine improved due to overhauling that means the proportion of defective articles is less after overhauling. If P_1 and P_2 denote the proportion defectives before and after the overhauling the machine so our claim is $P_1 > P_2$ and its complement $P_1 \leq P_2$. Since the complement contains the equality sign so we can take the complement as the null hypothesis and claim as the alternative hypothesis. Thus,

$$H_0 : P_1 \leq P_2 \text{ and } H_1 : P_1 > P_2$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Since P is unknown, so the pooled estimate of proportion is given by

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{60 + 30}{400 + 300} = \frac{90}{700} = \frac{9}{70} \text{ and } \hat{q} = 1 - \hat{p} = 1 - \frac{9}{70} = \frac{61}{70}.$$

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore n_1 p_1 = 400 \times 0.15 = 60 > 5, n_1 q_1 = 400 \times 0.85 = 340 > 5$$

$$n_2 p_2 = 300 \times 0.10 = 30 > 5, n_2 q_2 = 300 \times 0.90 = 270 > 5$$

We see that condition of normality meets, so we can go for Z-test.

For testing the null hypothesis, the statistic is given by

$$\begin{aligned} Z &= \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{0.15 - 0.10}{\sqrt{\frac{9}{70} \times \frac{61}{70} \left(\frac{1}{400} + \frac{1}{300}\right)}} = \frac{0.05}{0.0256} = 1.95 \end{aligned}$$

The critical value for right-tailed test at 1% level of significance is $z_\alpha = z_{0.01} = 2.33$.

Since calculated value of Z ($= 1.95$) is less than the critical value ($= 2.33$) that means calculated value of Z lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Decision according to p-value:

Since the test is right-tailed, therefore,

$$p\text{-value} = P[Z \geq z] = P[Z \geq 1.95]$$

$$= 0.5 - P[0 \leq Z \leq 1.95] = 0.5 - 0.4744 = 0.0256$$

Since p-value (= 0.0256) is greater than α (= 0.01) so we do not reject the null hypothesis at 1% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so the machine has not been improved after overhauling.

Now, you can try the following exercises.

-
- E10)** The proportions of literates between groups of people of two districts A and B are tested. Out of the 100 persons selected at random from each district, 50 from district A and 40 from district B are found literates. Test whether the proportion of literate persons in two districts A and B is same at 1% level of significance?
- E11)** In a large population 30% of a random sample of 1200 persons had blue-eyes and 20% of a random sample of 900 persons had the same blue-eyes in another population. Test the proportion of blue-eyes persons is same in two populations at 5% level of significance.
-

10.7 TESTING OF HYPOTHESIS FOR POPULATION VARIANCE USING Z-TEST

In Section 10.3, we have discussed testing of hypothesis for population mean but when analysing quantitative data, it is often important to draw conclusion about the average as well as the variability of a characteristic of under study. For example, a company manufactured the electric bulbs and the manager of the company would probably be interested in determining the average life of the bulbs and also determining whether or not the variability in the life of bulbs is within acceptable limits, the product controller of a milk company may be interested to know variance of the amount of fat in the whole milk processed by the company is no more than the specified level, etc. So we require a test for this purpose.

The procedure of testing a hypothesis for population variance or standard deviation is similar to the testing of population mean.

For testing a hypothesis about the population variance, we draw a random sample $X_1, X_2, \dots, X_n$ of size $n > 30$ from the population with mean μ and variance σ^2 where, μ be known or unknown.

We know that by central limit theorem that sample variance is asymptotically normally distributed with mean σ^2 and variance $2\sigma^4/n$ whether parent population is **normal or non-normal**. That is, if S^2 is the sample variance of the random sample then

$$E(S^2) = \sigma^2 \text{ and } \text{Var}(S^2) = \frac{2\sigma^4}{n} \quad \dots (9)$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore \text{SE}(S^2) = \sqrt{\text{Var}(S^2)} = \sqrt{\frac{2}{n}} \sigma^2 \quad \dots (10)$$

The general procedure of this test is explained in the next page.

As we are doing so far in all tests, first Step in hypothesis testing problems is to setup null and alternative hypotheses. Here, we want to test the hypothesis specified value σ_0^2 of the population variance σ^2 so we can take our null and alternative hypotheses as

$$\begin{aligned} H_0 : \sigma^2 = \sigma_0^2 \text{ and } H_1 : \sigma^2 \neq \sigma_0^2 & \quad [\text{for two-tailed test}] \\ \text{or} \quad \left. \begin{aligned} H_0 : \sigma^2 \leq \sigma_0^2 \text{ and } H_1 : \sigma^2 > \sigma_0^2 \\ H_0 : \sigma^2 \geq \sigma_0^2 \text{ and } H_1 : \sigma^2 < \sigma_0^2 \end{aligned} \right\} & \quad [\text{for one-tailed test}] \end{aligned}$$

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{S^2 - E(S^2)}{SE(S^2)} \sim N(0,1) \\ Z &= \frac{S^2 - \sigma_0^2}{\sigma_0^2 \sqrt{\frac{2}{n}}} \quad \left[\begin{array}{l} \text{Using equations(9) and(10) and} \\ \text{under } H_0 : \sigma^2 = \sigma_0^2 \end{array} \right] \end{aligned}$$

After that, we calculate the value of test statistic and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2.

Note 2: When population under study is normal then for testing the hypothesis about population variance or population standard deviation we use chi-square test which will be discussed in Unit 12 of this course. Whereas when the distribution of the population under study is not known and sample size is large then we apply Z-test as discussed above.

Now, it is time to do example based on above test.

Example 9: A random sample of size 65 screws is taken from a population of big box of screws and measured their length (in mm) which gives sample variance 9.0. Test that the two years old population variance 10.5 is still maintained at present at 5% level of significance.

Solution: We are given that

$$n = 65, \quad S^2 = 9.0, \quad \sigma_0^2 = 10.5$$

Here, we want to test that the two years old screw length population variance (σ^2) is still maintained at 10.5. So our claim is $\sigma^2 = 10.5$ and its complement is $\sigma^2 \neq 10.5$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \sigma^2 = \sigma_0^2 = 10.5 \text{ and } H_1 : \sigma^2 \neq 10.5$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, the distribution of population under study is not known and sample size is large ($n > 30$) so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{S^2 - \sigma_0^2}{\sigma_0^2 \sqrt{\frac{2}{n}}} \sim N(0,1)$$

$$= \frac{9.0 - 10.5}{10.5 \sqrt{\frac{2}{65}}} = \frac{-1.5}{10.5 \times 0.175} = \frac{-1.5}{1.84} = -0.81$$

The critical values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= -0.81) is less than critical value (= 1.96) and greater than the critical value (= -1.96), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that sample fails to provide us sufficient evidence against the claim so we may assume that two years old screw length population variance is still maintained at 10.5mm.

Now, you can try the following exercise.

E12) A random sample of size 120 bulbs is taken from a lot which gives the standard deviation of the life of electric bulbs 7 hours. Test the standard deviation of the life of bulbs of the lot is 6 hours at 5% level of significance.

10.8 TESTING OF HYPOTHESIS FOR TWO POPULATION VARIANCES USING Z-TEST

In previous section, we have discussed testing of hypothesis about the population variance. But there are so many situations where we want to test the hypothesis about equality of two population variances or standard deviations. For example, an economist may want to test whether the variability in incomes differ in two populations, a quality controller may want to test whether the quality of the product is changing over time, etc.

Let there be two populations, say, population-I and population-II under study. Also let μ_1, μ_2 and σ_1^2, σ_2^2 denote the means and variances of population-I and population-II respectively where both σ_1^2 and σ_2^2 are unknown but μ_1 and μ_2 may be known or unknown. For testing the hypothesis about equality of two population variances or standard deviations, we draw a random sample of large size n_1 from population-I and a random sample of large size n_2 from population-II. Let S_1^2 and S_2^2 be the sample variances of the samples selected from population-I and population-II respectively.

These two populations **may or may not be normal** but according to the central limit theorem, the sampling distribution of difference of two large sample variances asymptotically normally distributed with mean $(\sigma_1^2 + \sigma_2^2)$ and variance $(2\sigma_1^4/n_1 + 2\sigma_2^4/n_2)$.

Thus,

$$E(S_1^2 - S_2^2) = E(S_1^2) - E(S_2^2) = \sigma_1^2 - \sigma_2^2 \quad \dots (11)$$

and

$$\text{Var}(S_1^2 - S_2^2) = \text{Var}(S_1^2) + \text{Var}(S_2^2) = \frac{2\sigma_1^4}{n_1} + \frac{2\sigma_2^4}{n_2}$$

But we know that standard error = $\sqrt{\text{Variance}}$

$$\therefore \text{SE}(S_1^2 - S_2^2) = \sqrt{\text{Var}(S_1^2 - S_2^2)} = \sqrt{\frac{2\sigma_1^4}{n_1} + \frac{2\sigma_2^4}{n_2}} \quad \dots (12)$$

Now, follow the same procedure as we have discussed in Section 10.2, that is, first of all we have to setup null and alternative hypothesis. Here, we want to test the hypothesis about the two population variances, so we can take our null and alternative hypotheses as

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma^2 \text{ and } H_1 : \sigma_1^2 \neq \sigma_2^2 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \sigma_1^2 \leq \sigma_2^2 \text{ and } H_1 : \sigma_1^2 > \sigma_2^2 \\ H_0 : \sigma_1^2 \geq \sigma_2^2 \text{ and } H_1 : \sigma_1^2 < \sigma_2^2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

For testing the null hypothesis, the test statistic Z is given by

$$Z = \frac{(S_1^2 - S_2^2) - E(S_1^2 - S_2^2)}{\text{SE}(S_1^2 - S_2^2)} \sim N(0,1)$$

$$\text{or} \quad Z = \frac{(S_1^2 - S_2^2) - (\sigma_1^2 - \sigma_2^2)}{\sqrt{\frac{2\sigma_1^4}{n_1} + \frac{2\sigma_2^4}{n_2}}} \quad [\text{using equations (11) and (12)}]$$

Since under null hypothesis we assume that $\sigma_1^2 = \sigma_2^2 = \sigma^2$, therefore, we have

$$Z = \frac{S_1^2 - S_2^2}{\sigma^2 \sqrt{2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$$

Generally, population variances σ_1^2 and σ_2^2 are unknown, so we estimate them by their corresponding sample variances S_1^2 and S_2^2 as

$$\hat{\sigma}_1^2 = S_1^2 \quad \text{and} \quad \hat{\sigma}_2^2 = S_2^2$$

Thus, the test statistic Z is given by

$$Z = \frac{S_1^2 - S_2^2}{\sqrt{\left(\frac{2S_1^4}{n_1} + \frac{2S_2^4}{n_2}\right)}} \sim N(0,1)$$

After that, we calculate the value of test statistic as may be the case and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2.

Note 3: When populations under study are normal then for testing the hypothesis about equality of population variances we use F- test which will be discussed in Unit 12 of this course. Whereas when the form of the populations under study is not known and sample sizes are large then we apply Z-test as discussed above.

Now, it is time to do an example based on above test.

Example 10: A comparative study of variation in weights (in pound) of Army-soldiers and Navy- sailors was made. The sample variance of the weight of 120 soldiers was 60 pound² and the sample variance of the weight of 160 sailors was 70 pound². Test whether the soldiers and sailors have equal variation in their weights. Use 5% level of significance.

Solution: Given that

$$n_1 = 120, S_1^2 = 60, n_2 = 160, S_2^2 = 70$$

We want to test that the Army-soldiers and Navy-sailors have equal variation in their weights. If σ_1^2 and σ_2^2 denote the variances in the weight of Army-soldiers and Navy-sailors so our claim is $\sigma_1^2 = \sigma_2^2$ and its complement is $\sigma_1^2 \neq \sigma_2^2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \left[\begin{array}{l} \text{Army-soldiers and Navy-sailors} \\ \text{have equal variation in their weights} \end{array} \right]$$

and the alternative hypothesis as

$$H_1 : \sigma_1^2 \neq \sigma_2^2 \quad \left[\begin{array}{l} \text{Army-soldiers and Navy-sailors} \\ \text{have different variation in their weights} \end{array} \right]$$

Here, the distributions of populations under study are not known and sample sizes are large ($n_1 = 120 > 30$, $n_2 = 160 > 30$) so we can go for Z-test.

Since population variances are unknown so for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{S_1^2 - S_2^2}{\sqrt{\left(\frac{2S_1^4}{n_1} + \frac{2S_2^4}{n_2} \right)}} \\ &= \frac{60 - 70}{\sqrt{\frac{2 \times (60)^2}{120} + \frac{2 \times (70)^2}{160}}} \\ &= \frac{-10}{\sqrt{60.0 + 61.25}} = \frac{-10}{11.01} = -0.91 \end{aligned}$$

The critical values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= -0.91) is less than critical value (= 1.96) and greater than the critical value (= -1.96), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that the Army-soldiers and Navy-sailors have equal variation in their weights.

Now, you can try the following exercise.

E 13) Two sources of raw materials of bulbs are under consideration by a bulb manufacturing company. Both sources seem to have similar characteristics but the company is not sure about their respective uniformity. A sample of 52 lots from source A yields variance 25 and a sample of 40 lots from source B yields variance of 12. Test whether the variance of source A significantly differs to the variances of source B at $\alpha = 0.05$?

We now end this unit by giving a summary of what we have covered in it.

10.9 SUMMARY

In this unit we have covered the following points:

1. How to judge a given situation whether we should go for large sample test or not.
 2. Applying the Z-test for testing the hypothesis about the population mean and difference of two population means.
 3. Applying the Z-test for testing the hypothesis about the population proportion and difference of two population proportions.
 4. Applying the Z-test for testing the hypothesis about the population variance and two population variances.
-

10.10 SOLUTIONS / ANSWERS

E1) Since we have a rule that if the observed value of test statistic lies in rejection region then we reject the null hypothesis and if calculated value of test statistic lies in non-rejection region then we do not reject the null hypothesis. Therefore in our case, we do not reject the null hypothesis. So (iii) is the correct answer. Remember always on the basis of the one sample we never accept the null hypothesis.

E2) Since the test is two-tailed therefore rejection region will lie under both tails.

E3) Since test is two-tailed, therefore,

$$\begin{aligned} \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 2.42] \\ &= 2[0.5 - P[0 \leq Z \leq 2.42]] = 2(0.5 - 0.4922) = 0.0156 \end{aligned}$$

E4) We are given that

$$n = 900, \bar{X} = 3.4 \text{ cm}, \mu_0 = 3.25 \text{ cm and } \sigma = 2.61 \text{ cm}$$

Here, we wish to test that the sample comes from a large population of bolts with mean (μ) 3.25cm. So our claim is $\mu = 3.25\text{cm}$ and its complement is $\mu \neq 3.25\text{cm}$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 3.25 \text{ and } H_1 : \mu \neq 3.25$$

Since the alternative hypothesis is two-tailed, so the test is two-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown, so we should use t-test if the population of bolts known to be normal. But it is not the case. Since the sample size is large ($n > 30$) so we can go for Z-test instead of t-test as an approximate. So test statistic is given by

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

$$= \frac{3.40 - 3.25}{2.61 / \sqrt{900}} = \frac{0.15}{0.087} = 1.72$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of test statistic $Z (= 1.72)$ is less than the critical value ($= 1.96$) and greater than critical value ($= -1.96$), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that sample does not provide us sufficient evidence against the claim so we may assume that the sample comes from the population of bolts with mean 3.25cm.

E5) Here, we given that

$$\mu_0 = 1200, n = 100, \bar{X} = 1220, S = 90$$

Here, the company may accept the new CFL light when average life of CFL light is greater than 1200 hours. So the company wants to test that the new brand CFL light has an average life greater than 1200 hours. So our claim is $\mu > 1200$ and its complement is $\mu \leq 1200$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu \leq \mu_0 = 1200 \text{ and } H_1 : \mu > 1200$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown, so we should use t-test if the distribution of life of bulbs known to be normal. But it is not the case. Since the sample size is large ($n > 30$) so we can go for Z-test instead of t-test. Therefore, test statistic is given by

$$Z = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

$$= \frac{1220 - 1200}{90 / \sqrt{100}} = \frac{20}{9} = 2.22$$

The critical values for right-tailed test at 5% level of significance is $z_{\alpha} = z_{0.05} = 1.645$.

Since calculated value of test statistic $Z (= 2.22)$ is greater than critical value ($= 1.645$), that means it lies in rejection region so we reject the null hypothesis and support the alternative hypothesis i.e. we support our claim at 5% level of significance.

Thus, we conclude that sample does not provide us sufficient evidence against the claim so we may assume that the company accepts the new brand of bulbs.

E6) Given that

$$n_1 = 200, \quad \bar{X} = 1300, \quad S_1 = 41;$$

$$n_2 = 200, \quad \bar{Y} = 1280, \quad S_2 = 46$$

Here, we want to test that there is significant difference in the mean duration of their lives of two brands of electric bulbs. If μ_1 and μ_2 denote the mean lives of two brands of electric bulbs respectively then our claim is $\mu_1 \neq \mu_2$ and its complement is $\mu_1 = \mu_2$. Since the complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

We want to test the null hypothesis regarding equality of two population means. The standard deviations of both populations are unknown so we should go for t-test if population of difference is known to be normal. But it is not the case. Since sample sizes are large (n_1 , and $n_2 > 30$) so we go for Z-test.

So for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \\ &= \frac{1300 - 1280}{\sqrt{\frac{(41)^2}{200} + \frac{(46)^2}{200}}} = \frac{20}{\sqrt{8.41 + 10.58}} = \frac{20}{4.36} = 4.59 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of test statistic Z (= 4.59) is greater than the critical values (= ± 2.58), that means it lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support the claim at 1% level of significance.

Thus, we conclude that samples do not provide us sufficient evidence against the claim so there is significant difference in the mean duration of their lives of two brands of electric bulbs.

E7) Given that

$$n_1 = 50, \quad \bar{X} = 8.3, \quad S_1 = 1.2;$$

$$n_2 = 100, \quad \bar{Y} = 8.0, \quad S_2 = 1.5$$

Here, we want to test that the first drug provides a significant longer period of relief than the other. If μ_1 and μ_2 denote the mean relief time due to first and second drugs respectively then our claim is $\mu_1 > \mu_2$ and

its complement is $\mu_1 \leq \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 \leq \mu_2 \text{ and } H_1 : \mu_1 > \mu_2$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

We want to test the null hypothesis regarding equality of two population means. The standard deviations of both populations are unknown. So we should go for t-test if population of difference is known to be normal. But it is not the case. Since sample sizes are large (n_1 , and $n_2 > 30$) so we go for Z-test.

So for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \\ &= \frac{8.3 - 8.0}{\sqrt{\frac{(1.2)^2}{50} + \frac{(1.5)^2}{100}}} = \frac{0.3}{\sqrt{0.0288 + 0.0255}} \\ &= \frac{0.3}{0.2265} = 1.32 \end{aligned}$$

The critical (tabulated) value for right-tailed test at 5% level of significance is $z_\alpha = z_{0.05} = 1.645$.

Since calculated value of test statistic Z (= 1.32) is less than the critical value (=1.645), that means it lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so the first drug does not has longer period of relief than the other.

- E8)** Here, students are classified as Science background and other. We define our success and failure as getting student of Science background and other respectively. We are given that

n = Total number of students in the sample = 100

X = Number of students from Science background = 54

p = Sample proportion of students from Science background

$$= \frac{54}{100} = 0.54$$

We want to test whether 50% of the students are from Science background in MSc. If P denotes the proportion of first year Science background students in the University. So our claim is $P = P_0 = 0.5$ and its complement is $P \neq 0.5$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

Testing of Hypothesis

$$H_0 = P = P_0 = 0.5 (= 50\%)$$

$$H_1: P \neq 0.5 \text{ [Science background differs to 50\%]}$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore np = 100 \times 0.54 = 54 > 5$$

$$nq = 100 \times (1 - 0.54) = 100 \times 0.46 = 46 > 5$$

We see that condition of normality meets, so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.54 - 0.50}{\sqrt{0.5 \times 0.5 / 100}} \quad [\because Q_0 = 1 - P_0] \\ &= \frac{0.04}{0.05} = 0.80 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of test statistic Z (= 0.80) is less than the critical value (= 2.58) and greater than critical value (= -2.58), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that sample fails to provide us sufficient evidence against the claim so we may assume that 50% of the first year students in MSc. Economics in the University are from Science background.

E9) We define our success and failure as a patient is survived and not survived. Here, we are given that

$$n = 200$$

$$\begin{aligned} X &= \text{Number of survived patients who are given a particular injection} \\ &= 180 \end{aligned}$$

p = Sample proportion of survived patients who are given a particular injection

$$= \frac{X}{n} = \frac{180}{200} = 0.9$$

$$P_0 = 80\% = \frac{80}{100} = 0.80 \Rightarrow Q_0 = 1 - P_0 = \frac{80}{100} = 0.20$$

Here, we want to test that the survival rate of the patients is more than 80%. If P denotes the proportion of survival patients then our claim is $P > 0.80$ and its complement is $P \leq 0.80$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : P \leq P_0 = 0.80 \text{ and } H_1 : P > 0.80$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore np = 200 \times 0.9 = 180 > 5$$

$$nq = 200 \times (1 - 0.9) = 200 \times 0.1 = 20 > 5$$

We see that condition of normality meets, so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \\ &= \frac{0.9 - 0.8}{\sqrt{\frac{(0.8)(0.2)}{200}}} = \frac{0.1}{0.0283} = 3.53 \end{aligned}$$

The critical (tabulated) value for right-tailed test at 5% level of significance is $z_{\alpha} = z_{0.05} = 1.645$.

Since calculated value of test statistic Z (= 3.53) is greater than the critical value (=1.645), that means it lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support our claim at 5% level of significance.

Thus, we conclude that sample fails to provide us sufficient evidence against the claim so we may assume that the survival rate is greater than 80% in the population using that injection.

E10) Let X_1 and X_2 stand for number of literates in districts A and B respectively. Therefore, we are given that

$$n_1 = 100, X_1 = 50 \Rightarrow p_1 = \frac{X_1}{n_1} = \frac{50}{100} = 0.50$$

$$n_2 = 100, X_2 = 40 \Rightarrow p_2 = \frac{X_2}{n_2} = \frac{40}{100} = 0.40$$

Here, we want to test whether the proportion of literate persons in two districts A and B is same. If P_1 and P_2 denote the proportions of literate persons in two districts A and B respectively then our claim is $P_1 = P_2$ and its complement $P_1 \neq P_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : P_1 = P_2 = P \text{ and } H_1 : P_1 \neq P_2$$

Since the alternative hypothesis is two-tailed so the test is two tailed test.

The estimate of the combined proportion (P) of literates in districts A and B is given by

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{50 + 40}{100 + 100} = 0.45$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.45 = 0.55$$

Testing of Hypothesis

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore n_1 p_1 = 100 \times 0.50 = 50 > 5, n_1 q_1 = 100 \times 0.50 = 50 > 5$$

$$n_2 p_2 = 100 \times 0.40 = 40 > 5, n_2 q_2 = 100 \times 0.60 = 60 > 5$$

We see that condition of normality meets, so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{0.50 - 0.40}{\sqrt{0.45 \times 0.55\left(\frac{1}{100} + \frac{1}{100}\right)}} = \frac{0.10}{0.0704} = 1.42 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of test statistic Z (= 1.42) is less than the critical value (= 2.58) and greater than critical value (= -2.58), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support our claim at 1% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that the proportion of literates in districts A and B is equal.

E11) Here, we are given that

$$n_1 = 1200, \quad p_1 = 30\% = 0.30$$

$$n_2 = 900, \quad p_2 = 20\% = 0.20$$

Here, we want to test that the proportion of blue-eye persons in both the population is same. If P_1 and P_2 denote the proportions of blue-eye persons in two populations respectively then our claim is $P_1 = P_2$ and its complement $P_1 \neq P_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : P_1 = P_2 = P \text{ and } H_1 : P_1 \neq P_2$$

Since the alternative hypothesis is two-tailed so the test is two tailed test.

The estimate of the combined proportion (P) of literates in districts A and B is given by

$$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{1200 \times 0.30 + 900 \times 0.20}{1200 + 900} = 0.257$$

$$\hat{Q} = 1 - \hat{P} = 1 - 0.257 = 0.743$$

Before proceeding further, first we have to check whether the condition of normality meets or not.

$$\therefore n_1 p_1 = 1200 \times 0.30 = 360 > 5, n_1 q_1 = 1200 \times 0.70 = 840 > 5$$

$$n_2 p_2 = 900 \times 0.20 = 180 > 5, n_2 q_2 = 900 \times 0.80 = 720 > 5$$

We see that condition of normality meets, so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{0.30 - 0.20}{\sqrt{0.257 \times 0.743\left(\frac{1}{1200} + \frac{1}{900}\right)}} = \frac{0.10}{0.019} = 5.26 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 1.96$.

Since calculated value of test statistic Z (= 5.26) is greater than critical values (= ± 1.96), that means it lies in rejection region, so we reject the null hypothesis i.e. we reject our claim at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so the proportion of blue-eyed person in both the population is not same.

E12) Here, we are given that

$$n = 120, \quad S = 7, \quad \sigma_0 = 6$$

Here, we want to test that standard deviation (σ) of the life of bulbs of the lot is 6 hours. So our claim is $\sigma = 6$ and its complement is $\sigma \neq 6$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \sigma = \sigma_0 = 6 \text{ and } H_1 : \sigma \neq 6$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, the distribution of population under study is not known and sample size is large ($n > 30$) so we can go for Z-test.

For testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{S^2 - \sigma_0^2}{\sigma_0^2 \sqrt{\frac{2}{n}}} \sim N(0,1) \\ &= \frac{(7)^2 - (6)^2}{(6)^2 \sqrt{\frac{2}{120}}} = \frac{13}{36 \times 0.129} = \frac{13}{4.64} = 2.80 \end{aligned}$$

The critical values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= 2.8) is greater than critical values (= ± 1.96), that means it lies in rejection region, so we reject the null hypothesis i.e. we reject our claim at 5% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so standard deviation of the life of bulbs of the lot is not 6.0 hours.

E13) Here, we are given that

$$n_1 = 52, \quad S_1^2 = 25$$

$$n_2 = 40, \quad S_2^2 = 12$$

Here, we want to test that variance of source A significantly differs to the variances of source B. If σ_1^2 and σ_2^2 denote the variances in the raw materials of sources A and B respectively so our claim is $\sigma_1^2 \neq \sigma_2^2$ and its complement is $\sigma_1^2 = \sigma_2^2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ and } H_1 : \sigma_1^2 \neq \sigma_2^2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, the distributions of populations under study are not known and sample sizes are large ($n_1 = 52 > 30$, $n_2 = 40 > 30$) so we can go for Z-test.

Since population variances are unknown so for testing the null hypothesis, the test statistic Z is given by

$$\begin{aligned} Z &= \frac{S_1^2 - S_2^2}{\sqrt{\left(\frac{2S_1^4}{n_1} + \frac{2S_2^4}{n_2}\right)}} \\ &= \frac{25 - 12}{\sqrt{\frac{2(25)^2}{52} + \frac{2(12)^2}{40}}} = \frac{13}{5.5} = 2.36 \end{aligned}$$

The critical values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= 2.36) is greater than critical values (= ± 1.96), that means it lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. we support our claim at 5% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so variance of source A significantly differs to the variance of source B.

UNIT 11 SMALL SAMPLE TESTS

Structure

- 11.1 Introduction
 - Objectives
- 11.2 General Procedure of t-Test for Testing a Hypothesis
- 11.3 Testing of hypothesis for Population Mean Using t-Test
- 11.4 Testing of Hypothesis for Difference of Two Population Means Using t-Test
- 11.5 Paired t-Test
- 11.6 Testing of Hypothesis for Population Correlation Coefficient Using t-Test
- 11.7 Summary
- 11.8 Solutions /Answers

11.1 INTRODUCTION

In previous unit, we have discussed the testing of hypothesis for large samples in details. Recall that throughout the unit, we were making an assumption that “if sample size is sufficiently large then test statistic follows approximately standard normal distribution”. Also recall two points highlighted in this course, i.e.

- Cost of our study increases as sample size increases.
- Sometime nature of the units in the population under study is such that they destroyed under investigation.

If there are limited recourses in terms of money then first point listed above force us not to go for large sample size when items /units under study are very costly such as airplane, computer, etc. Second point listed above give an alarm for not to go for large sample if population units are destroyed under investigation.

So, we need an alternative technique which is used to test the hypothesis based on small sample(s). Small sample tests do this job for us. But in return they demand one basic assumption that population under study should be normal as you will see when you go through the unit. t , χ^2 and F -tests are some commonly used small sample tests.

In this unit, we will discuss t-test in details which is based on the t-distribution described in Unit 3 of this course. And χ^2 and F -tests will be discussed in next unit which are based on χ^2 and F -distributions described in Unit 3 and Unit 4 of this course respectively.

This unit is divided into eight sections. Section 11.1 is described the need of small sample tests. The general procedure of t-test for testing a hypothesis is described in Section 11.2. In Section 11.3, we discuss testing of hypothesis for population mean using t-test. Testing of hypothesis for difference of two population means when samples are independent is described in Section 11.4 whereas in Section 11.5, the paired t-test for difference of two population means when samples are dependent(paired) is discussed. In Section 11.6 testing of hypothesis for population correlation coefficient is explained. Unit

ends by providing summary of what we have discussed in this unit in Section 11.7 and solution of exercises in Section 11.8.

Before moving further a humble suggestion to you that please revise what you have learned in previous two units. The concepts discussed there will help you a lot to better understand the concepts discussed in this unit.

Objectives

After studying this unit, you should be able to:

- realize the importance of small sample tests;
- know the procedure of t-test for testing a hypothesis;
- describe testing of hypothesis for population mean for using t-test;
- explain the testing of hypothesis for difference of two population means when samples are independent using t-test;
- describe the procedure for paired t-test for testing of hypothesis for difference of two population means when samples are dependent or paired; and
- explain the testing of hypothesis for population correlation coefficient using t-test.

11.2 GENERAL PROCEDURE OF t-TEST FOR TESTING A HYPOTHESIS

The general procedure of t-test for testing a hypothesis is similar as Z-test already explained in Unit 10. Let us give you similar details here.

For this purpose, let $X_1, X_2, \dots, X_n$ be a random sample of **small size n (< 30)** selected from a **normal population** (recall the demand of small sample tests pointed out in previous Section 11.1) having parameter of interest, say, θ which is actually unknown but its hypothetical value, say, θ_0 estimated from some previous study or some other way is to be tested. t-test involves following steps for testing this hypothetical value:

Step I: First of all, we setup null and alternative hypotheses. Here, we want to test the hypothetical value θ_0 of parameter θ so we can take the null and alternative hypotheses as

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta \neq \theta_0 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0 \\ H_0 : \theta \geq \theta_0 \text{ and } H_1 : \theta < \theta_0 \end{array} \right\} \quad [\text{for one-tailed test}]$$

In case of comparing same parameter of two populations of interest, say, θ_1 , and θ_2 then our null and alternative hypotheses would be

$$H_0 : \theta_1 = \theta_2 \text{ and } H_1 : \theta_1 \neq \theta_2 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \theta_1 \leq \theta_2 \text{ and } H_1 : \theta_1 > \theta_2 \\ H_0 : \theta_1 \geq \theta_2 \text{ and } H_1 : \theta_1 < \theta_2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

Step II: After setting the null and alternative hypotheses our next step is to decide a criteria for rejection or non-rejection of null hypothesis i.e.

decide the level of significance α , at which we want to test our null hypothesis. We generally take $\alpha = 5\%$ or 1% .

Step III: The third step is to determine an appropriate test statistic, say, t for testing the null hypothesis. Suppose T_n is the sample statistic (may be sample mean, sample correlation coefficient, etc. depending upon θ) for the parameter θ then test-statistic t is given by

$$t = \frac{T_n - E(T_n)}{SE(T_n)}$$

Step IV: As we know, t -test is based on t -distribution and t -distribution is described with the help of its degrees of freedom, therefore, test statistic t follows t -distribution with specified degrees of freedom as the case may be.

By putting the values of T_n , $E(T_n)$ and $SE(T_n)$ in above formula, we calculate the value of test statistic t . Let t_{cal} be the calculated value of test statistic t after putting these values.

Step V: After that, we obtain the critical (cut-off or tabulated) value(s) in the sampling distribution of the test statistic t corresponding to α assumed in Step II. The critical values for t -test are given in **Table-II (t-table)** of the Appendix at the end of Block 1 of this course corresponding to different level of significance (α). After that, we construct rejection (critical) region of size α in the probability curve of the sampling distribution of test statistic t .

Step VI: Take the decision about the null hypothesis based on calculated and critical value(s) of test statistic obtained in Step IV and Step V respectively. Since critical value depends upon the nature of the test that it is one-tailed test or two-tailed test so following cases arise:

In case of one-tailed test:

Case I: When $H_0 : \theta \leq \theta_0$ and $H_1 : \theta > \theta_0$ (right-tailed test)

In this case, the rejection (critical) region falls under the right tail of the probability curve of the sampling distribution of test statistic t . Suppose $t_{(v),\alpha}$ is the critical value at α level of significance then entire region greater than or equal to $t_{(v),\alpha}$ is the rejection region and less than $t_{(v),\alpha}$ is the non-rejection region as shown in Fig. 11.1.

If $t_{cal} \geq t_{(v),\alpha}$, that means calculated value of test statistic t lies in the rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that sample data provides us sufficient evidence against the null hypothesis and there is a significant difference between hypothesized value and observed value of the parameter.

If $t_{cal} < t_{(v),\alpha}$, that means calculated value of test statistic t lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that the sample data fails to provide us sufficient evidence against the null hypothesis and the difference between hypothesized value and observed value of the parameter due to fluctuation of sample.

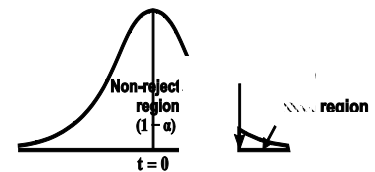


Fig. 11.1

In this case, the rejection (critical) region falls under the left tail of the probability curve of the sampling distribution of test statistic t .

Suppose $-t_{(v),\alpha}$ is the critical value at α level of significance then entire region less than or equal to $-t_{(v),\alpha}$ is the rejection region and greater than $-t_{(v),\alpha}$ is the non-rejection region as shown in Fig. 11.2.

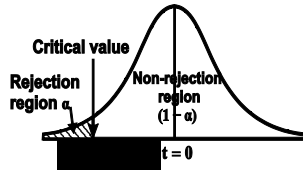


Fig. 11.2

If $t_{cal} \leq -t_{(v),\alpha}$, that means calculated value of test statistic t lies in the rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

If $t_{cal} > -t_{(v),\alpha}$, that means calculated value of test statistic t lies in the non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

In case of two-tailed test:

That is, when $H_0 : \theta = \theta_0$ and $H_1 : \theta \neq \theta_0$

In this case, the rejection region falls under both tails of the probability curve of sampling distribution of the test statistic t . Half the area (α) i.e. $\alpha/2$ will lie under left tail and other half under the right tail. Suppose $-t_{(v),\alpha/2}$ and $t_{(v),\alpha/2}$ are the two critical values at the left-tailed and right-tailed respectively. Therefore, entire region less than or equal to $-t_{(v),\alpha/2}$ and greater than or equal to $t_{(v),\alpha/2}$ are the rejection regions and between $-t_{(v),\alpha/2}$ and $t_{(v),\alpha/2}$ is the non-rejection region as shown in Fig. 11.3.

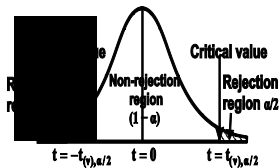


Fig. 11.3

If $t_{cal} \geq t_{(v),\alpha/2}$, or $t_{cal} \leq -t_{(v),\alpha/2}$, that means calculated value of test statistic t lies in the rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

And if $-t_{(v),\alpha/2} < t_{cal} < t_{(v),\alpha/2}$, that means calculated value of test statistic t lies in the non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Procedure of taking the decision about the null hypothesis on the basis of p-value:

To take the decision about the null hypothesis on the basis of p-value, the p-value is compared with given level of significance (α). And if p-value is less than or equal to α then we reject the null hypothesis and if p-value is greater than α then we do not reject the null hypothesis at α level of significance.

Since the distribution of test statistic t follows t-distribution with v df and we also know that t-distribution is symmetrical about $t = 0$ line therefore, if t_{cal} represents calculated value of test statistic t then p-value can be defined as:

For one-tailed test:

For $H_1 : \theta > \theta_0$ (right-tailed test)

$$p\text{-value} = P[t \geq t_{cal}]$$

For $H_1 : \theta < \theta_0$ (left-tailed test)

$$p\text{-value} = P[t \leq t_{cal}]$$

For two-tailed test: For $H_1 : \theta \neq \theta_0$

$$p\text{-value} = 2P[t \geq |t_{\text{cal}}|]$$

These p-values for t-test can be obtained with the help of **Table-II (t-table)** given in the Appendix at the end of Block 1 of this course. But this table gives the t-values corresponding to the standard values of α such as 0.10, 0.05, 0.025, 0.01 and 0.005 only, therefore, the exact p-values are not obtained with the help of this table and we can approximate the p-value for this test.

For example, if test is right-tailed and calculated (observed) value of test statistic t is 2.94 with 9 df then p-value is obtained as:

Since calculated value of test statistic t is based on the 9 df therefore, we use row for 9 df in the t-table and move across this row to find the values in which calculated t-value falls. Since calculated t-value falls between 2.821 and 3.250, which are corresponding to the values of one-tailed area $\alpha = 0.01$ and 0.005 respectively, therefore, p-value will lie between 0.005 and 0.01, that is,

$$0.005 < p\text{-value} < 0.01$$

If in the above example, the test is two-tailed then the two values 0.01 and 0.005 would be doubled for p-value, that is,

$$0.005 \times 2 = 0.01 < p\text{-value} < 0.02 = 2 \times 0.01$$

Note 1: With the help of computer packages and softwares such as SPSS, SAS, MINITAB, EXCEL, etc. we can find the exact p-values for t-test.

Now, you can try the following exercise.

E1) If test is two-tailed and calculated value of test statistic t is 2.42 with 15 df then find the p-value for t-test.

11.3 TESTING OF HYPOTHESIS FOR POPULATION MEAN USING t-TEST

In Section 10.3 of the previous unit, we have discussed Z-test for testing the hypothesis about population mean when population variance σ^2 is known and unknown.

Recall from these, we have already pointed out that one basic difference between Z-test and t-test is that Z-test is used when population SD is known whether sample size is large or small and t-test is used when population SD is unknown whether sample size is small or large. But in case of large sample size Z-test is an appropriate of t-test as we did in previous unit. But in practice standard deviation of population is not known and sample size is small so in this situation, we use t-test provided population under study is normal.

Assumptions

Virtually every test has some assumptions which must be met prior to the application of the test. This t-test needs following assumptions to work:

- (i) The characteristic under study follows normal distribution. In other words, populations from which random sample is drawn should be normal with respect to the characteristic of interest.
- (ii) Sample observations are random and independent.
- (iii) Population variance σ^2 is unknown.

Testing of Hypothesis

For describing this test, let $X_1, X_2, \dots, X_n$ be a random sample of **small size n (< 30)** selected from a **normal population** with mean μ and unknown variance σ^2 .

Now, follow the same procedure as we have discussed in previous section, that is, first of all we setup the null and alternative hypotheses. Here, we want to test the claim about the specified value μ_0 of population mean μ so we can take the null and alternative hypotheses as

$$\begin{aligned} & H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0 \quad \left[\text{for two-tailed test} \right] \quad \left[\begin{array}{l} \text{Here, } \theta = \mu \text{ and } \theta_0 = \mu_0 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right] \\ \text{or} \quad & \left. \begin{array}{l} H_0 : \mu \leq \mu_0 \text{ and } H_1 : \mu > \mu_0 \\ H_0 : \mu \geq \mu_0 \text{ and } H_1 : \mu < \mu_0 \end{array} \right\} \quad \left[\text{for one-tailed test} \right] \end{aligned}$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \sim t_{(n-1)} \quad \text{under } H_0$$

where, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance.

For computational simplicity, we may use the following formulae for $\bar{X}$, S^2 :

$$\bar{X} = a + \frac{1}{n} \sum d \text{ and } S^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

where, $d = (X - a)$, 'a' being the assumed arbitrary value.

Here, the test statistic t follows t-distribution with $(n - 1)$ degrees of freedom as we discussed in Unit 3 of this course.

After substituting values of $\bar{X}$, S and n, we get calculated value of test statistic t. Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the t-table. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in previous section.

Let us do some examples of testing of hypothesis about population mean using t-test.

Example 1: A manufacturer claims that a special type of projector bulb has an average life 160 hours. To check this claim an investigator takes a sample of 20 such bulbs, puts on the test, and obtains an average life 167 hours with standard deviation 16 hours. Assuming that the life time of such bulbs follows normal distribution, does the investigator accept the manufacturer's claim at 5% level of significance?

Solution: Here, we are given that

$$\mu_0 = 160, \quad n = 20, \quad \bar{X} = 167 \quad \text{and} \quad S = 16$$

Here, we want to test the manufacturer claims that a special type of projector bulb has an average life (μ) 160 hours. So claim is $\mu = 160$ and its complement is $\mu \neq 160$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 160 \text{ and } H_1 : \mu \neq 160$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. Also sample size is small $n = 20 (n < 30)$ and population under study is normal, so we can go for t-test for testing the hypothesis about population mean.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

$$= \frac{167 - 160}{16 / \sqrt{20}} = \frac{7}{3.58} = 1.96$$

The critical value of the test statistic t for various df and different level of significance α are given in **Table II** of the Appendix at the end of the Block 1 of this course.

The critical (tabulated) values of test statistic for two-tailed test corresponding $(n-1) = 19$ df at 5% level of significance are $\pm t_{(n-1), \alpha/2} = \pm t_{(19), 0.025} = \pm 2.093$.

Since calculated value of test statistic $t (= 1.96)$ is greater than the critical value $(= -2.093)$ and is less than critical value $(= 2.093)$, that means calculated value of test statistic lies in non-rejection region as shown in Fig. 11.4. So we do not reject the null hypothesis i.e. we support the manufacture's claim at 5% level of significance.

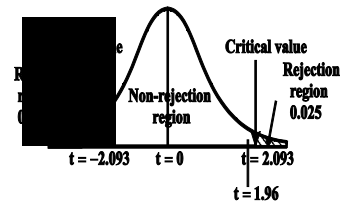


Fig. 11.4

Decision according to p-value:

Since calculated value of test statistic t is based on 19 df therefore, we use row for 19 df in the t -table and move across this row to find the values in which calculated t -value falls. Since calculated t -value falls between 1.729 and 2.093 corresponding to one-tailed area $\alpha = 0.05$ and 0.025 respectively therefore p -value lies between 0.025 and 0.05, that is,

$$0.025 < p\text{-value} < 0.05$$

Since test is two-tailed so

$$2 \times 0.025 = 0.05 < p\text{-value} < 0.10 = 0.05 \times 2$$

Since p -value is greater than $\alpha (= 0.05)$ so we do not reject the null hypothesis at 5% level of significance.

Thus, we conclude that sample fails to provide us sufficient evidence against the null hypothesis so we may assume that the manufacture's claim is true so the investigator may accept the manufacturer's claim at 5% level of significance.

Example 2: The mean share price of companies of Pharma sector is Rs.70. The share prices of all companies were changed time to time. After a month, a sample of 10 Pharma companies was taken and their share prices were noted as below:

70, 76, 75, 69, 70, 72, 68, 65, 75, 72

Assuming that the distribution of share prices follows normal distribution, test whether mean share price is still the same at 1% level of significance?

Testing of Hypothesis

Solution: Here, we wish to test that the mean share price (μ) of companies of Pharma sector is still Rs.70 besides all changes. So our claim is $\mu = 70$ and its complement is $\mu \neq 70$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu = \mu_0 = 70 \quad [\text{mean share price of companies is still Rs. 70}]$$

$$H_1 : \mu \neq \mu_0 = 70 \quad [\text{mean share price of companies is not still Rs. 70}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. Also sample size is small $n = 10 (n < 30)$ and population under study is normal, so we can go for t-test for testing the hypothesis about population mean.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \sim t_{(n-1)} \quad \dots (1)$$

Calculation for $\bar{X}$ and S:

S. No.	Sample value (X)	Deviation $d = (X - a), a = 70$	d^2
1	70	0	0
2	76	6	36
3	75	5	25
4	69	-1	1
5	70	0	0
6	72	2	4
7	68	-2	4
8	65	-5	25
9	75	5	25
10	72	2	4
Total		12	124

The assumed value of a is 70.

From the above calculation, we have

$$\bar{X} = a + \frac{1}{n} \sum d = 70 + \frac{1}{10} \times 12 = 71.2$$

$$\begin{aligned} S^2 &= \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right] \\ &= \frac{1}{10-1} \left[124 - \frac{(12)^2}{10} \right] = \frac{1}{9} \left[124 - \frac{144}{10} \right] = 12.18 \end{aligned}$$

$$\Rightarrow S = \sqrt{12.18} = 3.49$$

Putting the values in equation (1), we have

$$t = \frac{71.2 - 70}{3.49 / \sqrt{10}} = \frac{1.2}{1.10} = 1.09$$

The critical (tabulated) values of test statistic for two-tailed test corresponding $(n-1) = 9$ df at 1% level of significance are $\pm t_{(n-1), \alpha/2} = \pm t_{(9), 0.005} = \pm 3.250$.

Since calculated value of test statistic $t (= 1.09)$ is less than the critical value $(= 3.250)$ and greater than the critical value $(= -3.250)$, that means calculated value of t lies in non-rejection region as shown in Fig. 11.5. So we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Decision according to p-value:

Since calculated value of test statistic t is based on 9 df therefore, we use row for 9 df in the t -table and move across this row to find the values in which calculated t -value falls. Since all values in this row are greater than calculated t -value 1.09 and the smallest value is 1.383 corresponding to one-tailed area $\alpha = 0.10$ therefore p -value is greater than 0.10, that is,

$$p\text{-value} > 0.10$$

Since test is two-tailed so

$$p\text{-value} > 2 \times 0.10 = 0.20$$

Since p -value $(= 0.20)$ is greater than $\alpha (= 0.01)$ so we do not reject the null hypothesis at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so may assume that the mean share price is still Rs. 70.

Now, you can try the following exercises.

- E2)** A tyre manufacturer claims that the average life of a particular category of his tyre is 18000 km when used under normal driving conditions. A random sample of 16 tyres was tested. The mean and SD of life of the tyres in the sample were 20000 km and 6000 km respectively. Assuming that the life of the tyres is normally distributed, test the claim of the manufacturer at 1% level of significance using appropriate test.
- E3)** It is known that the average weight of cadets of a centre follows normal distribution. Weights of 10 randomly selected cadets from the same centre are as given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Can we say that average weight of all cadets of the centre from which the above sample was taken is equal to 60 kg at 5% level of significance?

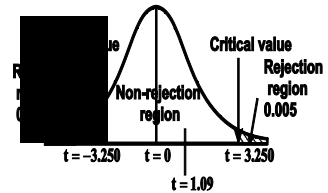


Fig. 11.5

11.4 TESTING OF HYPOTHESIS FOR DIFFERENCE OF TWO POPULATION MEANS USING t -TEST

In Section 10.4 of the previous unit, we have discussed Z -test for testing the hypothesis about difference of two population means under different possibility of population variances σ_1^2 and σ_2^2 . Recall from there, we have pointed out that one basic difference between Z -test and t -test is that, Z -test is used when standard deviations of both populations are known and t -test is used when standard deviations of both populations are unknown. But in practice standard

deviations of both populations are not known, so in real life problems t-test is more suitable compared to Z-test.

Assumptions

This test works under following assumptions:

- (i) The characteristic under study follows normal distribution in both the populations. In other words, both populations from which random samples are drawn should be normal with respect to the characteristic of interest.
- (ii) Samples and their observations both are independent to each other.
- (iii) Population variances σ_1^2 and σ_2^2 are both unknown but equal.

For describing this test, let there be two normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ under study. And we have to draw two independent random samples, say, $X_1, X_2, \dots, X_{n_1}$ and $Y_1, Y_2, \dots, Y_{n_2}$ of sizes n_1 and n_2 from these normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively. Let $\bar{X}$ and $\bar{Y}$ be the means of first and second sample respectively. Further, suppose the variances of both the populations are unknown but are equal, i.e., $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (say). In this case, σ^2 is estimated by value of pooled sample variance S_p^2 where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]$$

and

$$S_1^2 = \frac{1}{(n_1 - 1)} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \quad S_2^2 = \frac{1}{(n_2 - 1)} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

This can also be written as

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2 \right]$$

For computational simplicity, use the following formulae for $\bar{X}$, $\bar{Y}$ and S_p^2 :

$$\bar{X} = a + \frac{1}{n_1} \sum d_1, \quad \bar{Y} = b + \frac{1}{n_2} \sum d_2 \text{ and}$$

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\left\{ \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} \right\} + \left\{ \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} \right\} \right]$$

where, $d_1 = (X - a)$ and $d_2 = (Y - b)$, 'a' and 'b' are the assumed arbitrary values.

Now, follow the same procedure as we have discussed in Section 11.2, that is, first of all we have to setup null and alternative hypotheses. Here, we want to test the hypothesis about the difference of two population means so we can take the null hypothesis as

$$H_0 : \mu_1 = \mu_2 \text{ (no difference in means)} \quad \left[\begin{array}{l} \text{Here, } \theta_1 = \mu_1 \text{ and } \theta_2 = \mu_2 \\ \text{if we compare it with} \\ \text{general procedure.} \end{array} \right]$$

$$\text{or} \quad H_0 : \mu_1 - \mu_2 = 0 \text{ (difference in two means is 0)}$$

and the alternative hypothesis as

$$\begin{aligned}
 &H_1 : \mu_1 \neq \mu_2 \quad \text{[for two-tailed test]} \\
 \text{or} \quad &\left. \begin{aligned} &H_0 : \mu_1 \leq \mu_2 \text{ and } H_1 : \mu_1 > \mu_2 \\ &H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2 \end{aligned} \right\} \quad \text{[for one-tailed test]}
 \end{aligned}$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)} \quad \text{under } H_0$$

After substituting values of $\bar{X}$, $\bar{Y}$, S_p , n_1 and n_2 , we get calculated value of test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the **t-table**. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in Section 11.2.

Let us do some examples to become more user friendly with the test explained above.

Example 3: In a random sample of 10 pigs fed by diet A, the gain in weights (in pounds) in a certain period were

12, 8, 14, 16, 13, 12, 8, 14, 10, 9

In another random sample of 10 pigs fed by diet B, the gain in weights (in pounds) in the same period were

14, 13, 12, 15, 16, 14, 18, 17, 21, 15

Assuming that gain in the weights due to both foods follows normal distributions with equal variances, test whether diets A and B differ significantly regarding their effect on increase in weight at 5% level of significance.

Solution: Here, we can test that diets A and B differ significantly regarding their effect on increase in weight of pigs. If μ_1 and μ_2 denote the average gain in weights due to diet A and diet B respectively then our claim is $\mu_1 \neq \mu_2$ and its complement is $\mu_1 = \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Since it is given that the increase in the weight due to both foods follows normal distributions and population variances are equal and unknown. And other assumptions of t-test for testing a hypothesis about difference of two population means also meet. So we can go for this test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)} \quad \text{under } H_0 \quad \dots (2)$$

Calculation for $\bar{X}$, $\bar{Y}$ and S_p :

Diet A			Diet B		
X	$d_1 = (X-a)$ a = 12	d_1^2	Y	$d_2 = (Y-b)$ b = 16	d_2^2
12	0	0	14	-2	4
8	-4	16	13	-3	9
14	2	4	12	-4	16
16	4	16	15	-1	1
13	1	1	16	0	0
12	0	0	14	-2	4
8	-4	16	18	2	4
14	2	4	17	1	1
10	-2	4	21	5	25
9	-3	9	15	-1	1
Total	-4	70		-5	65

Here, a = 12, b = 16 are assumed values.

From above calculations, we have

$$\bar{X} = a + \frac{1}{n_1} \sum d_1 = 12 + \frac{(-4)}{10} = 11.6,$$

$$\bar{Y} = b + \frac{1}{n_2} \sum d_2 = 16 + \frac{(-5)}{10} = 15.5$$

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\left\{ \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} \right\} + \left\{ \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} \right\} \right]$$

$$= \frac{1}{10+10-2} \left[\left\{ 70 - \frac{(-4)^2}{10} \right\} + \left\{ 65 - \frac{(-5)^2}{10} \right\} \right]$$

$$= \frac{1}{18} (68.4 + 62.5) = 7.27$$

$$\Rightarrow S_p = \sqrt{7.27} = 2.70$$

Putting the values in equation (2), we have

$$t = \frac{11.6 - 15.5}{2.70 \sqrt{\frac{1}{10} + \frac{1}{10}}}$$

$$= \frac{-3.90}{2.70 \times 0.45} = \frac{-3.90}{1.215} = -3.21$$

The critical values of test statistic t for two-tailed test corresponding $(n_1 + n_2 - 2) = 18$ df at 5% level of significance are

$$\pm t_{(n_1+n_2-2), \alpha/2} = \pm t_{(18), 0.025} = \pm 2.101.$$

Since calculated value of test statistic t (= -3.21) is less than critical values (± 2.101) that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support our claim at 5% level of significance.

Thus, we conclude that samples do not provide us sufficient evidence against the claim so diets A and B differ significantly in terms of gain in weights of pigs.

Example 4: The means of two random samples of sizes 10 and 8 drawn from two normal populations are 210.40 and 208.92 respectively. The sum of squares of the deviations from their means is 26.94 and 24.50 respectively. Assuming that the populations are normal with equal variances, can samples be considered to have been drawn from normal populations having equal mean.

Solution: In usual notations, we are given that

$$n_1 = 10, n_2 = 8, \bar{X} = 210.40, \bar{Y} = 208.92,$$

$$\sum (X - \bar{X})^2 = 26.94, \quad \sum (Y - \bar{Y})^2 = 24.50$$

Therefore,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (X - \bar{X})^2 + \sum (Y - \bar{Y})^2 \right]$$

$$= \frac{1}{10 + 8 - 2} [26.94 + 24.50] = \frac{1}{16} \times 51.44 = 3.215$$

$$\Rightarrow S_p = \sqrt{3.215} = 1.79$$

We wish to test that both the samples are drawn from normal populations having the same means. If μ_1 and μ_2 denote the means of both normal populations respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \quad [\text{mean of both populations is equal}]$$

$$H_1 : \mu_1 \neq \mu_2 \quad [\text{mean of both populations is not equal}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Since it is given that two populations are normal with equal and unknown variances and other assumptions of t-test for testing a hypothesis about difference of two population means also meet. So we can go for this test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1 + n_2 - 2)} \quad \text{under } H_0$$

$$= \frac{210.40 - 208.92}{1.79 \sqrt{\frac{1}{10} + \frac{1}{8}}} = \frac{1.48}{1.79 \times 0.47} = \frac{1.48}{0.84} = 1.76$$

The critical values of test statistic t for two-tailed test corresponding $(n_1 + n_2 - 2) = 16$ df at 5% level of significance are

$$\pm t_{(n_1 + n_2 - 2), \alpha/2} = \pm t_{(16), 0.025} = \pm 2.12.$$

Since calculated value of test statistic t (= 1.76) is less than the critical value (= 2.12) and greater than the critical value (= -2.12), that means calculated value of test statistic t lies in non-rejection region so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that both samples are taken from normal populations having equal means.

Now, you can try the following exercises.

- E4)** Two different types of drugs A and B were tried on some patients for increasing their weights. Six persons were given drug A and other 7 persons were given drug B. The gain in weights (in ponds) is given below:

Drug A	5	8	7	10	9	6	–
Drug B	9	10	15	12	14	8	12

Assuming that increment in the weights due to both drugs follows normal distributions with equal variances, do the both drugs differ significantly with regard to their mean weights increment at 5% level of significance?

- E5)** To test the effect of fertilizer on wheat production, 26 plots of land with equal areas were chosen. Half of these plots were treated with fertilizer and the other half were untreated. Other conditions were the same. The mean yield of wheat on the untreated plots was 4.6 quintals with a standard deviation of 0.5 quintals, while the mean yield of the treated plots was 5.0 quintals with standard deviations of 0.3 quintals. Assuming that yields of wheat with and without fertilizer follow normal distributions with equal variances, can we conclude that there is significant improvement in wheat production due to effect of fertilizer at 1% level of significance?

11.5 PAIRED t-TEST

In the previous section, we have discussed t-test for equality of two population means in case of independent samples. However, there are so many situations where two samples are not independent and observations are recorded on the same individuals or items. Generally, such types of observations are recorded to assess the effectiveness of a particular training, diet, treatment, medicine, etc. In such situations, the observations are recorded “**before and after**” the insertion of training, treatment, etc. as the case may be. For example, if we wish to test a new diet on, say, 15 individuals then the weight of the individuals recorded before diet and after the diet will form two different samples in which observations will be paired as per each individual. Similarly, in the test of blood-sugar in human body, fasting sugar level before meal and sugar level after meal, both are recorded for a patient as paired observations, etc. The parametric test designed for this type of situation is known as paired t-test.

Now, come to the working principle of this test. This test first of all converts the two populations into a single population by taking the difference of paired observations.

Now, instead of two populations, we are left with one population, the population of differences. And the problem of testing equality of two population mean reduces to test the hypothesis that mean of the population of differences is equal to zero.

Assumptions

Small Sample Tests

This test works under following assumptions:

- (i) The population of differences follows normal distribution.
- (ii) Samples are not independent.
- (iii) Size of both the samples is equal.
- (iv) Population variances are unknown but not necessarily equal.

Let $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ be a paired random sample of size n and the difference between paired observations X_i & Y_i be denoted by D_i , that is,

$$D_i = X_i - Y_i \quad \text{for all } i=1, 2, \dots, n$$

Hence, we can assume that $D_1, D_2, \dots, D_n$ be a random sample from normal population of differences with mean μ_D and unknown variance σ_D^2 . This is same as the case of testing of hypothesis for population mean when population variance is unknown which is described in Section 11.3 of this unit.

Here, we want to test that there is an effect of a diet, training, treatment, medicine, etc. So we can take the null hypothesis as

$$H_0: \mu_1 = \mu_2 \text{ or } H_0: \mu_D = \mu_1 - \mu_2 = 0$$

and the alternative hypothesis

$$H_1: \mu_1 \neq \mu_2 \text{ or } H_1: \mu_D \neq 0 \quad [\text{for two-tailed test}]$$

$$\text{or } \left. \begin{array}{l} H_0: \mu_1 \leq \mu_2 \text{ and } H_1: \mu_1 > \mu_2 \\ H_0: \mu_1 \geq \mu_2 \text{ and } H_1: \mu_1 < \mu_2 \end{array} \right\} [\text{for one-tailed test}]$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{D}}{S_D / \sqrt{n}} \sim t_{(n-1)} \quad \text{under } H_0$$

$$\text{where, } \bar{D} = \frac{1}{n} \sum_{i=1}^n D_i \text{ and } S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n D_i^2 - \frac{\left(\sum_{i=1}^n D_i \right)^2}{n} \right]$$

After substituting values of $\bar{D}$, S_D and n we get calculated value of test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the **t-table**. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in Section 11.2.

Let us do some examples to become more user friendly with paired t-test.

Example 5: A group of 12 children was tested to find out how many digits they would repeat from memory after hearing them once. They were given practice session for this test. Next week they were retested. The results obtained were as follows:

Child Number	1	2	3	4	5	6	7	8	9	10	11	12
Recall Before	6	4	5	7	6	4	3	7	8	4	6	5
Recall After	6	6	4	7	6	5	5	9	9	7	8	7

Assuming that the memories of the children before and after the practice session follow normal distributions, is the memory practice session improve the performance of children?

Solution: Here, we want to test that memory practice session improve the performance of children. If μ_1 and μ_2 denote the mean digit repetition before and after the practice so our claim is $\mu_1 < \mu_2$ and its complement is $\mu_1 \geq \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

It is a situation of before and after. Also, it is given that the memories of the children before and after the practice session follow normal distributions. So, population of differences will also be normal. Also all the assumptions of paired t-test meet so we can go for paired t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{D}}{S_D / \sqrt{n}} \sim t_{(n-1)} \text{ under } H_0 \quad \dots (3)$$

where, $\bar{D}$ and S_D are mean and standard deviation of the population of differences.

Calculation for $\bar{D}$ and S_D :

Child Number	Digit recall		D = (X-Y)	D ²
	Before (X)	After (Y)		
1	6	6	0	0
2	4	6	-2	4
3	5	4	1	1
4	7	7	0	0
5	6	6	0	0
6	4	5	-1	1
7	3	5	-2	4
8	7	9	-2	4
9	8	9	-1	1
10	4	7	-3	9
11	6	8	-2	4
12	5	7	-2	4
			$\sum D = -14$	$\sum D^2 = 32$

From above calculations, we have

$$\bar{D} = \frac{1}{n} \sum D = \frac{1}{12}(-14) = -1.17$$

$$S_D^2 = \frac{1}{n-1} \left\{ \sum D^2 - \frac{(\sum D)^2}{n} \right\}$$

$$= \frac{1}{11} \left[32 - \frac{(-14)^2}{12} \right] = \frac{1}{11} \times 15.67 = 1.42$$

$$\Rightarrow S_D = \sqrt{1.42} = 1.19$$

Substituting these values in equation (3), we have

$$t = \frac{-1.17}{1.19/\sqrt{12}} = \frac{-1.17}{0.34} = -3.44$$

The critical value of test statistic t for left-tailed test corresponding $(n-1) = 11$ df at 5% level of significance is $-t_{(n-1), \alpha} = -t_{(11), 0.05} = -1.796$.

Since calculated value of test statistic t ($= -3.44$) is less than the critical value ($= -1.796$), that means calculated value of t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support the claim at 5% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that memory practice session improves the performance of children.

Example 6: Ten students were given a test in Statistics and after one month's coaching they were again given a test of the similar nature and the increase in their marks in the second test over the first are shown below:

Roll No.	1	2	3	4	5	6	7	8	9	10
Increase in Marks	6	-2	8	-4	10	2	5	-4	6	0

Assuming that increment in marks follows normal distribution. Do the data indicate that students have gained knowledge from the coaching at 1% level of significance?

Solution: Here, we want to test that students have gained knowledge from the coaching. If μ_D denotes the average increment in the marks due to one month's coaching then our claim is $\mu_D < 0$ but here we are given increment $D_i = (Y_i - X_i)$ instead of $D_i = (X_i - Y_i)$ so we take our claim is $\mu_D > 0$ and its complement is $\mu_D \leq 0$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_D \leq 0 \text{ and } H_1 : \mu_D > 0$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

It is given that increment in marks after one month coaching follows normal distribution and population variance is unknown. Also participants are same in both situations before and after the coaching. And all the assumption of paired t-test meet so we can go for paired t-test.

For testing H_0 , the test statistic is given by

$$t = \frac{\bar{D}}{S_D/\sqrt{n}} \sim t_{(n-1)} \text{ under } H_0 \quad \dots (4)$$

Calculation for $\bar{D}$ and S_D :

Roll No.	1	2	3	4	5	6	7	8	9	10	Total
D	6	-2	8	-4	10	2	5	-4	6	0	$\sum D = 27$
D ²	36	4	64	16	100	4	25	16	36	0	$\sum D^2 = 301$

From above calculations, we have

$$\bar{D} = \frac{1}{n} \sum D = \frac{1}{10} \times 27 = 2.7$$

$$S_D^2 = \frac{1}{n-1} \left\{ \sum D^2 - \frac{(\sum D)^2}{n} \right\}$$

$$= \frac{1}{10-1} \left[301 - \frac{(27)^2}{10} \right] = \frac{1}{9} \times 228.1 = 25.34$$

$$\Rightarrow S_D = \sqrt{25.34} = 5.03$$

Substituting these values in equation (4), we have

$$t = \frac{2.7}{5.03/\sqrt{10}} = \frac{2.7}{1.59} = 1.70$$

The critical value of test statistic t for right-tailed test corresponding $(n-1) = 9$ df at 1% level of significance is $t_{(n-1), \alpha} = t_{(9), 0.01} = 2.821$.

Since calculated value of test statistic $t (= 1.70)$ is less than the critical value $(= 2.821)$, that means calculated value of test statistic t lies in non-rejection region, so we do not reject null hypothesis and reject the alternative hypothesis i.e. we reject our claim at 1% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so students are not gained knowledge from the coaching.

Now, you can try the following exercises.

-
- E6)** To verify whether the programme “Post Graduate Diploma in Applied Statistics (PGDAST)” improved performance of the graduate students in Statistics, a similar test was given to 10 participants both before and after the programme. The original marks out of 100 (before course) recorded in an alphabetical order of the participants are 42, 46, 50, 36, 44, 60, 62, 43, 70 and 53. After the course the marks in the same order are 45, 46, 60, 42, 60, 72, 63, 43, 80 and 65. Assuming that marks of the students before and after the course follow normal distribution. Test whether the programme PGDAST has improved the performance of the graduate students in Statistics at 5% level of significance?
- E7)** A drug is given to 8 patients and the increments in their blood pressure are recorded to be 4, 0, 7, -2, 0, -3, 2, 0. Assume that increment in their blood pressure follows normal distribution. Is it reasonable to believe that the drug has no effect on the change of blood pressure at 5% level of significance?
-

11.6 TESTING OF HYPOTHESIS FOR POPULATION CORRELATION COEFFICIENT USING t-TEST

In Unit 6 of MST-002, we have discussed the concept of correlation. Where, we studied that if two variables are related in such a way that change in the value of one variable affects the value of another variable then the variables are said to be correlated or there is a correlation between these two variables. Correlation can be positive, which means the variables move together in the same direction, or negative, which means they move in opposite directions. And correlation coefficient is used to measure the intensity or degree of linear relationship between two variables. The value of correlation coefficient varies

between -1 and $+1$, where -1 representing a perfect negative correlation, 0 representing no correlation, and $+1$ representing a perfect positive correlation.

Sometime, the sample data indicate for non-zero correlation but in population they are uncorrelated ($\rho = 0$).

For example, price of tomato in Delhi (X) and in London (Y) are not correlated in population ($\rho = 0$). But paired sample data of 20 days of prices of tomato at both places may show correlation coefficient (r) $\neq 0$. In general, in sample data $r \neq 0$ does not ensure in population $\rho \neq 0$ holds.

In this section, we will know how we test the hypothesis that population correlation coefficient is zero.

Assumptions

This test works under following assumptions:

- (i) The characteristic under study follows normal distribution in both the populations. In other words, both populations from which random samples are drawn should be normal with respect to the characteristic of interest.
- (ii) Samples observations are random.

Let us consider a random sample $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ of size n taken from a bivariate normal population. Let ρ and r be the correlation coefficients of population and sample data respectively.

Here, we wish to test the hypothesis about population correlation coefficient (ρ), that is, linear correlation between two variables X and Y in the population, so we can take the null hypothesis as

$$H_0 : \rho = 0 \text{ and } H_1 : \rho \neq 0 \text{ [for two-tailed test] } \left[\begin{array}{l} \text{Here, } \theta = \rho \text{ and } \theta_0 = 0 \text{ if} \\ \text{we compare it with general} \\ \text{procedure given in Section 11.2.} \end{array} \right]$$

or

$$\left. \begin{array}{l} H_0 : \rho \leq 0 \text{ and } H_1 : \rho > 0 \\ H_0 : \rho \geq 0 \text{ and } H_1 : \rho < 0 \end{array} \right\} \text{ [for one-tailed test]}$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{(n-2)}$$

which follows t -distribution with $n - 2$ degrees of freedom.

After substituting values of r and n , we find out calculated value of test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the **t-table**. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in Section 11.2.

Let us do some examples of testing of hypothesis that population correlation coefficient is zero.

Example 7: A random sample of 18 pairs of observations from a normal population gave a correlation coefficient of 0.7. Test whether the population correlation coefficient is zero at 5% level of significance.

Solution: Given that

$$n = 18, r = 0.7$$

Here, we wish to test that population correlation coefficient (ρ) is zero so our claim is $\rho = 0$ and its complement is $\rho \neq 0$. Since the claim contains the

equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \rho = 0 \text{ and } H_1 : \rho \neq 0$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding population correlation coefficient is zero and the populations under study follow normal distributions, so we can go for t-test.

For testing the null hypothesis, the test statistic t is given by

$$\begin{aligned} t &= \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \\ &= \frac{0.7\sqrt{18-2}}{\sqrt{1-(0.7)^2}} = \frac{0.7 \times 4}{\sqrt{0.51}} = \frac{2.8}{0.71} = 3.94 \end{aligned}$$

The critical value of test statistic t for two-tailed test corresponding $(n-2) = 16$ df at 5% level of significance are $\pm t_{(n-2), \alpha/2} = \pm t_{(16), 0.025} = \pm 2.120$.

Since calculated value of test statistic t (= 3.94) is greater than the critical values (= ± 2.120), that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis. i.e. we reject the claim at 5% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so there exists a relationship between two variables.

Example 8: A random sample of 15 married couples was taken from a population consisting of married couples between the ages of 30 and 40. The correlation coefficient between the IQs of husbands and wives was found to be 0.68. Assuming that the IQs of husbands and wives follow normal distributions then test that IQs of husbands and wives in the population are positively correlated at 1% level of significance.

Solution: Given that

$$n = 15, r = 0.68$$

Here, we wish to test that IQs of husbands and wives in the population are positively correlated. If ρ denote the correlation coefficient between IQs of husbands and wives in the population then the claim is $\rho > 0$ and its complement is $\rho \leq 0$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \rho \leq 0 \text{ and } H_1 : \rho > 0$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis regarding population correlation coefficient is zero and the populations under study follow normal distributions, so we can go for t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$= \frac{0.68\sqrt{15-2}}{\sqrt{1-(0.68)^2}} = \frac{0.68 \times 3.61}{0.73} = 3.36$$

The critical value of test statistic t for right-tailed test corresponding $(n-2) = 13$ df at 1% level of significance is $t_{(n-2), \alpha} = t_{(13), 0.01} = 2.650$.

Since calculated value of test statistic t ($= 3.36$) is greater than the critical value ($= 2.650$), that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. we support our claim at 1% level of significance.

Thus, we conclude that sample fail to provide us sufficient evidence against the claim so we may assume that the correlation between IQs of husbands and wives in the population is positive.

In the same way, you can try the following exercise.

E8) Twenty families were selected randomly from a colony to determine that correlation exists between family income and the amount of money spent per family member on food each month. The sample correlation coefficient was computed as $r = 0.40$. Assuming that the family income and the amount of money spent per family member on food each month follow normal distributions then test that there is a positive linear relationship between the family income and the amounts of money spent per family member on food each month in colony at 1% level of significance.

We now end this unit by giving a summary of what we have covered in it.

11.7 SUMMARY

In this unit, we have discussed the following points:

1. Need of small sample tests.
 2. Procedure of testing a hypothesis for t-test.
 3. Testing of hypothesis for population mean using t-test.
 4. Testing of hypothesis for difference of two population means when samples are independent using t-test.
 5. The procedure of paired t-test for testing of hypothesis for difference of two population means when samples are dependent or paired.
 6. Testing of hypothesis for population correlation coefficient using t-test.
-

11.8 SOLUTIONS /ANSWERS

E1) Since calculated value of test statistic t is based on 15 df therefore, we use row for 15 df in the **t-table** and move across this row to find the values in which calculated t -value lies. Since calculated t -value falls between 2.131 and 2.602, which are corresponding to the values of one-tailed area $\alpha = 0.025$ and 0.01 respectively, therefore p -value will lie between 0.01 and 0.025, that is, $0.01 < p\text{-value} < 0.025$

Since test is two-tailed, therefore, the values are doubled, so

$$0.02 = 2 \times 0.01 < p\text{-value} < 2 \times 0.025 = 0.05$$

E2) Here, we are given that

Testing of Hypothesis

$$n = 16, \mu_0 = 18000, \bar{X} = 20000, S = 6000$$

Here, we want to test that manufacturer's claim is true that the average life (μ) of tyres is 18000 km. So claim is $\mu = 18000$ and its complement is $\mu \neq 18000$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0: \mu = \mu_0 = 18000 \quad [\text{average life of tyres is 18000 km}]$$

$$H_1: \mu \neq 18000 \quad [\text{average life of tyres is not 18000 km}]$$

Here, population SD is unknown and population under study is given to be normal. So we can go for t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} = \frac{20000 - 18000}{6000 / \sqrt{16}} = \frac{2000}{1500} = 1.33$$

The critical value of test statistic t for two-tailed test corresponding $(n-1) = 15$ df at 1% level of significance are $\pm t_{(15), 0.005} = \pm 2.947$.

Since calculated value of test statistic t (= 1.33) is less than the critical (tabulated) value (= 2.947) and greater than critical value (= -2.947), that means calculated value of test statistic lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the manufacturer's claim at 1% level of significance.

Thus, we conclude that sample fails to provide sufficient evidence against the claim so we may assume that manufacturer's claim is true.

E3) Here, we want to test that the average weight (μ) of all cadets of the centre is 60 kg. So our claim is $\mu = 60$ and its complement is $\mu \neq 60$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0: \mu = \mu_0 = 60 [\text{average weight of all cadets is 60 kg}]$$

$$H_1: \mu \neq 60 \quad [\text{average weight of all cadets is not 60 kg}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, population SD is unknown and population under study is given to be normal. So we can go for t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \sim t_{(n-1)} \text{ under } H_0 \quad \dots (5)$$

Before moving further, first we have to calculate value of $\bar{X}$ and S.

Calculation for $\bar{X}$ and S:

Sample value (X)	$(x - \bar{X})$	$(x - \bar{X})^2$
48	-15	225
50	-13	169
62	-1	1
75	12	144
80	17	289
60	-3	9
70	7	49

56	-7	49
52	-11	121
77	14	196
$\sum X = 630$		$\sum (X - \bar{X})^2 = 1252$

From the above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{10} \times 630 = 63$$

$$S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2$$

$$= \frac{1}{10-1} \times 1252 = 139.11$$

$$\Rightarrow S = \sqrt{139.11} = 11.79$$

Putting the values in equation (5), we have

$$t = \frac{63 - 60}{11.79 / \sqrt{10}}$$

$$= \frac{3}{3.73} = 0.80$$

The critical values of test statistic t for two-tailed test corresponding $(n-1) = 9$ df at 5% level of significance are $\pm t_{(9), 0.025} = \pm 2.262$.

Since calculated value of test statistic t ($= 0.80$) is less than the critical value ($= 2.262$) and greater than the critical value ($= -2.262$), that means calculated value of test statistic t lies in non-rejection region so we do not reject H_0 i.e. we support the claim at 5% level of significance.

Thus, we conclude that sample fails to provide sufficient evidence against the claim so we may assume that the average weight of all the cadets of given centre is 60 kg.

- E4)** Here, we want to test that there is no difference between drugs A and B with regard to their mean weight increment. If μ_1 and μ_2 denote the mean weight increment due to drug A and drug B respectively then our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_1 = \mu_2 \quad [\text{effect of both drugs is same}]$$

$$H_1 : \mu_1 \neq \mu_2 \quad [\text{effect of both drugs is not same}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Since it is given that increments in the weight due to both drugs follow normal distributions with equal and unknown variances and other assumptions of t-test for testing a hypothesis about difference of two population means also meet. So we can go for this test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)} \text{ under } H_0 \quad \dots (6)$$

Assume, $a = 8$, $b = 12$ and use short-cut method to find $\bar{X}$, $\bar{Y}$ and S_p .

Calculation for $\bar{X}$, $\bar{Y}$ and S_p :

Drug A			Drug B		
X	$d_1 = (X-a)$ $a = 8$	d_1^2	Y	$d_2 = (Y-b)$ $b = 12$	d_2^2
5	-3	9	9	-3	9
8	0	0	10	-2	4
7	-1	1	15	3	9
10	2	4	12	0	0
9	1	1	14	2	4
6	-2	4	8	-4	16
			12	0	0
	$\sum d_1 = -3$	$\sum d_1^2 = 19$		$\sum d_2 = -4$	$\sum d_2^2 = 42$

From above calculation, we have

$$\bar{X} = a + \frac{1}{n_1} \sum d_1 = 8 + \frac{1}{6}(-3) = 7.5,$$

$$\bar{Y} = b + \frac{1}{n_2} \sum d_2 = 12 + \frac{1}{7}(-4) = 11.43$$

$$\begin{aligned}
 S_p^2 &= \frac{1}{n_1 + n_2 - 2} \left[\left\{ \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} \right\} + \left\{ \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} \right\} \right] \\
 &= \frac{1}{6 + 7 - 2} \left[\left\{ 19 - \frac{(-3)^2}{6} \right\} + \left\{ 42 - \frac{(-4)^2}{7} \right\} \right] \\
 &= \frac{1}{11} (17.5 + 39.71) = 5.20
 \end{aligned}$$

$$\Rightarrow S_p = \sqrt{5.20} = 2.28$$

Putting these values in equation (6), we have

$$t = \frac{7.5 - 11.43}{2.28 \sqrt{\frac{1}{6} + \frac{1}{7}}} = \frac{-3.93}{2.28 \times 0.56} = \frac{-3.93}{1.28} = -3.07$$

The critical values of test statistic t for two-tailed test corresponding $(n_1 + n_2 - 2) = 11$ df at 5% level of significance are $\pm t_{(11), 0.025} = \pm 2.201$.

Since calculated value of test statistic t ($= -3.07$) is less than the critical values ($= \pm 2.201$) that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so drugs A and B differ significantly. Any one of them is better than other.

E5) Here, we are given that

$$n_1 = 13, \bar{X} = 4.6, S_1 = 0.5,$$

$$n_2 = 13, \bar{Y} = 5.0, S_2 = 0.3$$

Therefore, the pooled variance S_p^2 can be calculated as

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} [(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]$$

$$= \frac{1}{13 + 13 - 2} [12 \times (0.5)^2 + 12 \times (0.3)^2]$$

$$= \frac{1}{24} (3.00 + 1.08) = 0.17$$

$$\Rightarrow S_p = \sqrt{0.17} = 0.41$$

We want to test that there is significant improvement in wheat production due to fertilizer. If μ_1 and μ_2 denote the mean wheat productions without and with the fertilizer respectively then our claim is $\mu_1 < \mu_2$ and its complement is $\mu_1 \geq \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

Since it is given that yield of wheat with and without fertilizer follow normal distributions with equal and unknown variances and other assumptions of t-test for testing a hypothesis about difference of two population means also meet. So we can go for this test.

For testing the null hypothesis, the test statistic t is given by

$$\begin{aligned} t &= \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1 + n_2 - 2)} \quad \text{under } H_0 \\ &= \frac{4.6 - 5.0}{0.41 \sqrt{\frac{1}{13} + \frac{1}{13}}} = \frac{-0.4}{0.41 \times 0.39} = \frac{-0.4}{0.16} = -2.5 \end{aligned}$$

The critical values of test statistic t for left-tailed test corresponding $(n_1 + n_2 - 2) = 24$ df at 1% level of significance is $-t_{(24), 0.01} = -2.492$.

Since calculated value of test statistic t ($= -2.5$) is less than the critical value ($= -2.492$), that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. we support our claim at 1% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that there is significant improvement in wheat production due to fertilizer.

- E6)** Here, we want to test whether the programme PGDAST has improved the performance of the graduate students in Statistics. If μ_1 and μ_2 denote the average marks before and after the programme so our claim is $\mu_1 < \mu_2$ and its complement is $\mu_1 \geq \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \mu_1 \geq \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

It is a situation of before and after. Also, the marks of the students before and after the programme PGDAST follow normal distributions. So, population of differences will also be normal. Also all the assumptions of paired t-test meet. So we can go for paired t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{D}}{S_D / \sqrt{n}} \sim t_{(n-1)} \text{ under } H_0 \quad \dots (7)$$

Calculation for $\bar{D}$ and S_D :

Participant	Marks		D = (X-Y)	D ²
	Before (X)	After (Y)		
1	42	45	-3	9
2	46	46	0	0
3	50	60	-10	100
4	36	42	-6	36
5	44	60	-16	256
6	60	72	-12	144
7	62	63	-1	1
8	43	43	0	0
9	70	80	-10	100
10	53	65	-12	144
			$\sum D = -70$	$\sum D^2 = 790$

From above calculation, we have

$$\bar{D} = \frac{1}{n} \sum D = \frac{1}{10}(-70) = -7$$

$$S_D^2 = \frac{1}{n-1} \left[\sum D^2 - \frac{(\sum D)^2}{n} \right]$$

$$= \frac{1}{9} \left[790 - \frac{(-70)^2}{10} \right] = \frac{1}{9} \times 300 = 33.33$$

$$\Rightarrow S_D = \sqrt{33.33} = 5.77$$

Putting the values in equation (7), we have

$$t = \frac{-7.0}{5.77/\sqrt{10}} = -3.83$$

The critical value of test statistic t for left-tailed test corresponding $(n-1) = 9$ df at 5% level of significance is $-t_{(9), 0.05} = -1.833$.

Since calculated value of test statistic $t (= -3.83)$ is less than the critical (tabulated) value $(= -1.833)$, that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. we support our claim at 5% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that the participants have significant improvement after the programme “Post Graduate Diploma in Applied Statistics (PGDAST)”.

- E7)** Here, we want to test that the drug has no effect on change in blood pressure. If μ_D denotes the average increment in the blood pressure before drug then our claim is $\mu_D = 0$ and its complement is $\mu_D \neq 0$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \mu_D = \mu_1 - \mu_2 = 0 \text{ [the drug has no effect]}$$

$$H_1 : \mu_D \neq 0 \text{ [the drug has an effect]}$$

It is given that increment in the blood pressure follows normal distribution. Also patients are same in both situations before and after the drug. And all the assumption of paired t-test meet. So we can go for paired t-test.

For testing H_0 , the test statistic is given by

$$t = \frac{\bar{D}}{S_D / \sqrt{n}} \sim t_{(n-1)} \text{ under } H_0 \quad \dots (8)$$

Calculation for $\bar{D}$ and S_D :

Patient Number	1	2	3	4	5	6	7	8	Total
D	4	0	7	-2	0	-3	2	0	$\sum D = 8$
(D - $\bar{D}$)	3	-1	6	-3	-1	-4	1	-1	
(D - $\bar{D}$)²	9	1	36	9	1	16	1	1	$\sum (D - \bar{D})^2 = 74$

From above calculation, we have

$$\bar{D} = \frac{1}{n} \sum D = \frac{1}{8} \times 8 = 1$$

$$S_D^2 = \frac{1}{n-1} \sum (D - \bar{D})^2 = \frac{1}{7} \times 74 = 10.57$$

$$\Rightarrow S_D = \sqrt{10.57} = 3.25$$

Putting the values in test statistic, we have

$$t = \frac{1}{3.25/\sqrt{8}} = \frac{1}{1.15} = 0.87$$

The critical value of test statistic t for two-tailed test corresponding $(n-1) = 7$ df at 5% level of significance are $\pm t_{(7), 0.025} = \pm 2.365$.

Since calculated value of test statistic t ($= 0.87$) is less than the critical value ($= 2.365$) and greater than the critical value ($= -2.365$) that means calculated value of test statistic t lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that the drug has no effect on the change of blood pressure of patients.

E8) We are given that

$$n = 20, r = 0.40$$

and we wish to test that there is a positive linear relationship between the family income and the amounts of money spent per family member on food each month in colony. If ρ denote the correlation coefficient between the family income and the amounts of money spent per family member then the claim is $\rho > 0$ and its complement is $\rho \leq 0$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \rho \leq 0 \text{ and } H_0 : \rho > 0$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

For testing the null hypothesis, the test statistic t is given by

$$\begin{aligned} t &= \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \\ &= \frac{0.40\sqrt{20-2}}{\sqrt{1-(0.40)^2}} = \frac{0.40 \times 4.24}{0.92} = 1.84 \end{aligned}$$

The critical value of test statistic t for right-tailed test corresponding $(n-2) = 18$ df at 1% level of significance is $t_{(n-2), \alpha} = t_{(18), 0.01} = 2.552$.

Since calculated value of test statistic t ($= 1.84$) is less than the critical value ($= 2.552$), that means calculated value of test statistic t lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject our claim at 1% level of significance.

Thus, we conclude that sample provide us sufficient evidence against the claim so there is no positive linear correlation between the family income and the amounts of money spent per family member on food each month in colony.

UNIT 12 CHI-SQUARE AND F-TESTS

Structure

- 12.1 Introduction
 - Objectives
- 12.2 Testing of Hypothesis for Population Variance Using χ^2 -Test
- 12.3 Testing of Hypothesis for Two Population Variances Using F-Test
- 12.4 Summary
- 12.5 Solutions / Answers

12.1 INTRODUCTION

Recall from the previous unit when we test the hypothesis about the difference of means of two populations, t-test needs an assumption of equality of variances of two populations under study. Other than this there are situations where we want to test the hypothesis about the variances of two populations. For example, an economist may want to test whether the variability in incomes differ in two population. In such situations, we use F-test when the populations under study follow the normal distributions.

Similarly, there are many other situations where we need to test the hypothesis about the hypothetical or specified value of the variance of the population under study. For example, the manager of the electric bulbs company would probably be interested whether or not the variability in the life of bulbs is within acceptable limits, the product controller of a milk company may be interested in the variance of the amount of fat in the whole milk processed by the company is no more than the specified level. In such situations, we use χ^2 -test when the population under study follows the normal distribution.

This unit is divided into five sections. Section 12.1 is described the need of χ^2 and F-tests. χ^2 -test for testing the hypothesis about the population variance is discussed in Section 12.2. And F-test for equality of variances of two populations is discussed in Section 12.3. Unit ends by providing summary of what we have discussed in this unit in Section 12.4 and solution of exercises in Section 12.5.

Objectives

After studying this unit, you should be able to:

- describe the testing of hypothesis for population variance; and
- explain the testing of hypothesis for two population variances.

12.2 TESTING OF HYPOTHESIS FOR POPULATION VARIANCE USING χ^2 -TEST

In Section 11.3 of previous unit, we have discussed testing of hypothesis for population mean when the characteristic under study follows the normal distribution but when analysing quantitative data, it is often important to draw conclusion about the average as well as the variability of a characteristic under study. For example, if a company manufactured the electric bulbs then the

manager of the company would probably be interested in determining the average life of the bulbs and also determining whether or not the variability in the life of bulbs is within acceptable limits, the product controller of a milk company may be interested in the variance of the amount of fat in the whole milk processed by the company is no more than the specified level, etc.

The procedure of testing a hypothesis for population variance or standard deviation is similar to the testing of population mean. The basic difference is that here we use chi-square test instead of t-test because here the sampling distribution of test statistic follows the chi-square distribution.

Assumptions

This test works under the following assumptions:

- (i) The characteristic under study follows normal distribution. In other words, populations from which random sample is drawn should be normal with respect to the characteristic of interest.
- (ii) Sample observations are random and independent.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n drawn from a normal population with mean μ and variance σ^2 , where μ and σ^2 are unknown.

The general procedure of this test is explained below in detail:

As we are doing so far in all tests, first step in hypothesis testing problems is to setup null and alternative hypotheses. Here, we want to test the claim about the hypothesized or specified value σ_0^2 of population variance σ^2 so we can take our null and alternative hypotheses as

$$\begin{aligned} &H_0 : \sigma^2 = \sigma_0^2 \text{ and } H_1 : \sigma^2 \neq \sigma_0^2 \quad \left[\text{for two-tailed test} \right] \\ \text{or} \quad &\left. \begin{aligned} &H_0 : \sigma^2 \leq \sigma_0^2 \text{ and } H_1 : \sigma^2 > \sigma_0^2 \\ &H_0 : \sigma^2 \geq \sigma_0^2 \text{ and } H_1 : \sigma^2 < \sigma_0^2 \end{aligned} \right\} \quad \left[\text{for one-tailed test} \right] \end{aligned}$$

For testing the null hypothesis, the test statistic χ^2 is given by

$$\chi^2 = \frac{\sum (X_i - \bar{X})^2}{\sigma_0^2} = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{(n-1)} \text{ under } H_0 \quad \dots (1)$$

$$\text{where, } S^2 = \frac{1}{n-1} \sum (X - \bar{X})^2$$

Here, the test statistic χ^2 follows chi square distribution with $(n-1)$ degrees of freedom as we have discussed in Unit 3 of this course.

After substituting values of n , S and σ_0^2 , we get calculated value of test statistic. Let χ_{cal}^2 be the calculated value of test statistic χ^2 .

Obtain the critical value(s) or cut-off value(s) in the sampling distribution of the test statistic χ^2 and construct rejection (critical) region of size α . The critical value of the test statistic χ^2 for various df and different level of significance α are given in **Table III** of the Appendix given at the end of the Block 1 of this course.

After doing all the calculation discussed above, we have to take the decision about rejection or non-rejection of the null hypothesis. The procedure of taking the decision about the null hypothesis is explained in the next page:

For one-tailed test:

Case I: When $H_0 : \sigma^2 \leq \sigma_0^2$ and $H_1 : \sigma^2 > \sigma_0^2$ (right-tailed test)

In this case, the rejection (critical) region falls under the right tail of the probability curve of the sampling distribution of test statistic χ^2 .

Suppose $\chi_{(v),\alpha}^2$ is the critical value at α level of significance where, $v = n - 1$, so entire region greater than or equal to $\chi_{(v),\alpha}^2$ is the rejection region and less than $\chi_{(v),\alpha}^2$ is the non-rejection region as shown in Fig. 12.1.

If $\chi_{cal}^2 \geq \chi_{(v),\alpha}^2$, that means calculated value of test statistic lies in the rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that sample data provides us sufficient evidence against the null hypothesis and there is a significant difference between hypothesized value and observed value of the population variance σ^2 .

If $\chi_{cal}^2 < \chi_{(v),\alpha}^2$, that means calculated value of test statistic lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that the sample data fails to provide us sufficient evidence against the null hypothesis and the difference between hypothesized value and observed value of the population variance σ^2 due to fluctuation of sample.

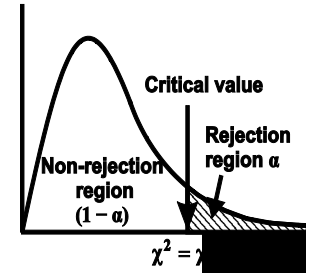


Fig. 12.1

Case II: When $H_0 : \sigma^2 \geq \sigma_0^2$ and $H_1 : \sigma^2 < \sigma_0^2$ (left-tailed test)

In this case, the rejection (critical) region falls under the left tail of the probability curve of the sampling distribution of test statistic χ^2 .

Suppose $\chi_{(v),(1-\alpha)}^2$ is the critical value at α level of significance then entire region less than or equal to $\chi_{(v),(1-\alpha)}^2$ is the rejection (critical) region and greater than $\chi_{(v),(1-\alpha)}^2$ is the non-rejection region as shown in Fig. 12.2.

If $\chi_{cal}^2 \leq \chi_{(v),(1-\alpha)}^2$, that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

For $\chi_{cal}^2 > \chi_{(v),(1-\alpha)}^2$, that means calculated value of test statistic lies in the non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

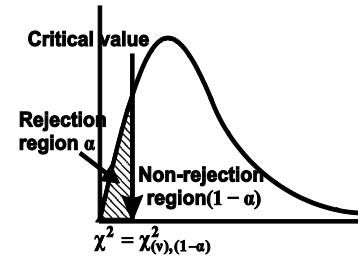


Fig. 12.2

For two-tailed test:

When $H_0 : \sigma^2 = \sigma_0^2$ and $H_1 : \sigma^2 \neq \sigma_0^2$

In this case, the rejection region falls under both tails of the probability curve of sampling distribution of the test statistic χ^2 and half the area (α) i.e. $\alpha/2$ of rejection (critical) region lies at left tail and other half on the right tail. Suppose $\chi_{(v),(1-\alpha/2)}^2$ and $\chi_{(v),\alpha/2}^2$ are the two critical values at the left-tailed and right-tailed respectively on pre-fixed α -level of significance. Therefore, entire region less than or equal to $\chi_{(v),(1-\alpha/2)}^2$ and greater than or equal to $\chi_{(v),\alpha/2}^2$ are the rejection

Testing of Hypothesis

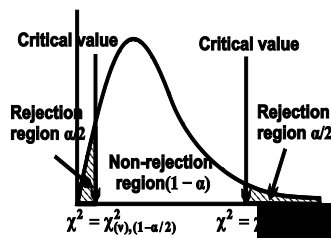


Fig. 12.3

(critical) regions and between $\chi^2_{(v), (1-\alpha/2)}$ and $\chi^2_{(v), \alpha/2}$ is the non-rejection region as shown in Fig. 12.3.

If $\chi^2_{\text{cal}} \geq \chi^2_{(v), \alpha/2}$ or $\chi^2_{\text{cal}} \leq \chi^2_{(v), (1-\alpha/2)}$, that means calculated value of test statistic χ^2 lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

If $\chi^2_{(v), (1-\alpha/2)} < \chi^2_{\text{cal}} < \chi^2_{(v), \alpha/2}$, that means calculated value of test statistic χ^2 lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Procedure of taking the decision about the null hypothesis on the basis of p-value:

You have done it in so many test, it has become a routine thing for you. You know that to take the decision about the null hypothesis on the basis of p-value, the p-value is compared with level of significance (α) and if p-value is less than or equal to α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

For χ^2 -test, p-value is defined as:

For one-tailed test:

For $H_1 : \sigma^2 > \sigma_0^2$ (right-tailed test)

$$\text{p-value} = P[\chi^2 \geq \chi^2_{\text{cal}}]$$

For $H_1 : \sigma^2 < \sigma_0^2$ (left-tailed test)

$$\text{p-value} = P[\chi^2 \leq \chi^2_{\text{cal}}]$$

For two-tailed test: $H_1 : \sigma^2 \neq \sigma_0^2$

For two-tailed test the p-value is approximated as

$$\text{p-value} = 2P[\chi^2 \geq \chi^2_{\text{cal}}]$$

The p-value for χ^2 -test can be obtained with the help of the **Table-III (χ^2 -table)** given in the Appendix at the end of Block 1 of this course. Similar to t-test, this table gives the χ^2 values corresponding to the standard values of α such as 0.995, 0.99, 0.10, 0.05, 0.025, 0.01, etc only. therefore, the exact p-value is not obtained with the help of this table and we can approximate the p-value for this test.

For example, if test is right-tailed and calculated (observed) value of test statistic χ^2 is 25.10 with 12 df then p-value is calculated as:

Since test statistic is based on the 12 df therefore, we use row for 12 df in the χ^2 -table and move across this row to find the values in which calculated χ^2 -value falls. Since calculated χ^2 -value falls between 23.24 and 26.22, corresponding to one-tailed area $\alpha = 0.025$ and 0.01 respectively, therefore p-value lies between 0.01 and 0.025, that is,

$$0.01 < \text{p-value} < 0.025$$

If in the above example the test is two-tailed then the two values 0.01 and 0.005 would be doubled for p-value, that is,

$$2 \times 0.01 = 0.02 < \text{p-value} < 0.05 = 2 \times 0.025$$

Note 1: With the help of computer packages and softwares such as SPSS, SAS, MINITAB, EXCEL, etc. we can find the exact p-value for χ^2 -test.

Let us do some examples to become more user friendly with the test explained above.

Example 1: The variance of a certain dimension article produced by a machine is 7.2 over a long period. A random sample of 20 articles gave a variance 8. Is it justifiable to conclude that variability has increased at 5% level of significance assuming that the measurement of dimension article is normally distributed?

Solution: Here, we are given that

Sample size = $n = 20$

Sample variance = $S^2 = 8$

Specified value of population variance under test = $\sigma_0^2 = 7.2$

Here, we want to test that variability of dimension article produced by a machine has increased. Since variability is measured in terms of variance (σ^2) so our claim is $\sigma^2 > 7.2$ and its complement is $\sigma^2 \leq 7.2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \sigma^2 \leq \sigma_0^2 = 7.2$$

$$H_1 : \sigma^2 > 7.2 \quad [\text{variability of dimension article has increased}]$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis about the population variance and sample size is small $n = 20 (< 30)$. Also we are given that measurement of dimension article follows normal distribution so we can go for χ^2 test for population variance.

So, test statistic is given by

$$\begin{aligned} \chi^2 &= \frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2 \text{ under } H_0 \\ &= \frac{19 \times 8}{7.2} = 21.11 \end{aligned}$$

The critical (tabulated) value of test statistic χ^2 for right-tailed test corresponding $(n-1) = 19$ df at 5% level of significance is

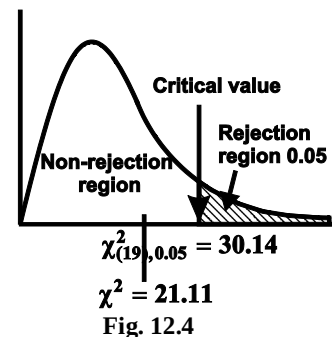
$$\chi_{(n-1), \alpha}^2 = \chi_{(19), 0.05}^2 = 30.14.$$

Since calculated value of test statistic (= 21.11) is less than the critical (tabulated) value (= 30.14), that means calculated value of test statistic lies in non-rejection region as shown in Fig. 12.4, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject our claim at 5% level of significance.

Decision according to p-value:

Since test statistic is based on 19 df therefore, we use row for 19 df in the χ^2 -table and move across this row to find the values in which calculated χ^2 -value falls. Since calculated χ^2 -value falls between 11.65 and 27.20, corresponding to one-tailed area $\alpha = 0.90$ and 0.10 respectively therefore p-value lies between 0.10 and 0.90, that is,

$$0.10 < \text{p-value} < 0.90$$



Since p-value is greater than $\alpha (= 0.05)$ so we do not reject the null hypothesis at 5% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so we may assume that the variability of dimension article produced by a machine is not increased.

Example 2: The 12 measurements of the same object on an instrument are given below:

1.6, 1.5, 1.3, 1.5, 1.7, 1.6, 1.5, 1.4, 1.6, 1.3, 1.5, 1.5

If the measurement of the instrument follows normal distribution then carry out the test at 1% level of significance that variance in the measurement of the instrument is less than 0.016.

Solution: Here, we are given

Sample size = $n = 12$

Specified value of population variance under test $\sigma_0^2 = 0.016$

Here, we want to test that the variance (σ^2) in the measurements of the instrument is less than 0.016. So our claim is $\sigma^2 < 0.016$ and its complement is $\sigma^2 \geq 0.016$. Since complement contains the equality sign so we can take complement as null hypothesis and claim as the alternative hypothesis. Thus,

$$H_0 : \sigma^2 \geq \sigma_0^2 = 0.016 \text{ and } H_1 : \sigma^2 < 0.016$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

Here, we want to test the hypothesis about the population variance and sample size is small $n = 12 (< 30)$. Also we are given that the measurement of the instrument follows normal distribution so we can go for χ^2 -test for population variance.

For testing the null hypothesis, the test statistic χ^2 is given by

$$\chi^2 = \frac{\sum (X_i - \bar{X})^2}{\sigma_0^2} \sim \chi^2_{(n-1)} \text{ under } H_0 \quad \dots (2)$$

Calculation for $\sum (X_i - \bar{X})^2$:

X	$(X - \bar{X})$	$(X - \bar{X})^2$
1.6	0.1	0.01
1.5	0	0
1.3	-0.2	0.04
1.5	0	0
1.7	0.2	0.04
1.6	0.1	0.01
1.5	0	0
1.4	-0.1	0.01
1.6	0.1	0.01
1.3	-0.2	0.04
1.5	0	0
1.5	0	0
$\sum X = 18$	0	$\sum (X - \bar{X})^2 = 0.16$

From above calculation, we have

$$\bar{X} = \frac{1}{n} \sum X = \frac{1}{12} \times 18 = 1.5$$

Putting the values of $\sum (X - \bar{X})^2$ and σ_0^2 in equation (2), we have

$$\chi^2 = \frac{\sum (X_i - \bar{X})^2}{\sigma_0^2} = \frac{0.16}{0.016} = 10$$

The critical value of test statistic χ^2 for left-tailed test corresponding $(n-1) = 11$ df at 5% level of significance is $\chi_{(n-1), (1-\alpha)}^2 = \chi_{(11), 0.95}^2 = 4.57$.

Since calculated value of test statistic (= 10) is greater than the critical value (= 4.57), that means calculated value of test statistic lies in non-rejection region as shown in Fig. 12.5 so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that sample provide sufficient evidence against the claim so the variance in the measurement of the instrument is not less than 0.016.

In the same way, you can try the following exercises.

-
- E1)** An ambulance agency claims that the standard deviation in the length of serving times is less than 15 minutes. Investigator suspects that this claim is wrong and takes a random sample of 20 serving times which has a standard deviation of 17 minutes. Assume that the service time of the ambulance follows normal distribution. Test at $\alpha = 0.01$, is there enough evidence to reject the agency's claim?
- E2)** A cigarette manufacturer claims that the variance of nicotine content of its cigarettes is 0.62. Nicotine content is measured in milligrams and is normally distributed. A sample of 25 cigarettes has a variance of 0.65. Test the manufacturer's claim at 5% level of significance.
-

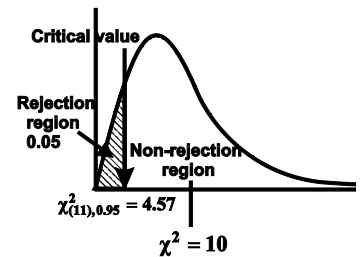


Fig. 12.5

12.5 TESTING OF HYPOTHESIS FOR TWO POPULATION VARIANCES USING F-TEST

In Section 12.1 we have already mentioned that before applying t-test for difference of two population means, one of the requirements is to check the equality of variances of two populations. This assumption can be checked with the help of F-test for two population variances. This F-test is also important in a number of contexts. For example, an economist may want to test whether the variability in incomes differ in two populations, a quality controller may want to test whether the quality of the product is changing over time, etc.

Assumptions

The assumptions for F-test for testing the variances of two populations are:

1. The populations from which the samples are drawn must be normally distributed.
2. The samples must be independent of each other.

Now, we come to the general procedure of this test.

Some author uses this test as the name "homoscedastic test". Because two populations with equal variances are called homoscedastic. This word is derived from two words, homo means "the same" and scedastic means "variability".

Testing of Hypothesis

Let $X_1, X_2, \dots, X_{n_1}$ be a random sample of size n_1 from a normal population with mean μ_1 and variance σ_1^2 . Similarly, $Y_1, Y_2, \dots, Y_{n_2}$ be a random sample of size n_2 from another normal population with mean μ_2 and variance σ_2^2 . Here, we want to test the hypothesis about the two population variances so we can take our alternative null and hypotheses as

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma^2 \text{ and } H_1 : \sigma_1^2 \neq \sigma_2^2 \quad [\text{for two-tailed test}]$$

$$\text{or} \quad \left. \begin{array}{l} H_0 : \sigma_1^2 \leq \sigma_2^2 \text{ and } H_1 : \sigma_1^2 > \sigma_2^2 \\ H_0 : \sigma_1^2 \geq \sigma_2^2 \text{ and } H_1 : \sigma_1^2 < \sigma_2^2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

For testing the null hypothesis, the test statistic F is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)} \quad \text{under } H_0 \quad \dots (3)$$

$$\text{where, } S_1^2 = \frac{1}{n_1 - 1} \sum (X - \bar{X})^2 \text{ and } S_2^2 = \frac{1}{n_2 - 1} \sum (Y - \bar{Y})^2.$$

For computational simplicity, we can also write

$$S_1^2 = \frac{1}{n_1 - 1} \left(\sum X^2 - \frac{(\sum X)^2}{n_1} \right) \text{ and } S_2^2 = \frac{1}{n_2 - 1} \left(\sum Y^2 - \frac{(\sum Y)^2}{n_2} \right)$$

The test statistic F follows F-distribution with $v_1 = (n_1 - 1)$ and $v_2 = (n_2 - 1)$ degrees of freedom as discussed in Unit 4 of this course.

After substituting the values of S_1^2 and S_2^2 , we get calculated value of test statistic. Let F_{cal} be calculated value of test statistic F.

Obtain the critical value(s) or cut-off value(s) in the sampling distribution of the test statistic F and construct rejection (critical) region of size α . The critical values of the test statistic F for various df and different level of significance α are given in **Table IV (F-table)** of the Appendix at the end of Block 1 of this course.

After doing all calculations discussed above, we have to take the decision about rejection or non rejection of the null hypothesis. This is explained below:

In case of one-tailed test:

Case I: When $H_0 : \sigma_1^2 \leq \sigma_2^2$ and $H_1 : \sigma_1^2 > \sigma_2^2$ (right-tailed test)

In this case, the rejection (critical) region falls at the right side of the probability curve of the sampling distribution of test statistic F.

Suppose $F_{(v_1, v_2), \alpha}$ is the critical value of test statistic F with $(v_1 = n_1 - 1, v_2 = n_2 - 1)$ df at α level of significance so entire region greater than or equal to $F_{(v_1, v_2), \alpha}$ is the rejection (critical) region and less than $F_{(v_1, v_2), \alpha}$ is the non-rejection region as shown in Fig. 12.6.

If $F_{cal} \geq F_{(v_1, v_2), \alpha}$, that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that samples data

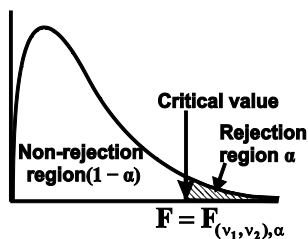


Fig. 12.6

provide us sufficient evidence against the null hypothesis and there is a significant difference between population variances.

If $F_{cal} < F_{(v_1, v_2), \alpha}$, that means calculated value of test statistic lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that the samples data fail to provide us sufficient evidence against the null hypothesis and the difference between population variances due to fluctuation of sample.

Case II: When $H_0 : \sigma_1^2 \geq \sigma_2^2$ and $H_1 : \sigma_1^2 < \sigma_2^2$ (left-tailed test)

In this case, the rejection (critical) region falls at the left side of the probability curve of the sampling distribution of test statistic F .

Suppose $F_{(v_1, v_2), (1-\alpha)}$ is the critical value at α level of significance then entire region less than or equal to $F_{(v_1, v_2), (1-\alpha)}$ is the rejection (critical) region and greater than $F_{(v_1, v_2), (1-\alpha)}$ is the non-rejection region as shown in Fig. 12.7.

If $F_{cal} \leq F_{(v_1, v_2), (1-\alpha)}$, that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

If $F_{cal} > F_{(v_1, v_2), (1-\alpha)}$, that means calculated value of test statistic lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Note 2: F-table mentioned above gives only the right tailed critical values with different degrees of freedom at different level of significance. The left tailed critical value of F-test can always be obtained by the formula given below (as described in Unit 4 of this course):

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}}$$

In case of two-tailed test:

When $H_0 : \sigma_1^2 = \sigma_2^2$ and $H_1 : \sigma_1^2 \neq \sigma_2^2$

In this case, the rejection (critical) region falls at both sides of the probability curve of the sampling distribution of test statistic F and half the area (α) i.e. $\alpha/2$ of rejection (critical) region lies at left tail and other half on the right tail.

Suppose $F_{(v_1, v_2), (1-\alpha/2)}$ and $F_{(v_1, v_2), \alpha/2}$ are the two critical values at the left-tailed and right-tailed respectively on pre-fixed α level of significance. Therefore, entire region less than or equal to $F_{(v_1, v_2), (1-\alpha/2)}$ and greater than or equal to $F_{(v_1, v_2), \alpha/2}$ are the rejection (critical) regions and between $F_{(v_1, v_2), (1-\alpha/2)}$ and $F_{(v_1, v_2), \alpha/2}$ is the non-rejection region as shown in Fig. 12.8.

If $F_{cal} \geq F_{(v_1, v_2), \alpha/2}$ or $F_{cal} \leq F_{(v_1, v_2), (1-\alpha/2)}$, that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

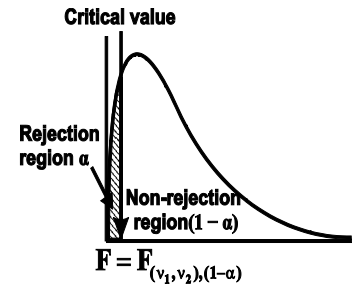


Fig. 12.7

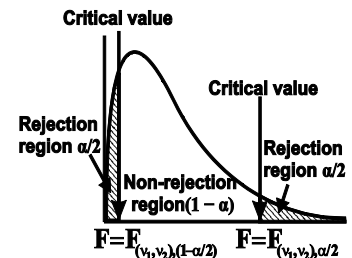


Fig. 12.8

If $F_{(v_1, v_2), (1-\alpha/2)} < F_{\text{cal}} < F_{(v_1, v_2), \alpha/2}$, that means calculated value of test statistic F lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Procedure of taking the decision about the null hypothesis on the basis of p-value:

To take the decision about the null hypothesis on the basis of p-value, the p-value is compared with level of significance (α) and if p-value is less than or equal to α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

For F-test, p-value can be defined as:

For one-tailed test:

For $H_1 : \sigma_1^2 > \sigma_2^2$ (right-tailed test)

$$\text{p-value} = P[F \geq F_{\text{cal}}]$$

For $H_1 : \sigma_1^2 < \sigma_2^2$ (left-tailed test)

$$\text{p-value} = P[F \leq F_{\text{cal}}]$$

For two-tailed test: $H_1 : \sigma_1^2 \neq \sigma_2^2$

For two-tailed test the p-value is approximated as

$$\text{p-value} = 2P[F \geq F_{\text{cal}}]$$

The p-value for F-test can be obtained with the help of **Table-IV (F-table)** given in the Appendix at the end of Block 1 of this course. Similar to t-test or χ^2 -test, this table gives F values corresponding to the standard values of α such as 0.10, 0.05, 0.025 and 0.01 only. Therefore with the help of F-table, we cannot find the exact p-value. So we can approximate p-value for this test.

For example, if test is right-tailed and calculated (observed) value of test statistic F is 2.65 with 24 degrees of freedom of numerator and 14 degrees of freedom of the denominator then we can find the p-value as:

Since test statistic is based on (24, 14) df therefore, we move across all the values of tabulated F corresponding to (24, 14) df at α such as 0.10, 0.05, 0.025 and 0.01 and find the values in which calculated F -value falls. Since we have F tabulated with (24, 14) df at $\alpha = 0.10, 0.05, 0.025$ and 0.01 as

$\alpha =$	0.10	0.05	0.025	0.01
$F_{(24, 14), \alpha}$	1.94	2.35	2.79	3.43

Since calculated F -value (= 2.65) falls between 2.35 and 2.79, corresponding to one-tailed area $\alpha = 0.05$ and 0.025 respectively therefore p-value lies between 0.0025 and 0.05, that is,

$$0.025 < \text{p-value} < 0.05$$

If in the above example the test is two-tailed then the two values 0.025 and 0.05 would be doubled for p-value, that is,

$$0.05 < \text{p-value} < 0.10$$

Note 3: With the help of computer packages and software such as SPSS, SAS, MINITAB, EXCEL, etc. we can find the exact p-value for F-test.

Let us do some examples based on this test.

Example 3: The following data relate to the number of items produced in a shift by two workers A and B for some days:

A	26	37	40	35	30	30	40	26	30	35	45
B	19	22	24	27	24	18	20	19	25		

Assuming that the parent populations are normal, can it be inferred that B is more stable (or consistent) worker compared to A?

Solution: Here, we want to test that worker B is more stable than worker A. As we know that stability of data is related to variance of the data. Smaller value of the variance implies data that it is more stable. Therefore, to compare stability of two workers, it is enough to compare their variances. If σ_1^2 and σ_2^2 denote the variances of worker A and worker B respectively then our claim is $\sigma_1^2 > \sigma_2^2$ and its complement is $\sigma_1^2 \leq \sigma_2^2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \sigma_1^2 \leq \sigma_2^2$$

$$H_1 : \sigma_1^2 > \sigma_2^2 \text{ [worker B is more stable than worker A]}$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis about two population variances and sample sizes $n_1 = 11 (< 30)$ and $n_2 = 9 (< 30)$ are small. Also populations under study are normal and both samples are independent so we can go for F-test for two population variances.

For testing the null hypothesis, test statistic is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)} \quad \dots (4)$$

$$\text{where, } S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \text{ and } S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

Calculation for S_1^2 and S_2^2 :

Items Produced by A (Variable X)	$(X - \bar{X})$ $= (X - 34)$	$(X - \bar{X})^2$	Items Produced by B (Variable Y)	$(Y - \bar{Y})$ $= (Y - 22)$	$(Y - \bar{Y})^2$
26	-8	64	19	-3	9
37	3	9	22	0	0
40	6	36	24	2	4
35	1	1	27	5	25
30	-4	16	24	2	4
30	-4	16	18	-4	16
40	6	36	20	-2	4
26	-8	64	19	-3	9
30	-4	16	25	3	9
35	1	1			
45	11	121			
Total = 374	0	380	198	0	80

Testing of Hypothesis

Therefore, we have

$$\bar{X} = \frac{1}{n_1} \sum X = \frac{1}{11} \times 374 = 34$$

and

$$\bar{Y} = \frac{1}{n_2} \sum Y = \frac{1}{9} \times 198 = 22$$

Thus,

$$S_1^2 = \frac{1}{n_1 - 1} \sum (X - \bar{X})^2 = \frac{1}{10} \times 380 = 38$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (Y - \bar{Y})^2 = \frac{1}{8} \times 80 = 10$$

Putting the value of S_1^2 and S_2^2 in equation (4), we have

$$F = \frac{38}{10} = 3.8$$

The critical (tabulated) value of test statistic F for right-tailed test corresponding $(n_1 - 1, n_2 - 1) = (10, 8)$ df at 1% level of significance is

$$F_{(n_1-1, n_2-1), \alpha} = F_{(10, 8), 0.01} = 5.81.$$

Since calculated value of test statistic (= 3.8) is less than the critical value (= 5.81), that means calculated value of test statistic lies in rejection region as shown in Fig. 12.9, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so worker B is not more stable (or consistent) worker compared to A.

Example 4: Two random samples drawn from two normal populations gave the following results:

Sample	Size	Mean	Sum of Squares of Deviation from the Mean
Sample I	9	59	26
Sample II	11	60	32

Test whether both samples are from the same normal populations?

Solution: Since we have to test whether both the samples are from same normal population, therefore, we will test two hypotheses separately:

(i) Two population means are equal, i.e. $H_0 : \mu_1 = \mu_2$

(ii) Two population variances are equal, i.e. $H_0 : \sigma_1^2 = \sigma_2^2$

Since sample sizes are small and populations under study are normal so two means will be tested using t-test whereas two variances will be tested using F-test. But t-test is based on the prior assumption that both population variances are same, therefore, first we apply F-test and later the t-test (when F-test accepts equality hypothesis).

Given that

$$n_1 = 9, \quad \bar{X} = 59, \quad \sum (X - \bar{X})^2 = 26$$

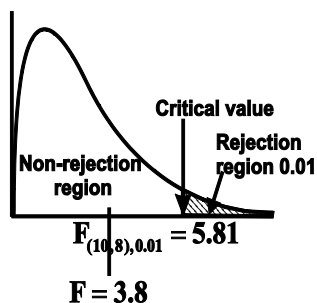


Fig. 12.9

$$n_2 = 11, \quad \bar{Y} = 60, \quad \sum (Y - \bar{Y})^2 = 32$$

Therefore,

$$S_1^2 = \frac{1}{n_1 - 1} \sum (X - \bar{X})^2 = \frac{1}{9 - 1} \times 26 = 3.25$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (Y - \bar{Y})^2 = \frac{1}{11 - 1} \times 36 = 3.60$$

First we want to test that the variances of both normal populations are equal so our claim is $\sigma_1^2 = \sigma_2^2$ and its complement is $\sigma_1^2 \neq \sigma_2^2$. Thus, we can take the null and alternative hypotheses as

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ and } H_1 : \sigma_1^2 \neq \sigma_2^2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

For testing this, the test statistic F is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{(n_1 - 1, n_2 - 1)}$$

$$= \frac{3.20}{3.65} = 0.88$$

The critical (tabulated) value of test statistic F for two-tailed test corresponding $(n_1 - 1, n_2 - 1) = (8, 11)$ df at 5% level of significance are $F_{(n_1 - 1, n_2 - 1), \alpha/2} =$

$$F_{(8, 11), 0.025} = 3.66 \text{ and } F_{(n_1 - 1, n_2 - 1), (1 - \alpha/2)} = \frac{1}{F_{(n_2 - 1, n_1 - 1), \alpha/2}} = \frac{1}{F_{(11, 8), 0.025}} = \frac{1}{4.24} = 0.24.$$

Since calculated value of test statistic (= 0.88) is less than the critical value (= 3.66) and greater than the critical value (= 0.24), that means calculated value of test statistic F lies in non-rejection region so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that both samples may be taken from normal populations having equal variances.

Now, we test that the means of two normal populations are equal so our claim is $\mu_1 = \mu_2$ and its complement is $\mu_1 \neq \mu_2$. Thus, we can take the null and alternative hypotheses as

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 \neq \mu_2$$

The test statistic is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \dots (5)$$

$$\text{where, } S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (X - \bar{X})^2 + \sum (Y - \bar{Y})^2 \right]$$

$$= \frac{1}{9 + 11 - 2} (26 + 32) = \frac{1}{18} \times 58 = 3.22$$

$$S_p = \sqrt{3.22} = 1.79$$

Putting the values of $\bar{X}, \bar{Y}, S_p, n_1$ and n_2 in equation (5), we have,

$$t = \frac{59 - 60}{1.79 \sqrt{\frac{1}{9} + \frac{1}{11}}} = \frac{-1}{1.79 \times 0.45} = \frac{-1}{0.81} = -1.23$$

The critical values of test statistic t for $(n_1 + n_2 - 2) = 18$ df at 5% level of significance for two-tailed test are $\pm t_{(18), 0.025} = \pm 2.101$.

Since calculated value of test statistic t ($= -1.23$) is less than the critical value ($= 2.101$) and greater than the critical value ($= -2.101$), that means calculated value of test statistic t lies in non-rejection region so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that both samples may be taken from normal populations having equal means.

Hence, overall we conclude that both samples may come from the same normal populations.

Now, you can try the following exercises.

-
- E3)** Two sources of raw materials are under consideration by a bulb manufacturing company. Both sources seem to have similar characteristics but the company is not sure about their respective uniformity. A sample of 12 lots from source A yields a variance of 125 and a sample of 10 lots from source B yields a variance of 112. Is it likely that the variance of source A significantly differs to the variance of source B at significance level $\alpha = 0.01$?
- E4)** A laptop computer maker uses battery packs of two brands, A and B. While both brands have the same average battery life between charges (LBC), the computer maker seems to receive more complaints about shorter LBC than expected for battery packs of brand A. The computer maker suspects that this could be caused by higher variance in LBC for brand A. To check that, ten new battery packs from each brand are selected, installed on the same models of laptops, and the laptops are allowed to run until the battery packs are completely discharged. The following are the observed LBCs in hours:

Brand A	3.2	3.7	3.1	3.3	2.5	2.2	3.2	3.1	3.2	4.3
Brand B	3.4	3.6	3.0	3.2	3.2	3.2	3.0	3.1	3.2	3.2

Assuming that the LBCs of both brands follows normal distribution, test the LBCs of brand A have a larger variance than those of brand B at 5% level of significance.

We now end this unit by giving a summary of what we have covered in it.

12.4 SUMMARY

In this unit, we have discussed the following points:

1. Testing of hypothesis for population variance using χ^2 -test.
2. Testing of hypothesis for two population variances using F-test.

E1) Here, we are given that

$$\sigma_0 = 15, \quad n = 20, \quad S = 17$$

Here, we want to test the agency's claim that the standard deviation (σ) of the length of serving times is less than 15 minutes. So our claim is $\sigma < 15$ and its complement is $\sigma \geq 15$. Since complement contains the equality sign so we can take complement as null hypothesis and claim as the alternative hypothesis. Thus,

$$H_0 : \sigma \geq \sigma_0 = 15$$

$$H_1 : \sigma < 15 \quad \left[\begin{array}{l} \text{SD of the length of serving} \\ \text{times is less than 15 minutes} \end{array} \right]$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

Here, we want to test the hypothesis about the population standard deviation and sample size is small $n = 20 (< 30)$. Also we are given that the service time of the ambulance follows normal so we can go for χ^2 test for population variance.

The test statistic is given by

$$\begin{aligned} \chi^2 &= \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)} \\ &= \frac{19 \times (17.0)^2}{(15.0)^2} = 24.40 \end{aligned}$$

The critical value of test statistic χ^2 for left-tailed test corresponding $(n-1) = 19$ df at 1% level significance is $\chi^2_{(n-1), (1-\alpha)} = \chi^2_{(19), 0.99} = 7.63$.

Since calculated value of test statistic (= 24.40) is greater than the critical value (= 7.63), that means calculated value of test statistic lies in non-rejection region so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject our claim at 1% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so agency's claim that the standard deviation (σ) of the length of serving times is less than 15 minutes is not true.

E2) Here, we are given that

$$\sigma_0^2 = 0.62, \quad n = 25, \quad S^2 = 0.65$$

Here, we want to test the cigarette manufacturer's claims that the variance (σ^2) of nicotine content of its cigarettes is 0.62 milligrams. So claim is $\sigma^2 = 0.62$ and its complement is $\sigma \neq 0.62$. Since claim contains the equality sign so we can take claim as null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \sigma^2 = \sigma_0^2 = 0.62 \text{ and } H_1 : \sigma^2 \neq 0.62$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis about the population variance and sample size is small $n = 25 (< 30)$. Also we are given that the nicotine content of its cigarettes follows normal distribution so we can go for χ^2 test for population variance.

The test statistic is given by

$$\begin{aligned}\chi^2 &= \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)} \\ &= \frac{24 \times 0.65}{0.62} = 25.16\end{aligned}$$

The critical (tabulated) values of test statistic χ^2 for two-tailed test corresponding $(n-1) = 24$ df at 5% level of significance are

$$\chi^2_{(n-1), \alpha/2} = \chi^2_{(24), 0.025} = 39.36 \text{ and } \chi^2_{(n-1), (1-\alpha/2)} = \chi^2_{(24), 0.975} = 12.40.$$

Since calculated value of test statistic ($= 25.16$) is less than the critical value ($= 39.36$) and greater than the critical value ($= 12.40$), that means calculated value of test statistic lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that sample fails to provide sufficient evidence against the claim so we may assume that manufacturer's claim that the variance of the nicotine content of the cigarettes is 0.62 milligram is true.

E3) Here, we are given that

$$n_1 = 12, \quad S_1^2 = 125, \quad n_2 = 10, \quad S_2^2 = 112$$

Here, we want to test that variance of source A significantly differs to the variances of source B. If σ_1^2 and σ_2^2 denote the variances in the raw materials of sources A and B respectively so our claim is $\sigma_1^2 \neq \sigma_2^2$ and its complement is $\sigma_1^2 = \sigma_2^2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ and } H_1 : \sigma_1^2 \neq \sigma_2^2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis about two population variances and sample sizes $n_1 = 12 (< 30)$ and $n_2 = 10 (< 30)$ are small. Also populations under study are normal and both samples are independent so we can go for F-test for two population variances.

For testing this, the test statistic is given by

$$\begin{aligned}F &= \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)} \\ &= \frac{125}{112} = 1.11\end{aligned}$$

The critical (tabulated) value of test statistic F for two-tailed test corresponding $(n_1 - 1, n_2 - 1) = (11, 9)$ df at 5% level of significance are

$$F_{(n_1-1, n_2-1), \alpha/2} = F_{(11, 9), 0.025} = 3.91 \text{ and}$$

$$F_{(n_1-1, n_2-1), (1-\alpha/2)} = \frac{1}{F_{(n_2-1, n_1-1), \alpha/2}} = \frac{1}{F_{(9, 11), 0.025}} = \frac{1}{3.59} = 0.28.$$

Since calculated value of test statistic (= 1.11) is less than the critical value (= 3.91) and greater than the critical value (= 0.28), that means calculated value of test statistic lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so we may assume that the variances of source A and B is differ.

- E4)** Here, we want to test that the LBCs of brand A have a larger variance than those of brand B. If σ_1^2 and σ_2^2 denote the variances in the LBCs of brands A and B respectively so our claim is $\sigma_1^2 > \sigma_2^2$ and its complement is $\sigma_1^2 \leq \sigma_2^2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \sigma_1^2 \leq \sigma_2^2 \text{ and } H_1 : \sigma_1^2 > \sigma_2^2$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis about two population variances and sample sizes $n_1 = 10 (< 30)$ and $n_2 = 10 (< 30)$ are small. Also populations under study are normal and both samples are independent so we can go for F-test for two population variances.

For testing the null hypothesis, test statistic is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)} \quad \dots (6)$$

$$\text{where, } S_1^2 = \frac{1}{n_1 - 1} \sum (X - \bar{X})^2 = \frac{1}{n_1 - 1} \left[\sum X^2 - \frac{(\sum X)^2}{n_1} \right] \text{ and}$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (Y - \bar{Y})^2 = \frac{1}{n_2 - 1} \left[\sum Y^2 - \frac{(\sum Y)^2}{n_2} \right].$$

Calculation for S_1^2 and S_2^2 :

LBCs of Brand A (X)	X ²	LBCs of Brand A (Y)	Y ²
3.7	13.69	3.6	12.96
3.2	10.24	3.2	10.24
3.3	10.89	3.2	10.24
3.1	9.61	3.0	9.00
2.5	6.25	3.0	9.00
2.2	4.84	3.2	10.24
3.1	9.61	3.2	10.24
3.2	10.24	3.1	9.61
4.3	18.49	3.2	10.24
3.2	10.24	3.1	9.61
Total = 3.18	104.1	31.8	101.38

From the calculation, we have

$$\bar{X} = \frac{1}{n_1} \sum X = \frac{1}{10} \times 31.8 = 3.18,$$

$$\bar{Y} = \frac{1}{n_2} \sum Y = \frac{1}{10} \times 31.8 = 3.18$$

Thus,

$$\begin{aligned} S_1^2 &= \frac{1}{n_1 - 1} \left[\sum X^2 - \frac{(\sum X)^2}{n_1} \right] = \frac{1}{9} \left[104.10 - \frac{(31.8)^2}{10} \right] \\ &= \frac{1}{9} \times 2.98 = 0.33 \end{aligned}$$

$$\begin{aligned} S_2^2 &= \frac{1}{n_2 - 1} \left[\sum Y^2 - \frac{(\sum Y)^2}{n_2} \right] = \frac{1}{9} \left[101.38 - \frac{(31.8)^2}{10} \right] \\ &= \frac{1}{9} \times 0.26 = 0.03 \end{aligned}$$

Putting the values of S_1^2 and S_2^2 in equation (6), we have

$$F = \frac{0.33}{0.03} = 1.1$$

The critical (tabulated) value of test statistic F for right-tailed test corresponding $(n_1 - 1, n_2 - 1) = (9, 9)$ df at 1% level of significance is $F_{(n_1-1, n_2-1), \alpha} = F_{(9,9), 0.01} = 5.35$.

Since calculated value of test statistic (= 1.1) is less than the critical value (= 5.35), that means calculated value of test statistic lies in non-rejection region so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so variance in LBCs of brand A is not greater than variance in LBCs of brand B.

Block

4

NON-PARAMETRIC TESTS

UNIT 13

One-Sample Tests	5
-------------------------	----------

UNIT 14

Two-Sample Tests	41
-------------------------	-----------

UNIT 15

k-Sample Tests	71
-----------------------	-----------

UNIT 16

Analysis of Frequencies	87
--------------------------------	-----------

Appendix	109
-----------------	------------

Curriculum and Course Design Committee

Prof. K. R. Srivathasan
Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Rahul Roy
Math. and Stat. Unit
Indian Statistical Institute, New Delhi

Prof. Parvin Sinclair
Pro-Vice Chancellor
IGNOU, New Delhi

Dr. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Geeta Kaicker
Director, School of Sciences
IGNOU, New Delhi

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. R. M. Pandey
Department of Bio-Statistics
All India Institute of Medical Sciences
New Delhi

Dr. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Faculty members of School of Sciences, IGNOU

Statistics

Dr. Neha Garg
Dr. Nitin Gupta
Mr. Rajesh Kaliraman
Dr. Manish Trivedi

Mathematics

Dr. Deepika
Prof. Poornima Mital
Prof. Sujatha Varma
Dr. S. Venkataraman

Block Preparation Team

Prof. G. K. Shukla (**Editor**)
Decision Science Group
Indian Institute of Management
Lucknow (UP)

Prof. Vyas Dubey (**Units 13 & 14**)
Department of Statistics
Pt. Ravishankar Shukla University, Raipur

Dr. Parmod Kumar (**Language Editor**)
School of Humanities, IGNOU

Mr. Prabhat Kumar Sangal (**Units 13 - 16**)
School of Sciences, IGNOU

Course Coordinator: Mr. Prabhat Kumar Sangal

Programme Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P), School of Sciences, IGNOU
CRC prepared by Mr. Prabhat Kumar Sangal, School of Sciences, IGNOU

Acknowledgement: I gratefully acknowledge my colleague Mr. Rajesh Kaliraman, Statistics Discipline, School of Sciences for their great support.

July, 2013

© Indira Gandhi National Open University, 2013

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University.

Further information on the Indira Gandhi National Open University may be obtained from University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Printed at:

NON-PARAMETRIC TESTS

In the previous block, we described the procedure for testing various hypotheses involving population parameter(s) such as mean(s), proportion(s), variance(s), etc. But in many real life problems particularly in Social and Behavioural Sciences where the requirement of parametric tests cannot be satisfied, that is, neither one can know the form of population nor the observations measured in quantitative form. In such situations, the parametric tests are not applicable. Thus, statisticians discovered various tests and methods which are independent of population distribution and also applicable when the observations are not measured in numerical scale that is in ordinal scale or nominal scale. These tests are known as “Non-parametric tests” or “Distribution Free Tests”.

Unit 13: One-Sample Tests

This unit explains the need of non-parametric tests and application of non-parametric tests in various fields with their advantages and disadvantages over parametric tests. In this unit, some of the frequently used non-parametric tests for one sample such as sign test, Wilcoxon signed-rank test, run test and Kolmogorov-Smirnov goodness of fit test are discussed.

Unit 14: Two-Sample Tests

This unit explores two samples tests such as paired sign test, Wilcoxon matched-pair signed-rank test, Mann-Whitney U test and Kolmogorov-Smirnov test.

Unit 15: k-Sample Tests

This unit provides the brief discussion on the procedures for testing for the significance of differences among three or more populations. Kruskal -Wallis test and Friedman test for $k(> 2)$ samples are discussed.

Unit 16: Analysis of Frequencies

Last unit of this block is devoted to describe the procedure of chi-square tests for categorical data such as goodness of fit test and test for independence of two attributes.

Notations and Symbols

$X_1, X_2, \dots, X_n$	:	Random sample of size n
H_0	:	Null hypothesis
H_1 or H_A	:	Alternative hypothesis
α	:	Size of critical region or type-I error or level of significance
$\tilde{\mu}$	:	Median
$ d $	:	Absolute value of d
S^+ and S^-	:	Number of plus and minus signs
T^+ and T^-	:	Sum of positive and negative ranks
T_α	:	Critical value of Wilcoxon signed-rank test at α level of significance
R	:	Number of runs
R_L and R_U	:	Lower and upper critical values of run test at α level of significance
$F(x)$	:	Cumulative distribution function of the population
$S(x)$	:	Empirical or observed or sample cumulative distribution function
$\sup_x d(x) $	:	Supreme over all x , of the absolute value of $d(x)$
$U_{L,\alpha}$ and $U_{U,\alpha}$	:	Lower and upper critical values of Mann-Whitney U test at α level of significance
$\chi^2_{(v), \alpha}$	:	Critical value of χ^2 -test with v degrees of freedom at α level of significance

UNIT 13 ONE-SAMPLE TESTS

Structure

- 13.1 Introduction
 - Objectives
- 13.2 Advantages, Disadvantages and Types of Non-parametric Tests
- 13.3 Sign Test
- 13.4 Wilcoxon Signed-Rank Test
- 13.5 Run Test
- 13.6 Kolmogorov-Smirnov Goodness of Fit Test
- 13.7 Summary
- 13.8 Solutions /Answers

13.1 INTRODUCTION

In the previous block, we have described the procedure for testing various hypotheses involving population parameter(s) such as mean(s), proportion(s), variance(s), etc., together with a large number of examples. What we were doing there can be summarised in following two points:

- We were making an assumption regarding the form of the distribution function of the parent population from which the sample has been drawn is known to us, and
- We were testing a statistical hypothesis regarding population parameters under study such as mean(s) or variance(s) or proportion(s), etc.

Those types of tests where we deal above two points are known as **“parametric tests”**.

But in many real life problems particularly in social and behavioural sciences the requirements of parametric tests do not meet. Therefore, in such situations, the parametric tests are not applicable. Thus, statisticians discovered various tests and methods which are independent of the form of population distribution and also applicable when the observations are not measured in interval scale that is observations are measured in ordinal scale or nominal scale. These tests are known as **“Non-parametric tests”** or **“Distribution Free Tests”**. The name distribution free test is suggested, since the distribution of the test statistic is free (independent) from the form of population distribution. In addition to it the non-parametric test is suggested, since these tests are developed for such hypothesis testing which are not involving the population parameter(s).

These tests are based on the **“order statistics”** theory which has the property that “The distribution of the area under the density function between any two ordered observations is independent of the form of the density function”. Therefore, in these tests we use median, range, quartile, etc., in our hypothesis.

Assumptions Associated with Non-parametric Tests:

Although the non-parametric tests are not based on the assumption that the form of parent population is known but these tests require some basic assumptions which are listed below:

- (i) Sample observations are independent.

A parametric test is a test which is based on the fundamental assumption that the form of parent population is known and involved parameter(s).

A non-parametric test is a test that does not make any assumption regarding the form of the parent population from which the sample is drawn.

(ii) The variable under study is continuous.

(iii) Lower order moments exist.

Obviously, these assumptions are fewer and weaker than those associated with parametric tests.

This unit is divided into eight sections. Section 13.1 is described the need of non-parametric tests. The advantages and disadvantages of the non-parametric tests over parametric tests are described in Section 13.2. In Section 13.3 and 13.4, we discuss sign test and Wilcoxon signed-rank test for one-sample which are generally used when assumption(s) of t-test is (are) not fulfilled. To test the randomness of the data we use run test so in Section 13.5 we explore the run test. In Section 13.6, we describe the Kolmogorov-Smirnov test for goodness of fit. Unit ends by providing summary of what we have discussed in this unit in Section 13.7 and solution of exercises in Section 13.8.

Objectives

After completion of this unit, you should be able to:

- describe the need of non-parametric tests;
- describe the applications of non-parametric tests in various fields with their advantages and disadvantages over parametric tests;
- describe the one-sample non-parametric tests;
- describe an appropriate one-sample non-parametric test for different situations;
- perform the test which is used in place of t-test for mean when assumption(s) of the t-test is (are) not fulfilled, that is, sign test and Wilcoxon signed-rank test;
- perform the test for testing the randomness of the data, that is, run test; and
- perform the Kolmogorov-Smirnov test for goodness of fit.

13.2 ADVANTAGES, DISADVANTAGES AND TYPES OF NON-PARAMETRIC TESTS

In previous section, we have already said that non-parametric tests do the job for us when assumptions of parametric tests do not meet. With this advantage and some other advantages, non parametric test also have some disadvantages. Both are listed in the following table:

Advantages	Disadvantages
The non-parametric tests do not make any assumption regarding the form of the parent population from which the sample is drawn.	When the assumptions of parametric tests are fulfilled then parametric tests are more powerful than non-parametric tests. For example, Wilcoxon test has approximately 95% power compare to t-test. It means that if null hypothesis is wrong then Wilcoxon test requires 5% more observations in the sample compare to t-test to achieve same power. As an example, Wilcoxon test requires 20 observations compare to t-test which requires 19 observations to achieve same power.
Non-parametric tests have	They do not use all the information available in terms of

simple calculation compare to parametric tests because they generally use ranks rather than actual values of the sample observations.	sample observations because they generally use ranks rather than actual values of the sample observations.
Because of using ranks they are also applicable even if data is available in ordinal measurement scale.	In non-parametric tests the required sample size reduces if samples observations are tied as we will see during calculation.

Now, you can answer the following exercise.

E1) Write any three advantages and disadvantages of non-parametric tests compare to parametric tests?

Types of Non-parametric Tests

After discussing the advantages and disadvantages of the non-parametric tests, now, you are interesting to know type of non-parametric tests. Broadly, the non-parametric tests are classified into three categories as:

1. One-sample Tests
2. Two-sample Tests
3. k-sample Tests.

In this unit, we will discuss one-sample tests and other type of tests will be discussed in subsequent units of this block.

Types of One-sample Tests

Some of the commonly used one-sample tests are listed below:

1. Sign Test
2. Wilcoxon Signed-Rank Test
3. Run Test
4. Kolmogorov-Smirnov Goodness of Fit Test

Let us discuss these tests one by one in subsequent sections.

13.3 SIGN TEST

The sign test is the simplest of the non-parametric tests. It is called the sign test because this test is based on signs as plus and minus as you will see when we proceed through an example. The sign test is used as an alternative of the t-test for testing population median instead of population mean under the circumstances when the parent population is not normal. Further it also works if data is available in ordinal scale while for t-test we needed data at least in interval scale.

Assumptions

This test works under following assumptions:

- (i) The sample is selected from the population with unknown median.
- (ii) The variable under study is continuous.
- (iii) The variable under study is measured on at least ordinal scale.

The hypothesis concerning the median cannot be considered as non-parametric testing problem in strict sense (median is itself a parameter) but it is taken as a valid non-parametric testing problem since the distribution of the test statistic does not involve the parent population

Let us discuss general procedure of this test:

Let $X_1, X_2, \dots, X_{n'}$ be a set of n' random observations arranged in the order in which they occur taken from the parent population having unknown median $\tilde{\mu}$. Suppose, we wish to test the hypothesis about the specified value $\tilde{\mu}_0$ of population median $\tilde{\mu}$. So we can take the null and alternative hypotheses as

$$\begin{aligned}
 &H_0 : \tilde{\mu} = \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} \neq \tilde{\mu}_0 \quad \left[\text{for two-tailed test} \right] \\
 \text{or} \quad &\left. \begin{aligned} &H_0 : \tilde{\mu} \leq \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} > \tilde{\mu}_0 \\ &H_0 : \tilde{\mu} \geq \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} < \tilde{\mu}_0 \end{aligned} \right\} \quad \left[\text{for one-tailed test} \right]
 \end{aligned}$$

After setting null and alternative hypotheses, sign test involves following steps:

Step 1: First of all, we convert the given observations into a sequence of plus and minus signs. For this, we subtract the postulated value of median ($\tilde{\mu}_0$) from each observation, that is, we obtain the differences $X_i - \tilde{\mu}_0$ for all observations and check their signs. Another way, we compare each observation X_i with $\tilde{\mu}_0$. If X_i is greater than $\tilde{\mu}_0$, that is, $X_i > \tilde{\mu}_0$, then we replace the observation X_i by a plus sign and if X_i is less than $\tilde{\mu}_0$, that is, $X_i < \tilde{\mu}_0$, then we replace X_i by a minus sign. But when the observation X_i which is equal to $\tilde{\mu}_0$ give no information in terms of plus or minus signs so we exclude all such observations from the analysis part. Due to such observations our sample size reduces and let reduced sample size be denoted by n .

Step 2: After that count the number of plus signs and number of minus signs. Suppose they are denoted by S^+ and S^- respectively.

Step 3: When null hypothesis is true and the population is dichotomised on the basis of postulated value of median $\tilde{\mu}_0$ then we expect that the number of plus signs (success) and number of minus signs (failure) approximately equal. And number of plus signs or number of minus signs follows binomial distribution ($n, p = 0.5$). For convenient consider the smaller number of plus or minus signs. If plus sign (S^+) is less than minus sign (S^-) then we will take plus sign (S^+) as success and minus sign (S^-) as failure. Similarly, if minus sign (S^-) is less than plus sign (S^+) then we will assume minus sign (S^-) as success and plus sign (S^+) as failure.

Step 4: To take the decision about the null hypothesis, we use concept of p-value. We have discussed p-value in Unit 9 of this course. For p-value we determine the probability that test statistic is less than or equal to the calculated value of test statistic i.e. actually observed plus or minus signs. Since distribution of number of plus or minus signs is binomial ($n, p = 0.5$) therefore, this probability can be obtained with the help of **Table I** given in Appendix at the end of this block which provide the cumulative binomial probability and compare this probability with the level of significance (α). Here, test statistic depends upon the alternative hypothesis so the following cases arise:

The critical values of this test are not generally in tabular form and slightly difficult to obtain whereas p-value is easy to obtained with the help of cumulative binomial table so we use concept of p-value for take the decision about the null hypothesis.

For one-tailed test:**Case I:** When $H_0 : \tilde{\mu} \leq \tilde{\mu}_0$ and $H_1 : \tilde{\mu} > \tilde{\mu}_0$ (right-tailed test)

In this case, we expect that number of minus signs (S^-) is smaller than number of plus signs (S^+) therefore, the test statistic(S) is the number of minus signs (S^-). The p-value is determined as

$$\text{p-value} = P[S \leq S^-]$$

If p-value is less than or equal to α , that is, $P[S \leq S^-] \leq \alpha$ then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

Case II: When $H_0 : \tilde{\mu} \geq \tilde{\mu}_0$ and $H_1 : \tilde{\mu} < \tilde{\mu}_0$ (left-tailed test)

In this case, we expect that number of plus signs (S^+) is smaller than number of minus signs (S^-) therefore, the test statistic(S) is the number of plus signs (S^+). The p-value is determined as

$$\text{p-value} = P[S \leq S^+]$$

If p-value is less than or equal to α , that is, $P[S \leq S^+] \leq \alpha$ then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

Here, we take p-value as the probability less than or equal to observed value of the test statistic in each case because in this test we consider the smaller sign so critical region lies in lower tail.

For two-tailed test: When $H_0 : \tilde{\mu} = \tilde{\mu}_0$ and $H_1 : \tilde{\mu} \neq \tilde{\mu}_0$

For two tailed test, the test statistic(S) is the smaller of number of plus signs (S^+) and minus signs (S^-), that is,

$$S = \min\{S^+, S^-\}$$

and approximate p-value is determined as

$$\text{p-value} = 2P[S \leq S^+]; \quad \text{if } S^+ \text{ is small}$$

$$\text{p-value} = 2P[S \leq S^-]; \quad \text{if } S^- \text{ is small}$$

If p-value is less than or equal to α , then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

For large sample ($n > 20$):

For a large sample size n greater than 20, we use normal approximation to binomial distribution with mean

$$E(S) = np = n \times \frac{1}{2} = \frac{n}{2} \quad \dots (1)$$

and variance

$$\text{Var}(S) = npq = n \times \frac{1}{2} \times \frac{1}{2} = \frac{n}{4} \quad \dots (2)$$

Therefore in this case, we use normal test (i.e. Z-test which is described in Unit 10 of Block 3 of this course). The test statistic of Z-test is given by

$$Z = \frac{S - E(S)}{SE(S)}$$

$$= \frac{S - \frac{n}{2}}{\sqrt{\frac{n}{4}}} \sim N(0,1) \quad [\text{Using equations (1) and (2)}] \quad \dots (3)$$

After that, we calculate the value of test statistic Z and compare it with the critical value given in **Table 10.1** of the Unit 10 of this course at prefixed level of significance α . Take the decision about the null hypothesis as described in the Section 10.2 of Unit 10 of this course.

Note 1: Here, you may shock on the point that we used $n > 20$ for large sample instead of $n > 30$, because binomial distribution with $p = 0.5$ approximated by normal distribution less than 30.

Let us do some examples to become more user friendly with this test.

Example 1: The breaking strength (in pounds) of a random sample of 10 ropes made by a manufacturer is given by

163 165 165 160 171 158 151 162 169 172

Use the sign test to test the manufacturer's claim that the average breaking strength of a rope is greater than 160 pounds at 5% level of significance.

Solution: Here, distribution of the population of the breaking strengths of the ropes is not given. So the assumption of normality for t-test is not fulfilled. Also sample size is small so we can not use Z-test. So we go for sign test.

Here, we want to test the manufacturer's claim that the average (median) breaking strength ($\tilde{\mu}$) of a rope is greater than 160 pounds. So the claim is $\tilde{\mu} > 160$ and its complement is $\tilde{\mu} \leq 160$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} \leq \tilde{\mu}_0 = 160 \text{ and } H_1 : \tilde{\mu} > 160$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

For applying sign test, we compare each observation with $\tilde{\mu}_0 (= 160)$ and replacing each observation greater than 160 with a plus sign and each observation less than 160 with a minus sign and discarding the one observation which equals to 160, we get

+ + + + - - + + +

By counting, we have

$$S^+ = \text{number of plus signs} = 7$$

$$S^- = \text{number of minus signs} = 2$$

$$n = \text{total number of plus and minus signs} = 9$$

Since alternative hypothesis is right-tailed so the test statistic (S) is the number of minus signs (S^-), that is,

$$S = \text{number of minus signs } (S^-) = 2$$

Here, $n = 9 (< 20)$ so it is a case of small sample. Thus, to take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block. Here, $n = 9$, $p = 0.5$ and $r = 2$. Thus, we have

$$\begin{aligned} \text{p-value} &= P[S \leq 2] \\ &= 0.0899 \end{aligned}$$

Since $\text{p-value} = 0.0899 > 0.05 (= \alpha)$. So we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the manufacturer's claim at 5% level of significance.

Thus, we conclude that the sample provide us sufficient evidence against the claim so manufacturer's claim that the breaking strength of a rope is greater than 160 pounds is not true.

Example 2: An economist believes that the median starting salary for a computer programmer in a certain city is Rs 12,000. To verify this claim, 12 computer programmers with similar backgrounds who recently got the jobs were randomly selected. Their starting salaries (in Rs) were 18000, 15000, 12000, 10000, 13000, 12000, 10000, 16000, 11000, 9000, 10000 and 9000. At 1% level of significance give your conclusion where you reached after using sign test.

Solution: Here, distribution of the salaries of the computer programmers is not given. So the assumption of normality for t-test is not fulfilled. Also sample size is small so we can not use Z-test. So we go for sign test.

Here, we want to test that the median starting salary ($\tilde{\mu}$) for a computer programmer in a certain city is Rs 12000. So our claim is $\tilde{\mu} = 12000$ and it complement is $\tilde{\mu} \neq 12000$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 12000 \text{ and } H_1 : \tilde{\mu} \neq 12000$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

For applying sign test, we replacing each value greater than $\tilde{\mu}_0 (= 12000)$ with a plus sign and each value less than 12000 with a minus sign and discarding the two values which are equal to 12000, we get

+ + - + - + - - - -

By counting, we have

$$S^+ = \text{number of plussigns} = 4$$

$$S^- = \text{number of minus signs} = 6$$

$$n = \text{total number of plus and minus signs} = 10$$

Since alternative hypothesis is two-tailed so the test statistic(S) is the minimum of number of plus signs (S^+) and minus signs (S^-), that is,

$$S = \min \{S^+, S^-\} = \min \{4, 6\} = 4$$

Here, $n = 10 (< 20)$, so it is a case of small sample. Thus, to take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block. Here, $n = 10$, $p = 0.5$ and $r = 4$. Thus, we have

$$\text{p-value} = P[S \leq 4] = 0.3770$$

Since $p\text{-value} = 0.37707 > 0.01(\alpha)$ so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that median salary of the newly appointed computer programmers is Rs 12000.

Example 3: The following data give the milk production (in thousand kg) in full cream by 40 different dairies:

17	15	20	29	19	19	22	25	27	9
24	20	17	6	24	14	15	23	24	26
19	23	28	19	16	22	24	17	20	13
19	10	23	18	31	13	20	17	24	14

Use the sign test to test that median ($\tilde{\mu}$) production of milk in dairies is 21.5 thousand kg at 1% level of significance.

Solution: Here, distribution of the milk production of the different dairies is not given. So the assumption of normality for t-test is not satisfied. So we go for sign test.

Here, we want to test that median production ($\tilde{\mu}$) of milk in dairies is 21.5 thousand kg. So our claim is $\tilde{\mu} = 21.5$ and its complement is $\tilde{\mu} \neq 21.5$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 21.5 \text{ and } H_1 : \tilde{\mu} \neq 21.5$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

For applying sign test, we replace each value greater than 21.5 (median value) with a plus sign and each value less than 21.5 with a minus sign, we get

–	–	–	+	–	–	+	+	+	–
+	–	–	–	+	–	–	+	+	+
–	+	+	–	–	+	+	–	–	–
–	–	+	–	+	–	–	–	+	–

By counting, we have

$$S^+ = \text{number of plus signs} = 16$$

$$S^- = \text{number of minus signs} = 24$$

$$n = \text{total number of plus and minus signs} = 40$$

Since alternative hypothesis is two-tailed so the test statistic (S) is given by

$$S = \min \{S^+, S^-\} = \min \{16, 24\} = 16$$

Here, $n = 40 (> 20)$, so it is a case of large sample. In this case, we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{S - \frac{n}{2}}{\sqrt{\frac{n}{4}}} \sim N(0,1)$$

$$\begin{aligned}
 &= \frac{16 - \frac{40}{2}}{\sqrt{\frac{40}{4}}} \\
 &= \frac{16 - 20}{\sqrt{10}} = -1.27
 \end{aligned}$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of $Z (= -1.27)$ is greater than the critical value $(= -2.58)$ and less than the critical value $(= 2.58)$, that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Decision according to p-value:

Since test is two-tailed, therefore,

$$\begin{aligned}
 \text{p-value} &= 2P[Z \geq |z|] = 2P[Z \geq 1.27] \\
 &= 2[0.5 - P[0 \leq Z \leq 1.27]] \\
 &= 2(0.5 - 0.3980) = 0.204
 \end{aligned}$$

Since p-value $(= 0.204)$ is greater than $\alpha (= 0.01)$ so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the median production of milk in dairies is 21.5 thousand kg.

Now, you can try the following exercises.

E2) Give one difference between t-test and sign test for one-sample.

E3) A Metropolitan Area Road Transport Authority claims that the average (median) waiting time on a well travelled rout is 20 minutes. A random sample of 12 passengers showed weighting times of 22, 13, 17, 14, 25, 26, 19, 20, 22, 30, 10 and 15 minutes. Test the Metropolitan Area Road Transport Authority (MARTA) claim's that average (median) waiting time on a well traveled rout is 20 minutes by sign test at 1% level of significance.

E4) The following data shows the weight (in kg) of a random sample of 30 cadets of a centre:

49	46	57	40	45	57	50	65	34	50
58	46	47	42	53	67	40	52	49	66
53	48	68	40	53	49	61	54	48	38

Use sign test to examine whether the median weight of all the cadets of the centre is 50 kg at 5% level of significance.

13.4 WILCOXON SIGNED-RANK TEST

In the previous section, we discussed sign test. We recall that the sign test utilizes only information whether the difference between each observation and the postulated value of the median is positive or negative, that is, it considers only sign of differences. This test is fine if the information about the observation sample is available in ordinal scale only. But if the measurement of the observation is available interval or ratio scales then choice of sign test is not recommended. Because this test do not take into account the information available in terms of the magnitude of the differences. To overcome this drawback of the sign test Wilcoxon signed-rank test do the job for us, which takes into account the information of signs as well as of magnitude of differences. Since Wilcoxon signed-rank test use more information than sign test so it is more powerful than the sign test. The Wilcoxon signed-rank test is also used as an alternative of the t-test.

Assumptions

Wilcoxon signed-rank test requires following assumptions to work:

- (i) The sample is selected from the population with unknown median.
- (ii) The sampled population is symmetric about its median.
- (iii) The variable under study is continuous.
- (iv) The variable under study is measured on at least interval scale.

Let us discuss general procedure of this test:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n' from a continuous symmetric population. Let $\tilde{\mu}$ be median of the population. Here, we wish to test the hypothesis about the specified value $\tilde{\mu}_0$ of population median $\tilde{\mu}$. So we can take the null and alternative hypotheses as

$$\begin{aligned} H_0 : \tilde{\mu} = \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} \neq \tilde{\mu}_0 & \quad [\text{for two-tailed test}] \\ \text{or} \quad \left. \begin{aligned} H_0 : \tilde{\mu} \leq \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} > \tilde{\mu}_0 \\ H_0 : \tilde{\mu} \geq \tilde{\mu}_0 \text{ and } H_1 : \tilde{\mu} < \tilde{\mu}_0 \end{aligned} \right\} & \quad [\text{for one-tailed test}] \end{aligned}$$

After setting null and alternative hypotheses, Wilcoxon signed-rank test involves following steps:

Step 1: We subtract $\tilde{\mu}_0$ from each observation and obtain the difference d_i with their plus and minus sign as

$$d_i = X_i - \tilde{\mu}_0 \text{ for all observations}$$

But when the observation X_i equal to $\tilde{\mu}_0$ give no information in terms of signs as well as magnitude so we exclude all such observations from the analysis part. Due to such observations our sample size reduces and let reduced sample size be denoted by n .

Step 2: After that, we find the absolute value of these d_i 's obtained in Step 1 as $|d_1|, |d_2|, \dots, |d_n|$.

Step 3: In this step, we are ranked $|d_i|$'s (obtained in Step 2) with respect to their magnitudes from smallest to largest, that is, the rank 1 is given to the smallest of $|d_i|$'s, rank 2 is given to the second smallest and so

on up to the largest $|d_i|$'s. If several values are same (tied), we assign each the average of ranks they would have received if there were no repetition.

Step 4: Now assign the signs to the ranks which the original differences have.

Step 5: Finally, we calculate the sum of the positive ranks (T^+) and sum of negative ranks (T^-) separately.

Under H_0 , we expect approximately equal number of positive and negative ranks. And under the assumption that population under study is symmetric about its median we expect that sum of the positive ranks (T^+) and sum of negative ranks (T^-) are equal.

Step 6: Decision Rule:

To take the decision about the null hypothesis, the test statistic is the smaller of T^+ and T^- . And the test statistic is compared with the critical (tabulated) value for a given level of significance (α) under the condition that the null hypothesis is true. **Table II** given in Appendix at the end of this block provides the critical values of test statistic at α level of significance for both one-tailed and two-tailed tests. Here, test statistic depends upon the alternative hypothesis so the following cases arise:

For one-tailed test:

Case I: When $H_0 : \mu \leq \mu_0$ and $H_1 : \mu > \mu_0$ (right-tailed test)

In this case, we expect that sum of negative ranks (T^-) is smaller than sum of positive ranks (T^+) therefore, the test statistic (T) is the sum of negative ranks (T^-).

If computed value of test statistic (T) is less than or equal to the critical value T_α at α level of significance, that is, $T \leq T_\alpha$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

Case II: When $H_0 : \mu \geq \mu_0$ and $H_1 : \mu < \mu_0$ (left-tailed test)

In this case, we expect that sum of positive ranks (T^+) is smaller than sum of negative ranks (T^-) therefore, the test statistic (T) is the sum of positive ranks (T^+).

If computed value of test statistic (T) is less than or equal to the critical value T_α at α level of significance, that is, $T \leq T_\alpha$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

For two-tailed test:

When $H_0 : \mu = \mu_0$ and $H_1 : \mu \neq \mu_0$

In this case, the test statistic (T) is the smaller of sum of positive ranks (T^+) and sum of negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

If computed value of test statistic (T) is less than or equal to the critical value $T_{\alpha/2}$ at α level of significance, that is, $T \leq T_{\alpha/2}$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

One-Sample Tests

For more detailed about repeated (tied) ranks please go through Section 7.4 of the Unit 2 of MST-002.

Here, we consider that if test statistic less than or equal to critical value of the test statistic then we reject null hypothesis in each case because in this test we consider the smaller sum so critical region lies in lower tail.

For large sample ($n > 25$):

For a large sample size n greater than 25, the distribution of test statistic (T) approximated by a normal distribution with mean

$$E(T) = \frac{n(n+1)}{4} \quad \dots (4)$$

and variance

$$\text{Var}(T) = \frac{n(n+1)(2n+1)}{24} \quad \dots (5)$$

The proof of mean and variance of T is beyond the scope of this course.

Therefore in this case, we use normal test (Z-test). The test statistic of Z-test is given by

$$\begin{aligned} Z &= \frac{T - E(T)}{\text{SE}(T)} = \frac{T - E(T)}{\sqrt{\text{Var}(T)}} \\ &= \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}} \sim N(0,1) \left[\begin{array}{l} \text{Using equations} \\ (4) \text{ and } (5) \end{array} \right] \quad \dots (6) \end{aligned}$$

After that, we calculate the value of test statistic Z and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in the Section 10.2 of Unit 10 of this course.

Now, it is time to do some examples based on above test.

Example 4: A random sample of 15 children of one month or older shows the following pulse rates (beats per minute):

119, 120, 125, 122, 118, 117, 126, 114, 115, 126, 121, 120, 124, 127, 126

Assuming that the distribution of pulse rate is symmetric about its median and continuous, is there evidence to suggest that the median pulse rate of one month or older children is 120 beats per minute at 5% level of significance?

Solution: Here, distribution of pulse rate is not given. So the assumption of normality for t-test is not fulfilled although all the assumptions of Wilcoxon signed-rank test hold. So we go for this test.

Here, we want to test that median pulse rate ($\tilde{\mu}$) of children of one month or older is 120 beats per minute. So our claim is $\tilde{\mu} = 120$ and its complement is $\tilde{\mu} \neq 120$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 120 \quad [\text{median pulse rate is equal to 120}]$$

$$H_1 : \tilde{\mu} \neq 120 \quad [\text{median pulse rate is not equal to 120}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed and the test statistic is the smaller of sum of positive ranks (T^+) and sum of negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

S. No.	Beats per Minute (X)	Difference $d = X - \tilde{\mu} = X - 120$	Absolute Value of Difference $ d $	Rank of $ d $	Signed Rank
1	119	-1	1	1.5	-1.5
2	120	Tie	---	---	---
3	125	5	5	7.5	7.5
4	122	2	2	3.5	3.5
5	118	-2	2	3.5	-3.5
6	117	-3	3	5	-5
7	126	6	6	10.5	10.5
8	114	-6	6	10.5	-10.5
9	115	-5	5	7.5	-7.5
10	126	6	6	10.5	10.5
11	121	1	1	1.5	1.5
12	120	Tie	---	---	---
13	124	4	4	6	6
14	127	7	7	13	13
15	126	6	6	10.5	10.5

From the above calculations, we have

$$T^+ = 7.5 + 3.5 + 10.5 + 10.5 + 1.5 + 6 + 13 + 10.5 = 63.0$$

$$T^- = 1.5 + 3.5 + 5 + 10.5 + 7.5 = 28.0$$

n = number of non-zero d_i 's = 13

Putting the values in test statistic, we have

$$T = \min\{T^+, T^-\} = \min\{63.0, 28.0\} = 28.0$$

The critical (tabulated) value for two-tailed test corresponding $n = 13$ at 5% level of significance is 18.

Since calculated value of test statistic T ($= 28.0$) is greater than the critical value ($= 18$) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that median pulse rate of children of one month or older is 120 beats per minute.

Example 5: The following data show the weight (in kg) of a random sample of 30 cadets of a college:

49	46	57	37	45	57	50	65	34	50	58
46	47	42	53	67	40	52	49	66	53	48
68	40	53	49	61	54	48	38			

Assume that distribution of the weight of the cadets is symmetric about its median, test to examine whether the median weight of all the cadets of the college is 50 kg by using Wilcoxon signed-rank at 5% level of significance.

Solution: Here, distribution of weight of the cadets is not given. So the assumption of normality for t-test is not fulfilled. We are given to us that distribution of the weight of the cadets is symmetric about its median and assumption of continuity holds because the characteristic weight is continuous in nature. So we go for Wilcoxon signed-rank test.

Here, we want to test that median weight ($\tilde{\mu}$) of all the cadets of a college is 50 kg. So our claim is $\tilde{\mu} = 50$ and its complement is $\tilde{\mu} \neq 50$. Since the claim contains the equality sign, so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 50 \quad [\text{median weight is equal to 50 kg}]$$

$$H_1 : \tilde{\mu} \neq 50 \quad [\text{median weight is not equal to 50 kg}]$$

Since the alternative hypothesis is two-tailed, so the test is two-tailed test and the test statistic is the smaller of sum of positive ranks (T^+) and sum of negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

Calculation for T:

S. No.	Weight (in kg) (X)	Difference $d = X - \tilde{\mu} = X - 50$	Absolute Value of Difference $ d $	Rank of $ d $	Signed Rank
1	49	-1	1	2	-2
2	46	-4	4	12	-12
3	57	7	7	15.5	15.5
4	37	-13	13	23	-23
5	45	-5	5	14	-14
6	57	7	7	15.5	15.5
7	50	Tie	---	---	---
8	65	15	15	24	24
9	34	-16	16	25.5	-25.5
10	50	Tie	---	---	---
11	58	8	8	17.5	17.5
12	46	-4	4	12	-12
13	47	-3	3	8.5	-8.5
14	42	-8	8	17.5	-17.5
15	53	3	3	8.5	8.5
16	67	17	17	27	27
17	40	-10	10	19.5	-19.5
18	52	2	2	5	5
19	49	-1	1	2	-2
20	66	16	16	25.5	25.5
21	53	3	3	8.5	8.5
22	48	-2	2	5	-5
23	68	18	18	28	28
24	40	-10	10	19.5	-19.5
25	53	3	3	8.5	8.5
26	49	-1	1	2	-2
27	61	11	11	21	21
28	54	4	4	12	12
29	48	-2	2	5	-5
30	38	-12	12	22	-22

From the above calculations, we have

$$\begin{aligned} T^+ &= 15.5 + 15.5 + 24 + 17.5 + 8.5 + 27 + 5 + 25.5 + 8.5 + 28 + 8.5 + 21 + 12 \\ &= 216.5 \end{aligned}$$

$$T^- = 2 + 12 + 23 + 14 + 25.5 + 12 + 8.5 + 17.5 + 19.5 + 2 + 5 + 19.5 + 2 + 5 + 22$$

$$= 189.5$$

n = number of non-zero d_i 's = 28

Putting the values in test statistic, we have

$$T = \min\{T^+, T^-\} = \min\{216.5, 189.5\} = 189.5$$

Also $n = 28 (> 25)$ therefore, it is the case of large sample. So in this case, we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{T - E(T)}{SE(T)} \sim N(0,1)$$

$$\text{where, } E(T) = \frac{n(n+1)}{4} = \frac{28(28+1)}{4} = 203 \text{ and}$$

$$SE(T) = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{28(28+1)(2 \times 28+1)}{24}} = 43.91$$

Putting the values in test statistic Z, we have

$$Z = \frac{189.5 - 203}{43.91} = -0.31$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of Z (= -0.31) is greater than the critical value (= -1.96) and less than the critical value (= 1.96), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the median weight of all the cadets of the college is 50 kg.

Now, you can try the following exercises in the same manner.

E5) Give one main difference between sign test and Wilcoxon signed-rank test.

E6) The observations of a random sample of size 10 from a distribution which is continuous and symmetric about its median are given below:

20.2, 24.1, 21.3, 17.2, 19.8, 16.5, 21.8, 18.7, 17.1, 19.9

Use Wilcoxon test to test the hypothesis that the sample is taken from a population having median greater than 18 at 5% level of significance.

E7) The breaking strength (in pounds) of a random sample of 29 rods made by a manufacturer is given as follows:

19	30	28	35	23	37	36	32	40	24
31	33	21	36	30	32	26	40	38	28
30	41	22	43	25	28	24	37	17	

On the basis of this sample test the manufacturer's claim that the average (median) breaking strength of the rope is 30 pounds by Wilcoxon signed-rank test at 1% level of significance. Assume that breaking strength of the rods is symmetric about its median.

A run is a succession of identical letters or symbols that is followed and preceded by a different letter or by no letter at all.

13.5 RUN TEST

One of the fundamental assumptions of the parametric test is that the observed data are random and test statistic and the subsequent analysis are based on this assumption. It is always better to check whether this assumption is true or not. A very simple tool for checking this assumption is run test. This section is devoted to throw light on the run test. Before discussing the run test first we have to explain what we mean by a “run”.

A run in observations, is defined as a sequence of letters or symbols of one kind, immediately preceded and succeeded by letters of other kind or no letters. For example a sequence of two letters H and T as given below:

HHTHTTTHTHHHTTT

In this sequence, we start with first letter H and go up to other kind of letter, that is, T. In this way, we get first run of two H's. Then we start with this T and go up to other kind of letter, that is, H. Then we get a run of one T and so on and finally a run of three T's. In all, we see that there are eight runs as shown below:

HHTHTTTHTHHHTTT
 1 2 3 4 5 6 7 8

Under run test, randomness of observations is judged by number of runs in the observed sequence. Too few runs indicate that there is some clustering or trend and too large runs indicate that there is some kind of repeated or cycles according to some pattern.

For example, the following sequence of H's and T's is obtained when tossing a coin 20 times:

HHHHHHHHHH TTTTTTTTTT
 1 2

In this sequence, we have 2 runs—a run of 10 heads followed by a run of 10 tails. Since similar items tend to cluster together, therefore, such a sequence could not be considered random even though as would theoretically be expected 10 heads and 10 tails out of 20 tosses. In another example, suppose the following sequence has been obtained:

HTHTHTHTHTHTHTHTHTHT
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

In this sequence there are 20 runs—10 runs of one head each and 10 runs of one tail each. This sequence also could not be considered random because too many runs indicate a pattern.

On the other hand, if the sequence of H's and T's is thoroughly mixed, such as,

HTTTHHTTTHTHTTHHHTHH
 1 2 3 4 5 6 7 8 9 10 11

where neither the number of runs too small nor too large, this type of sequence may be considered random. Therefore, in testing for randomness, **ordering** or **positioning** of the items in the sequence is essential, **not** the frequency of the items of each type.

Assumptions

Run test make the following assumptions:

- (i) Observed data should be such that we can categorise the observations into two mutually exclusive types.

(ii) The variable under study is continuous.

Let $X_1, X_2, \dots, X_{n'}$ be a set of n' observations arranged in the order in which they occur. Generally, we are interested to test whether a population or a sample or a sequence is random or not. So here we consider only two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0 : The observations are random

H_1 : The observations are not random [two-tailed test]

Let us discuss the general procedure of this test:

Step 1: First of all, we check the form of the given data that the given data are in symbolical form such as sequence of H and T, A and B, etc. or in the numeric form. If the data in symbolical form then it is ok, but if data in numeric form then first we convert numeric data in symbolical form. For this, we calculate median of the given observations by using either of the following formula (discussed in Unit 1 of the course MST-002.) given below:

$$\text{Median} = \left(\frac{n' + 1}{2} \right)^{\text{th}} \text{ observation; if } n' \text{ is odd}$$

$$\text{Median} = \frac{1}{2} \left[\left(\frac{n'}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{n'}{2} + 1 \right)^{\text{th}} \text{ observation} \right]; \text{ if } n' \text{ is even}$$

provided observations should be either in ascending or descending order of magnitude.

After that, we replace the observations which are above the median by a symbol 'A' (say) and the observation which are below the median by a symbol 'B' (say) without altering the observed order. The observations which are equal to median are discarded from the analysis and let reduced size of the sample denoted by n .

Step 2: Counts number of times first symbol (A) occurs and denote it by n_1 .

Step 3: Counts number of times second symbol (B) occurs and denote it by n_2 where, $n_1 + n_2 = n$.

Step 4: For testing the null hypothesis, the test statistic is the total number of runs so in this step we count total number of runs in the sequence of symbols and denote it by R .

Step 5: Obtain critical values of test statistic corresponding n_1, n_2 at α % level of significance under the condition that null hypothesis is true. Part I and Part II of **Table III** in the Appendix respectively provide lower (R_L) and upper (R_U) critical values of the number of runs for a given combination of n_1 and n_2 at 5% level of significance.

Note1: Generally, critical values for run test are available at 5% level of significance so we test our hypotheses for 5% level of significance.

Step 6: Decision Rule:

To take the decision about null hypothesis, the test statistic is compared with the critical (tabulated) values.

If the observed number of runs(R) is either less than or equal to the lower critical value (R_L) or greater than or equal to the upper critical value (R_U), that is, if $R \leq R_L$ or $R \geq R_U$ then we reject the null hypothesis at 5% level of significance.

If R lies between R_L and R_U , that is, $R_L < R < R_U$, then we do not reject the null hypothesis at 5% level of significance.

For large sample ($n > 40$):

For a sample size n greater than 40 or either n_1 or n_2 exceeds 20, the statistic R is approximately normally distributed with mean

$$E(R) = \frac{2n_1n_2}{n} + 1 \quad \dots (7)$$

and variance

$$\text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)} \quad \dots (8)$$

The proof of mean and variance of R is beyond the scope of this course.

Therefore in this case, we use normal test (Z-test). The test statistic of Z-test is given by

$$Z = \frac{R - E(R)}{\sqrt{\text{Var}(R)}} \sim N(0,1)$$

$$Z = \frac{R - \frac{2n_1n_2}{n} - 1}{\sqrt{\frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}}} \quad [\text{Using equations (7) and (8)}] \quad \dots (9)$$

After that, we calculate the value of test statistic Z and compare it with the critical value given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in the Section 10.2 of Unit 10 of this course.

Now, it is time to do some examples based on above test.

Example 6: A sequence of Heads (H) and Tails (T) in tossing of a coin 16 times is given below:

HTTTHTHTHHTHTTTHH

- (i) Count the number of runs.
- (ii) Test whether the Heads and Tails occur in random order.

Solution: Here, we are given a sequence of Heads of Tails as

HTTTHTHTHHTHTTTHH

- (i) For counting the number of runs, we start with first letter H and go up to other kind of letter, that is, T. In this way, we get first run of one H. Then we start with this T and go up to other kind of letter, that is, H then we get a run of three T's and so on in the last there is a run of two H's. In this way, the total number of runs is 11 as shown below:

H TTT H T HT HH T HT TT HH
 1 2 3 4 5 6 7 8 9 10 11

- (ii) Here, we want to test the randomness of the sequence so we go for run test.

We want to test that the given sequence of H and T is in random order. So our claim is “the given sequence of H and T is in random order” and its complement is “the given sequence of H and T is not in random order”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Given sequence of Heads (H) and Tails (T) is in random order

H_1 : Given sequence of Heads (H) and Tails (T) is not in random order

Here by counting, we have

n_1 = number of times H occurs = 8

n_2 = number of times T occurs = 8

R = total number of runs in given sequence = 11

For run test, the test statistic is the total number of runs in the given sequence.

Since $n_1 = 8 (< 20)$ and $n_2 = 8 (< 20)$, so it is a small sample case.

The lower and upper critical values of runs corresponding to $n_1 (= 8)$ and $n_2 (= 8)$ at 5% level of significance are $R_L = 4$ and $R_U = 14$.

Since observed number of runs ($R = 11$) lies between critical values of runs, $R_L = 4$ and $R_U = 14$, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the given sequence of H and T is in random order.

Example 7: A machine produced a number of defective items per day. To check the randomness of the defective items, a sample of defective items produced per day over a period of 20 days is taken and found following results:

13	17	14	20	18	17	16	14	19	21
18	20	17	13	14	19	20	15	19	17

Test at 5% level of significance that the machine produced defective items in random order.

Solution: Here, we want to test the randomness of the defective items produced per day so we go for run test.

We want to test that machine produced defective items in random order. So our claim is “the machine produced defective items are in random order” and its complement is “the machine produced defective items are not in random order”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The machine produced defective items are in random order

H_1 : The machine produced defective items are not in random order

For applying run test, we need given observations in a sequence of two symbols. In order to convert these observations into a sequence of two symbols first of all, we have to calculate median of the given observations. For

obtaining median, we have to arrange the given observations in ascending or descending order of magnitude. So arrange the observations in ascending order of magnitude, we have

13 13 14 14 14 15 16 17 17 17 17 18 18 19 19 19 20
20 20 21

Number of observations (n') = 20 (even) therefore the median can be obtained as

$$\begin{aligned}\text{Median} &= \frac{1}{2} \left[\left(\frac{n'}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{n'}{2} + 1 \right)^{\text{th}} \text{ observation} \right] \\ &= \frac{1}{2} \left[\left(\frac{20}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{20}{2} + 1 \right)^{\text{th}} \text{ observation} \right] \\ &= \frac{1}{2} [(10)^{\text{th}} \text{ observation} + (11)^{\text{th}} \text{ observation}] \\ &= \frac{1}{2} (17 + 17) = 17\end{aligned}$$

Now, we replace the given observations by the symbol 'A' if it is above the median value (= 17) and by symbol 'B' if it is below the median value and discard the observation equal to median value. In this way, the sequence obtained by discarding 4 observations which are equal to median value 17 is given below:

BBAABBAAAABBAABA

For applying run test, we count the number of runs. We see that there is a first run of two B's then there is a run of two A's and so on in the last there is a run of single A as shown below:

$\underbrace{BB}_1 \underbrace{AA}_2 \underbrace{BB}_3 \underbrace{AAAA}_4 \underbrace{BB}_5 \underbrace{AA}_6 \underbrace{BA}_7 \underbrace{A}_8$

By counting, we have

n_1 = number of symbol A occurs = 9

n_2 = number of symbol B occurs = 7

R = total number of runs in above sequence = 8

For run test, the test statistic is the total number of runs in the given sequence.

Since $n_1 = 9 (< 20)$ and $n_2 = 7 (< 20)$ so it is a small sample case.

The lower and upper critical values of runs corresponding to $n_1 = 9$ and $n_2 = 7$ and at 5% level of significance are $R_L = 4$ and $R_U = 14$.

Since observed number of runs ($R = 9$) lies between critical values of runs, $R_L = 4$, and $R_U = 14$ so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence

against the claim so we may assume that the machine produced defective items are in random order.

Example 8: A sample of 36 tools produced by a machine shows the following sequence of good (G) and bad (B) tools:

GGGBGGGGGBBBGGGGGGBBGGGGGGGGBBGGGGG

Test the machine produced good (G) and bad (B) tools in random order at 5% level of significance.

Solution: Here, we want to test the randomness of the sequence so we go for run test.

We want to test that machine produced good (G) and bad (B) tools in random order. So our claim is “the machine produced good and bad tools are in random order” and its complement is “the machine produced good and bad tools are not in random order”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The machine produced good and bad tools are in random order

H_1 : The machine produced good and bad tools are not in random order

For applying run test, we count the number of runs.

We see that there is a first run of three G's then there is a run of single B and so on in the last there is a run of five G's as shown below:

GGG B GGGGG BBB GGGGGG BB GGGGGGGGG BB GGGGG

1 2 3 4 5 6 7 8 9

By counting, we have

n_1 = number of symbol A occurs = 28

n_2 = number of symbol B occurs = 8

R = total number of runs in above sequence = 9

Since $n_1 = 28$ (>20) therefore, it is the case of large sample so we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{R - E(R)}{SE(R)} \sim N(0,1)$$

where, $E(R) = \frac{2n_1n_2}{n} + 1 = \frac{2 \times 28 \times 8}{36} + 1 = 13.44$ and

$$SE(R) = \sqrt{\frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}} = \sqrt{\frac{2 \times 28 \times 8(2 \times 28 \times 8 - 36)}{36^2(36-1)}}$$

$$= \sqrt{4.07} = 2.02$$

Putting the values in test statistic Z, we have

$$Z = \frac{9 - 13.44}{2.02} = -2.20$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 1.96$.

Since calculated value of $Z (= -2.20)$ is less than critical values $(= \pm 1.96)$, that means it lies in rejection region so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so the machine produced good and bad tools are not in random order.

Now, you can try the following exercises.

E8) The following observations were taken from a table of random numbers from a distribution $F(x)$ with median 0:

0.464	0.137	2.455	- 0.323	- 0.068
0.907	-0.513	- 0.525	0.595	0.886
- 0.482	1.678	- 0.057	-1.229	- 0.486
-1.787	-0.261	1.237	1.046	0.962

Test whether the data are random at 5% level of significance?

E9) The cars that enter in a parking are classified either Foreign-made (F) or Indian-made (I). To check that the car enter in the parking are in random order, a sample of first 40 cars that entered in the parking is taken and found the following sequence:

I I F F F I F I F F F F I I F I I I F I F F I F F F I I F F F I F F I I I I

Test that the cars enter in the parking are in random order at 1% level of significance.

13.6 KOLMOGOROV-SMIRNOV GOODNESS OF FIT TEST

This test has its name on the names of its discoverers A. N. Kolmogorov and N.V. Smirnov. It is a simple non-parametric test for testing whether data follow a specified or assumed distribution or sample has come from a specified or assumed distribution or there is a significant difference between an observed distribution and a theoretical distribution. Therefore, it is a measure of goodness of fit to a theoretical distribution. We also use chi-square test for goodness of fit (will be described in Unit 16 of this block). The main difference between chi-square test and Kolmogorov-Smirnov (K-S) test is that the chi-square test is designed for categorical data whereas K-S test is designed for the continuous data.

Assumptions

This test works under the following assumptions:

- (i) The sample is randomly selected from some unknown distribution.
- (ii) The observations are independent.
- (iii) The variable under study is continuous.
- (iv) The variable under study is measured on at least ordinal scale.

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with unknown continuous distribution function $F(x)$. Generally, we are interested to test whether data follow a specified distribution $F_0(x)$ or a sample has come from a specified or assumed distribution or not. So here we consider only two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0 : Data follow a specified distribution

H_1 : Data do not follow a specified distribution

In symbolical form

$$H_0 : F(x) = F_0(x) \quad \text{for all values of } x$$

$$H_1 : F(x) \neq F_0(x) \quad \text{for at least one value of } x \quad [\text{two-tailed test}]$$

We summarise the Kolmogorov-Smirnov goodness of fit test in the following steps:

Step 1: K-S goodness of fit test is based on the comparison of empirical (observed or sample) cumulative distribution function with the theoretical (specified or population) cumulative distribution function. Therefore first of all, we compute the empirical cumulative distribution function, say, $S(x)$ from the sample data which is defined as the proportion of sample observations less than or equal to some value x , that is,

$S(x)$ = proportion of sample observations less than or equal to some value x

$$S(x) = \frac{\text{The number of sample observation less than or equal to } x}{\text{Total number of observations}}$$

Step 2: In this Step, we find the theoretical cumulative distribution function $F_0(x)$ for all possible values of x on the basis of specified distribution.

Step 3: After finding the empirical and theoretical cumulative distribution functions for all possible values of x , we find the deviation between the empirical and theoretical cumulative distribution functions for all x . That is,

$$S(x) - F_0(x) \quad \text{for all } x$$

Step 4: Since $S(x)$ is the statistical image of the population distribution function $F(x)$ so if null hypothesis is true then the difference between $S(x)$ and $F(x)$ should be small for all values of x . Thus, in all deviations obtained in Step 3, we find the point(s) at which the two functions show the maximum deviation.

The test statistic is denoted by D_n and it is the greatest vertical deviation between $S(x)$ and $F_0(x)$, that is,

$$D_n = \sup_x |S(x) - F_0(x)|$$

which is read as “ D_n equals the supreme over all x , of the absolute value of the difference $S(x) - F_0(x)$ ”

Step 5: Obtain critical value of test statistic at α % level of significance under the condition that null hypothesis is true. **Table IV** in Appendix provides the critical values at α level of significance.

Step 6: Decision Rule:

To take the decision about the null hypothesis, the test statistic (calculated in Step 4) is compared with the critical (tabulated) value (obtained in Step 5) for a given level of significance (α).

If computed value of D_n is greater than or equal to critical value $D_{n,\alpha}$ at α level of significance, that is, if $D_n \geq D_{n,\alpha}$ then we reject the null hypothesis at α level of significance, otherwise we do not reject H_0 .

For large sample ($n > 40$):

For a sample size n greater than 40, the critical value of test statistic at given α level of significance, is approximated by the formula given in last row of **Table IV** in the Appendix. As the critical value for two-tailed test at 5% level of significance is calculated by the formula given by

$$D_{n,0.05} = \frac{1.36}{\sqrt{n}}$$

where, n is the sample size.

Let us do some examples to become more user friendly with the calculations explained above:

Example 9: A coin is tossed 400 times in sets of four. The frequencies of getting 0 to 4 tails are:

Number of Tails	0	1	2	3	4	Total
Frequency	23	95	164	92	26	400

Examine the fairness of coin using K-S test at 5% level of significance.

Solution: Here, we want to test that the coin is fair. We know that if a coin is fair then number of tails follows the binomial distribution with parameter n and $p = 1/2$. So we can test that the number of tails follows the binomial distribution. If $F_0(x)$ denotes the cumulative distribution function of binomial distribution with parameter $n = 4$ and $p = 1/2$ then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \neq F_0(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of } x$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of } x$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x |S(x) - F_0(x)|$$

where, $S(x)$ is the empirical distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function $S(x)$, first we obtain cumulative frequencies and then by the definition of empirical distribution function we obtain $S(x)$ as

$$S(x) = \frac{\text{The number of sample observations } \leq x}{\text{Total number of observations}}$$

Therefore, at $x = 0, 1, 2, 3, 4$, we have

$$S(0) = \frac{23}{400} = 0.0575, S(1) = \frac{118}{400} = 0.2950, S(2) = \frac{282}{400} = 0.705,$$

$$S(3) = \frac{374}{400} = 0.935, S(4) = \frac{400}{400} = 1$$

The theoretical distribution can be obtained by the definition of cumulative distribution function as

$$F_0(x) = P[X \leq x]$$

In this case, our theoretical distribution is binomial distribution with parameter $n = 4$ and $p = 1/2$. Therefore,

$$\begin{aligned} \text{i.e. } F_0(x) &= P[X \leq x] = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x} \\ &= {}^4C_x \left(\frac{1}{2}\right)^4, x = 0, 1, 2, 3, 4 \end{aligned}$$

With the help of **Table I** given in Appendix at the end of this block we can find the values of $F_0(x)$ at different values of x .

Thus, at $x = 0$, we have

$$F_0(0) = P[X \leq 0] = 0.0625$$

Similarly,

$$\text{At } x = 1, F_0(1) = P[X \leq 1] = 0.3125$$

$$\text{At } x = 2, F_0(2) = P[X \leq 2] = 0.6875$$

$$\text{At } x = 3, F_0(3) = P[X \leq 3] = 0.9375$$

$$\text{At } x = 4, F_0(4) = P[X \leq 4] = 1$$

Calculation for $|S(x) - F_0(x)|$:

Number of Tails (x)	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function $S(x) = (3)/n$	Theoretical Distribution Function	Absolute Difference $ S(x) - F_0(x) $
(1)	(2)	(3)	(4)	(5)	(6)
0	23	23	0.0575	0.0625	0.0050
1	95	118	0.2950	0.3125	0.0175
2	164	282	0.7050	0.6875	0.0175
3	92	374	0.9350	0.9375	0.0025
4	26	400	1	1	0

From the above calculation, we have

$$D_n = \sup_{0 \leq x \leq 4} |S(x) - F_0(x)| = 0.0175$$

Here, $n = 400 > 40$, therefore, critical value of test statistic can be obtained by the formula given by

$$\begin{aligned} D_{n,0.05} &= \frac{1.36}{\sqrt{n}} \\ &= \frac{1.36}{\sqrt{400}} = 0.068 \end{aligned}$$

The Kolmogorov-Smirnov test for goodness of fit is applied only when all the parameters of the fitting distribution are known. If all parameter (s) of the fitted distribution is (are) not known then we use chi-square goodness of fit

Since calculated value of test statistic $D_n (= 0.0175)$ is less than critical value $D_{n,0.05} (= 0.068)$ so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that number of tails follows the binomial distribution or the coin is fair or unbiased.

Example 10: The following data were obtained from a table of random numbers of normal distribution with mean 3 and standard deviation 1:

2.1, 1.9, 3.2, 2.8, 1.0, 5.1, 4.2, 3.6, 3.9, 2.7

Test by K-S test that the data have come from the same normal distribution at 5% level of significance.

Solution: Here, we want to test that the data have come from normal distribution with mean 3 and variance 1. If $F_0(x)$ is the cumulative distribution function of $N(3,1)$ then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \neq F_0(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of } x$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of } x$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x |S(x) - F_0(x)|$$

where, $S(x)$ is the empirical distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function $S(x)$, first we obtain cumulative frequencies and then by the definition of empirical distribution function we obtain $S(x)$ as

$$S(x) = \frac{\text{The number of sample observations } \leq x}{\text{Total number of observations}}$$

Here, theoretical cumulative distribution function $F_0(x)$ can be obtained with the help of the method as described in Unit 4 of MST-003. Therefore, by the definition of distribution function, we have

$$F_0(x) = P[X \leq x] = P\left[\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right] = P[Z \leq z]$$

where, $P[Z \leq z]$ can be obtained with the help of **Table I** given in Appendix at the end of the Block 1 of this course.

$$\text{Thus, when } x = 1.0, z = \frac{x - \mu}{\sigma} = \frac{1.0 - 3}{1} = -2.0 \text{ and}$$

$$\begin{aligned} F_0(1.0) &= P[Z \leq z] = P[Z \leq -2.0] = 0.5 - P[-2.0 < Z < 0] \\ &= 0.5 - P[0 < Z < -2.0] \\ &= 0.5 - 0.4772 = 0.0228 \end{aligned}$$

Similarly, when $x = 3.2$, $z = \frac{3.2 - 3}{1} = 0.2$ and

$$\begin{aligned} F_0(3.2) &= P[Z \leq z] = P[Z \leq 0.2] \\ &= 0.5 + P[0 < Z \leq 0.2] \\ &= 0.5 + 0.0793 = 0.5793 \text{ and so on.} \end{aligned}$$

Calculation for $|S(x) - F_0(x)|$:

X	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function $S(x)$	$Z = \frac{X - \mu}{\sigma} = \frac{X - 3}{1}$	Theoretical Distribution Function $F_0(x)$	Absolute Difference $ S(x) - F_0(x) $
1.0	1	1	1/10 = 0.1	-2.0	0.0228	0.0772
1.9	1	2	2/10 = 0.2	-1.1	0.1562	0.0438
2.1	1	3	3/10 = 0.3	-0.9	0.1841	0.1159
2.7	1	4	4/10 = 0.4	-0.3	0.3821	0.0179
2.8	1	5	5/10 = 0.5	-0.2	0.4207	0.0793
3.2	1	6	6/10 = 0.6	0.2	0.5793	0.0207
3.6	1	7	7/10 = 0.7	0.6	0.7257	0.0257
3.9	1	8	8/10 = 0.8	0.9	0.8159	0.0159
4.2	1	9	9/10 = 0.9	1.2	0.8849	0.0151
5.1	1	10	10/10 = 1.0	2.1	0.9821	0.0179

From the above calculations, we have

$$D_n = \sup_x |S(x) - F_0(x)| = 0.1159$$

The critical value of test statistic at 5% level of significance is 0.409.

Since calculated value of test statistic $D_n (= 0.1159)$ is less than critical value ($= 0.409$) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the data have come from the normal distribution with mean 3 and standard deviation 1.

Now, you can try the following exercises.

E10) Write the main difference between chi-square test and Kolmogorov-Smirnov (K-S) test for goodness of fit.

E11) The following data were obtained from a table of random numbers of normal distribution with mean 5.6 and standard deviation 1.2:

6.0	6.6	5.9	5.0	6.2	4.8
5.5	6.3	5.9	5.8	6.6	6.2

Test the hypothesis by K-S test that the data have come from the same normal distribution at 5% level of significance.

We now end this unit by giving a summary of what we have covered in it.

13.7 SUMMARY

In this unit, we have discussed following points:

1. Need of non-parametric tests.
 2. Applications of non-parametric tests in various fields.
 3. Advantages and disadvantages of non-parametric tests over parametric tests.
 4. The one-sample non-parametric tests.
 5. The suitable non-parametric test is appropriate for a particular situation.
 6. Test the hypothesis when we cannot make any assumption about the distribution from which we are sampling.
 7. The tests which are used in place of t-test for mean when assumption(s) of the t-test is (are) not fulfilled, that is, sign test and Wilcoxon signed-rank test.
 8. The test for randomness of data, that is, run test.
 9. The Kolmogorov-Smirnov test for goodness of fit.
-

13.8 SOLUTIONS /ANSWERS

E1) Refer Section 13.2.

E2) The main difference between t-test and sign test is that the t-test based on the assumption that the parent population is normal where as sign test is not based on such assumption.

E3) Here, distribution of the waiting time on a well travelled rout is not given. So the assumption of normality for t-test is not fulfilled. Also sample size is small so we can not use Z-test. So we go for sign test.

Here, we wish to test the claim of Metropolitan Area Road Transport Authority that the average (median) waiting time ($\tilde{\mu}$) on a well travelled rout is 20 minutes. So our claim is $\tilde{\mu} = 20$ and it complement is $\tilde{\mu} \neq 20$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 20 \text{ and } H_1 : \tilde{\mu} \neq 20$$

For applying sign test, we replacing each observation greater than 20 (median value) with a plus sign and each observation less than 20 with a minus sign and discarding the one observation which equal to 20, we get

+ - - - + + - + + - -

By counting, we have

$$S^+ = \text{number of plus signs} = 5$$

$$S^- = \text{number of mines signs} = 6$$

$$n = \text{total number of plus and minus signs} = 11$$

Since alternative hypothesis is two-tailed so the test statistic (S) is given by

$$S = \min\{S^+, S^-\}$$

$$= \min\{5, 6\} = 5$$

Here, $n = 11 (< 20)$ so it is a case of small sample. Thus, to take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block. Here, $n = 11$, $p = 0.5$ and $r = 5$. Thus, we have

$$\text{p-value} = P[S \leq 5] = 0.50$$

Since $\text{p-value} = 0.50 > 0.01 (\alpha)$. So we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the claim of Metropolitan Area Road Transport Authority that the average waiting time is 20 minutes is true.

- E4)** Here, distribution of the weight of the cadets is not given. So the assumption of normality for t-test is not fulfilled. So we go for sign test.

Here, we want to test that median weight ($\tilde{\mu}$) of all the cadets of a college is 50 kg. So our claim is $\tilde{\mu} = 50$ and its complement is $\tilde{\mu} \neq 50$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 50 \text{ [median weight is equal to 50 kg]}$$

$$H_1 : \tilde{\mu} \neq 50 \text{ [median weight is not equal to 50 kg]}$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

For applying sign test, we replace each observation greater than 50 (median value) with a plus sign and each observation less than 50 with a minus sign and discard the two observations which are equal to 50, we get

–	–	+	–	–	+	+	–	+	–
–	–	+	+	–	+	–	+	+	–
+	–	+	–	+	+	–	–		

By counting, we have

$$S^+ = \text{number of plus signs} = 13$$

$$S^- = \text{number of minus signs} = 15$$

$$n = \text{total number of plus and minus signs} = 28$$

Since alternative hypothesis is two-tailed so the test statistic (S) is the minimum of number of plus signs (S^+) and minus signs (S^-), that is,

$$S = \min\{S^+, S^-\} = \min\{13, 15\} = 13$$

Here, $n = 28 (> 20)$, so it is a case of large sample. In this case, we use Z-test. The test-statistic of Z-test is given by

$$Z = \frac{S - \frac{n}{2}}{\sqrt{\frac{n}{4}}} = \frac{13 - \frac{28}{2}}{\sqrt{\frac{28}{4}}}$$

$$= \frac{13 - 14}{\sqrt{7}} = 0.38$$

The critical (tabulated) values for two-tailed test at 5% level of significance are $\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96$.

Since calculated value of test statistic $Z (= 0.38)$ is greater than the critical value $(= -1.96)$ and less than the critical value $(= 1.96)$, that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the samples do not provide us sufficient evidence against the claim so we may assume that the median weight of all the cadets of the centre is 50 kg.

- E5)** The main difference between sign test and Wilcoxon signed-rank test is that the sign test utilizes only information whether the difference between observation and the postulated value of the median is positive or negative, that is, it considers only sign of differences but ignores the magnitude of these differences whereas the Wilcoxon signed-rank test uses both the signs as well as the magnitude of the differences.
- E6)** We are given that characteristic under study is continuous and population is symmetric about its median and other assumptions of Wilcoxon signed-rank test hold. So we can apply this test.

Here, we want to test that sample is taken from the population having median ($\tilde{\mu}$) greater than 18. So our claim is $\tilde{\mu} > 18$ and its complement is $\tilde{\mu} \leq 18$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} \leq \tilde{\mu}_0 = 18 \text{ and } H_1 : \tilde{\mu} > 18$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test and the test statistic (T) is the sum negative ranks (T^-).

Calculation for T :

S. No.	Sample (X)	Difference $d = X - \tilde{\mu} = X - 18$	Absolute Value of Difference $ d $	Rank of $ d $	Signed Rank
1	20.2	2.2	2.2	7	7
2	24.1	6.1	6.1	10	10
3	21.3	3.3	3.3	8	8
4	17.2	-0.8	0.8	2	-2
5	19.8	1.8	1.8	5	5
6	16.5	-1.5	1.5	4	-4
7	21.8	3.8	3.8	9	9
8	18.7	0.7	0.7	1	1
9	17.1	-0.9	0.9	3	-3
10	19.9	1.9	1.9	6	6

Therefore, we have

$$T^+ = 7 + 10 + 8 + 5 + 9 + 1 + 6 = 46$$

$$T^- = 2 + 4 + 3 = 9$$

$$n = \text{number of non-zero } d_i\text{'s} = 10$$

So the value of test statistic $T = 9$

The critical value of test statistic T for right-tailed test corresponding $n = 10$ at 5% level of significance is 11.

Since calculated value of test statistic $T (= 9)$ is less than the critical value ($= 11$) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the sample is taken from the population with median greater than 18.

E7) Here all the assumptions of Wilcoxon signed-rank test hold. So we go for this test.

Here, we want to test the manufacturer's claim that the average (median) breaking strength ($\tilde{\mu}$) of the rods is 30 pounds. So our claim is $\tilde{\mu} = 30$ and its complement is $\tilde{\mu} \neq 30$. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 = 30 \quad [\text{median breaking strength is 30}]$$

$$H_1 : \tilde{\mu} \neq 30 \quad [\text{median breaking strength is not 30}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test and the test statistic is the smaller of sum of positive ranks (T^+) and sum of negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

Calculation for T :

S. No.	Breaking Strength (X)	Difference $d = X - \tilde{\mu} = X - 30$	Absolute Value of Difference $ d $	Rank of $ d $	Signed Rank
1	19	-11	11	23.5	-23.5
2	30	Tie	---	---	---
3	28	-2	2	4	-4
4	35	5	5	9.5	9.5
5	23	-7	7	16	-16
6	37	7	7	16	16
7	36	6	6	12.5	12.5
8	32	2	2	4	4
9	40	10	10	21.5	21.5
10	24	-6	6	11.5	-11.5
11	31	1	1	1	1
12	33	3	3	7	7

13	21	-9	9	20	-20
14	36	6	6	12.5	12.5
15	30	Tie	---	---	---
16	32	2	2	4	4
17	26	-4	4	8	-8
18	40	10	10	21.5	21.5
19	38	8	8	18.5	18.5
20	28	-2	2	4	4
21	30	Tie	---	---	---
22	41	11	11	23.5	23.5
23	22	-8	8	18.5	-18.5
24	43	13	13	25.5	25.5
25	25	-5	5	9.5	-9.5
26	28	-2	2	4	-4
27	24	-6	6	12.5	-12.5
28	37	7	7	16	16
29	17	-13	13	25.5	-25.5

From the above calculation, we have

$$T^+ = 9.5 + 16 + 12.5 + 4 + 21.5 + 1 + 7 + 12.5 + 4 + 21.5 + 18.5 + 23.5 + 25.5 + 16 = 193.0$$

$$T^- = 23.5 + 4 + 16 + 12.5 + 20 + 8 + 4 + 18.5 + 9.5 + 4 + 12.5 + 25.5 = 158.0$$

n = number of non-zero d_i 's = 29

Putting the values in test statistic, we have

$$T = \min\{T^+, T^-\} = \min\{193.0, 158.0\} = 158.0$$

Also $n = 29 (> 20)$, therefore, it is the case of large sample. So in this case, we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{T - E(T)}{SE(T)} \sim N(0,1)$$

$$\text{where, } E(T) = \frac{n(n+1)}{4} = \frac{29(29+1)}{4} = 217.5 \text{ and}$$

$$SE(T) = \sqrt{\frac{n(n+1)(2n+1)}{24}} \\ = \sqrt{\frac{29(29+1)(2 \times 29+1)}{24}} = 46.25$$

Putting the values in test statistic Z , we have

$$Z = \frac{158.0 - 217.5}{46.25} = -1.29$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of test statistic $Z (= -1.29)$ is greater than the critical value $(= -2.58)$ and less than the critical value $(= 2.58)$, that means it lies in non-rejection region so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the manufacturer's claim that the average (median) breaking strength of the rods is 30 pounds is true.

- E8)** Here, we want to test the randomness of the given data so we go for run test.

We want to test that given data are in random order. So our claim is "the data are in random order" and its complement is "the data are not in random order". So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The data are in random order

H_1 : The data are not in random order

For applying run test, we need given observations in a sequence of two symbols. In order to convert given observations into a sequence of two symbols, first of all we replace the given observations by the symbol A if it is above the median value $(= 0)$ and by symbol B if it is below the median value and discard the observation equal to median value. In this way, we get a sequence of symbols A's and B's which is given below:

$\underbrace{AAA}_{1} \underbrace{BB}_{2} \underbrace{A}_{3} \underbrace{BB}_{4} \underbrace{AA}_{5} \underbrace{B}_{6} \underbrace{A}_{7} \underbrace{BBBBB}_{8} \underbrace{AAA}_{9}$

Here, we see that there is a first run of three A's then there is a run of two B's and so on. In the last there is a run of three A's.

Here by counting, we have

n_1 = number of symbol A occurs = 10

n_2 = number of symbol B occurs = 10

R = total number of runs in above sequence = 9

For run test the test statistic is the total number of runs in the given sequence.

Since $n_1 = 10 (< 20)$ and $n_2 = 10 (< 20)$, so it is a small sample case.

The lower and upper critical values of runs corresponding to n_1 and n_2 at 5% level of significance are $R_L = 6$, $R_U = 16$.

Since observed number of runs ($R = 9$) lies between critical values of runs, $R_L = 6$ and $R_U = 16$ so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the given data are in random order.

- E9)** Here, we want to test the randomness of the cars entered in the parking so we go for run test.

Also we wish to test that the cars entered in the parking are in random order. So our claim is "the cars that enter in the parking are in random order" and its complement is "the cars that enter in the parking are not

in random order”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The cars that enter in the parking are in random order

H_1 : The cars that enter in the parking are not in random order

For applying run test, we count the number of runs in the given sequence.

$\underbrace{II}_{1\ 2} \underbrace{FFF}_{3\ 4\ 5} \underbrace{IF}_{6} \underbrace{FFFF}_{7\ 8\ 9\ 10} \underbrace{IIF}_{11\ 12} \underbrace{FF}_{13} \underbrace{I}_{14} \underbrace{FFFF}_{15\ 16\ 17} \underbrace{IIF}_{18} \underbrace{FF}_{19} \underbrace{IIII}_{20}$

We see that there is a first run of two I's then there is a run of three F's and so on in the last there is a run of four I's. By counting, we have

n_1 = number of symbol I occurs = 18

n_2 = number of symbol F occurs = 22

R = total number of runs in above sequence = 19

Since $n_2 = 22(>20)$ therefore, we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{R - E(R)}{SE(R)} \sim N(0,1)$$

$$\text{where } E(R) = \frac{2n_1n_2}{n} + 1$$

$$= \frac{2 \times 18 \times 22}{40} + 1 = 20.8 \text{ and}$$

$$SE(R) = \sqrt{\frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}} \\ = \sqrt{\frac{2 \times 18 \times 22(2 \times 18 \times 22 - 40)}{40^2(40-1)}} = \sqrt{9.54} = 3.08$$

Putting the values in test statistic Z, we have

$$Z = \frac{19 - 20.8}{3.08} = -0.58$$

The critical (tabulated) values for two-tailed test at 1% level of significance are $\pm z_{\alpha/2} = \pm z_{0.005} = \pm 2.58$.

Since calculated value of Z ($= -0.58$) is greater than critical value ($= -2.58$) and less than critical value ($= 2.58$), that means it lies in non-rejection region, so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the cars enter in the parking lot are in random order.

E10) The main difference between chi-square test and Kolmogorov-Smirnov (K-S) test is that the chi-square test is designed for categorical data whereas K-S test is designed for the continuous data.

- E11)** Here, we want to test that the data have come from normal distribution with mean 5.6 and standard deviation 1.2. If $F_0(x)$ is the cumulative distribution function of this normal distribution then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \neq F_0(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of } x$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of } x$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x |S(x) - F_0(x)|$$

where, $S(x)$ is the empirical cumulative distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function $S(x)$, first we obtain cumulative frequencies and then by the definition of empirical cumulative distribution function we obtain $S(x)$ as

$$S(x) = \frac{\text{The number of sample observations} \leq x}{\text{Total number of observations}}$$

Here, theoretical cumulative distribution function $F_0(x)$ can be obtained with the help of the method as described in Unit 4 of Block 4 of the course MST-003. Therefore, by the definition of distribution function, we have

$$F_0(x) = P[X \leq x] = P\left[\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right] = P[Z \leq z]$$

where, $P[Z \leq z]$ can be obtained with the help of **Table I** given in Appendix at the end of the Block 1 of this course.

Thus, when $x = 4.8$, $z = \frac{x - \mu}{\sigma} = \frac{4.8 - 5.6}{1.2} = -0.67$ and

$$\begin{aligned} F_0(4.8) &= P[Z \leq z] = P[Z \leq -0.67] \\ &= 0.5 - P[0 < Z < 0.67] \\ &= 0.5 - 0.2486 \\ &= 0.2514 \end{aligned}$$

Similarly, when $x = 5.8$, $z = \frac{5.8 - 5.6}{1.2} = 0.17$ and

$$\begin{aligned} F_0(5.8) &= P[Z \leq z] = P[Z \leq 0.17] \\ &= 0.5 + P[0 < Z \leq 0.17] \\ &= 0.5 + 0.0675 = 0.5675 \text{ and so on.} \end{aligned}$$

Calculation for $|S(x) - F_0(x)|$:

X	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function $S(x)$	$Z = \frac{X - \mu}{\sigma} = \frac{X - 5.6}{1.2}$	Theoretical Distribution Function $F_0(x)$	Absolute Difference $ S(x) - F_0(x) $
4.8	1	1	$1/12 = 0.0833$	-0.67	0.2514	0.1681
5.0	1	2	$2/12 = 0.1667$	-0.50	0.3085	0.1418
5.5	1	3	$3/12 = 0.2500$	-0.08	0.4681	0.2181
5.8	1	4	$4/12 = 0.3333$	0.17	0.5675	0.2342
5.9	2	6	$6/12 = 0.5000$	0.25	0.5987	0.0987
6.0	1	7	$7/12 = 0.5833$	0.33	0.6293	0.0460
6.2	2	9	$9/12 = 0.7500$	0.50	0.6915	0.0585
6.3	1	10	$10/12 = 0.8333$	0.58	0.7190	0.1143
6.6	2	12	$12/12 = 1.0000$	0.83	0.7967	0.2033

From the above calculations, we have

$$D_n = \sup_x |S(x) - F_0(x)| = 0.2342$$

The critical value of test statistic D_n corresponding $n = 12$ at 5% level of significance is 0.375.

Since calculated value of test statistic $D_n (= 0.2342)$ is less than critical value ($= 0.375$) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that data have come from the normal distribution with mean 5.6 and standard deviation 1.2.

UNIT 14 TWO-SAMPLE TESTS

Structure

- 14.1 Introduction
 - Objectives
- 14.2 Paired Sign Test
- 14.3 Wilcoxon Matched-Pair Signed-Rank Test
- 14.4 Mann-Whitney U Test
- 14.5 Kolmogorov-Smirnov Two-Sample Test
- 14.6 Summary
- 14.7 Solutions / Answers

14.1 INTRODUCTION

In previous unit, we have already highlighted the main difference between parametric and non-parametric tests. Also we have discussed some well know/popular/commonly used one-sample non-parametric tests in the same unit. But comparison of a characteristic in two populations is a common problem faced by statisticians. We have already addressed this problem when distributions of the two populations are known to us. But what will do if assumption(s) of the test is (are) not fulfilled as distributions of the two populations are not known to us. This unit is devoted to address this problem. In this unit, we describe some of the frequently used two-sample tests based on non-parametric methods.

This unit is divided into seven sections. Section 14.1 is described the need of two-sample non-parametric tests. The paired sign test and Wilcoxon matched-pair signed-rank test for two samples are described in Sections 14.2 and 14.3 respectively. The Mann-Whitney U test which is used to test the difference in median of two independent populations is discussed in Section 14.4. In Section 14.5, we describe the Kolmogorov-Smirnov two-sample test. Unit ends by providing summary of what we have discussed in this unit in Section 14.6 and solution of exercises in Section 14.7.

Objectives

After completion of this unit, you should be able to:

- describe the need of two-sample non-parametric tests;
- identify which of the two-sample non-parametric tests is appropriate for a given situation;
- perform the test which is used in place of paired t-test when assumption(s) of the paired t-test is (are) not fulfilled, that is, paired sign test and Wilcoxon matched-pair signed-rank test;
- describe the Mann-Whitney U test which is used when the assumption(s) of two sample t-test is (are) not fulfilled;
- describe the Kolmogorov-Smirnor two-sample test for testing the hypothesis that two samples come from the populations having the same distribution.

14.2 PAIRED SIGN TEST

Paired sign test is the non-parametric version of the paired t-test. It is used when the assumption(s) of paired t-test is (are) not fulfilled.

In social sciences and in marketing research, generally two related groups are paired and we are interested to examine differences among two related groups. This often happens when the observations are recorded as before and after. Here, the paired observations are recorded on the same individuals or items. In these situations, we use paired t-test. If the assumptions of paired t-test is (are) not fulfilled then we use paired sign test, that means paired sign test is the non parametric version of the paired t-test. For example, if a college administrator is interested in knowing how a particular three-weeks training on basic Statistics affects student's grades in Statistics paper. In this situation, paired t-test is not applicable because of grades, that is, observations are given in ordinal scale as A, A⁺, B, B⁺, etc. But paired sign test can be applied because it needs only to convert the observations into plus and minus signs.

Assumptions

This test needs following assumptions to work:

- (i) The pairs of measurements are independent.
- (ii) The measurement scale is at least ordinal within each pair.
- (iii) The variable under study is continuous.

Let us discuss the general procedure of this test:

Let $(X_1, Y_1), (X_2, Y_2), \dots, (X_{n'}, Y_{n'})$ be a random sample of n' independent items / units / individuals, each with two measurements as before and after the training, diet, treatment, medicine, etc. Also let $\tilde{\mu}_1$ and $\tilde{\mu}_2$ be the medians of the populations before and after the training, diet, treatment, medicine, etc. Here, we want to test whether there is an effect of a diet, training, treatment, medicine, etc. or not. So we can take the null and alternative hypotheses as

$$\begin{aligned}
 &H_0: \tilde{\mu}_1 = \tilde{\mu}_2 \text{ or } H_0: \tilde{\mu}_1 - \tilde{\mu}_2 = 0 \\
 &H_1: \tilde{\mu}_1 \neq \tilde{\mu}_2 \text{ or } H_1: \tilde{\mu}_1 - \tilde{\mu}_2 \neq 0 \quad [\text{for two-tailed test}] \\
 \text{or} \quad &\left. \begin{aligned} H_0: \tilde{\mu}_1 \leq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 > \tilde{\mu}_2 \\ H_0: \tilde{\mu}_1 \geq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 < \tilde{\mu}_2 \end{aligned} \right\} \quad [\text{for one-tailed test}]
 \end{aligned}$$

The procedure for paired sign test is similar to sign test for one sample as discussed in pervious unit. After setting null and alternative hypothesis, paired sign test involves following steps:

Step 1: First of all, we convert the given observations into a sequence of plus and minus signs. For this, we find the difference $X_i - Y_i$ for all observations and consider only the sign of difference as plus or minus or compare X_i with Y_i for all observations. If $X_i > Y_i$ then we replace the pair of observation (X_i, Y_i) by a plus sign and if $X_i < Y_i$ then we replace (X_i, Y_i) by minus sign. But when $X_i = Y_i$, then corresponding pair of observations gives no information in terms of plus or minus signs so we exclude all such pairs from the analysis part. Due to such observations our common sample size reduces and let reduced common sample size be denoted by n .

Step 2: After that, we count the number of plus signs and number of minus signs. Suppose they are denoted by S^+ and S^- respectively.

Step 3: When null hypothesis is true and the population is dichotomised on the basis of sign of the difference then we expect that the number of plus signs (success) and number of minus signs (failure) approximately equal. And number of plus signs or number of minus signs follows binomial distribution ($n, p = 0.5$). For convenient consider the smaller number of plus or minus signs. If S^+ is less than S^- then we will take plus sign as success and minus sign as failure. Similarly, if S^- is less than S^+ then we will assume minus sign as success and plus sign as failure.

Step 4: To take the decision about the null hypothesis, we use concept of p-value as we have discussed in sign test for one sample. For p-value we determine the probability that test statistic is less than or equal to the actually observed of plus or minus signs. Since distribution of number of plus or minus signs is binomial ($n, p = 0.5$) therefore, this probability can be obtained with the help of **Table I** given in Appendix at the end of this block which provide the cumulative binomial probability and compare this probability with the level of significance (α). Here, test statistic depends upon the alternative hypothesis so the following cases arise:

The critical values of this test are not generally in tabular form and slightly difficult to obtain whereas p-value is easy to obtained with the help of cumulative binomial table so we use concept of p-value for take the decision about the null hypothesis.

For one-tailed test:

Case I: When $H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 > \tilde{\mu}_2$ (right-tailed test)

In this case, we expect that number of minus signs (S^-) is smaller than number of plus signs (S^+) therefore, the test statistic(S) is the number of minus signs (S^-). The p-value is determined as

$$\text{p-value} = P[S \leq S^-]$$

If p-value is less than or equal to α , that is, $P[S \leq S^-] \leq \alpha$ then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

Case II: When $H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 < \tilde{\mu}_2$ (left-tailed test)

In this case, we expect that number of plus signs (S^+) is smaller than number of minus signs (S^-) therefore, the test statistic(S) is the number of plus signs (S^+). The p-value is determined as

$$\text{p-value} = P[S \leq S^+]$$

If p-value is less than or equal to α , that is, $P[S \leq S^+] \leq \alpha$ then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

For two-tailed test: When $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$

For two tailed test, the test statistic (S) is the smaller of number of plus signs (S^+) and minus signs (S^-), that is,

$$S = \min \{S^+, S^-\}$$

and approximate p-value is determined as

$$\text{p-value} = 2P[S \leq S^+]; \quad \text{if } S^+ \text{ is small}$$

$$\text{p-value} = 2P[S \leq S^-]; \quad \text{if } S^- \text{ is small}$$

If p-value is less than or equal to α then we reject the null hypothesis at α level of significance and if the p-value is greater than α then we do not reject the null hypothesis.

For large sample ($n > 20$):

For a large sample size n greater than and equal to 20, we use normal approximation to binomial distribution with mean

$$E(S) = np = n \times \frac{1}{2} = \frac{n}{2} \quad \dots (1)$$

and variance

$$\text{Var}(S) = npq = n \times \frac{1}{2} \times \frac{1}{2} = \frac{n}{4} \quad \dots (2)$$

Therefore, we use normal test (Z-test). The test statistic of Z-test is given by

$$\begin{aligned} Z &= \frac{S - E(S)}{SE(S)} \\ &= \frac{S - \frac{n}{2}}{\sqrt{\frac{n}{4}}} \sim N(0,1) \quad [\text{Using equations (1) and (2)}] \quad \dots (3) \end{aligned}$$

After that, we calculate the value of test statistic Z and compare it with the critical value(s) given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2 of Unit 10 of this course.

Now, let us do some examples based on paired sign test.

Example 1: To compare the softness of the paper of two types A and B, a sample of each product of A and B is given to ten judges. Each judge rates the softness of each product on a scale from 1 to 10 with higher ratings implying a softer product. The results of the experiment are shown below. The first rating in the pair is of product A and the second is of product B.

(4, 5) (6, 4) (8, 5) (9, 8) (4, 1) (7, 9) (6, 2) (5, 3) (8, 2) (6, 7)

Apply the paired sign test to test the hypothesis that the products A and B are equally soft at 1% level of significance.

Solution: Here, the ratings are given in pair form so it can be considered as the case of dependent samples. But assumption of normality is not given also the given data are in the form of ranks so we cannot use the paired t-test in this case. So we go for paired sign test.

Here, we wish to test that the product A and B are equally soft. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average rank of softness of products A and B respectively then our claim is $\tilde{\mu}_1 = \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \neq \tilde{\mu}_2$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 \text{ and } H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test and the test statistic (S) is the minimum of plus signs (S^+) and minus signs (S^-), that is,

$$S = \min\{S^+, S^-\}$$

Calculation for S:

Judge	Product A	Product B	Sign of Difference (A-B)
1	4	5	-
2	6	4	+
3	8	5	+
4	9	8	+
5	4	1	+
6	7	9	-
7	6	2	+
8	5	3	+
9	8	2	+
10	6	7	-

From the above, we have

n = number of non zero differences = 10

S^+ = number of plus signs = 7

S^- = number of minus signs = 3

Therefore,

$$S = \min\{S^+, S^-\} = \min\{7, 3\} = 3$$

To take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block. Here, $n = 10$, $p = 0.5$ and $r = 3$. Thus, we have

$$p\text{-value} = 2P[S \leq 3] = 2 \times 0.1719 = 0.3438$$

Since $p\text{-value} = 0.3438 > 0.01 (= \alpha)$ so we do not reject the null hypothesis i.e. support the claim.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that products A and B are equally soft.

Example 2: The following are the weight (in pounds) before and after a dieting programme of 18 persons for one month:

Before	148.0	181.5	150.5	230.2	196.5	206.3	170.0	214.4	140.6
After	139.2	172.4	154.2	212.6	193.2	200.0	163.0	218.0	140.0
Before	166.8	150.3	197.2	168.0	172.4	160.3	210.0	172.4	180.2
After	166.8	156.2	185.4	158.5	167.2	162.3	202.3	166.2	170.5

Using the paired sign test to test whether the dieting programme is effective or not at 5% level of significance.

Solution: It is case of before and after study, so either we can go for paired t-test or paired sign test. But t-test requires assumptions of normality which is not given. So we go for paired sign test.

If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the median weights of the two depending populations before and after the dieting programme respectively. Here, we want to test whether the dieting programme is effective or not. So we can take our claim that the dieting programme is effective that means average (median) weight before dieting programme is greater than average (median) weight after the programme, that is, $\tilde{\mu}_1 > \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \leq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2 \text{ and } H_1 : \tilde{\mu}_1 > \tilde{\mu}_2$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Since alternative hypothesis is right-tailed so the test statistic (S) is the number of minus signs (S^-)

Calculation for S:

S. No	Weight before diet (X)	Weight after diet (Y)	Sign of Difference (X – Y)
1	148.0	139.2	+
2	181.5	172.4	+
3	150.5	154.2	–
4	230.2	212.6	+
5	196.5	193.2	+
6	206.3	200.0	+
7	170.0	163.0	+
8	214.4	218.0	–
9	140.6	140.0	+
10	166.8	166.8	Tie
11	150.3	156.2	–
12	197.2	185.4	+
13	168.0	158.5	+
14	172.4	167.2	+
15	160.3	162.3	–
16	210.0	202.3	+
17	172.4	166.2	+
18	180.2	170.5	+

From the above, we have

n = number of non-zero difference = 17

S^+ = number of plus signs = 13

S^- = number of minus signs = 4

Therefore, value of test statistic(S) is $S^- = 4$

To take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block. Here, $n = 17$, $p = 0.5$ and $r = 4$. Thus, we have

$$p\text{-value} = P[S \leq 4] = 0.0245$$

Since $p\text{-value} = 0.0245 < 0.05$ (α) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so may assume that the dieting programme is effective.

Now, you can try the following exercises.

-
- E1)** Write one difference between paired sign test and paired t-test.
- E2)** Students are given a course designed to improve their IQ on a standard test. Their scores before and after the course are recorded as follows:

Before	95	112	128	96	97	117	105	95	99	86	75	82
After	99	115	133	96	99	120	106	106	100	89	90	85

With the help of paired sign test, examine whether IQ is affected due to course at 1% level of significance?

E3) A randomly chosen group of 15 consumers was asked to rate the two products A and B on a scale of 1 to 5 (5 being the highest). For each consumer, the first rating in the pair is of product A and the second is of product B. The data are:

(5, 2) (2, 3) (4, 1) (4, 2) (5, 3) (4, 4) (5, 4) (5, 1) (2, 4)
 (3, 2) (3, 2) (4, 3) (3, 3) (3, 4) (2, 3)

The aim of study was to show that product A is favoured over product B. Do these data present evidence to justify this claim at 1% level of significance?

14.3 WILCOXON MATCHED-PAIR SIGNED-RANK TEST

We have already discussed Wilcoxon signed-rank test for one-sample in Unit 13. The Wilcoxon signed-rank test is also useful in comparing two populations for which we have paired observations. In this case, it is known as Wilcoxon matched-pair signed rank-test. As such, the test is a good alternative to the paired t-test in cases where the differences between paired observations are not believed to be normally distributed. Paired sign test discussed in previous section of this unit is also for the same situation. The paired sign test uses only the information whether X is larger or smaller than Y. This test is fine if the information about the observation sample is available in ordinal scale only. But if the measurement of the observation is available interval or ratio scales then choice of paired sign test is not recommended. Because this test do not take into account the information available in terms of the magnitude of the differences. To overcome this drawback of the paired sign test Wilcoxon matched-pair signed-rank test do the job for us, which takes into account the information of signs as well as of magnitude of differences. The Wilcoxon matched-pair signed-rank test is exactly the same as Wilcoxon signed-rank test described in Section 13.4 of previous unit of this course when $d_i = X_i - \tilde{\mu}_0$ is replaced by $d_i = X_i - Y_i$.

Assumptions

This test needs following assumptions to work:

- (i) The distribution of differences d_i 's is symmetric.
- (ii) The differences d_i 's are mutually independent.
- (iii) The measurement scale is at least interval within each pair.
- (iv) The variable under study is continuous.

Let us discuss the general procedure of this test:

Let $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ be a random sample of n independent items / units / individuals each with two measurements as before and after the training, diet, treatment, medicine, etc. Also let $\tilde{\mu}_1$ and $\tilde{\mu}_2$ be the medians of the populations before and after the training, diet, treatment, medicine, etc. Here, we want to test whether there is an effect of a diet, training, treatment, medicine, etc. or not. So we can take the null and alternative hypotheses as

$$H_0: \tilde{\mu}_1 = \tilde{\mu}_2 \text{ or } H_0: \tilde{\mu}_1 - \tilde{\mu}_2 = 0$$

$$H_1: \tilde{\mu}_1 \neq \tilde{\mu}_2 \text{ or } H_1: \tilde{\mu}_1 - \tilde{\mu}_2 \neq 0 \quad [\text{for two-tailed test}]$$

$$\text{or } \left. \begin{array}{l} H_0: \tilde{\mu}_1 \leq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 > \tilde{\mu}_2 \\ H_0: \tilde{\mu}_1 \geq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 < \tilde{\mu}_2 \end{array} \right\} \quad [\text{for one-tailed test}]$$

The procedure of this test is similar to Wilcoxon signed-rank test for one-sample as discussed in pervious unit. After setting null and alternative hypotheses, this test involves following steps:

Step 1: First of all, we obtain the difference d_i for all observations with their plus and minus sign as

$$d_i = X_i - Y_i \text{ for all observations}$$

But when the observation X_i equal to Y_i then the difference $d_i = X_i - Y_i$ gives no information in terms of signs as well as in terms of magnitude of the difference. So we exclude all such observations from the analysis part. Due to these observations our sample size reduces and let reduced sample size be denoted by n .

Step 2: After that, we find the absolute value of these d_i 's obtained in Step1 as $|d_1|, |d_2|, \dots, |d_n|$.

Step 3: In this step, we rank all $|d_i|$'s (obtained in Step 2) with respect to their magnitudes from smallest to largest, that is, the rank 1 is given to the smallest of $|d_i|$'s, rank 2 is given to the second smallest and so on up to the largest $|d_i|$'s. If several values are same (tied), we assign each the average of ranks they would have received if there were no repetition.

Step 4: Now, assign the signs to the ranks which the original differences have.

Step 5: Finally, we calculate the sum of the positive ranks (T^+) and sum of negative ranks (T^-) separately.

Under H_0 , we expect approximately equal number of positive and negative ranks. And under the assumption that distribution of differences is symmetric about its median, we expect that sum of the positive ranks (T^+) and sum of negative ranks (T^-) are equal.

Step 6: Decision Rule:

To take the decision about the null hypothesis, the test statistic (T) is the smaller of T^+ and T^- . And the test statistic is compared with the critical (tabulated) value for a given level of significance (α) under the condition that the null hypothesis is true. **Table II** given in Appendix at the end of this block provides the critical values of test statistic at α level of significance for both one-tailed and two-tailed tests. Here, test statistic depends upon the alternative hypothesis so the following cases arise:

For one-tailed test:

Case I: When $H_0: \tilde{\mu}_1 \leq \tilde{\mu}_2$ and $H_1: \tilde{\mu}_1 > \tilde{\mu}_2$ (right-tailed test)

In this case, we expect that sum of negative ranks (T^-) is smaller than sum of positive ranks (T^+) therefore, the test statistic (T) is the sum of negative ranks (T^-).

For more detailed about repeated (tied) rank please go through Section 7.4 of the Unit 2 of MST-002

If computed value of test statistic (T) is less than or equal to the critical value T_α at α level of significance, that is, $T \leq T_\alpha$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

Case II: When $H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 < \tilde{\mu}_2$ (left-tailed test)

In this case, we expect that sum of positive ranks (T^+) is smaller than sum of negative ranks (T^-) therefore, the test statistic (T) is the sum of positive ranks (T^+).

If computed value of test statistic (T) is less than or equal to the critical value T_α at α level of significance, that is, $T \leq T_\alpha$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

For two-tailed test:

When $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$

In this case, the test statistic (T) is the smaller of sum of positive ranks (T^+) and sum of negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

If computed value of test statistic (T) is less than or equal to the critical value $T_{\alpha/2}$ at α level of significance, that is, $T \leq T_{\alpha/2}$ then we reject the null hypothesis at α level of significance, otherwise we do not reject the null hypothesis.

For large sample ($n > 25$):

For a large sample size n greater than 25, the distribution of test statistic (T) approximated by a normal distribution with mean

$$E(T) = \frac{n(n+1)}{4} \quad \dots (4)$$

and variance

$$\text{Var}(T) = \frac{n(n+1)(2n+1)}{24} \quad \dots (5)$$

The proof of mean and variance of test statistic T is beyond the scope of this course.

Therefore in this case, we use normal test (Z-test). The test statistic of Z-test is given by

$$\begin{aligned} Z &= \frac{T - E(T)}{SE(T)} = \frac{T - E(T)}{\sqrt{\text{Var}(T)}} \\ &= \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}} \sim N(0,1) \left[\begin{array}{l} \text{Using equations} \\ (4) \text{ and } (5) \end{array} \right] \quad \dots (6) \end{aligned}$$

After that, we calculate the value of test statistic Z and compare it with the critical value(s) given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2 of Unit 10 of this course.

Now, it is time to do some examples based on above test.

Example 3: A group of 12 children was tested to find out how many digits they would repeat from memory after hearing them once. They were given practice session for this test. Next week they were retested. The results obtained were as follows:

Child Number	1	2	3	4	5	6	7	8	9	10	11	12
Recall Before	6	4	5	7	6	4	3	7	8	4	6	5
Recall After	6	6	4	7	6	5	5	9	9	7	8	7

Assuming that the distribution of the differences of the scores before and after the practice session is symmetrical about its median, can the memory practice session improve the performance of children at 5% level of significance using Wilcoxon matched-pair signed-rank?

Solution: It is case of before and after study so we can go for paired t-test or paired sign test or Wilcoxon matched-pair signed-rank. But t-test requires assumptions of normality which is not given. Also measurement of observations is available in interval scale, so we will go for Wilcoxon matched-pair signed-rank test.

Here, we wish to test that memory practice session improves the performance of children. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) score before and after the practice session so our claim is $\tilde{\mu}_1 < \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \geq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 < \tilde{\mu}_2 \quad [\text{memory practice session improves the performance}]$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test and the test statistic (T) is the sum of positive ranks (T^+).

Calculation for T:

Child Number	Recall Before (X)	Recall After (Y)	Difference (X-Y)	Absolute Difference d	Rank of d	Signed Rank
1	6	6	Tie	---	----	----
2	4	6	-2	2	6	-6
3	5	4	-1	1	2	-2
4	7	7	Tie	---	----	----
5	6	6	Tie	---	----	----
6	4	5	1	1	2	2
7	3	5	-2	2	6	-6
8	7	9	-2	2	6	-6
9	8	9	-1	1	2	-2
10	4	7	-3	3	9	-9
11	6	8	-2	2	6	-6
12	5	7	-2	2	6	-6

From the above calculations, we have

$$T^+ = \text{sum of positive ranks} = 2$$

$$n = \text{number of non-zero } d_i\text{'s} = 9$$

So the value of test statistic (T) is $T^+ = 2$

The critical (tabulated) value of test statistic for one-tailed test corresponding $n = 9$ at 5% level of significance is 9.

Since calculated value of test statistic $T (= 2)$ is less than the critical value ($= 9$) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that samples fail to provide us sufficient evidence against the claim so we may assume that memory practice session improves the performance of children.

Example 4: A drug is given to 30 patients and the increment in their blood pressures were recorded to be 2, 0, -4, -1, 2, 3, -2, 5, -6, 6, 2, 4, 3, 5, 4, 2, 2, -1, -3, -5, 4, 6, -8, -4, 2, 4, 0, -6, 3, 1. Assuming that the distribution of the increment in the blood pressures is symmetrical about its median, is it reasonable to believe that the drug is effective to reduce the blood pressure at 1% level of significance?

Solution: We are given that the increment in blood pressures of 30 patients after given a drug to the patients. The assumption of normality for increment in blood pressure is not given so we will not go for paired t-test. Also measurement of observations is available in interval scale, so we will go for Wilcoxon matched pair signed-rank test.

If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote medians of blood pressures of two populations before and after the drug respectively and we want to test that drug is effective to reduce the blood pressure so our claim is $\tilde{\mu}_1 > \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \leq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 > \tilde{\mu}_2 \quad [\text{The drug is effective}]$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Since in this exercise the increment in the blood pressures is given that means we are given that the difference of after on before that is $Y - X$ instead of $X - Y$ therefore, for testing the null hypothesis, the test statistic is the sum of positive ranks (T^+) instead of sum of negative ranks (T^-).

Calculation for T:

Patient Number	Increment ($d = X - Y$)	Absolute Difference $ d $	Rank of $ d $	Signed Rank
1	2	2	7	7
2	0	Tie	---	---
3	-4	4	17.5	-17.5
4	-1	1	2	-2
5	2	2	7	7
6	3	3	12.5	12.5
7	-2	2	7	-7
8	5	5	22	22
9	-6	6	25.5	-25.5
10	6	6	25.5	25.5
11	2	2	7	7
12	4	4	17.5	17.5
13	3	3	12.5	12.5

14	5	5	22	22
15	4	4	17.5	17.5
16	2	2	7	7
17	2	2	7	7
18	-1	1	2	-2
19	-3	3	12.5	-12.5
20	-5	5	22	-22
21	4	4	17.5	17.5
22	6	6	25.5	25.5
23	-8	8	28	-28
24	4	4	17.5	17.5
25	2	2	7	7
26	4	4	17.5	17.5
27	0	Tie	---	---
28	-6	6	25.5	-25.5
29	3	3	12.5	12.5
30	1	1	2	2

From the above calculations, we have

$$T^+ = 264$$

n = number of non-zero d_i 's = 28

Therefore, the value of test statistic (T) is $T^+ = 264$

Also, $n = 28 (> 25)$ therefore, it is the case of large sample. So in this case, we use Z-test. The test statistic of Z-test is given by

$$Z = \frac{T - E(T)}{SE(T)} \sim N(0,1)$$

$$\text{where, } E(T) = \frac{n(n+1)}{4} = \frac{28(28+1)}{4} = 203 \text{ and}$$

$$SE(T) = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{28(28+1)(2 \times 28+1)}{24}} = 43.91$$

Putting the values in test statistic Z , we have

$$Z = \frac{264 - 203}{43.91} = 1.38$$

The critical (tabulated) value of test statistic for right-tailed test at 1% level of significance is $z_\alpha = z_{0.01} = 1.645$.

Since calculated value of $Z (= 1.38)$ is less than the critical value ($= 1.645$), that means it lies in non-rejection region, so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the sample provide us sufficient evidence against the claim so we can say that drug is not effective to reduce the blood pressure.

Now, you can try the following exercises.

-
- E4)** Write one assumption of the Wilcoxon matched-pair signed-rank test which is not required in paired sign test.
- E5)** A group of 10 students were given a test in Statistics and after one month's coaching they were given another test of the similar nature. The following table gives the increment in their marks in the second test over the first:

Roll No.	1	2	3	4	5	6	7	8	9	10
Increase in Marks	6	-2	8	-4	10	2	5	-4	6	0

Assuming that the distribution of the increment in the marks is symmetrical about its median, do the marks indicate that students have benefited by coaching. Test it at 1% level of significance?

- E6)** To compare the softness of the paper products, a sample of two products A and B is given to ten judges. Each judge rates the softness of each product on a scale from 1 to 10 with higher ratings implying a softer product. The results of the experiment are shown below. The first rating in the pair is of product A and the second is of product B.

(4, 5) (6, 4) (8, 5) (9, 8) (4, 1) (7, 9) (6, 2) (5, 3) (8, 2)
(6, 7)

Assume that the distribution of the differences of the ranks of products A and B is symmetrical about its median. Apply the Wilcoxon matched pair signed-rank test to test the hypothesis that the products A and B are equally soft at 1% level of significance.

14.4 Mann-Whitney U Test

In Section 14.2 and 14.3, we have discussed the paired sign test and Wilcoxon matched-pair signed-rank test respectively which are used when the observations are dependent, that is, observations are paired. When we are interested in testing of difference in means of two independent populations then we use two sample t-test. To use the t-test however, it is necessary to make a set of assumptions (as described in Unit 11 of this course). In particular, it is necessary that two independent samples be randomly drawn from normal populations having equal variances and the data be measured in at least of an interval scale. But in studies of consumer behaviour, marketing research, experiment of psychology, etc. generally the data are collected in ordinal scale and the form of the population is not known. Since the parametric t-test could not be used in such situation, an appropriate non-parametric technique is needed. In such a circumstance a very simple non-parametric test known as Mann-Whitney U test may be used.

This test was developed jointly by Mann, Whitney and Wilcoxon. Wilcoxon considered only the case of equal sample sizes while Mann and Whitney seem to have been the first to treat the case of unequal sample sizes. Therefore, it is sometimes called the Mann-Whitney U test and sometimes the Wilcoxon ranks sum test. Thus, this test may also be viewed as a substitute for the parametric t-test for the difference between two population means. When assumptions of the two-sample t-test are fulfilled then this test is slightly weaker than t-test.

Assumptions

This test work under the following assumptions:

- (i) The two samples are randomly and independently drawn from their respective populations.
- (ii) The variable under study is continuous.
- (iii) The measurement scale is at least ordinal.
- (iv) The distributions of two populations differ only with respect to location parameter.

Let us discuss the general procedure of this test:

Let us suppose that we have two independent random samples $X_1, X_2, \dots, X_{n_1}$ and $Y_1, Y_2, \dots, Y_{n_2}$ drawn from two populations having medians $\tilde{\mu}_1$ and $\tilde{\mu}_2$ respectively. Here, we want to test the hypothesis about the medians of two populations so we can take the null and alternative hypotheses as

$$\begin{aligned} &H_0: \tilde{\mu}_1 = \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 \neq \tilde{\mu}_2 \quad [\text{for two-tailed test}] \\ \text{or} \quad &\left. \begin{aligned} &H_0: \tilde{\mu}_1 \leq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 > \tilde{\mu}_2 \\ &H_0: \tilde{\mu}_1 \geq \tilde{\mu}_2 \text{ and } H_1: \tilde{\mu}_1 < \tilde{\mu}_2 \end{aligned} \right\} \quad [\text{for one-tailed test}] \end{aligned}$$

We can also form the null and alternative hypotheses in the form of distribution functions. If $F_1(x)$ and $F_2(x)$ are the distribution functions of the first and second populations respectively and we want to test that the two independent samples come from the populations that are identical with respect to location, that is, median.

Therefore, we can take the null and alternative hypotheses as

$$\begin{aligned} &\left. \begin{aligned} &H_1: F_1(x) = F_2(x) \text{ for at least one value of } x \\ &H_1: F_1(x) \neq F_2(x) \text{ for at least one value of } x \end{aligned} \right\} \quad [\text{for two-tailed test}] \\ \text{or} \quad &\left. \begin{aligned} &H_0: F_1(x) \leq F_2(x) \text{ for at least one value of } x \\ &H_1: F_1(x) > F_2(x) \text{ for at least one value of } x \end{aligned} \right\} \quad [\text{for right-tailed test}] \\ \text{or} \quad &\left. \begin{aligned} &H_0: F_1(x) \geq F_2(x) \text{ for at least one value of } x \\ &H_1: F_1(x) < F_2(x) \text{ for at least one value of } x \end{aligned} \right\} \quad [\text{for left-tailed test}] \end{aligned}$$

After setting null and alternative hypotheses, this test involves following steps:

Step 1: First of all, we combine the observations of two samples.

Step 2: After that, ranking all these combined observations from smallest to largest, that is, the rank 1 is given to the smallest of the combined observations, rank 2 is given to the second smallest and so on up to the largest observation. If several values are same (tied), we assign each the average of ranks they would have received if there were no repetition.

Step 3: If null hypothesis is true then we can expect that the sum of ranks of the two samples are equal. Now for convenience, we consider the smaller sized sample and calculate the sum of ranks of the observations of this sample. Let S be the sum of the ranks assigned to the sample observations of smaller sized sample then for testing the null hypothesis the test statistic is given by

$$U = S - \frac{n_1(n_1 + 1)}{2}; \quad \text{if } n_1 \text{ is small}$$

$$U = S - \frac{n_2(n_2 + 1)}{2}; \quad \text{if } n_2 \text{ is small}$$

Step 4: Obtain critical value(s) of test statistic U at given level of significance under the condition that null hypothesis is true. **Table VII** in the Appendix at the end of this block provides the lower and upper critical values for a given combination of n_1 and n_2 at α level of significance for one-tailed and two-tailed test.

Step 5: Decision rule:

To take the decision about the null hypothesis, the calculated value of test statistic U (computed in Step 3) is compared with the critical (tabulated) value (obtained in Step 4) at a given level of significance (α) under the condition that null hypothesis is true. Since test may be one-tailed or two-tailed so following cases arise:

For two-tailed test: When $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$

For two-tailed test, we see critical values at $\alpha/2$ for α level of significance. If calculated value of test statistic U is either less than or equal to the lower critical value ($U_{L,\alpha/2}$) or greater than or equal to the upper critical value ($U_{U,\alpha/2}$), that is, $U \leq U_{L,\alpha/2}$ or $U \geq U_{U,\alpha/2}$ then we reject the null hypothesis at α level of significance. However, if computed U lies between these critical values, that is, $U_{L,\alpha/2} < U < U_{U,\alpha/2}$, then we do not reject the null hypothesis at $\alpha\%$ level of significance.

For one-tailed test:

For one-tailed test, we see the critical value at α for α level of significance.

Case I: When $H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 > \tilde{\mu}_2$ (right-tailed test)

If calculated value of test statistic U is greater than or equal to the upper critical value ($U_{U,\alpha}$), that is, $U \geq U_{U,\alpha}$ then we reject the null hypothesis at α level of significance. However, if computed U is less than upper critical value ($U_{U,\alpha}$), that is, $U < U_{U,\alpha}$ then we do not reject the null hypothesis at $\alpha\%$ level of significance.

Case II: When $H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$ and $H_1 : \tilde{\mu}_1 < \tilde{\mu}_2$ (left-tailed test)

If calculated value of test statistic U is less than or equal to the lower critical value ($U_{L,\alpha}$), that is, $U \leq U_{L,\alpha}$ then we reject the null hypothesis at α level of significance. However, if computed U is greater than lower critical value ($U_{L,\alpha}$), that is, $U > U_{L,\alpha}$ then we do not reject the null hypothesis at $\alpha\%$ level of significance.

For large (n_1 or $n_2 > 20$):

When either n_1 or n_2 exceeds 20, the statistic U is approximately normally distributed with mean

$$E(U) = \frac{n_1 n_2}{2} \quad \dots (7)$$

and variance

$$\text{Var}(U) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12} \quad \dots (8)$$

The proof of mean and variance of test statistic(T) is beyond the scope of this course.

Therefore in this case, we use normal test (Z-test) (described in Unit 10 of Block 3 of this course.). The test statistic of Z-test is given by

$$Z = \frac{U - E(U)}{\text{SE}(U)} = \frac{U - E(U)}{\sqrt{\text{Var}(U)}} \sim N(0, 1)$$

$$= \frac{U - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \sim N(0,1) \left[\begin{array}{l} \text{Using equations} \\ (7) \text{ and } (8) \end{array} \right] \quad \dots (9)$$

After that, we calculate the value of test statistic Z and compare it with the critical value(s) given in **Table 10.1** at prefixed level of significance α . Take the decision about the null hypothesis as described in Section 10.2 of Unit 10 of this course.

Let us do some examples to become more user friendly with the test explained above:

Example 5: A Statistics professor taught two special sections of a basic course in which students in each section were considered outstanding. He used a “traditional” method of instruction (T) in one section and an “experimental” method of instruction (E) in the other. At the end of the semester, he ranked the students based on their performance from 1 (worst) to 20 (best).

T	1	2	3	5	8	10	12	13	14	15
E	4	6	7	9	11	16	17	18	19	20

Test whether there is any evidence of a difference in performances based on the two methods at 5% level of significance.

Solution: It is case of two independent populations and the assumption of normality of both the populations is not given also the given data are in the form of ranks so we cannot use two-sample t-test in this case. Also sample sizes are small so we cannot use the Z-test. Therefore, we go for Mann-Whitney U test.

Here, we want to test that performance of the students based on the two methods is different. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) ranks of the students who taught by traditional” and “experimental” methods respectively so our claim is $\tilde{\mu}_1 \neq \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 = \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2 \quad [\text{both methods are different}]$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, combined ranks are given. Also $n_1 = n_2$ so we take any sum of ranks(S) of first sample or second sample. We consider first sample. Thus,

$$S = 1 + 2 + 3 + 5 + 8 + 10 + 12 + 13 + 14 + 15 = 83$$

Here, we consider the first sample so the test statistic is given by

$$U = S - \frac{n_1(n_1 + 1)}{2} = 83 - \frac{10 \times (10 + 1)}{2} = 28$$

Since test is two-tailed so the lower and upper critical values of test statistic for two-tailed test corresponding $n_1 = 10$ and $n_2 = 10$ at 5% level of significance are

$$U_{L, \alpha/2} = U_{L, 0.025} = 79 \text{ and } U_{U, \alpha/2} = U_{U, 0.025} = 131.$$

Since calculated value of test statistic U (= 28) is less than critical values (= 79 and 131) so we reject the null hypothesis and support the alternative

hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we can say that performances based on two methods are different.

Example 6: The following data show the median amount to be spent on a birthday gift by students who are unemployed and employed:

Unemployed	200	350	500	250	150	275			
Employed	550	500	1000	450	700	200	500	600	1000

Examine that the median amount to be spent on a birthday gift by students who are unemployed is lower than for students who are employed by Mann-Whitney U test at 1% level of significance.

Solution: It is case of two independent populations and the assumption of normality of both the populations is not given so we cannot use t-test in this case. Also sample sizes are small so we cannot use the Z-test. Therefore, we go for Mann-Whitney U test.

Here, we want to test that the median amount to be spent on a birthday gift by students who are unemployed is lower than for students who are employed. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) amount to be spent on a birthday gift by students who are unemployed and employed respectively so our claim is $\tilde{\mu}_1 < \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \geq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 < \tilde{\mu}_2 \left[\begin{array}{l} \text{median amount to be spent by students who are} \\ \text{unemployed is lower than the students who are employed} \end{array} \right]$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

To perform the Mann-Whitney U test, we combine the observations of two samples and for convenient, we arrange the data in ascending order. After that we rank all these combined observations from smallest to largest as:

Unemployed	Ranks	Unemployed	Ranks
150	1	200	2.5
200	2.5	450	7
250	4	500	9
275	5	500	9
350	6	550	11
500	9	600	12
		700	13
		1000	14.5
		1000	14.5
Total	27.5		

Since $n_1 (= 6)$ is small than $n_2 (= 9)$ so we take sum of the ranks(S) assigned to the sample of unemployed students. Thus, $S = 27.5$.

Here, we consider the first sample so the test statistic is given by

$$U = S - \frac{n_1(n_1 + 1)}{2}$$

$$= 27.5 - \frac{6 \times (6 + 1)}{2} = 6.5$$

To decide about the null hypothesis, the calculated value of test statistic U is compared with the lower critical (tabulated) value at 1% level of significance.

Since test is left-tailed so the lower critical value of test statistic for left-tailed corresponding to $n_1 = 6$ and $n_2 = 9$ at 1% level of significance is

$$U_{L,\alpha} = U_{L, 0.01} = 29.$$

Since calculated value of test statistic U ($= 6.5$) is less than lower critical value ($= 29$) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that the median amount to be spent on a birthday gift by students who are unemployed is lower than for students who are employed.

Now, you can do following exercises in same manner.

-
- E7)** Write one difference between two-sample t-test and Mann-Whitney U test.
- E8)** Write one difference between Wilcoxon matched-pair signed-rank test and Mann-Whitney U test.
- E9)** The senior class in a particular high school had 25 boys. Twelve boys lived in villages and other thirteen lived in a town. A test was conducted to see that village boys in general were physically fit than the town boys. Each boy in the class was given a physical fitness test in which a low score indicates poor physical condition. The scores of the village boys (V) and the town boys (T) are as follows:

Village Boys(V)	15.7	8.2	6.5	7.2	9.0	4.5	10.6
	12.4	16.2	12.9	11.4	5.6		
Town Boys(T)	12.7	3.2	11.8	7.9	5.6	6.7	12.6
	7.9	2.7	6.1	3.6	6.5	2.8	

Test whether the village boys are more fit than town boys at 5% level of significance.

- E10)** The following data represent lifetime (in hours) of batteries for two different brands A and B:

Brand A	40	30	55	40	40	35	30	40	50	45	40	35
Brand B	45	60	50	60	35	50	55	60	50	50	40	55

Examine, is the average life of two brands is same at 1% level of significance?

14.5 KOLMOGOROV-SMIRNOV TWO-SAMPLE TEST

In Mann-Whitney U test, we have tested the hypothesis that the two independent samples come from the populations that are identical with respect to location whereas Kolmogorov-Smirnov two-sample test is sensitive to differences of all types that may exist between two distributions, that is, location, dispersion, skewness, etc. Therefore, it is referred as a general test. This test was developed by Smirnov. This test also carries the name of Kolmogorov because of its similarity to the one-sample test developed by Kolmogorov. In Kolmogorov-Smirnov one sample test, the observed (sample or empirical) cumulative distribution function is compared with the hypothesized cumulative distribution function whereas in two-sample case the comparison is made between the empirical cumulative distributions functions of the two samples.

Assumptions

The assumptions necessary for this test are:

- (i) The two samples are randomly and independently drawn from their respective populations.
- (ii) The variable under study is continuous.
- (iii) The measurement scale is at least ordinal.

Let us discuss the general procedure of this test:

Let us suppose that we have two independent random samples $X_1, X_2, \dots, X_{n_1}$ and $Y_1, Y_2, \dots, Y_{n_2}$ drawn from first and second populations with distribution functions $F_1(x)$ and $F_2(x)$ respectively. Also let $S_1(x)$ and $S_2(x)$ be the sample or empirical cumulative distribution functions of samples drawn from first and second populations respectively.

Generally, we want to test whether independent random samples come from populations having the same distribution functions in all respect or the distribution functions of two populations differ with respect to location, dispersion, skewness, etc. So we can take the null and alternative hypotheses as

$$H_0 : F_1(x) = F_2(x) \quad \text{for all values of } x$$

$$H_1 : F_1(x) \neq F_2(x) \quad \text{for at least one value of } x$$

After setting null and alternative hypotheses, the procedure of Kolmogorov - Smirnov two-sample test summarise in the following steps:

Step 1: This test is based on the comparison of the empirical (sample) cumulative distribution functions, therefore, first of all we compute sample cumulative distribution functions $S_1(x)$ and $S_2(x)$ from the sample data as the proportion of the number of sample observations less than or equal to some number x to the number of observations, that is,

$$S_1(x) = \frac{\text{The number of sample observations less than or equal to } x}{n_1}$$

and

$$S_2(x) = \frac{\text{The number of sample observations less than or equal to } x}{n_2}$$

Step 2: After finding the empirical cumulative distribution functions $S_1(x)$ and $S_2(x)$ for all possible values of x , we find the deviation between the empirical cumulative distribution functions for all x . That is,

$$S_1(x) - S_2(x) \quad \text{for all } x$$

Step 3: If the two samples have been drawn from identical populations then $S_1(x)$ and $S_2(x)$ should be fairly close for all value of x . Therefore, we find the point at which the two functions show the maximum deviation. So we take the test statistic which calculate the greatest vertical deviation between $S_1(x)$ and $S_2(x)$, that is,

$$D = \sup_x |S_1(x) - S_2(x)|$$

which is read as “D equals the supreme over all x , of the absolute value of the difference $S_1(x) - S_2(x)$ ”

Step 4: Obtain critical value of test statistic at α % level of significance under the condition that null hypothesis is true. **Table V and VI** in Appendix provides the critical values of test statistic for equal and unequal sample sizes respectively at different level of significance.

Step 5: Decision Rule:

To take the decision about the null hypothesis, the test statistic (calculated in Step 3) is compared with the critical (tabulated) value (obtained in Step 4) for a given level of significance (α).

If computed value of D is greater than or equal to critical value (D_α), that is, $D \geq D_\alpha$ then we reject the null hypothesis at α level of significance, otherwise we do not reject H_0 .

For large sample ($n_1 = n_2 = n > 40$ and $n_1 > 16$ for unequal):

For a sample size $n_1 = n_2 = n > 40$ and $n_1 > 16$ for unequal sample sizes, the critical value of test statistic at given level of significance is approximated by the formula given in last row of **Table V & VI** in the Appendix.

Now, it is time to do some examples based on above test.

Example 7: The following data represent lifetime (in hours) of batteries for two different brands A and B:

Brand A	40	30	55	40	40	35	30	40	50	45	40	35
Brand B	45	60	50	60	35	50	55	60	50	50	40	55

Examine by the Kolmogorov-Smirnov two-sample test, is the average life of two brands same at 5% level of significance?

Solution: Here, we want to test that the average life of batteries of two brands is same. If $F_1(x)$ and $F_2(x)$ are cumulative distribution functions of the life of batteries of brand A and brand B respectively then our claim is $F_1(x) = F_2(x)$ and its complement is $F_1(x) \neq F_2(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : F_1(x) = F_2(x) \text{ for all values of } x$$

$$H_1 : F_1(x) \neq F_2(x) \text{ for at least one value of } x$$

So the test statistic is given by

$$D = \sup_x |S_1(x) - S_2(x)|$$

where, $S_1(x)$ and $S_2(x)$ are the empirical cumulative distribution functions of the lives of the batteries of brand A and brand B drawn in the samples respectively which are defined as:

$$S_1(x) = \frac{\text{number of batteries of brand A whose average life} \leq x}{n_1}$$

$$S_2(x) = \frac{\text{number of batteries of brand B whose average life} \leq x}{n_2}$$

Calculation for $|S_1(x) - S_2(x)|$:

Average Life	Frequency (Brand A)	C.F. Brand A	$S_A(x)$	Frequency (Brand B)	C.F. Brand B	$S_B(x)$	$ S_1(x) - S_2(x) $
30	2	2	2/12	0	0	0	2/12
35	2	4	4/12	1	1	1/12	3/12
40	5	9	9/12	1	2	2/12	7/12
45	1	10	10/12	1	3	3/12	7/12
50	1	11	11/12	4	7	7/12	4/12
55	1	12	12/12	2	9	9/12	3/12
60	0	12	12/12	3	12	12/12	0
Total	$n_1 = 12$			$n_2 = 12$			

From the above calculation, we have

$$D = \sup_x |S_1(x) - S_2(x)| = \frac{7}{12} = 0.58$$

The critical value of test statistic for equal sample sizes $n_1 = n_2 = 12$ at 5% level of significance is $6/12 = 0.5$.

Since calculated value of test statistic (= 0.58) is greater than critical value (= 0.5) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so the average life of batteries of two brands is different.

Example 8: The following are the marks in Statistics of B.Sc. students taken randomly from two colleges A and B:

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
College (A)	2	2	4	6	3	3	4	8	7	5
College (B)	1	1	2	5	7	3	3	2	6	6

Apply Kolmogorov-Smirnov test to examine that the distribution of marks in college A and college B is same at 1% level of significance.

Solution: Here, we wish to test that the distribution of marks in Statistics of B.Sc. students in college A and college B is same. If $F_1(x)$ and $F_2(x)$ are distribution functions of the marks in college A and college B respectively then

our claim is $F_1(x) = F_2(x)$ and its complement is $F_1(x) \neq F_2(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : F_1(x) = F_2(x) \quad \text{for all values of } x$$

$$H_1 : F_1(x) \neq F_2(x) \quad \text{for at least one value of } x$$

So the test statistic is given by

$$D = \sup_x |S_1(x) - S_2(x)|$$

where, $S_1(x)$ and $S_2(x)$ are the empirical distribution functions of samples drawn from college A and college B respectively which are defined as:

$$S_1(x) = \frac{\text{number of students of college A whose marks} \leq x}{n_1}$$

$$S_2(x) = \frac{\text{number of students of college B whose marks} \leq x}{n_2}$$

Calculation for $|S_1(x) - S_2(x)|$:

Marks	Frequency (College A)	C.F. (College A)	$S_A(x)$	Frequency (College B)	C. F. (College B)	$S_B(x)$	$ S_A(x) - S_B(x) $
0-10	2	2	0.0455	1	1	0.0278	0.0177
10-20	2	4	0.0909	1	2	0.0556	0.0354
20-30	4	8	0.1818	2	4	0.1111	0.0707
30-40	6	14	0.3182	5	9	0.2500	0.0682
40-50	3	17	0.3864	7	16	0.4444	0.0581
50-60	3	20	0.4545	3	19	0.5278	0.0732
60-70	4	24	0.5455	3	22	0.6111	0.0657
70-80	8	32	0.7273	2	24	0.6667	0.0606
80-90	7	39	0.8864	6	30	0.8333	0.0530
90-100	5	44	1	6	36	1	0
Total	$n_1 = 44$			$n_2 = 36$			

From the above calculation, we have

$$D = \sup_x |S_1(x) - S_2(x)|$$

$$= 0.0732$$

Since $n_1 = 44 > 16$ so we can calculate the critical value of test statistic for unequal sample sizes $n_1 = 44$ and $n_2 = 36$ at 1% level of significance by the formula

$$D_{n,\alpha} = 1.63 \sqrt{\frac{n_1 + n_2}{n_1 n_2}} = 1.63 \sqrt{\frac{44 + 36}{44 \times 36}} = 0.366$$

Since calculated value of test statistic $D (= 0.0732)$ is less than critical value $(= 0.366)$ so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that the distribution of marks in Statistics of B.Sc. students in college A and college B are same.

Now, you can try the following exercises.

E11) What is the main difference between Mann-Whitney U test and Kolmogorov-Smirnov two-sample tests?

E12) The following are the grades in Mathematics of High school students taken randomly from two School I and School II:

Grades	A	A ⁺	B	B ⁺	C	C ⁺	D	D ⁺
Number of Students in School- I	2	2	4	6	3	3	0	0
Number of Students in School- II	1	1	2	3	4	2	4	3

Apply Kolmogorov -Smirnov test to examine that the performance in Mathematics of High school students in School-I and School-II is same at 5% level of significance.

We now end this unit by giving a summary of what we have covered in it.

14.6 SUMMARY

In this unit, we have discussed following points:

1. Need of two-sample non-parametric tests.
2. Which non-parametric two-sample test is appropriate for a particular situation.
3. The test which is used in place of paired t-test when assumption(s) of the test is (are) not fulfilled, that is, paired sign test and Wilcoxon matched-pair signed-rank tests.
4. The Mann-Whitney U test which is used when the assumption(s) of two sample t-test is (are) not fulfilled.
5. The Kolmogorov-Smirnov two-sample test for testing the hypothesis that two samples come from the populations having the same distribution.

14.7 SOLUTIONS / ANSWERS

E1) The main difference between paired t-test and paired sign test is that the paired t-test based on normality assumption where as sign test is not based on such assumption.

E2) Here, the given data are in the form of before and after study but assumption of normality is not given so we cannot use paired t-test. So we go for paired sign test instead of paired t-test.

Here, we wish to test that IQ is improved due to course designed that means average score before the course is less than average score after the course. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) score before and after the course so our claim is $\tilde{\mu}_1 < \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \geq \tilde{\mu}_2$.

Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 < \tilde{\mu}_2 \quad [\text{course improves the IQ}]$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test and the test statistic (S) is the number of plus signs (S^+).

Calculation for S:

S. No	IQ Before (X)	IQ After (Y)	Sign of Difference (X – Y)
1	95	99	–
2	112	115	–
3	128	133	–
4	96	96	Tie
5	97	99	–
6	117	120	–
7	105	106	–
8	95	106	–
9	99	100	–
10	86	89	–
11	75	90	–
12	82	85	–

From the above calculation, we have

n = number of non-zero difference = 11

S^+ = number of plus signs = 0

S^- = number of minus signs = 11

Therefore, the value of the test statistic is

$$S = \text{number of plus signs } (S^+) = 0$$

To take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block.

Here, $n = 11$, $p = 0.5$ and $r = 0$. Thus, we have

$$p\text{-value} = P[S \leq 0] = 0.0005$$

Since $p\text{-value} = 0.0005 < 0.05 (= \alpha)$ so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that IQ of students is improved due to course.

- E3)** Here, the ratings are given in the form of pairs so it can be considered as the case of dependent samples. But assumption of normality is not given also the given data are in the form of ranks so we cannot use the paired t-test in this case. So we go for paired sign test.

Here, we want to test that the product A is favoured over product B that means average (median) rating consumption of product A is greater than B. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) rating consumption of product A and product B respectively then our claim is $\tilde{\mu}_1 > \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \leq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 > \tilde{\mu}_2 \quad [\text{the product A is favoured over product B}]$$

Since the alternative hypothesis is right-tailed so the test is right-tailed test so the test statistic (S) is the number of minus signs (S^-).

Calculation for S:

S. No	Favour of Product A (X)	Favour of Product B (Y)	Sign of Difference (X – Y)
1	5	2	+
2	2	3	–
3	4	1	+
4	4	2	+
5	5	3	+
6	4	4	Tie
7	5	4	+
8	5	1	+
9	2	4	–
10	3	2	+
11	3	2	+
12	4	3	+
13	3	3	Tie
14	3	4	–
15	2	3	–

From the above calculation, we have

n = number of non-zero differences = 13

S^+ = number of plus signs = 9

S^- = number of minus signs = 4

Therefore, value of test statistic is

S = number of minus signs (S^-) = 4

To take the decision about the null hypothesis, we determine p-value with the help of **Table I** given in Appendix at the end of this block.

Here, $n = 13$, $p = 0.5$ and $r = 4$. Thus, we have

$$p\text{-value} = P[S \leq 4] = 0.1334$$

Since $p\text{-value} = 0.1334 > 0.01 (= \alpha)$ so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so product A is not favoured over product B.

- E4)** Wilcoxon matched-pair signed-rank test is based on the assumption that the distribution of differences is symmetric whereas paired sign test does not require this assumption.
- E5)** It is case of two dependent populations and the assumption of normality for the differences is not given so we will not go for paired t-test. Also measurement of observations is available in interval scale so we will go for Wilcoxon matched pair signed-rank test.

Here, we wish to test that the students have benefited by coaching that means average score before the coaching is less than average score after the coaching. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) score before and after the coaching then our claim is $\tilde{\mu}_1 < \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \geq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \geq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 < \tilde{\mu}_2 \quad [\text{students have benefited by coaching}]$$

Since the alternative hypothesis is left-tailed so the test is left-tailed test.

Since in this exercise the increment in marks is given that means we are given that the difference of after on before, that is, $Y - X$ instead of $X - Y$ therefore, for testing the null hypothesis, the test statistic is the sum of negative ranks (T^-) instead of sum of positive ranks (T^+).

Calculation for T:

Roll No.	Difference (d)	Absolute Difference d	Rank of d	Signed Rank
1	6	6	6.5	6.5
2	-2	2	1.5	-1.5
3	8	8	8	8
4	-4	4	3.5	-3.5
5	10	10	9	9
6	2	2	1.5	1.5
7	5	5	5	5
8	-4	4	3.5	-3.5
9	6	6	6.5	6.5
10	Tie	---	---	---

From the above calculations, we have

$$T^- = 8.5$$

$$n = \text{number of non-zero } d_i\text{'s} = 9$$

So the value of test statistic(T) is $T^- = 8.5$

The critical (tabulated) value of test statistic for one-tailed test corresponding to $n = 9$ at 5% level of significance is 9.

Since calculated value of test statistic $T (= 8.5)$ is less than the critical value ($= 9$) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the coaching is effective.

- E6)** Here, the data are given in the form of pairs so it can be considered as the case of dependent samples. But assumption of normality is not given also the given data are in the form of ranks so we cannot use the paired t-test in this case. So we go for Wilcoxon matched-pair signed-rank test.

Here, we wish to test that the product A and B are equally soft. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average rank of softness of products A and B respectively then our claim is $\tilde{\mu}_1 = \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \neq \tilde{\mu}_2$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 \text{ and } H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test so the test statistic is the minimum of the sum of positive ranks (T^+) and negative ranks (T^-), that is,

$$T = \min\{T^+, T^-\}$$

Calculation for T:

Judge	Product A	Product B	Difference (d)	Absolute Difference d	Rank of d	Signed Rank
1	4	5	-1	1	2	-2
2	6	4	2	2	5	5
3	8	5	3	3	7.5	7.5
4	9	8	1	1	2	2
5	4	1	3	3	7.5	7.5
6	7	9	-2	2	5	-5
7	6	2	4	4	9	9
8	5	3	2	2	5	5
9	8	2	6	6	10	10
10	6	7	-1	1	2	-2

From the above calculations, we have

$$T^+ = 46 \text{ and } T^- = 9$$

n = number of non-zero d_i 's = 10

Putting the values in test statistic, we have

$$T = \min\{T^+, T^-\} = \min\{46, 9\} = 9$$

The critical value of test statistic for two-tailed test corresponding $n = 10$ at 1% level of significance is 4.

Since calculated value of test statistic $T (= 9)$ is greater than the critical value ($= 4$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fail to provide us sufficient evidence against the claim so we may assume that the products A and B are equally soft.

- E7)** Two-sample t-test is based on the assumptions that populations under study are normally distributed and the observations are measured in at least of an interval scale whereas Mann-Whitney U test is non required such assumptions.
- E8)** Wilcoxon matched-pair signed-rank test is used when observations are paired whereas the Mann-Whitney U test is used when two independent samples are randomly drawn two populations.

E9) It is case of two independent populations and the assumption of normality of both the populations is not given so we cannot use t-test in this case. So we go for Mann-Whitney U test.

Here, we want to test that village boys are more fit than town boys that means average (median) score of village boys is greater than average (median) score of town boys. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) score of the village boys and town boys respectively then our claim is $\tilde{\mu}_1 > \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \leq \tilde{\mu}_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0 : \tilde{\mu}_1 \leq \tilde{\mu}_2$$

$$H_1 : \tilde{\mu}_1 > \tilde{\mu}_2 \quad [\text{village boys are more fit than town boys}]$$

To perform the Mann-Whitney U test, we combine the observations of both samples and for convenient, we arrange the observations in ascending order. Then rank all these combined observations from smallest to largest as:

Village Boys(V)	Ranks	Town Boys (T)	Ranks
4.5	5	2.7	1
5.6	6.5	2.8	2
6.5	9.5	3.2	3
7.2	12	3.6	4
8.2	15	5.6	6.5
9.0	16	6.1	8
10.6	17	6.5	9.5
11.4	18	6.7	11
12.4	20	7.9	13.5
12.9	23	7.9	13.5
15.7	23	11.8	19
16.2	25	12.6	21
		12.7	22
Total	190		

Here, $n_1 = 12$ and $n_2 = 13$. Since n_1 is less than n_2 therefore we take sum of ranks corresponding to village boys.

Since we consider the first sample so test statistic U is given by

$$U = S - \frac{n_1(n_1 + 1)}{2} = 190 - \frac{12 \times (12 + 1)}{2} = 112$$

The upper critical value of test statistic for right-tailed corresponding to $n_1 = 12$ and $n_2 = 13$ at 5% level of significance is $U_{U,\alpha} = U_{U,0.05} = 198$.

Since calculated value of test statistic (=112) is less than upper critical value (= 198) so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so village boys are not more fit than town boys.

- E10)** It is case of two independent populations and the assumption of normality of both the populations is not given so we cannot use two sample t-test in this case. So we go for Mann-Whitney U test.

Here, we want to test that average life of two brands of batteries is same. If $\tilde{\mu}_1$ and $\tilde{\mu}_2$ denote the average (median) life of batteries of brands A and B respectively then our claim is $\tilde{\mu}_1 = \tilde{\mu}_2$ and its complement is $\tilde{\mu}_1 \neq \tilde{\mu}_2$. Since claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 \text{ [average life of two brands of batteries is same]}$$

$$H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2 \text{ [average life of two brands of batteries is not same]}$$

To perform the Mann-Whitney U test, we combine the observations of both samples and for convenient, we arrange the observations in ascending order. Then rank all these combined observations from smallest to largest as:

Brand A	Ranks	Brand B	Ranks
30	1.5	35	4
30	1.5	40	8.5
35	4	45	12.5
35	4	50	16
40	8.5	50	16
40	8.5	50	16
40	8.5	50	16
40	8.5	55	20
40	8.5	55	20
45	12.5	60	23
50	16	60	23
55	20	60	23
Total	102		

Here, $n_1 = n_2$ so we can take any the sum of ranks(S) of first sample or second sample. We consider first sample. Therefore, the test statistic U is given by

$$U = S - \frac{n_1(n_1 + 1)}{2} = 102 - \frac{12 \times (12 + 1)}{2} = 24$$

The critical values of test statistic for two-tailed test corresponding $n_1 = n_2 = 12$ at 1% level of significance are $U_{L, \alpha/2} = U_{L, 0.005} = 106$ and $U_{U, \alpha/2} = U_{U, 0.005} = 194$.

Since calculated value of test statistic U (= 24) is less than both critical values (= 106 and 194) so we reject the null hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so average life of two brands of batteries is not same.

- E11)** Mann-Whitney U test is used to test the hypothesis that two independent samples come from the populations follow distributions that are identical with respect to location whereas Kolmogorov-Smirnov two-sample test is sensitive to differences of all types that may

exist between two distributions that is, location, dispersion, skewness, etc.

- E12)** Here, we want to test that performance of students in School-I and School-II are same. If $F_1(x)$ and $F_2(x)$ are distribution functions of grades in Mathematics of High school students in School-I and School-II then our claim is $F_1(x) = F_2(x)$ and its complement is $F_1(x) \neq F_2(x)$. Since the claim contains the equality sign so we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0 : F_1(x) = F_2(x) \quad \text{for all values of } x$$

$$H_1 : F_1(x) \neq F_2(x) \quad \text{for at least one value of } x$$

For testing the null hypothesis, the test statistic is given by

$$D = \sup_x |S_1(x) - S_2(x)|$$

where, $S_1(x)$ and $S_2(x)$ are the empirical cumulative distribution functions of samples drawn from School-I and School-II respectively which are defined as:

$$S_1(x) = \frac{\text{number of students of School-I whose grade} \leq x}{n_1}$$

and

$$S_2(x) = \frac{\text{number of students of School-II whose grade} \leq x}{n_2}$$

Calculation for $|S_1(x) - S_2(x)|$:

Grade	Frequency (School-I)	C.F. (School-I)	$S_A(x)$	Frequency (School-II)	C. F. (School-II)	$S_B(x)$	$ S_A(x) - S_B(x) $
A	2	2	0.10	1	1	0.05	0.05
A ⁺	2	4	0.20	1	2	0.10	0.10
B	4	8	0.40	2	4	0.2	0.20
B ⁺	6	14	0.70	3	7	0.35	0.35
C	3	17	0.85	4	11	0.55	0.30
C ⁺	3	20	1.00	2	13	0.65	0.35
D	0	20	1.00	4	17	0.85	0.15
D ⁺	0	20	1.00	3	20	1	0.00
Total	$n_1 = 20$			$n_2 = 20$			

From the above calculation, we have

$$D = \sup_x |S_1(x) - S_2(x)| = 0.35$$

The critical value of test statistic for equal sample sizes $n_1 = n_2 = 20$ at 5% level of significance is $8/20 = 0.4$.

Since calculated value of test statistic (= 0.35) is less than critical value (= 0.4) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that the performance of students in School-I and School-II is same.

UNIT 15 k-SAMPLE TESTS

Structure

- 15.1 Introduction
 - Objectives
- 15.2 Kruskal-Wallis Test
- 15.3 Friedman Test
- 15.4 Summary
- 15.5 Solutions /Answers

15.1 INTRODUCTION

In Unit 13 of this block, we have studied some one-sample non-parametric tests for testing the hypothesis about the population median and in Unit 14, we have studied some two-sample non-parametric tests for testing the hypothesis about the equality of two population medians. If there is a situation where three or more populations are under study and we are interested to test the equality of their mean or medians. In such situations, generally, we use analysis of variance (ANOVA) technique that will describe in Block 2 of MST-005. But ANOVA procedure requires the assumption that the populations being compared are all normally distributed with equal variances. When the populations under study are not normally distributed then we cannot use the ANOVA procedure. Therefore, we require non-parametric tests that not based on any assumption about the shape (form) of the population distributions. This unit is devoted to discuss some such tests that use in these situations as Kruskal-Wallis test and Friedman test.

This unit is divided into five sections. Section 15.1 is described the need of k-sample non-parametric tests. The Kruskal-Wallis test is discussed in Section 15.2 in details whereas Friedman test is discussed in Section 15.3. Unit ends by providing summary of what we have discussed in this unit in Section 15.4 and solution of exercises in Section 15.5.

Objectives

After studying this unit, you should be able to:

- describe the need of k-sample test;
- perform the Kruskal-Wallis test for testing the hypothesis about equality of more than two population medians; and
- perform the Friedman test.

15.3 KRUSKAL-WALLIS TEST

As we have discussed in previous section that analysis of variance (ANOVA) test whether three or more independent samples have been drawn from the populations having the same mean or not. Also it is considered as the extension of two-sample t-test for equality of two means. But ANOVA procedure requires the assumption that the populations being compared are all normally distributed with equal variances. When the populations under study are not normally distributed then we cannot use the ANOVA procedure. Therefore, we require a non-parametric test that not based on any assumption about the shape

Kruskal-Wallis test is the non-parametric alternative to the one way analysis of variance or completely randomised design. Also this test represents a generalisation of the two-sample Mann-Whitney U test.

of the population distribution and use in testing of such situations. Such a test is the Kruskal-Wallis test. This test is a non-parametric alternative to the one-way analysis of variance or completely randomised design. Kruskal-Wallis test also a generalisation of the two-sample Mann-Whitney U test.

Assumptions

This test works under the following assumptions:

- (i) k samples are randomly and independently drawn from their respective populations.
- (ii) The variable under study is continuous.
- (iii) The measurement scale is at least ordinal.
- (iv) The populations are identical in shape except for a possible difference in location.

Let us suppose that we have k independent random samples of sizes $n_1, n_2, \dots, n_k$ taken from k populations respectively which are shown as below:

Sample (Treatment) Number	Sample Observation			
1	X_{11}	X_{12}	...	X_{1n_1}
2	X_{21}	X_{22}	...	X_{2n_2}
.	.	.	...	.
.	.	.	...	.
.	.	.	...	.
k	X_{k1}	X_{k2}	...	X_{kn_k}

Generally, we are interested to test whether all k populations have same median or not so we can take the null and alternative hypotheses as

H_0 : All k populations have same median

H_1 : All k populations have not same median

In ANOVA generally, we test the effect of treatments so we can also take the null and alternative hypotheses as

H_0 : All k treatments have same effect

H_1 : All k treatments have not same effect

A treatment is an object or a factor which is allocated to one or more units to estimate its effect. For example, if some fertilizers are applied on experimental fields to know the effect to these then fertilizers are known as treatments.

After setting the null and alternative hypotheses, this test involves following steps:

Step 1: First of all, we combine the observations of all samples.

Step 2: After that, ranking all these combined observations from smallest to largest, that is, the rank 1 is given to the smallest of the combined observations, rank 2 is given to the second smallest and so on up to the largest observation. If several values are same (tied), we assign each the average of ranks they would have received if there were no repetition.

Step 3: If the null hypothesis is true, we expect that sum of ranks of all samples to be equal. Therefore, in this step we find the sum of ranks for each sample. Let $R_1, R_2, \dots, R_k$ represent the sum of ranks of 1st sample, 2nd sample and so on kth sample respectively.

Step 4: To test the null hypothesis, the Kruskal-Wallis test statistic is defined as:

$$H = \frac{12}{n(n+1)} \left(\sum_{i=1}^k \frac{R_i^2}{n_i} \right) - 3(n+1)$$

where, $n = n_1 + n_2 + \dots + n_k$

If tie occurs, an adjustment in Kruskal-Wallis test statistic has to be made. The adjustment factor is

$$C = 1 - \frac{1}{(n^3 - n)} \sum_{i=1}^r (t_i^3 - t_i)$$

where, r is the number of groups of different tied ranks, and t_i is the number of tied values within i^{th} group that are tied at a particular value. Therefore, adjusted Kruskal-Wallis test statistic is

$$H_C = H / C$$

Step 5: Obtain critical value of Kruskal-Wallis test statistic at given level of significance under the condition, that null hypothesis is true. **Table VIII** in the Appendix at the end of this block provides critical values for three samples ($k = 3$) and each sample has five or fewer observations ($n_i \leq 5$) at α level of significance.

Step 6: Decision rule:

To take the decision about the null hypothesis, the calculated value of Kruskal-Wallis test statistic (computed in Step 4) is compared with the critical (tabulated) value (obtain in Step 5) at given level of significance (α) under the condition that null hypothesis is true.

If calculated value of test statistic is greater than critical (tabulated) value at α level of significance then we reject the null hypothesis at α level of significance otherwise we do not reject the null hypothesis H_0 .

For large samples

For large number of samples ($k > 3$) or for large sample sizes ($n_i > 5$), the distribution of the test statistic under the null hypothesis is well approximated by the chi-square distribution with $(k-1)$ degrees of freedom. So when number of samples or observations per sample are such that we cannot use **Table VIII** then we use χ^2 -table (given in the Appendix of the Block 1 of this course) for critical value of the test statistic.

Now, it is time to do some examples based on above test.

Example 1: Three different feeds are to be compared to determine that they have same distribution of weight gains on experimental animals (such as pig). Suppose 12 animals are divided at random into three groups of four animals and each group at different feed. The following results are obtained:

Feed	Weight Gains			
1	104	110	106	102
2	112	117	115	114
3	120	126	121	128

Test that the median weight gains due to three feeds are same by Kruskal-Wallis test at 5% level of significance.

Solution: Here, we want to test that the median weight gains due to three feeds are same. Therefore, our claim is “the median weight gains due to three feeds are same” and its complement is “the median weight gains due to three feeds are not same”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_1 : The median weight gains due to three feeds are same

H_1 : The median weight gains due to three feeds are not same

Here, the distributions of weight gains due to different feeds are not given so the assumption of normality for one-way ANOVA is not fulfilled. Therefore, we go for Kruskal-Wallis test.

To perform the Kruskal-Wallis test, the test statistic is given by

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(n+1) \quad \dots (1)$$

where, R_i is the sum of ranks of i^{th} group.

Therefore, first we rank all the observations of three groups and then find the sum of ranks (R_i) of each group.

Calculation for R_i :

Weight Gains					
Feed I	Rank	Feed II	Rank	Feed III	Rank
104	2	112	5	120	9
110	4	117	8	126	11
106	3	115	7	121	10
102	1	114	6	128	12
Total	$R_1 = 10$		$R_2 = 26$		$R_3 = 42$

Putting the values in equation (1), we have

$$\begin{aligned}
 H &= \frac{12}{12(12+1)} \left[\frac{(10)^2}{4} + \frac{(26)^2}{4} + \frac{(42)^2}{4} \right] - 3(12+1) \\
 &= \frac{12}{12 \times 13} \left[\frac{100}{4} + \frac{676}{4} + \frac{1764}{4} \right] - 39 \\
 &= 48.85 - 39 = 9.85
 \end{aligned}$$

The critical (tabulated) value corresponding to $k = 3$, $n_1 = 4$, $n_2 = 4$, $n_3 = 4$ at 5% level of significance is 5.692.

Since calculated value of test statistic $H (= 9.85)$ is greater than critical value ($= 5.692$) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that samples provide us sufficient evidence against the claim so the median weight gains due to feeds are not same.

Example 2: An analyst in the publishing industry wants to test whether the price of newspaper advertisement of a given size is about the same in four large newspaper groups. Random sample of seven newspapers from each group is

selected and the price (in thousand rupees) of an ad is recorded which are given below:

Group A:	65	62	45	70	48	57	50
Group B:	81	64	72	55	90	68	75
Group C:	42	58	48	59	61	60	64
Group D:	85	92	69	82	94	64	73

Do you believe that there is significant difference in the price of an ad across the four groups by Kruskal-Wallis test at 1% level of significance?

Solution: Here, we want to test that price of an ad across the four groups is different. Therefore, our claim is “the price of an ad across the four groups is not same” and its complement is “the price of an ad across the four groups is same”. Therefore, we can take complement as the null hypothesis and claim as the alternative hypothesis. Thus,

H_0 : The price of an ad across the four newspapers is same

H_1 : The price of an ad across the four newspapers is not same

Here the assumption of normality is not fulfilled, so we cannot use one-way ANOVA. Therefore, we go for Kruskal-Wallis test.

To perform the Kruskal-Wallis test, the test statistic is given by

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(n+1) \quad \dots (2)$$

where, R_i is the sum of ranks of i^{th} group.

Therefore, first we rank all the observations of three groups. For convenient, we arrange the data of each group in ascending order and then find the sum of ranks (R_i) of each group.

Calculation for R_i :

Group A	Rank	Group B	Rank	Group C	Rank	Group D	Rank
45	2	55	6	42	1	64	14
48	3.5	64	14	48	3.5	69	18
50	5	68	17	58	8	73	21
57	7	72	20	59	9	82	24
62	12	75	22	60	10	85	25
65	16	81	23	61	11	92	27
70	19	90	26	64	14	94	28
Total	$R_1 = 64.5$		$R_2 = 128$		$R_3 = 56.5$		$R_4 = 157$

Putting the values in equation (2), we have

$$\begin{aligned}
 H &= \frac{12}{28(28+1)} \left(\frac{(64.5)^2}{7} + \frac{(128)^2}{7} + \frac{(56.5)^2}{7} + \frac{(157)^2}{7} \right) - 3(28+1) \\
 &= \frac{12}{28 \times 29} \left(\frac{4160.25 + 16384 + 3192.25 + 24649}{7} \right) - 87 \\
 &= \frac{12}{28 \times 29} \times 6912.21 - 87 = 102.15 - 87 = 15.15
 \end{aligned}$$

From the above calculations, we see that there are tied ranks therefore, we have

Number of groups of tied ranks = $r = 2$ (3.4 and 14)

Number of times group 3.4 of tied ranks occurs = $t_1(3.4) = 2$

Number of times group 14 of tied ranks occurs = $t_2(14) = 3$

Since ties occur, therefore, the correction factor is

$$C = 1 - \frac{1}{(n^3 - n)} \sum_{i=1}^r (t_i^3 - t_i) = 1 - \frac{1}{(28)^3 - 28} [(2^3 - 2) - (3^3 - 3)]$$

$$= 1 - \frac{1}{21924} \times 30 = 1 - 0.0013 = 0.9987$$

Therefore, adjusted Kruskal-Wallis test statistic is

$$H_c = H/C = 15.15/0.9987 = 15.17$$

Since all sample sizes $n_1 = 7$, $n_2 = 7$, $n_3 = 7$ exceed 5 therefore, the test statistic H_c is approximated by chi square with $k - 1 = 4 - 1 = 3$ degrees of freedom. Thus, we compare calculated test statistic H_c with critical value given in χ^2 -table.

The critical (tabulated) value of χ^2 with 3 degrees of freedom at 1% level of significance is 11.34.

Since calculated value of test statistic $H_c (= 15.17)$ is greater than critical value ($= 11.34$) so we reject the null hypothesis and support the alternative hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that the price of an ad across the four newspapers is different.

Now, you can try the following exercises in same manner.

- E1)** Suppose dairy farmer wishes to compare the effect of four different diets on the amount of milk produced by cattle. He randomly selects 12 cows and assigns four cows randomly to each diet for eight week. The following table gives the rank assigned to each cows.

Diet	Rank				Total R_i
A	2	5	4	9	20
B	7	1	8	3	19
C	6	10	12	11	39

Test that the diets are equally productive by Kruskal-Wallis test at 1% level of significance.

- E2)** A manufacturer tried three alternative manufacturing processes and recorded the number of defective articles in each process as follows:

Number of Defectives per Batch		
Process I	Process II	Process III
12	10	13
5	15	11
2	2	6
6	7	16
0	8	14
4	6	20

Assuming that same number of articles is produced in each batch. Can you conclude that there is no difference in the three processes in terms of number of defective per batch by Kruskal-Wallis test at the 1% level of significance?

15.4 FRIEDMAN TEST

In the previous section, we have studied Kruskal-Wallis test which is a non-parametric version of one-way ANOVA or completed randomised design and also an extension of Mann-Whitney U test. The Friedman test is a non-parametric version of two-way ANOVA or randomised block design with one item per call that will be discussed in Block 2 of MST-005. This test is also based on ranks and may be viewed as an extension of the Wilcoxon signed-rank test for two samples.

Assumptions

This test work under the following assumptions:

- (i) The observations are independent within and between samples.
- (ii) The treatments are randomly assigned to experimental units within each block.
- (iii) The measurement scale is at least ordinal.
- (iv) The variable under study is continuous.

Suppose, we have k mutually independent random samples of equal size n , that is $X_{i1}, X_{i2}, \dots, X_{in}$ where, $i = 1, 2, \dots, k$. In other words, we can say that there are n treatments and k blocks of each of size n and each treatment occurs once in each block. The data for k blocks and n treatments can be presented in the following two-way table:

Block (Sample)	Treatment					
	1	2	...	j	...	n
1	X_{11}	X_{12}	...	X_{1j}	...	X_{1n}
2	X_{21}	X_{22}	...	X_{2j}	...	X_{2n}
.	.	.	...	.	...	.
.	.	.	...	.	...	.
.	.	.	...	.	...	.
i	X_{i1}	X_{i2}	...	X_{ij}	...	X_{in}
.	.	.	...	.	...	.
.	.	.	...	.	...	.
.	.	.	...	.	...	.
k	X_{k1}	X_{k2}	...	X_{kj}	...	X_{kn}

Generally, we are interested to test whether all treatments have same effect or not so we can take the null and alternative hypotheses as

H_0 : All treatments have same effect

H_1 : All treatments have same not effect

After setting null and alternative hypotheses, this test involves following steps:

Step 1: To perform the Friedman test, we rank the observations within each block (sample) independently from smallest to largest. The rank 1 is given to the smallest observation of a block, rank 2 is given to the second smallest and so on up to the largest observation of the block (sample). If several values are same (tied) we assign each the average of ranks they would have received if there were no repetition. Then we replace each observation of the block (sample) by corresponding rank. This process is repeated for each block (sample).

Kruskal-Wallis test is a non-parametric version of one way ANOVA or completely randomised design whereas the Friedman test is a non-parametric version of the randomised block design or two way ANOVA with one item per call.

In Kruskal-walls test, we rank all observations which are combined from all samples whereas in Friedman test we rank the observations within each sample or block.

Step 2: After that, we sum all the ranks for each treatment. Let R_1 be the sum of all ranks for 1st treatment, R_2 be the sum of all ranks for 2nd treatment and so on, R_n be the sum of all ranks for nth treatment.

Step 3: If null hypothesis is true, that is, all treatments have same effects then we expect that the sum of the ranks for each treatment would be approximately equal. The sum of squares of differences among the sums of the ranks will be indicative of the difference in the treatment effects. These sum of squares are measured by the Friedman and give in the simple form which is known as Friedman test statistic (F) and given by

$$F = \frac{12}{kn(n+1)} \sum_{j=1}^n R_j^2 - 3k(n+1)$$

The distribution of Friedman test statistic F is approximated by chi-square distribution with $(n-1)$ degrees of freedom. In order to the reasonably good approximation, we require that either number of blocks (k) or number of treatments (n) exceed 5.

If tie occurs within blocks, an adjustment in Friedman F-statistic has to be made. The adjustment factor is

$$C = 1 - \frac{1}{k(n^2-1)} \sum_{i=1}^k (t_i^3 - t_i)$$

where t_i is the number of tied observations in the ith block (sample). Therefore, adjusted Friedman statistic is

$$F_c = F / C$$

Step 4: Obtain critical value of Friedman test statistic at given level of significance under the condition that null hypothesis is true. **Table IX** in the Appendix at the end of this block provides critical values for 3 to 6 ($k = 3$ to 6) samples and each sample has $n = 20$ or fewer observations at α level of significance.

Step 5: Decision rule:

To take the decision about the null hypothesis, the calculated value of Friedman test statistic (computed in Step 3) is compared with the critical (tabulated) value (obtain in Step 4) at a given level of significance (α) under the condition that null hypothesis is true.

If calculated value of test statistic is greater than critical (tabulated) value at α level of significance then we reject the null hypothesis at α level of significance otherwise we do not reject the null hypothesis H_0 .

For large samples

For large number of treatments or for large sample sizes the distribution of the test statistic under the null hypothesis is well approximated by the chi-square distribution with $(n-1)$ degrees of freedom. So when number of samples and observations per sample are such that we cannot use **Table IX** then we use χ^2 -table (given in the Appendix of the Block 1 of this course) for critical values of the test statistic.

Now, it is time to do some examples based on Friedman test.

Example 3: The table given below shows reaction time (in second) data from 6 objects (treatments) each of which was tested under three conditions I, II and III:

Object \ Condition	A	B	C	D	E	F
I	356	590	665	450	540	500
II	401	564	667	560	570	525
III	455	570	650	575	560	550

Apply the Friedman test to test, whether reaction time of each object is same or not at 5% level of significance?

Solution: Here, we want to test that reaction time of each object is same. Therefore, our claim is “reaction time of each object is same” and its complement is “reaction time of each object is not same”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Reaction time of each object is same

H_1 : Reaction time of each object is not same

Here, two factors (object and condition) are under study so we can use two-way ANOVA if the distributions of both factors follow normal distributions. But it is not the case so we go for Friedman test.

For testing the null hypothesis, we have Friedman test statistic as

$$F = \frac{12}{kn(n+1)} \sum_{j=1}^n R_j^2 - 3k(n+1) \quad \dots (3)$$

To perform the Friedman test, we rank the observations within each condition (sample or block) and then calculate the sum of ranks of each object (treatment) as:

Condition	Object					
	A	B	C	D	E	F
I	1	5	6	2	4	3
II	1	4	6	3	5	2
III	1	4	6	5	3	2
Sum	3	13	18	10	12	7

Here, $k = 3$ and $n = 6$, therefore, by putting the values in equation (3), we have

$$\begin{aligned}
 F &= \frac{12}{3 \times 6 \times (6+1)} \left[3^2 + 13^2 + 18^2 + 10^2 + 12^2 + 7^2 \right] - 3 \times 3 \times (6+1) \\
 &= \frac{2}{21} \times 795 - 63 = 75.71 - 63 = 12.71
 \end{aligned}$$

The critical (tabulated) value of test statistic F corresponding $k = 3$ and $n = 6$ at 5% level of significance is 7.0.

Since calculated value of test statistic $F (= 12.71)$ is greater than critical value ($= 7.0$) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the samples provide us sufficient evidence against the claim so reaction time of each object is not same.

Example 4: An experiment is conducted to investigate the toxic effect of three chemicals A, B and C on the skin of rats. Three adjacent 1-inch squares are marked on the backs of eight rats and each of the three chemicals is applied to each rat. The squares of skin are the scored from 0 to 10 depending on the degree of imitation. The data are given in the following table:

Rat \ Chemical	A	B	C
1	6	5	3
2	9	8	4
3	6	9	6
4	5	8	6
5	7	8	9
6	5	7	6
7	6	7	5
8	6	5	7

Test there is sufficient evidence that effect of all chemicals is same by the Freidman test at 1% level of significance.

Solution: Here, we want to test that effect of all chemicals is same. So our claim is “effect of all chemicals is same” and its complement is “effect of all chemicals is not same”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Effect of all chemicals is same

H_1 : Effect of all chemicals is not same

Here, two factors (chemical and rat) are under study, so we can use two-way ANOVA if the distributions of both factors follow normal distributions. But it is not the case, so we go for Friedman test.

For testing the null hypothesis, we have Friedman test statistic as

$$F = \frac{12}{kn(n+1)} \sum_{j=1}^n R_j^2 - 3k(n+1) \quad \dots (4)$$

To perform the Friedman test, we rank the observations within each rat (block) and then calculate the sum of ranks of each chemical (treatment) as:

Rat \ Chemical	A	B	C
1	3	2	1
2	3	2	1
3	1.5	3	1.5
4	1	3	2
5	1	2	3
6	1	3	2
7	2	3	1
8	2	1	3
Total	14.5	19	14.5

Here, $k = 8$ and $n = 3$, therefore, by putting the values in equation (4), we have

k-Sample Tests

$$\begin{aligned} F &= \frac{12}{8 \times 3 \times (3+1)} \left[(14.5)^2 + (19)^2 + (14.5)^2 \right] - 3 \times 8 \times (3+1) \\ &= \frac{1}{8} [210.25 + 361 + 210.25] - 96 \\ &= 97.69 - 96 = 1.69 \end{aligned}$$

Since tie occurs within 3rd rat (block) therefore, an adjustment in Friedman F statistic has to be made. The adjustment factor is

$$C = 1 - \frac{1}{k(n^2 - 1)} \sum_{i=1}^k (t_i^3 - t_i)$$

where, t_i is the number of tied observations in the i^{th} rat (sample or block).

Here, $t_1 = 2$ therefore,

$$C = 1 - \frac{(2^3 - 2)}{8(3^2 - 1)} = 1 - 0.094 = 0.91$$

Adjusted Friedman statistic is

$$F_c = F / C = 1.69 / 0.91 = 1.86$$

Since number of rats (samples or blocks) $k = 8 > 6$ so the distribution of the test statistic under the null hypothesis is well approximated by the chi-square distribution with $n - 1 = 2 - 1 = 1$ degree of freedom. Thus, we compare calculated test statistic F_c with critical value given in χ^2 -table.

The critical (tabulated) value of χ^2 with 1 degree of freedom at 1% level of significance is 6.63.

Since calculated value of test statistic $F_c (= 1.86)$ is less than critical value ($= 6.63$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that samples do not provide us sufficient evidence against the claim so we may assume that effect of all chemicals on rats is same.

Now, you can try the following exercises.

E3) An experiment was conducted using randomised block design with four treatments and six blocks. The ranks of the measurements within each block are shown in the following table:

Treatment \ Block	1	2	3	4	5	6
A	3	3	2	3	2	3
B	1	1	1	2	1	1
C	4	4	3	4	4	4
D	2	2	4	1	3	2

Use the Friedman test for a randomised block design to test that effect of all treatments is same at 5% level of significance.

- E4)** A random sample of 10 consumers asked to rank their preferences of four new smells that a perfume manufacturer wants to introduce to the market in the coming fall. The data are as follows where best is ranked by 1, second best by 2 and worst by 4.

Respondent	Smell I	Smell II	Smell III	Smell IV
1	1	2	4	3
2	1	3	4	2
3	1	3	2	4
4	2	1	3	4
5	1	3	4	2
6	1	3	2	4
7	2	1	4	3
8	1	3	4	2
9	1	3	2	4
10	1	4	3	2

Applying Freidman test, do you believe that all four smells are equally liked at 1% level of significance?

We now end this unit by giving a summary of what we have covered in it.

15.4 SUMMARY

In this unit, we have discussed following points:

1. Need of k-sample non-parametric tests.
2. The Kruskal-Wallis test for testing the hypothesis about equality of more than two population medians and is known as the non-parametric version of one-way ANOVA or completely randomised design.
3. Friedman test which is known as the non-parametric version of for randomised block design.

15.5 SOLUTIONS / ANSWERS

- E1)** Here, we want to test that diets are equally productive of milk. So our claim is “the diets are equally productive of milk” and its complement is “diets are not equally productive of milk”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The diets are equally productive of milk

H_1 : The diets are not equally productive of milk

Here, the assumption of normality is not fulfilled and the data are given in the form of ranks so we cannot use one-way ANOVA. Therefore, we go for Kruskal-Wallis test.

To perform the Kruskal-Wallis test, the test statistic is given by

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(n+1) \quad \dots (5)$$

where, R_i is the sum of ranks of i^{th} feed.

Since data are given in the form of ranks so first we find the sum of rank (R_i) of each feed.

Calculation for R_i :

k-Sample Tests

Feed A	Feed B	Feed C
2	7	6
5	1	10
4	8	12
9	3	11
$R_1 = 20$	$R_1 = 19$	$R_1 = 39$

Putting the values in equation (5), we have

$$H = \frac{12}{12(12+1)} \left[\frac{(20)^2}{4} + \frac{(19)^2}{4} + \frac{(39)^2}{4} \right] - 3(12+1)$$

$$= \frac{1}{13} \left(\frac{400}{4} + \frac{361}{4} + \frac{1521}{4} \right) - 39 = 43.88 - 39 = 4.88$$

The critical (tabulated) value corresponding $k = 3$, $n_1 = 4$, $n_2 = 4$, $n_3 = 4$ at 1% level of significance is 7.654.

Since calculated value of test statistic $H (= 4.88)$ is less than critical value ($= 7.654$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples do not provide us sufficient evidence against the claim so we may assume that the diets are equally productive of milk.

- E2)** Here, we want to test that there is no difference in the three processes in terms of number of defective articles per batch, that is, three manufacturing processes are identical. Therefore, our claim is “three manufacturing processes are identical” and its complement is “three manufacturing processes are not identical”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The three manufacturing processes are identical

H_1 : The three manufacturing processes are not identical

Here, the assumption of normality is not fulfilled so we cannot use one-way ANOVA. Therefore, we go for Kruskal-Wallis test.

To perform the Kruskal-Wallis test, the test statistic is given by

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(n+1) \quad \dots (6)$$

where, R_i is the sum of ranks of i^{th} processes.

Therefore, we first rank all the number of defectives articles manufactured by three processes and then find the sum of ranks (R_i) of each process.

Calculation for R_i :

Process I	Rank	Process II	Rank	Process III	Rank
12	13	10	11	13	14
5	5	15	16	11	12
2	2.5	2	2.5	6	7
6	7	7	9	16	17
0	1	8	10	14	15
4	4	6	7	20	18
Total	$R_1 = 32.5$		$R_2 = 55.5$		$R_3 = 83$

Putting the values in equation (6), we have

$$\begin{aligned}
 H &= \frac{12}{18(18+1)} \left(\frac{(32.5)^2}{6} + \frac{(55.5)^2}{6} + \frac{(83)^2}{6} \right) - 3(18+1) \\
 &= \frac{12}{18 \times 19} \left(\frac{1056.25 + 3080.25 + 6889}{6} \right) - 57 \\
 &= \frac{12}{18 \times 19} \times \frac{11025.5}{6} - 57 = 64.48 - 57 = 7.48
 \end{aligned}$$

From the above calculations, we see that there are tied ranks therefore, we have

Number of groups of tied ranks = $r = 2$ (2.5 and 7)

Number of times group 2.5 of tied ranks occurs = $t_1(2.5) = 2$

Number of times group 7 of tied ranks occurs = $t_2(7) = 3$

Since ties occur so the correction factor is given by

$$\begin{aligned}
 C &= 1 - \frac{1}{(n^3 - n)} \sum_{i=1}^r (t_i^3 - t_i) \\
 &= 1 - \frac{1}{(18)^3 - 18} [(2^3 - 2) + (3^3 - 3)] \\
 &= 1 - \frac{1}{5814} \times 30 = 0.995
 \end{aligned}$$

Therefore, adjusted Kruskal-Wallis test statistic is

$$H_c = H / C = 7.48 / 0.995 = 7.52$$

Since all sample sizes exceed 5, therefore, the test statistic H_c is approximated by chi square with $k-1 = 3-1 = 2$ degrees of freedom. Thus, we compare calculated test statistic H_c with critical value given in χ^2 -table with 2 degrees of freedom and 1% level of significance.

The critical (tabulated) value of χ^2 with 2 degrees of freedom at 1% level of significance is 9.21.

Since calculated value of test statistic $H_c (= 7.52)$ is less than critical value ($= 9.21$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that three manufacturing process are identical.

- E3)** Here, we want to test that the effect of four treatments is same. Therefore, our claim is “effect of four treatments is same” and its complement is “effect of four treatments is not same”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Effect of four treatments is same

H_1 : Effect of four treatments is not same

Here, two factors (treatment and block) are under study so we can use two-way ANOVA if the distributions of both factors follow normal distributions but it is not the case. Also data are given in the form of ranks so we go for Friedman test.

For testing the null hypothesis, we have Friedman test statistic as

$$F = \frac{12}{kn(n+1)} \sum_{j=1}^n R_j^2 - 3k(n+1) \quad \dots (7)$$

In this exercise, ranks are given therefore to perform the Friedman test we calculate sum of ranks of each treatment as:

Block \ Treatment	1	2	3	4	5	6	Total
A	3	3	2	3	2	3	16
B	1	1	1	2	1	1	7
C	4	4	3	4	4	4	23
D	2	2	4	1	3	2	14

Here $k = 6$ and $n = 4$, therefore, by putting the values in equation (7), we have

$$\begin{aligned}
 F &= \frac{12}{6 \times 4 \times (4+1)} (16^2 + 7^2 + 23^2 + 14^2) - 3 \times 6 \times (4+1) \\
 &= \frac{1}{10} (256 + 49 + 529 + 196) - 90 = 13
 \end{aligned}$$

The critical value of test statistic F corresponding $k = 6$ and $n = 4$ at 5% level of significance is 10.29.

Since calculated value of test statistic $F (= 13)$ is greater than critical value ($= 10.29$) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that data provide us sufficient evidence against the claim so effect of four treatments is not same.

- E4)** Here, we want to test that the four smells are equally liked. Therefore, our claim is “four smells are equally liked” and its complement is “four smells are not equally liked”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Four smells are equally liked

H_1 : Four smells are not equally liked

Here, two factors (consumer and smell) are under study so we can use two-way ANOVA if the distributions of both factors follow normal distributions but it is not the case. Also data are given in the form of ranks so we go for Friedman test.

For testing the null hypothesis, we have Friedman test statistic as

$$F = \frac{12}{kn(n+1)} \sum_{j=1}^n R_j^2 - 3k(n+1) \quad \dots (8)$$

In this exercise, ranks are given therefore to perform the Friedman test we calculate sum of ranks of each smell as:

Respondent	Smell I	Smell II	Smell III	Smell IV
1	1	2	4	3
2	1	3	4	2
3	1	3	2	4
4	2	1	3	4
5	1	3	4	2
6	1	3	2	4
7	2	1	4	3
8	1	3	4	2
9	1	3	2	4
10	1	4	3	2
Total	12	26	32	30

Here $k = 10$ and $n = 4$ so by putting the values in equation (8), we have

$$\begin{aligned}
 F &= \frac{12}{10 \times 4 \times (4+1)} (12^2 + 26^2 + 32^2 + 30^2) - 3 \times 10 \times (4+1) \\
 &= \frac{12}{200} (144 + 676 + 1024 + 900) - 150 \\
 &= \frac{12}{200} \times 2744 - 150 = 14.64
 \end{aligned}$$

Since $k > 6$, therefore, the Friedman test statistic F is approximated by chi square with $n - 1 = 4 - 1 = 3$ degrees of freedom. Thus, we compare calculated test statistic with critical value given in χ^2 -table at 3 degrees of freedom and 1% level of significance.

The critical value of χ^2 with 3 degrees of freedom at 1% level of significance is 11.34.

Since calculated value of test statistic ($= 14.64$) is less than critical value ($= 11.34$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the samples fail to provide us sufficient evidence against the claim so we may assume that four smells are equally liked.

UNIT 16 ANALYSIS OF FREQUENCIES

Structure

- 16.1 Introduction
 - Objectives
- 16.2 Chi-Square Test for Goodness of Fit
- 16.3 Chi-Square Test for Independence of Attributes
- 16.4 Summary
- 16.5 Solutions / Answers

16.1 INTRODUCTION

In testing of hypothesis and estimation of parameter(s), we generally assume that the random variable follows a distribution such as normal, binomial, Poisson distribution, etc., but often the need is felt to confirm whether our assumption is true or not. Therefore, on the basis of observational data for such testing we use Kolmogorov-Smirnov test (described in Unit 13). But in many real world situations, like in business and other areas, the data are collected in the form of counts. In some cases, the collected data are classified into different categories or groups according to one or more attributes. Such type of data is known as categorical data. For example, the number of people of a colony can be classified into different categories according to age, sex, income, job, etc. or the books of library can be classified according to their subjects such as books of Science, Commerce, Art, etc. Now, the question arises “how we tackle the inference problems arising out of categorical data?” The chi square test is usually used in such situations. In this unit, we shall discuss two most widely used applications of chi-square test as goodness of fit and independence of attributes.

When the data are classified into different categories or groups according to one or more attributes such type of data known as categorical data.

This unit is divided into five sections. Section 16.1 is described the need of tests which are applied for categorical data. The chi-square test for goodness of fit is discussed in Section 16.2 in details whereas chi-square test for independence of attributes is discussed in Section 16.3. Unit ends by providing summary of what we have discussed in this unit in Section 16.4 and solution of exercises in Section 16.5.

Objectives

After studying this unit, you should be able to:

- define the categorical data;
- describe the need of tests which are applied for categorical data.
- apply chi-square test for goodness of fit in different situations;
- define the contingency table; and
- apply chi-square test for independence of two attributes.

16.2 CHI-SQUARE TEST FOR GOODNESS OF FIT

In the previous block, we have discussed the parametric tests in which we first assume the form of the parent population and then perform a test about some parameter(s) of the population(s) such as mean, variance, proportion, etc.

Generally, the parametric tests are based on the assumption of a normal population. The suitability of a normal distribution or some other distribution may itself be verified by means of a goodness of fit test. The chi-square (χ^2) test for goodness of fit was given by Karl Pearson in 1900. It is the oldest non-parametric test. With the help of this test, we test whether the random variable under study follows a specified distribution such as binomial, Poisson, normal or any other distribution when the data are in categorical form. Here, we compare the actual or observed frequencies in each category with theoretically expected frequencies that would have occurred if the data followed a specified or assumed or hypothesized probability distribution. This test is known as “**goodness of fit test**” because we test how well an observed frequency distribution (distribution from which the sample is drawn) fit to the theoretical distribution such as normal, uniform, binomial, etc.

Assumptions

This test works under the following assumptions:

- (i) The sample observations are random and independent.
- (ii) The sample size is large.
- (iii) The observations may be classified into non-overlapping categories.
- (iv) The expected frequency of each class is greater than five.
- (v) Sum of observed frequencies is equal to sum of expected frequencies, i.e.,

$$\sum O = \sum E.$$

As usual first step in testing of hypothesis is to setup null and alternative hypotheses, so our null and alternative hypotheses are setup, below in Step 1.

Step 1: Generally, we are interested to test whether data follow a specified or assumed or hypothesized distribution $F_0(x)$ or a sample has come from a specified distribution or not. So here we consider only two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0 : Data follow a specified distribution

H_1 : Data does not follow a specified distribution

In symbolical form

$H_0 : F(x) = F_0(x) \text{ for all values of } x$

$H_1 : F(x) \neq F_0(x) \text{ for at least one value of } x$

Here, we take the notation for sample size is ‘N’ instead of ‘n’ since in this test generally we deal with frequencies of the observations and to represent the sum of frequencies we take notation ‘N’.

Step 2: After setting null and alternative hypotheses, our next set up is to draw a random sample. So, let a random sample of size N be drawn from a population with unknown distribution function $F(x)$ and the data are categorized into k groups or classes. Also let $O_1, O_2, \dots, O_k$ are the observed frequencies and $E_1, E_2, \dots, E_k$ are the corresponding expected frequencies. If the parameter (s) of assumed distribution is (are) unknown then in this step, we estimate the value of each parameter of the assumed distribution with the help of sample data by calculating sample mean, variance, proportion, etc as may be the case.

Step 3: After that, we find the probability of each category or group in which an observation falls with the help of the assumed probability distribution.

Step 4: If p_i ($i = 1, 2, \dots, k$) is the probability that an observation falls in i^{th}

category then we find the expected frequency by the formula given below

$$E_i = Np_i; \quad \text{for all } i = 1, 2, \dots, k$$

Note 1: Sometimes, (generally, when the expected frequencies are come in the form of decimal) it is observed that sum of expected frequencies not come equal to sum of observed frequencies yet it is necessary condition for this test so in such cases, last expected frequency is obtained by subtracting sum of all expected frequencies except last expected frequency from the sum of observed frequencies, that is,

$$E_k = N - (E_1 + E_2 + \dots + E_{k-1})$$

Step 5: Test statistic:

Since this test compares observed frequencies with the corresponding expected frequencies, therefore, we are interested in the magnitudes of the differences between the observed and expected frequencies. Specifically, we wish to know whether the differences are small enough to be attributed to chance or they are large due to some other factors. With the help of observed and expected frequencies we may compute a test statistic that reflects the magnitudes of differences between these two quantities when H_0 is true. The test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)} \text{ under } H_0$$

where, k represents the number of classes. If any expected frequency is less than 5 then it is pooled or combined with the preceding or succeeding class then k represents the number of classes that remain after the combining classes.

The χ^2 -statistic follows approximately chi-square distribution with $(k - 1)$ degrees of freedom.

If the parameter (s) of the distribution to be fitted is (are) unknown, that is, not specified in null hypothesis then test statistic χ^2 follows approximately chi-square distribution with $(k - r - 1)$ degrees of freedom, that is,

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-r-1)} \text{ under } H_0$$

where, r is the number of unknown parameters which are estimated from the sample.

Step 6: Obtain critical value of test statistic at given level of significance under the condition that null hypothesis is true. **Table III** in the Appendix at the end of Block I of this course provides critical values of the test statistic χ^2 for various degrees of freedom and different level of significance.

Step 7: Take the decision about the null hypothesis as:

To take the decision about the null hypothesis, the test statistic (calculated in Step 5) is compared with chi-square critical (tabulated) value (observed in Step 6) for a given level of significance (α) under the condition that the null hypothesis is true.

If calculated value of test statistic is greater than or equal to critical value with $(k - r - 1)$ degrees of freedom at α level of significance then we reject the null hypothesis H_0 at α level of significance, otherwise we do not reject H_0 .

Let us do some example based on above test.

Example 1: The following data are collected during a test to determine consumer preference among five leading brands of bath soaps:

Brand Preferred	A	B	C	D	E	Total
Number of Customers	194	205	204	196	201	1000

Test that the preference is uniform over the five brands at 5% level of significance.

Solution: Here, we want to test that the preference of customers over five brands is uniform. So our claim is “the preference of customers over five brands is uniform” and its complement is “the preference of customers over five brands is not uniform”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The preference of customers over the five brands of bath soap is uniform

H_1 : The preference of customers over the five brands of bath soap is not uniform

In other words, we can say that

H_0 : The probability distribution is uniform

H_1 : The probability distribution is not uniform

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi- square goodness of fit test.

If X denotes preference of customers over the five brands of bath soap that follows uniform distribution then the probability mass function of uniform distribution is given by

$$P[X = x] = \frac{1}{N}; \quad x = 0, 1, 2, \dots, N$$

The uniform distribution has a parameter N which is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)} \text{ under } H_0$$

where, O_i and E_i are the observed and expected frequencies of i^{th} brand of bath soap respectively.

Here, we want to test the null hypothesis that the preference of the customers is uniform i.e. follows uniform distribution. Uniform distribution is one in which all outcomes have equal (or uniform) probability. Therefore, the probability that the customers prefer one of any brand is same. Thus,

$$p_1 = p_2 = p_3 = p_4 = p_5 = p = \frac{1}{5}$$

The theoretical or expected number of customers or frequency for each brand is obtained by multiplying the appropriate probability by total number of customers, that is, sample size N . Therefore,

$$E_1 = E_2 = E_3 = E_4 = E_5 = Np = 1000 \times \frac{1}{5} = 200$$

Calculations for $\frac{(O - E)^2}{E}$:

Soap Brand	Observed Frequency (O)	Expected Frequency (E)	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
A	194	200	-6	36	0.18
B	205	200	5	25	0.13
C	204	200	4	16	0.08
D	196	200	-4	16	0.08
E	201	200	1	1	0.01
Total	1000	1000			0.48

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} = 0.48$$

The critical value of chi-square with $k - 1 = 5 - 1 = 4$ degrees of freedom at 5% level of significance is 9.49.

Since calculated value of test statistic (= 0.48) is smaller than critical value (= 9.49) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the preference of customers over the five brands of bath soap is uniform.

Example 2: The following data give the number of weekly accidents occurring on a mile stretch of a particular road:

Number of Accidents	0	1	2	3	4	5	6 or more
Frequency	10	12	12	9	5	3	1

A highway engineer wants to know whether the data follow Poisson distribution or not at 5% level of significance.

Solution: Here, highway engineer wants to test that the number of accidents follows Poisson distribution. So our claim is “the number of accidents follows Poisson distribution” and its complement is “the number of accidents does not follow Poisson distribution”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The number of accidents follows Poisson distribution

H_1 : The number of accidents does not follow Poisson distribution

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi-square goodness of fit test.

Here, the parameter of Poisson distribution is unknown so testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-r-1}^2 \text{ under } H_0$$

where, k = number of classes

r = number of parameters estimated from the sample data

Here, we can not use the Kolmogorov-Smirnov test for goodness of fit because this test is applied only when all the parameters of the fitting distribution are known.

If X denotes the number of accidents per week that follows Poisson distribution then the Poisson probability mass function is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots$$

where, λ is the mean number of accidents per week.

The difficulty here is that the parameter λ is unknown (it is not specified in null hypothesis), therefore, it is estimated from sample data. Since λ represents the mean of the population so it can be estimated by value of sample mean. Thus, first we find the sample mean and the value of sample mean would be taken as the estimate of the mean of the Poisson distribution.

S. No.	Number of Accidents(X)	Frequency(f)	fX
1	0	10	0
2	1	12	12
3	2	12	24
4	3	9	27
5	4	5	20
6	5	3	15
7	6	1	6
		N = 52	$\sum fX = 104$

The formula for calculating mean is

$$\bar{X} = \frac{1}{N} \sum fX = \frac{1}{52} \times 104 = 2$$

Therefore, $\hat{\lambda} = \bar{X} = 2$ and Poisson distribution is

$$P[X = x] = \frac{e^{-2} 2^x}{x!}; \quad x = 0, 1, 2, \dots \quad \dots (1)$$

Now, to find the expected or theoretical number of accidents, we first find the probability of each class ($X = 0, 1, 2, \dots, 6$) by putting $X = 0, 1, \dots, 6$ in equation (1) respectively. **Table X** in the Appendix given at the end of this block provides the Poisson probabilities at a specified value of X and at various values of λ . Therefore, with the help of this table, we can also find these probabilities by taking $X = 0, 1, \dots, 6$ and $\lambda = 2$ as:

$$p_1 = P[X = 0] = 0.1353, p_2 = P[X = 1] = 0.2707, p_3 = P[X = 2] = 0.2707,$$

$$p_4 = P[X = 3] = 0.1804, p_5 = P[X = 4] = 0.0902, p_6 = P[X = 5] = 0.0361,$$

$$p_7 = P[X = 6 \text{ or more}] = 1 - P[X < 6] = 1 - [P[X = 0] + \dots + P[X = 5]]$$

$$= 1 - (0.1353 + 0.2707 + 0.2707 + 0.1804 + 0.0902 + 0.0361) = 0.0166$$

Thus, expected number of accidents is obtained by multiplying the appropriate probability by total number of accidents, that is, N . Therefore,

$$E_1 = Np_1 = 52 \times 0.1353 = 7.0356$$

Similarly,

$$E_2 = Np_2 = 14.0764, E_3 = Np_3 = 14.0764, E_4 = Np_4 = 9.3808,$$

$$E_5 = Np_5 = 4.6904, E_6 = Np_6 = 1.8772,$$

$$E_7 = N - (E_1 + E_2 + \dots + E_6)$$

$$= 52 - (7.0356 + 14.0764 + \dots + 1.8772) = 0.8632$$

Since the expected frequencies in the last three classes are < 5 , therefore, we combine the last three classes in order to realize a cell total of at least 5.

Calculations for $\frac{(O - E)^2}{E}$:

S.No.	Observed Frequency	Expected Frequency	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
1	10	7.0356	2.9644	8.7877	1.2490
2	12	14.0764	-2.0764	4.3114	0.3063
3	12	14.0764	-2.0764	4.3114	0.3063
4	9	9.3808	-0.3808	0.1450	0.0155
5	9	7.4308	1.5692	2.4624	0.3314
Total	52	52			2.2085

From above calculations, we have

$$\chi^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 2.2085$$

Here, we combined the classes so

k = number of classes that remain after the combining classes = 5

r = number of parameters estimated from the sample data = 1

The critical value of χ^2 with $k - r - 1 = 5 - 1 - 1 = 3$ degrees of freedom at 5% level of significance is 7.81.

Since calculated value of test statistic (= 2.2085) is less than critical value (= 7.81) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so the number of accidents follows Poisson distribution.

Example 3: A random sample of 400 car batteries revealed the following distribution of battery life (in years):

Life	0-1	1-2	2-3	3-4	4-5	5-6
Frequency	16	90	145	112	27	10

Do these data follow a normal distribution at 1% level of significance?

Solution: Here, we want to test that life of car batteries follows a normal distribution. So our claim is “the life of car batteries follows normal distribution” and its complement is “the life of car batteries does not follow normal distribution”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The life of car batteries follows normal distribution

H_1 : The life of car batteries does not follow normal distribution

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi- square goodness of fit test.

Here, the parameters of normal distribution is not given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-r-1}^2 \text{ under } H_0$$

where, k = number of classes

r = number of parameters estimated from the sample data

If X represents the life of car battery that follows normal distribution then the probability density function of normal distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

There are two parameters (mean μ and standard deviation σ) in the normal distribution that can be estimated from the sample data as:

Life	Mid Value (X)	Frequency (f)	fX	fX ²
0-1	0.5	16	8.0	4.00
1-2	1.5	90	135.0	202.50
2-3	2.5	145	362.5	906.25
3-4	3.5	112	392.0	1372.00
4-5	4.5	27	121.5	546.75
5-6	5.5	10	55.0	302.50
Total		N = 400	1074	3330.00

The formula for calculating mean is

$$\hat{\mu} = \bar{X} = \frac{1}{N} \sum fX = \frac{1}{400} \times 1074 = 2.68$$

The formula for calculating standard deviation is

$$\begin{aligned} \hat{\sigma} = S &= \sqrt{\frac{1}{N-1} \left\{ \sum fX^2 - \frac{(\sum fX)^2}{N} \right\}} \\ &= \sqrt{\frac{1}{399} \left\{ 3330 - \frac{(1074)^2}{400} \right\}} = \sqrt{1.199} = 1.06 \end{aligned}$$

The next step is to find the expected frequency for each class under the assumption of normally distributed population. The expected frequencies of normal distribution can be obtained with the help of the method as described in Unit 4 of MST-003. Therefore, first we find the standard normal variate

$$Z = \frac{X - \hat{\mu}}{\hat{\sigma}} \text{ corresponding to each lower limit of the class and then find}$$

$F(x) = P[X \leq x] = P[Z \leq z]$ for each value with the help of **Table I** given in the Appendix at the end of Block 1 of this course.

$$\text{Thus, when } x = 0, z = \frac{x - \hat{\mu}}{\hat{\sigma}} = \frac{0 - 2.68}{1.06} = -2.53 \text{ and}$$

$$\begin{aligned} F(0) &= P[Z \leq z] = P[Z \leq -2.53] = 0.5 - P[-2.53 < Z < 0] \\ &= 0.5 - P[0 < Z < 2.53] = 0.5 - 0.4943 = 0.0057 \end{aligned}$$

$$\text{Similarly, when } x = 3, z = \frac{3 - 2.68}{1.06} = 0.30 \text{ and}$$

$$\begin{aligned} F(3) &= P[Z \leq z] = P[Z \leq 0.30] = 0.5 + P[0 < Z < 0.30] \\ &= 0.5 + 0.1179 = 0.6179 \end{aligned}$$

Therefore, expected frequencies are calculated as follows:

Life	Lower Limit	Standard Normal Variate $Z = \frac{X - \hat{\mu}}{\hat{\sigma}}$	Area Under Normal Curve to the Left of Z i.e. $P[Z \leq z]$	Difference between Successive Areas	Expected Frequency $= 400 \times \text{col.5}$
Below 0	$-\infty$	$-\infty$	0	$0.0057 - 0 = 0.0057$	2.28
0-1	0	-2.53	0.0057	$0.0571 - 0.0057 = 0.0514$	20.56
1-2	1	-1.58	0.0571	$0.2611 - 0.0571 = 0.2040$	81.60
2-3	2	-0.64	0.2611	$0.6179 - 0.2611 = 0.3568$	142.72
3-4	3	0.30	0.6179	$0.8944 - 0.6179 = 0.2765$	110.6
4-5	4	1.25	0.8944	$0.9706 - 0.8944 = 0.0762$	30.48
5-6	5	1.89	0.9706	$0.9991 - 0.9706 = 0.0285$	11.76
6 and above	6	3.13	0.9991	--	--

Here, last cell expected frequency is obtained by the formula given by

$$E_7 = N - (E_1 + E_2 + \dots + E_6) = 11.76$$

Since expected frequency in the first cell is < 5 , therefore, we combine the first two classes in order to realize a cell total of at least 5.

Calculations for $\frac{(O - E)^2}{E}$:

Life	Observed Frequency	Expected Frequency	(O-E)	(O-E) ²	$\frac{(O - E)^2}{E}$
0-1	16	22.84	-6.84	46.7856	2.0484
1-2	90	81.60	8.40	70.5600	0.8647
2-3	145	142.72	2.28	5.1984	0.0364
3-4	112	110.60	1.40	1.9600	0.0177
4-5	27	30.48	-3.48	12.1104	0.3973
5-6	10	11.76	-1.76	3.0976	0.2634
Total	400	400			3.6279

Therefore, from above calculations, we have

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = 3.6279$$

Here, we combined the classes so

k = number of classes that remain after the combining classes = 6

r = number of parameters estimated from the sample data = 2

The critical value of χ^2 with $k - r - 1 = 6 - 2 - 1 = 3$ degrees of freedom at 1% level of significance is 11.34.

Since calculated value of test statistic (= 3.6279) is less than critical value (= 11.34) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the life of car batteries follows normal distribution.

Example 4: The following data represent the results of an investigation of the sex distribution of the children of 30 families containing 4 children each:

Number of Sons	0	1	2	3	4
Number of Families	4	8	8	7	3

Apply the chi-square test to see whether the number of sons in a family follows binomial distribution with probability that a child to be a son is 0.5 at 5% level of significance.

Solution: Here, we want to test whether the number of sons in a family follows binomial distribution with $p = 0.5$. So our claim is “the number of sons in a family follows binomial distribution with $p = 0.5$ ” and its complement is “the number of sons in a family does not follow binomial distribution with $p = 0.5$ ”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The number of sons in a family follows binomial distribution with $p = 0.5$

H_1 : The number of sons in a family does not follow binomial distribution with $p = 0.5$

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi-square goodness of fit test.

Here, the parameter p of binomial distribution is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-1}^2$$

where, k = number of classes

If X denotes the number of sons in a family that follows binomial distribution then the binomial probability mass function is given by

$$P[X = x] = {}^nC_x p^x (1-p)^{n-x}; \quad x = 0, 1, 2, \dots, n$$

Here, total number of children is 4 and $p = 0.5$. Therefore, we have

$$\begin{aligned} P[X = x] &= {}^4C_x (0.5)^x (1-0.5)^{4-x}; \quad x = 0, 1, 2, 3, 4 \\ &= {}^4C_x (0.5)^x (0.5)^{4-x} = {}^4C_x (0.5)^4; \quad x = 0, 1, 2, 3, 4 \end{aligned} \quad \dots(2)$$

Now, to find the expected or theoretical number of families, we first find the probability of each class by putting $X = 0, 1, 2, 3, 4$ in equation (2) respectively. Thus,

$$p_1 = P[X = 0] = {}^4C_0 (0.5)^4 = 0.0625$$

$$p_2 = P[X = 1] = {}^4C_1 (0.5)^4 = 4 \times 0.0625 = 0.2500$$

$$p_3 = P[X = 2] = {}^4C_2 (0.5)^4 = 6 \times 0.0625 = 0.3750$$

$$p_4 = P[X = 3] = {}^4C_3 (0.5)^4 = 4 \times 0.0625 = 0.2500$$

$$p_5 = P[X = 4] = {}^4C_4 (0.5)^4 = 1 \times 0.0625 = 0.0625$$

The expected number of families is obtained by multiplying the appropriate probability by total number of families, that is, sample size N . Therefore,

$$E_1 = Np_1 = 30 \times 0.0625 = 1.875$$

Similarly,

$$E_2 = Np_2 = 7.5, E_3 = Np_3 = 11.25, E_4 = Np_4 = 7.5,$$

$$E_5 = N - (E_1 + E_2 + E_3 + E_4) = 30 - (1.875 + 7.5 + 11.25 + 7.5) = 1.875$$

Since the expected frequency in the last cell is less than 5 therefore, we combine the last two classes in order to realize a cell total of at least 5.

Calculation for $\frac{(O-E)^2}{E}$:

S. No.	Observed Frequency	Expected Frequency	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
1	4	1.875	2.125	4.5156	2.4083
2	8	7.5	0.500	0.2500	0.0333
3	8	11.25	-3.250	10.5625	0.9389
4	10	9.375	0.625	0.3906	0.0417
Total	30	30			3.4222

From above calculations, we have

$$\chi^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 3.4222$$

Here, we combined the classes so

k = number of classes that remain after the combining classes = 4

The tabulated value of χ^2 with $k-1 = 4-1 = 3$ degrees of freedom and at 5% level of significance is 7.81.

Since calculated value of test statistic (= 3.4222) is less than critical value (= 7.81) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the number of sons in a family follows binomial distribution with probability that a child to be a son is 0.5.

Now, you can try the following exercises.

E1) Write one difference between chi-square test and Kolmogorov-Smirnov test for goodness of fit.

E2) The following table gives the numbers of road accidents that occurred during the various days of the week:

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Number of Accidents	14	15	8	20	11	9	14

Test whether the accidents are uniformly distributed over the week by chi-square test at 1% level of significance.

E3) The number of customers waiting for service on the checkout counter line of a large supermarket is examined at random on 64 occasions during a period. The results are as follows:

Waiting Time (in minutes)	0 or 1	2	3	4	5	6	7	8 or more
Frequency	5	8	10	11	10	9	7	4

Does the number of customers waiting for service per occasion follows Poisson distribution with mean 8 minutes at 5% level of significance?

16.3 CHI-SQUARE TEST FOR INDEPENDENCE OF ATTRIBUTES

The chi-square test for independence of attributes can be used in the situation in which the data classified according to two attributes or characteristics.

A contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.

There are many situations where we need to test the independency of two characteristic or attributes of categorical data. For example, a sociologist may wish to know whether level of formal education is independent with income, whether height of sons depending on height of their fathers or not, etc. If there is no association between two variables, we say that they are independent. In other words, we can say that two variables are independent if the distribution of one is not depending on the distribution of other. To test the independence of two variables when observations in a population are classified according to some attributes we may use chi-square test for independence. This test will indicate only whether or not any association exists between attributes.

To conduct the test, a sample is drawn from the population and the observed frequencies are cross-classified according to the two characteristics so that each observation belongs to one and only one level of each characteristic. The cross-classification can be conveniently displayed by mean of a table called a contingency table. Therefore, a contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.

Assumptions

This test work under the following assumptions:

- (i) The sample observations are random and independent.
- (ii) The observations may be classified into non-overlapping categories.
- (iii) The observed and expected frequencies of each class are greater than 5.
- (iv) Sum of observed frequencies is equal to sum of expected frequencies, i.e., $\sum O = \sum E$.
- (v) Each observation in the sample may be classified according to two characteristics so that each observation belongs to one and only one level of each characteristic.

Let us discuss the procedure of this test:

Step 1: As usual, first step is to setup null and alternative hypotheses. Generally, we are interested to test whether two characteristics or attributes, say, A and B are independent or not. So here we consider only two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0 : The two characteristics of classification, say, A and B are independent

H_1 : They are not independent

Step 2: After setting null and alternative hypotheses, next step is to draw a random sample. So, let a sample of N observations is drawn from a population and they are cross-classified according to two characteristics, say, A and B. Also let the characteristic A be assumed to have 'r' categories $A_1, A_2, \dots, A_r$ and characteristic B be assumed to have 'c' categories $B_1, B_2, \dots, B_c$. The various observed frequencies in different classes can be expressed in the form of a table known as contingency table.

B A	B₁	B₂	... B_j ...	B_c	Total
A₁	O ₁₁	O ₁₂	... O _{1j} ...	O _{1c}	R₁
A₂	O ₂₁	O ₂₂	... O _{2j} ...	O _{2c}	R₂
·	·	·	·	·	·
·	·	·	·	·	·
·	·	·	·	·	·
A_i	O _{i1}	O _{i2}	... O _{ij} ...	O _{ic}	R_i
·	·	·	·	·	·
·	·	·	·	·	·
·	·	·	·	·	·
A_r	O _{r1}	O _{r2}	... O _{ri} ...	O _{rc}	R_r
Total	C₁	C₂	... C_j ...	C_c	N

Here, O_{ij} represents the number of observations corresponding to i^{th} level of characteristic A and j^{th} level of characteristic B. R_i and C_j represent the sum of number of observations corresponding to i^{th} level of characteristic A, i.e. i^{th} row and j^{th} level of characteristic B, i.e. j^{th} column respectively.

Step 3: The chi-square test of independence of attributes compares observed frequencies with frequencies that are expected when H_0 is true. To calculate the expected frequencies, we first find the probabilities that an observation lies in a particular cell for all $i = 1, 2, \dots, r$ and $j = 1, 2, \dots, c$. These probabilities are calculated according to the multiplication law of probability (described in Section 3.4 of Unit 3 of the course MST-003). According to this law, if two events are independent then the probability of their joint occurrence is equal to the product of their individual probabilities. Therefore, the probability that an observation falls in cell (i, j) is equal to the probability that observation falls in the i^{th} row multiplied by the probability of falling in the j^{th} column. We estimate these probabilities from the sample data by R_i/N and C_j/N , respectively. Thus,

$$P[A_i B_j] = \frac{R_i}{N} \cdot \frac{C_j}{N}$$

Step 4: To obtain expected frequency E_{ij} for (i, j) cells, we multiply this estimated probability by the total sample size. Thus,

$$E_{ij} = N \cdot \frac{R_i}{N} \cdot \frac{C_j}{N}$$

This equation reduces to

$$E_{ij} = \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{\text{th}} \text{ row} \times \text{Sum of } j^{\text{th}} \text{ column}}{\text{Total sample size}}$$

This form of the equation indicates that we can easily compute an expected cell frequency for each cell by multiplying together the appropriate row and column totals and dividing the product by the total sample size.

Step 5: Test statistic:

Since this test also compares the observed cell frequencies with the corresponding expected cell frequencies, therefore, we are interested in the magnitudes of the differences between the observed and expected cell frequencies. Specifically, we wish to know whether the differences are small enough to be attributed to chance or they are

large due to some other factors. With the help of observed and expected frequencies, we may compute a test statistic that reflects the magnitudes of differences between these two quantities. When H_0 is true, the test statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

The test statistic follows as a chi-square distribution with $(r-1)(c-1)$ degrees of freedom.

Step 6: Obtain critical value of test statistic corresponding $(r-1)(c-1)$ df at given level of significance under the condition that null hypothesis is true. **Table III** in the Appendix at the end of Block I of this course provides critical values of the test statistic χ^2 for various df and different level of significance.

Step 7: Take the decision about the null hypothesis as:

To take the decision about the null hypothesis, the test statistic (calculated in Step 5) is compared with chi-square critical (tabulated) value (observed in Step 6) for a given level of significance (α) under the condition that the null hypothesis is true.

If calculated value of test statistic is greater than or equal to tabulated value with $(r-1)(c-1)$ degrees of freedom at α level of significance, we reject the null hypothesis at α level of significance otherwise we do not reject it.

Let us do some examples to become more user friendly with this test.

Example 5: 1000 students at college level were graded according to their IQ level and the economic condition of their parents.

Economic Condition	IQ level		
	High	Low	Total
Poor	240	160	400
Rich	460	140	600
Total	700	300	1000

Test that IQ level of students is independent of the economic condition of their parents at 5% level of significance.

Solution: Here, we want to test that IQ level of students is independent of the economic condition of their parents. So our claim is “IQ level and economic condition are independent” and its complement is “IQ level and economic condition are not independent”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : IQ level and economic condition are independent

H_1 : IQ level and economic condition are not independent

Here, we are interesting to test the independence of two characteristics IQ and economic condition, so we go for chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0 , the expected frequencies can be obtained as:

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{\text{th}} \text{ row} \times \text{Sum of } j^{\text{th}} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{400 \times 700}{1000} = 280, E_{12} = \frac{R_1 \times C_2}{N} = \frac{400 \times 300}{1000} = 120$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{600 \times 700}{1000} = 420, E_{22} = \frac{R_2 \times C_2}{N} = \frac{600 \times 300}{1000} = 180$$

Calculations for $\frac{(O-E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
240	280	-40	1600	5.71
160	120	40	1600	13.33
460	420	40	1600	3.81
140	180	-40	1600	8.89
Total = 1000	1000			31.74

Therefore, from above calculations, we have

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 31.74$$

The degrees of freedom will be $(r-1)(c-1) = (2-1)(2-1) = 1$.

The critical value of χ^2 with 1 degree of freedom at 5% level of significance is 3.84.

Since calculated value of test statistic (= 31.74) is greater than critical value (= 3.84) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so IQ level of students is not independent of the economic condition of their parents.

Example 6: Calculate the expected frequencies for the following data presuming the two attributes and check that condition of home and condition of the child are independent at 5% level of significance.

Condition of Child	Condition of Home	
	Clean	Dirty
Clear	70	50
Fairly Clean	80	20
Dirty	35	45

Solution: Here, we want to test that condition of home and condition of the child are independent. So our claim is “condition of home and condition of child are independent” and its complement is “condition of home and condition of child are not independent”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Condition of home and condition of child are independent

H_1 : Condition of home and condition of child are not independent

Here, we are interesting to test the independence of two characteristics, condition of home and condition of the child, so we go for chi-square test of independence.

For testing the null hypothesis, test statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0 , the expected frequencies can be obtained as:

Condition of Child	Condition of Home		Total
	Clean	Dirty	
Clear	70	50	120
Fairly Clean	80	20	100
Dirty	35	45	80
Total	185	115	300

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{\text{th}} \text{ row} \times \text{Sum of } j^{\text{th}} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{120 \times 185}{300} = 74; \quad E_{12} = \frac{R_1 \times C_2}{N} = \frac{120 \times 115}{300} = 46;$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{100 \times 185}{300} = 61.67; \quad E_{22} = \frac{R_2 \times C_2}{N} = \frac{100 \times 115}{300} = 38.33;$$

$$E_{31} = \frac{R_3 \times C_1}{N} = \frac{80 \times 185}{300} = 49.33; \quad E_{32} = \frac{R_3 \times C_2}{N} = \frac{80 \times 115}{300} = 30.67$$

Calculations for $\frac{(O - E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
70	74.00	-4.00	16.00	0.22
50	46.00	4.00	16.00	0.35
80	61.67	18.33	335.99	5.45
20	38.33	-18.33	335.99	8.77
35	49.33	-14.33	205.35	4.16
45	30.67	14.33	205.35	6.70
Total = 300	300			25.64

Therefore, from above calculations, we have

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 25.64$$

The degrees of freedom will be $(r-1)(c-1) = (3-1)(2-1) = 2$.

The critical value of χ^2 with 2 degrees of freedom at 5% level of significance is 5.99.

Since calculated value of test statistic (= 25.64) is greater than critical value (= 5.99) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so condition of home and condition of the child are not independent.

- E4)** 1500 families were selected at random in a city to test the belief that high income families usually send their children to public schools and low income families often send their children to government schools. The following results were obtained in the study conducted.

Income	Public School	Government School	Total
Low	300	600	900
High	435	165	600
Total	735	765	1500

Use chi-square test at 1% level of significance to test whether the two attributes are independent.

- E5)** The following contingency table presents the analysis of 300 persons according to hair colour and eye colour:

Hair Colour	Eye Colour			
	Blue	Grey	Brown	Total
Fair	30	10	40	80
Brown	40	20	40	100
Black	50	30	40	120
Total	120	60	120	300

Test the hypothesis that there is an association between hair colour and eye colour at 1% level of significance.

We now end this unit by giving a summary of what we have covered in it

16.4 SUMMARY

In this unit, we have discussed following points:

1. When the data are classified into different categories or groups according to one or more attributes such type of data known as categorical data.
2. Needs of tests which are applied for categorical data.
3. The chi-square test for goodness of fit.
4. A contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.
5. The chi-square test for independence of two attributes.

16.5 SOLUTIONS / ANSWERS

- E1)** The main difference between chi-square test and Kolmogorov-Smirnov (K-S) test for goodness of fit is that the chi-square test is designed for categorical data whereas K-S test is designed for the continuous data.
- E2)** Here, we want to test that road accidents are uniformly distributed over the week. So our claim is “the accidents are uniformly distributed over the week” and its complement is “the accidents are not uniformly distributed over the week”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The accidents are uniformly distributed over the week

H_1 : The accidents are not uniformly distributed over the week

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi-square goodness of fit test.

If X denotes the number of accidents per day that follows uniform distribution then the probability mass function of uniform distribution is given by

$$P[X = x] = \frac{1}{N}; \quad x = 0, 1, 2, \dots, N$$

The uniform distribution has a parameter N which is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)}$$

Since the uniform distribution is one in which all outcomes considered have equal or uniform probability. Therefore, the probability that the accident occurs in any day is same. Thus,

$$p_1 = p_2 = p_3 = p_4 = p_5 = p = \frac{1}{7}$$

The theoretical or expected frequency for each day is obtained by multiplying the appropriate probability by the total number of accidents, that is, sample size N . Therefore,

$$E_1 = E_2 = E_3 = E_4 = E_5 = Np = 91 \times \frac{1}{7} = 13$$

Calculations for $\frac{(O - E)^2}{E}$:

Days	Observed Frequency (O)	Expected Frequency (E)	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
Mon	14	13	1	1	0.0769
Tue	15	13	2	4	0.3077
Wed	8	13	-5	25	1.9231
Thu	20	13	7	49	3.7692
Fri	11	13	-2	4	0.3077
Sat	9	13	-4	16	1.2308
Sun	14	13	1	1	0.0769
Total	91	91			7.6923

From the above calculation, we have

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} = 7.6923$$

The critical value of chi-square with $k - 1 = 7 - 1 = 6$ degrees of freedom at 1% level of significance is 16.81.

Since calculated value of test statistic ($= 7.6923$) is less than critical value ($= 16.81$) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the accidents are uniformly distributed over the week.

- E3)** Here, we want to test that the number of costumers waiting for servicing follow a Poisson distribution with mean waiting time 8 minutes. So our claim is “the number of costumers waiting for servicing follow a Poisson distribution with mean waiting time 8 minutes” and its complement is “the number of costumers waiting for servicing does not follow a Poisson distribution with mean waiting time 8 minutes”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The number of costumers waiting for servicing follows Poisson distribution with mean waiting time 8 minutes

H_1 : The number of costumers waiting for servicing does not follow Poisson distribution with mean waiting time 8 minutes

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for chi- square goodness of fit test.

Here, the parameter λ of Poisson distribution is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)}$$

where, k = number of classes

If X denotes the number of costumers waiting for servicing per occasion that follows the Poisson distribution then the Poisson probability mass function is given by

$$P[X = x] = \frac{\bar{e}^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots$$

where, λ is the mean number of costumers waiting for servicing per occasion. Since it is specified 8 minutes in null hypothesis therefore, the Poisson probability mass function is,

$$P[X = x] = \frac{\bar{e}^{-8} 8^x}{x!}; \quad x = 0, 1, 2, \dots \quad \dots (3)$$

Now, to find the expected number of costumers waiting for servicing per occasion, we first find the probability of each class by putting $X = 0, 1, \dots$ in equation (3). With the help of **Table X** in the Appendix at the end of this block, we can also find these probabilities by taking $X = 0, 1, \dots$ and $\lambda = 8$ as:

$$\begin{aligned} p_1 &= P[X = 0 \text{ or } 1] = P[X = 0] + P[X = 1] = 0.0003 + 0.0027 = 0.0030, \\ p_2 &= P[X = 2] = 0.0107, \quad p_3 = P[X = 3] = 0.0286, \quad p_4 = P[X = 4] = 0.0573, \\ p_5 &= P[X = 5] = 0.0916, \quad p_6 = P[X = 6] = 0.1221, \quad p_7 = P[X = 7] = 0.1396, \\ p_8 &= P[X = 8 \text{ or more}] = 1 - P[X < 8] = 1 - [P[X = 0] + \dots + P[X = 7]] \\ &= 1 - (0.0003 + 0.0027 + 0.0107 + 0.0286 + 0.0573 + 0.0916 + 0.1221 + 0.1396) = 0.5471 \end{aligned}$$

The expected number of costumers waiting for servicing is obtained by multiplying the appropriate probability by total number of accidents, that is, N . Therefore,

$$E_1 = Np_1 = 64 \times 0.0030 = 0.1920$$

$$E_2 = Np_2 = 0.6848, E_3 = Np_3 = 1.8304, E_4 = Np_4 = 3.6672,$$

$$E_5 = Np_5 = 5.8624, E_6 = Np_6 = 7.8144, E_7 = Np_7 = 8.9344,$$

$$E_8 = N - (E_1 + E_2 + \dots + E_7) = 35.0144$$

Since the expected frequencies in the first four classes are less than 5 therefore, we combine the first four classes in order to realize a cell total of at least 5.

Calculations for $\frac{(O - E)^2}{E}$:

S. No.	Observed Frequency	Expected Frequency	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
1	34	6.3744	27.6256	763.1738	119.7248
2	10	5.8624	4.1376	17.1197	2.9203
3	9	7.8144	1.1856	1.4056	0.1799
4	7	8.9344	1.9344	3.7419	0.4188
5	4	35.0144	-31.0144	961.8930	27.4714
	64	64			150.7151

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 150.7151$$

Here, we combined the classes so

k = number of classes that remain after the combining classes = 5

The critical value of chi-square with $k - 1 = 5 - 1 = 4$ degrees of freedom at 5% level of significance is 9.49.

Since calculated value of test statistic (= 150.7151) is greater than critical value (= 9.49) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so the number of costumers waiting for servicing per occasion does not follow Poisson distribution.

- E4)** Here, we want to test that the family income and selection of school are independent. So our claim is “the family income and selection of school are independent” and its complement is “the family income and selection of school are not independent”. So we can take claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : Family income and selection of school are independent

H_1 : Family income and selection of school are not independent

Here, we are interesting to test the independence of two attributes family income and selection of school, so we go for chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(r-1)(c-1)}^2 \text{ under } H_0$$

Here,
param
we tak

Now, under H_0 , the expected frequencies can be obtained as:

$$E_{ij} = \text{Expected frequency of the } i^{\text{th}} \text{ row and } j^{\text{th}} \text{ column} \\ = \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{\text{th}} \text{ row} \times \text{Sum of } j^{\text{th}} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{900 \times 735}{1500} = 441, E_{12} = \frac{R_1 \times C_2}{N} = \frac{900 \times 765}{1500} = 459,$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{600 \times 735}{1500} = 294, E_{22} = \frac{R_2 \times C_2}{N} = \frac{600 \times 765}{1500} = 306$$

Calculations for $\frac{(O - E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
300	441	-141	19881	45.08
600	459	141	19881	43.31
435	294	141	19881	67.62
165	306	-141	19881	64.97
Total = 1500	1500			220.98

Therefore, from above calculations, we have

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 220.98$$

The degrees of freedom will be $(r-1)(c-1) = (2-1)(2-1) = 1$.

The critical value of chi square with 1 df at 1% level of significance is 6.63.

Since calculated value of test statistic (= 220.98) is greater than critical value (= 6.63) so we reject the null hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so two attributes family income and selection of school are not independent.

- E5)** Here, we want to test that there is an association between hair colour and eye colour that means we want to test that hair colour and eye colour are not independent. So our claim is “hair colour and eye colour are not independent” and its complement is “hair colour and eye colour are independent”. So we can take complement as the null hypothesis and claim as the alternative hypothesis. Thus,

H_0 : Hair and eye colour are independent

H_1 : Hair and eye colour are associated

Here, we are interesting to test the independence of two attributes hair colour and eye colour, so we go for chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(r-1)(c-1)}^2 \text{ under } H_0$$

Now, under H_0 , the expected frequencies can be obtained as:

$$E_{ij} = \text{Expected frequency of the } i^{\text{th}} \text{ row and } j^{\text{th}} \text{ column} \\ = \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{\text{th}} \text{ row} \times \text{Sum of } j^{\text{th}} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{80 \times 120}{300} = 32; \quad E_{12} = \frac{R_1 \times C_2}{N} = \frac{80 \times 60}{300} = 16;$$

$$E_{13} = \frac{R_1 \times C_3}{N} = \frac{80 \times 120}{300} = 32; \quad E_{21} = \frac{R_2 \times C_1}{N} = \frac{100 \times 120}{300} = 40;$$

$$E_{22} = \frac{R_2 \times C_2}{N} = \frac{100 \times 60}{300} = 20; \quad E_{23} = \frac{R_2 \times C_3}{N} = \frac{100 \times 120}{300} = 40;$$

$$E_{31} = \frac{R_3 \times C_1}{N} = \frac{120 \times 120}{300} = 48; \quad E_{32} = \frac{R_3 \times C_2}{N} = \frac{120 \times 60}{300} = 24;$$

$$E_{33} = \frac{R_3 \times C_3}{N} = \frac{120 \times 120}{300} = 48$$

Calculations for $\frac{(O - E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
30	32	-2	4	0.13
10	16	-6	36	2.25
40	32	8	64	2
40	40	0	0	0
20	20	0	0	0
40	40	0	0	0
50	48	2	4	0.08
30	24	6	36	1.50
40	48	-8	64	1.33
Total = 300	300			7.29

Therefore, from above calculations, we have

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 7.29$$

The degrees of freedom will be $(r-1)(c-1) = (3-1)(3-1) = 4$.

The critical value of chi square with 4 df at 1% level of significance is 13.28.

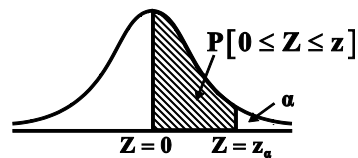
Since calculated value of test statistic chi-square (= 7.29) is less than critical value (= 13.28) so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the sample provide us sufficient evidence against the claim so hair colour is independent of eye colour.

APPENDIX

Table-I: Standard Normal Distribution (Z-table)

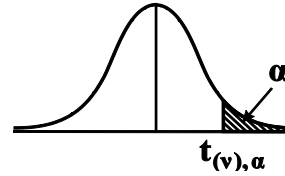
The first column and first row of the table indicate the values of standard normal variate Z at first and second place of decimal. The entry represents the upper tail area under the curve or probability i.e. $P[0 \leq Z \leq z]$ for different values of Z.

[illegible]

Sampling Distributions

Table-II: Student's t-Distribution (t-table)

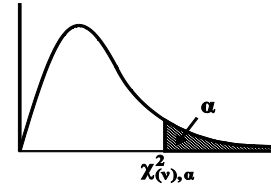
The first column of this table indicates the degrees of freedom and first row a specified upper tail area (α). The entry represents the value of t-statistic such that the area under the curve of t-distribution to its upper tail is equal to α .



One- tail $\alpha =$	0.10	0.05	0.025	0.01	0.005
v =1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Table-III: The Chi-square Distribution (χ^2 -table)

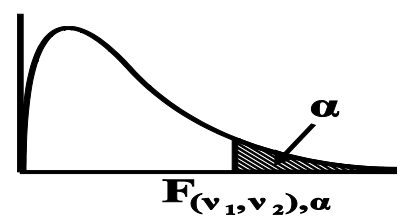
The first column of this table indicates the degrees of freedom and first row a specified upper tail area (α). The entry represents the value of chi-square statistic such that the area under the curve of chi square distribution to its upper tail is equal to α .



$\alpha =$	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
v = 1	---	---	---	---	0.02	2.71	3.84	5.02	6.63	7.88
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21	10.60
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34	12.84
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28	14.86
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	21.05	23.68	26.12	29.14	31.32
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96	49.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	107.56	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81	140.17
120	83.85	86.92	91.58	95.70	100.62	140.23	146.57	152.21	158.95	163.64

Table-IV: F-Distribution (F-table)

F-table contains the values of F-statistic for different set of degrees of freedom (v_1, v_2) of numerator and denominator such that the area under the curve of F-distribution to its right (upper tail) is equal to α .



F values for $\alpha = 0.1$

Degrees of Freedom for Denominator (v ₂)	Degrees of Freedom for Numerator(v ₁)																										
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞	
1	39.86	49.50	53.60	55.83	57.23	58.21	58.91	59.44	59.86	60.20	60.47	60.70	60.91	61.07	61.22	61.35	61.47	61.57	61.66	61.74	62.00	62.26	62.53	62.79	63.06	63.33	
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.40	9.41	9.41	9.42	9.42	9.43	9.43	9.44	9.44	9.45	9.46	9.47	9.47	9.48	9.48	9.49	
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.22	5.21	5.20	5.20	5.20	5.19	5.19	5.19	5.18	5.18	5.17	5.16	5.15	5.14	5.13	
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.96	3.94	3.92	3.91	3.90	3.89	3.88	3.87	3.86	3.86	3.85	3.85	3.84	3.83	3.82	3.80	3.79	3.78	3.76	
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.28	3.27	3.26	3.25	3.24	3.23	3.22	3.22	3.21	3.21	3.19	3.17	3.16	3.14	3.12	3.11	
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.92	2.90	2.89	2.88	2.87	2.86	2.86	2.85	2.84	2.84	2.82	2.80	2.78	2.76	2.74	2.72	
7	3.59	3.26	3.07	2.96	2.88	2.83	2.79	2.75	2.72	2.70	2.68	2.67	2.65	2.64	2.63	2.62	2.61	2.61	2.60	2.59	2.58	2.56	2.54	2.51	2.49	2.47	
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.52	2.50	2.49	2.48	2.46	2.45	2.45	2.44	2.43	2.42	2.40	2.38	2.36	2.34	2.32	2.29	
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.40	2.38	2.36	2.35	2.34	2.33	2.32	2.31	2.31	2.30	2.28	2.25	2.23	2.21	2.18	2.16	
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.30	2.28	2.27	2.26	2.24	2.23	2.22	2.22	2.21	2.20	2.18	2.16	2.13	2.11	2.08	2.06	
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.23	2.21	2.19	2.18	2.17	2.16	2.15	2.14	2.13	2.12	2.10	2.08	2.05	2.03	2.00	1.97	
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.17	2.15	2.13	2.12	2.10	2.09	2.08	2.08	2.07	2.06	2.04	2.01	1.99	1.96	1.93	1.90	
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.12	2.10	2.08	2.07	2.05	2.04	2.03	2.02	2.01	2.01	1.98	1.96	1.93	1.90	1.88	1.85	
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.07	2.05	2.04	2.02	2.01	2.00	1.99	1.98	1.97	1.96	1.94	1.91	1.89	1.86	1.83	1.80	
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.95	1.94	1.93	1.92	1.90	1.87	1.85	1.82	1.79	1.76	
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	2.01	1.99	1.97	1.95	1.94	1.93	1.92	1.91	1.90	1.89	1.87	1.84	1.81	1.78	1.75	1.72	
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.98	1.96	1.94	1.93	1.91	1.90	1.89	1.88	1.87	1.86	1.84	1.81	1.78	1.75	1.72	1.69	
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.95	1.93	1.92	1.90	1.89	1.87	1.86	1.85	1.85	1.84	1.81	1.78	1.75	1.72	1.69	1.66	
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.93	1.91	1.89	1.88	1.86	1.85	1.84	1.83	1.82	1.81	1.79	1.76	1.73	1.70	1.67	1.63	
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.91	1.89	1.87	1.86	1.84	1.83	1.82	1.81	1.80	1.79	1.77	1.74	1.71	1.68	1.64	1.61	
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.90	1.88	1.86	1.84	1.83	1.81	1.80	1.79	1.78	1.78	1.75	1.72	1.69	1.66	1.62	1.59	
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.88	1.86	1.84	1.83	1.81	1.80	1.79	1.78	1.77	1.76	1.73	1.70	1.67	1.64	1.60	1.57	
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.87	1.85	1.83	1.81	1.80	1.78	1.77	1.76	1.75	1.74	1.72	1.69	1.66	1.62	1.59	1.55	
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.85	1.83	1.81	1.80	1.78	1.77	1.76	1.75	1.74	1.73	1.70	1.67	1.64	1.61	1.57	1.53	
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.84	1.82	1.80	1.79	1.77	1.76	1.75	1.74	1.73	1.72	1.69	1.66	1.63	1.59	1.56	1.52	
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.83	1.81	1.79	1.77	1.76	1.75	1.73	1.72	1.71	1.71	1.68	1.65	1.61	1.58	1.54	1.50	
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.82	1.80	1.78	1.76	1.75	1.74	1.72	1.71	1.70	1.70	1.67	1.64	1.60	1.57	1.53	1.49	
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.81	1.79	1.77	1.75	1.74	1.73	1.71	1.70	1.69	1.69	1.66	1.63	1.59	1.56	1.52	1.48	
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.80	1.78	1.76	1.75	1.73	1.72	1.71	1.69	1.68	1.68	1.65	1.62	1.58	1.55	1.51	1.47	
30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82	1.79	1.77	1.75	1.74	1.72	1.71	1.70	1.69	1.68	1.67	1.64	1.61	1.57	1.54	1.50	1.46	
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.74	1.71	1.70	1.68	1.66	1.65	1.64	1.62	1.61	1.61	1.57	1.54	1.51	1.47	1.42	1.38	
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.68	1.66	1.64	1.62	1.60	1.59	1.58	1.56	1.55	1.54	1.51	1.48	1.44	1.40	1.35	1.29	
120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.63	1.60	1.58	1.56	1.55	1.53	1.52	1.50	1.49	1.48	1.45	1.41	1.37	1.32	1.26	1.19	
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.57	1.55	1.52	1.51	1.49	1.47	1.46	1.44	1.43	1.42	1.38	1.34	1.30	1.24	1.17	1.00	

F values for $\alpha = 0.05$

Degrees of Freedom for Denominator (v ₂)	Degrees of freedom for numerator(v ₁)																										
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞	
1	161	199	216	225	230	234	237	239	240	242	243	244	245	245	246	246	247	247	248	248	249	250	251	252	253	254	
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.39	19.40	19.40	19.41	19.42	19.43	19.43	19.43	19.44	19.44	19.44	19.45	19.45	19.46	19.47	19.48	19.49	19.50	
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.72	8.70	8.69	8.68	8.67	8.67	8.66	8.64	8.62	8.59	8.57	8.55	8.53	
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.86	5.84	5.83	5.82	5.81	5.80	5.77	5.75	5.72	5.69	5.66	5.63	
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70	4.68	4.66	4.64	4.62	4.60	4.59	4.58	4.57	4.56	4.53	4.50	4.46	4.43	4.40	4.37	
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.94	3.92	3.91	3.90	3.88	3.87	3.84	3.81	3.77	3.74	3.70	3.67	
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.51	3.49	3.48	3.47	3.46	3.44	3.41	3.38	3.34	3.30	3.27	3.23	
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.22	3.20	3.19	3.17	3.16	3.15	3.12	3.08	3.04	3.01	2.97	2.93	
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	3.01	2.99	2.97	2.96	2.95	2.94	2.90	2.86	2.83	2.79	2.75	2.71	
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.85	2.83	2.81	2.80	2.79	2.77	2.74	2.70	2.66	2.62	2.58	2.54	
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.69	2.67	2.66	2.65	2.61	2.57	2.53	2.49	2.45	2.40	
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.62	2.60	2.58	2.57	2.56	2.54	2.51	2.47	2.43	2.38	2.34	2.30	
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.53	2.51	2.50	2.48	2.47	2.46	2.42	2.38	2.34	2.30	2.25	2.21	
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.46	2.44	2.43	2.41	2.40	2.39	2.35	2.31	2.27	2.22	2.18	2.13	
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48	2.45	2.42	2.40	2.38	2.37	2.35	2.34	2.33	2.29	2.25	2.20	2.16	2.11	2.07	
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46	2.42	2.40	2.37	2.35	2.33	2.32	2.30	2.29	2.28	2.24	2.19	2.15	2.11	2.06	2.01	
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.41	2.38	2.35	2.33	2.31	2.29	2.27	2.26	2.24	2.23	2.19	2.15	2.10	2.06	2.01	1.96	
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.31	2.29	2.27	2.25	2.23	2.22	2.20	2.19	2.15	2.11	2.06	2.02	1.97	1.92	
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.34	2.31	2.28	2.26	2.23	2.21	2.20	2.18	2.17	2.16	2.11	2.07	2.03	1.98	1.93	1.88	
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.18	2.17	2.15	2.14	2.12	2.08	2.04	1.99	1.95	1.90	1.84	
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28	2.25	2.22	2.20	2.18	2.16	2.14	2.12	2.11	2.10	2.05	2.01	1.96	1.92	1.87	1.81	
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26	2.23	2.20	2.17	2.15	2.13	2.11	2.10	2.08	2.07	2.03	1.98	1.94	1.89	1.84	1.78	
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.24	2.20	2.18	2.15	2.13	2.11	2.09	2.08	2.06	2.05	2.01	1.96	1.91	1.86	1.81	1.76	
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.22	2.18	2.15	2.13	2.11	2.09	2.07	2.05	2.04	2.03	1.98	1.94	1.89	1.84	1.79	1.73	
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.20	2.16	2.14	2.11	2.09	2.07	2.05	2.04	2.02	2.01	1.96	1.92	1.87	1.82	1.77	1.71	
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.12	2.09	2.07	2.05	2.03	2.02	2.00	1.99	1.95	1.90	1.85	1.80	1.75	1.69	
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.17	2.13	2.10	2.08	2.06	2.04	2.02	2.00	1.99	1.97	1.93	1.88	1.84	1.79	1.73	1.67	
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.91	1.87	1.82	1.77	1.71	1.65	
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.14	2.10	2.08	2.05	2.03	2.01	1.99	1.97	1.96	1.94	1.90	1.85	1.81	1.75	1.70	1.64	
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13	2.09	2.06	2.04	2.01	1.99	1.98	1.96	1.95	1.93	1.89	1.84	1.79	1.74	1.68	1.62	
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04	2.00	1.97	1.95	1.92	1.90	1.89	1.87	1.85	1.84	1.79	1.74	1.69	1.64	1.58	1.51	
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.89	1.86	1.84	1.82	1.80	1.78	1.76	1.75	1.70	1.65	1.59	1.53	1.47	1.39	
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.87	1.83	1.80	1.78	1.75	1.73	1.71	1.69	1.67	1.66	1.61	1.55	1.50	1.43	1.35	1.25	
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.79	1.75	1.72	1.69	1.67	1.64	1.62	1.60	1.59	1.57	1.52	1.46	1.39	1.32	1.22	1.00	

F values for $\alpha = 0.025$

Degrees of Freedom for Denominator (v_2)	Degrees of freedom for numerator(v_1)																									
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞
1	648	800	864	900	922	937	948	957	963	969	973	977	980	983	985	987	989	990	992	993	997	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.41	39.42	39.43	39.43	39.44	39.44	39.44	39.45	39.45	39.46	39.47	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.37	14.34	14.30	14.28	14.25	14.23	14.21	14.20	14.18	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.79	8.75	8.72	8.68	8.66	8.63	8.61	8.59	8.58	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.57	6.52	6.49	6.46	6.43	6.40	6.38	6.36	6.34	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.41	5.37	5.33	5.30	5.27	5.24	5.22	5.20	5.18	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.71	4.67	4.63	4.60	4.57	4.54	4.52	4.50	4.48	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.24	4.20	4.16	4.13	4.10	4.08	4.05	4.03	4.02	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.91	3.87	3.83	3.80	3.77	3.74	3.72	3.70	3.68	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.66	3.62	3.58	3.55	3.52	3.50	3.47	3.45	3.44	3.42	3.37	3.31	3.26	3.20	3.14	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.47	3.43	3.39	3.36	3.33	3.30	3.28	3.26	3.24	3.23	3.17	3.12	3.06	3.00	2.94	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.32	3.28	3.24	3.21	3.18	3.15	3.13	3.11	3.09	3.07	3.02	2.96	2.91	2.85	2.79	2.73
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.20	3.15	3.12	3.08	3.05	3.03	3.00	2.98	2.96	2.95	2.89	2.84	2.78	2.72	2.66	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.09	3.05	3.01	2.98	2.95	2.92	2.90	2.88	2.86	2.84	2.79	2.73	2.67	2.61	2.55	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	3.01	2.96	2.92	2.89	2.86	2.84	2.81	2.79	2.77	2.76	2.70	2.64	2.59	2.52	2.46	2.40
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.93	2.89	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.68	2.63	2.57	2.51	2.45	2.38	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.87	2.82	2.79	2.75	2.72	2.70	2.67	2.65	2.63	2.62	2.56	2.50	2.44	2.38	2.32	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.81	2.77	2.73	2.70	2.67	2.64	2.62	2.60	2.58	2.56	2.50	2.45	2.38	2.32	2.26	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.76	2.72	2.68	2.65	2.62	2.59	2.57	2.55	2.53	2.51	2.45	2.39	2.33	2.27	2.20	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.72	2.68	2.64	2.60	2.57	2.55	2.52	2.50	2.48	2.46	2.41	2.35	2.29	2.22	2.16	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.68	2.64	2.60	2.56	2.53	2.51	2.48	2.46	2.44	2.42	2.37	2.31	2.25	2.18	2.11	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.65	2.60	2.56	2.53	2.50	2.47	2.45	2.43	2.41	2.39	2.33	2.27	2.21	2.15	2.08	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.62	2.57	2.53	2.50	2.47	2.44	2.42	2.39	2.37	2.36	2.30	2.24	2.18	2.11	2.04	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.59	2.54	2.50	2.47	2.44	2.41	2.39	2.36	2.35	2.33	2.27	2.21	2.15	2.08	2.01	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.56	2.51	2.48	2.44	2.41	2.38	2.36	2.34	2.32	2.30	2.24	2.18	2.12	2.05	1.98	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.54	2.49	2.45	2.42	2.39	2.36	2.34	2.31	2.29	2.28	2.22	2.16	2.09	2.03	1.95	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.51	2.47	2.43	2.39	2.36	2.34	2.31	2.29	2.27	2.25	2.19	2.13	2.07	2.00	1.93	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.49	2.45	2.41	2.37	2.34	2.32	2.29	2.27	2.25	2.23	2.17	2.11	2.05	1.98	1.91	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.46	2.48	2.43	2.39	2.36	2.30	2.27	2.25	2.23	2.21	2.15	2.09	2.03	1.96	1.89	1.81
30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.33	2.41	2.37	2.34	2.31	2.28	2.26	2.23	2.21	2.20	2.14	2.07	2.01	1.94	1.87	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.26	2.29	2.25	2.21	2.18	2.15	2.13	2.11	2.09	2.07	2.01	1.94	1.88	1.80	1.72	1.64
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.22	2.17	2.13	2.09	2.06	2.03	2.01	1.98	1.96	1.94	1.88	1.82	1.74	1.67	1.58	1.48
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.10	2.05	2.01	1.98	1.94	1.92	1.89	1.87	1.84	1.82	1.76	1.69	1.61	1.53	1.43	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.99	1.94	1.90	1.87	1.83	1.80	1.78	1.75	1.73	1.71	1.64	1.57	1.48	1.39	1.27	1.00

F values for $\alpha = 0.01$

Degrees of Freedom for Denominator (v_2)	Degrees of freedom for numerator (v_1)																											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞		
1	4063	4992	5404	5637	5760	5890	5890	6025	6025	6025	6167	6167	6167	6167	6167	6167	6167	6167	6167	6167	6235	6261	6287	6313	6339	6366		
2	98.50	99.00	99.15	99.27	99.30	99.34	99.34	99.38	99.38	99.38	99.42	99.42	99.42	99.42	99.42	99.42	99.46	99.46	99.46	99.46	99.46	99.47	99.47	99.48	99.49	99.50		
3	34.11	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.34	27.23	27.13	27.05	26.98	26.92	26.87	26.83	26.79	26.75	26.72	26.69	26.60	26.51	26.41	26.32	26.22	26.13		
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.45	14.37	14.31	14.25	14.20	14.15	14.11	14.08	14.05	14.02	13.93	13.84	13.75	13.65	13.56	13.46		
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.96	9.89	9.83	9.77	9.72	9.68	9.64	9.61	9.58	9.55	9.47	9.38	9.29	9.20	9.11	9.02		
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.66	7.60	7.56	7.52	7.48	7.45	7.42	7.40	7.31	7.23	7.14	7.06	6.97	6.88		
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54	6.47	6.41	6.36	6.31	6.28	6.24	6.21	6.18	6.16	6.07	5.99	5.91	5.82	5.74	5.65		
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.73	5.67	5.61	5.56	5.52	5.48	5.44	5.41	5.38	5.36	5.28	5.20	5.12	5.03	4.95	4.86		
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18	5.11	5.05	5.01	4.96	4.92	4.89	4.86	4.83	4.81	4.73	4.65	4.57	4.48	4.40	4.31		
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77	4.71	4.65	4.60	4.56	4.52	4.49	4.46	4.43	4.41	4.33	4.25	4.17	4.08	4.00	3.91		
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46	4.40	4.34	4.29	4.25	4.21	4.18	4.15	4.12	4.10	4.02	3.94	3.86	3.78	3.69	3.60		
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22	4.16	4.10	4.05	4.01	3.97	3.94	3.91	3.88	3.86	3.78	3.70	3.62	3.54	3.45	3.36		
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.91	3.86	3.82	3.78	3.75	3.72	3.69	3.66	3.59	3.51	3.43	3.34	3.26	3.17		
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.75	3.70	3.66	3.62	3.59	3.56	3.53	3.51	3.43	3.35	3.27	3.18	3.09	3.00		
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73	3.67	3.61	3.56	3.52	3.49	3.45	3.42	3.40	3.37	3.29	3.21	3.13	3.05	2.96	2.87		
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62	3.55	3.50	3.45	3.41	3.37	3.34	3.31	3.28	3.26	3.18	3.10	3.02	2.93	2.85	2.75		
17	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52	3.46	3.40	3.35	3.31	3.27	3.24	3.21	3.19	3.16	3.08	3.00	2.92	2.84	2.75	2.65		
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43	3.37	3.32	3.27	3.23	3.19	3.16	3.13	3.10	3.08	3.00	2.92	2.84	2.75	2.66	2.57		
19	8.19	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36	3.30	3.24	3.19	3.15	3.12	3.08	3.05	3.03	3.00	2.93	2.84	2.76	2.67	2.58	2.49		
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29	3.23	3.18	3.13	3.09	3.05	3.02	2.99	2.96	2.94	2.86	2.78	2.70	2.61	2.52	2.42		
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.24	3.17	3.12	3.07	3.03	2.99	2.96	2.93	2.90	2.88	2.80	2.72	2.64	2.55	2.46	2.36		
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.07	3.02	2.98	2.94	2.91	2.88	2.85	2.83	2.75	2.67	2.58	2.50	2.40	2.31		
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.14	3.07	3.02	2.97	2.93	2.89	2.86	2.83	2.80	2.78	2.70	2.62	2.54	2.45	2.35	2.26		
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09	3.03	2.98	2.93	2.89	2.85	2.82	2.79	2.76	2.74	2.66	2.58	2.49	2.40	2.31	2.21		
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	3.06	2.99	2.94	2.89	2.85	2.81	2.78	2.75	2.72	2.70	2.62	2.54	2.45	2.36	2.27	2.17		
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02	2.96	2.90	2.86	2.82	2.78	2.75	2.72	2.69	2.66	2.59	2.50	2.42	2.33	2.23	2.13		
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.99	2.93	2.87	2.82	2.78	2.75	2.71	2.68	2.66	2.63	2.55	2.47	2.38	2.29	2.20	2.10		
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96	2.90	2.84	2.79	2.75	2.72	2.68	2.65	2.63	2.60	2.52	2.44	2.35	2.26	2.17	2.06		
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.93	2.87	2.81	2.77	2.73	2.69	2.66	2.63	2.60	2.57	2.50	2.41	2.33	2.23	2.14	2.03		
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91	2.84	2.79	2.74	2.70	2.66	2.63	2.60	2.57	2.55	2.47	2.39	2.30	2.21	2.11	2.01		
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73	2.66	2.61	2.56	2.52	2.48	2.45	2.42	2.39	2.37	2.29	2.20	2.11	2.02	1.92	1.81		
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.44	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.12	2.03	1.94	1.84	1.73	1.60		
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.40	2.34	2.28	2.23	2.19	2.15	2.12	2.09	2.06	2.03	1.95	1.86	1.76	1.66	1.53	1.38		
∞	6.64	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.25	2.19	2.13	2.08	2.04	2.00	1.97	1.94	1.91	1.88	1.79	1.70	1.59	1.47	1.33	1.00		

APPENDIX

Table I: Cumulative Binomial Probabilities, $\sum_{x=0}^r p^x (1-p)^{n-x}; r=0,1,2,...,n-1$

n	r	p = 0.5	n	r	p = 0.5	n	r	p = 0.5	n	r	p = 0.5
1	0	0.5000		3	0.1719		11	0.9935		3	0.0038
	1	1.0000		4	0.3770		12	0.9991		4	0.0154
2	0	0.2500		5	0.6230		13	0.9999		5	0.0481
	1	0.7500		7	0.9453		14	1.0000		6	0.1189
3	2	1.0000		8	0.9893	15	0	0.0000		7	0.2403
	0	0.1250		9	0.9990		2	0.0037		8	0.4073
	1	0.5000	11	10	1.0000		3	0.0176		9	0.5927
	2	0.8750		0	0.0005		4	0.0592		10	0.7597
	3	1.0000		1	0.0059		5	0.1509		11	0.8811
	0	0.0625		2	0.0327		6	0.3036		12	0.9519
4	1	0.3125		3	0.1133		7	0.5000		13	0.9846
	2	0.6875		4	0.2744		8	0.6964		14	0.9962
	3	0.9375		5	0.5000		9	0.8491		15	0.9993
	4	1.0000		6	0.7256		10	0.9408		16	0.9999
5	0	0.0312		7	0.8867		11	0.9824		17	1.0000
	1	0.1875		8	0.9673		12	0.9963	19	18	1.0000
	2	0.5000		9	0.9941		13	0.9995		0	0.0000
	3	0.8125		10	0.9995		14	1.0000		1	0.0000
	4	0.9688		11	1.0000	16	15	1.0000		2	0.0004
	5	1.0000	12	0	0.0002		0	0.0000		3	0.0022
6	0	0.0156		1	0.0032		1	0.0003		4	0.0096
	1	0.1096		2	0.0193		2	0.0021		5	0.0318
	2	0.3438		3	0.0730		3	0.0106		6	0.0835
	3	0.6562		4	0.1938		4	0.0384		7	0.1796
	4	0.8606		5	0.3872		5	0.1051		8	0.3238
	5	0.9844		6	0.6128		6	0.2272		9	0.5000
	6	1.0000		7	0.8062		7	0.4018		10	0.6762
7	0	0.0078		8	0.9270		8	0.5982		11	0.8204
	1	0.0625		9	0.9807		9	0.7728		12	0.9165
	2	0.2266		10	0.9968		10	0.8949		13	0.9682
	3	0.5000		11	0.9998		11	0.9616		14	0.9904
	4	0.7734	13	12	1.0000		12	0.9894		15	0.9978
	5	0.9375		0	0.0001		13	0.9979		16	0.9996
	6	0.9922		1	0.0017		14	0.9997		17	1.0000
	7	1.0000		2	0.0112		15	1.0000		18	1.0000
8	0	0.0039		3	0.0461		16	1.0000	20	19	1.0000
	1	0.0352		4	0.1334	17	0	0.0000		0	0.0000
	2	0.1445		5	0.2905		1	0.0001		1	0.0000
	3	0.3633		6	0.5000		2	0.0012		2	0.0002
	4	0.6367		7	0.7095		3	0.0064		3	0.0013
	5	0.8555		8	0.8666		4	0.0245		4	0.0059
	6	0.9648		9	0.9539		5	0.0717		5	0.0207
	7	0.9961		10	0.9888		6	0.1662		6	0.0577
	8	1.0000		11	0.9983		7	0.3145		7	0.1316
	0	0.0020		12	0.9999		8	0.5000		8	0.2517
9	1	0.0195		13	1.0000		9	0.6855		9	0.4119
	2	0.0898		0	0.0001		10	0.8338		10	0.5881
	3	0.2539	14	1	0.0009		11	0.9283		11	0.7483
	4	0.5000		2	0.0065		12	0.9755		12	0.8684
	5	0.7461		3	0.0287		13	0.9936		13	0.9423
	6	0.9102		4	0.0898		14	0.9989		14	0.9793
	7	0.9805		5	0.2120		15	0.9999		15	0.9941
	8	0.9980		6	0.3953		16	1.0000		16	0.9987
	9	1.0000		7	0.6047		17	1.0000		17	0.9998
	0	0.0010		8	0.7880	18	0	0.0000		18	1.0000
10	1	0.0107		9	0.9102		1	0.0001		19	1.0000
	2	0.0547		10	0.9713		2	0.0007		20	1.0000

TABLE II: Critical Values for Wilcoxon Test

One-Tailed Test:					
$\alpha =$	0.005	0.01	0.025	0.05	0.10
Two-Tailed Test:					
$\alpha =$	0.01	0.02	0.05	0.10	0.20
n = 3	0	0	0	0	0
4	0	0	0	0	1
5	0	0	0	1	3
6	0	0	1	3	4
7	0	1	3	4	6
8	1	2	4	6	9
9	2	4	6	9	11
10	4	6	9	11	15
11	6	8	11	14	18
12	8	10	14	18	22
13	10	13	18	22	27
14	13	16	22	26	32
15	16	20	26	31	37
16	20	24	30	36	43
17	24	28	35	42	49
18	28	33	41	48	56
19	33	38	47	54	63
20	38	44	53	61	70
21	44	50	59	68	78
22	49	56	67	76	87
23	55	63	74	84	95
24	62	70	82	92	105
25	69	77	90	101	114

Table III: Critical Values of Runs Test

Appendix

Part-I (Lower Critical Values)
 $(\alpha = 0.025)$

$n_2 \backslash n_1$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2											2	2	2	2	2	2	2	2	2
3					2	2	2	2	2	2	2	2	2	3	3	3	3	3	3
4				2	2	2	3	3	3	3	3	3	3	3	4	4	4	4	4
5			2	2	3	3	3	3	3	4	4	4	4	4	4	4	5	5	5
6		2	2	3	3	3	3	4	4	4	4	5	5	5	5	5	5	6	6
7		2	2	3	3	3	4	4	5	5	5	5	5	6	6	6	6	6	6
8		2	3	3	3	4	4	5	5	5	6	6	6	6	6	7	7	7	7
9		2	3	3	4	4	5	5	5	6	6	6	7	7	7	7	8	8	8
10		2	3	3	4	5	5	5	6	6	7	7	7	7	8	8	8	8	9
11		2	3	4	4	5	5	6	6	7	7	7	8	8	8	9	9	9	9
12	2	2	3	4	4	5	6	6	7	7	7	8	8	8	9	9	9	10	10
13	2	2	3	4	5	5	6	6	7	7	8	8	9	9	9	10	10	10	10
14	2	2	3	4	5	5	6	7	7	8	8	9	9	9	10	10	10	1	11
15	2	3	3	4	5	6	6	7	7	8	8	9	9	10	10	11	11	11	12
16	2	3	4	4	5	6	6	7	8	8	9	9	12	10	11	1	11	12	12
17	2	3	4	4	5	6	7	7	8	9	9	10	10	11	11	1	12	12	13
18	2	3	4	5	5	6	7	8	8	9	9	10	10	11	11	12	12	13	13
19	2	3	4	5	6	6	7	8	8	9	10	10	11	11	12	12	13	13	13
20	2	3	4	5	6	6	7	8	9	9	10	10	11	12	12	13	13	13	14

Part-II (Upper Critical Values)
 $(\alpha = 0.025)$

$n_2 \backslash n_1$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2																			
3																			
4				9	9														
5			9	10	10	11	11												
6			9	10	11	12	12	13	13	13	13								
7				11	12	13	13	14	14	14	14	15	15	15					
8				11	12	13	14	14	15	15	16	16	16	16	17	17	17	17	17
9					13	14	14	15	16	16	16	17	17	18	18	18	18	18	18
10					13	14	15	16	16	17	17	18	18	18	19	19	19	20	20
11					13	14	15	16	17	17	18	19	19	19	20	20	20	21	21
12					13	14	16	16	17	18	19	19	20	20	21	21	21	22	22
13						15	16	17	18	19	19	20	20	21	21	22	22	23	23
14						15	16	17	18	19	20	20	21	22	22	23	23	23	24
15						15	16	18	18	19	20	21	22	22	23	23	24	24	25
16							17	18	19	20	21	21	22	23	23	24	25	25	25
17							17	18	19	20	21	22	23	23	24	25	25	26	26
18							17	18	19	20	21	22	23	24	25	25	26	26	27
19							17	18	20	21	22	23	23	24	25	26	26	27	27
20							17	18	20	21	22	23	24	25	25	26	27	27	27

One-Tailed Test:											
$\alpha =$	0.10	0.05	0.025	0.01	0.005	$\alpha =$	0.10	0.05	0.025	0.01	0.005
Two-Tailed Test:											
$\alpha =$	0.20	0.10	0.05	0.02	0.01	$\alpha =$	0.20	0.10	0.05	0.02	0.01
n = 1	0.900	0.950	0.975	0.990	0.995	n = 21	0.226	0.259	0.287	0.321	0.344
2	0.684	0.776	0.842	0.900	0.929	22	0.221	0.253	0.281	0.314	0.337
3	0.565	0.636	0.708	0.785	0.829	23	0.216	0.247	0.275	0.307	0.330
4	0.493	0.565	0.624	0.689	0.734	24	0.212	0.242	0.269	0.301	0.323
5	0.447	0.509	0.563	0.627	0.669	25	0.208	0.238	0.264	0.295	0.317
6	0.410	0.468	0.519	0.577	0.617	26	0.204	0.233	0.259	0.290	0.311
7	0.381	0.436	0.483	0.538	0.576	27	0.200	0.229	0.254	0.284	0.305
8	0.358	0.410	0.454	0.507	0.542	28	0.197	0.225	0.250	0.279	0.300
9	0.339	0.387	0.430	0.480	0.513	29	0.193	0.221	0.246	0.275	0.295
10	0.323	0.369	0.409	0.457	0.489	30	0.190	0.218	0.242	0.270	0.290
11	0.308	0.352	0.391	0.437	0.468	31	0.187	0.214	0.238	0.266	0.285
12	0.296	0.338	0.375	0.419	0.449	32	0.184	0.211	0.234	0.262	0.281
13	0.285	0.325	0.361	0.404	0.432	33	0.182	0.208	0.231	0.258	0.277
14	0.275	0.314	0.349	0.390	0.418	34	0.179	0.205	0.227	0.254	0.273
15	0.266	0.304	0.338	0.377	0.404	35	0.177	0.202	0.224	0.251	0.269
16	0.258	0.295	0.327	0.366	0.392	36	0.174	0.199	0.221	0.247	0.265
17	0.250	0.286	0.318	0.355	0.381	37	0.172	0.196	0.218	0.244	0.262
18	0.244	0.279	0.309	0.346	0.371	38	0.170	0.194	0.215	0.241	0.258
19	0.237	0.271	0.301	0.337	0.361	39	0.168	0.191	0.213	0.238	0.255
20	0.232	0.265	0.294	0.329	0.352	40	0.165	0.189	0.210	0.235	0.252
Approximation											
for n > 40							$\frac{1.07}{\sqrt{n}}$	$\frac{1.22}{\sqrt{n}}$	$\frac{1.36}{\sqrt{n}}$	$\frac{1.52}{\sqrt{n}}$	$\frac{1.63}{\sqrt{n}}$

Table V: Critical Values of Kolmogorov-Smirnov Test for Two Samples of Equal Size

Appendix

One-Tailed Test:											
$\alpha =$	0.10	0.05	0.025	0.01	0.005	$\alpha =$	0.10	0.05	0.025	0.01	0.005
Two-Tailed Test:											
$\alpha =$	0.20	0.10	0.05	0.02	0.01	$\alpha =$	0.20	0.10	0.05	0.02	0.01
n = 3	2/3	2/3				n = 20	6/20	7/20	8/20	9/20	10/20
4	3/4	3/4	3/4			21	6/21	7/21	8/21	9/21	10/21
5	3/5	3/5	4/5	4/5	4/5	22	7/22	8/22	8/22	10/22	10/22
6	3/6	4/6	4/6	5/6	5/6	23	7/23	8/23	9/23	10/23	10/23
7	4/7	4/7	5/7	5/7	5/7	24	7/24	8/24	9/24	10/24	11/24
8	4/8	4/8	5/8	5/8	6/8	25	7/25	8/25	9/25	10/25	11/25
9	4/9	5/9	5/9	6/9	6/9	26	7/26	8/26	9/26	10/26	11/26
10	4/10	5/10	6/10	6/10	7/10	27	7/27	8/27	9/27	11/27	11/27
11	5/11	5/11	6/11	7/11	7/11	28	8/28	9/28	10/28	11/28	12/28
12	5/12	5/12	6/12	7/12	7/12	29	8/29	9/29	10/29	11/29	12/29
13	5/13	6/13	6/13	7/13	8/13	30	8/30	9/30	10/30	11/30	12/30
14	5/14	6/14	7/14	7/14	8/14	31	8/31	9/31	10/31	11/31	12/31
15	5/15	6/15	7/15	8/15	8/15	32	8/32	9/32	10/32	12/32	12/32
16	6/16	6/16	7/16	8/16	9/16	34	8/34	10/34	11/34	12/34	13/34
17	6/17	7/17	7/17	8/17	9/17	36	9/36	10/36	11/36	12/36	13/36
18	6/18	7/18	8/18	9/18	9/18	38	9/38	10/38	11/38	13/38	14/38
19	6/19	7/19	8/19	9/19	9/19	40	9/40	10/40	12/40	13/40	14/40
Approximation											
for $n > 40$							1.52	1.73	1.92	2.15	2.30
							$\frac{1.52}{\sqrt{n}}$	$\frac{1.73}{\sqrt{n}}$	$\frac{1.92}{\sqrt{n}}$	$\frac{2.15}{\sqrt{n}}$	$\frac{2.30}{\sqrt{n}}$

Table VI: Critical Values of K-S Test for Two Samples of Unequal Size

One-Sided Test:		$\alpha = 0.10$	0.05	0.25	0.01	0.005
Two-Sided Test:		$\alpha = 0.20$	0.1	0.05	0.25	0.01
$n_1 = 1$	$n_2 = 9$	17/18				
	10	9/10				
$n_1 = 2$	$n_2 = 3$	5/6				
	4	3/4				
	5	4/5	4/5			
	6	5/6	5/6			
	7	5/7	6/7			
	8	3/4	7/8	7/8		
	9	7/9	8/9	8/9		
	10	7/10	4/5	9/10		
	$n_2 = 4$	3/4	3/4			
	5	2/3	4/5	4/5		
$n_1 = 3$	6	2/3	2/3	5/6		
	7	2/3	5/7	6/7	6/7	
	8	5/8	3/4	3/4	7/8	
	9	2/3	2/3	7/9	8/9	8/9
	10	3/5	7/10	4/5	9/10	9/10
	12	7/12	2/3	3/4	5/6	11/12
	$n_2 = 5$	3/5	3/4	4/5	4/5	
	6	7/12	2/3	3/4	5/6	5/6
	7	17/28	5/7	3/4	6/7	6/7
	8	5/8	5/8	3/4	7/8	7/8
	9	5/9	2/3	3/4	7/9	8/9
	10	11/20	13/20	7/10	4/5	4/5
$n_1 = 4$	12	7/12	2/3	2/3	3/4	5/6
	16	9/16	5/8	11/16	3/4	13/16
	$n_2 = 6$	3/5	2/3	2/3	5/6	5/6
	7	4/7	23/35	5/7	29/35	6/7
	8	11/20	5/8	47/40	4/5	4/5
	9	5/9	3/5	31/45	7/9	4/5
	10	1/2	3/5	7/10	7/10	4/5
	15	8/15	3/5	2/3	11/15	11/15
	20	1/2	11/20	3/5	7/10	3/4
	$n_2 = 7$	23/42	4/7	29/42	5/7	5/6
$n_1 = 5$	8	1/2	7/12	2/3	3/4	3/4
	9	1/2	5/9	2/3	13/18	7/9
	10	1/2	17/30	19/30	7/10	11/15
	12	1/2	7/12	7/12	2/3	3/4
	18	4/9	5/9	11/18	2/3	13/18
	24	11/24	1/2	7/12	5/8	2/3
	$n_2 = 8$	27/56	33/56	5/8	41/56	3/4
	9	31/63	5/9	40/63	5/7	47/63
	10	33/70	39/70	43/70	7/10	5/7
	14	3/7	1/2	4/7	9/14	5/7
$n_1 = 6$	28	3/7	13/28	15/28	17/28	9/14
	$n_2 = 9$	4/9	13/24	5/8	2/3	3/4
	10	19/40	21/40	23/40	27/40	7/10
	12	11/24	1/2	7/12	5/8	2/3
	16	7/16	1/2	9/16	5/8	5/8
	32	13/32	7/16	1/2	9/16	19/32
$n_1 = 7$	$n_2 = 10$	7/15	1/2	26/45	2/3	31/45
	12	4/9	1/2	5/9	11/18	2/3
	15	19/45	22/45	8/15	3/5	29/45
	18	7/18	4/9	1/2	5/9	11/18
	36	13/36	5/12	17/36	19/36	5/9
	$n_2 = 15$	2/5	7/15	1/2	17/30	19/30
$n_1 = 8$	20	2/5	9/20	1/2	11/20	3/5
	40	7/20	2/5	9/20	1/2	
	$n_2 = 16$	23/60	9/20	1/2	11/20	7/12
	16	3/8	7/16	23/48	13/24	7/12
$n_1 = 9$	18	13/36	5/12	17/36	19/36	5/9
	20	11/30	5/12	7/15	31/60	17/30
	$n_2 = 20$	7/20	2/5	13/30	29/60	31/60
	$n_2 = 20$	27/80	33/80	17/40	19/40	41/80
Large-sample approximation		$1.07 \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$	$1.07 \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$	$1.36 \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$	$1.52 \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$	$1.63 \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$

Table-VII: Critical Values of Mann-Whiney U Test (Lower Critical Values)

Appendix

n_1	α	$n_2 = 2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	0.001	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	0.005	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	4	4
	0.01	3	3	3	3	3	3	3	3	3	3	3	4	4	4	4	4	4	5	5
	0.025	3	3	3	3	3	3	4	4	4	5	5	5	5	5	5	6	6	6	6
	0.05	3	3	4	4	4	4	5	5	5	5	6	6	7	7	7	7	8	8	8
	0.1	3	4	6	5	5	5	6	6	7	7	8	8	8	9	9	10	10	11	11
3	0.001	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	7	7	7	7
	0.005	6	6	6	6	6	6	6	7	7	7	8	8	8	9	9	9	9	10	10
	0.01	6	6	6	6	6	7	7	8	8	8	9	9	9	10	10	11	11	11	12
	0.025	6	6	7	7	8	8	9	9	10	10	11	11	12	12	13	13	14	14	15
	0.05	6	7	7	8	9	9	10	11	11	12	12	13	14	14	15	16	16	17	18
	0.1	7	8	8	9	10	11	12	12	13	14	15	16	17	17	18	19	20	21	22
4	0.001	10	10	10	10	10	10	10	10	11	11	11	12	12	12	13	13	14	14	14
	0.005	10	10	10	10	11	11	12	12	13	13	14	14	15	16	16	17	17	18	19
	0.01	10	10	10	11	12	12	13	14	14	15	16	16	17	18	18	19	20	20	21
	0.025	10	10	11	12	13	14	15	15	16	17	18	19	20	21	22	22	23	24	25
	0.05	10	11	12	13	14	15	16	17	18	19	20	21	22	23	25	26	27	28	29
	0.1	11	12	14	15	16	17	18	20	21	22	23	24	26	27	28	29	31	32	33
5	0.001	15	15	15	15	15	15	16	17	17	18	18	19	19	20	21	21	22	23	23
	0.005	15	15	15	16	17	17	18	19	20	21	22	23	23	24	25	26	27	28	29
	0.01	15	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
	0.025	15	16	17	18	19	21	22	23	24	25	27	28	29	30	31	33	34	35	36
	0.05	16	17	18	20	21	22	24	25	27	28	29	31	32	34	35	36	38	39	41
	0.1	17	18	20	21	23	24	26	28	29	31	33	34	36	38	39	41	43	44	46
6	0.001	21	21	21	21	21	21	23	24	25	26	27	28	29	30	31	32	33	34	34
	0.005	21	21	22	23	24	25	26	27	28	29	31	32	33	34	35	37	38	39	40
	0.01	21	21	23	24	25	26	28	29	30	31	33	34	35	37	38	40	41	42	44
	0.025	21	23	24	25	27	28	30	32	33	35	36	38	39	41	43	44	46	47	49
	0.05	22	24	25	27	29	30	32	34	36	38	39	41	43	45	47	48	50	52	54
	0.1	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51	53	56	58	60
7	0.001	28	28	28	28	29	30	31	32	34	35	36	37	38	39	40	42	43	44	45
	0.005	28	28	29	30	32	33	35	36	38	39	41	42	44	45	47	48	50	51	53
	0.01	28	29	30	32	33	35	36	38	40	41	43	45	46	48	50	52	53	55	57
	0.025	28	30	32	34	35	37	39	41	43	45	47	49	51	53	55	57	59	61	63
	0.05	29	31	33	35	37	40	42	44	46	48	50	53	55	57	59	62	64	66	68
	0.1	30	33	35	37	40	42	45	47	50	52	55	57	60	62	65	67	70	72	75
8	0.001	36	36	36	37	38	39	41	42	43	45	46	48	49	51	52	54	55	57	58
	0.005	36	36	38	39	41	43	44	46	48	50	52	54	55	57	59	61	63	65	67
	0.01	36	37	39	41	43	44	46	48	50	52	54	56	59	61	63	65	67	69	71
	0.025	37	39	41	43	45	47	50	52	54	56	59	61	63	66	68	71	73	75	78
	0.05	38	40	42	45	47	50	52	55	57	60	63	65	68	70	73	76	78	81	84
	0.1	39	42	44	47	50	53	53	59	61	64	67	70	73	76	79	82	85	88	91
9	0.001	45	45	45	47	48	49	51	53	54	56	58	60	61	63	65	67	69	71	72
	0.005	45	46	47	49	51	53	55	57	59	62	64	66	68	70	73	75	77	79	82
	0.01	45	47	49	51	53	55	57	60	62	64	67	69	72	74	77	79	82	84	86
	0.025	46	48	50	53	56	58	61	63	66	69	72	74	77	80	83	85	88	91	94
	0.05	47	50	52	55	58	61	64	67	70	73	76	79	82	85	88	91	94	97	100
	0.1	48	51	55	58	61	64	68	71	74	77	81	84	87	91	94	98	101	104	108
10	0.001	55	55	56	57	59	61	62	64	66	68	70	73	75	77	79	81	83	85	88
	0.005	55	56	58	60	62	65	67	69	72	74	77	80	82	85	87	90	93	95	98
	0.01	55	57	59	62	64	67	69	72	75	78	80	83	86	89	92	94	97	100	103
	0.025	56	59	61	64	67	70	73	76	79	82	85	89	92	95	98	101	104	108	111
	0.05	57	60	63	67	70	73	76	80	83	87	90	93	97	100	104	107	111	114	118
	0.1	59	62	66	69	73	77	80	84	88	92	95	99	103	107	110	114	118	122	126

Non-Parametric Tests

n_1	α	$n_2 = 2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
11	0.001	66	66	67	69	71	73	75	77	79	82	84	87	89	91	94	96	99	101	104
	0.005	66	67	69	72	74	77	80	83	85	88	91	94	97	100	103	106	109	112	115
	0.01	66	68	71	74	76	79	82	85	89	92	95	98	101	104	108	111	114	117	120
	0.025	67	70	73	76	80	83	86	90	93	97	100	104	107	111	114	118	122	125	129
	0.05	68	72	75	79	83	86	90	94	98	101	105	109	113	117	121	124	128	132	136
	0.1	70	74	78	82	86	90	94	98	103	107	111	115	119	124	128	132	136	140	145
12	0.001	78	78	79	81	83	86	88	91	93	96	98	102	104	106	110	113	116	118	121
	0.005	78	80	82	85	88	91	94	97	100	103	106	110	113	116	120	123	126	130	133
	0.01	78	81	84	87	90	93	96	100	103	107	110	114	117	121	125	128	132	135	139
	0.025	80	83	86	90	93	97	101	105	108	112	116	120	124	128	132	136	140	144	148
	0.05	81	84	88	92	96	100	105	109	111	117	121	126	130	134	139	143	147	151	156
	0.1	83	87	91	96	100	105	109	114	118	123	128	132	137	142	146	151	156	160	165
13	0.001	91	91	93	95	97	100	103	106	109	112	115	118	121	124	127	130	134	137	140
	0.005	91	93	95	99	102	105	109	112	116	119	123	126	130	134	137	141	145	149	152
	0.01	92	94	97	101	104	108	112	115	119	123	127	131	135	139	143	147	151	155	159
	0.025	93	96	100	104	108	112	116	120	125	129	133	137	142	146	151	155	159	164	168
	0.05	94	98	102	107	111	116	120	125	129	134	139	143	148	153	157	162	167	172	176
	0.1	96	101	105	110	115	120	125	130	135	140	145	150	155	160	166	171	176	181	186
14	0.001	105	105	107	109	112	115	118	121	125	128	131	135	138	142	145	149	152	156	160
	0.005	105	107	110	113	117	121	124	128	132	136	140	144	148	152	156	160	164	169	173
	0.01	106	108	112	116	119	123	128	132	136	140	144	149	153	157	162	166	171	175	179
	0.025	107	111	115	119	123	128	132	137	142	146	151	156	161	165	170	175	180	184	189
	0.05	109	113	117	122	127	132	137	142	147	152	157	162	167	172	177	183	188	193	198
	0.1	110	116	121	126	131	137	142	147	153	158	164	169	175	180	186	191	197	203	208
15	0.001	120	120	122	125	128	133	135	138	142	145	149	153	157	161	164	168	172	176	180
	0.005	120	123	126	129	133	137	141	145	150	154	158	163	167	172	176	181	185	190	194
	0.01	121	124	128	132	136	140	145	149	154	158	163	168	172	177	182	187	191	196	201
	0.025	122	126	131	135	140	145	150	155	160	165	170	175	180	185	191	196	201	206	211
	0.05	124	128	133	139	144	149	154	160	165	171	176	182	187	193	198	204	209	215	221
	0.1	126	131	137	143	148	154	160	166	172	178	184	189	195	201	207	213	219	225	231
16	0.001	136	136	139	142	145	148	152	156	160	164	168	172	176	180	185	189	193	197	202
	0.005	136	139	142	146	150	155	159	164	168	173	178	182	187	192	197	202	207	211	216
	0.01	137	140	144	149	153	158	163	168	173	178	183	188	193	198	203	208	213	219	224
	0.025	138	143	148	152	158	163	168	174	179	184	190	196	201	207	212	218	223	229	235
	0.05	140	145	151	156	162	167	173	179	185	191	197	202	208	214	220	226	232	238	244
	0.1	142	148	154	160	166	173	179	185	191	198	204	211	217	223	230	236	243	249	256
17	0.001	153	154	156	159	163	167	171	175	179	183	188	192	197	201	206	211	215	220	224
	0.005	153	156	158	164	169	173	178	183	188	193	198	203	208	214	219	224	229	235	240
	0.01	154	158	160	167	172	177	182	187	192	198	203	209	214	220	225	231	236	242	247
	0.025	156	160	163	171	176	182	188	193	199	205	211	217	223	229	235	241	247	253	259
	0.05	157	163	166	174	180	187	193	199	205	211	218	224	231	237	243	250	256	263	269
	0.1	160	166	172	179	185	192	199	206	212	219	226	233	239	246	253	260	267	274	281
18	0.001	171	172	174	178	182	186	190	195	199	204	209	214	218	223	228	233	238	243	248
	0.005	171	174	176	183	188	193	198	203	209	214	219	225	230	236	242	247	253	259	264
	0.01	172	176	179	186	191	196	202	208	213	219	225	231	237	242	248	254	260	266	272
	0.025	174	179	181	190	196	202	208	214	220	227	233	239	246	252	258	265	271	278	284
	0.05	176	181	185	194	200	207	213	220	227	233	240	247	254	260	267	274	281	288	295
	0.1	178	185	191	199	206	213	220	227	234	241	249	256	263	270	278	285	292	300	307
19	0.001	190	191	194	198	202	206	211	216	220	225	231	236	241	246	251	257	262	268	273
	0.005	191	194	195	203	208	213	219	224	230	236	242	248	254	260	265	272	278	284	290
	0.01	192	195	198	206	211	217	223	229	235	241	247	254	260	266	273	279	285	292	298
	0.025	193	198	201	210	216	223	229	236	243	249	256	263	269	276	283	290	297	304	310
	0.05	195	201	205	214	221	228	235	242	249	256	263	271	278	285	292	300	307	314	321
	0.1	198	205	211	219	227	234	242	249	257	264	272	280	288	295	303	311	319	326	334
20	0.001	210	211	214	218	223	227	232	237	243	248	253	259	265	270	276	281	287	293	299
	0.005	211	214	219	224	229	235	241	247	253	259	265	271	278	284	290	297	303	310	316
	0.01	212	216	221	227	233	239	245	251	258	264	271	278	284	291	298	304	311	318	325
	0.025	213	219	225	231	238	245	251	259	266	273	280	287	294	301	309	316	323	330	338
	0.05	215	222	229	236	243	250	258	265	273	280	288	295	303	311	318	326	334	341	349
	0.1	218	226	233	241	249	257	265	273	281	289	297	305	313	321	330	338	346	354	362

(Upper Critical Values)

Appendix

n_1	α	$n_2=2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	0.001	10	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43
	0.005	10	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	40	42
	0.01	10	9	11	13	15	17	19	21	23	25	27	28	30	32	34	36	38	39	41
	0.025	10	9	11	13	15	17	18	20	22	23	25	27	29	31	33	34	36	38	40
	0.05	10	9	10	12	14	16	17	19	21	23	24	26	27	29	31	33	34	36	38
	0.1	10	8	8	11	13	15	16	18	19	21	22	24	26	27	29	30	32	33	35
3	0.001	18	15	18	21	24	27	30	33	36	39	42	45	48	51	54	56	59	62	65
	0.005	18	15	18	21	24	27	30	32	35	38	40	43	46	48	51	54	57	59	62
	0.01	18	15	18	21	24	26	29	31	34	37	39	42	45	47	50	52	55	58	60
	0.025	18	15	17	20	22	25	27	30	32	35	37	40	42	45	47	50	52	55	57
	0.05	18	14	17	19	21	24	26	28	31	33	36	38	40	43	45	47	50	52	54
	0.1	18	13	16	18	20	22	24	27	29	31	33	35	37	40	42	44	46	48	50
4	0.001	28	22	26	30	34	38	42	46	49	53	57	60	64	68	71	75	78	82	86
	0.005	28	22	26	30	33	37	40	44	47	51	54	58	61	64	68	71	75	78	81
	0.01	28	22	26	29	32	36	39	42	46	49	52	56	59	62	66	69	72	76	79
	0.025	28	22	25	28	31	34	37	41	44	47	50	53	56	59	62	66	69	72	75
	0.05	28	21	24	27	30	33	36	39	42	45	48	51	54	57	59	62	65	68	71
	0.1	28	20	22	25	28	31	34	36	39	42	45	48	50	53	56	59	61	64	67
5	0.001	40	30	35	40	45	50	54	58	63	67	72	76	81	85	89	94	98	102	107
	0.005	40	30	35	39	43	48	52	56	60	64	68	72	77	81	85	89	93	97	101
	0.01	40	30	34	38	42	46	50	54	58	62	66	70	74	78	82	86	90	94	98
	0.025	40	29	33	37	41	44	48	52	56	60	63	67	71	75	79	82	86	90	94
	0.05	40	28	32	35	39	43	46	50	53	57	61	64	68	71	75	79	82	86	89
	0.1	40	27	30	34	37	41	44	47	51	54	57	61	64	67	71	74	77	81	84
6	0.001	54	39	45	51	57	63	67	72	77	82	88	93	98	103	108	113	118	123	128
	0.005	54	39	44	49	54	59	64	69	74	79	83	88	93	98	103	107	112	117	122
	0.01	54	39	43	48	53	58	62	67	72	77	81	86	91	95	100	104	109	114	118
	0.025	54	37	42	47	51	56	60	64	69	73	78	82	87	91	95	100	104	109	113
	0.05	54	36	41	45	49	54	58	62	66	70	75	79	83	87	91	96	100	104	108
	0.1	54	35	39	43	47	51	55	59	63	67	71	75	79	83	87	91	94	98	102
7	0.001	70	49	56	63	69	75	81	87	92	98	104	110	116	122	128	133	139	145	151
	0.005	70	49	55	61	66	72	77	83	88	94	99	105	110	116	121	127	132	138	143
	0.01	70	48	54	59	65	70	76	81	86	92	97	102	108	113	118	123	129	134	139
	0.025	70	47	52	57	63	68	73	78	83	88	93	98	103	108	113	118	123	128	133
	0.05	70	46	51	56	61	65	70	75	80	85	90	94	99	104	109	113	118	123	128
	0.1	70	44	49	54	58	63	67	72	76	81	85	90	94	99	103	108	112	117	121
8	0.001	88	60	68	75	82	89	95	102	109	115	122	128	135	141	148	154	161	167	174
	0.005	88	60	66	73	79	85	92	98	104	110	116	122	129	135	141	147	153	159	165
	0.01	88	59	65	71	77	84	90	96	102	108	114	120	125	131	137	143	149	155	161
	0.025	88	57	63	69	75	81	86	92	98	104	109	115	121	126	132	137	143	149	154
	0.05	88	56	62	67	73	78	84	89	95	100	105	111	116	122	127	132	138	143	148
	0.1	88	54	60	65	70	75	83	85	91	96	101	106	111	116	121	126	131	136	141
9	0.001	108	72	81	88	96	104	111	118	126	133	140	147	155	162	169	176	183	190	198
	0.005	108	71	79	86	93	100	107	114	121	127	134	141	148	155	161	168	175	182	188
	0.01	108	70	77	84	91	98	105	111	118	125	131	138	144	151	157	164	170	177	184
	0.025	108	69	76	82	88	95	101	108	114	120	126	133	139	145	151	158	164	170	176
	0.05	108	67	74	80	86	92	98	104	110	116	122	128	134	140	146	152	158	164	170
	0.1	108	66	71	77	83	89	94	100	106	112	117	123	129	134	140	145	151	157	162
10	0.001	130	85	94	103	111	119	128	136	144	152	160	167	175	183	191	199	207	215	222
	0.005	130	84	92	100	108	115	123	131	138	146	153	160	168	175	183	190	197	205	212
	0.01	130	83	91	98	106	113	121	128	135	142	150	157	164	171	178	186	193	200	207
	0.025	130	81	89	96	103	110	117	124	131	138	145	151	158	165	172	179	186	192	199
	0.05	130	80	87	93	100	107	114	120	127	133	140	147	153	160	166	173	179	186	192
	0.1	130	78	84	91	97	103	110	116	122	128	135	141	147	153	160	166	172	178	184

Non-Parametric Tests

n_1	α	$n_2 = 2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
11	0.001	154	99	109	118	127	136	145	154	163	171	180	188	197	206	214	223	231	240	248
	0.005	154	98	107	115	124	132	140	148	157	165	173	181	189	197	205	213	221	229	237
	0.01	154	97	105	113	122	130	138	146	153	161	169	177	185	193	200	208	216	224	232
	0.025	154	95	103	111	118	126	134	141	149	156	164	171	179	186	194	201	208	216	223
	0.05	154	93	101	108	115	123	130	137	144	152	159	166	173	180	187	195	202	209	216
	0.1	154	91	98	105	112	119	126	133	139	146	153	160	167	173	180	187	194	201	207
12	0.001	180	114	125	135	145	154	164	173	183	192	202	210	220	230	238	247	256	266	275
	0.005	180	112	122	131	140	149	158	167	176	185	194	202	211	220	228	237	246	254	263
	0.01	180	111	120	129	138	147	156	164	173	181	190	198	207	215	223	232	240	249	257
	0.025	180	109	118	126	135	143	151	159	168	176	184	192	200	208	216	224	232	240	248
	0.05	180	108	116	124	132	140	147	155	165	171	179	186	194	202	209	217	225	233	240
	0.1	180	105	113	120	128	135	143	150	158	165	172	180	187	194	202	209	216	224	231
13	0.001	208	130	141	152	163	173	183	193	203	213	223	233	243	253	263	273	282	292	302
	0.005	208	128	139	148	158	168	177	187	196	206	215	225	234	243	253	262	271	280	290
	0.01	208	127	137	146	156	165	174	184	193	202	211	220	229	238	247	256	265	274	283
	0.025	208	125	134	143	152	161	170	179	187	196	205	214	222	231	239	248	257	265	274
	0.05	208	123	132	140	149	157	166	174	183	191	199	208	216	224	233	241	249	257	266
	0.1	208	120	129	137	145	153	161	169	177	185	193	201	209	217	224	232	240	248	256
14	0.001	238	147	159	171	182	193	204	215	225	236	247	257	268	278	289	299	310	320	330
	0.005	238	145	156	167	177	187	198	208	218	228	238	248	258	268	278	288	298	307	317
	0.01	238	144	154	164	175	185	194	204	214	224	234	243	253	263	272	282	291	301	311
	0.025	238	141	151	161	171	180	190	199	208	218	227	236	245	255	264	273	282	292	301
	0.05	238	139	149	158	167	176	185	194	203	212	221	230	239	248	257	265	274	283	292
	0.1	238	136	145	154	163	171	180	189	197	206	214	223	231	240	248	257	265	273	282
15	0.001	270	165	178	190	202	212	225	237	248	260	271	282	293	304	316	327	338	349	360
	0.005	270	162	174	186	197	208	219	230	240	251	262	272	283	293	304	314	325	335	346
	0.01	270	161	172	183	194	205	215	226	236	247	257	267	278	288	298	308	319	329	339
	0.025	270	159	169	180	190	200	210	220	230	240	250	260	270	280	289	299	309	319	329
	0.05	270	157	167	176	186	196	206	215	225	234	244	253	263	272	282	291	301	310	319
	0.1	270	154	163	172	182	191	200	209	218	227	236	246	255	264	273	282	291	300	309
16	0.001	304	184	197	210	223	236	248	260	272	284	296	308	320	332	343	355	367	379	390
	0.005	304	181	194	206	218	229	241	252	264	275	286	298	309	320	331	342	353	365	376
	0.01	304	180	192	203	215	226	237	248	259	270	281	292	303	314	325	336	347	357	368
	0.025	304	177	188	200	210	221	232	242	253	264	274	284	295	305	316	326	337	347	357
	0.05	304	175	185	196	206	217	227	237	247	257	267	278	288	298	308	318	328	338	348
	0.1	304	172	182	192	202	211	221	231	241	250	260	269	279	289	298	308	317	327	336
17	0.001	340	203	218	232	245	258	271	284	297	310	322	335	347	360	372	384	397	409	422
	0.005	340	201	216	227	239	252	264	276	288	300	312	324	336	347	359	371	383	394	406
	0.01	340	199	214	224	236	248	260	272	284	295	307	318	330	341	353	364	376	387	399
	0.025	340	197	211	220	232	243	254	266	277	288	299	310	321	332	343	354	365	376	387
	0.05	340	194	208	217	228	238	249	260	271	282	292	303	313	324	335	345	356	366	377
	0.1	340	191	202	212	223	233	243	253	264	274	284	294	305	315	325	335	345	355	365
18	0.001	378	224	240	254	268	282	296	309	323	336	349	362	376	389	402	415	428	441	454
	0.005	378	222	238	249	262	275	288	301	313	326	339	351	364	376	388	401	413	425	438
	0.01	378	220	235	246	259	272	284	296	309	321	333	345	357	370	382	394	406	418	430
	0.025	378	217	233	242	254	266	278	290	302	313	325	337	348	360	372	383	395	406	418
	0.05	378	215	229	238	250	261	273	284	295	307	318	329	340	352	363	374	385	396	407
	0.1	378	211	223	233	244	255	266	277	288	299	309	320	331	342	352	363	374	384	395
19	0.001	418	246	262	277	292	307	321	335	350	364	377	391	405	419	433	446	460	473	517
	0.005	418	243	261	272	286	300	313	327	340	353	366	379	392	405	419	431	444	457	470
	0.01	418	242	258	269	283	296	309	322	335	348	361	373	386	399	411	424	437	449	462
	0.025	418	239	255	265	278	290	303	315	327	340	352	364	377	389	401	413	425	437	450
	0.05	418	236	251	261	273	285	297	309	321	333	345	356	368	380	392	403	415	427	439
	0.1	418	232	245	256	267	279	290	302	313	325	336	347	358	370	381	392	403	415	426
20	0.001	460	269	286	302	317	333	348	363	377	392	407	421	435	450	464	479	493	507	521
	0.005	460	266	281	296	311	325	339	353	367	381	395	409	422	436	450	463	477	490	504
	0.01	460	264	279	293	307	321	335	349	362	376	389	402	416	429	442	456	469	482	495
	0.025	460	261	275	289	302	315	330	341	354	367	380	393	406	419	431	444	457	470	482
	0.05	460	258	271	284	297	310	322	335	347	360	372	385	397	409	422	434	446	459	471
	0.1	460	254	267	279	291	303	315	327	339	351	363	375	387	399	410	422	434	446	458

Table VIII: Critical Values of Kruskal-Wallis Test

Appendix

n_1	n_2	n_3	$\alpha = 0.1$	0.05	0.025	0.01
2	2	2	4.571	—	—	—
3	2	1	4.286	—	—	—
3	2	2	4.500	4.714	—	—
3	3	1	4.571	5.143	—	—
3	3	2	4.556	5.361	5.556	—
3	3	3	4.622	5.600	5.956	7.200
4	2	1	4.500	—	—	—
4	2	2	4.458	5.333	5.500	—
4	3	1	4.056	5.208	5.833	—
4	3	2	4.511	5.444	6.000	6.444
4	3	3	4.709	5.791	6.155	6.745
4	4	1	4.167	4.967	6.167	6.667
4	4	2	4.555	5.455	6.327	7.036
4	4	3	4.545	5.598	6.394	7.144
4	4	4	4.654	5.692	6.615	7.654
5	2	1	4.200	5.000	—	—
5	2	2	4.373	5.160	6.000	6.533
5	3	1	4.018	4.960	6.044	—
5	3	2	4.651	5.251	6.004	6.909
5	3	3	4.533	5.648	6.315	7.079
5	4	1	3.987	4.985	5.858	6.955
5	4	2	4.541	5.273	6.068	7.205
5	4	3	4.549	5.656	6.410	7.445
5	4	4	4.668	5.657	6.673	7.760
5	5	1	4.109	5.127	6.000	7.309
5	5	2	4.623	5.338	6.346	7.338
5	5	3	4.545	5.705	6.549	7.578
5	5	4	4.523	5.666	6.760	7.823
5	5	5	4.560	5.780	6.740	8.000

Notes: 1. The critical values in above table are approximated values.

2. When this table is not applicable then we use chi-square table for critical values of Kruskal-Wallis test which is given in the Appendix at the end of the block 1 of this course.

Table IX: Critical Values of Freidman Test

n	k = 3		k = 4		k = 5		k = 6	
	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$
2	—	—	—	6.00	—	—	9.14	9.71
3	—	6.00	7.40	9.00	7.60	8.00	9.86	11.76
4	6.50	8.00	7.80	9.60	8.53	10.13	10.29	12.71
5	6.40	8.40	7.80	9.96	8.80	11.20	10.49	13.23
6	7.00	9.00	7.60	10.20	8.96	11.68	10.57	13.62
7	7.14	8.86	7.80	10.54	9.07	11.87	10.67	13.86
8	6.25	9.00	7.65	10.50	9.14	12.11	10.71	14.00
9	6.22	9.56	7.67	10.73	9.20	13.20	10.78	14.14
10	6.20	9.60	7.68	10.68	9.24	12.44	10.80	14.23
11	6.55	9.46	7.69	10.75	9.28	12.48	10.84	14.32
12	6.50	9.50	7.70	10.80	9.31	12.58	10.86	14.38
13	6.62	9.39	7.80	10.85	9.33	12.60	10.89	14.45
14	6.14	9.14	7.71	10.89	9.35	12.68	10.90	14.49
15	6.40	8.93	7.72	10.92	9.37	12.74	10.92	14.54
16	6.50	9.38	7.80	10.95	9.39	12.80	10.96	14.57
17	6.12	9.29	7.80	10.05	9.40	12.85	10.95	14.61
18	6.33	9.00	7.73	10.93	9.42	12.89	10.95	14.63
19	6.42	9.58	7.86	11.02	9.43	12.88	11.00	14.67
20	6.30	9.30	7.80	11.80	9.40	12.92	11.00	14.66

Note: 1. When this table is not applicable then we use chi-square table for critical values of Freidman test which is given in the Appendix at the end of the block 1 of this course.

Table X: Poisson Probability

Appendix

For a given value of λ entry indicates the probability of obtaining a specified value of X

X	$\lambda = 0.1$	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066	0.3679
1	0.0905	0.1637	0.2222	0.2681	0.3033	0.3293	0.3476	0.3595	0.3659	0.3679
2	0.0045	0.0164	0.0333	0.0536	0.0758	0.0988	0.1217	0.1438	0.1647	0.1839
3	0.0002	0.0011	0.0033	0.0072	0.0126	0.0198	0.0284	0.0383	0.0494	0.0613
4	0	0.0001	0.0003	0.0007	0.0016	0.003	0.005	0.0077	0.0111	0.0153
5	0	0	0	0.0001	0.0002	0.0004	0.0007	0.0012	0.002	0.0031
6	0	0	0	0	0	0	0.0001	0.0002	0.0003	0.0005
7	0	0	0	0	0	0	0	0	0	0.0001
X	$\lambda = 1.1$	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	0.3329	0.3012	0.2725	0.2466	0.2231	0.2019	0.1827	0.1653	0.1496	0.1353
1	0.3662	0.3614	0.3543	0.3452	0.3347	0.323	0.3106	0.2975	0.2842	0.2707
2	0.2014	0.2169	0.2303	0.2417	0.251	0.2584	0.264	0.2678	0.27	0.2707
3	0.0738	0.0867	0.0998	0.1128	0.1255	0.1378	0.1496	0.1607	0.171	0.1804
4	0.0203	0.026	0.0324	0.0395	0.0471	0.0551	0.0636	0.0723	0.0812	0.0902
5	0.0045	0.0062	0.0084	0.0111	0.0141	0.0176	0.0216	0.026	0.0309	0.0361
6	0.0008	0.0012	0.0018	0.0026	0.0035	0.0047	0.0061	0.0078	0.0098	0.012
7	0.0001	0.0002	0.0003	0.0005	0.0008	0.0011	0.0015	0.002	0.0027	0.0034
8	0	0	0.0001	0.0001	0.0001	0.0002	0.0003	0.0005	0.0006	0.0009
9	0	0	0	0	0	0	0.0001	0.0001	0.0001	0.0002
X	$\lambda = 2.1$	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
0	0.1225	0.1108	0.1003	0.0907	0.0821	0.0743	0.0672	0.0608	0.055	0.0498
1	0.2572	0.2438	0.2306	0.2177	0.2052	0.1931	0.1815	0.1703	0.1596	0.1494
2	0.27	0.2681	0.2652	0.2613	0.2565	0.251	0.245	0.2384	0.2314	0.224
3	0.189	0.1966	0.2033	0.209	0.2138	0.2176	0.2205	0.2225	0.2237	0.224
4	0.0992	0.1082	0.1169	0.1254	0.1336	0.1414	0.1488	0.1557	0.1622	0.168
5	0.0417	0.0476	0.0538	0.0602	0.0668	0.0735	0.0804	0.0872	0.094	0.1008
6	0.0146	0.0174	0.0206	0.0241	0.0278	0.0319	0.0362	0.0407	0.0455	0.0504
7	0.0044	0.0055	0.0068	0.0083	0.0099	0.0118	0.0139	0.0163	0.0188	0.0216
8	0.0011	0.0015	0.0019	0.0025	0.0031	0.0038	0.0047	0.0057	0.0068	0.0081
9	0.0003	0.0004	0.0005	0.0007	0.0009	0.0011	0.0014	0.0018	0.0022	0.0027
10	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003	0.0004	0.0005	0.0006	0.0008
11	0	0	0	0	0	0.0001	0.0001	0.0001	0.0002	0.0002
12	0	0	0	0	0	0	0	0	0	0.0001
X	$\lambda = 3.1$	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4.0
0	0.045	0.0408	0.0369	0.0334	0.0302	0.0273	0.0247	0.0224	0.0202	0.0183
1	0.1397	0.1304	0.1217	0.1135	0.1057	0.0984	0.0915	0.085	0.0789	0.0733
2	0.2165	0.2087	0.2008	0.1929	0.185	0.1771	0.1692	0.1615	0.1539	0.1465
3	0.2237	0.2226	0.2209	0.2186	0.2158	0.2125	0.2087	0.2046	0.2001	0.1954
4	0.1734	0.1781	0.1823	0.1858	0.1888	0.1912	0.1931	0.1944	0.1951	0.1954
5	0.1075	0.114	0.1203	0.1264	0.1322	0.1377	0.1429	0.1477	0.1522	0.1563
6	0.0555	0.0608	0.0662	0.0716	0.0771	0.0826	0.0881	0.0936	0.0989	0.1042
7	0.0246	0.0278	0.0312	0.0348	0.0385	0.0425	0.0466	0.0508	0.0551	0.0595
8	0.0095	0.0111	0.0129	0.0148	0.0169	0.0191	0.0215	0.0241	0.0269	0.0298
9	0.0033	0.004	0.0047	0.0056	0.0066	0.0076	0.0089	0.0102	0.0116	0.0132
10	0.001	0.0013	0.0016	0.0019	0.0023	0.0028	0.0033	0.0039	0.0045	0.0053
11	0.0003	0.0004	0.0005	0.0006	0.0007	0.0009	0.0011	0.0013	0.0016	0.0019
12	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003	0.0003	0.0004	0.0005	0.0006
13	0	0	0	0	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002
14	0	0	0	0	0	0	0	0	0	0.0001
X	$\lambda = 4.1$	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	5.0
0	0.0166	0.015	0.0136	0.0123	0.0111	0.0101	0.0091	0.0082	0.0074	0.0067
1	0.0679	0.063	0.0583	0.054	0.05	0.0462	0.0427	0.0395	0.0365	0.0337
2	0.1393	0.1323	0.1254	0.1188	0.1125	0.1063	0.1005	0.0948	0.0894	0.0842
3	0.1904	0.1852	0.1798	0.1743	0.1687	0.1631	0.1574	0.1517	0.146	0.1404
4	0.1951	0.1944	0.1933	0.1917	0.1898	0.1875	0.1849	0.182	0.1789	0.1755
5	0.16	0.1633	0.1662	0.1687	0.1708	0.1725	0.1738	0.1747	0.1753	0.1755
6	0.1093	0.1143	0.1191	0.1237	0.1281	0.1323	0.1362	0.1398	0.1432	0.1462
7	0.064	0.0686	0.0732	0.0778	0.0824	0.0869	0.0914	0.0959	0.1002	0.1044
8	0.0328	0.036	0.0393	0.0428	0.0463	0.05	0.0537	0.0575	0.0614	0.0653
9	0.015	0.0168	0.0188	0.0209	0.0232	0.0255	0.028	0.0307	0.0334	0.0363
10	0.0061	0.0071	0.0081	0.0092	0.0104	0.0118	0.0132	0.0147	0.0164	0.0181
11	0.0023	0.0027	0.0032	0.0037	0.0043	0.0049	0.0056	0.0064	0.0073	0.0082
12	0.0008	0.0009	0.0011	0.0014	0.0016	0.0019	0.0022	0.0026	0.003	0.0034
13	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009	0.0011	0.0013
14	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003	0.0003	0.0004	0.0005
15	0	0	0	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002

Non-Parametric Tests

X	$\lambda = 5.1$	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	6.0
0	0.0061	0.0055	0.005	0.0045	0.0041	0.0037	0.0033	0.003	0.0027	0.0025
1	0.0311	0.0287	0.0265	0.0244	0.0225	0.0207	0.0191	0.0176	0.0162	0.0149
2	0.0793	0.0746	0.0701	0.0659	0.0618	0.058	0.0544	0.0509	0.0477	0.0446
3	0.1348	0.1293	0.1239	0.1185	0.1133	0.1082	0.1033	0.0985	0.0938	0.0892
4	0.1719	0.1681	0.1641	0.16	0.1558	0.1515	0.1472	0.1428	0.1383	0.1339
5	0.1753	0.1748	0.174	0.1728	0.1714	0.1697	0.1678	0.1656	0.1632	0.1606
6	0.149	0.1515	0.1537	0.1555	0.1571	0.1584	0.1594	0.1601	0.1605	0.1606
7	0.1086	0.1125	0.1163	0.12	0.1234	0.1267	0.1298	0.1326	0.1353	0.1377
8	0.0692	0.0731	0.0771	0.081	0.0849	0.0887	0.0925	0.0962	0.0998	0.1033
9	0.0392	0.0423	0.0454	0.0486	0.0519	0.0552	0.0586	0.062	0.0654	0.0688
10	0.02	0.022	0.0241	0.0262	0.0285	0.0309	0.0334	0.0359	0.0386	0.0413
11	0.0093	0.0104	0.0116	0.0129	0.0143	0.0157	0.0173	0.019	0.0207	0.0225
12	0.0039	0.0045	0.0051	0.0058	0.0065	0.0073	0.0082	0.0092	0.0102	0.0113
13	0.0015	0.0018	0.0021	0.0024	0.0028	0.0032	0.0036	0.0041	0.0046	0.0052
14	0.0006	0.0007	0.0008	0.0009	0.0011	0.0013	0.0015	0.0017	0.0019	0.0022
15	0.0002	0.0002	0.0003	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009
16	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002	0.0003	0.0003
17	0	0	0	0	0	0	0.0001	0.0001	0.0001	0.0001
X	$\lambda = 6.1$	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	7.0
0	0.0022	0.002	0.0018	0.0017	0.0015	0.0014	0.0012	0.0011	0.001	0.0009
1	0.0137	0.0126	0.0116	0.0106	0.0098	0.009	0.0082	0.0076	0.007	0.0064
2	0.0417	0.039	0.0364	0.034	0.0318	0.0296	0.0276	0.0258	0.024	0.0223
3	0.0848	0.0806	0.0765	0.0726	0.0688	0.0652	0.0617	0.0584	0.0552	0.0521
4	0.1294	0.1249	0.1205	0.1162	0.1118	0.1076	0.1034	0.0992	0.0952	0.0912
5	0.1579	0.1549	0.1519	0.1487	0.1454	0.142	0.1385	0.1349	0.1314	0.1277
6	0.1605	0.1601	0.1595	0.1586	0.1575	0.1562	0.1546	0.1529	0.1511	0.149
7	0.1399	0.1418	0.1435	0.145	0.1462	0.1472	0.148	0.1486	0.1489	0.149
8	0.1066	0.1099	0.113	0.116	0.1188	0.1215	0.124	0.1263	0.1284	0.1304
9	0.0723	0.0757	0.0791	0.0825	0.0858	0.0891	0.0923	0.0954	0.0985	0.1014
10	0.0441	0.0469	0.0498	0.0528	0.0558	0.0588	0.0618	0.0649	0.0679	0.071
11	0.0245	0.0265	0.0285	0.0307	0.033	0.0353	0.0377	0.0401	0.0426	0.0452
12	0.0124	0.0137	0.015	0.0164	0.0179	0.0194	0.021	0.0227	0.0245	0.0264
13	0.0058	0.0065	0.0073	0.0081	0.0089	0.0098	0.0108	0.0119	0.013	0.0142
14	0.0025	0.0029	0.0033	0.0037	0.0041	0.0046	0.0052	0.0058	0.0064	0.0071
15	0.001	0.0012	0.0014	0.0016	0.0018	0.002	0.0023	0.0026	0.0029	0.0033
16	0.0004	0.0005	0.0005	0.0006	0.0007	0.0008	0.001	0.0011	0.0013	0.0014
17	0.0001	0.0002	0.0002	0.0002	0.0003	0.0003	0.0004	0.0004	0.0005	0.0006
18	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002
19	0	0	0	0	0	0	0	0.0001	0.0001	0.0001
X	$\lambda = 7.1$	7.2	7.3	7.4	7.5	7.6	7.7	7.8	7.9	8.0
0	0.0008	0.0007	0.0007	0.0006	0.0006	0.0005	0.0005	0.0004	0.0004	0.0003
1	0.0059	0.0054	0.0049	0.0045	0.0041	0.0038	0.0035	0.0032	0.0029	0.0027
2	0.0208	0.0194	0.018	0.0167	0.0156	0.0145	0.0134	0.0125	0.0116	0.0107
3	0.0492	0.0464	0.0438	0.0413	0.0389	0.0366	0.0345	0.0324	0.0305	0.0286
4	0.0874	0.0836	0.0799	0.0764	0.0729	0.0696	0.0663	0.0632	0.0602	0.0573
5	0.1241	0.1204	0.1167	0.113	0.1094	0.1057	0.1021	0.0986	0.0951	0.0916
6	0.1468	0.1445	0.142	0.1394	0.1367	0.1339	0.1311	0.1282	0.1252	0.1221
7	0.1489	0.1486	0.1481	0.1474	0.1465	0.1454	0.1442	0.1428	0.1413	0.1396
8	0.1321	0.1337	0.1351	0.1363	0.1373	0.1382	0.1388	0.1392	0.1395	0.1396
9	0.1042	0.107	0.1096	0.1121	0.1144	0.1167	0.1187	0.1207	0.1224	0.1241
10	0.074	0.077	0.08	0.0829	0.0858	0.0887	0.0914	0.0941	0.0967	0.0993
11	0.0478	0.0504	0.0531	0.0558	0.0585	0.0613	0.064	0.0667	0.0695	0.0722
12	0.0283	0.0303	0.0323	0.0344	0.0366	0.0388	0.0411	0.0434	0.0457	0.0481
13	0.0154	0.0168	0.0181	0.0196	0.0211	0.0227	0.0243	0.026	0.0278	0.0296
14	0.0078	0.0086	0.0095	0.0104	0.0113	0.0123	0.0134	0.0145	0.0157	0.0169
15	0.0037	0.0041	0.0046	0.0051	0.0057	0.0062	0.0069	0.0075	0.0083	0.009
16	0.0016	0.0019	0.0021	0.0024	0.0026	0.003	0.0033	0.0037	0.0041	0.0045
17	0.0007	0.0008	0.0009	0.001	0.0012	0.0013	0.0015	0.0017	0.0019	0.0021
18	0.0003	0.0003	0.0004	0.0004	0.0005	0.0006	0.0006	0.0007	0.0008	0.0009
19	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002	0.0003	0.0003	0.0003	0.0004
20	0	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002
21	0	0	0	0	0	0	0	0	0	0.0001

X	$\lambda = 8.1$	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	9.0
0	0.0003	0.0003	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0001	0.0001
1	0.0025	0.0023	0.0021	0.0019	0.0017	0.0016	0.0014	0.0013	0.0012	0.0011
2	0.01	0.0092	0.0086	0.0079	0.0074	0.0068	0.0063	0.0058	0.0054	0.005
3	0.0269	0.0252	0.0237	0.0222	0.0208	0.0195	0.0183	0.0171	0.016	0.015
4	0.0544	0.0517	0.0491	0.0466	0.0443	0.042	0.0398	0.0377	0.0357	0.0337
5	0.0882	0.0849	0.0816	0.0784	0.0752	0.0722	0.0692	0.0663	0.0635	0.0607
6	0.1191	0.116	0.1128	0.1097	0.1066	0.1034	0.1003	0.0972	0.0941	0.0911
7	0.1378	0.1358	0.1338	0.1317	0.1294	0.1271	0.1247	0.1222	0.1197	0.1171
8	0.1395	0.1392	0.1388	0.1382	0.1375	0.1366	0.1356	0.1344	0.1332	0.1318
9	0.1256	0.1269	0.128	0.129	0.1299	0.1306	0.1311	0.1315	0.1317	0.1318
10	0.1017	0.104	0.1063	0.1084	0.1104	0.1123	0.114	0.1157	0.1172	0.1186
11	0.0749	0.0776	0.0802	0.0828	0.0853	0.0878	0.0902	0.0925	0.0948	0.097
12	0.0505	0.053	0.0555	0.0579	0.0604	0.0629	0.0654	0.0679	0.0703	0.0728
13	0.0315	0.0334	0.0354	0.0374	0.0395	0.0416	0.0438	0.0459	0.0481	0.0504
14	0.0182	0.0196	0.021	0.0225	0.024	0.0256	0.0272	0.0289	0.0306	0.0324
15	0.0098	0.0107	0.0116	0.0126	0.0136	0.0147	0.0158	0.0169	0.0182	0.0194
16	0.005	0.0055	0.006	0.0066	0.0072	0.0079	0.0086	0.0093	0.0101	0.0109
17	0.0024	0.0026	0.0029	0.0033	0.0036	0.004	0.0044	0.0048	0.0053	0.0058
18	0.0011	0.0012	0.0014	0.0015	0.0017	0.0019	0.0021	0.0024	0.0026	0.0029
19	0.0005	0.0005	0.0006	0.0007	0.0008	0.0009	0.001	0.0011	0.0012	0.0014
20	0.0002	0.0002	0.0002	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0006
21	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002	0.0002	0.0003
22	0	0	0	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
X	$\lambda = 9.1$	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10.0
0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0
1	0.001	0.0009	0.0009	0.0008	0.0007	0.0007	0.0006	0.0005	0.0005	0.0005
2	0.0046	0.0043	0.004	0.0037	0.0034	0.0031	0.0029	0.0027	0.0025	0.0023
3	0.014	0.0131	0.0123	0.0115	0.0107	0.01	0.0093	0.0087	0.0081	0.0076
4	0.0319	0.0302	0.0285	0.0269	0.0254	0.024	0.0226	0.0213	0.0201	0.0189
5	0.0581	0.0555	0.053	0.0506	0.0483	0.046	0.0439	0.0418	0.0398	0.0378
6	0.0881	0.0851	0.0822	0.0793	0.0764	0.0736	0.0709	0.0682	0.0656	0.0631
7	0.1145	0.1118	0.1091	0.1064	0.1037	0.101	0.0982	0.0955	0.0928	0.0901
8	0.1302	0.1286	0.1269	0.1251	0.1232	0.1212	0.1191	0.117	0.1148	0.1126
9	0.1317	0.1315	0.1311	0.1306	0.13	0.1293	0.1284	0.1274	0.1263	0.1251
10	0.1198	0.121	0.1219	0.1228	0.1235	0.1241	0.1245	0.1249	0.125	0.1251
11	0.0991	0.1012	0.1031	0.1049	0.1067	0.1083	0.1098	0.1112	0.1125	0.1137
12	0.0752	0.0776	0.0799	0.0822	0.0844	0.0866	0.0888	0.0908	0.0928	0.0948
13	0.0526	0.0549	0.0572	0.0594	0.0617	0.064	0.0662	0.0685	0.0707	0.0729
14	0.0342	0.0361	0.038	0.0399	0.0419	0.0439	0.0459	0.0479	0.05	0.0521
15	0.0208	0.0221	0.0235	0.025	0.0265	0.0281	0.0297	0.0313	0.033	0.0347
16	0.0118	0.0127	0.0137	0.0147	0.0157	0.0168	0.018	0.0192	0.0204	0.0217
17	0.0063	0.0069	0.0075	0.0081	0.0088	0.0095	0.0103	0.0111	0.0119	0.0128
18	0.0032	0.0035	0.0039	0.0042	0.0046	0.0051	0.0055	0.006	0.0065	0.0071
19	0.0015	0.0017	0.0019	0.0021	0.0023	0.0026	0.0028	0.0031	0.0034	0.0037
20	0.0007	0.0008	0.0009	0.001	0.0011	0.0012	0.0014	0.0015	0.0017	0.0019
21	0.0003	0.0003	0.0004	0.0004	0.0005	0.0006	0.0006	0.0007	0.0008	0.0009
22	0.0001	0.0001	0.0002	0.0002	0.0002	0.0002	0.0003	0.0003	0.0004	0.0004
23	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002
24	0	0	0	0	0	0	0	0.0001	0.0001	0.0001
X	$\lambda = 11$	12	13	14	15	16	17	18	19	20.0
0	0	0	0	0	0	0	0	0	0	0
1	0.0002	0.0001	0	0	0	0	0	0	0	0
2	0.001	0.0004	0.0002	0.0001	0	0	0	0	0	0
3	0.0037	0.0018	0.0008	0.0004	0.0002	0.0001	0	0	0	0
4	0.0102	0.0053	0.0027	0.0013	0.0006	0.0003	0.0001	0.0001	0	0
5	0.0224	0.0127	0.007	0.0037	0.0019	0.001	0.0005	0.0002	0.0001	0.0001
6	0.0411	0.0255	0.0152	0.0087	0.0048	0.0026	0.0014	0.0007	0.0004	0.0002
7	0.0646	0.0437	0.0281	0.0174	0.0104	0.006	0.0034	0.0018	0.001	0.0005
8	0.0888	0.0655	0.0457	0.0304	0.0194	0.012	0.0072	0.0042	0.0024	0.0013
9	0.1085	0.0874	0.0661	0.0473	0.0324	0.0213	0.0135	0.0083	0.005	0.0029
10	0.1194	0.1048	0.0859	0.0663	0.0486	0.0341	0.023	0.015	0.0095	0.0058
11	0.1194	0.1144	0.1015	0.0844	0.0663	0.0496	0.0355	0.0245	0.0164	0.0106
12	0.1094	0.1144	0.1099	0.0984	0.0829	0.0661	0.0504	0.0368	0.0259	0.0176
13	0.0926	0.1056	0.1099	0.106	0.0956	0.0814	0.0658	0.0509	0.0378	0.0271
14	0.0728	0.0905	0.1021	0.106	0.1024	0.093	0.08	0.0655	0.0514	0.0387

Non-Parametric Tests

X	$\lambda = 11$	12	13	14	15	16	17	18	19	20.0
15	0.0534	0.0724	0.0885	0.0989	0.1024	0.0992	0.0906	0.0786	0.065	0.0516
16	0.0367	0.0543	0.0719	0.0866	0.096	0.0992	0.0963	0.0884	0.0772	0.0646
17	0.0237	0.0383	0.055	0.0713	0.0847	0.0934	0.0963	0.0936	0.0863	0.076
18	0.0145	0.0256	0.0397	0.0554	0.0706	0.083	0.0909	0.0936	0.0911	0.0844
19	0.0084	0.0161	0.0272	0.0409	0.0557	0.0699	0.0814	0.0887	0.0911	0.0888
20	0.0046	0.0097	0.0177	0.0286	0.0418	0.0559	0.0692	0.0798	0.0866	0.0888
21	0.0024	0.0055	0.0109	0.0191	0.0299	0.0426	0.056	0.0684	0.0783	0.0846
22	0.0012	0.003	0.0065	0.0121	0.0204	0.031	0.0433	0.056	0.0676	0.0769
23	0.0006	0.0016	0.0037	0.0074	0.0133	0.0216	0.032	0.0438	0.0559	0.0669
24	0.0003	0.0008	0.002	0.0043	0.0083	0.0144	0.0226	0.0328	0.0442	0.0557
25	0.0001	0.0004	0.001	0.0024	0.005	0.0092	0.0154	0.0237	0.0336	0.0446
26	0	0.0002	0.0005	0.0013	0.0029	0.0057	0.0101	0.0164	0.0246	0.0343
27	0	0.0001	0.0002	0.0007	0.0016	0.0034	0.0063	0.0109	0.0173	0.0254
28	0	0	0.0001	0.0003	0.0009	0.0019	0.0038	0.007	0.0117	0.0181
29	0	0	0.0001	0.0002	0.0004	0.0011	0.0023	0.0044	0.0077	0.0125
30	0	0	0	0.0001	0.0002	0.0006	0.0013	0.0026	0.0049	0.0083
31	0	0	0	0	0.0001	0.0003	0.0007	0.0015	0.003	0.0054
32	0	0	0	0	0.0001	0.0001	0.0004	0.0009	0.0018	0.0034
33	0	0	0	0	0	0.0001	0.0002	0.0005	0.001	0.002
34	0	0	0	0	0	0	0.0001	0.0002	0.0006	0.0012
35	0	0	0	0	0	0	0	0.0001	0.0003	0.0007
36	0	0	0	0	0	0	0	0.0001	0.0002	0.0004
37	0	0	0	0	0	0	0	0	0.0001	0.0002
38	0	0	0	0	0	0	0	0	0	0.0001
39	0	0	0	0	0	0	0	0	0	0.0001